

MATHEMATICAL METHODS
OF GAME
AND ECONOMIC THEORY

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MASSACHUSETTS INSTITUTE OF TECHNOLOGY

OXFORD UNIVERSITY PRESS, NEW YORK, 1950

OXFORD UNIVERSITY PRESS, LONDON, 1950

OXFORD UNIVERSITY PRESS, NEW DELHI, 1950

OXFORD UNIVERSITY PRESS, CALCUTTA, 1950

OXFORD UNIVERSITY PRESS, SINGAPORE, 1950

OXFORD UNIVERSITY PRESS, KARACHI, 1950

OXFORD UNIVERSITY PRESS, RANGOON, 1950

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OXFORD UNIVERSITY PRESS, PUNE, 1950

OXFORD UNIVERSITY PRESS, RAIPUR, 1950

OXFORD UNIVERSITY PRESS, DELHI, 1950

OXFORD UNIVERSITY PRESS, LUCKNOW, 1950

OXFORD UNIVERSITY PRESS, AGRA, 1950

OXFORD UNIVERSITY PRESS, JAIPUR, 1950

OXFORD UNIVERSITY PRESS, UJAIN, 1950

STUDIES IN MATHEMATICS AND ITS APPLICATIONS

VOLUME 7

Editors:

J. L. LIONS, *Paris*

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發行人：廖文遠

出版者：楓城出版社

新竹市武昌街42號

劃撥帳號104063

發行所：台北市和平東路二段113號2樓

電話：(02)7036554

印刷所：聯和印製廠

行政院新聞局登記證局版台業字第1322號

中華民國七十二年六月

MATHEMATICAL METHODS OF GAME AND ECONOMIC THEORY

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1979

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North-Holland ISBN 0 444 85184 4

Published by:

NORTH-HOLLAND PUBLISHING COMPANY
AMSTERDAM · NEW YORK · OXFORD

Sole distributors for the U.S.A. and Canada:

Elsevier North-Holland, Inc.
52 Vanderbilt Avenue
New York, NY 10017

Library of Congress Cataloging in Publication Data

Aubin, Jean-Pierre.
Mathematical methods of game and economic theory.

(Studies in mathematics and its applications; v. 7)

Bibliography: p. 590

Includes index.

1. Game theory. 2. Economics, Mathematical.
3. Functional analysis. 4. Mathematical optimization.

I. Title. II. Series.

QA269.A87 515'.7 78-18762

ISBN 0-444-85184-4

PRINTED IN HUNGARY

This book is dedicated to N.

PREFACE

It is said that the preface is that part of a book written last, placed first and read least. I do hope, however, that the reader will glance through it to allay any fears that might otherwise be raised by the length of this book. The book has been written with two audiences in mind.

(a) *Mathematical economists and operations research specialists.* Workers in these areas will find that the book provides a solid foundation in non-linear functional analysis at a level just beyond that attained in most mathematical texts intended for economists.

(b) *Mathematicians.* Mathematicians with an interest in non-linear functional analysis will find that the applications offered in optimization, game theory and mathematical economics provide valuable insight into the general structure of the theory.

Non-linear functional analysis is the central theme of this book. It not only provides powerful and versatile tools for solving specific problems in economics and the social sciences, but also serves as a unifying theme in the mathematical theory of these subjects as well as in pure mathematics itself.

It will be clear that this is a *book of mathematics* in which formal definitions are given and theorems are proved. However, the interpretation of these theorems in various contexts is given equal weight. The central application treated is *the fundamental economic problem of allocating scarce resources among competing agents*. This leads us to consider simultaneously the interrelated applications in *game theory* and the *theory of optimization*. No attempt is made in this book at a critical appraisal of the behavioural assumptions implicit in the mathematical models considered nor is any attempt made to justify their importance as objects of study. In particular, no pretence is made that the models studied are immediately applicable and, in many cases, the level of idealization involved in the models is very high indeed. Readers interested in these interpretive questions will have to look elsewhere. Fortunately there are many good texts which study these problems.

Applications are introduced as early as possible in order to stimulate the reader and to motivate the theorems. There would have been advantages in

gathering the applications together in one place, but I felt that a lively treatment was to be preferred. In the beginning, of course, the applications offered tend to be rather flimsy since they inevitably consist essentially of semantic re-interpretations of mathematical definitions. In spite of this, they can still often be of considerable help to one's intuition. As the book proceeds, the applications given naturally become deeper and of more interest.

The reader will find that much of the mathematics in the book will be accessible without the necessity of having a sophisticated mathematical background. However, certain more difficult passages have been marked with an asterisk (*) and may be omitted at a first reading.

The book is organized into three parts. The first part constitutes a course in optimisation theory and convex analysis. The second covers a number of topics in game theory and mathematical economics. The third part provides an introduction to non-linear analysis and control theory.

Part I (Chapters 1–5)

Linear and convex analysis is developed in the framework of optimization theory. The treatment includes results on the existence and stability of solutions to optimization problems as well as an introduction to duality theory. Numerical results and solution algorithms are not discussed at all.

Part II (Chapters 6–12)

Two-person games are considered first. These prove to be the right framework within which to study the later theorems of non-linear analysis. The treatment continues with Walrasian models for the allocation of scarce resources and with an introduction to n -person non-cooperative games.

Most (but not all) important solution concepts for n -person cooperative games are considered in conjunction with their use in Walrasian models. Of course, many interesting ideas have had to be omitted.

Part III (Chapters 13–15)

In this part, more advanced issues are considered from a primarily mathematical point of view. A class of functions is introduced which shares the properties of the ordinary convex functions. The treatment continues with an introduction to monotone and pseudo-monotone operators. Duality theory is extended to the case of infinite-dimensional spaces and this allows a short introduction to the calculus of variations and optimal control theory. Impulsive control theory, introduced for solving inventory problems, is also briefly discussed for the deterministic case. In the last chapter, an array of various fixed point and surjectivity theorems are given. These are powerful tools in proving existence theorems. The chapter continues with a short account

of quasi-variational inequalities and concludes with some further results on correspondences (i.e. "multi-valued functions").

In developing the material presented in this book, I have benefited from the active collaboration of my friends at the Centre de Recherches de Mathématiques de la Décision de l'Université Paris-Dauphine. I am particularly grateful to Ivar Ekeland (who introduced me to game theory), Alain Bensoussan, Frank Clarke, Bernard Cornet, Jean-Michel Lasry and Hervé Moulin. I also wish to thank the Mathematics Research Center of the University of Wisconsin for the support it has granted me during the last six summers and, in particular, to Richard H. Day (who introduced me to economics). I would also like to point out that the book owes much to the students of the Université Paris-Dauphine and the Ecole Polytechnique whose reactions helped me in improving its pedagogical aspects. Finally it is my pleasure to thank Ken Binmore for correcting the manuscript and thereby drastically improving the text.

SUMMARY OF RESULTS: A GUIDELINE FOR THE READER

We summarize the main concepts and results presented in this book. Of course, this summary is not really meant to be understood all at once, but rather to supply the reader with guidelines and to help him locate the main results in their true perspective. We hope that this summary will provide a rough idea of what this book is about and also that it will indicate the place in the book where a given result is presented.

We have tried to compose the chapters in such a way that they can be read independently of each other in so far as this is possible.

Part I. *Optimization and Convex Analysis*

We devote the first part of this book to the study of optimization problems (i.e. one-player games) of the form

$$\alpha = \inf_{x \in X} f(x),$$

where X represents the strategy set and f the loss function. These together describe the behavior of a player. Most of the results presented in this first part are used later on.

(1) We begin by studying the *existence* of a minimal solution. This holds whenever we assume that f is lower semi-continuous (i.e. has closed lower sections) and lower semi-compact (i.e. has relatively compact lower sections) (see Section 2.1). We devote the third chapter to a rather comprehensive study of these required continuity and compactness properties. The important particular case of quadratic loss functions on Hilbert spaces is studied in Sections 2.2 and 2.3.

(2) We continue by studying *stability* properties, i.e. the smoothness of the behavior of the minimal value

$$\alpha(y) = \inf_{x \in X(y)} f(x, y)$$

when the loss function $f(\cdot, y)$ and/or the strategy set $X(y)$ depend upon perturbations y . The most important example is obtained when we perturb a loss function f defined on a subset X of a topological vector space U by the “simplest” functions, which are the continuous linear functions $p \in U^*$. The function f^* defined on U^* by

$$-f^*(p) = \inf_{x \in X} [f(x) - \langle p, x \rangle]$$

is obviously a convex lower semi-continuous function. It is called the “conjugate” function of f . Convex lower semi-continuous functions are precisely those for which $f = f^{**}$ (see Section 2) and, consequently, this class of functions plays an important role throughout this book.

More generally, we show in Section 4.9 that smoothness requirements on the function $y \rightarrow \alpha(y) = \inf_{x \in X} f(x, y)$ guarantee the existence and uniqueness of a minimal solution of the minimization problem $\alpha(y)$ for “almost all” perturbations y . This approach allows us to study a whole family of problems rather than a (possibly unstable) problem. When we allow the strategy set to depend upon y , we are led to define and study continuity properties of *correspondences* $y \rightarrow X(y)$. We initiate this study in Section 2.5 and we continue it in Section 15.3. When the strategy sets $X(y)$ are represented by families of inequality constraints, the study of stability properties leads to the “duality theory” for minimization problems.

✓(3) We begin to treat *duality theory* in Chapter 5 (in the case of finite dimensional spaces) and we continue its study in Chapter 14 (in the case of infinite dimensional spaces). Roughly speaking, we can devise dual problems whenever a minimization problem $\alpha = \inf_{x \in X} f(x)$ can be written in the form

$$\alpha = \inf_{x \in U} \sup_{p \in V^*} l(x, p)$$

where l is a function defined on $U \times V^*$ called the “Lagrangian”. The question arises as to whether it is possible to find $\bar{p} \in V^*$ (called a Lagrange multiplier) such that the initial problem is equivalent to the simpler problem $\alpha = \inf_{x \in U} l(x, \bar{p})$. The dual problem amounts to finding such Lagrange multipliers.

(4) It is also useful to ~~characterize~~ minimal solutions. This will be done in Chapter 4, by using several different concepts of *differentiability*. For instance, we prove that $\bar{x}$ is a minimal solution if and only if $f(\bar{x}) = f^{**}(\bar{x})$ and $\bar{x} \in \partial f^*(0)$, where $\partial f^*(p)$ is the “subdifferential of f^* at p ” defined by $\partial f^*(p) = \{x \in U \text{ such that } f^*(p) - f^*(q) \leq \langle p - q, x \rangle \text{ for all } q\}$. We devote Section 4.1 to subdifferentiability and show in Section 4.2 and 4.3 how it is related to the usual concepts of differentiability (convexity will play an important role).

In particular we prove the following “variational principle”. If f is differentiable on a convex subset X and if $\bar{x}$ is a minimal solution, then

$$\langle Df(\bar{x}), \bar{x} - y \rangle \leq 0 \quad \text{for any } y \in X.$$

Such problems are called “variational inequalities”. (They will be solved in Section 13.2.)

(5) *Convexity* plays a fundamental role in solving the problems mentioned above. The reason lies primarily in the fact that we implicitly or explicitly perturb minimization problems on topological vector spaces by continuous linear functionals (since the pointwise supremum of linear functions is convex). A secondary reason is that convexity allows the use of the Hahn–Banach separation theorem. We prove the fundamental results of convex analysis (in particular, the fundamental minimax theorem) by using the separation theorem together with the following fundamental property of families of convex functions: if $f_1, \dots, f_n$ are n convex functions defined on a convex set X and if we set $F(x) = \{f_1(x), \dots, f_n(x)\} \in \mathbf{R}^n$, then $F(X) + \mathbf{R}_+^n$ is convex.

In Sections 1.3 and 1.4 we recall elementary properties of convex sets and functions and we devote Section 13.3 to the study of convex cones of functions isomorphic to cones of convex functions. The characterization of convex lower semi-continuous functions is presented in Section 2.4. The minimax theorem for convex-concave functions is proved in Section 7.1.1, Section 7.1.7, Section 7.1.8 and Section 7.1.9 and generalized in Section 13.1.

Part II. Game Theory and the Walras Model of Allocation of Resources

The second part of this book deals with game theory. For simplicity, we present in Chapter 6 the main solution concepts in the case of two-person games and illustrate them with some examples (finite games, Cournot’s duopoly and Edgeworth’s economic game).

(1) In Chapter 7 we study two-person zero-sum games. We state and prove two fundamental results, which will be used as basic tools in the rest of the book.

Theorem A (minisup theorem). *If X is a convex compact subset, Y is a convex subset and $f: X \times Y \rightarrow \mathbf{R}$ is convex and lower semi-continuous with respect to x and concave with respect to y , then there exists $\bar{x} \in X$ such that*

$$\sup_{y \in Y} f(\bar{x}, y) = \sup_{y \in Y} \inf_{x \in X} f(x, y).$$

Note that this implies that $\sup_{y \in Y} \inf_{x \in X} f(x, y) = \inf_{x \in X} \sup_{y \in Y} f(x, y)$ and also the Von Neumann theorem for the existence of saddle points.

Theorem B (Ky-Fan's inequality). *If X is a convex compact subset and if $f: X \times X \rightarrow \mathbb{R}$ is lower semi-continuous with respect to X and concave with respect to y , then there exists $\bar{x} \in X$ such that*

$$\sup_{y \in X} f(\bar{x}, y) \leq \sup_{y \in X} f(y, y).$$

Theorem B is actually equivalent to the Brouwer fixed point theorem which asserts that any continuous map from a convex compact subset into itself has a fixed point. But it happens that the Ky-Fan inequality is much more useful and versatile as a tool for proving the main results of game theory and non-convex analysis. We use *the Ky-Fan inequality systematically in proving all subsequent existence theorems.*

These results are proved in Section 7.1 and are extended and improved in Chapter 13. The rest of Chapter 7 deals with other issues concerning two-person zero-sum games.

(2) We devote Chapter 8 to the description of the Walras model of allocation of scarce commodities among competing consumers and to the proof of the Debreu theorem for the existence of a Walras equilibrium (by using the Ky-Fan inequality). This model will be used as the main example of an n -person game.

(3) In Chapter 9, we study the non-cooperative concept of equilibrium and prove the Nash theorem on the existence of a non-cooperative equilibrium. More generally, we prove the Arrow-Debreu-Nash theorem in the more general case when the strategy set of each player i depends upon the choice of the multi-strategy implemented by the coalition $\bar{i}$ of the other players j .

Again, these theorems are deduced from the Ky-Fan inequality. As a by-product, we also prove the Brouwer and Kakutani fixed-point theorems (in Section 9.3). We devote Section 9.2 to the case of quadratic loss functions, in which we need only elementary results. We apply these results to obtain the existence of equilibria in economic models (Section 9.2.5 and Section 9.4).

(4) Chapters 10, 11 and 12 deal with the *cooperative* concepts of n -person games. The ideas are introduced in Chapter 10. Difficult results are proved in Chapter 11 (for games with side-payments) and in Chapter 12 (for games without side-payments). In the case of non-cooperative games, we assume that the players do not "communicate" among themselves. In cooperative games, the players can participate in coalitions. Such participation will, of course, influence their behavior.

In other words, cooperative games are described by the behavior of a given family of coalitions instead of by the behavior of the n players regarded as individuals. The main solution concept is the *core* of the game. To define it, we will describe how a coalition *rejects* a strategy. The core of the game is then the subset of strategies which are not rejected by any coalition which is allowed to form. If the only coalition allowed to form is the whole set of players, the core is called the set of *Pareto minima*. The larger the family of coalitions allowed to form, the smaller is the core. In order to have a core as small as possible, we extend the family of ordinary coalitions of players (described as subsets A of the set N of players) to the family of "fuzzy coalitions" (described as fuzzy subsets τ of N). In other words, a fuzzy coalition τ is a vector $\tau = \{\tau_1, \dots, \tau_i, \dots, \tau_n\}$ of $[0, 1]^n$ where $\tau_i \in [0, 1]$ represents the rate of participation of the i^{th} player in the fuzzy coalition. (For conventional coalitions, the rate of participation τ_i is either 0 or 1.) By describing the behavior of fuzzy coalitions, we define the concept of fuzzy core of a game. We prove that the fuzzy core is non-empty under reasonable assumptions (again using the Ky-Fan inequality). Furthermore, we prove that, in the case of games with side-payments, the fuzzy core coincides with the "fuzzy value". This is another "solution concept" defined and studied in Sections 11.3 and 11.4.

In the case of games *without* side payments, we prove that the fuzzy core coincides with the set of "cooperative equilibria" defined in Chapter 12. In particular, we associate a cooperative game with the Walras model of allocation of scarce resources and define the fuzzy core of an economy as the set of allocations which are not "rejected" by any coalition. We prove that this fuzzy core coincides with the set of Walras equilibria. This result allows an evaluation of the concept of a Walras equilibrium from a normative point of view. (We will not study the alternative approach in which, instead of a finite number of players forming fuzzy coalitions, one considers a continuum of players forming only ordinary coalitions.)

Part III. *Non-linear Analysis and Optimal Control Theory*

As we have mentioned in the preface, the last three chapters are devoted to improving and completing earlier results.

(1) Chapter 13 presents a rather comprehensive study of minimax type inequalities. We use them to define pseudo-monotone and monotone operators and to study associated variational inequalities. In the last section of this chapter, we characterize classes of functions which share the same properties as convex functions.

(2) Chapter 14 is devoted to duality theory in infinite dimensional spaces. In the first section, we extend the duality theory devised in Chapter 5. Section 14.2 devoted to minimization problems for (non-convex) integral criteria. Section 14.3 is a short introduction to the calculus of variations in the framework of convex analysis and to the Pontriagin principle for the optimal control problem. In Section 14.4 we present a short introduction to the dynamic programming approach for optimal control problems as well as to stopping time and impulsive control problems.

We devote the first section of Chapter 15 to the proof of a collection of fixed point and surjectivity theorems using the Ky-Fan inequality. We begin by proving the existence of critical points x , i.e. solutions x of multivalued equations $0 \in S(x)$, where S maps a space X into a vector space U . In section 15.2 we study quasi-variational inequalities. We continue this chapter by giving examples of lower semi-continuous correspondences. Finally, we prove the Michael continuous selection theorem and characterize semi-continuous correspondences with convex images.

CONTENTS OF OTHER POSSIBLE COURSES

A) *Convex analysis*

- (a) Optimization and convexity: Chapter 1.
- (b) Continuity of convex functions: Sections 2.1, 2.4, 3.3 and 3.4.
- (c) Differentiability of convex functions: Sections 4.1, 4.2 and 4.3.
- (d) Minimax Theorem: Sections 13.1.1 and 13.1.2: Existence of a conservative solution; Sections 7.1.7, 7.1.8 and 7.1.9: The minisup theorem; Section 6.5.3: perturbation by linear functions; Section 13.1.3: Existence of a minisup under weaker compactness assumptions.
- (e) Duality for optimization problems: Sections 5.1, 5.2 and 14.1.
- (f) γ -convex functions: Section 13.3.
- (g) Applications: Sections 14.2 and 14.3.

(B) *Correspondences*

- (a) Upper semi-continuous correspondences: Sections 2.5.1, 4.1.4, 2.5.3 and 2.5.4.
- (b) Lower semi-continuous correspondences: Sections 2.4.2, 15.3.1, 15.3.2, and 15.3.3.
- (c) Semi- and hemi-continuity: Sections 2.5.1, 15.1.4 and 15.1.5.
- (d) Monotone correspondences: Sections 13.2.6 and 13.2.7.

(C) *Non-convex analysis* (prerequisite: course B: correspondences)

- (a) The Ky-Fan inequality: Sections 13.1.1, 13.1.2, 7.1.2, 7.1.3, 7.1.5, 7.1.6, 13.2.1 and 13.2.3.
- (b) Existence of critical points: Sections 15.1.1, 15.1.3 and 15.1.4.
- (c) Fixed point and surjectivity theorems: Section 15.1.2.
- (d) Variational inequalities: Sections 13.2.4 and 13.2.5.
- (e) Quasi-variational inequalities: Sections 9.3.1 and 15.2.

Table of economic illustrations and applications

Production sets, profit, cost and production functions: Sections 1.4.5, 1.4.6 and 2.4.5.

Marginal profit: Sections 4.2.5 and 5.1.4.

Analysis of the duopoly and oligopoly: Sections 6.2, 9.2.4 and 9.2.5. The Edgeworth box: Section 6.4.

The Walras model: Chapter 8, Sections 9.4, 10.3.5, 10.4.7, 11.4.1 and 12.3.

NOTATIONS

(1) The reader who prefers to avoid the intricacies of infinite dimensional spaces can assume in the two first parts that the vector spaces are finite dimensional. Most of the examples used in economic theory or game theory are, of course, finite dimensional spaces. Otherwise, Appendix A provides a short summary of what one needs to know in linear functional analysis.

We denote by $\mathbf{R}_+^n$ the cone of *non-negative* vectors $a = \{a_1, \dots, a_n\}$ (i.e. satisfying $a_i \geq 0 \ \forall i$), by $\mathbf{R}_{0+}^n$ the cone of *positive* vectors a (i.e. satisfying $a \in \mathbf{R}_+^n$ and $a \neq 0$) and by $\dot{\mathbf{R}}^n$ the cone of *strongly positive* vectors a (i.e. satisfying $a_i > 0$ for all i).

We set $a \geq b$ if $a - b \in \mathbf{R}_+^n$ (a is *not smaller* than b), $a > b$ if $a - b \in \mathbf{R}_{0+}^n$ (a is *greater* than b) and $a \gg b$ if $a - b \in \dot{\mathbf{R}}_+^n$ (a is *strongly greater* than b). If J is a subset of $\mathbf{R}^n$, we set

$$A_+ = A + \mathbf{R}_+^n \quad \text{and} \quad \dot{A}_+ = A + \dot{\mathbf{R}}_+^n.$$

We denote by $\mathbf{R}^* = \mathcal{L}(\mathbf{R}^n, \mathbf{R})$ the dual of $\mathbf{R}^n$ and by $\mathbf{R}_+^*$ and $\dot{\mathbf{R}}_+^*$ the cone of non-negative and strongly positive linear forms.

The duality pairing on $\mathbf{R}^{n*} \times \mathbf{R}^n$ is denoted by $\langle p, x \rangle = \sum_{i=1}^n p^i x_i$.

(2) A *correspondence* (or *multivalued map*) from X into Y associates with any $x \in X$ a subset $S(x)$ of Y , called an image or value of the correspondence S .

(3) If a topological space is not metric (or not metrizable), we have to replace the use of *sequences* $\{x_n\}_{n \in \mathbf{N}}$ of elements x_n of X by the use of *generalized sequences* (or *nets*) $\{x_\mu\}_{\mu \in \mathcal{M}}$ of elements x_μ of X . These are defined as maps $\mu \in M \mapsto x_\mu \in X$, where $\mathcal{M}$ is a set of *indices* with a preorder $\geq$ such that we can associate with any pair $\{\mu_1, \mu_2\}$ another μ such that $\mu \geq \max(\mu_1, \mu_2)$.

If $\mathcal{N}$ is another set of indices, a *generalized subsequence* $\{x_{\mu_\nu}\}_{\nu \in \mathcal{N}}$ of the generalized sequence is the map $\nu \mapsto x_{\mu_\nu}$ where the map $\nu \mapsto \mu_\nu$ satisfies the property: $\forall \mu, \exists \nu_0$ such that $\nu \geq \nu_0$ implies $\mu_\nu \geq \mu$.

Recall that compact spaces are those with the property that we can extract from any generalized sequence a *convergent generalized subsequence*.

(4) We list below other symbols we shall use.

A, B	coalitions (or finite subsets)
$ A $	number of elements of the coalition A
C, D	decision rules
U, V, W, E, F, G	vector spaces
J	duality operator
L, M	linear operators
$N = \{1, 2, \dots, n\}$	set of players
P, Q, R	subsets
S, T	correspondences
X, Y, Z	strategy or commodity sets
a, b, c, d	multilosses
f, g, h	loss functions
$F = \{f_1, \dots, f_n\}$	multiloss operator
l	Lagrangian
p, q, r	linear functionals
x, y, z	strategies or commodities
$\langle p, x \rangle$	duality pairing
d	semi-distance
p	semi-norm
$\mathcal{A}$	family of coalitions
$\mathcal{C}, \mathcal{D}$	families of decision rules
$\mathcal{C}(X)$	space of continuous functions
$\mathcal{F}(X, Y)$	space of maps from X into Y
$\mathcal{L}(U, V)$	space of continuous linear operators from U into V
$\mathcal{M}(X)$	subset of discrete probabilities on X
$\mathcal{S}$	family of finite subsets,
$\mathcal{S}(X) = \mathbf{R}^X(\mathcal{S}_+(X) = \mathbf{R}_+^X)$	space (cone) of all (positive) functions on X
$\mathcal{T}$	family of fuzzy coalitions
$\mathcal{U}(X)$	space of bounded functions
L^p	Lebesgue space
$\beta(U, F), \tau(U, F), \sigma(U, F)$	strong, Mackey and weak topologies
β	barycentric operator
δ	Dirac operator
$\gamma(x, p)$	constraints
τ, σ	fuzzy coalitions
μ, ν	measures or linear functionals on functions spaces

$\phi(\tau)$	fuzzy value (Shapley value)
$\pi(X, \cdot)$	gauge of X
$\sigma^{\#}(X; \cdot), \sigma^b(X; \cdot)$	upper and lower support functions of X
$\psi(X; \cdot)$	indicator of X
∂	subdifferential
U^*	topological dual of U
L^*	transpose of L
f^*	conjugate function of f
$f^{\#}(X) = \sup_{y \in Y} f(x, y)$	(f -sharp)
$v^{\#} = \inf_{x \in X} \sup_{y \in Y} f(x, y)$	
$f_b(y) = \inf_{x \in X} f(x, y)$	(f -flat)
$v^b = \sup_{y \in Y} \inf_{x \in X} f(x, y)$	
$v^{\square} = \sup_{K \in \mathcal{K}} \inf_{x \in X} \sup_{y \in K} f(x, y)$	
f^{i*}	loss function of the dual problem
f^{Δ}	linear extension of f
$S(f, \lambda)$	lower section of f
$Df(x)$	gradient of f
$M^b(f) (M^{\#}(f))$	minimal (maximal) set of f
$L f, g \ L$	product of f by L
$f \square g$	inf convolution of f and g
$\text{co}(X), \overline{\text{co}}(X), \overline{\text{cos}}(X)$	(convex, closed convex, symmetric closed convex hull)
$P(X)$	cone spanned by X
$P_{\infty}(X)$	recession cone of X
$X^{\#}(X^b)$	upper (lower) polar set of X
$X^+(X^-)$	positive (negative) polar cone of X
$X^{\perp}$	orthogonal (annihilator) of X
$\hat{A} = [A \text{ complement of } A, \text{adverse coalition}, i = N - i]$	
$\bar{A} = \text{closure of } A, \hat{A} \text{ interior of } A$	
$A_+ = A + \mathbf{R}_+^n, \hat{A}_+ = A + \hat{\mathbf{R}}_+^n \text{ if } A \subset \mathbf{R}^n$	
$X^A = \prod_{i \in A} X^i, X^N = \prod_{i \in N} X^i = X^A \times X^{\hat{A}} = X^i \times X^i$	
$X(A)$	strategy set of a coalition A
$F^A = \{f_1^A, \dots, f_n^A\}$	multiloss operator of a coalition A
$\{X(N), F\}$	game described in strategic form
$\{X(A), F^A\}_{A \in \mathcal{A}}$	cooperative game described in strategic form
$(\mathcal{A}, J) [\mathcal{A}, v]$	cooperative game (with side-payments) described in characteristic form

$(\mathcal{C}, J) [(\mathcal{C}, v)]$	fuzzy cooperative game (without side payment)
$\mathcal{C}(\mathcal{A}, J) [\mathcal{C}(\mathcal{T}, J)]$	core of a game (of a fuzzy game)
$\{R^i, Y(i), f_i\}_{i \in N}$	(exchange) economy
$\{R^i, f_p, w^i, \theta_j^i, g_j\}_i$	economy with producers

CHAPTER 1

MINIMIZATION PROBLEMS AND CONVEXITY

This chapter is largely concerned with generalities about optimization problems and includes an introduction to convexity. In Section 1.1 we recall the general form of an optimization problem $\inf_{x \in X} f(x)$ and give some examples of strategy sets X and loss functions f .

In Section 1.2 we introduce the “product from the left” of a function. We use the notation Lf for this concept and define.

$$(Lf)(y) = \inf_{Lx=y} f(x).$$

We also discuss the related concept of the inf-convolution of functions. The next topic is the identity

$$\inf_{Lx \in Y} [f(x) + g(Lx)] = \inf_{y \in Y} [Lf(y) + g(y)]$$

which yields some useful general decomposition principles for optimization problems.

It will come as no surprise that *convexity plays a key role throughout this book*. In Section 1.3 we motivate the introduction of convexity ideas by drawing attention to the usefulness of replacing an initial minimization problem

$$\alpha = \inf_{x \in X} f(x)$$

by an *extended minimization* problem

$$\alpha = \inf_{m \in \mathcal{M}(X)} f^A(m),$$

in which X is embedded in the convex subset $\mathcal{M}(X)$ of probability measures (called “*mixed strategies*”) and where $f^A(m)$ is the corresponding expected loss.

In defining convex functions by the usual barycentric inequalities, we observe that the initial and the extended minimization problems are equivalent for convex functions. In the remainder of Section 1.3 we state some elementary properties of convex sets and functions. In Section 1.4, we introduce the *indicator* function ψ_X , the *support* function $\sigma_X^\#$ and the *gauge* function π_X for a closed

convex subset $X \subset U$ which contains 0. The indicator function is defined by $\psi_X(x) = 0$ if $x \in X$ and $+\infty$ otherwise. The support function σ_X^* is defined on the dual U^* by $\sigma_X^*(p) = \sup_{x \in X} \langle p, x \rangle$. The gauge function π_X is defined by $\pi_X(x) = \inf \{ \lambda > 0 \text{ such that } \lambda^{-1}x \in X \}$. These three functions are all convex. The indicator and the gauge characterize X in the sense that

$$X = \{x \in U \text{ such that } \psi_X(x) = 0\}$$

$$\text{and } X = \{x \in U \text{ such that } \pi_X(x) \leq 1\}.$$

In Section 1.4.2, we reformulate the fundamental Hahn–Banach theorem as the assertion that $X = \{x \in U \text{ such that } \langle p, x \rangle \leq \sigma_X^*(p) \text{ for all } p \in U^*\}$. This representation of a closed convex set will play a fundamental role at a later stage. Finally, we mention an economic interpretation of these concepts in the framework of production sets.

1.1. Strategy sets and loss functions

We shall define a minimization problem $\alpha = \inf_{x \in X} f(x)$ in which X is regarded as a strategy set and f as a loss function. In economic models, X often describes a set of available commodities. The minimization problem can then be viewed as a method for selecting a commodity from those. Usually such minimization problems are given more explicitly. We shall comment on the following special case in which $X = \{x \in U \text{ such that } Lx \in Y\}$ (where U and V are vector spaces, $L \in \mathcal{L}(U, V)$ and $Y \subset V$) and $f = gM$ where $M \in \mathcal{L}(U, W)$ and g is defined W .

For the sake of simplicity, we shall allow loss functions to take the value $+\infty$. If $f: U \mapsto]-\infty, +\infty]$ and $X = \text{Dom } f = \{x \in U \text{ such that } f(x) < +\infty\}$, the restriction to X is an ordinary function from X into $\mathbb{R}$. In other words, a loss function from U into $]-\infty, +\infty]$ involves the description of both the strategy set X and the actual loss function f .

We end this section by recalling the definition of sections and epigraphs of functions.

1.1.1. Optimization problem

We begin by considering the case of a single player game (or a single decision-maker decision problem). In fact a player can represent a team of players. In this case, the “normative rule” for selecting a strategy is to solve an optimization problem defined by the following items:

(a) a set U , representing a set of *unconstrained strategies* (or *unconstrained decisions*).

(b) a subset X of U , representing a set of (feasible) *strategies* (or *decisions*). In many cases, X is defined by a set of constraints.

(c) a function $f : X \rightarrow \mathbf{R}$, associating with a strategy (or decision) x a *loss* $f(x)$ (or a *cost* or *disutility*). We call $-f(x)$ a *payoff* (or a *profit* or *utility*).

Thus, the *optimization problem* amounts to selecting a strategy $\bar{x} \in X$ which minimizes the loss function $f(x)$ as x ranges over the strategy set X .

In other words, we have to find $\bar{x}$ such that

$$(1) \quad \begin{cases} (i) \ \bar{x} \in X, \\ (ii) \ f(\bar{x}) = \inf_{x \in X} f(x). \end{cases}$$

We shall say that $\bar{x}$ is an *optimal strategy* (or *decision*). It is clear that such a general optimization problem can “model” many instances. We mentioned the vocabulary used in (parlor) games or in optimal decision problems, but the implications range far beyond, to mathematics, economics, politics, etc.

1.1.2. Allocation of available commodities

In particular, we consider the *basic problem of economics* by which we mean the *problem of allocating available commodities among competing ends*. This may be modelled in various ways of which we give some examples. In these examples

- U describes the set of “commodities”,
- $Y \subset U$ describes the set of “available commodities”.

An optimization problem can be viewed in this context as a *selection procedure for choosing a commodity* from those available which minimizes a given loss function f (provided such minimizing commodities exist).

Example. Commodity space $\mathbf{R}^l$. In most examples, we describe the *commodity space* U by a finite dimensional space $\mathbf{R}^l$. If the l goods are labelled by $i = 1, \dots, l$, a *commodity vector* $x = \{x_1, \dots, x_l\} \in \mathbf{R}^l$ represents x_1 units of the first good, $\dots$, x_l units of the l^{th} good.

The elements $e^i = \{0, \dots, 0, 1, 0, \dots, 0\}$ of the canonical basis of $\mathbf{R}$ are the “unitary commodities”, since e^i represents one unit of the i^{th} good. Note that we can write the commodity vector $x = \sum_{i=1}^l x_i e^i$ as the sum of x_1 units of the first unitary commodity, $\dots$, of x_l units of the l^{th} unitary commodity.

In most examples, the real line represents a set of values, where the basis $\{1\}$ represents the “unit of account” of the model (for instance, dollar, franc, $\dots$).

Thus the dual $\mathbf{R}^{I*}$ of $\mathbf{R}^I$ can be regarded as the space of *prices* p associating (linearly) with any commodity $x \in \mathbf{R}^I$ its value $\langle p, x \rangle = p(x)$.

The dual canonical basis of $\mathbf{R}^{I*}$ is spanned by the forms e_i^* defined by

$$(2) \quad e_i^*(e^k) = \begin{cases} 0 & \text{if } k \neq i, \\ 1 & \text{if } k = i. \end{cases}$$

For any $x \in \mathbf{R}$, $e_i^*(x) = x_i$. Any $p \in \mathbf{R}^{I*}$ can be written

$$(3) \quad p = \sum_{i=1}^I p^i e_i^* \quad \text{where } p^i = \langle p, e^i \rangle.$$

In other words, the i^{th} component p^i denotes the value of one unit of the i^{th} good, i.e. what is usually called “the price of the i^{th} good”, or “unitary price the i^{th} good”.

The duality pairing can be written

$$(4) \quad \langle p, x \rangle = p(x) = \sum_{i=1}^I p^i x_i.$$

This formula speaks for itself.

1.1.3. Resource and service operators

A more explicit general form for an optimization problem can be described in the following way.

We introduce the following sets:

- (a) the *unconstrained strategy set* U ;
- (b) a set V of *resources*;
- (c) a set W of *services* (or “outcomes”, “attributes”, “acts”, “criteria”, “characteristics”, “observations”, etc.);
- (d) the *resource operator* $L : U \rightarrow V$ which associates with any strategy $x \in U$ the resource $L(x) \in V$ needed to implement x ;
- (e) the *service operator* $M : U \rightarrow W$ which associates with any strategy $x \in U$ a “service” $M(x) \in W$ produced by x ;
- (f) a subset $Y \subset V$ of “available resources” (for instance, Y is the set of resources smaller than or equal to y for a convenient ordering);
- (g) a function d mapping W into $\mathbf{R}$ which associates with any service $v \in W$ a “distance” $d(v, u)$ to an “objective” $u \in W$.

We then define the strategy set X by

$$X = \{x \in U \text{ such that } L(x) \in Y\}$$

(i.e. X is the subset of strategies whose required resources are feasible) and the loss function by

$$f(x) = d(M(x), u)$$

where the loss $f(x)$ associated with the strategy x is the “distance” to u of the service $M(x)$ performed by x .

In particular, we shall restrict our attention to the case where

— U, V, W are finite dimensional vector spaces (or, more generally, topological vector spaces)

— L, M are linear operators (or continuous linear operators.)

A special treatment will be offered in the case when

$$\begin{cases} \text{(i) } W \text{ is a Hilbert space,} \\ \text{(ii) } d(v, u) = \frac{1}{2} \|v - u\|^2 = \frac{1}{2} ((v - u, v - u)), \end{cases}$$

where $((v, w))$ denotes the scalar product on W (see Section 2.3.1).

Example. Take $U = \mathbf{R}^l$ to be a commodity space as in the above example. We shall regard U as the “output” space.

Take $V = \mathbf{R}^k$ and interpret this as a space of “resources” (or an “input space”, space of “primary goods”, etc.). The map $L \in \mathcal{L}(U, V)$ describes a “linear production operator”, which associates with any output $x \in U$ the input $Lx \in V$ which is used to produce x .

If $x = \sum_{j=1}^l x_j e^j$, the input $Lx = \sum_{j=1}^l x_j L e^j$ is the linear combination of inputs $L e^j$ used to produce the unitary commodities e^j . Then the entries

$$(5) \quad a_i^j = \langle e_i^*, L e^j \rangle$$

of the matrix of L represent the amounts of units of the i^{th} input used to produce one unit of the j^{th} output. They are called the “*technical coefficients*” of the production process.

Usually, the “production operator” L is assumed to be surjective, i.e. any input $y \in \mathbf{R}^k$ can produce at least one output $x \in \mathbf{R}^l$.

If $p \in V^* = \mathbf{R}^{k*}$ is an *input price* and if L^* is the transpose of L , then $L^* p$ is the output price such that the value $\langle L^* p, x \rangle$ of any output x is equal to the value $\langle p, Lx \rangle$ of the input Lx needed to produce x . Note that

$$(6) \quad \forall j = 1, \dots, l, \quad q^j = \sum_{i=1}^k a_i^j p^i.$$

The unitary price q^j of the j^{th} output is the sum of the values $a_i^j p^i$ of the a_i^j units of input i used to produce j .

Since we assumed L to be surjective, L^* is injective. If the input price $p \in \mathbb{R}^{k*}$ is non-zero, the output price $L^*p \in \mathbb{R}^{l*}$ is also non-zero.

Example. We shall regard

$$W = \mathbb{R}^m \text{ as a space of "services"}$$

and $M \in \mathcal{L}(U, W)$ as a linear map associating with any commodity $x \in \mathbb{R}$ the service $Mx \in \mathbb{R}^m$ produced by x .

The entries $b_i^j = \langle e_i^*, Me^j \rangle$ of the matrix of M represent the numbers of units of the i^{th} service produced by a unit of the j^{th} commodity. Usually, it is assumed that a service operator M is injective, i.e. that any non-zero commodity $x \in \mathbb{R}^l$ produces a non-zero service Mx .

If $q \in \mathbb{R}^{m*}$ is a service price, then the price $M^*q \in \mathbb{R}^{l*}$ is the price such that the value $\langle M^*q, x \rangle$ of x equals the value $\langle q, Mx \rangle$ of the service Mx produced by x .

Since we assumed a service operator M to be injective, the operator M^* is surjective. If $p \in \mathbb{R}^{l*}$ is a commodity price, there exists at least one service price $q \in \mathbb{R}^{m*}$ such that $q = M^*p$.

The problem of choosing a service price q satisfying the equation $q = M^*p$ is called a "pricing problem".

1.1.4. Extension of loss functions

It will be useful to extend f to a function f_X mapping the set U of unconstrained strategies into $]-\infty, +\infty]$ by setting

$$(7) \quad f_X(x) = \begin{cases} f(x) & \text{if } x \in X, \\ +\infty & \text{if } x \notin X. \end{cases}$$

The optimization problem (1) is then equivalent to finding $\bar{x} \in U$ such that

$$(8) \quad f_X(\bar{x}) = \alpha = \inf_{x \in U} f_X(x).$$

In other words, the *extended function* f_X "involves" both the *strategy subset* $X \subset U$ and the *original real-valued loss function* f .

Definition 1. Let U be an unconstrained strategy set and f (resp. g) a function mapping U into $]-\infty, +\infty] = \mathbb{R} \cup \{+\infty\}$ (resp. $[-\infty, +\infty[= \mathbb{R} \cup \{-\infty\}$). We define the "domain" $\text{Dom } f$ of f (resp. $\text{Dom } g$ of g) by

$$(9) \quad \begin{aligned} \text{Dom } f &= \{x \in U \text{ such that } f(x) < +\infty\}. \\ (\text{resp. } \text{Dom } g &= \{x \in U \text{ such that } g(x) > -\infty\}. \end{aligned}$$

We exclude once and for all extended functions f or g with $\text{Dom } f = \emptyset$ or $\text{Dom } g = \emptyset$. Occasionally we emphasize this exclusion by referring to our extended functions as “proper” extended functions.

If f is a loss function defined on the strategy subset X of U , f_X is called its “extension” to U .

It is clear that

$$(10) \quad \text{Dom } f_X = X.$$

The use of a finite loss function defined on a strategy subset X is equivalent to the use of its extension mapping U into $] - \infty, + \infty]$, where the domain $\text{Dom } f$ of f represents the strategy subset X .

It will be useful to use the second approach when, for example, a given loss function is constructed as a pointwise supremum of functions. More generally, whenever a function f is constructed according to certain formulas, these specify the domain of f implicitly, by specifying whether $f(x)$ is $+\infty$ or not. In the first approach, one would always need to describe the domain of f explicitly before the values of f on that domain could be given.

Another instance where it is useful to consider functions mapping U into $] - \infty, + \infty]$ occurs when the strategy set $X(y)$ depends upon a parameter y and we have to study the function

$$\alpha(y) = \inf_{x \in X(y)} f(x).$$

We shall adopt the following rule for defining $\alpha(y)$ when $X(y)$ is empty. We write

$$(11) \quad \inf_{x \in \emptyset} f(x) = +\infty, \quad \sup_{x \in \emptyset} f(x) = -\infty.$$

(The loss becomes infinite when it is impossible to select a strategy.)

We extend the usual arithmetic operations to $] - \infty, + \infty]$ by setting

$$\begin{aligned} a + \infty &= \infty + a = \infty \quad \text{for } -\infty < a \leq +\infty, \\ a - \infty &= -\infty + a = -\infty \quad \text{for } -\infty \leq a < \infty, \\ a \cdot \infty &= \infty \cdot a = \infty; \quad a(-\infty) = (-\infty)a = -\infty \quad \text{for } 0 < a \leq \infty, \\ a \cdot \infty &= \infty \cdot a = -\infty; \quad a(-\infty) = (-\infty)a = \infty \quad \text{for } -\infty \leq a < 0, \\ 0 \cdot \infty &= \infty \cdot 0 = 0 = 0(-\infty) = (-\infty)0; \quad -(-\infty) = +\infty, \\ \inf \emptyset &= +\infty; \quad \sup \emptyset = -\infty. \end{aligned}$$

The combinations $\infty - \infty$ and $-\infty + \infty$ are undefined and *forbidden*. We can use the usual laws.

$$\begin{aligned}(a_1 + a_2) + a_3 &= a_1 + (a_2 + a_3); & a_1 + a_2 &= a_2 + a_1, \\ (a_1 a_2) a_3 &= a_1 (a_2 a_3); & a_1 a_2 &= a_2 a_1; \\ a(a_1 + a_2) &= aa_1 + aa_2\end{aligned}$$

provided the sums $a+b$ are not the forbidden $\infty - \infty$ or $-\infty + \infty$.

1.1.5. Sections and epigraphs

Definition 2. Let $f: U \mapsto]-\infty, +\infty]$ be a function. The subsets

$$(12) \quad S(f, \lambda) = \{x \in U \text{ such that } f(x) \leq \lambda\}$$

are called the “(lower) sections” of f . The subsets $\hat{S}(f, \lambda) = \{x \in U : f(x) < \lambda\}$, $S^*(f, \lambda) = \{x \in U : f(x) \geq \lambda\}$ and $\hat{S}^*(f, \lambda) = \{x \in U : f(x) > \lambda\}$ are called respectively the (lower) open sections, the upper sections and the upper open sections of f . The subset

$$(13) \quad \mathcal{E}p(f) = \{ \{x, \lambda\} \in U \times \mathbf{R} \text{ such that } f(x) \leq \lambda \}$$

is said to be the “epigraph” of f .

Proposition 1. Let $f = \sup_{i \in I} f_i$ be the pointwise supremum of a family $\{f_i\}_{i \in I}$ of functions $f_i: U \mapsto]-\infty, +\infty]$. Then

$$(14) \quad \begin{cases} \text{(i)} & \text{Dom } f \subset \bigcap_{i \in I} \text{Dom } f_i, \\ \text{(ii)} & S(f, \lambda) = \bigcap_{i \in I} S(f_i, \lambda), \\ \text{(iii)} & \mathcal{E}p(f) = \bigcap_{i \in I} \mathcal{E}p(f_i). \end{cases}$$

Proof. This is left as an exercise. $\square$

Remark. If we are only concerned with minimization problems, loss functions are only used via their lower sections. Indeed, only the subsets $S(f, f(x))$ consisting of those $y \in X$ such that $f(y) \leq f(x)$ are relevant.

These sections are nothing other than the lower sections of the total preordering “ y is preferred to x ” if and only if $f(y) \leq f(x)$, i.e. $y \in S(f, f(x))$. It is clear that the sections $S(f, f(x))$ are invariant when f is replaced by $\varphi \cdot f$ where φ is a (strictly) increasing function from $\mathbf{R}$ into itself.

*1.2. Decomposition principle

Let U and V be two vector spaces and let L belong to $\mathcal{L}(U, V)$. We can associate with any function g defined on V the composition product gL defined on U . Symmetrically, we will associate with any function f defined on U the product (from the left) Lf defined on V by $Lf(y) = \inf_{Lx=y} f(x)$.

Such products yield ways of constructing new loss functions. For instance, the inf-convolution of functions can be constructed in this way.

The identity

$$\inf_{Lx \in Y} (f(x) + g(Lx)) = \inf_{y \in Y} (Lf(y) + g(y))$$

is obvious but yields some useful general decomposition principles. We deduce for instance that $(ML)f = M(Lf)$ and that

$$\inf_{Ax+By=z} [f(x) + g(y)] = \inf_y [Af(z-By) + g(y)].$$

1.2.1. Product of a loss function by a linear operator

Let $g : V \mapsto]-\infty, +\infty]$ be a function defined on V . If $L \in \mathcal{L}(U, V)$ is a map from U into V , we shall denote by gL the function from U into $]-\infty, +\infty]$ defined by $x \mapsto g[L(x)]$.

It is clear that

$$(1) \quad \text{Dom } [gL] = L^{-1}[\text{Dom } g]$$

It will be convenient to refer to gL as the “*product from the right*” of g by the map L .

We now define the “*product from the left*” Lf of f by a map L as follows.

Definition 1. Let L be a linear operator mapping U into V . Let $f : U \mapsto]-\infty, +\infty]$ be a function defined on U . We shall associate with f and L the function $Lf : V \rightarrow]-\infty, +\infty]$ defined by

$$(2) \quad \forall y \in V; \quad Lf(y) = \inf_{Lx=y} f(x).$$

This is called the “*product from the left*” of f by L .

Since the infimum over the empty set is $+\infty$, we shall set

$$(3) \quad Lf(y) = +\infty \quad \text{if } y \notin L \text{ Dom } (f).$$

Then

$$(4) \quad \text{Dom } Lf = L \text{ Dom } f.$$

Remark. We can of course define Lf even for non-linear operators but all the examples we shall study involve only linear operators and so we consider only this case.

1.2.2. Example: Inf-convolution of functions

Consider n functions f_i mapping U into $]-\infty, +\infty]$.

We associate with these functions a function $f: U^n \rightarrow]-\infty, +\infty]$ defined by $\tilde{f}(x) = f_1(x^1) + f_2(x^2) + \dots + f_n(x^n)$ and we let the operator L be the operator Σ from U^n into U defined by $\Sigma x = \sum_{i=1}^n x^i$.

Then

$$(5) \quad \Sigma \tilde{f}(y) = \inf_{\sum_{i=1}^n x^i = y} \sum_{i=1}^n f_i(x^i).$$

Definition 2. We shall say that the function $\Sigma \tilde{f}$ defined by (5) is the *inf-convolution* of the functions f_i . We shall set

$$(6) \quad \Sigma \tilde{f} = f_1 \square f_2 \square \dots \square f_n = \bigcap_{i=1}^n f_i.$$

Notice that we can write the inf-convolution of two functions f and g in the following way

$$\begin{aligned} (f \square g)(x) &= \inf_{y+z=x} (f(y) + g(z)) \\ &= \inf_{y \in U} (f(y) + g(x-y)) \\ &= \inf_{z \in U} [f(x-z) + g(z)]. \end{aligned}$$

Examples. Consider the case when

$$U = U_1 \times U_2, \quad L = L_1 + L_2 \quad \text{where } L_i \in \mathcal{L}(U_i, V) \quad (i = 1, 2).$$

Then

$$(7) \quad ((L_1 + L_2)f)(y) = \inf_{L_1 x_1 + L_2 x_2 = y} f(x_1, x_2).$$

In particular, consider the case

$$U_1 = U_2 = V, \quad L_1 = 1 \quad \text{and} \quad L_2 = -1.$$

Then

$$(8) \quad ((L_1 + L_2)f)(y) = \inf_{x \in V} f(x, y + x).$$

Taking

$$V = U_1, \quad L_1 = 1, \quad L_2 = 0,$$

we say that

$$(9) \quad (\pi_{U_1}f)(x_1) = \inf_{x_2 \in U_2} f(x_1, x_2) = (L_1 + L_2)(x_1)$$

is the “projection of f onto U_1 .”

Interpretation. We have already met the product from the right fM when M is a service operator.

Consider the case when $L \in \mathcal{L}(U, V)$ is a resource operator from U into a resource space V . If f is a loss function associating with any strategy $x \in U$ its loss $f(x)$, the loss function Lf associates with any resource $y \in V$ the smallest possible loss which can result from the implementation of a strategy x available for the resource y (i.e. $Lx = y$).

The inf-convolution can be regarded in the following way. We suppose that n players i share a resource y , i.e. choose strategies x^i such that $\sum_{i=1}^n x^i = y$. The inf-convolution $\square_{i=1}^n f_i$ associates with any y the smallest total loss occurred during the sharing.

1.2.3. Decomposition principle

Any minimization problem of the form

$$(P): \quad \inf_{Lx \in Y} (f(x) + g(Lx))$$

can be decomposed into two minimization problems, i.e.

$$(Q): \quad \inf_{y \in Y} (Lf(y) + g(y))$$

and

$$(R(y)): \quad \inf_{Lx=y} f(x)$$

for a convenient y .

Proposition 1. *The identity*

$$(10) \quad \inf_{Lx \in Y} [f(x) + g(Lx)] = \inf_{y \in Y} [Lf(y) + g(y)]$$

holds and the two following statements are equivalent:

$$(11) \quad \begin{cases} (a) \ \bar{x} \text{ minimizes } f(x) + g(Lx) \text{ on the set } X = L^{-1}(Y) \\ (b) \ \bar{y} = L\bar{x} \text{ minimizes } Lf(y) + g(y) \text{ on } Y \text{ and } \bar{x} \text{ minimizes} \\ \quad f(x) \text{ on the set } L^{-1}(\bar{y}). \end{cases}$$

Proof. Since the set $X = \{x \in U \text{ such that } Lx \in Y\}$ is the union of the sets $L^{-1}(y) = \{x \in U \text{ such that } Lx = y\}$ as y ranges over Y , we deduce that

$$\begin{aligned} \inf_{Lx \in Y} [f(x) + g(Lx)] &= \inf_{y \in Y} \inf_{\substack{x \in U \\ Lx = y}} [f(x) + g(Lx)] \\ &= \inf_{y \in Y} \inf_{\substack{x \in U \\ Lx = y}} [f(x) + g(y)] = \inf_{y \in Y} [Lf(y) + g(y)]. \end{aligned}$$

It is clear that, if $\bar{x} \in X$ minimizes $f(x) + g(Lx)$ on X , then $\bar{y} = L\bar{x} \in Y$ minimizes $Lf(y) + g(y)$ over Y and $f(\bar{x}) = (Lf)(\bar{y})$.

Conversely, suppose that $\bar{y}$ minimizes $Lf(y) + g(y)$ over Y and that $\bar{x}$ minimizes $f(x)$ over $L^{-1}(\bar{y})$. Then $\bar{x}$ minimizes $f(x) + g(Lx)$ on X . To see this, observe that

$$\begin{aligned} f(\bar{x}) + g(L\bar{x}) &= Lf(\bar{y}) + g(\bar{y}) \\ &\leq Lf(Lx) + g(Lx) \leq f(x) + g(Lx) \end{aligned}$$

when $x \in X$. $\square$

From this general principle we deduce the following consequence.

Proposition 2.

$$(12) \quad (M \cdot L)(f) = M \cdot [Lf]$$

Furthermore, the two following statements are equivalent:

$$(13; a) \quad \bar{x} \text{ minimizes } f \text{ under the constraint } MLx = z$$

and

$$(13; b) \quad \begin{cases} (i) \ \bar{y} \text{ minimizes } Lf \text{ under the constraint } My = z, \\ (ii) \ \bar{x} \text{ minimizes } f \text{ under the constraint } Lx = \bar{y}. \end{cases}$$

Proof. The result follows from Proposition 1 with $g = 0$ and $Y = M^{-1}z$. $\square$

In particular, consider the problem

$$(P): \quad \inf_{\sum_{i=1}^n L_i(x^i) = y} \sum_{i=1}^n f_i(x^i).$$

We shall decompose this problem into a "centralized problem" for the allocation of y

$$(Q): \inf_{\sum_{i=1}^n y^i = y} \sum_{i=1}^n (L_i f_i)(y^i)$$

and n "decentralized problems"

$$(R_i(y^i)): \inf_{L_i x^i = y^i} f_i(x^i).$$

Proposition 3.

$$(14) \quad \inf_{\sum_{i=1}^n L_i x^i = y} \sum_{i=1}^n f_i(x^i) = (L_1 f_1 \square L_2 f_2 \square \dots \square L_n f_n)(y).$$

The two following statements are equivalent

$$(15; a) \quad \bar{x} = (\bar{x}^1, \dots, \bar{x}^n) \text{ minimizes } \sum_{i=1}^n f_i(x^i) \text{ under the constraints } \sum_{i=1}^n L_i \bar{x}^i = y$$

and

$$(16; b) \quad \begin{aligned} &\bar{y}^i = L_i \bar{x}^i \text{ minimizes } \sum_{i=1}^n (L_i f_i)(y^i) \\ &\text{under the constraints } \sum_{i=1}^n y^i = y \text{ and, for any } i = 1, \dots, n, \\ &\bar{x}^i \text{ minimizes } f_i(x) \text{ under the constraints } L_i x^i = \bar{y}^i. \end{aligned}$$

Interpretation. The above proposition can be interpreted in the following way. We seek strategies x^i which minimize the loss function $\sum_{i=1}^n f_i(x^i)$ under the constraint that the sum of the resources $L_i(x^i)$ needed to implement x^i is equal to the scarce resource y . This problem is equivalent to

• *first*, allocating the resource $y = \sum_{i=1}^n \bar{y}^i$ among the n players by minimizing the loss function $\sum_{i=1}^n (L_i f_i)(y^i)$

• *and then*, each player implementing the strategy $\bar{x}^i$ which minimizes his loss function f_i over the set of strategies x^i which can be implemented with the resource $\bar{y}^i$.

1.2.4. Another decomposition principle

Let U , V and W be three vector spaces. Let $A \in \mathcal{L}(U, W)$, $B \in \mathcal{L}(V, W)$, and let f and g be two functions defined on U and V respectively.

Consider the problem

$$(P): \inf_{Ax + By = z} (f(x) + g(y)).$$

This can be decomposed into the two following problems

$$(Q): \inf_y (Af(z - By) + g(y))$$

and

$$(R(y)): \inf_{Ax = z - By} f(x)$$

Proposition 4.

$$(17) \quad \inf_{Ax + By = z} (f(x) + g(y)) = \inf_y (Af(z - By) + g(y)).$$

The two following statements are equivalent

$$(18; a) \quad \{\bar{x}, \bar{y}\} \text{ minimizes (P)}$$

and

$$(18; b) \quad \begin{cases} \text{(i) } \bar{y} \text{ minimizes (Q)} \\ \text{(ii) } \bar{x} \text{ minimizes } R(\bar{y}) \end{cases}$$

Proof.

$$\begin{aligned} \inf_{Ax + By = z} (f(x) + g(y)) &= \inf_{z_1 + z_2 = z} (Af(z_1) + Bg(z_2)) \\ &= \inf_{z_2} (Af(z - z_2) + Bg(z_2)) \\ &= \inf_y (Af(z - By) + g(y)). \quad (\text{By (10)}) \end{aligned}$$

It is clear that, if $\bar{x}$ and $\bar{y}$ are solutions of the first problem, then $\bar{y}$ is a solution of the second problem and $\bar{x}$ minimizes f under the constraint $Ax = z - B\bar{y}$. The converse is also obvious. $\square$

Interpretation. We can regard U and V as the consumption space and the production space of a consumer and of a producer respectively. The consumer and the producer have to share a (scarce) resource z by minimizing the sum of their losses $f(x)$ and $g(y)$. This problem can be solved in two steps. In the first place, the producer chooses his production y to be optimal for a new loss function. This new loss function is the sum of his first loss function $g(y)$ and of the loss $Af(z - By)$. The latter loss represents the smallest of the losses possible for the consumer when his available resource is $z - By$, i.e. that which remains once the producer has implemented y .

Secondly, given that $\bar{y}$ is the actual production implemented, the consumer chooses x to minimize his loss under the constraint $Ax = z - B\bar{y}$. In following this program, the producer and consumer achieve the optimal consumption-production pair.

Remark. We can use several of the above decomposition principles successively. For instance, consider the following problem involving n consumers and m producers. We take $U = \prod_{i=1}^n U^i$, $V = \prod_{j=1}^m V^j$, $A = \sum_{i=1}^n A_i$, $B = \sum_{j=1}^m B_j$. We then have the problem:

$$(P): \inf_{\sum_{i=1}^n A_i x^i + \sum_{j=1}^m B_j y^j = z} \left(\sum_{i=1}^n f_i(x^i) + \sum_{j=1}^m g_j(y^j) \right).$$

This can be decomposed into the following problems:

$$(Q): \inf_{y^j \in V^j} \left[\sum_{j=1}^m [\sum g_j(y^j) + [A_1 f_1 \square \dots \square A_n f_n] \left(z - \sum_{j=1}^m B_j y^j \right)] \right]$$

$$R(y^1, \dots, y^m): \inf_{\sum_{i=1}^n z^i = z - \sum_{j=1}^m B_j y^j} \sum_{i=1}^n (A_i f_i)(z^i)$$

and

$$S_i(z^i): \inf_{A_i x^i = z^i} f_i(x^i).$$

In other words, the optimal solution $\{\bar{x}^1, \dots, \bar{x}^n, \bar{y}^1, \dots, \bar{y}^m\}$ of problem (P) can be obtained by finding the optimal production $\{\bar{y}^1, \dots, \bar{y}^m\}$ as a solution of problem (Q), then by allocating the rest of available resources in resources $\bar{z}^1, \dots, \bar{z}^n$ as a solution of problem $R(\bar{y}^1, \dots, \bar{y}^m)$ and finally, by letting each consumer i choose (in a decentralized way) his optimal consumption by solving the problem $S_i(\bar{z}^i)$.

1.3. Mixed strategies and convexity

(informative)
 We introduce mixed strategies via the following extension problem. Can we replace the minimization problem $\alpha = \inf_{x \in X} f(x)$ by an "extended minimization problem" $\tilde{\alpha} = \inf_{m \in \mathcal{M}(X)} f^d(m)$ in which $\alpha = \tilde{\alpha}$ and X is embedded in $\mathcal{M}(X)$ by an injective map δ ? Since any optimal solution $\bar{x}$ of the initial minimization problem is mapped by δ into an optimal solution $\delta \bar{x}$ of the extended minimization problem, it is advantageous to extend a given minimization problem to a minimization problem which *always has an optimal solution*, especially if we can take $\mathcal{M}(X)$ to be convex and f^d to be linear.

This desirable result can be achieved by taking $\mathcal{M}(X)$ to be the set of discrete probability measures.

In terms of game theory, these are called mixed strategies.

Note that there are other extension possibilities which will be examined in more detail in the framework of two-person games (see Section 7.2). We shall only check in this chapter that, for finite strategy sets, the subset of mixed strategies is equal to the subset of infinite sequences of pure strategies.

It is natural to devise sufficient conditions for the initial and the extended minimization problems to be equivalent. This can be done by associating with any map $\beta: \mathcal{M}(X) \rightarrow X$, such that $\beta\delta = 1$, the subset of functions f such that $f(\beta m) \leq f^\delta(m)$ for all $m \in \mathcal{M}(X)$. The main example is provided by the barycentric operator β when X is a convex subset. The functions satisfying the above inequalities are the convex functions. Other classes of functions satisfying analogous properties are introduced in Section 13.3.

We end this section by recalling some elementary properties of convex subsets and functions. We mention here only the following property which will play a fundamental role throughout this book.

Let $f_1, \dots, f_n$ be n convex functions defined on a convex subset X and let F be the associated operator from X into $\mathbb{R}^n$ defined by $F(x) = \{f_1(x), \dots, f_n(x)\}$. Then $F(x) + \mathbb{R}_+^n$ is a convex subset of $\mathbb{R}^n$.

1.3.1. Motivation: extension of strategy sets and loss functions

It is often the case that an optimization problem with no solution can be converted into one with a solution by enlarging the strategy set. In formal terms, we propose to “enlarge” the strategy set X (by embedding it in a larger space $\mathcal{M}(X)$) and to “extend” the loss function f defined on X to a new loss function f^δ defined on $\mathcal{M}(X)$ whenever there is no solution of the initial problem.

A minimum requirement for such an extension is that the minimal and maximal values be invariant, i.e.

$$(1) \quad \inf_{x \in X} f(x) = \inf_{m \in \mathcal{M}(X)} f^\delta(m); \quad \sup_{x \in X} f(x) = \sup_{m \in \mathcal{M}(X)} f^\delta(m)$$

Indeed, this property obviously implies the following result.

Proposition 1. *If $\delta: X \rightarrow \mathcal{M}(X)$ is the injective map embedding X into $\mathcal{M}(X)$ and if property (1) is satisfied, then any minimal (resp. maximal) solution $\bar{x} \in X$ of f is mapped onto a minimal (resp. maximal) solution $\delta(\bar{x})$ of f^δ .*

The converse holds whenever the following property is satisfied by the loss function.

$$(2) \quad \text{There exists a map } \beta \text{ from } \mathcal{M}(X) \text{ into } X \text{ such that } f(\beta m) \leq f^\delta(m) \text{ for any } m \in \mathcal{M}(X).$$

We therefore obtain the following result.

Proposition 2. *If properties (1) and (2) are satisfied, then any minimal (resp. maximal) solution $m \in \mathcal{M}(X)$ of f^Δ is mapped onto a minimal (resp. maximal) solution βm of f^Δ .*

Among all possible extensions satisfying properties (1) and (2), we single out the most important example in which

- (a) $\mathcal{M}(X)$ is the *convex* subset of discrete probability measures;
- (b) δ is the *Dirac* operator;
- (c) f^Δ is the *linear extension* of f to $\mathcal{M}(X)$;
- (d) β is the *barycentric* operator;
- (e) functions satisfying property (2) are *convex* functions.

1.3.2. Mixed strategies and linearized loss functions

Definition 1. Let X be a set and $\mathcal{S}(X) = \mathbf{R}^X$ be the vector space of real-valued functions f defined on X (supplied with the topology of pointwise convergence).

We shall denote by $\delta(x)$ the “*Dirac measure*”

$$(3) \quad \delta(x): f \mapsto f(x)$$

which is a (continuous) linear functional on $\mathcal{S}(X)$.

We shall call any linear combination

$$(4) \quad m = \sum_{i=1}^n \lambda^i \delta(x_i): f \mapsto \langle m, f \rangle = \sum_{i=1}^n \lambda^i f(x_i)$$

a “*discrete measure*” on X . We shall say that

$$(5) \quad m = \sum_{i=1}^n \lambda^i \delta(x_i) \text{ is “positive” if, for any } i, \lambda^i \geq 0$$

and that

$$(6) \quad m \text{ is a “discrete probability measure” if } m \text{ is positive and } \sum_{i=1}^n \lambda^i = 1.$$

We shall often use the word “*probability*” as a synonym for “*probability measure*”.

We denote by

$$(7) \quad \left\{ \begin{array}{l} \mathcal{S}^*(X) \text{ the vector space of discrete measures,} \\ \mathcal{S}_+^*(X) \text{ the convex cone of positive discrete measures,} \\ \mathcal{M}(X) = \mathcal{S}_1^*(X) \text{ the convex set of discrete probabilities,} \\ \delta: X \rightarrow \mathcal{M}(X) \text{ the injective map associating with } x \in X \text{ the} \\ \text{Dirac measure } \delta(x) \in \mathcal{M}(X). \end{array} \right.$$

In other words, the *Dirac operator* δ is an embedding from a set X into a convex subset $\mathcal{M}(X)$.

Following the terminology of game theory, we make the following definition.

Definition 2. We shall say that the convex subset $\mathcal{M}(X)$ of discrete probabilities is the set of “mixed strategies” on X .

The map $m \mapsto \langle f, m \rangle$ is a linear functional defined on $\mathcal{S}(X)$ and, in particular, on $\mathcal{M}(X)$.

Definition 3. Let $f : X \rightarrow \mathbb{R}$ be a loss function defined on a strategy set X . We shall define the “linearized loss” function $f^d : \mathcal{S}^*(X) \rightarrow \mathbb{R}$ by

$$(8) \quad \forall m = \sum_{i=1}^n \lambda_i \delta(x_i) \in \mathcal{S}^*(X), \quad f^d(m) = \langle m, f \rangle = \sum_{i=1}^n \lambda_i f(x_i).$$

The minimum requirement (property (1)) is clearly fulfilled.

Proposition 3. Let f be a loss function defined on a strategy set X and f^d be the linearized loss function defined on the mixed strategy set $\mathcal{M}(X)$. Then

$$(1) \quad \inf_{x \in X} f(x) = \inf_{m \in \mathcal{M}(X)} f^d(m); \quad \sup_{x \in X} f(x) = \sup_{m \in \mathcal{M}(X)} f^d(x).$$

Proof. It suffices to prove the first statement. Since δ embeds X in $\mathcal{M}(X)$ and $f(x) = f^d(\delta(x))$,

$$\inf_{m \in \mathcal{M}(X)} f^d(m) \leq \inf_{x \in X} f(x).$$

On the other hand, for any

$$m = \sum \lambda_i \delta(x_i) \in \mathcal{M}(X),$$

we have

$$f^d(m) = \sum_{i=1}^n \lambda_i f(x_i) \geq \left(\sum_{i=1}^n \lambda_i \right) \inf_{x \in X} f(x)$$

and thus,

$$\inf_{x \in X} f(x) \leq \inf_{m \in \mathcal{M}(X)} f^d(m). \quad \square$$

Remark. In fact when X is compact, we shall embed $\mathcal{M}(X)$ in the convex subset $\tilde{\mathcal{M}}(X)$ of Radon probabilities (which is compact for a suitable topology) and extend f (when continuous on X) to a function f^d which is continuous on $\tilde{\mathcal{M}}(X)$. Since this construction involves several results which will be proved later, the construction is postponed until Section 3.1.6.

1.3.3 Interpretation of mixed strategies

The usual interpretation of a mixed strategy is that of a “random strategy” or “lottery”

$$m = \sum_{i=1}^n \lambda^i \delta(x_i)$$

where λ^i is the probability of selecting the i^{th} strategy. In this case, $\langle m, f \rangle = \sum_{i=1}^n \lambda^i f(x_i)$ is the “expected loss”.

With the above interpretation, a “mixed strategy” would be implemented by a probability mechanism (for instance, toss of coin, roll of a die; “wheel of fortune”, table of random numbers).

Instead, we shall use a more operational interpretation, by assuming that the game can be “repeated”.

In a large number of repetitions of the game, the coefficients λ^i represent the proportion of times that x_i is expected to be played. Assume in the first place that the coefficients $\lambda^i = p_i/p$ are rational. In other words, we assume that the game is repeated p times. In this case we can interpret the mixed strategy

$m = \sum_{i=1}^n (p_i/p) \delta(x_i)$ as the strategy x_1 played p_1 times, . . . , the strategy x_i played p_i times, . . . , the strategy “ x_n ” played p_n times in a game repeated p times.

Since any real number λ_i can be approximated by rational numbers, we shall regard a mixed strategy $m = \sum_{i=1}^n \lambda^i \delta(x_i)$ as the strategy x_1 played “ λ^1 times” . . . , the strategy x_n played “ λ^n times” in an infinitely repeated game.

Now, if f is a loss function defined on X , and if $m = \sum_{i=1}^n \lambda^i \delta(x_i)$ is a mixed strategy, we associate with m the loss $\sum_{i=1}^n \lambda^i f(x_i) = \langle m, f \rangle$ equal to “ λ^1 times” the loss $f(x_1)$ plus . . . plus “ λ^n times” the loss $f(x_n)$. When $\lambda^i = p^i/q$ is a rational number, the loss $\sum_{i=1}^n \lambda^i f(x_i) = (1/p) \sum_{i=1}^n p_i f(x_i)$ is nothing other than the average loss.

1.3.4. Case of finite strategy sets

Suppose that

$$X = \{1, \dots, n\}$$

is a finite set of n elements.

Since any function $f \in \mathcal{S}(X)$ is defined by $i \in X \rightarrow f(i) \in \mathbf{R}$, then $\mathcal{S}(X) = \mathbf{R}^X$ is isomorphic to $\mathbf{R}^n$ and the cone $\mathcal{S}_+(X)$ to the positive orthant $\mathbf{R}_+^n$.

The space $\mathcal{S}^*(X)$ of discrete measures $m = \sum_{i=1}^n \lambda^i \delta(x_i)$ is isomorphic to the dual $\mathbf{R}^{n*}$ of $\mathbf{R}^n$. In this case

$$f^A(m) = \langle m, f \rangle = \sum_{i=1}^n \lambda^i f(i)$$

is the familiar duality pairing on $\mathbf{R}^{n*} \times \mathbf{R}^n$. Furthermore, the cone $\mathcal{S}_+^*(X)$ is isomorphic to the positive orthant $\mathbf{R}_+^{n*}$ of $\mathbf{R}^{n*}$ and the set $\mathcal{M}(X)$ is isomorphic to the probability simplex $\mathcal{M}^n$ of $\mathbf{R}^n$ defined by

$$(9) \quad \mathcal{M}^n = \{\lambda = \{\lambda^1, \dots, \lambda^n\} \text{ such that } \lambda^i \geq 0 \text{ and } \sum_{i=1}^n \lambda^i = 1\}$$

which is compact.

In summary, when $X = \{1, \dots, n\}$, we shall make the following identifications

$$(10) \quad \mathcal{S}(X) = \mathbf{R}^n, \quad \mathcal{S}^*(X) = \mathbf{R}^{n*}, \quad \mathcal{S}_+^*(X) = \mathbf{R}_+^{n*}, \quad \mathcal{M}(X) = \mathcal{M}^n$$

and $f^A(m) = \langle m, f \rangle = \sum_{i=1}^n \lambda^i f_i$ whenever $m = \sum \lambda^i \delta(i)$ and $f = \{f_i\}$.

* 1.3.5. Representation by infinite sequences of pure strategies

We shall prove that any mixed strategy $m \in \mathcal{M}(X) = \mathcal{M}^n$ of a finite set X of n elements can be written

$$(11) \quad m = \sum_{i=1}^{\infty} a_i \delta(x_i) \quad \text{where } x_i \in X \text{ and } a_i = \frac{1}{n} \left(\frac{n-1}{n} \right)^{i-1}.$$

In other words, any mixed strategy can be implemented by playing an infinite sequence $\{x_i\}$ of pure strategies and the expectation $\langle m, f \rangle = \sum_{i=1}^{\infty} a_i f(x_i)$ is a “generalized discounted” summation of the losses $f(x_i)$ of pure strategies x_i .

Notice first of all that $a_i \geq 0$ and that

$$(12) \quad \sum_{i=1}^{\infty} a_i = \sum_{i=1}^{\infty} \frac{(n-1)^{i-1}}{n^i} = 1.$$

We denote by $X^{\mathbf{N}}$ the set of all sequences of elements of X .

Proposition 4. Let X be a finite set of n elements and π_a^* the map from $X^{\mathbf{N}}$ into $\mathcal{M}(X)$ defined by

$$(13) \quad \pi_a^* (\{x_i\}_i) = \sum_{i=1}^{\infty} a_i \delta(x_i) \in \mathcal{M}(X).$$

Then the map π_a^* is surjective.

Proof. Let $m = \sum_{k=1}^n \alpha_k \delta(k)$ be a probability on the strategy set $X = \{1, \dots, k, \dots, n\}$. We define recursively a sequence of pure strategies x_t and non-negative measures $m_t = \sum_{k=1}^n \alpha'_k \delta(x_t)$ as follows. For $t = 1$,

$$(14) \quad \begin{cases} x_1 = \inf \left\{ k \text{ such that } \alpha_k \geq \frac{1}{n} = a_1 \right\}, \\ m_1 = m - \frac{1}{n} \delta(x_1) = \sum_{k=1}^n \alpha'_k \delta(x_k) \end{cases}$$

and, for $t+1$

$$(15) \quad \begin{cases} x_{t+1} = \inf \{k \text{ such that } \alpha'_k \geq a_{t+1}\}, \\ m_{t+1} = m_t - a_{t+1} \delta(x_{t+1}) = \sum_{k=1}^n \alpha_k^{t+1} \delta(x_k). \end{cases}$$

Note that x_1 is well-defined since $\alpha_k \geq 0$, $\sum \alpha_k = 1$ and at least one of the α_k is not smaller than $1/n$.

Therefore, m_1 is a non-negative discrete measure such that

$$\sum_{k=1}^n \alpha'_k = \frac{n-1}{n}.$$

Similarly, if a positive measure m_t satisfies

$$\sum_{k=1}^n \alpha'_k = \frac{(n-1)^t}{n^t},$$

then at least one of the α'_k 's is not smaller than $(1/n) ((n-1)/n)^t = a_{t+1}$ and so x_{t+1} is well-defined. Thus m_{t+1} is a non-negative measure satisfying

$$\sum_{k=1}^n \alpha_k^{t+1} = n a_{t+1} - a_{t+1} = \frac{(n-1)^{t+1}}{t+1}.$$

Since we can write

$$m_{t+1} = m - (a_1 \delta(x_1) + a_2 \delta(x_2) + \dots + a_t \delta(x_t))$$

we deduce that

$$m_{t+1} = m - \left(\sum_{s=1}^t a_s \delta(x_s) \right) \geq 0.$$

Therefore $\sum_{s=1}^{\infty} a_s \delta(x_s) \leq m$. Since both $\sum_{s=1}^{\infty} a_s \delta(x_s)$ and m are probability measures, they are equal. $\square$

1.3.6. Linearized extension of maps and the barycentric operator

We can also extend a map F from X into a vector space U to a linear operator F mapping $\mathcal{S}^*(X)$ into U .

Definition 4. Let F be a map from X into a vector space U . We shall say that the linear operator F^d mapping $\mathcal{S}^*(X)$ into U defined by

$$(16) \quad \forall m = \sum_{i=1}^n \lambda^i \delta(x_i), \quad F^d(m) = \sum_{i=1}^n \lambda^i F(x_i)$$

is the “linearized extension of F ”.

It is clear that this linear operator F^d is the unique operator F^d mapping $\mathcal{S}^*(X)$ into U such that

$$(17) \quad F = F^d \delta$$

and that the map $F \rightarrow F^d$ is a one to one correspondence between the space $\mathcal{S}(X, U)$ of maps from X into U and the space $\mathcal{L}(\mathcal{S}^*(X), U)$ of linear operators mapping $\mathcal{S}^*(X)$ into U .

In particular, we shall study the linearized extension of the canonical injection from a subset X of a vector space U into U .

We assume now that

$$(18) \quad X \text{ is a subset of a vector space } U.$$

Definition 5. We shall say that the map β from the space $\mathcal{S}^*(X)$ of discrete measures on X into U defined by

$$(19) \quad \forall m = \sum_{i=1}^n \lambda^i \delta(x_i) \in \mathcal{S}^*(X), \quad \beta m = \sum_{i=1}^n \lambda^i x_i \in U$$

is the “barycentric operator”.

In other words, β is the linearized extension i^d of the canonical injection i from X into U .

The following proposition, together with propositions 1 and 2, explain the importance of convex subsets and convex functions in optimization theory.

Proposition 5. Let X be a subset of a vector space U . Then the “convex hull” $\text{co}(X)$ of X is equal to the image under β of the set of mixed strategies, i.e.

$$(20) \quad \beta(\mathcal{M}(X)) = \text{co}(X).$$

Thus

(21) X is “convex” if and only if β maps $\mathcal{M}(X)$ into X .

A function f defined on a convex subset X is “convex” if and only if

$$(22) \quad \forall m \in \mathcal{M}(X), \quad f(\beta m) \leq \langle m, f \rangle = f^A(m)$$

and f is affine if and only if .

$$(23) \quad \forall m \in \mathcal{M}(X), \quad f(\beta m) = f^A(m).$$

A function $f : U \rightarrow]-\infty, +\infty]$ is “convex” if and only if

$$(24) \quad \forall m = \sum_{i=1}^n \lambda^i \delta(x_i) \in \mathcal{M}(U), \quad f(\beta m) \leq f^A(m) = \sum_{i=1}^n \lambda^i f(x_i).$$

A function $f : U \rightarrow [-\infty, +\infty[$ is “concave” if $-f$ is convex.

It is clear that the domain $\text{Dom } f$ of a convex function is convex. Also, if $X \subset U$ is convex and $f : X \rightarrow \mathbf{R}$ is convex, its extension $f_X : U \rightarrow]-\infty, +\infty]$ is convex.

1.3.7. Interpretation of convex functions in terms of risk aversion

We regard

$$(25) \quad \begin{cases} \text{(i) } X \text{ as a strategy set,} \\ \text{(ii) } \mathcal{M}(X) \text{ as the set of mixed strategies } m = \sum_{i=1}^n \lambda^i \delta(x_i) \\ \text{(iii) } \beta m = \sum_{i=1}^n \lambda^i x_i \text{ as the expected value of the mixed strategy } m. \end{cases}$$

We consider the following equivalence relation. Two mixed strategies m_1 and m_2 are *equivalent* if and only if they yield the same expected value $\beta m_1 = \beta m_2$.

We call βm the “certainty equivalent” of the mixed strategy m . In this framework, a player with a convex loss function f is said to be “risk averse”, in the sense that he prefers the certainty equivalent βm of a mixed strategy m to m itself because $f(\beta m) \leq \langle m, f \rangle$

1.3.8. Elementary properties of convex subsets and functions

Proposition 6. *The image and the pre-image of any convex subset X by a linear operator L are convex.*

The intersection of any family of convex subsets is convex.

Any product of convex subsets is convex.

If X_1 and X_2 are convex subsets, the set

$$\alpha_1 X_1 + \alpha_2 X_2 = \{\alpha_1 x_1 + \alpha_2 x_2\}_{x_1 \in X_1, x_2 \in X_2}$$

is also convex.

Any linear subspace of a vector space is convex.

Proof. This is left as an exercise. $\square$

Definition 6. A subset X of a vector space U is called a cone if $\lambda X \subset X$ for any $\lambda \geq 0$. A convex cone is a cone which is convex.

Proposition 7. A subset X is a convex cone if and only if

$$(26) \quad \beta \mathcal{S}_+^*(X) \subset X$$

i.e. if and only if $\beta m = \sum_{i=1}^k \lambda^i x_i \in X$ for all positive discrete measures $m = \sum_{i=1}^k \lambda^i \delta(x_i) \in \mathcal{S}_+^*(X)$.

Proof. This is left as an exercise. $\square$

Next we recall some of the main properties of convex functions.

Proposition 8. A function $f: U \rightarrow]-\infty, +\infty]$ is convex if and only if its epigraph $\mathcal{E}p(f)$ is convex.

The pointwise supremum of convex functions is convex.

The lower sections and the lower open sections of a convex function are convex.

Proof. This is left as an exercise. $\square$

Proposition 9. The products from the right $g L$ and from the left $L f$ of convex functions f and g by a linear map L are convex.

Proof. The first statement is obvious. We prove that $L f$ is convex by estimating $L f(\lambda y_1 + (1-\lambda)y_2)$.

We assume that both $L f(y_1)$ and $L f(y_2)$ are finite (if not, the barycentric inequality is trivial). This implies that we can associate with any $\varepsilon > 0$ elements x_1 and x_2 such that

$$L x_k = y_k \quad \text{and} \quad f(x_k) \leq L f(y_k) + \varepsilon \quad (k = 1, 2).$$

Therefore $x = \lambda x_1 + (1-\lambda)x_2$ satisfies the constraint $L x = \lambda y_1 + (1-\lambda)y_2$.

Hence

$$\begin{aligned} Lf(\lambda y_1 + (1-\lambda)y_2) &\leq f(x) = f(\lambda_1 x + (1-\lambda)x_2) \\ &\leq \lambda f(x_1) + (1-\lambda)f(x_2) \leq \lambda Lf(y_1) + (1-\lambda)Lf(y_2) + \varepsilon. \end{aligned}$$

Letting ε tend to 0, we obtain the inequality

$$Lf(\lambda y_1 + (1-\lambda)y_2) \leq \lambda Lf(y_1) + (1-\lambda)Lf(y_2). \quad \square$$

Definition 7. Let X be a convex subset of a vector space U and let

$$F: x \in X \mapsto F(x) = \{f_1(x), \dots, f_n(x)\} \in \mathbf{R}^n$$

be a map from X into $\mathbf{R}^n$. We say that F is a “convex operator” if and only if its components $f_i: X \rightarrow \mathbf{R}$ are convex.

Proposition 10. Let X be a convex subset of U and let $F: X \rightarrow \mathbf{R}^n$ be a convex operator. Then

$$(27) \quad F(X) + \mathbf{R}_+^n \text{ is a convex subset of } \mathbf{R}^n.$$

Proof. Let $c = \sum_{k=1}^m \alpha_k (F(x^k) + b^k) \in \text{co}(F(X) + \mathbf{R}_+^n)$ be a convex combination of elements of $F(X) + \mathbf{R}_+^n$. Consider $x = \sum_{k=1}^m \alpha_k x^k \in X$ (which is convex) and write

$$c = F(x) + d \quad \text{where } d = \sum_{k=1}^m \alpha_k b^k + \sum_{k=1}^m \alpha_k F(x^k) - F(x).$$

The components d_j of d are non-negative since

$$d_j = \sum_{k=1}^m \alpha_k b_j^k + \sum_{k=1}^m \alpha_k f_j(x^k) - f_j\left(\sum_{k=1}^m \alpha_k x^k\right)$$

and the b^k 's belong to $\mathbf{R}_+^n$ while the functions f_j are convex. $\square$

1.4. Indicators, support functions and gauges

The aim of this section is to associate with any closed convex subset X containing 0 some convex lower semi-continuous functions which characterize it. We first introduce the indicator ψ_X of X defined by $\psi_X(X) = 0$ if $x \in X$ and $\psi_X(x) = +\infty$ if $x \notin X$. The domain of ψ_X is X . The second function is the support function $\sigma_X^\#$ defined on the dual U^* of U by

$$\sigma_X(p) = \sup_{x \in X} \langle p, x \rangle = \sup_{x \in U} [\langle p, x \rangle - \psi_X(x)].$$

We associate with it the *polar subset* X^* of X defined by

$$X^* = \{p \in U^* \text{ such that } \sigma_X^*(p) \leq 1\}.$$

We shall see in the second part of this section that the Hahn–Banach theorem (see Appendix A) amounts to saying that

$$X = \{x \in U \text{ such that } \langle p, x \rangle \leq \sigma_X^*(p) \text{ for all } p \in U^*\}.$$

We continue by proving the *bipolar theorem*, which states that $X^{**} = \overline{\text{co}}(X \cup \{0\})$. In particular, we obtain the following generalization of the Farkas lemma: If X is a closed convex subset containing 0 and if $L(X)$ is closed, then $L(X) = [L^{*-1}(X^*)]^*$. We pursue this point in the fourth part. We associate with X its recession cone $P_\infty(X) = \bigcap_{\lambda \geq 0} \lambda X$ and its barrier cone $P(X^*) = \bigcup_{\lambda \geq 0} \lambda X^*$ and prove that $P_\infty(X) = P(X^*)^*$. We also show that $P(X^*)$ is closed when $P_\infty(X)$ has a non-empty interior.

In the fifth part, we interpret the above results when X is regarded as a production set. We state the main assumptions usually made and note that the indicator is the net cost function and that the support function is the profit function.

We end this section by introducing the gauge $\pi_X(x)$ of X defined by

$$\pi_X(x) = \inf \{\lambda > 0 \text{ such that } \lambda^{-1} x \in X\}.$$

This is a third function characterizing X : We have that

$$X = \{x \in U \text{ such that } \pi_X(x) \leq 1\}.$$

1.4.1. Indicators and support functions

Definition 1. Let X be a subset of U . We shall say that the function $\psi_X = \psi(X; \cdot)$ from U into $[0, +\infty]$ defined by

$$(1) \quad \psi(X; x) = \begin{cases} 0 & \text{if } x \in X, \\ +\infty & \text{if } x \notin X. \end{cases}$$

is the “*indicator*” of the subset X .

The following proposition is obvious.

Proposition 1. *The indicator of X is convex (resp. lower semicontinuous) whenever X is convex (resp. closed).*

It satisfies

$$(2) \quad \psi(X; x) + \psi(Y; x) = \psi(X \cap Y; x).$$

Definition 2. Let U and U^* be two paired spaces and let X be a non-empty subset of U . With any $p \in U^*$, we shall associate

$$(3) \quad \sigma_X^*(p) = \sigma^*(X; p) = \sup_{x \in X} \langle p, x \rangle \in]-\infty, +\infty]$$

and

$$(4) \quad \sigma_X^b(p) = \sigma^b(X; p) = \inf_{x \in X} \langle p, x \rangle \in [-\infty, +\infty[.$$

We shall say that the functions $p \mapsto \sigma^*(X; p)$ and $p \mapsto \sigma^b(X; p)$ are respectively the “upper and lower support functions” of X .

The upper support function σ_X^* is clearly *lower semi-continuous, convex and positively homogeneous*, since it is the pointwise supremum of continuous linear functions. It is *non-negative* if $0 \in X$.

It is related to the indicator of X by the fundamental relation

$$(5) \quad \sigma^*(X; p) = \sup_{x \in U} [\langle p, x \rangle - \psi(X; x)].$$

Proposition 2. *The upper and lower support functions are related by*

$$(6) \quad \sigma^*(X; p) = -\sigma^b(X; -p)$$

The upper support function satisfies the following properties. If λ and μ are positive, then

$$(7) \quad \sigma^*(\lambda X + \mu Y; p) = \lambda \sigma^*(X; p) + \mu \sigma^*(Y; p).$$

Also,

$$(8) \quad \sigma^*(X - Y; p) = \sigma^*(X; p) - \sigma^b(Y; p).$$

If $X \subset Y$, then

$$(9) \quad \sigma^*(X, p) \leq \sigma^*(Y, p).$$

If $\{X_i\}_{i \in I}$ is a family of subsets X_i , then

$$(10) \quad \sigma^*\left(\bigcup_{i \in I} X_i; p\right) = \sup_{i \in I} \sigma^*(X_i; p).$$

Proof. The first statements are obvious. We prove (17).

Since $X_i \subset X = \bigcup_{i \in I} X_i$, $\sigma^*(X_i; p) \leq \sigma^*(X; p)$ and thus,

$$\sup_{i \in I} \sigma^*(X_i; p) \leq \sigma^*(X; p)$$

Conversely, let $x \in X = \bigcup_{i \in I} X_i$. Then $x \in X_i$ for some $i \in I$.

Thus $\langle p, x \rangle \leq \sigma^*(X_i; p) \leq \sup_{i \in I} \sigma^*(X_i; p)$. Hence $\sigma^*(X; p) = \sup_{x \in X} \langle p, x \rangle \leq \sup_{i \in I} \sigma^*(X_i; p)$. $\square$

Definition 3. We shall say that the closed convex subsets of U defined

$$(11) \quad \begin{cases} \text{(i) } X^* = \{p \in U^* \text{ such that } \sigma^*(X; p) \leq 1\} \\ \text{(ii) } X^b = \{p \in U^* \text{ such that } \sigma^b(X; p) \geq -1\} \end{cases}$$

are the “upper and lower polar subsets” of X respectively and that

$$(12) \quad P(X^*) = \{p \in U^* \text{ such that } \sigma^*(X; p) < +\infty\} = \text{Dom } \sigma_X^*$$

is the “barrier cone” of X .

Definition 4. Let X be a subset of U . We shall say that

$$(13) \quad \begin{cases} \text{(i) } X^+ = \{p \in U^* \text{ such that } \langle p, x \rangle \geq 0 \text{ for all } x \in X\}, \\ \text{(ii) } X^- = \{p \in U^* \text{ such that } \langle p, x \rangle \leq 0 \text{ for all } x \in X\} \end{cases}$$

are the “positive and negative polar cones of X ” and that the closed subspace

$$(14) \quad X^\perp = \{p \in U^* \text{ such that } \langle p, x \rangle = 0 \text{ for all } x \in X\}$$

is the “orthogonal complement of X ” or “annihilator of X ”.

The following relations are trivial:

$$(15) \quad \begin{cases} \text{(i) } X^\perp \subset X^- \subset X^* \\ \text{(ii) } X^b = -X^*, \quad X^- = -X^+, \quad X^\perp = X^- \cap X^+. \end{cases}$$

It is also clear that, if $X \subset Y$, then

$$(16) \quad Y^* \subset X^*, \quad Y^+ \subset X^+ \quad \text{and} \quad Y^\perp \subset X^\perp.$$

Proposition 3. Suppose that both X and Y contain 0. Then

$$(17) \quad (X+Y)^+ = X^+ \cap Y^+.$$

Proof. Let $p \in X^+ \cap Y^+$ and $z = x+y \in X+Y$, where $x \in X$ and $y \in Y$. Then $\langle p, x+y \rangle = \langle p, x \rangle + \langle p, y \rangle \geq 0$ i.e., $p \in (X+Y)^+$.

Conversely, if $p \in (X+Y)^+$, then $\langle p, x+y \rangle \geq 0$ whenever $x \in X$ and $y \in Y$. By taking $y = 0$, we deduce that $p \in X^+$ and by taking $x = 0$, that $p \in Y^+$. Thus $p \in X^+ \cap Y^+$. $\square$

Remark. If X is a cone, then clearly $X^- = X^*$ and if X is a vector space, then $X^\perp = X^- = X^*$.

Proposition 4. *Suppose that X is a convex cone in U . Then*

$$(18) \quad \sigma^*(X; p) = \psi(X^-; p) = \begin{cases} 0 & \text{if } p \in X^-, \\ +\infty & \text{if } p \notin X^-. \end{cases}$$

Proof. In the first place, $\sigma^*(X, p) = \sup_{x \in X} \langle p, x \rangle \geq 0$ since 0 belongs to X .

If $p \in X^-$, then $\langle p, x \rangle \leq 0$ for any $x \in X$ and thus, $\sigma^*(X, p) \leq 0$. Therefore, $\sigma^*(X, p) = 0$ whenever $p \in X^-$.

If $p \notin X^-$, then there exists $x \in X$ such that $\langle p, x \rangle = \theta > 0$. Since X is a cone, $\lambda x \in X$ for any $\lambda > 0$. Hence

$$\sigma^*(X, p) \geq \sup_{\lambda > 0} \langle p, \lambda x \rangle = \sup_{\lambda > 0} \lambda \theta = +\infty. \quad \square$$

Proposition 5. *Suppose that X is a convex cone and that Y is any subset. Then*

$$(19) \quad \sigma^*(Y+X; p) = \begin{cases} \sigma^*(Y; p) & \text{if } p \in X^-, \\ +\infty & \text{if } p \notin X^- \end{cases}$$

and

$$(20) \quad \sigma^b(Y+X; p) = \begin{cases} \sigma^b(Y; p) & \text{if } p \in X^+, \\ -\infty & \text{if } p \notin X^+. \end{cases}$$

1.4.2. Reformulation of the Hahn-Banach theorem

Since p is a linear form, $\sup_{x \in X} \langle p, x \rangle = \sup_{x \in \text{co}(X)} \langle p, x \rangle$. Since p is continuous, $\sup_{x \in X} \langle p, x \rangle = \sup_{x \in \bar{X}} \langle p, x \rangle$, where $\bar{X}$ denotes the closure of X . Therefore,

$$(21) \quad \sigma^*(X; p) = \sigma^*(\overline{\text{co}}(X); p)$$

where $\overline{\text{co}}(X)$ denotes the *closed convex hull* of X .

The Hahn-Banach separation theorem amounts to characterizing closed convex subsets by their support functions.

Theorem 1. *Let X be a subset of a Hausdorff locally convex space U . Then*

$$(22) \quad \overline{\text{co}}(X) = \begin{cases} \{x \in U \text{ such that } \langle p, x \rangle \leq \sigma^*(X; p) \text{ for all } p \in U^*\}, \\ \{x \in U \text{ such that } \langle p, x \rangle \geq \sigma^b(X; p) \text{ for all } p \in U^*\}. \end{cases}$$

Proof. Denote by $M = \bigcap_{p \in U} K_p$ the intersection of the subsets

$$(23) \quad K_p = \{x \in U \text{ such that } \langle p, x \rangle \leq \sigma^*(X, p)\}.$$

We notice that the subsets K_p are closed and convex. Therefore, M is also a closed convex subset. Since X is obviously contained in K_p for each p , we deduce that $X \subset M$. Thus, $\overline{\text{co}}(X) \subset M$. Assume that $\overline{\text{co}}(X) \neq M$, i.e. that there exists $x \in M$ which does not belong to $\overline{\text{co}}(X)$.

The Hahn-Banach separation theorem states that there exist a non-zero $p \in U^*$ and $\varepsilon > 0$ such that

$$(24) \quad \forall y \in \overline{\text{co}}(X), \quad \langle p, y \rangle \leq \langle p, x \rangle - \varepsilon$$

This implies that $\sigma^*(X; p) \leq \langle p, x \rangle - \varepsilon \leq \sigma^*(X; p) - \varepsilon$ which is a contradiction. $\square$

Remark. *Closed convex subsets in infinite dimensional spaces.*

Let U be a Hausdorff locally convex space and let $U^* = \mathcal{L}(U, \mathbf{R})$ be its dual. We know that U^* is also isomorphic to the space $\mathcal{L}(U_\sigma, \mathbf{R})$ when U is supplied with the weakened topology $\sigma(U, U^*)$ (see Appendix A, Theorem 6). In other words, the space of continuous linear forms on U coincides with the space of weakly-continuous linear forms on U . Theorem 1 implies the following proposition.

Proposition 6. *Let U be a Hausdorff locally convex space. A convex subset X of U is closed if and only if it is weakly closed.*

Now consider the case when U and U^* are paired spaces. The Mackey theorem states that U^* is isomorphic to the space $\mathcal{L}(U, \mathbf{R})$ of continuous linear forms whenever U is supplied with the Mackey topology $\tau(U, U^*)$ (see Appendix A, Theorem 9). Theorem 1 implies the following result.

Proposition 7. *Let U and U^* be two paired vector spaces. A convex subset X of U is closed in the Mackey topology if and only if it is weakly closed.*

* 1.4.3. The bipolar theorem

We now prove another characterization of closed convex subsets.

Theorem 2. *Let X be a subset of a Hausdorff locally convex space U . The bipolar X^{**} of X is the closed convex hull of $X \cup \{0\}$. The bipolar cone X^{++} of X is the closed convex cone spanned by X . The biorthogonal $X^{\perp\perp}$ of X is the closed vector space spanned by X .*

Proof. Since $X^\perp \subset X^- \subset X^*$, (see (15)), we deduce that $X^{**} \subset (X^-)^* = X^{--} \subset (X^\perp)^* = X^{\perp\perp}$. On the other hand, $X \cup \{0\}$ is clearly contained in X^{**} .

(a) We have that $\overline{\text{co}}(X \cup \{0\})$ is contained in the closed convex subset X^{**} . We shall assume that there exists $x \in X^{**}$ which does not belong to $\overline{\text{co}}(X \cup \{0\})$ and obtain a contradiction. By the Hahn-Banach separation theorem, there exists $p \in U^*$, $p \neq 0$, and $\varepsilon > 0$ such that

$$(25) \quad \langle p, y \rangle \leq \langle p, x \rangle - 2\varepsilon \quad \forall y \in \overline{\text{co}}(X \cup \{0\}).$$

On taking $y = 0$, this implies that $\langle p, x \rangle \geq 2\varepsilon > \varepsilon$. Put $q = p/(\langle p, x \rangle - \varepsilon)$. We deduce from (25) that

$$(26) \quad \langle q, y \rangle \leq \frac{\langle p, x \rangle - 2\varepsilon}{\langle p, x \rangle - \varepsilon} \leq 1 \quad \text{for any } y \in X.$$

Therefore, taking the supremum over X , we deduce that

$$\sup_{y \in X} \langle q, y \rangle \leq 1, \quad \text{i.e. } q \in X^*.$$

Since $x \in X^{**}$, $\langle q, x \rangle \leq 1$.

But $\langle q, x \rangle = \langle p, x \rangle / (\langle p, x \rangle - \varepsilon) > 1$ by the very definition of q . This is the required contradiction.

(b) The closed convex cone P spanned by X is contained in X^{--} . We assume that there exists $x \in X^{--}$ which does not belong to P and obtain a contradiction. By the Hahn-Banach theorem, there exists $p \in U^*$, $p \neq 0$ and $\varepsilon > 0$ such

$$(27) \quad \langle p, y \rangle \leq \langle p, x \rangle - \varepsilon \quad \forall y \in P.$$

Taking $y = 0$, we deduce that $0 < \varepsilon \leq \langle p, x \rangle$. On the other hand, $\sup_{y \in P} \langle p, y \rangle = 0$ since it is bounded above (see Proposition 4). Thus $p \in X^-$ and $\langle p, x \rangle \leq 0$ because $x \in X^{--}$. This is the required contradiction.

(c) We prove the last statement of the theorem as in (b) above except that P is taken to be the closed subspace spanned by X and X^{--} is replaced by $X^{\perp\perp}$. $\square$

As a consequence, we have the following result.

Proposition 8. *Let L belong to $\mathcal{L}(U, V)$ and let X be a convex subset of U containing 0. Then*

$$(28) \quad L(X)^* = L^{*-1}(X^*).$$

Furthermore, if $L(X)$ is a closed convex subset containing 0, we obtain that

$$(29) \quad L(X) = [(L^{*-1}(X^*))]^*.$$

Proof. To say that p belongs to $L(X)^*$ amounts to saying that

$$\forall x \in X, \quad \langle p, Lx \rangle = \langle L^*p, x \rangle \leq 1,$$

i.e. that L^*p belongs to X^* , or that p belongs to $L^{*-1}(X^*)$. This implies that $[L^{*-1}(X^*)]^* = L(X)^{**} = \overline{L(X)}$. $\square$

Remark. We can restate eq. (29) by saying that *there exists* $x \in X$ such that $Lx = y$ if and only if

$$(30) \quad \langle p, y \rangle \leq 1 \quad \text{whenever } L^*p \in X^*.$$

*1.4.4. *Recession cones and barrier cones*

Definition 5. Let X be a subset of U containing 0. We shall say that

$$(31) \quad \begin{cases} \text{(i) } P(X) = \bigcup_{\lambda \geq 0} \lambda X \text{ is the "cone spanned by } X", \\ \text{(ii) } P_\infty(X) = \bigcap_{\lambda \geq 0} \lambda X \text{ is the "recession cone" of } X, \\ \text{(iii) } P(X^*) = \bigcup_{\lambda \geq 0} \lambda X^* \text{ is the "barrier cone" of } X. \end{cases}$$

Proposition 9. Suppose that X is a closed convex subset such that $0 \in X$. Then

$$(32) \quad P_\infty(X) = P(X^*)^-.$$

Proof. Let $x \in P_\infty(X)$ and $p \in P(X^*)$. Since $\lambda x \in X$ for all $\lambda > 0$, we deduce that $\lambda \langle p, x \rangle \leq \sigma^*(X, p) < +\infty$ for any $\lambda > 0$. This implies that $\langle p, x \rangle \leq 0$.

Conversely, let $x \in P(X^*)^-$. We prove that $\lambda x \in X$ for any $\lambda > 0$. Since $P(X^*)$ is a cone, then $\lambda x \in P(X^*)^-$ for any $\lambda > 0$. Thus $\lambda x \in X^{**}$ since $P(X^*)^- \subset (X^*)^- \subset X^{**}$.

Since $0 \in X$ and X is closed, $X^{**} = X$. Thus, $\lambda x \in X$ for any $\lambda > 0$. $\square$

The converse is true under additional assumptions. We need that U is either a finite dimensional space or a topological vector space supplied with the Mackey topology $\tau(U, U^*)$. For this purpose, we use the bipolar theorem:

Theorem 3. Suppose that X is a closed convex set containing 0 and that the interior of its recession cone is non-empty (in the Mackey topology). Then its barrier cone $P(X^*)$ is closed and

$$(33) \quad P(X^*) = P_\infty(X)^-.$$

Finally, $P(X^*)$ is spanned by the compact subsets

$$(34) \quad P = \{p \in P(X^*) \text{ such that } \langle p, \tilde{x} \rangle = -1\}$$

where $\tilde{x} \in \text{Int } P_\infty(X)^-$.

Proof. Suppose that $\tilde{x}$ belongs to the interior of the recession cone $P_\infty(X)$. Then there exists a neighborhood K^* of $\tilde{x}$ such that $\tilde{x} + K^* \subset P_\infty(X)$. Such a neighborhood is the polar subset of a ball of positive radius K (which is compact) when U is a finite dimensional space or the polar subset of a weak compact convex subset containing 0 when U is supplied with the Mackey topology.

Firstly, we prove that

$$(35) \quad \langle p, \tilde{x} \rangle < 0 \quad \text{whenever } p \in P(X^*), \quad p \neq 0.$$

If not, there exists $p_0 \in P(X^*)$, $p_0 \neq 0$ such that $\langle p_0, \tilde{x} \rangle = 0$. This implies that

$$\sup_{x \in K^*} \langle p_0, x \rangle \leq \sup_{x \in P_\infty(X)} \langle p_0, x \rangle \leq 0$$

since $P_\infty(X) = P(X^*)^-$. But this is impossible because $\sup_{x \in K^*} \langle p_0, x \rangle > 0$.

Secondly, we check that P defined by (34) spans $P(X^*)$. If $p \in P(X^*)$, $\langle p, \tilde{x} \rangle = -\lambda$ where $\lambda > 0$ and thus, $p = \lambda \bar{p}$ where $\bar{p} = p/\langle p, \tilde{x} \rangle$ belongs to P .

Thirdly, we prove that P is weakly compact. For this purpose, we notice that the inclusion $\tilde{x} + K^* \subset P_\infty(X)$ implies that, if $p \in P$, then

$$-1 + \sigma^*(K^*; p) = \sigma^*(\tilde{x} + K^*; p) \leq \sigma^*(P_\infty(X); p) \leq 0$$

since $\langle p, \tilde{x} \rangle = -1$ and $p \in P_\infty(X) = P(X^*)^-$.

Thus $\sigma^*(K^*; p) \leq 1$, i.e. $p \in K^{**} = K$ by the bipolar Theorem 2 since K is a (weakly) compact convex subset.

Finally, we prove that $P(X^*)$ is closed. Let $\{p_\mu\}$ be a generalized sequence of elements $p_\mu \in P(X^*)$ converging to p . We prove that p belongs to $P(X^*)$. Either $p = 0 \in P(X^*)$ or $p \neq 0$. In this case, we can suppose that $p_\mu \neq 0$ for all μ . Then $q_\mu = -p_\mu/\langle p_\mu, \tilde{x} \rangle$ belongs to the compact subset P . Hence a subsequence q_ν converges to an element $q \in P$. On the other hand, the subsequence $\langle p_\nu, \tilde{x} \rangle$ converges to $\langle p, \tilde{x} \rangle \leq 0$. Therefore the subsequence $p_\nu = \langle p_\nu, \tilde{x} \rangle q_\nu$ converges to $p = -\langle p, \tilde{x} \rangle q$ which belongs to $P(X^*)$. $\square$

1.4.5. Interpretation: production sets and profit functions

Let U be a commodity space.

We shall represent each particular realization of the production of a firm by a pair $\{x, y\}$ consisting of an output vector $x \in U$ and an input vector $y \in U$. Such a pair $\{x, y\}$ is called a *production process*.

Thus the “*production technology*” set T of a given firm is the subset of $U \times U$ consisting of all possible production processes. If T is a given production

set, we define the “*production correspondence*” which associates with any output $x \in U$ the subset

$$(36) \quad L(x) = \{y \in U \text{ such that } \{x, y\} \in T\}$$

of inputs which can be used to produce x .

Conversely, we can describe a firm by a production correspondence L mapping U into itself and define T to be the set of pairs $\{x, y\}$ where $y \in L(x)$ and x ranges over U (T is said to be the *graph* of the correspondence). (We have already introduced linear operators mapping U into itself to describe a firm (see Section 1.1.3)). Actually, we will use only the image Z of T under the map $\langle x, y \rangle \mapsto x - y$ to describe a firm. We shall say that Z is the *production set* of the firm. When $U = \mathbb{R}^l$, we agree that, if the h^{th} component z_h of z is positive, good h is an output, if $z_h = 0$, good h is not used in the production process and if $z_h < 0$, good h is used as an input.

The *first assumption* we usually make is that

$$(37) \quad 0 \in Z,$$

i.e. it is always possible for the firm to engage in no production.

The *second assumption* which is usually made is that

$$(38) \quad Z \text{ is closed.}$$

This amounts to assuming that, if a given production z is arbitrarily close to possible productions, then the production Z is also possible.

The *third assumption* we will often need is that

$$(39) \quad Z \text{ is convex.}$$

This is a much more restrictive assumption. It is related to the more elementary assumption of additivity and decreasing returns to scale. *Additivity* means that, whenever z^1 and z^2 belong to Z (i.e. are possible), $z^1 + z^2$ also belongs to Z . We say that Z exhibits *decreasing* (resp. *constant*) *returns to scale* if $\lambda z \in Z$ whenever $z \in Z$ and $\lambda \in]0, 1[$ (resp. $\lambda > 0$). The assumption of additivity is inevitable. Decreasing return to scale is not always realistic, since activities are not necessarily divisible even if we assume that the commodities are divisible.

Note that additivity and decreasing return to scale imply that Z is a convex cone (and thus, that Z exhibits constant returns to scale).

The *fourth assumption* which is often required is that

$$(40) \quad -\mathbb{R}_+^l \text{ is the recession cone of } Z.$$

This implies in particular that $Z - \mathbf{R}_+^I = Z$ i.e. Z “admits free disposal”. This means that, if the production z is possible, then the net production of smaller quantities $z' \leq z$ is also possible. This can be done by disposing of (or throwing away) some commodities. This assumption implies that

$$(41) \quad \text{Int } Z = \dot{Z} = Z - \dot{\mathbf{R}}_+^I$$

and that the *barrier cone* of Z is the cone $\mathbf{R}_+^{I*}$ of non-negative prices p (see Theorem 3). Two other assumptions are sometimes made. The first of these is that

$$(42) \quad Z \cap \mathbf{R}_+^I = \{0\}$$

which means that it is impossible to produce positive amounts of *all* goods without inputs (the “land of cockaigne” does not exist). Finally, the property

$$(43) \quad Z \cap -Z = \{0\}$$

represents the *irreversibility of the production possibilities*, i.e. the impossibility of producing the original input as output by consuming the original output as input.

(Note that one important source of irreversibility is the existence of labor or other indispensable factor of production).

Net cost function. — We shall interpret the indicator ψ_Z of Z as the “*net cost function*” of the production set X . The cost $\psi_Z(z)$ of z is either 0 if z is possible or infinite if z is impossible to produce.

Profit function. — Since we interpreted the indicator ψ_Z as a loss function, we shall say that its support function σ_Z^* is the *profit function*, associating with any price $p \in \mathbf{R}^{I*}$ the maximum profit

$$(44) \quad \sigma_Z^*(p) = \sup_{z \in Z} \langle p, z \rangle$$

which can be obtained when the price p prevails.

The profit function is lower semi-continuous, positively homogeneous and convex. If $0 \in Z$, it is non-negative. If we assume that Z is convex and closed, then the *production set* Z can be described by *budgetary constraints*, i.e.

$$(45) \quad Z = \{z \in \mathbf{R}^I \text{ such that } \langle p, z \rangle \leq \sigma_Z^*(p) \text{ for all } p \in \mathbf{R}^{I*}\}$$

i.e. a net production z is possible if and only if, for each price p , its value is not larger than the maximum profit.

Remark. Suppose that Z is the image of the graph T of a production correspondence L . If T is closed and convex, it is characterized by its support function $\sigma_T^\#$ defined by

$$(46) \quad \sigma_T^\#(p, q) = \sup_{\{x, y\} \in T} [\langle p, x \rangle + \langle q, y \rangle] \quad \text{where } p, q \in U^*.$$

The profit function is equal to

$$(47) \quad \sigma_Z^\#(p) = \sup_{\{x, y\} \in T} \langle p, x - y \rangle = \sigma_T^\#(p, -p).$$

We shall say that

$$(48) \quad C(x, p) = \inf_{y \in L(x)} \langle p, y \rangle = \sigma^b(L(x); p)$$

is *the cost function of x* , i.e. the smallest value of the inputs y which can be used to produce x .

Therefore, we can write

$$(49) \quad \begin{aligned} \sigma_Z^\#(p) &= \sup_{x \in U} \sup_{y \in L(x)} (\langle p, x \rangle - \langle p, y \rangle) \\ &= \sup_{x \in U} [\langle p, x \rangle - C(x, p)]. \end{aligned}$$

* 1.4.6. Gauges

Definition 6. Let X be a convex subset of a vector space U . We assume that $0 \in X$. Write

$$(50) \quad \pi_X(x) = \pi(X; x) = \inf \{ \lambda < 0 \text{ such that } \lambda^{-1}x \in X \} \in [0, \infty]$$

and

$$(51) \quad \begin{cases} \text{(i) } B_X(\lambda) = \{x \in U \text{ such that } \pi(X, x) \leq \lambda\}, \\ \text{(ii) } \dot{B}_X(\lambda) = \{x \in U \text{ such that } \pi(X, x) < \lambda\}. \end{cases}$$

We shall say that $\pi_X = \pi(X, \cdot) : U \rightarrow [0, +\infty]$ is the “*gauge of X* ”.

Proposition 10. Let X be a convex subset such that $0 \in X$. The gauge $\pi(X, \cdot)$ satisfies the following properties.

$$(52) \quad \begin{cases} \text{(i) } P(X) = \{x \in U \text{ such that } \pi(X, x) < +\infty\}, \\ \text{(ii) } P_\infty(X) = \{x \in U \text{ such that } \pi(X, x) = 0\}. \end{cases}$$

If U is a topological vector space, then

$$(53) \quad \text{Int}(X) \subset \dot{B}(1) \subset X \subset B_X(1) \subset \bar{X}$$

and

$$(54) \quad \begin{cases} \text{(i)} \quad \pi(X; \lambda x) = \lambda \pi(X; x), \\ \text{(ii)} \quad \pi(X; x+y) \leq \pi(X; x) + \pi(X; y). \end{cases}$$

Moreover, the subsets $B_X(1)$ and $\dot{B}_X(1)$ are convex.

Therefore, if X is a closed convex subset containing 0, we can represent it in the form

$$(55) \quad X = \{x \in U \text{ such that } \pi(X; x) \leq 1\}.$$

Proof. Consider the subset

$$I(x) = \{\lambda > 0 \text{ such that } \lambda^{-1}x \in X\}$$

If $x \notin P(X)$, then $I(x) = \{+\infty\}$ and $\pi(X; x) = +\infty$.

If $x \in P(X)$, then $I(x)$ is a half-line because, if $\lambda \in I(x)$ and $\mu > \lambda$, then $\mu \in I(x)$ since

$$\mu^{-1}x = (\mu^{-1}\lambda)\lambda^{-1}x = (1 - \mu^{-1}\lambda)0 + (\mu^{-1}\lambda)\lambda^{-1}x \in X.$$

Thus

$$\pi(X; x) = \inf_{\lambda \in I(x)} \lambda.$$

(a) To say that $\pi(X; x) < +\infty$ amounts to saying that there exists $\lambda > 0$ such that $\lambda^{-1}x \in X$, i.e. $x \in \lambda X \subset P(X)$.

(b) To say that $\pi(X; x) = 0$ amounts to saying that for all $\lambda > 0$, $\lambda^{-1}x \in X$, i.e. $x \in \lambda X$ for all $\lambda > 0$.

(c) Suppose that $\text{Int}(X) = \dot{X} \neq \emptyset$ and choose $x \in \dot{X}$. There exists $\eta > 0$ such that $x + \eta x = (1/(1+\eta))x$ belongs to X . Thus $\pi(X; x) \leq 1/(1+\eta) < 1$, i.e. $X \in \dot{B}_X(1)$.

(d) Let x belong to $\dot{B}_X(1)$. Then $\pi(X; x) < 1$ and thus, $1 \in I(x)$. Therefore $x \in X$.

(e) If $x \in X$, then $1 \in I(x)$ and thus, $\pi(X; x) \leq 1$, i.e. $x \in B_X(1)$.

(f) If $x \in B_X(1)$, then $\pi(X; x) \leq 1$ and thus, $\lambda^{-1}x \in X$ for all $\lambda > 1$. Since $x = \lim_{\lambda \rightarrow 1} \lambda^{-1}x$, we deduce that x belongs to the closure $\bar{X}$ of X .

(g) Since $I(\theta x) = \theta I(x)$, we deduce that

$$\pi(X; \theta x) = \theta \pi(X; x) \quad \text{for all } \theta > 0.$$

(h) We prove that $\pi(X; x+y) \leq \pi(X; x) + \pi(X; y)$. If either x or y does not belong to $P(X)$, then the right-hand side is infinite and the inequality holds.

Suppose that both x and y belong to $P(X)$. For all $\varepsilon > 0$, there exist $\lambda > 0$ and $\mu > 0$ such that $x \in \lambda X$ and $y \in \mu X$ and $\lambda \leq \pi(X; x) + \frac{1}{2}\varepsilon$, $\mu \leq \pi(X; y) + \frac{1}{2}\varepsilon$.

Therefore, $x+y \in (\lambda+\mu)X$ and thus, $\pi(X; x+y) \leq \lambda+\mu \leq \pi(X; x)+\pi(X; y) + 2 \cdot \frac{1}{2} \varepsilon$ for all $\varepsilon > 0$.

(i) Finally, the gauge $\pi(X; \cdot)$ is convex and thus the subsets $B_X(1) = S(\pi(X; \cdot), 1)$ and $\hat{B}_X(1) = \hat{S}(\pi(X; \cdot), 1)$ are convex. $\square$

Interpretation. Production functions. Suppose that the production set Z satisfies

(56) Z is closed, convex and contains 0.

Then its gauge π_Z defines a "production function" $f_Z = \pi_Z - 1$ in the sense that

(57) $Z = \{z \in \mathbf{R}^1 \text{ such that } f_Z(z) = \pi_Z(z) - 1 \leq 0\}.$

Productions z such that $f_Z(z) = 0$ (or $\pi_Z(z) = 1$) are called "efficient".

The formula

(58) $\pi_Z(z) = \inf \{\lambda > 0 \text{ such that } z \in \lambda Z\}.$

has an obvious interpretation.

The number $\pi_Z(z)$ is the smallest scalar λ by which the production set Z must be "increased" in order to produce z . $\square$

We also show that certain properties of convex subsets induce properties on their gauges.

Proposition 11. *Let X be a convex subset such that $0 \in X$.*

If X is symmetric, then $\pi(X; x) = \pi(X; -x)$. If $\pi(X; x) = \pi(X; -x)$ for all x , then $B_X(1)$ is symmetric.

If X is closed, then $\pi(X; \cdot)$ is lower semi-continuous.

If $\pi(X; \cdot)$ is lower semi-continuous, then $B_X(1)$ is closed.

If $0 \in \overset{\circ}{X}$, then $\pi(X; \cdot)$ is uniformly continuous and there exists a semi-norm $\mathbf{P}$ and a constant M such that

(59) $|\pi(X; x) - \pi(X; y)| \leq M\mathbf{P}(x-y) \text{ for all } x, y.$

Proof. If X is symmetric, then $I(x) = I(-x)$ since, if $\lambda^{-1}x \in X$, then $\lambda^{-1}(-x) \in -X = X$. Thus $\pi(X; -x) = \pi(X; x)$. If X is closed, then $X = B_X(1)$ and thus the sections $B_X(\lambda) = S(\pi(X; \cdot), \lambda)$ are closed. This amounts to saying that $\pi(X; \cdot)$ is lower semi-continuous. If 0 belongs to the interior of X , there exists a semi-norm $\mathbf{P}$ for the topology of U and a constant $M > 0$ such that $\mathbf{P}(z)/M \in X$ for all z such that $\mathbf{P}(z) > 0$. Thus

$$\pi\left(X; \frac{1}{M} \frac{z}{\mathbf{P}(z)}\right) = \frac{1}{M\mathbf{P}(z)} \pi(X; z) \leq 1.$$

If x and y are given and if $p(x-y) > 0$, then

$$||\pi(X; x) - \pi(X; y)|| \leq \pi(X; x-y) \leq MP(x-y). \quad \square$$

Proposition 12. *Let X be a closed convex subset such that $0 \in X$. Then*

$$(60) \quad \forall x \in U, \forall p \in U^*, \quad \langle p, x \rangle \leq \pi(X; x) \sigma^*(X; p).$$

Proof. This inequality holds trivially if either $x \notin P(X)$ or $p \notin P(X^*)$, since the right-hand side is infinite. Assume therefore that $x \in P(X)$ and $p \in P(X^*)$.

If $\pi(X; x) > 0$, then $x/\pi(X; x) \in X$.

Thus $\langle p, x/\pi(X; x) \rangle \leq \sigma^*(X; p)$ and (60) holds.

If $\pi(X; x) = 0$, then $\langle p, x \rangle \leq 0$ for all p because, if there exists $p \in U^*$ such that $\langle p, x \rangle = \theta > 0$, then $\lambda x \in X$ for all $\lambda > 0$ (since $\pi(X; \lambda x) = \lambda \pi(X; x) = 0$).

Thus $\sigma^*(X; p) \geq \sup_{\lambda > 0} [\langle p, \lambda x \rangle] = \sup_{\lambda > 0} \lambda \theta = +\infty$. This contradicts the assumption that $\sigma^*(X; p) < +\infty$. $\square$

CHAPTER 2

EXISTENCE, UNIQUENESS AND STABILITY OF OPTIMAL SOLUTIONS

This chapter deals with general results concerning the existence and uniqueness of an optimal solution (and the structure of the optimal set). The stability properties of optimal solutions are also considered. By this is meant their regularity with respect to perturbations of the loss function and/or the strategy sets.

We begin by noticing that the optimal set of the minimization problem $\alpha = \inf_{x \in X} f(x)$ is equal to $\bigcap_{\lambda > \alpha} S(f, \lambda)$ where $S(f, \lambda) = \{x: f(x) \leq \lambda\}$ is a lower section of f . Hence any property of the lower sections which is preserved under intersection holds for the optimal set. For instance, if f is convex, its sections $S(f, \lambda)$ are convex and so the optimal set is convex.

Since a decreasing sequence of non-empty compact subsets has a non-empty intersection, an optimal solution exists whenever all the sections $S(f, \lambda)$ are compact, i.e. closed and relatively compact. A function with closed sections is said to be lower semi-continuous. We shall say that a function with relatively compact sections is lower semi-compact (Section 2.1). Another instance of a decreasing sequence of sets with non-empty intersection occurs when the sets are closed, non-empty subsets of a complete metric space whose diameters converge to 0. This property holds when f is a quadratic function defined by $f(x) = \frac{1}{2} \|x - u\|^2$, where $\|\cdot\|$ is the norm of a Hilbert space U and X is a closed, convex subset of U . We deduce the existence of projectors of best approximation (Section 2.2).

In Section 2.3, we consider the case when $X = \{x \in U \text{ such that } Lx = w\}$, where w belongs to a Hilbert space V and $L \in \mathcal{L}(U, V)$. The optimal solutions $\bar{x}$ are given by a nice formula: $\bar{p} = u - J^{-1} L^* \bar{p}$, where $\bar{p} = (LJ^{-1}L^*)^{-1}(Lu - w)$ and $J \in \mathcal{L}(U, U^*)$ is the duality operator from U onto U^* .

In Section 2.4, we study the perturbation of the loss function by the "simplest" loss functions (i.e. the continuous linear forms $p: x \rightarrow \langle p, x \rangle$) as p ranges over U^* . If f is a proper loss function, we set $\inf_{x \in U} [f(x) - \langle p, x \rangle] = -f^*(p)$ so that $f^*: p \rightarrow f^*(p)$ is a lower semi-continuous convex function from U^* into $]-\infty, +\infty]$, called the *conjugate function* of f . If we consider the conjugate $f^{**} = (f^*)^*$ of f^* , we obtain a lower semi-continuous convex

function smaller than f . We prove that $f = f^{**}$ if and only if f is lower semi-continuous. Thus $f \leftrightarrow f^*$ defines a symmetric, one to one correspondence between lower semi-continuous convex functions defined on U and U^* respectively.

This conjugacy operation will play a useful role throughout this book. In many situations it proves to be the key which allows us to take advantage of Duality Theory.

Finally in Section 2.5, we study more general perturbations. We examine the continuity properties of the function α defined by $\alpha(y) = \inf_{x \in S(y)} f(x, y)$ in terms of the continuity properties of the perturbed loss function $\{x, y\} \mapsto f(x, y)$ and of the correspondence $y \mapsto S(y)$. For that purpose, we give a short introduction to correspondences. Further properties of correspondences are described in Sections 8.2, 8.3, 9.3, 13.2 and in Chapter 15.

2.1. Existence and uniqueness of an optimal solution

From the fact that $\bigcap_{\lambda > \alpha} S(f, \lambda)$ is the minimal set of $\inf_{x \in U} f(x)$, we deduce the existence of an optimal solution when the proper function $f: U \rightarrow]-\infty, +\infty]$ is lower semi-continuous (i.e. $S(f, \lambda)$ is closed) and lower semi-compact (i.e., $S(f, \lambda)$ is relatively compact). We notice that these two requirements strain in opposite directions in the sense that the number of lower semi-continuous functions increases as the topology is strengthened while the number of lower semi-compact functions decreases.

We also mention that, if n functions $f_i: X \rightarrow \mathbf{R}$ are lower semi-continuous and at least one is lower semi-compact, then $F(X) + \mathbf{R}_+^n$ is closed, where $F(x) = \{f_1(x), \dots, f_n(x)\} \in \mathbf{R}^n$. We have already pointed out that the optimal set is convex when f is a convex function. This implies that there exists at most one solution when the function f is assumed to be strictly convex. A function f has the *non-satiation property* if, for any $x \in \text{Dom } f$, there always exists $y \in \text{Dom } f$ satisfying $f(y) < f(x)$. If f is also a convex function, we show that the minimal set is contained in the boundary of the strategy set X on which f is minimized.

2.1.1. Structure of the optimal set

Let $f: U \rightarrow]-\infty, +\infty]$ be a function with domain X .

We shall denote by

$$(1) \quad \alpha = \inf_{x \in U} f(x) = \inf_{x \in X} f(x)$$

the „minimal value of f ” and by

$$(2) \quad M^b(f) = \{x \in U \text{ such that } f(x) = \alpha\}$$

the “*minimal set*” of f , i.e. the subset of *minimal solutions of the minimization problem* (1).

In the same way, if $f : U \mapsto [-\infty, +\infty]$, we denote by

$$(3) \quad \beta = \sup_{x \in U} f(x) = \sup_{x \in X} f(x)$$

the “*maximal value of f* ” and by

$$(4) \quad M^*(f) = \{x \in U \text{ such that } f(x) = \beta\}$$

the “*maximal set of f* ”, i.e. the subset of “*maximal solutions*” of the maximization problem (3).

It is clear that the minimal set of f is equal to

$$(5) \quad M^b(f) = \bigcap_{\lambda > \alpha} S(f, \lambda).$$

Therefore, any property of subsets which is preserved under intersection satisfied by the sections of f is also satisfied by the minimal subset $M^b(f)$.

Definition 1. Let f be a function mapping U into $]-\infty, +\infty]$ (resp. $[-\infty, +\infty[$). If U is a topological space, we shall say that

(a) f is “*lower semi-continuous*” (resp. “*upper semi-continuous*”) if its sections $S(f, \lambda)$ (resp. $S^*(f, \lambda)$) are closed.

(b) f is “*lower semi-compact*” (resp. “*upper semi-compact*”) if its sections $S(f, \lambda)$ (resp. $S_*(f, \lambda)$) are relatively compact.

If U is a vector space, we shall say that

(c) f is “*quasi-convex*” (resp. “*quasi concave*”) if its sections $S(f, \lambda)$ (resp. $S^*(f, \lambda)$) are convex.

The following are some obvious results concerning the extensions f_X of real valued functions defined on X .

Proposition 1. *Let X be a subset of a topological space U , f be a real valued function on X and f_X its extension to U .*

If X is relatively compact, then f is lower semi-compact. If X is closed and f is lower semi-continuous on X , then f_X is lower semi-continuous on U .

Proof. The sections $S(f, \lambda)$ are contained in $X = \text{Dom } f_X$ and thus are relatively compact when X is relatively compact.

When f is a lower semi-continuous function on the topological subspace X of U , then the sections $S(f, \lambda) = S(f_X, \lambda)$ are closed in X , and, since X is closed, are closed in U . $\square$

Of course, there are examples where f_X is lower semi-continuous on U when

X is *not* closed in U and examples where f_X is lower semi-compact when X is *not* relatively compact.

2.1.2. Existence of an optimal solution

From (5) and Definition 1 we deduce the following theorem.

Theorem 1. (Weierstrass). *Let U be a topological space and let f be a function mapping U into $]-\infty, +\infty]$. If f is lower semi-continuous and lower semi-compact, then the minimal subset M^b of f is non-empty and compact.*

Proof. Since f is both lower semi-continuous and lower semi-compact, the sections $S(f, \lambda)$ are compact subsets for $\lambda \in]\alpha, \lambda_0]$. Since $\bigcap_{i=1}^n S(f, \lambda_i) = S(f, \min_{i=1, \dots, n} \lambda_i)$, any finite intersection of sections $S(f, \lambda)$ is non-empty. Therefore the intersection $M^b(f) = \bigcap_{\alpha < \lambda \leq \lambda_0} S(f, \lambda)$ is non-empty and compact. $\square$

In view of the above theorem, we shall study sufficiency conditions for f to be lower semi-continuous and/or lower semi-compact. A sufficiency condition for the former will be called a “*continuity condition*” and a sufficiency condition for the latter, a “*compactness condition*”.

2.1.3. Continuity versus compactness

The two requirements of continuity and compactness pull in opposite directions. The stronger the topology defined on U , the *more* lower semi-continuous functions there are and the *fewer* lower semi-compact functions.¹

When U is a subset of a finite dimensional vector space, there is no problem, since all vector space topologies are equivalent. Otherwise, if we are to use the above Theorem 1, *the problem of the existence of an optimal solution amounts to finding a topology on the unconstrained strategy set U strong enough to ensure the lower semi-continuity of the loss function f and weak enough to ensure the lower semi-compactness of f .*

* 2.1.4. Lower semi-continuity of convex functions in infinite dimensional spaces

Let U be a Hausdorff locally convex space and let $U^* = \mathcal{L}(U, \mathbf{R})$ be its dual. A weakly lower semi-continuous function is one whose sections are weakly

¹ In other words, let $U_i (i = 1, 2)$ be a set U supplied with two topologies $i = 1, 2$, where the first is stronger than the second. This amounts to saying that the identity map is continuous from U_1 onto U_2 . Therefore any relatively compact subset of U_1 is relatively compact in U_2 and any closed subset of U_2 is closed in U_1 .

closed. Any weakly lower semi-continuous proper function on U is lower semi-continuous.

Proposition 1.2.1 implies that the converse is true for convex and quasi-convex functions.

Proposition 2. *A quasi-convex function $f: U \rightarrow]-\infty, +\infty]$ is lower semi-continuous if and only if it is weakly lower semi-continuous.*

Proof. The sections $S(f, \lambda)$ are convex subsets and so they are closed if and only if they are weakly closed by Proposition 1.2.1. $\square$

The following result for paired spaces is proved similarly but using Proposition 1.2.2.

Proposition 3. *Let U and U^* be two paired vector spaces and f be a quasi-convex function. It is lower semi-continuous for the weak topology if and only if it is lower semi-continuous for the Mackey topology.*

Remark. We will devote Section 3.1 to the study of lower semi-compactness of functions.

2.1.5. Fundamental property of lower semi-continuous and compact functions

The following property of a family of lower semi-continuous and lower semi-compact functions is worth mentioning.

Proposition 4. *Consider n functions $f_i: U \rightarrow]-\infty, +\infty]$ satisfying*

$$(6) \quad \begin{cases} (i) \forall i, f_i \text{ is lower semi-continuous,} \\ (ii) \exists i_0 \text{ such that } f_{i_0} \text{ is lower semi-compact.} \end{cases}$$

Let $X = \bigcap_{i=1}^n \text{Dom } f_i$ and let be the operator from X into $\mathbb{R}^n$ defined by $F(x) = \{f_1(x), \dots, f_n(x)\}$. Then

$$(7) \quad F(X) + \mathbb{R}_+^n \text{ is closed in } \mathbb{R}^n.$$

Proof. Let $\{d^k\}_{k \in \mathbb{N}}$ be a sequence of elements of $F(X) + \mathbb{R}_+^n$ which converges to $d \in \mathbb{R}^n$. Write $d^k = F(x^k) + c^k$. For $k \geq k_0$,

$$(8) \quad f_{i_0}(x^k) \leq d_{i_0}^k \leq d_{i_0} + 1$$

and since f_{i_0} is lower semi-compact, we deduce that x^k stays in a relatively compact subset of X . Hence a subsequence x^l converges to an element x of U .

Since f_i is lower semi-continuous, we deduce that

$$(9) \quad f_i(x) \leq \liminf_i f_i(x^i) \leq \lim d_i^i = d_i \quad \text{for all } i.$$

Hence $d \in F(X) + \mathbf{R}^n$. $\square$

2.1.6. Uniqueness of an optimal solution

We now consider how convexity assumptions can imply uniqueness of the minimal solution. First, we deduce from (5) and Definition 1 the following result.

Proposition 5. *If f is quasi-convex, then the minimal set $M^b(f)$ is convex.*

Convex functions are, of course, quasi-convex.

Definition 2. We shall say that a real valued function f defined on a convex subset X of a vector space is “*strictly convex*” if

$$(10) \quad \forall x, y \in X, \quad f\left(\frac{1}{2}(x+y)\right) < f\left(\frac{1}{2}x\right) + f\left(\frac{1}{2}y\right) \text{ unless } x = y.$$

Proposition 6. *If f is a strictly convex function mapping a convex subset X into $\mathbf{R}$, then there exists at most one minimal solution.*

Proof. If x and y are minimal solutions, then $f(x) = f(y) = \alpha$. Since f is strictly convex, then $f\left(\frac{1}{2}(x+y)\right) < \frac{1}{2}(f(x) + f(y)) = \alpha$. But $M^b(f)$ is convex. Thus $\frac{1}{2}(x+y) \in M^b(f)$. Therefore, $f\left(\frac{1}{2}(x+y)\right) = \alpha$ and we obtain a contradiction unless $x = y$. $\square$

Proposition 7. *Let g and h be two convex functions and let f be defined by $f(x) = [g(x)]^2 + h(x)$. Then g and h are constant on the minimal set $M^b(f)$ of f .*

Proof. Let x_1 and x_2 be two elements of $M^b(f)$. Since the function $a \rightarrow a^2$ is strictly convex, either $g(x_1) = g(x_2)$ or else

$$\begin{aligned} \alpha &= \left[g\left(\frac{x_1+x_2}{2}\right) \right]^2 + h\left(\frac{x_1+x_2}{2}\right) \leq \left[\left(\frac{g(x_1)+g(x_2)}{2} \right) \right]^2 \\ &\quad + \frac{1}{2}(h(x_1) + h(x_2)) < \frac{1}{2}g(x_1)^2 + \frac{1}{2}g(x_2)^2 + \frac{1}{2}h(x_1) + \\ &\quad + \frac{1}{2}h(x_2) = \frac{1}{2}(f(x_1) + f(x_2)) = \alpha \end{aligned}$$

and this is a contradiction. $\square$

2.1.7. *Non-satiation property*

We shall point out a property of f which implies that the minimal set is contained in the boundary of the strategy set X .

Definition 3. We shall say that f has the “*non-satiation*” property if

$$(11) \quad \forall x \in \text{Dom } f, \text{ there exists } \xi \in \text{Dom } f \text{ such that } f(\xi) < f(x).$$

✓ **Proposition 8.** *Suppose that*

$$(12) \quad f \text{ is convex}$$

and that

$$(13) \quad f \text{ has the non-satiation property.}$$

If $\bar{x}$ minimizes f on X , then $\bar{x}$ belongs to the boundary of X .

Proof. Suppose that $\bar{x}$ belongs to the interior of X . We shall obtain a contradiction. For that purpose, let $\xi \in \text{Dom } f$ satisfy $f(\xi) < f(\bar{x})$. Since $\bar{x} \in \text{Int } X$, there exists $\theta > 0$ small enough to ensure that $y = \bar{x} + \theta(\xi - \bar{x}) = (1 - \theta)\bar{x} + \theta\xi$ belongs to X . The contradiction follows from the fact that

$$\begin{aligned} f(y) &= f((1 - \theta)\bar{x} + \theta\xi) \leq (1 - \theta)f(\bar{x}) + \theta f(\xi) \\ &< (1 - \theta)f(\bar{x}) + \theta f(\bar{x}) = f(\bar{x}). \quad \square \end{aligned}$$

Remark. We could replace the convexity assumption (12) by the weaker assumption

$$(14) \quad \begin{aligned} &\text{Whenever } f(\xi) < f(x) \text{ and } \theta \in]0, 1], \text{ then} \\ &f((1 - \theta)x + \theta\xi) < f(x). \end{aligned}$$

Such a function is said to be *semi-strictly quasi-convex*.

2.2. *Minimization of quadratic functionals on convex sets*

We shall prove that a quadratic minimization problem has a unique solution which we shall characterize. Then we define the projectors of best approximation and study them.

2.2.1. Hilbert spaces

Let U be a Hilbert space. This is a vector space supplied with a bilinear form $\{x, y\} \in U \times U \mapsto ((x, y)) \in \mathbb{R}$ called the *scalar product* which satisfies

$$(1) \quad \begin{cases} \forall x, y \in U, ((x, y)) = ((y, x)) & (\text{symmetry}), \\ \forall x = 0, ((x, x)) \geq 0 \end{cases}$$

and whose topology is defined by the norm

$$(2) \quad \|x\| = \sqrt{((x, x))}$$

and which is complete for this topology.

The fact that $\|\cdot\|$ is a norm follows from the *Cauchy-Schwarz* inequality

$$(3) \quad \forall x, y \in U, ((x, y)) \leq \|x\| \cdot \|y\|.$$

We shall prove the existence of a unique optimal solution of the “quadratic minimization problem” (or quadratic program)

$$(4) \quad \alpha = \inf_{x \in X} f(x)$$

where

$$(5) \quad \begin{cases} (i) X \text{ is a } \underline{\text{closed convex subset}} \text{ of } U \\ (ii) f(x) = \frac{1}{2} \|x - u\|^2. \end{cases}$$

The problem amounts to minimizing the distance between the set X and an objective $u \in U$.

2.2.2. Existence and uniqueness of the minimal solution

For this purpose we use the fact that, in complete metric spaces, the intersection of a decreasing sequence of closed, non-empty subsets whose diameters converge to 0 is equal to one point.

Proposition 1. *Let U be a Hilbert space supplied with the norm $\|x\|$, let X be a closed convex subset of U and let f be the quadratic functional defined by*

$$(6) \quad f(x) = \frac{1}{2} \|x - u\|^2 \quad \text{where } u \text{ is given in } U.$$

Then there exists a unique minimal solution $\bar{x} \in X$.

Proof. Let $\lambda_n = \alpha + 1/n$ and let $s_n = \sup_{x, y \in S(f, \lambda_n)} \|x - y\|$. The sections $S(f, \lambda)$ are closed subsets of the complete metric space X and so to prove that

$$(7) \quad M^b(f) = \bigcap_n S(f, \lambda_n) = \{\bar{x}\},$$

(i.e. that $\bar{x}$ is the unique minimal solution) it is enough to show that s_n converges to zero.

Let x, y belong to $S(f, \lambda)$. Adding the equations

$$\|x \pm y\|^2 = \|x\|^2 + \|y\|^2 \pm 2((x, y))$$

we obtain that

$$\begin{aligned} \|x - y\|^2 &= 2\|x - u\|^2 + 2\|y - u\|^2 - \|x + y - 2u\|^2 \\ &= 4 \left[f(x) + f(y) - 2f\left(\frac{x+y}{2}\right) \right]. \end{aligned}$$

Since X is convex, $\frac{1}{2}(x+y)$ belongs to X and thus $f(\frac{1}{2}(x+y)) \geq \alpha$. Therefore, if x and y belong to $S(f, \lambda)$, then $\|x - y\|^2 \leq 8(\lambda - \alpha)$. Thus the diameter δ_n of $S(f, \lambda_n)$ is smaller than or equal to

$$2\sqrt{2} \sqrt{\lambda_n - \alpha} = 2\sqrt{2} \frac{1}{\sqrt{n}}$$

and converges to 0. $\square$

2.2.3. Characterization of the minimal solution

The following characterization of the optimal solution plays a fundamental role.

Proposition 2. *An element $\bar{x} \in X$ is an optimal solution if and only if it is a solution of the variational inequalities*

$$(8) \quad ((\bar{x} - u, \bar{x} - y)) \leq 0 \quad \text{for all } y \in X.$$

Proof. Let y belong to X and $\theta \in [0, 1]$. Then $(1 - \theta)\bar{x} + \theta y = \bar{x} + \theta(y - \bar{x})$ belongs to X because X is convex. Thus

$$\frac{1}{2} \frac{\|\bar{x} - u\|^2 - \|\bar{x} + \theta(y - \bar{x}) - u\|^2}{\theta} \leq 0 \quad \text{for any } \theta > 0.$$

Since $\|\bar{x} - u\|^2 - \|\bar{x} + \theta(y - \bar{x}) - u\|^2 = 2\theta((\bar{x} - u, \bar{x} - y)) + \theta^2\|y - \bar{x}\|^2$, we obtain (8) by letting θ converge to 0.

Conversely, if $\bar{x}$ is a solution of (8), it satisfies $\|\bar{x} - u\|^2 - ((\bar{x} - u, y - u)) = ((\bar{x} - u, \bar{x} - u + (y - u))) = ((\bar{x} - u, \bar{x} - y)) \leq 0$ for all $y \in X$. Therefore, applying the Cauchy-Schwarz inequality, we obtain $\|\bar{x} - u\|^2 \leq ((\bar{x} - u, y - u)) \leq \|\bar{x} - u\| \|y - u\|$ and thus, $f(\bar{x}) \leq f(y)$. $\square$

Proposition 3. *Let X be a closed convex cone. Then $\bar{x} \in X$ is an optimal solution if and only if*

$$(9) \quad \begin{cases} \text{(i)} ((\bar{x}-u, x)) = 0, \\ \text{(ii)} ((\bar{x}-u, y)) \geq 0 \text{ for all } y \in X. \end{cases}$$

Let X be a closed subspace. Then $\bar{x} \in X$ is an optimal solution if and only if

$$(10) \quad ((\bar{x}-u, y)) = 0 \text{ for all } y \in X.$$

Proof. Let X be a closed convex cone. If $\bar{x}$ satisfies (9 (i) and (ii)), it satisfies (8) (by subtracting (9 (ii)) from (9 (i))).

Conversely, let $\bar{x}$ be a solution of (8). Since X is a cone, 0 and $2\bar{x}$ belong to X : we obtain $((\bar{x}-u, \bar{x})) \leq 0$ and $-((\bar{x}-u, \bar{x})) \leq 0$, and thus $((\bar{x}-u, \bar{x})) = 0$. Hence $((\bar{x}-u, y)) \geq 0$.

If X is a closed subspace, equations (10) are clearly equivalent to inequalities (9(ii)). $\square$

2.2.4. Projectors of best approximation

Definition 1. The map $u \rightarrow \bar{x} = t_X(u) \in X$ is called the “*projector of best approximation on X* .”

It is a continuous map. In fact, it satisfies the following properties

Proposition 4. *The projector of best approximation t_X satisfies*

$$(11) \quad \|t_X(x) - t_X(y)\| \leq \|x - y\|.$$

If X is a closed convex cone, t_X also satisfies

$$(12) \quad \begin{cases} \text{(i)} t_X(\lambda x) = \lambda t_X(x) \text{ for any } x \in X, \lambda \geq 0, \\ \text{(ii)} \|x\|^2 = \|t_X(x)\|^2 + \|(1 - t_X)(x)\|^2, \\ \text{(iii)} \|t_X(x)\| \leq \|x\|. \end{cases}$$

If X is a closed subspace, then t_X is linear.

Proof. Since $t_X(x)$ and $t_X(y)$ belong to X , we deduce from the inequalities (8) characterizing $t_X(x)$ and $t_X(y)$ that $((t_X(x) - x, t_X(x) - t_X(y))) \leq 0$ and $((t_X(y) - y, t_X(y) - t_X(x))) \leq 0$. Adding, we obtain that $((x - y - (t_X(x) - t_X(y)), t_X(x) - t_X(y))) \geq 0$. This implies that $\|t_X(x) - t_X(y)\|^2 \leq ((x - y, t_X(x) - t_X(y))) \leq \|x - y\| \|t_X(x) - t_X(y)\|$ by the Cauchy-Schwarz inequality.

If X is a cone and $\lambda > 0$ we deduce from inequalities (9(i) and (ii)) characterizing $t_X(\lambda x)$ that

$$\left(\left(x - \frac{1}{\lambda} t_X(\lambda x), \frac{1}{\lambda} t_X(\lambda x) \right) \right) = \frac{1}{\lambda^2} ((\lambda x - t_X(\lambda x), t_X(\lambda x))) = 0$$

and that, for any $y \in X$,

$$\left(\left(x - \frac{1}{\lambda} t_X(\lambda x), y \right) \right) = \frac{1}{\lambda} ((\lambda x - t_X(\lambda x), y)) \leq 0.$$

This implies that $t_X(x) = (1/\lambda) t_X(\lambda x)$. Furthermore,

$$\begin{aligned} \|x\|^2 &= ((x - t_X(x) + t_X(x), x - t_X(x) + t_X(x))) \\ &= \|x - t_X(x)\|^2 + \|t_X(x)\|^2 + 2((x - t_X(x), t_X(x))) \\ &= \|x - t_X(x)\|^2 + \|t_X(x)\|^2. \end{aligned}$$

Therefore,

$$\|t_X(x)\|^2 \leq \|x\|^2; \quad \|x - t_X(x)\|^2 \leq \|x\|^2.$$

Finally, if X is a subspace, equations

$$\alpha_i((x_i - t_X(x_i), y)) = 0 \quad (\text{where } y \in X)$$

imply that

$$((\sum \alpha_i x_i - \sum \alpha_i t_X(x_i), y)) = 0 \quad \text{for all } y \in X.$$

Thus $t_X(\sum \alpha_i x_i) = \sum \alpha_i t_X(x_i)$.

Remark. When X is a cone, we can show that $(1 - t_X)$ is the projector of best approximation onto the closed convex cone

$$X^\ominus = \{y \in U \text{ such that } \sup_{x \in X} ((x, y)) \leq 0\}.$$

When X is subspace, $1 - t_X$ is the projector onto the closed subspace $X^\oplus = \{y \in U \text{ such that } ((x, y)) = 0 \text{ for all } x \in X\}$.

2.2.5. The duality map from an Hilbert space onto its dual

We shall denote by U^* either the space $\mathcal{L}(U, \mathbf{R})$ of continuous linear forms or a Banach space isomorphic to $\mathcal{L}(U, \mathbf{R})$.

Recall that a Banach space U^* is isometric to $\mathcal{L}(U, \mathbf{R})$ if and only if there exists a bilinear form $\{p, x\} \in U^* \times U \mapsto \langle p, x \rangle \in \mathbf{R}$ satisfying

$$(13) \quad \begin{cases} \text{(i) if } \langle p, x \rangle = 0 \text{ for any } x \in U, \text{ then } p = 0, \\ \text{(ii) if } \langle p, y \rangle = 0 \text{ for any } p \in U^*, \text{ then } x = 0, \\ \text{(iii) } \forall p \in U^*, \|p\|_{U^*} = \sup_{x \in U} |\langle p, x \rangle| / \|x\|. \end{cases}$$

The bilinear form $\langle \cdot, \cdot \rangle$ on $U^* \times U$ is called *the duality pairing*. The isometry j from U^* into $\mathcal{L}(U, \mathbf{R})$ is defined by $p \in U^* \mapsto j(p) \in \mathcal{L}(U, \mathbf{R})$ where

$$j(p) : x \mapsto jp(x) = \langle p, x \rangle.$$

Note that the duality pairing on $\mathcal{L}(U, \mathbf{R}) \times U$ is defined by $\{p, x\} \mapsto p(x)$.

It is often useful to choose for the dual a space U^* isometric to $\mathcal{L}(U, \mathbf{R})$ instead of $\mathcal{L}(U, \mathbf{R})$ (when U^* “looks like” U for instance). Once such a space U^* is chosen (explicitly or implicitly), we shall say that it is the dual of U . If U is a Hilbert space, there exists an isomorphism $J \in \mathcal{L}(U, U^*)$ from U onto U^* defined by

$$(14) \quad \forall x, y \in U, \quad ((x, y)) = \langle Jx, y \rangle$$

called the *duality map*.

This means that for any $x \in X$, Jx is the continuous linear form $y \mapsto \langle Jx, y \rangle = ((x, y))$.

It satisfies

$$(15) \quad \langle Jx, y \rangle = \lim_{\theta \rightarrow 0} \frac{\frac{1}{2} \|x + \theta y\|^2 - \frac{1}{2} \|x\|^2}{\theta}.$$

We can supply the dual U^* with the “dual scalar product”

$$(16) \quad ((p, q))_* = ((J^{-1}p, J^{-1}q)) = \langle p, J^{-1}q \rangle.$$

The associated norm coincides with the dual norm

$$(17) \quad \|p\|_* = \sqrt{((p, p))_*} = \sup_{x \in U} \frac{|\langle p, x \rangle|}{\|x\|}$$

thanks to the Cauchy-Schwarz inequality.

Example. Let $U = \mathbf{R}^I$ be finite dimensional. We shall denote its dual by $U^* = \mathbf{R}^{I^*}$.

Let $e^i = \{0, \dots, 0, 1, 0, \dots, 0\}$ denote the i^{th} element of the canonical basis.

Since any bilinear form $((x, y))$ can be written

$$(18) \quad ((x, y)) = \sum_{i,j=1}^l x_i y_j ((e^i, e^j)) = \sum_{j=1}^l \left(\sum_{i=1}^l ((e^i, e^j)) x_i \right) y_j,$$

it is characterized by a matrix $J = \{((e^i, e^j))\}_{i,j=1, \dots, l}$ from $\mathbf{R}^l$ into $\mathbf{R}^{l*}$.

The bilinear form $((x, y))$ is a scalar product if and only if

$$(19) \quad \begin{cases} \text{(i)} \quad ((e^i, e^j)) = ((e^j, e^i)), \\ \text{(ii)} \quad \forall x \neq 0, \sum_{i,j=1}^l ((e^i, e^j)) x_i x_j > 0. \end{cases}$$

The matrix J is said to be *symmetric* and *positive definite*. Therefore, we can write

$$(20) \quad ((x, y)) = \langle Jx, y \rangle \quad \text{where } J = \{((e^i, e^j))\}_{i,j=1, \dots, l}.$$

A scalar product on $\mathbf{R}^l$ is thus characterized by a symmetric positive definite matrix from $\mathbf{R}^l$ into $\mathbf{R}^{l*}$.

2.3. Minimization of quadratic functionals on subspaces

We shall prove that the optimal solution of $\alpha = \inf_{Lx=w} \frac{1}{2} \|x - u\|^2$ is defined by $\bar{x} = u - J^{-1}L^* \bar{p}$ and $\bar{p} = (LJ^{-1}L^*)^{-1}(Lu - w)$. This implies that $L^+ = J^{-1}L^*(LJ^{-1}L^*)^{-1}$ is a linear continuous right-inverse of the surjective operator $L \in \mathcal{L}(U, V)$ and that $M^- = (M^*JM)^{-1}M^*J$ is a linear continuous left-inverse of an isomorphism M from U into W . We deduce that a problem $\alpha = \inf_{Lx=w} \|Mx - u\|$ is equivalent to the problem $\alpha = \inf_{LM^-z=w} \|z - u\|$.

2.3.1. The fundamental formula

Let

$$(1) \quad L \in \mathcal{L}(U, V) \text{ be a surjective continuous linear operator from a Hilbert space } U \text{ onto a Hilbert space } V.$$

and let

$$(2) \quad u \in U, w \in V \text{ be fixed.}$$

Theorem 1. *Given (1) and (2) the unique solution $\bar{x}$ of the minimization problem*

$$(3) \quad \min_{Lx=w} \frac{1}{2} \|x - u\|^2$$

is defined by

$$(4) \quad \begin{cases} \text{(i)} \quad \bar{x} = u - J^{-1} L^* \bar{p}, \\ \text{(ii)} \quad \bar{p} = (LJ^{-1}L^*)^{-1}(Lu - w) \in V^*. \end{cases}$$

As a corollary, we mention:

Proposition 1. *The unique solution $\bar{x}$ of the minimization problem*

$$(5) \quad \min_{\langle p, x \rangle = r} \frac{1}{2} \|x - u\|^2$$

is defined by

$$(6) \quad \bar{x} = u - \frac{\langle p, u \rangle - r}{\|p\|_*^2} J^{-1} p.$$

Proof. We leave this as an exercise. $\square$

Proof of Theorem 1. If $\bar{x}$ minimizes $\frac{1}{2} \|x - u\|^2$ on the subset $X = \{x \in U \text{ such that } Lx = w\}$, we deduce from (1-8) that $\bar{x} \in X$ satisfies

$$(7) \quad ((\bar{x} - u, \bar{x} - x)) \leq 0 \quad \text{for any } x \in X.$$

But $x \in X$ if and only if $y = \bar{x} - x \in \text{Ker } L$. Thus $\bar{x}$ is the optimal solution if and only if

$$(8) \quad \langle J\bar{x} - Ju, y \rangle = ((\bar{x} - u, y)) = 0 \quad \text{for any } y \in \text{Ker } L,$$

i.e., if and only if

$$(9) \quad J\bar{x} - Ju \in (\text{Ker } L)^\perp = \text{Im } L^*.$$

[The closed range theorem implies that $\text{Im } L^*$ is closed and equal to $(\text{Ker } L)^\perp$. Therefore, there exists (a unique) $\bar{p} \in V^*$ such that $J\bar{x} - Ju = -L^* \bar{p}$, i.e. such that

$$(10) \quad \bar{x} = u - J^{-1} L^* \bar{p}.$$

By applying L to both sides of this equation and recalling that $L\bar{x} = w$, we deduce that $\bar{p}$ is solution of the equation

$$(11) \quad (LJ^{-1}L^*)\bar{p} = Lu - L\bar{x} = Lu - w.$$

It remains to prove that $LJ^{-1}L^*$ is invertible. This follows from Lemma 1 below.

Lemma 1. *If (I) holds, then $LJ^{-1}L^*$ is an isomorphism.*

Proof. We use the Lax–Milgram Proposition 9.2.1 below which asserts that $LJ^{-1}L^*$ is an isomorphism when it is V^* -elliptic, i.e. when there exists a positive constant c such that

$$(12) \quad \forall p \in V^*, \quad \langle (LJ^{-1}L^*)p, p \rangle \geq c \|p\|_{V^*}^2.$$

$$\text{But } \langle LJ^{-1}L^*p, p \rangle = \langle J^{-1}L^*p, L^*p \rangle = \|L^*p\|_{U^*}^2.$$

On the other hand, L^* is a bijective continuous map from V^* onto its closed range $\text{Im } L^*$ which is a Hilbert space. The Banach open mapping principle implies that there exists a positive constant c such that $\|L^*p\|_{U^*} \geq \sqrt{c} \|p\|_{V^*}$. The V^* -ellipticity is therefore proved. $\square$

*2.3.2. Orthogonal right inverse

Proposition 2. Suppose that (I) holds. Then the map $L^+ = J^{-1}L^*(LJ^{-1}L^*)^{-1} \in \mathcal{L}(V, U)$ is a continuous right inverse of L . The product from the left $L\|\cdot\|$ of the norm of U by L is equal to the product from the right of the norm by L^+ , i.e.

$$(13) \quad \forall y \in V, \quad (L\|\cdot\|)(y) = \|L^+y\| = \inf_{Lx=y} \|x\|.$$

Furthermore,

$$(14) \quad \lim_{\theta \rightarrow 0} \frac{\frac{1}{2} \|L^+(y+\theta z)\|^2 - \frac{1}{2} \|L^+y\|^2}{\theta} = \langle (LJ^{-1}L^*)^{-1}y, z \rangle.$$

Proof. We have that $LL^+ = (LJ^{-1}L^*)(LJ^{-1}L^*)^{-1} = 1$. Thus L^+ is a continuous right inverse of L . We can write the optimal solution $\bar{x}$ of (3) in the form

$$(15) \quad \bar{x} = u - L^+Lu + L^+w.$$

In particular, by taking $u = 0$, we deduce that L^+w minimizes $x \mapsto \|x\|$ under the constraint $Lx = w$.

Therefore

$$(L\|\cdot\|)(w) = \|L^+w\|.$$

Finally,

$$\begin{aligned} \lim_{\theta \rightarrow 0} \frac{\frac{1}{2} \|L^+(y+\theta z)\|^2 - \frac{1}{2} \|L^+y\|^2}{\theta} &= \langle (L^+y, L^+z) \rangle \\ &= \langle (L^+)^* JL^*y, z \rangle = \langle (LJ^{-1}L^*)^{-1}y, z \rangle \end{aligned}$$

since we can easily check that

$$(16) \quad (L^+)^* JL^+ = (LJ^{-1}L^*)^{-1}. \quad \square$$

* 2.3.3. Orthogonal left inverse

Let

- (17) $M \in \mathcal{L}(U, W)$ be an *injective* continuous linear operator from a Hilbert space U onto its *closed range*

where

- (18) W is a Hilbert space for the scalar product $((\cdot, \cdot))$
and $K \in \mathcal{L}(W, W^*)$ denotes the duality map.

We consider the case of a loss function f defined by $f(x) = \frac{1}{2} \|Mx - u\|^2$. The optimization problem may be interpreted as that of minimizing the distance from a “service” Mx produced by x to an objective u in the “service space” W .

Proposition 3. *Suppose that (17) holds. Then the unique solution $\bar{x}$ of the minimization problem*

$$\alpha = \min_{x \in U} \frac{1}{2} \|Mx - u\|^2$$

is given by

$$(19) \quad \bar{x} = (M^*KM)^{-1}M^*Ku.$$

Proof. We have that $M\bar{x}$ minimizes $\frac{1}{2} \|z - u\|^2$ on the range $M(U)$. Therefore,

$$\begin{aligned} ((M\bar{x} - u, Mx)) &= \langle KM\bar{x} - Ku, Mx \rangle \\ &= \langle M^*KM\bar{x} - M^*Ku, x \rangle = 0 \quad \text{for any } x \in U. \end{aligned}$$

Hence $\bar{x}$ is the solution of equation

$$(M^*KM)\bar{x} = M^*Ku.$$

Lemma 1 with $L = M^*$ and $K = J^{-1}$ shows that $M^*KM = L^*J^{-1}L$ is invertible. Hence (19) follows. $\square$

Proposition 4. *Suppose that (17) holds. Then the map $M^- = (M^*KM)^{-1}M^*K \in \mathcal{L}(W, U)$ is a continuous left inverse of M . Furthermore*

$$(20) \quad \begin{cases} \text{(i) } (M^-)^* = (M^*)^+, \\ \text{(ii) } M = (M^-)^+. \end{cases}$$

Proof. We have that $M^-M = 1$. Also

$$\begin{aligned} (M^-)^* &= KM(M^*KM)^{-1} = [K^{-1}]^{-1}(M^*)^*[(M^*)[K^{-1}]^{-1}(M^*)^*]^{-1} = \\ &= (M^*)^+. \end{aligned}$$

This implies in particular that

$$M^-K^{-1}(M^-)^* = (M^*KM)^{-1}.$$

Therefore

$$\begin{aligned} (M^-)^+ &= K^{-1}(M^-)^*[M^-K^{-1}(M^-)^*]^{-1} = \\ &= M(M^*KM)^{-1}[(M^*KM)^{-1}]^{-1} = M. \end{aligned}$$

***2.3.4. Another decomposition property**

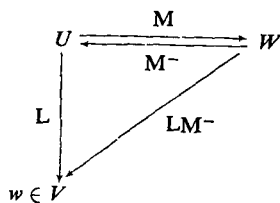
Consider the minimization problem

$$(21) \quad \min_{Lx=w} \frac{1}{2} \|Mx - u\|^2,$$

where U, V, W are Hilbert spaces and

$$(22) \quad \begin{cases} \text{(i) } L \in \mathcal{L}(U, V) \text{ is surjective} \\ \text{(ii) } M \in \mathcal{L}(U, W) \text{ is injective and has a closed range.} \end{cases}$$

Let $M^- = (M^*KM)^{-1}M^*K \in \mathcal{L}(W, U)$ be the left inverse of M . The position is illustrated below.



We have the following decomposition property.

Proposition 5. Suppose that (22) holds. Let $\bar{x} \in U$ be the unique solution of (21). It is defined by

$$(23) \quad \begin{cases} \text{(i) } \bar{x} = M^-u - (M^*KM)^{-1}L^*\bar{p}, \\ \text{(ii) } \bar{p} = [L(M^*KM)^{-1}L^*]^{-1}(LM^-u - w). \end{cases}$$

It is also defined by

$$(24) \quad \bar{x} = M^-z \text{ where } \bar{z} \text{ minimizes } \frac{1}{2} \|z - u\|^2 \text{ under the constraints } LM^-z = w,$$

i.e. by the formula

$$(25) \quad \bar{x} = M^-u - M^-(LM^-)^+LM^-u + M^-(LM^-)^+w.$$

Proof. Let $\bar{x}$ be a solution of (21). Since $Lx = w$ if and only if $\bar{x} - x \in \text{Ker } L$, we deduce that $\langle (M\bar{x} - u, Mx) \rangle = \langle KM\bar{x} - Ku, Mx \rangle = \langle M^*KM\bar{x} - M^*Ku, x \rangle = 0$ whenever $x \in \text{Ker } L$. Hence $M^*KM\bar{x} - M^*Ku$ belongs to $(\text{Ker } L)^\perp = \text{Im } L^*$ and so there exists $\bar{p}$ such that $M^*KM\bar{x} - M^*Ku = L^*\bar{p}$, i.e., such that $\bar{x} = (M^*KM)^{-1}M^*Ku - (M^*KM)^{-1}L^*\bar{p} = M^-u - (M^*KM)^{-1}L^*\bar{p}$. Writing $L\bar{x} = w$, we obtain the relation

$$[L(M^*KM)^{-1}L^*]\bar{p} = LM^-u - w.$$

Applying Lemma 1 to $J = M^*KM$, we deduce that $L(M^*KM)^{-1}L^*$ is invertible. Thus, formula (23) is proved.

Notice that $LM^- = L(M^*KM)^{-1}M^*K$ and $(LM^-)^* = KM(M^*KM)^{-1}L^*$. It follows that $(LM^-)K^{-1}(LM^-)^* = L(M^*KM)^{-1}L^*$. Therefore, we can write

$$(26) \quad \bar{p} = [(LM^-)K^{-1}(LM^-)^*]^{-1}(LM^-u - w).$$

Also,

$$M^-K^{-1}(LM^-)^* = (M^*KM)^{-1}L^*.$$

Therefore, we obtain

$$(27) \quad \begin{aligned} \bar{x} &= M^-(u - K^{-1}(LM^-)^*\bar{p}) \quad \text{where} \\ \bar{p} &= [(LM^-)K^{-1}(LM^-)^*]^{-1}(LM^-u - w). \end{aligned}$$

By Theorem 1, $\bar{z} = u - K^{-1}(LM^-)^*\bar{p}$ minimizes $\frac{1}{2} \|z - u\|^2$ under the constraints $LM^-z = w$. $\square$

*2.3.5. Interpretation

We can, of course, describe many economic problems by the minimization problem $\min_{Lx=w} \frac{1}{2} \|x - u\|^2$.

For instance, we can regard U as the output space, $u \in U$ as a “demand” in output, V as the input space, L as the production operator, $w \in V$ as the “supply” of input, $Lu \in V$ as the input necessary to produce the “demand” u , i.e., the “demand in input”. Hence $\bar{p} = (LJ^{-1}L^*)(Lu - w)$ can be regarded as an input-price, which depends linearly upon the difference between the demand and the supply of input (in other words, this relation describes the *law of supply and demand*). Then $\bar{q} = L^*\bar{p}$ is the associated output price. Therefore the production $\bar{x} = u - J^{-1}\bar{q}$ is the closest to the demand u either under the production constraint $Lx = w$ or the budgetary constraint $\langle \bar{q}, x \rangle = \langle \bar{q}, \bar{x} \rangle$.

2.4. Perturbation by linear forms: conjugate functions

Let U be a topological vector space. The class of "simplest" functions is the dual $U^* = \mathcal{L}(U, \mathbb{R})$, i.e. the space of continuous linear forms in U . Let f be a proper loss function mapping U into $] -\infty, +\infty]$. It is natural to associate with the minimization problem

$$(1) \quad \alpha = \inf_{x \in U} f(x)$$

the perturbed problems

$$(2) \quad -f^*(p) = \inf_{x \in U} [f(x) - \langle p, x \rangle] \in [-\infty, +\infty[$$

for the continuous linear forms $p \in U^*$.

We shall study the relations which exist between the problem and its perturbations. We begin by studying the properties of the function $p \mapsto f^*(p)$ defined on the dual U^* of U . A proof is given that a function f is *convex and lower semi-continuous* if and only if f is equal to $(f^*)^*$. Explicit forms for the conjugates of support functions and functions $\phi(\|\cdot\|)$ of norms are obtained. Finally, some other elementary properties are discussed.

2.4.1. Conjugate functions

First we notice that, $f: U \mapsto]-\infty, +\infty]$ and $x_0 \in \text{Dom } f$, then, for all $p \in U^*$,

$$-\infty < \langle p, x_0 \rangle - f(x_0) \leq \sup_{x \in U} [\langle p, x \rangle - f(x)].$$

Definition 1. We shall say that the function $f^*: U^* \mapsto]-\infty, +\infty]$ which associates with any $p \in U^*$

$$(3) \quad f^*(p) = \sup_{x \in U} [\langle p, x \rangle - f(x)] \in]-\infty, +\infty]$$

is the "*conjugate function*" of the function $f: U \rightarrow]-\infty, +\infty]$. In the same way, we define the "*conjugate function*" g^* of a function $g: U^* \rightarrow]-\infty, +\infty]$ by

$$(4) \quad g^*(x) = \sup_{p \in U^*} [\langle p, x \rangle - g(p)].$$

The "*biconjugate* f^{**} " of a function f is therefore defined on U by $f^{**} = (f^*)^*$.

First, notice that

$$(5) \quad -f^*(0) = \inf_{x \in U} f(x) = \alpha$$

and that the biconjugate f^{**} satisfies

$$(6) \quad f^{**}(x) \leq \sup_{p \in U^*} [\langle p, x \rangle - [\langle p, x \rangle - f(x)]] \leq f(x)$$

by (2) and (3). Also, we have the so-called *Fenchel inequality*

$$(7) \quad \langle p, x \rangle \leq f(x) + f^*(p) \quad \forall x \in U, \forall p \in U^*.$$

Since f^* is the pointwise supremum of a family of continuous affine functions $p \rightarrow \langle p, x \rangle - f(x)$, we have that

$$(8) \quad \text{the functions } f^* \text{ and } f^{**} \text{ are convex and lower semi-continuous.}$$

2.4.2. Characterization of lower semi-continuous convex functions

It follows from (8) that a necessary condition for $f = f^{**}$ is that f be a convex lower semi-continuous function. This is actually also sufficient.

Theorem 1. *A function f defined on a topological vector space U is convex and lower semi-continuous if and only if $f = f^{**}$.*

Remark. In other words, the conjugacy operation $f \rightarrow f^*$ induces a symmetric, one to one correspondence between the cones of convex lower semi-continuous functions defined on U and U^* respectively.

Proof. (a) We assume that $R < f(x)$. Since the pair $\{x, R\}$ does not belong to the epigraph $\mathcal{E}_p(f)$ of f , there exists a continuous linear form $\{p, \alpha\} \in U^* \times \mathbb{R}$ which strictly separates $\{x, R\}$ from the closed convex set $\mathcal{E}_p(f)$ (see Theorem 2.3.1): there exists $\varepsilon > 0$ such that

$$(9) \quad \forall y \in \text{Dom } f, \quad \forall \lambda \geq f(y), \quad \langle p, y \rangle - \alpha \lambda \leq \langle p, x \rangle - \alpha R - \varepsilon.$$

(b) We note that $\alpha \geq 0$; if not, by taking $y = x_0 \in \text{Dom } f$ (which is nonempty by assumption) and $\lambda = f(x_0) + n$, we would have $-\alpha n \leq \langle p, x \rangle - \alpha R - \langle p, x_0 \rangle + \alpha f(x_0) - \varepsilon$, which is impossible if $\alpha < 0$ and n is large enough.

(c) We consider the case where $x \in \text{Dom } f$. By taking $y = x$ and $\lambda = f(x)$, we deduce from (9) that $\alpha(f(x) - R) > \varepsilon$. Hence $\alpha > 0$; by dividing by $\alpha > 0$ and setting $\bar{p} = p/\alpha$, we obtain

$$(10) \quad \forall y \in \text{Dom } f, \quad \langle \bar{p}, y \rangle - f(y) \leq \langle \bar{p}, x \rangle - R - \varepsilon/\alpha.$$

By taking the supremum with respect to y , we deduce that

$$(11) \quad f^*(\bar{p}) \leq \langle \bar{p}, x \rangle - R - \varepsilon/\alpha.$$

This implies that $\bar{p} \in \text{Dom } f^*$ and also, that $R \leq f^{**}(x)$ for any $R < f(x)$. Hence $f(x) = f^{**}(x)$.

(d) We consider the case where $x \notin \text{Dom } f$. If $\alpha > 0$, assertions (10) and (11) still hold true and we deduce that $R \leq f^{**}(x)$ for all R . Hence $f^{**}(x) = +\infty$.

It remains to consider the case where $\alpha = 0$. Inequalities (9) imply that

$$(12) \quad \forall y \in \text{Dom } f, \quad \langle p, y - x \rangle + \varepsilon \leq 0.$$

Let us take $\bar{p} \in \text{Dom}(f^*)$ (which we proved to be non-empty). By multiplying inequality (12) by $n \geq 0$ and adding it to inequality $\langle \bar{p}, y \rangle - f^*(\bar{p}) - f(y) \leq 0$ we obtain

$$(13) \quad \forall y \in \text{Dom } f, \quad \langle \bar{p} + np, y \rangle - n\langle p, x \rangle + n\varepsilon - f^*(\bar{p}) - f(y) < 0.$$

It implies, by taking the supremum with respect to $y \in \text{Dom } f$,

$$(14) \quad f^*(\bar{p} + np) \leq n\langle p, x \rangle - n\varepsilon + f^*(\bar{p}).$$

Therefore,

$$\langle \bar{p}, x \rangle + n\varepsilon - f^*(\bar{p}) \leq \langle \bar{p} + np, x \rangle - f^*(\bar{p} + np) \leq f^{**}(x).$$

By letting $n \rightarrow \infty$, we deduce that $f^{**}(x) = +\infty$. $\square$

2.4.3. Examples of conjugate functions

We begin by pointing out that support functions and indicators are related by conjugacy relations.

Proposition 1. *Let X be a subset. Then its support function σ_X^* is the conjugate of its indicator, i.e. $\sigma_X^* = \psi_X^*$. Conversely, if X is closed and convex, $\psi_X = \sigma_X^{**}$.*

Proof. The fact that $\sigma_X^* = \psi_X^*$ follows immediately from their definitions (Section 1.4). Since ψ_X is convex and lower semi-continuous if and only if X is convex and closed, the converse result follows from Theorem 1. $\square$

We next show that the conjugate function of $\phi(\|\cdot\|)$ is $\phi^*(\|\cdot\|_*)$. We begin with a function

$$(15) \quad \varphi : \mathbf{R}_+ \rightarrow \mathbf{R}_+ \text{ which is continuous, strictly increasing and satisfies } \varphi(0) = 0 \text{ and } \lim_{\lambda \rightarrow \infty} \varphi(\lambda) = \infty.$$

We associate with $\bar{x}$ the convex function Φ defined by

$$(16) \quad \Phi(t) = \begin{cases} \int_0^t \varphi(\lambda) d\lambda & \text{if } t \geq 0, \\ +\infty & \text{if } t < 0. \end{cases}$$

We shall consider the convex function f defined on the Banach space U by

$$(17) \quad f(x) = \Phi(\|x\|).$$

Proposition 2. *The conjugate function of f is defined by*

$$(18) \quad f^*(x) = \Phi^*(\|x\|_*).$$

where Φ^* is the conjugate function of Φ and $\|p\|_* = \sup |\langle p, x \rangle| / \|x\|$ is the dual norm of U^* .

Proof. We have that

$$\begin{aligned} f^*(p) &= \sup_{x \in U} [\langle p, x \rangle - \Phi(\|x\|)] \\ &= \sup_{\lambda > 0} \sup_{\|x\|=\lambda} \left[\lambda \left\langle p, \frac{x}{\|x\|} \right\rangle - \Phi(\lambda) \right] \\ &= \sup_{\lambda > 0} \sup_{\|y\|=1} [\lambda \langle p, y \rangle - \Phi(\lambda)] \\ &= \sup_{\lambda > 0} [\lambda \|p\|_* - \Phi(\lambda)] \\ &= \Phi^*(\|p\|_*). \end{aligned}$$

Example. Consider the case when $\varphi(\lambda) = \lambda^{q-1}$ where $q > 1$. Then

$$\Phi(t) = \begin{cases} \frac{1}{q} t^q & \text{if } t \geq 0, \\ +\infty & \text{if } t < 0 \end{cases}$$

and

$$\Phi^*(t) = \begin{cases} \frac{1}{q^*} t^{q^*} & \text{if } t \geq 0, \\ +\infty & \text{if } t < 0, \end{cases}$$

where $1/q + 1/q^* = 1$.

This follows immediately from the well-known fact that, for any $a, b > 0$,

$$ab \leq \frac{1}{q} a^q + \frac{1}{q^*} b^{q^*}$$

and

$$ab = \frac{1}{q} a^q + \frac{1}{q^*} b^{q^*} \quad \text{if } b = a^{\frac{1}{q-1}}.$$

Therefore, the conjugate function of

$$(19) \quad f_q(x) = \frac{1}{q} \|x\|^q$$

is the function

$$(20) \quad f_{q^*}(p) = \frac{1}{q^*} \|p\|_*^{q^*} \quad \text{where } \frac{1}{q} + \frac{1}{q^*} = 1.$$

For $q = 2$, we find that

$$(21) \quad f_2(x) = \frac{1}{2} \|x\|^2; \quad f_2^*(p) = \frac{1}{2} \|p\|_*^2.$$

2.4.4. Elementary properties of conjugate functions

Proposition 3. If $f \leq g$, then $g^* \leq f^*$. If $g(x) = f(x - x_0) + \langle p_0, x \rangle + \alpha$, then

$$g^*(p) = f^*(p - p_0) + \langle p, x_0 \rangle - (\langle p_0, x_0 \rangle + \alpha).$$

Proof. The first statement is obvious. The second is obtained as follows.

$$\begin{aligned} \sup_{x \in U} [\langle p, x \rangle - g(x)] &= \sup_{x \in U} [\langle p - p_0, x \rangle - f(x - x_0)] - \alpha \\ &= \sup_{x \in U} [\langle p - p_0, x - x_0 \rangle - f(x - x_0)] - \alpha + \langle p - p_0, x_0 \rangle \\ &= f^*(p - p_0) + \langle p, x_0 \rangle - \alpha - \langle p_0, x_0 \rangle. \quad \square \end{aligned}$$

We can also prove the following property.

Proposition 4. Let $L \in \mathcal{L}(U, V)$ and let f be a function from U into $]-\infty, +\infty]$. Then

$$(22) \quad (Lf)^* = f^* L^*.$$

Proof. If $p \in U^*$, we have that

$$\begin{aligned} (Lf)^*(p) &= \sup_{y \in V} [\langle p, y \rangle - \inf_{Lx=y} f(x)] \\ &= \sup_{y \in V} \sup_{Lx=y} [\langle p, y \rangle - f(x)] \\ &= \sup_{x \in U} [\langle p, Lx \rangle - f(x)] = \sup_{x \in U} [\langle L^*p, x \rangle - f(x)] \\ &= f^*(L^*p). \quad \square \end{aligned}$$

Remark. Under suitable assumptions, we shall prove that $(gL)^* = L^*g^*$. (See Proposition 5.2.9.)

In particular, we can compute the conjugate function of an inf-convolution.

Proposition 5. *The conjugate function of the inf-convolution is the sum of the conjugate functions:*

$$(23) \quad (f \square g)^*(p) = f^*(p) + g^*(p)$$

Proof. If $p \in U^*$, we have that

$$\begin{aligned} (f \square g)^*(p) &= \sup_{x \in U} [\langle p, x \rangle - \inf_{y \in U} [f(y) + g(x - y)]] \\ &= \sup_{z \in U} \sup_{y \in U} [\langle p, z \rangle - g(z) + \langle p, y \rangle - f(y)] \\ &= g^*(p) + f^*(p). \quad \square \end{aligned}$$

Remark. Under suitable conditions, we shall prove that $(f+g)^*(p) = (f^* \square g^*)(p)$. (See Proposition 5.2.10.)

Proposition 6. *Let $U = U_1 \times U_2$ be the product of two spaces, $f : U_1 \times U_2 \rightarrow]-\infty, +\infty]$ and $L = L_1 + L_2 \in \mathcal{L}(U_1 \times U_2, V)$ where $L_i \in \mathcal{L}_i(U_i, V)$. Then*

$$(24) \quad [(L_1 + L_2)f]^*(p) = f^*(L_1^*p, L_2^*p).$$

In particular, if $V = U$, $L_1 = 1$, $L_2 = 0$, then

$$(25) \quad (\pi_{U_1}f)^*(p) = f^*(p, 0) \quad \text{where} \quad (\pi_{U_1}f)(x_1) = \inf_{x_2 \in U_2} f(x_1, x_2).$$

2.4.5. Interpretation: cost and profit functions

We may interpret function $g : U \rightarrow]-\infty, +\infty]$ as a “cost function” of a producer. The domain $Z = \text{Dom } g$ can then be viewed as the “production set”. In Example 1.5.5, we considered the particular case when $g = \psi_Z$ is the indicator of Z . If $p \in U^*$ is a price system, $\langle p, z \rangle - g(z)$ is the profit associated with z . Then the conjugate function g^* associates with any price system p the maximum profit $g^*(p) = \sup_{z \in Z} [\langle p, z \rangle - g(z)]$ obtained when price p prevails.

Now, consider m producers $j = 1, \dots, m$, described by their cost functions g_j . If they have to produce $z \in Z = \sum_{j=1}^m Z^j$ (where $Z^j = \text{Dom } g^j$ is the production set of the j^{th} firm), their minimal loss is equal to

$$g(z) = (g \square \dots \square g_m)(z) = \inf_{\sum_{j=1}^m z^j = z} \sum_{j=1}^m g_j(z^j).$$

Hence the profit function

$$g^*(p) = \sup_{z \in Z} [\langle p, z \rangle - g(z)] = \sum_{j=1}^m g_j^*(p)$$

is the sum of the profit functions of each producer by Proposition 5.

2.5. Stability properties: an introduction to correspondences

We investigate in this section the problem of stability of minimization problems, that is the regularity of the function α defined by

$$\alpha(y) = \inf_{x \in S(y)} f(x, y)$$

and of the correspondence M^b defined by

$$M^b(y) = \{x \in X \text{ such that } \alpha(y) = f(x, y)\}$$

when the loss function $f(\cdot, y)$ and the strategy set $S(y)$ depend upon a parameter y .

We are led to introduce “continuity” properties of correspondences S mapping elements $y \in Y$ into subsets $S(y)$ of a topological space R . These are the so-called “upper semi-continuity” and “lower semi-continuity” properties. Both generalize the concept of a continuous map. For instance, we shall prove that if f is lower semi-continuous and S upper semi-continuous with non-empty compact images, then α is lower semi-continuous. Also, we shall prove that if f is upper semi-continuous and S lower semi-continuous with non-empty images, then α is upper semi-continuous. Finally, in order to prove that the correspondence M^b is upper semi-continuous, we study the relations between closed correspondences (correspondences with closed graphs) and upper semi-continuous correspondences. We end this section by studying the construction of upper semi-continuous correspondences. We shall prove the upper semi-continuity of the subdifferential of a continuous convex functions in Section 4.1. A further study of correspondences is undertaken in Section 15.3.

2.5.1. Upper semi-continuous correspondences

✓ **Definition 1.** We shall say that a correspondence $S : Y \rightarrow R$ is “upper semi-continuous” at $y_0 \in Y$ if

- (1) for any neighbourhood $N(S(y_0))$ of $S(y_0)$,
there exists a neighbourhood $N(y_0)$ of y_0 such that
for all $y \in N(y_0)$, $S(y) \subset N(S(y_0))$.

A correspondence is upper semi-continuous on Y if it is upper semi-continuous at any point of Y .

It is clear that such a property generalizes the usual notion of the continuity of a map.

Theorem 1. *Suppose that*

- (i) *the map f is lower semi-continuous on $R \times Y$*
- (ii) *the correspondence $S : X \rightarrow R$ is upper semi-continuous at $y_0 \in Y$*
- ✓(iii) *$S(y_0)$ is compact and the images $S(y)$ are non-empty when $y \in Y$.*

Then the function $\alpha : y \mapsto \alpha(y) = \inf_{x \in S(y)} f(x, y)$ is lower semi-continuous at y_0 .

Proof. We have to prove that for any $\varepsilon > 0$, there exists a neighborhood $N(y_0)$ of y_0 such that

$$(3) \quad \alpha(y_0) - \varepsilon \leq \alpha(y) \quad \text{for all } y \in N(y_0).$$

Since f is lower semi-continuous at any pair $\{x, y_0\}$, we can associate with any x (open) neighborhoods $N(x)$ of x and $N_x(y_0)$ of y_0 such that

$$(4) \quad f(x, y_0) - \varepsilon \leq f(x, y) \quad \text{for all } x \in N(x), y \in N_x(y_0).$$

Since $S(y_0)$ is a non-empty compact subset of U , we can cover it by a finite number n of neighborhoods $N(x_i)$. Therefore,

$$(5) \quad N(S(y_0)) = \bigcup_{i=1}^n N(x_i) \text{ is a neighborhood of } S(y_0).$$

Since the correspondence S is upper semi-continuous at y_0 , there exists a neighborhood $N_0(y_0)$ such that

$$(6) \quad \forall y \in N_0(y_0), \quad S(y) \subset \bigcup_{i=1}^n N(x_i).$$

Consider

$$(7) \quad \text{the neighborhood } N(y_0) = N_0(y_0) \cap \bigcap_{i=1}^n N_{x_i}(y_0) \text{ of } y_0.$$

When $y \in N_0(y_0)$ and $x \in S(y)$, x belongs to a neighborhood $N(x_i)$ (by (6)) and thus, by (4), we obtain

$$f(x_i, y_0) - \varepsilon \leq f(x, y)$$

since $x \in N(x_i)$ and $y \in N_{x_i}(y_0)$.

Thus

$$\alpha(y_0) - \varepsilon \leq \min_{1 \leq i \leq n} f(x_i, y_0) - \varepsilon \leq f(x, y)$$

when $y \in N(y_0)$ and $x \in S(y)$. We may therefore deduce (3). $\square$

Definition 2. Let U and U^* be two paired spaces and let S be a correspondence from Y into U . We shall say that S is "upper hemi-continuous" if

$$(8) \quad \forall p \in U, \quad y \mapsto \sigma^*(S(y), p) \text{ is upper semi-continuous}$$

The following proposition will play an important role later (Chapters 8, 9 and 15).

Proposition 1. Let S be a correspondence with non-empty values from a topological space Y into a Hausdorff locally convex space U supplied with the weak topology. If S is upper semi-continuous, then it is upper hemi-continuous.

Proof. Let $p \in U^*$ and $B = \varepsilon\{-p, p\}^*$, which is a neighbourhood of 0 in U for the weak topology. Since S is upper semi-continuous, there exists a neighbourhood $N(y_0)$ of y_0 such that $S(y) \subset S(y_0) + B$ as y ranges over $N(y_0)$. Therefore

$$(9) \quad \sigma^*(S(y), p) \leq \sigma^*(S(y_0), p) + \sigma^*(B, p) \leq \sigma^*(S(y_0), p) + \varepsilon.$$

This shows that $y \mapsto \sigma^*(S(y), p)$ is upper semi-continuous. $\square$

2.5.2. Lower semi-continuous correspondences

The following concept of lower semi-continuous correspondences is also a generalization of the idea of a continuous map.

Definition 3. We shall say that a correspondence $S : Y \rightarrow R$ is "lower semi-continuous" at $y_0 \in Y$ if

$$(10) \quad \begin{array}{l} \text{for any } x \in S(y_0) \text{ and for any neighborhood } N(x), \\ \text{there exists a neighborhood } N(y_0) \text{ of } y_0 \text{ such that, for all } y \in N(y_0), \\ N(x) \cap S(y) \neq \emptyset. \end{array}$$

A correspondence is lower semi-continuous on Y if it is lower semi-continuous at any point of Y .

Definition 4. We shall say that a correspondence S from Y into R is “*continuous*” at y_0 (resp. on Y) if it is both upper and lower semi-continuous at y_0 (resp. on Y).

Theorem 2. Suppose that

- (11) (i) the map f is upper semi-continuous on $R \times Y$,
 (ii) the correspondence $S : Y \longrightarrow R$ is lower semi-continuous at $y_0 \in Y$,
 (iii) the images $S(y)$ of S are non-empty whenever $y \in Y$.

Then the function $y \mapsto \alpha(y) = \inf_{x \in S(y)} f(x, y)$ is upper semi-continuous at y_0 .

Proof. We have to prove that for any $\varepsilon > 0$, there exists a neighborhood $N(y_0)$ of y_0 such that

$$(12) \quad \alpha(y) \leq \alpha(y_0) + \varepsilon \quad \text{for all } y \in N(y_0).$$

Choose an element $x_0 \in S(y_0)$ such that

$$(13) \quad f(x_0, y_0) \leq \alpha(y_0) + \frac{\varepsilon}{2}.$$

Since f is upper semi-continuous, there exist open neighborhoods $N_0(y_0)$ and $N(x_0)$ such that

$$(14) \quad f(x, y) \leq f(x_0, y_0) + \frac{\varepsilon}{2} \quad \text{for all } x \in N(x_0), y \in N_0(y_0).$$

Since the correspondence S is lower semi-continuous, we can associate with the neighborhood $N(x_0)$ a neighborhood $N_1(y_0)$ such that

$$(15) \quad S(y) \cap N(x_0) \neq \emptyset \quad \text{for any } y \in N_1(y_0).$$

Consider

$$(16) \quad \text{the neighborhood } N(y_0) = N_0(y_0) \cap N_1(y_0).$$

If $y \in N(y_0)$, there exists $x \in S(y)$ which belongs to $N(x_0)$ (by (15)) and which therefore satisfies (14). We deduce that

$$\begin{aligned} \alpha(y) &= \inf_{u \in S(y)} f(u, y) \leq f(x, y) \\ &\leq f(x_0, y_0) + \frac{\varepsilon}{2} \leq \alpha(y_0) + \frac{2\varepsilon}{2} = \alpha(y_0) + \varepsilon. \quad \square \end{aligned}$$

2.5.3. Closed correspondences

These two results give sufficient conditions for the continuity of α . We shall prove further that the minimal sets

$$M^b(y) = \{x \in S(y) \text{ such that } f(x, y) = \alpha(y)\}$$

define a correspondence which is upper semi-continuous.

Theorem 3. *Suppose that*

- (17) (i) *the function f is continuous on $R \times Y$,*
 (ii) *the correspondence S is continuous on Y .*
 (iii) *the subsets $S(y)$ are non empty and compact.*

Then the function $y \mapsto \alpha(y) = \inf_{x \in S(y)} f(x, y)$ is continuous and the correspondence $y \mapsto M^b(y) = \{x \in S(y) \text{ such that } f(x, y) = \alpha(y)\}$ is upper semi-continuous.

Proof. The first statement follows immediately from Theorems 1 and 2. To prove the upper semi-continuity of the correspondence M^b , we write

$$(18) \quad M^b(y) = S(y) \cap T(y),$$

where

$$(19) \quad T(y) = \{x \in U \text{ such that } f(x, y) - \alpha(y) \leq 0\}.$$

Since the functions f and α are continuous on $R \times Y$ and Y respectively, the subset

$$(20) \quad G(T) = \{\{y, x\} \in Y \times R \text{ such that } x \in T(y)\}$$

is a closed subset of $Y \times R$ because, by (19), it can be written as

$$(21) \quad G(T) = \{\{y, x\} \in Y \times U \text{ such that } f(x, y) - \alpha(y) \leq 0\}.$$

Therefore, Theorem 3 is a direct consequence of the following Definition 5 and Proposition 2. $\square$

Definition 5. Let T be a correspondence from a topological space Y into a topological space R . The subset

$$(22) \quad G(T) = \{\{y, x\} \in Y \times R \text{ such that } x \in T(y)\}$$

is called the graph of T . A correspondence T is said to be “closed” if its graph $G(T)$ is a closed subset of $Y \times R$.

Proposition 2. *Let S and T be two correspondences mapping a topological space Y into a topological space R .*

Suppose that

- (23) *(i) the correspondence S is upper semi-continuous at y_0 ,
 (ii) the correspondence T is closed,
 (iii) $S(y_0)$ is compact and the subsets $S(y) \cap T(y)$ are non-empty as y ranges over Y .*

Then the correspondence $S \cap T : y \longrightarrow S(y) \cap T(y)$ is upper semi-continuous at y_0 .

Proof. We have to associate with any open neighborhood $N[S(y_0) \cap T(y_0)]$ of $S(y_0) \cap T(y_0)$ a neighborhood $N(y_0)$ of y_0 such that

$$(24) \quad S(y) \cap T(y) \subset N[S(y_0) \cap T(y_0)] \quad \text{when } y \in N(y_0).$$

If $N[S(y_0) \cap T(y_0)]$ contains $S(y_0)$, then (24) follows from the upper semi-continuity of S at y_0 . If not, we introduce the non-empty subset

$$(25) \quad K = S(y_0) \cap \complement N[S(y_0) \cap T(y_0)]$$

which is compact (since $S(y_0)$ is compact by assumption).

Now, since $K \cap T(y_0)$ is empty by the very definition of K , then the pairs $\{y_0, x\}$ does not belong to $G(T)$ for any x in K .

Therefore, since the graph $G(T)$ of T is closed, we can associate with any $x \in K$ open neighborhoods $N_x(y_0)$ and $N(x)$ of y_0 and x respectively such that $G(T) \cap (N_x(y_0) \times N(x)) = \emptyset$.

Hence, these neighborhoods satisfy

$$(26) \quad N(x) \cap T(y) = \emptyset \quad \text{for any } y \in N_x(y_0).$$

Since K is compact, there exists a finite sequence $\{x_1, \dots, x_n\}$ of elements x_i of K such that

$$(27) \quad K \subset N(K) = \bigcup_{i=1}^n N(x_i).$$

Therefore

$$(28) \quad T(y) \cap N(K) = \emptyset \quad \text{when } y \in \bigcap_{i=1}^n N_{x_i}(y_0).$$

On the other hand, $N(K) \cup N[S(y_0) \cap T(y_0)]$ is an open subset containing $S(y_0)$. Since S is upper semi-continuous at y_0 , there exists a neighborhood $N_0(y_0)$

such that

$$(29) \quad S(y) \subset N(K) \cup N[S(y_0) \cap T(y_0)] \quad \text{when } y \in N_0(y_0).$$

Define the neighborhood $N(y_0)$ by

$$(30) \quad N(y_0) = N_0(y_0) \cap \bigcap_{i=1}^n N_{x_i}(y_0).$$

Then for any $y \in N(y_0)$,

$$(31) \quad \begin{cases} \text{(i) } S(y) \subset N(K) \cup N[S(y_0) \cap T(y_0)], \\ \text{(ii) } T(y) \cap N(K) = \emptyset \end{cases}$$

by (29) and (28) respectively.

Thus, (31) implies (24) as required. $\square$

The latter proposition is very often used in the following particular case.

Proposition 3. *Any closed correspondence T mapping Y into a compact space R is upper semi-continuous.*

Proof. Take S to be the constant correspondence defined by $S(y) = R$ for all $y \in Y$. $\square$

The converse statement is also true.

Proposition 4. *Suppose that a correspondence S is upper semi-continuous with non-empty compact images. Then S is closed.*

Proof. We shall prove that the complement of the graph $G(S)$ is open. Let $\{x_0, y_0\} \notin G(S)$. Then x_0 does not belong to the non-empty compact $S(y_0)$. Thus there exist disjoint neighborhoods $N(x_0)$ and $N[S(y_0)]$ of x_0 and $S(y_0)$. Now, since S is upper semi-continuous, there exists a neighborhood $N(y_0)$ of y_0 such that $S(y) \subset N[S(y_0)]$ as y ranges over $N(y_0)$. This implies that the intersection of the neighborhood $N(x_0) \times N(y_0)$ of $\{x_0, y_0\}$ and of the graph $G(S)$ is empty. $\square$

We also mention the following property.

Proposition 5. *Suppose that S is upper semi-continuous with non-empty compact images. If Y is compact, then $S(Y) = \bigcup_{y \in Y} S(y)$ is also compact.*

Proof. We shall prove that any open covering $\{U_\lambda\}_{\lambda \in A}$ of $S(Y)$ contains a finite covering.

First, since $S(y)$ is compact, it can be covered by a finite number $n(y)$ of such U_{λ_i} . We write

$$(32) \quad S(y) \subset U_y = \bigcup_{1 \leq i \leq n(y)} U_{\lambda_i}$$

Now, since S is upper semi-continuous, there exist open neighborhoods $N(y)$ of y such that $S(N(y)) \subset U_y$.

Since Y is compact, it is contained in a finite union $\bigcup_{1 \leq j \leq p} N(y_j)$ of such open subsets.

Hence, $S(Y)$ can be covered by the finite number p of such $S(N(y_j)) \subset U_{y_j}$, i.e.

$$S(Y) \subset \bigcup_{1 \leq j \leq p} U_{y_j} = \bigcup_{1 \leq j \leq p} \bigcup_{1 \leq i \leq n(y_j)} U_{\lambda_i}.$$

*2.5.4. Construction of upper semi-continuous correspondences

We begin by showing that the product of upper semi-continuous correspondences is upper semi-continuous.

Proposition 6. *Let $S : Y \rightarrow R$ and $T : R \rightarrow W$ be two correspondences. We define their product $TS : Y \rightarrow W$ by*

$$(33) \quad TS(y) = \bigcup_{x \in S(y)} T(x).$$

If S and T are upper semi-continuous, their product is also upper semi-continuous.

Proof. This is trivial. If N is an open neighborhood of $TS(y_0)$, the subset $M = \{x \in R \text{ such that } T(x) \subset N\}$ is open since T is upper semi-continuous. Therefore, since S is also upper semi-continuous, there exists a neighborhood $N(y_0)$ such that $S(N(y_0)) \subset M$. Then, for any $y \in N(y_0)$, $S(y) \subset M$ and $TS(y) \subset N$. Hence TS is upper semi-continuous. $\square$

We now consider n correspondences $S_i : Y \rightarrow R^i$ with non-empty images.

Define $S = S_1 \times \cdots \times S_n$ to be the correspondence associating with any $y \in Y$ the product $S(y) = S_1(y) \times \cdots \times S_n(y) \subset R = \prod_{i=1}^n R^i$.

Proposition 7. *The product $S = S_1 \times \cdots \times S_n$ of n closed correspondences S_i is closed. If the correspondences S_i satisfy*

$$(34) \quad \begin{cases} \text{(i) } S_i \text{ is upper semi-continuous,} \\ \text{(ii) the images } S_i(y) \text{ are compact,} \end{cases}$$

then the product $S = S_1 \times \cdots \times S_n$ is upper semi-continuous.

Proof. The first statement is trivial. If y_μ converges to y and $x_\mu \in S(y_\mu)$ converges to $x \in R$, then $x_\mu^i \in S_i(y_\mu)$ converges to $x^i \in R^i$ for any i . Since S_i is closed, $x^i \in S_i(y)$. Hence $x = \{x^i\}_i$ belongs to $S(y) = \prod_{i=1}^n S_i(y)$.

We now prove the second statement. Let N be a neighborhood of the product $S(y_0)$ of compact subsets $S_i(y_0)$.

We can find n neighborhoods N^i of $S_i(y_0)$ such that

$$(35) \quad S(y_0) \subset \prod_{i=1}^n N^i \subset N.$$

Since the correspondences S_i are upper semi-continuous, there exist neighborhoods $N_i(y_0)$ of y_0 such that $S_i(N_i(y_0)) \subset N^i$. Thus the neighborhood $N_0(y_0) = \bigcap_{1 \leq i \leq n} N_i(y_0)$ of y_0 is mapped into $\prod_{i=1}^n N^i \subset N$. $\square$

Proposition 8. Consider n correspondences S_i mapping Y into a topological vector space U . If, for all i , S_i is upper semi-continuous with non-empty compact images, then the sum S

$$y \mapsto S(y) = S_1(y) + \cdots + S_n(y)$$

of the correspondences S_i is upper semi-continuous.

Proof. This is left as an exercise. $\square$

Finally, we mention the following extension property.

Proposition 9. Let

$$(36) \quad \begin{cases} \text{(i) } D \text{ be a dense subspace of a topological space } Y, \\ \text{(ii) } X \text{ be a compact subset,} \\ \text{(iii) } S : D \rightarrow X \text{ be a closed correspondence.} \end{cases}$$

Then there exists a closed correspondence $\hat{S}$ mapping Y into X satisfying

$$(37) \quad \forall x \in D, \quad S(x) = \hat{S}(x).$$

Proof. We have to construct $\hat{S}$ such that its graph $G(\hat{S})$ is closed. The graph $G(S) \subset D \times X$ is closed in $D \times X$, but not necessarily closed in $Y \times X$. Let $\overline{G(S)}$ be its closure in $Y \times X$. We define $\hat{S}$ by $G(\hat{S}) = \overline{G(S)}$, i.e., by

$$(38) \quad \forall x \in Y, \quad \hat{S}(y) = \{x \in X \text{ such that } \{x, y\} \in \overline{G(S)}\}.$$

Since $G(S)$ is closed in $D \times X$, we have

$$G(S) = \overline{G(S)} \cap (D \times Y) = G(\hat{S}) \cap (D \times Y).$$

This implies that $\hat{S}(y) = S(y)$ for any $y \in D$.

Furthermore, S is closed since its graph $G(S)$ is closed by construction. $\square$

CHAPTER 3

COMPACTNESS AND CONTINUITY PROPERTIES

We proved in the preceding chapter that functions which are lower semi-continuous and lower semi-compact achieve their minimum. We devote this chapter to a more extensive study of these compactness and continuity requirements.

We begin in Section 3.1 by constructing examples of lower semi-compact functions. We prove that semi-coercive functions f (satisfying $\lim_{\|x\| \rightarrow \infty} f(x)/\|x\| > 0$) and functions f such that f^* is continuous at 0 are lower semi-compact. We also study results of less immediate importance.

In Section 3.2 we study the compactness of strategy sets of the form $X = \{x \in R \text{ such that } Lx \in Y\}$. Such a set can be written as $X = \pi_R G$, where π_R is the projection of $R \times Y$ onto R and $G = \{[x, y] \in R \times Y \text{ such that } Lx - y = 0\}$ is the kernel of $L - 1$. Thus X is compact when G is compact. We are therefore led to consider maps M from A into B for which the pre-image of any point is compact.

A continuous map from a topological space A into a topological space B is said to be *proper* if the pre-images of points are compact and if it maps closed subsets.

After characterizing proper maps, we obtain several sufficient conditions for a map $\{x, y\} \mapsto Lx - y$ from $R \times Y$ into V to be proper.

We devote Section 3.3 and Section 3.4 to the continuity of convex functions (Section 3.4 can be passed over in a first reading). We begin by proving that a convex function f is lower semi-continuous at x_0 only if we can associate with any $\varepsilon > 0$ a semi-norm $\mathbf{p}$ and a constant M such that

$$f(x_0) \leq f(x) + M\mathbf{p}(x - x_0) + \varepsilon \quad \text{for all } x$$

and that we can take $\varepsilon = 0$ when f is continuous. In this case, we can even prove that

$$|f'(x) - f(x_0)| \leq M\mathbf{p}(x - x_0) \quad \text{when } (x - x_0) \leq \eta$$

for a convenient $\eta > 0$. We go on to prove that a convex function is continuous on the interior of its domain when it is bounded above on some non-empty

open subset. This implies that any convex function on $\mathbf{R}^n$ is continuous on the interior of its domain and that any convex lower semi-continuous function on a Fréchet space is continuous on the interior of its domain.

We continue in Section 3.4 by showing that any lower semi-continuous convex function is continuous on the *strong* interior of its domain. Using estimates which can be interesting in themselves, we prove that a convex function f is lower semi-compact (for the weak topology) *if and only if* its conjugate f^* is continuous at 0 for the Mackey topology. We end this section by proving that any convex continuous function defined on a compact subset achieves its maximum at an “extremal point” at least.

3.1. Lower semi-compact functions

We devote this section to the study of lower semi-compact functions. For the sake of simplicity, we begin with the case of finite dimensional spaces and prove that semi-coercive functions f (i.e. those which satisfy $\lim_{\|x\| \rightarrow \infty} f(x)/\|x\| > 0$) and functions f whose conjugate function f^* is continuous at 0 are lower semi-compact. Also, we introduce the so-called “constraint qualification hypothesis” $0 \in \text{Int}(L(R) - Y)$, where $R \subset U, Y \subset V$ and $L \in \mathcal{L}(U, V)$. This hypothesis will be met throughout the book. It implies that, if F is a function defined on $R \times Y$, then function $p \mapsto F^*(-L^*p, p)$ is lower semi-compact. We extend these results to the case of infinite dimensional spaces. Finally, we embed X in a convex compact subset $\overline{\mathcal{M}}(X)$ of mixed strategies and we extend a bounded function f on X into a continuous linear function f^A such that

$$\alpha = \inf_{x \in X} f(x) = \inf_{m \in \overline{\mathcal{M}}(X)} f^A(m) = f^A(\bar{m}).$$

3.1.1. Coercive and semi-coercive functions

We give examples of lower semi-compact functions. We begin with the simple case of finite dimensional spaces.

Definition 1. Let X be a subset of a normed space U and let f be a function defined on X .

We shall say that f is “coercive” if

$$(1) \quad \lim_{\|x\| \rightarrow \infty} \frac{f(x)}{\|x\|} = +\infty$$

and “semi-coercive” if

$$(2) \quad \lim_{\|x\| \rightarrow \infty} \frac{f(x)}{\|x\|} > 0.$$

Proposition 1. *Let X be a subset of $\mathbf{R}^n$. Then any semi-coercive function is lower semi-compact.*

Proof. We shall prove that the sections of a semi-coercive function are bounded and use the fact that bounded subsets are relatively compact in $\mathbf{R}^n$.

Suppose that

$$(3) \quad \lim_{\|x\| \rightarrow \infty} \frac{f(x)}{\|x\|} > a > 0.$$

Then, there exists R such that, for any $\|x\| \geq R$,

$$(4) \quad \frac{f(x)}{\|x\|} \geq a.$$

We obtain the inequality

$$\|x\| \leq \max \left(R, \frac{1}{a} f(x) \right).$$

If x belongs to $S(f, \lambda)$, then

$$(5) \quad \|x\| \leq \max \left(R, \frac{\lambda}{a} \right).$$

and so $S(f, \lambda)$ is bounded in $\mathbf{R}^n$. $\square$

Example. The main examples of coercive functions are those derived from norms.

Proposition 2. *The functions $x \mapsto \|x - u\|^p$ are coercive when $p > 1$ and semi-coercive when $p = 1$.*

Proof. We can write

$$\frac{f(x)}{\|x\|} = \|x\|^{p-1} \left\| \frac{x}{\|x\|} - \frac{u}{\|x\|} \right\|^p \geq \|x\|^{p-1} \left| 1 - \frac{\|u\|}{\|x\|} \right|^p.$$

Hence, $\|x\|^{p-1}(1 - \|u\|/\|x\|)^p$ converges to $+\infty$ when $p > 1$ and to a scalar no smaller than 1 if $p = 1$. $\square$

3.1.2. Functions such that f^* is continuous at 0

We now use continuity properties of the conjugate function f^* of f to deduce the lower semi-compactness of f .

Proposition 3. *Let $f: \mathbb{R}^n \mapsto]-\infty, +\infty]$ be a proper function. If its conjugate function f^* is continuous at $0 \in \mathbb{R}^{n*}$, then f is lower semi-compact.*

Proof. If f^* is continuous at 0, there exists a ball of radius η on which f^* is bounded by 1: $f^*(p) \leq 1$ when $\|p\|_* \leq \eta$. Therefore,

$$(6) \quad \begin{aligned} &\forall x \in S(f, \lambda), \forall p \text{ such that } \|p\|_* \leq \eta, \\ &\langle p, x \rangle \leq f(x) + f^*(p) \leq \lambda + 1. \end{aligned}$$

Thus, $\forall p \in \mathbb{R}^{n*}$,

$$(7) \quad \frac{\eta}{\|p\|_*} \langle p, x \rangle \leq (\lambda + 1).$$

Hence, $\forall x \in S(f, \lambda)$,

$$(8) \quad \|x\| = \sup_p \frac{\langle p, x \rangle}{\|p\|_*} \leq \frac{\lambda + 1}{\eta}.$$

The sections of f are bounded, and thus, relatively compact. $\square$

Remark. Proposition 3.4.5 below shows that the converse is true when f is convex and lower semi-continuous.

3.1.3. Lower semi-compactness of linear forms

Let $p \in \mathbb{R}^{n*}$ be a linear function on $\mathbb{R}^n$ and let X be a nonempty subset of $\mathbb{R}^n$. Let $p_X: \mathbb{R}^n \mapsto]-\infty, +\infty]$ be the function defined by

$$(9) \quad p_X(x) = \begin{cases} \langle p, x \rangle & \text{if } x \in X, \\ +\infty & \text{if } x \notin X. \end{cases}$$

Proposition 4. *If p belongs to the interior of the positive polar subset X^b of X , then p_X is lower semi-compact.*

Proof. Indeed, since $p \in \text{Int}(X^b)$, there exists a ball $B(\eta)$ of radius η such that $p + B(\eta) \subset X^b = -X^*$, i.e. such that

$$(10) \quad \forall q \in B(\eta), \quad q - p \in X^*.$$

This amounts to saying that

$$(11) \quad \forall q \in B(\eta), \quad \sigma_X^{\#}(q - p) \leq 1$$

and thus, that $q \mapsto \sigma_X^{\#}(q - p)$ is continuous at 0.

But, $\sigma_X^*(q-p) = p_X^*(q)$ because

$$(12) \quad \sigma_X^*(q-p) = \sup_{x \in X} (\langle q, x \rangle - \langle p, x \rangle) = \sup_{x \in \mathbf{R}^n} (\langle q, x \rangle - p_X(x)).$$

Therefore, p_X is lower semi-compact by Proposition 3. $\square$

Proposition 5. *Let p belong to the interior of the positive polar subset X^b of X . Then X is the union of the relatively compact subsets $H(X, p, \lambda)$ defined by*

$$(13) \quad H(X, p, \lambda) = \{x \in X \text{ such that } \langle p, x \rangle \leq \lambda\}.$$

Proof. The subset $H(X, p, \lambda)$ is nothing other than the section $S(p_X, \lambda)$. $\square$

3.1.4. Constraint qualification hypothesis

Consider also the following example which we shall study thoroughly in Chapter 5.

Let

$$(14) \quad U \text{ and } V \text{ be two topological vector spaces, } L \in \mathcal{L}(U, V).$$

and

$$(15) \quad \begin{cases} \text{(i) } R \subset U \text{ and } Y \subset V \text{ be two subsets of } U \text{ and } V, \\ \text{(ii) } F : R \times Y \rightarrow \mathbf{R} \text{ be a function defined on } R \times Y. \end{cases}$$

Proposition 6. *Suppose that*

$$(16) \quad 0 \in \text{Int} (L(R) - Y) \quad (\text{constraint qualification hypothesis})$$

and that

$$(17) \quad V \text{ is a finite dimensional space.}$$

Then the function $p \in V^* \mapsto F^*(-L^*p, p)$ is lower semi-compact, where F^* is the conjugate function of F defined by

$$(18) \quad F^*(q, p) = \sup_{x \in R} \sup_{y \in Y} [\langle q, x \rangle + \langle p, y \rangle - F(x, y)].$$

Proof. We have to prove that the subsets $S(\lambda) = \{p \in V^* \text{ such that } F^*(-L^*p, p) \leq \lambda\}$ are relatively compact, i.e. bounded, since V^* is a finite dimensional space. For any $z \in V$ assumption (16) implies that there exist $\varepsilon > 0$, $x \in R$ and $y \in Y$

such that $z = \varepsilon(y - Lx)$. Therefore

$$(19) \quad \left\langle p, \frac{z}{\varepsilon} \right\rangle = \langle p, y - Lx \rangle = \langle -L^*p, x \rangle + \langle p, y \rangle \\ \leq F(x, y) + F^*(-L^*p, p) \leq F(x, y) + \lambda$$

when $p \in S(\lambda)$.

This implies that, for any $z \in V$,

$$(20) \quad \sup_{p \in S(\lambda)} \langle p, z \rangle \leq \varepsilon(F(x, y) + \lambda) < +\infty$$

and thus, that $S(\lambda)$ is bounded. $\square$

We now give some examples for which the constraint qualification assumption is satisfied.

Proposition 7. *Suppose that*

$$(21) \quad L(R) \cap \text{Int } Y \neq \emptyset.$$

Then

$$(16) \quad 0 \in \text{Int } (L(R) - Y).$$

Proof. Assumption (21) implies that

$$0 \in L(R) - \text{Int } Y \subset \text{Int } (L(R) - Y). \quad \square$$

Remark. When

$$Y = y - \mathbb{R}_+^n \subset V = \mathbb{R}^n$$

assumption (21) is known as the “Slater condition”. It can then be written in the form

$$(22) \quad \exists \bar{x} \in R \text{ such that, } \forall j = 1, \dots, n, \quad L_j(\bar{x}) < y_j.$$

Proposition 8. *Suppose that R and Y are convex cones. Then $0 \in \text{Int } (L(R) - Y)$ if and only if*

$$(23) \quad V = L(R) - Y.$$

In particular, this assumption is satisfied when

$$(24) \quad X = U, \quad Y = \{0\} \quad \text{and} \quad L \text{ is surjective.}$$

Proof. This is left as an exercise. $\square$

The constraint qualification assumption will play a crucial role for proving the existence of a solution to dual problems (see Chapter 5 and Section 14.1). It implies also that the map $L^* + 1$ from $Y^* \times R^*$ into V^* is proper (see Proposition 3.2.10 below). Finally, it implies that the correspondence S defined by $S(y) = \{x \in R \text{ such that } Lx \in Y - y\}$ is lower semi-continuous (see Theorem 15.3.1).

*3.1.5. Case of infinite dimensional spaces

(a) Case of dual of Banach space

Suppose that

(25) $U = E^* = \mathcal{L}(E, \mathbf{R})$ is the dual of a Banach space.

This is the case when, for instance, U is a *Hilbert space*. We recall that any bounded subset $M \subset E^*$ is relatively compact for the weak topology of the dual E^* of E (see Corollary 4 of Appendix A). The proofs of Section 3.1 can therefore be applied.

Proposition 9. *Suppose that $U = E^*$ is the dual of a Banach space and that the function $f: U \rightarrow]-\infty, +\infty]$ is either semi-coercive or else has a conjugate f^* continuous at 0. Then f is lower semi-compact when $U = E^*$ is supplied with the weak topology.*

*(b) Case of locally convex spaces

Consider two paired spaces U and U^* . We can extend proposition 9 when U is supplied with the weak topology $\sigma(U, U^*)$ and U^* is supplied with the Mackey topology $\tau(U^*, U)$ (the strongest topology on U^* such that U is the dual of U^*).

Recall that the neighborhoods of U^* for the Mackey topology are the polars of the symmetric compact convex subsets of U (for the weak topology).

Proposition 10. *Let U and U^* be two paired spaces and let $f: U \rightarrow]-\infty, +\infty]$ be a proper function. If its conjugate function f^* is continuous at 0 on U^* supplied with the Mackey topology, then f is lower semi-compact on U supplied with the weak topology.*

In particular, if

$U = E^*$ is the dual of a Fréchet space or, more generally, a barreled space E , paired with $U^* = E$,

then the Mackey topology $\tau(U^*, U)$ is the initial topology on E and the weak topology $\sigma(U, U^*)$ on U is the weak topology of the dual E^* of E .

**3.1.6. Extension to compact subsets of mixed strategies*

If X is a compact space, the subset $\mathcal{M}(X)$ of mixed strategies we introduced in Section 1.3.2 is not necessarily compact. We shall embed it in a larger subset $\bar{\mathcal{M}}(X)$ which is compact (and remains convex). The construction of $\bar{\mathcal{M}}(X)$ is analogous to the construction of $\mathcal{M}(X)$ made in Section 1.3.2. The space $\mathcal{S}(x)$ of all functions on X is replaced by

(26) $\mathcal{C}(X)$, the vector space of continuous functions defined on X

which is a *Banach space* when supplied with the norm

$$(27) \quad \|f\| = \sup_{x \in X} |f(x)|.$$

We shall assume that

$$(28) \quad \begin{cases} \text{(i) } \mathcal{C}_+(X) \text{ is the cone of continuous positive functions,} \\ \text{(ii) } \theta \in \mathcal{C}(X) \text{ is the constant function equal to 1;} \end{cases}$$

and introduce

$$(29) \quad \begin{cases} \text{(i) the dual } \mathcal{C}^*(X) \text{ of the Banach space } \mathcal{C}(X) \text{ supplied with the} \\ \text{weak topology,} \\ \text{(ii) the polar cone } \mathcal{C}_+^*(X) \text{ of } \mathcal{C}_+(X): \mathcal{C}_+^*(X) = \{m \in \mathcal{C}^*(X) \text{ such} \\ \text{that } \langle m, f \rangle \geq 0 \text{ whenever } f \in \mathcal{C}_+(X)\}. \end{cases}$$

The elements of $\mathcal{C}^*(X)$ are called “*Radon measures*”.

Definition 2. We denote by $\bar{\mathcal{M}}(X)$ the subset of $\mathcal{C}_+^*(X)$ satisfying $\langle m, \theta \rangle = 1$. The set $\bar{\mathcal{M}}(X)$ is the set of all (*Radon*) *probability measures*.

Proposition 11. Let $\mathcal{C}^*(X)$ be supplied with the weak topology. Then $\bar{\mathcal{M}}(X)$ is a compact convex subset of $\mathcal{C}(X)$. The set $\mathcal{M}(X)$ is a dense subset of $\bar{\mathcal{M}}(X)$, and

$$(30) \quad \inf_{m \in \bar{\mathcal{M}}(X)} \langle m, f \rangle = \inf_{x \in X} f(x); \quad \sup_{m \in \bar{\mathcal{M}}(X)} \langle m, f \rangle = \sup_{x \in X} f(x).$$

Proof. The function θ clearly belongs to the interior of the convex cone $\mathcal{C}_+(X)$. Therefore, $\mathcal{C}_+(X)$ being the positive polar cone of $\mathcal{C}_+^*(X)$, Proposition 5 and Proposition 9 imply that the subset $\{m \in \mathcal{C}_+^*(X) \text{ such that } \langle m, \theta \rangle \geq 1\}$ is compact. Therefore, $\bar{\mathcal{M}}(X)$ is also compact, since it is a closed subset. It is obvious that $\bar{\mathcal{M}}(X)$ is convex.

Now any Dirac measure $\delta(x)$ defines a continuous linear form in $\bar{\mathcal{M}}(X)$, namely $f \in \mathcal{O}(X) \mapsto \langle \delta(x), f \rangle = f(x) \in \mathbb{R}$. Therefore $\mathcal{M}(X)$ is contained in $\bar{\mathcal{M}}(X)$.

Moreover, $\mathcal{M}(X)$ is dense in $\bar{\mathcal{M}}(X)$. If not, there exists $m_0 \in \bar{\mathcal{M}}(X)$ outside the closure of $\mathcal{M}(X)$ in $\mathcal{O}^*(X)$. By the Hahn-Banach extension theorem, there then exists a non-zero $f \in \mathcal{O}(X)$ [regarded as the dual of $\mathcal{O}^*(X)$ supplied with the weak topology] such that

$$(31) \quad \langle m_0, f \rangle \leq \inf_{m \in \mathcal{M}(X)} \langle m, f \rangle - \varepsilon \leq \inf_{x \in X} f(x) - \varepsilon.$$

We shall exhibit a contradiction by proving that

$$(32) \quad \inf_{x \in X} f(x) = \inf_{m \in \bar{\mathcal{M}}(X)} \langle m, f \rangle.$$

The inequality $\alpha = \inf_{x \in X} f(x) \geq \inf_{m \in \bar{\mathcal{M}}(X)} \langle m, f \rangle$ is obvious. It remains to prove that the inequality can be reversed. But $f - \alpha \theta \in \mathcal{O}_+(X)$ and $m \in \bar{\mathcal{M}}(X)$ is positive. Thus $\langle m, f - \alpha \theta \rangle = \langle m, f \rangle - \alpha \langle m, \theta \rangle = \langle m, f \rangle - \alpha \geq 0$ for all $m \in \bar{\mathcal{M}}(X)$. $\square$

3.2. Proper maps and preimages of compact subsets

This section is devoted to the compactness of strategy sets or subsets of allocations defined by

$$X = \{x \in R \text{ such that } Lx \in Y\}$$

where $R \subset U, Y \subset V$ are subsets of topological vector spaces and $L \in \mathcal{L}(U, V)$. If G denotes the graph of L defined by

$$G = \{\{x, y\}\} \in R \times Y \text{ such that } Lx - y = 0\},$$

we prove that X is the projection onto R of G and so it is compact whenever G is compact. Since G is the preimage of $\{0\}$ by the map $M = L - 1$ from $R \times Y$ into V , we are led to introduce "proper maps" M which are *closed continuous maps for which the preimages of points are compact*.

We begin with the following characterization: A map M is proper if and only if, whenever a generalized sequence $y_\mu = Mx_\mu$ converges to $y \in B$, we can extract a generalized subsequence x_ν converging to $x \in A$ such that Mx_ν converges to $y = Mx$. We then show that, if $R \subset \mathbb{R}^I$ is bounded below and $Y \subset \mathbb{R}^I$ is bounded above, then the map $\{x, y\} \in R \times Y \mapsto x - y \in \mathbb{R}^I$ is proper.

The above result can be generalized to the case where Y is a closed convex subset containing an element w such that $(Y-w) \cap \mathbf{R}_+^I = \{0\}$. This result can be improved when $Y = \sum_{j=1}^n Z^j$ is the sum of closed convex subsets. These results will be used throughout our study of the economic models of Chapters 8 and 9.

We end this section by showing that the *constraint qualification hypothesis* $0 \in \text{Int}(L(R)-Y)$ implies that the map $\{p, q\} \in Y^* \times R^* \rightarrow L^*p + q$ is proper.

3.2.1. Proper maps

Proposition 1. *Let X and Y be two topological spaces and let L be a continuous map from X into Y . The two following statements are equivalent.*

- (1) L is closed (i.e., maps closed subsets onto closed subsets) and, $\forall y \in Y, L^{-1}(y)$ is compact.
- (2) From any generalized sequence x_μ of X such that Lx_μ converges to y , we can extract a convergent generalized subsequence.

Proof. It is easy to prove that (2) implies (1). Let B be a closed subset of X . Let $\{L(x_\mu)\}$ be a generalized sequence of elements of $L(B)$ converging to y . We can extract a subsequence $x_{\mu_\nu} \in B$ converging to x , which belongs to B since B is closed. Then $y = L(x) \in L(B)$ since L is continuous. Hence $L(B)$ is closed. Next; let x_μ be any generalized sequence from a preimage $L^{-1}(y)$. Since $L(x_\mu) = y$, we can extract a subsequence x_{μ_ν} which converges to an element x , satisfying $L(x) = y$ because L is continuous. Hence $L^{-1}(y)$ is compact.

Conversely, suppose that (1) holds. Let $y = \lim L(x_\mu)$. Then $y \in \bigcap_\mu \overline{L(V_\mu)}$ where $V_\mu = \{x_\nu\}_{\nu \geq \mu}$. Since L is closed and continuous, $L(V_\mu) = \overline{L(V_\mu)}$. Since $y \in \bigcap_\mu \overline{L(V_\mu)}$, the subsets $L^{-1}(y) \cap V_\mu$ are non-empty compact subsets, which have the finite intersection property. Therefore $L^{-1}(y) \cap \bigcap_\mu V_\mu$ is not empty. Any element x belonging to this subset satisfies $Lx = y$ and is a cluster point of the sequence $\{x_\mu\}$. $\square$

Definition 1. Any continuous map L from X into Y satisfying one of the two above equivalent statements is said to be “proper”.

Proposition 2. *Let L be a proper map from X into Y . Then the pre-image $L^{-1}(K)$ of any compact subset K of Y is compact.*

Proof. Let $\{x_\mu\}$ be a generalized sequence of $L^{-1}(K)$. We can extract a convergent generalized subsequence of the sequence $L(x_\mu)$ which lies in the compact

subset K . Since L is proper, we can again extract a subsequence of $\{x_\mu\}$ which converges to an element x . This limit belongs to $L^{-1}(K)$, since $L^{-1}(K)$ is closed because L is continuous. $\square$

Proposition 3. *If X is compact, any continuous map defined on X is proper. If X is compact, the projection from $X \times Y$ onto Y is proper.*

Proof. This is left as an exercise. $\square$

We also mention the following obvious properties.

Proposition 4. *Let L be a continuous map from X into Y and M a continuous map from Y into Z . Then*

$$(3) \quad \begin{cases} \text{(i) if } L \text{ and } M \text{ are proper, } ML \text{ is proper,} \\ \text{(ii) if } ML \text{ is proper, then } L \text{ is proper,} \\ \text{(iii) if } ML \text{ is proper and } L \text{ is surjective, then } M \text{ is proper.} \end{cases}$$

Proof. This is left as an exercise. $\square$

We now study several examples of proper maps.

3.2.2. Compactness of some strategy sets

We shall study examples for which the map $\{x, y\} \in R \times S \mapsto x + y$ is proper, where R and S are subsets of a vector space V . We use these results to prove the compactness of various subsets of allocations.

Proposition 5. *Consider n subsets $R^i \subset \mathbf{R}^I$ satisfying*

$$(4) \quad \forall i = 1, \dots, n, R^i \text{ is closed and bounded from below.}$$

Then the map $x \in \prod_{1 \leq i \leq n} R^i \mapsto \sum_{1 \leq i \leq n} x^i \in \mathbf{R}^I$ is proper. In particular,

$$(5) \quad \sum_{1 \leq i \leq n} R^i \text{ is closed.}$$

Proof. Let $\{x_m^i\}_{m \in \mathbf{N}}$ be n sequences of elements $x_m^i \in R^i$ such that $y_m = \sum_{i=1}^n x_m^i$ converges to an element y of $\mathbf{R}^I$. Since $R^i \subset \xi^i + \mathbf{R}_+^I$, we obtain that, for all i ,

$$(6) \quad \xi^i \leq x_m^i = y_m - \sum_{j \neq i} x_m^j \leq y_m - \sum_{j \neq i} \xi^j.$$

Since y_m converges, it is bounded and thus, the sequence $\{x_m^i\}$ is bounded. We can therefore extract subsequences (also denoted by) x_m^i which converge

to elements x^i of R^i (since R^i is closed). Thus $y = \sum_{i=1}^n x^i$ and the map $\{x^i\}_{1 \leq i \leq n} \mapsto \sum_{1 \leq i \leq n} x^i$ is proper. $\square$

Proposition 6. *Let*

(7) *R be a closed subset of $\mathbf{R}^l$ which is bounded below*

and let Y be a subset of $V = \mathbf{R}^l$ satisfying

(8)
$$\begin{cases} \text{(i) } Y \text{ is convex and closed,} \\ \text{(ii) } \exists \omega \in Y \text{ such that } (Y - \omega) \cap \mathbf{R}_+^l = \{0\}. \end{cases}$$

Then the map $\{x, y\} \in R \times Y \rightarrow x - y \in \mathbf{R}_+^l$ is proper.

Proof. Let $u_k = x_k - y_k$ be a sequence converging to u in $\mathbf{R}^l$, where $x_k \in R$ and $y_k \in Y$. We begin by proving that a subsequence of y_k converges to y . If not, there would exist a subsequence (also denoted by) y_k such that $\|y_k\| \rightarrow \infty$.

Let $z_k = \frac{y_k}{\|y_k\|}$ belong to the unit sphere $S(1) \subset \mathbf{R}^l$. Since $S(1)$ is compact, there exists a subsequence (again denoted by) z_k which converges to an element $\bar{z} \in S(1)$. Since u_k converges to u , it is bounded and there exists $m \in \mathbf{R}^l$ such that $u_k \leq m$. Also, because R is bounded below, $R \subset \xi + \mathbf{R}_+^l$. Therefore, $y_k = x_k - u_k \geq \xi - m$. Hence, the inequalities $z_k \geq (\xi - m)/\|y_k\|$ imply that $\bar{z} \in \mathbf{R}_+^l$.

On the other hand, since $\|y_k\| \geq 1$ for k large enough, and since $\omega \in Y$, we have that $(1/\|y_k\|)y_k + (1 - 1/\|y_k\|)\omega \in Y$ (because Y is convex). Letting k tend to ∞ , we deduce that

$$\bar{z} + \omega \in Y \text{ (because } Y \text{ is closed).}$$

We have therefore proved that $\bar{z} \in (Y - \omega) \cap \mathbf{R}_+^l$. By assumption (8(ii)), we deduce that $\bar{z} = 0$ and this is a contradiction. Thus a subsequence $\{y_k\}$ converges to an element y of Y (which is closed). Therefore the subsequence $x_k = u_k + y_k$ converges to $x = u + y$ which belongs to X since X is closed. $\square$

Propositions 5 and 6 imply the following result.

Proposition 7. *Suppose that*

(9) $\forall i = 1, \dots, n, R^i \subset \mathbf{R}^l \text{ is closed and bounded below}$

and that Y satisfy properties (8).

Then the map $\{x^1, \dots, x^n, y\} \in \prod_{i=1}^n R^i \times Y \mapsto \sum_{i=1}^n x_i - y \in \mathbf{R}^l$ is proper. In particular, the subset $\sum_{i=1}^n R^i - Y$ is closed, the subset $X = \{x \in \prod_{i=1}^n R^i \text{ such that } \sum_{i=1}^n x^i \in Y\}$ is compact and its projections $\tilde{R}^i = R^i \cap (Y - \sum_{j \neq i} R^j)$ are compact.

Proposition 8. Consider m subsets Z^j of $\mathbf{R}^l$ satisfying

$$(10) \quad \begin{cases} \text{i) } \forall j = 1, \dots, m, Z^j \text{ is closed and } 0 \in Z^j, \\ \text{ii) } Z = \sum_{j=1}^m Z^j \text{ is closed and convex,} \\ \text{iii) } Z \cap -Z = \{0\} \text{ and } Z \cap \mathbf{R}_+^l = \{0\}. \end{cases}$$

Then the map $z = \{z^1, \dots, z^m\} \in \prod_{j=1}^m Z^j \mapsto \sum_{j=1}^m z^j \in \mathbf{R}^l$ is proper.

Proof. Let $u_k = \sum_{j=1}^m z_k^j$ be a sequence of $\sum_{j=1}^m Z^j$ converging to $u \in \mathbf{R}^l$. Then u_k is bounded and there exists a vector $d \in \mathbf{R}^l$ such that

$$(11) \quad \sum_{j=1}^m z_k^j = u_k \geq d.$$

We have to prove that for all j , there exists a convergent subsequence of z_k^j . Let us assume the contrary. There exists j_0 such that the sequence $z_k^{j_0}$ does not converge. Hence

$$(12) \quad \alpha_k = \max_{1 \leq j \leq m} \|z_k^j\| \text{ goes to } \infty$$

while the sequence $y_k^j = z_k^j / \alpha_k$ stay in the unit ball of $\mathbf{R}^l$. Therefore, there exists a subsequence (again denoted by) y_k^j such that, $\forall j = 1, \dots, m$, y_k^j converges to y^j . Then, we deduce that

$$(13) \quad \sum_{j=1}^m y^j \geq \lim_{k \rightarrow \infty} \frac{d}{\alpha_k} = 0.$$

Since 0 belongs to the convex subset Z , then $1/\alpha_k (\sum_{j=1}^m z_k^j) + (1 - 1/\alpha_k) 0 \in Z$. By taking the limit, we deduce that $\sum_{j=1}^m y^j \in Z$ since Z is closed. The assumption that $Z \cap \mathbf{R}_+^l = \{0\}$ implies that $\sum_{j=1}^m y^j = 0$. This can be written as

$$(14) \quad 0 + \sum_{j \neq 1} y^j = -\left(y^1 + \sum_{j \neq 1} 0\right).$$

Since 0 belongs to Z^j for all j , the left-hand side belongs to Z and the right-hand side belongs to $-Z$. Hence the assumption that $Z \cap -Z = \{0\}$ implies that $y^1 = -(\sum_{j \neq 1} y^j) \in Z \cap -Z = \{0\}$.

We have now proved that $\forall j$, $y_k^j = z_k^j / \alpha_k$ converges to 0. This implies that the equations $\alpha_k = \|z_k^j\|$ can hold for at most finitely many k for fixed j . But *this*

is a contradiction, since the very definition of α_k implies that such equations must hold for at least one j for each k , and hence, for infinitely many k for some j . $\square$

Propositions 5 and 8 imply the following consequence.

Proposition 9. Suppose that the subsets $R^i \subset \mathbf{R}^l$ satisfy

(15) $\forall i = 1, \dots, n, \quad R^i$ is closed and bounded below

and that the subsets $Z^j \subset \mathbf{R}^l$ satisfy

$$(16) \quad \begin{cases} \text{(i) } \forall j, Z^j \text{ is closed and } \sum_{j=1}^m Z^j = Z \text{ is closed and convex,} \\ \text{(ii) } Z \cap \mathbf{R}_+^l = \{0\}, \\ \text{(iii) } Z \cap -Z = \{0\}, \\ \text{(iv) } \forall j = 1, \dots, m, 0 \in Z^j \end{cases}$$

Then the map $\{x, z, p\} \in \prod_{i=1}^n R^i \times \prod_{j=1}^m Z^j \times \mathbf{R}_+^l \mapsto \sum_{i=1}^n x^i - \sum_{j=1}^m z^j + p$ is proper.

In particular, the subset

$$(17) \quad \sum_{i=1}^n R^i - \sum_{j=1}^m Z^j + \mathbf{R}_+^l \text{ is closed.}$$

Also, the subset

$$U_\omega = \left\{ \{x, z\} \text{ such that } \sum_{i=1}^n x^i - \sum_{j=1}^m z^j \leq \omega \right\}$$

is compact and is contained in $\prod_{i=1}^n \tilde{R}_\omega^i \times \prod_{j=1}^m \tilde{Z}_\omega^j$, where

$$(18) \quad \tilde{R}_\omega^i = R^i \cap \left(\omega - \mathbf{R}_+^l - Z - \sum_{j \neq i} R^j \right) \text{ is compact}$$

and

$$(19) \quad \tilde{Z}_\omega^j = Z^j \cap \left(\sum_{i=1}^n R^i + \mathbf{R}_+^l - \omega - \sum_{k \neq j} Z^k \right) \text{ is compact.}$$

3.2.3. Examples where the map $L^ + I$ is proper

Proposition 10. Suppose that

$$(20) \quad \begin{cases} \text{(i) } R \text{ is a subset of the dual } U = F^* \text{ of a Fréchet space } F \\ \text{(i.e. metrisable and complete)} \\ \text{(ii) } Y \text{ is a subset of a barreled space } V \end{cases}$$

and that

$$(21) \quad 0 \in \text{Int} (L(R) - Y) \text{ (constraint qualification hypothesis).}$$

Then the map $\{p, q\} \in Y^* \times R^* \mapsto L^*p + q \in F$ is proper (when $Y^* \times R^*$ is supplied with the weak topology and F with the initial topology). In particular, this implies that the subsets $L^*Y^* + R^*$ and the subsets $L^*Y^- + R^-$ are closed and that the subsets $\{p \in Y^+ \text{ such that } L^*p - q \in -R^+\}$ are compact.

Proof. Since F is a Fréchet space (i.e. metrisable and complete), we have to check that for any denumerable sequence $r_n = L^*p_n + q_n$ (where $p_n \in Y^*$ and $q_n \in R^*$) converging to r in F , we can extract subsequences of p_n and q_n converging to p and q .

Since p_n belongs to the dual V^* of a barreled space, we have to check that the sequence p_n is bounded and thus, relatively compact (see Corollary 4 of Appendix A). By assumption (21), we can associate with any $z \in V$ an $\varepsilon > 0$, $x \in R$ and $y \in Y$ such that $z = \varepsilon(y - Lx)$. Hence

$$\begin{aligned} \left\langle p_n, \frac{z}{\varepsilon} \right\rangle &= \langle p_n, y - Lx \rangle = \langle -L^*p_n, x \rangle + \langle p_n, y \rangle \\ &= \langle q_n, x \rangle + \langle p_n, y \rangle - \langle r_n, x \rangle \\ &\leq 2 - \langle r_n, x \rangle \leq 2 + c \end{aligned}$$

since $\langle q_n, x \rangle \leq 1$, $\langle p_n, y \rangle \leq 1$ and $|\langle r_n, x \rangle| \leq c$ because any convergent denumerable sequence is bounded.

Therefore, we can extract from p_n a generalized subsequence which converges to an element $p \in Y^*$. Hence, we can extract from $q_n = r_n - L^*p_n$ a generalized subsequence which converges to an element $q \in R^*$. $\square$

When U is no longer the dual of a Fréchet space, we have to strengthen the constraint qualification hypothesis (21).

Proposition 11. Let $\{U, U^*\}$ and $\{V, V^*\}$ be two paired spaces, $L \in \mathcal{L}(U, V)$ and $R \subset U$, $Y \subset V$. If we assume that

$$(22) \quad L(R) \cap \text{Int}(Y) \neq \emptyset,$$

then the map $\{p, q\} \in Y^* \times R^* \mapsto L^*p + q \in U^*$ is proper (when U^*, V^* are supplied with the weak topologies and V with the Mackey topology).

Proof. Let $r_\mu = L^*p_\mu + q_\mu$ (where $p_\mu \in Y^*$ and $q_\mu \in R^*$) be a generalized sequence converging to r in U^* .

Let $x \in R$ have the property that $Lx = y$ belongs to the interior of Y . We can write

$$\begin{aligned} \langle r_\mu, x \rangle &= \langle L^*p_\mu + q_\mu, x \rangle = \langle p_\mu, Lx \rangle + \langle q_\mu, x \rangle \\ &= \langle p_\mu, y \rangle + \langle q_\mu, x \rangle. \end{aligned}$$

On the other hand, there exists μ_0 such that, for any $\mu \geq \mu_0$, $|\langle r_\mu, x \rangle - \langle r, x \rangle| \leq \varepsilon$. Since $\langle q_\mu, x \rangle \leq 1$ (for $q_\mu \in R^*$ and $x \in R$), we deduce that

$$(23) \quad \langle p_\mu, -y \rangle \leq 1 - \langle r, x \rangle + \varepsilon \text{ whenever } \mu \geq \mu_0.$$

Since $-y$ belongs to the interior of $-Y \subset (Y^*)^b$, we infer from Proposition 1.5 that the subset of p_μ satisfying (23) is compact. We can therefore extract a subsequence (again denoted by) p_μ which converges to $p \in Y^*$. Thus $q_\mu = r_\mu - L^* p_\mu$ converges to $q \in R^*$. $\square$

*3.3. Continuous convex functions

Consider a topological vector space U whose topology is defined by a family Γ of semi-norms. A function f is lower semi-continuous at x_0 only if we can associate with any $\varepsilon > 0$ a semi-norm $\mathbf{p} \in \Gamma$ and a constant $\eta > 0$ such that

$$f(x_0) \leq f(x) + \varepsilon \text{ whenever } \mathbf{p}(x - x_0) \leq \eta.$$

If f is also convex, we deduce that

$$f(x_0) \leq f(x) + \frac{\varepsilon}{\eta} \mathbf{p}(x - x_0) \text{ whenever } \mathbf{p}(x - x_0) > \eta$$

When, in addition, f is continuous, we obtain that

$$f(x_0) \leq f(x) + \frac{\varepsilon}{\eta} \mathbf{p}(x - x_0) \text{ for all } x \in U$$

and

$$|f(x_0) - f(x)| \leq \frac{\varepsilon}{\eta} \mathbf{p}(x - x_0) \text{ whenever } \mathbf{p}(x - x_0) \leq \eta.$$

In fact, we prove that a convex function is continuous on the interior of its domain if and only if it is bounded from above over some non-empty open subset. This implies that a convex function f is continuous on the interior of its domain when $U = \mathbf{R}^l$ or when U is a Fréchet space and f is lower semi-continuous. We end this section by proving that, if V is supplied with the semi-norms $L\mathbf{p}$ (where $L \in \mathcal{L}(U, V)$), a function g is continuous on V if and only if gL is continuous on U . Also, if f is a continuous convex function, we prove that Lf is continuous and convex.

3.3.1. A characterization of lower semi-continuous convex functions

Proposition 1. *Suppose that a convex function $f : U \rightarrow]-\infty, +\infty]$ is lower semi-continuous at $x_0 \in \text{Dom } f$. Then, for any $\varepsilon > 0$, there exists a semi-norm $\mathbf{p}$*

on U and $\eta > 0$ such that

$$(1) \quad \begin{cases} (i) f(x_0) \leq f(x) + \varepsilon & \text{if } \mathbf{p}(x - x_0) \leq \eta, \\ (ii) f(x_0) \leq f(x) + \frac{\varepsilon}{\eta} \mathbf{p}(x - x_0) & \text{if } \mathbf{p}(x - x_0) > \eta. \end{cases}$$

Proof. We have that f is lower semi-continuous at x_0 if and only if (1(i)) is satisfied. In order to prove inequality (1(ii)), we assume that $\mathbf{p}(x - x_0) > \eta$ and consider $y = \theta x + (1 - \theta)x_0$ where $\theta = \eta/\mathbf{p}(x - x_0) < 1$. Then, $\mathbf{p}(y - x_0) = \theta\mathbf{p}(x - x_0) = \eta$. We deduce from (1(i)) and from the convexity of f that

$$f(x_0) \leq f(y) + \varepsilon \leq \theta f(x) + (1 - \theta)f(x_0) + \varepsilon.$$

Hence

$$f(x_0) \leq f(x) + \frac{\varepsilon}{\theta} = f(x) + \frac{\varepsilon}{\eta} \mathbf{p}(x - x_0). \quad \square$$

3.3.2. A characterization of continuous convex functions

In the case where f is continuous, we obtain the following inequalities.

Proposition 2. *A convex function f is continuous at a point $x_0 \in \text{Dom}(f)$ if and only if f is bounded above on a neighborhood of x_0 .*

In this case, there exists a semi-norm $\mathbf{p}$ and a constant $c > 0$ such that

$$(2) \quad \begin{cases} (i) f(x_0) \leq f(x) + c\mathbf{p}(x - x_0) & \text{for all } x \in U, \\ (ii) f(x) \leq f(x_0) + c\mathbf{p}(x - x_0) & \text{when } \mathbf{p}(x - x_0) \leq \eta. \end{cases}$$

Furthermore, f is continuous on the interior of its domain.

Proof. It is obvious that a function f continuous at x_0 is bounded above on an open neighborhood of x_0 . In particular, this implies that x_0 belongs to the interior of the domain of f .

Conversely, let us assume that there exists a semi-norm $\mathbf{p}$ and scalars $\eta > 0$ and $a > 0$ such that

$$(3) \quad f(y) \leq a \quad \text{whenever } \mathbf{p}(y - x_0) \leq \eta.$$

(a) In order to prove inequality (2(i)), we associate with any $x \in U$ the element

$$(4) \quad y = \frac{x_0 - (1 - \theta)x}{\theta} \quad \text{where } \theta = \frac{\mathbf{p}(x - x_0)}{(\mathbf{p}(x - x_0) + \eta)} < 1.$$

Thus $p(y - x_0) = \eta$ since $y - x_0 = ((1 - \theta)/\theta)(x - x_0) = (\eta/p(x - x_0))(x - x_0)$. We deduce from (3) that $f(y) \leq a$ and from the convexity of f that

$$(5) \quad \begin{aligned} f(x_0) &= f(\theta y + (1 - \theta)x) \leq \theta f(y) + (1 - \theta)f(x) \\ &\leq \theta a + (1 - \theta)f(x). \end{aligned}$$

Hence

$$(1 - \theta)f(x_0) \leq \theta(a - f(x_0)) + (1 - \theta)f(x).$$

Dividing by $(1 - \theta)$, we obtain that

$$(6) \quad f(x_0) \leq \frac{\theta}{1 - \theta}(a - f(x_0)) + f(x) = \frac{a - f(x_0)}{\eta} p(x - x_0) + f(x).$$

We have therefore proved (2(i)) with $c = (a - f(x_0))/\eta$.

(b) In order to prove inequality (2(ii)), we associate with any x satisfying $p(x - x_0) \leq \eta$ the element

$$(7) \quad y = \frac{x - (1 - \theta)x_0}{\theta} \quad \text{where } \theta = \frac{p(x - x_0)}{\eta} \leq 1.$$

Thus $p(y - x_0) = \eta$ since $y - x_0 = (x - x_0)/\theta = (\eta/p(x - x_0))(x - x_0)$. We deduce from (3) that $f(y) \leq a$. The convexity of f implies that

$$\begin{aligned} f(x) &= f(\theta y + (1 - \theta)x_0) \leq \theta f(y) + (1 - \theta)f(x_0) \\ &\leq \theta(a - f(x_0)) + f(x_0) = \frac{a - f(x_0)}{\eta} p(x - x_0) + f(x_0). \end{aligned}$$

We have therefore (2(ii)) with $c = (a - f(x_0))/\eta$. Inequalities (2(i) and (ii)) imply that

$$(8) \quad |f(x) - f(x_0)| \leq \frac{a - f(x_0)}{\eta} p(x - x_0) \quad \text{whenever } p(x - x_0) \leq \eta$$

and thus, that f is continuous at x_0 .

(c) We now prove that f is continuous on the interior $X = \text{Int Dom } (f)$ of the domain of f . Let x_1 belong to $\text{Int Dom } (f)$. To prove that f is continuous at x_1 it is enough to show that

$$(9) \quad f(y) \leq b \quad \text{whenever } p(y - x_1) \leq \lambda \eta \text{ and } \lambda > 0.$$

Since $x_1 \in \text{Int Dom } f$, there exists $\lambda < 1$ such that

$$(10) \quad x_2 = x_0 + \frac{1}{(1 - \lambda)}(x_1 - x_0) = \frac{x_1 - \lambda x_0}{1 - \lambda}$$

belongs to $\text{Dom } f$ since

$$(11) \quad p(x_2 - x_1) = \frac{\lambda}{1-\lambda} p(x_1 - x_0).$$

Let y satisfy $p(y - x_1) \leq \lambda\eta$. We associate with y the element

$$(12) \quad z = \frac{1}{\lambda}(y + \lambda x_0 - x_1) = \frac{1}{\lambda}(y - (1-\lambda)x_2).$$

Notice that

$$(13) \quad p(z - x_0) = \frac{1}{\lambda} p(y - x_1) \leq \eta.$$

By (3), we deduce that $f(z) \leq a$. Hence the convexity of f implies that

$$\begin{aligned} f(y) &= f(\lambda z + (1-\lambda)x_2) \leq \lambda f(z) + (1-\lambda)f(x_2) \\ &\leq \lambda a + (1-\lambda)f(x_2) = b. \end{aligned}$$

Thus property (9) is satisfied. $\square$

3.3.3. Examples of continuous convex functions

The last result has the following important consequences.

Proposition 3. *Any convex function f defined on an open subset X of $\mathbb{R}^n$ is continuous on X .*

Proof. Let x_0 belong to X and let $N(x_0) \subset X$ be a neighborhood of x_0 contained in X . We can find n elements $x_i \in N(x_0)$ such that the vectors $x_i - x_0$ are linearly independent. The function f is bounded above on the convex hull $K = \text{co}(x_0, x_1, \dots, x_n)$ because

$$f\left(\sum_{i=0}^n \lambda_i x_i\right) \leq \sum_{i=0}^n \lambda_i f(x_i) \leq \max_{0 \leq i \leq n} f(x_i)$$

Since the vectors $x_i - x_0$ are linearly independent, the interior of K is non-empty and thus contains a neighborhood of x_0 . By Proposition 2, the function f is continuous on X . $\square$

In the case of Fréchet spaces, we obtain the following result.

Proposition 4. *Suppose that*

$$(14) \quad U \text{ is a Fréchet (complete metrizable) space.}$$

Any convex lower semi-continuous function is continuous on the interior of its domain.

Proof. Suppose that the interior of the domain $X = \text{Dom } f$ is non-empty.

By *Proposition 2*, we have to prove that f is bounded on an open subset of X . For this purpose, we consider the closed subsets $F_n = \{x \in X \text{ such that } f(x) \leq n\}$. Then $X = \bigcup_{n=1}^{\infty} F_n$. If X has a non-empty interior, the Baire theorem¹ implies that at least one of the closed subsets F_n has a non-empty interior $\overset{\circ}{F}_n$. $\square$

The study of continuous convex functions is continued in the next section. We conclude this section with a study of the continuity of the functions gL and Lf .

3.3.4. Continuity of gL and Lf

Consider the case when $L \in \mathcal{L}(U, V)$ is surjective. If $\mathbf{p}$ is a semi-norm on U it is clear that $L\mathbf{p}$ is a semi-norm on V .

If U is a locally convex space whose topology is defined by a family Γ of semi-norms $\mathbf{p}$, we shall supply the space V with the topology defined by the family $L\Gamma$ of semi-norms $L\mathbf{p}$. This topology is called the image by L of the topology of U .

The operator L is clearly continuous for such a topology.

Proposition 5. *A map g from V into a topological space Z is continuous if and only if the map gL from U into Z is continuous.*

Proof. It is clear that, if g is continuous, then gL is also continuous. Conversely, assume that gL is continuous from U into Z . We prove that g is continuous at $y_0 \in V$, i.e. that we can associate with any neighborhood N of $g(y_0)$ in Z a semi-norm $\mathbf{p}$ of Γ and $\eta > 0$ such that

$$(15) \quad L\mathbf{p}(y - y_0) \leq \eta \quad \text{implies that } g(y) \in N.$$

But we know that gL is continuous at a point x_0 satisfying $Lx_0 = y_0$. Thus there exists $\mathbf{p} \in \Gamma$ and η such that

$$(16) \quad \mathbf{p}(x - x_0) \leq 2\eta \quad \text{implies that } g(Lx) \in N.$$

Now, we can associate with any y satisfying $L\mathbf{p}(y - y_0) \leq \eta$ an element x satisfying

$$(17) \quad Lx = y \quad \text{and} \quad \mathbf{p}(x - x_0) \leq L\mathbf{p}(y - y_0) + \eta \leq 2 \cdot \eta$$

Therefore inequality (15) follows from (16) and (17). $\square$

¹ The Baire theorem states that in a complete metrizable space (or a locally compact space), if the union of a denumerable sequence of closed subsets F_n has a non-empty interior, then the interior of at least one of the subsets F_n is non-empty.

Next we prove that Lf is convex and continuous whenever f is convex and continuous.

Proposition 6. *Suppose that f is convex and continuous at a point x_0 satisfying $Lx_0 = y_0$. Then Lf is convex and continuous at y_0 .*

Proof. By assumption, we can associate with any $\varepsilon > 0$ a semi-norm $\mathbf{p}$ on U and $\eta > 0$ such that

$$(18) \quad f(x) \leq f(x_0) + \varepsilon \quad \text{whenever } \mathbf{p}(x - x_0) \leq \eta.$$

On the other hand, we can associate with any $\bar{y}$ an element $\bar{x}$ satisfying

$$(19) \quad L\bar{x} = \bar{y}; \quad \mathbf{p}(\bar{x}) \leq L\mathbf{p}(\bar{y}) + \frac{1}{2}\eta.$$

Now, by taking $\bar{y} = y - y_0$ where $\mathbf{p}(y - y_0) \leq \frac{1}{2}\eta$, we can find an associated $x = \bar{x} + x_0$ which satisfies $Lx = y$ and $\mathbf{p}(x - x_0) \leq 2 \cdot \frac{1}{2}\eta = \eta$. Therefore,

$$Lf(y) \leq f(x) \leq f(x_0) + \varepsilon \quad \text{whenever } L\mathbf{p}(y - y_0) \leq \frac{1}{2}\eta.$$

Since Lf is convex, this implies that Lf is continuous at x_0 . $\square$

*3.4. Continuous convex functions (continuation)

In this section we continue the study of continuous convex functions. (The section can be passed over in a first reading.) Using the characterization of continuous convex functions of the preceding section, we deduce that any convex lower semi-continuous function is continuous on the interior of its domain for the strong topology $\beta(U, U^*)$ of U .

We shall prove the following important characterization: a convex function f is lower semi-compact if and only if its conjugate function f^* is continuous at 0. This implies, for instance, that a lower support function σ_K^b is continuous at p_0 if (and only if) the subsets $\{x \in K \text{ such that } \langle p_0, x \rangle \leq \lambda\}$ are compact.

To prove the above result, we obtain the following estimates for a lower semi-continuous convex function:

$$\begin{cases} f(x) \leq f(x_0) + c\pi(x - x_0) & \text{if } f(x) \leq f(x_0) + c, \\ f(x) \geq f(x_0) + c\pi(x - x_0) & \text{if } f(x) \geq f(x_0) + c, \end{cases}$$

where π is the gauge of the subset $\{x \in U \text{ such that } f(x_0 + x) \leq f(x_0) + c\}$.

We continue this section by defining the extremal points of convex subsets and by proving that a continuous convex function defined on a convex compact

set achieves its *maximum* at an extremal point (at least). Finally, we prove that any convex compact subset is the closed convex hull of the set of its extremal points.

3.4.1. Strong continuity of lower semi-continuous convex functions

Proposition 1. *Let U and U^* be two paired spaces. Then any lower semi-continuous function (for any topology compatible with the pairing) is strongly continuous on the strong interior of its domain $\text{Dom}(f)$ provided this is non-empty.*

Proof. Recall (see Section 4 of Appendix A) that strong neighborhoods of U are convex symmetric (weakly) closed and *absorbing* subsets of U . Therefore, if x_0 belongs to the strong interior of $\text{Dom}(f)$, then, for any $x \in U$, there exists $\theta_x > 0$ such that $x_0 + \theta x \in \text{Dom}(f)$ whenever $\theta \leq \theta_x$, i.e. such that

$$(1) \quad f(x_0 + \theta x) < +\infty \quad \text{whenever } \theta \leq \theta_x.$$

We set

$$(2) \quad M = \{y \in U \text{ such that } f(x_0 + y) \leq f(x_0) + c\}$$

where $c > 0$.

Note that M is a *closed convex subset which contains 0*, since f is convex and lower semi-continuous. If we prove that M is absorbing, then $M \cap -M$ is a strong neighborhood (being closed, convex symmetric and absorbing) on which f is bounded above. This implies that f is continuous at x_0 by Proposition 3.2.

To prove that M is absorbing, we have to associate with any $x \in U$ a scalar $\eta_x > 0$ such that

$$(3) \quad f(x_0 + \theta x) \leq f(x_0) + c \quad \text{whenever } \theta \leq \eta_x.$$

By (1), the function $\theta \mapsto f(x_0 + \theta x) - f(x_0)$ is defined on the subset $]-\theta_{-x}, \theta_x[$ of the real line and thus is continuous at 0 by Proposition 3.3. Thus there exists $\eta_x > 0$ such that $|f(x_0 + \theta x) - f(x_0)| \leq c$ whenever $|\theta| \leq \eta_x$, i.e. (3) holds. $\square$

Proposition 2. *Let U be a barreled space. Any lower semi-continuous convex function is continuous on the interior of its domain.*

Proof. If U is barreled, the strong topology $\beta(U, U^*)$ coincides with the initial topology (see Corollary 3 of Appendix A). Therefore, Proposition 2 is a consequence of Proposition 1. $\square$

3.4.2. Estimates of lower semi-continuous convex functions

We shall need the following result for characterizing continuous convex functions.

Proposition 3. *Let f be a lower semi-continuous convex function and $x_0 \in \text{Dom } f$. Let M be convex subset defined by*

$$(4) \quad M = \{x \in U \text{ such that } f(x_0 + x) \leq f(x_0) + c\}$$

where $c > 0$. Let $\pi = \pi_M$ be the gauge of M . Then the following inequalities hold.

$$(5) \quad \begin{cases} \text{(i) } f(x) \leq f(x_0) + c\pi(x - x_0) & \text{whenever } f(x) \leq f(x_0) + c, \\ \text{(ii) } f(x) \geq f(x_0) + c\pi(x - x_0) & \text{whenever } f(x) \geq f(x_0) + c. \end{cases}$$

Proof. By setting $\phi(x) = f(x + x_0) - f(x_0)$, we can write $M = \{x \in U \text{ such that } \phi(x) \leq c\}$. If $\phi(x) \leq c$, then $\pi(x) \leq 1$. Thus

$$(6) \quad \begin{cases} \phi(x) = \phi((1 - \pi(x))0 + \pi(x)\frac{x}{\pi(x)}) \\ \leq (1 - \pi(x))\phi(0) + \pi(x)\phi\left(\frac{x}{\pi(x)}\right) = 0 + \pi(x)c. \end{cases}$$

since $\phi(x/\pi(x)) = c$.

If $\phi(x) > c$, then $\pi(x) > 1$. Thus

$$(7) \quad \begin{aligned} c &= \phi\left(\frac{x}{\pi(x)}\right) = \phi\left(\left(1 - \frac{1}{\pi(x)}\right)0 + \frac{1}{\pi(x)}x\right) \\ &\leq \left(1 - \frac{1}{\pi(x)}\right)\phi(0) + \frac{1}{\pi(x)}\phi(x) = \frac{\phi(x)}{\pi(x)}. \end{aligned}$$

Therefore, $\phi(x) \geq c\pi(x)$ whenever $\phi(x) > c$. We obtain inequalities (5) by taking $x = y - x_0$ since $\phi(x) = f(y) - f(x_0)$. $\square$

Proposition 4. *Let f be a lower semi-continuous convex function which is bounded below. Let $\alpha = \inf_{x \in U} f(x)$ and x_0 belong to $\text{Dom } f$. Let*

$$(8) \quad M = \{x \in U \text{ such that } f(x_0 + x) \leq f(x_0) + c\}.$$

Then, if $\pi = \pi_M$ denotes the gauge of M , we have that

$$(9) \quad \text{for any } x \in U, \quad f(x) \geq c\pi(x - x_0) - c + \alpha.$$

Proof. We set $\phi(x) = f(x_0 + x) - f(x_0)$. Then $M = \{x \in U \text{ such that } \phi(x) \leq c\}$. If $\phi(x) \leq c$, then $c(\pi(x) - 1) = c\pi(x) - c \leq 0$. Since $\alpha - f(x_0) = \inf_{x \in U} \phi(x)$, we deduce that

$$c\pi(x) - c + \alpha - f(x_0) \leq \phi(x) \quad \text{if } \phi(x) \leq c.$$

If $\phi(x) > c$, it follows from (7) that $c\pi(x) \leq \phi(x)$. Since $\alpha - f(x_0) - c \leq \phi(x) - c \leq 0$, we conclude that

$$c\pi(x) - c + \alpha - f(x_0) \leq \phi(x) \quad \text{if } \phi(x) \leq c.$$

Thus, setting $x = y - x_0$, we obtain (9). $\square$

3.4.3. Characterization of continuous convex functions

Proposition 5. Let U and U^* be two paired spaces and let f be a lower semi-continuous convex function from U into $]-\infty, +\infty]$. The two following statements are equivalent.

$$(10) \quad \begin{cases} \text{(i) } f \text{ is continuous at } x_0 \text{ for the Mackey topology } \tau(U, U^*) \\ \text{(ii) the sections } S(f^*, x_0, \lambda) = \{p \in U^* \text{ such that } f^*(p) - \langle p, x_0 \rangle \leq \lambda\} \text{ are weakly compact.} \end{cases}$$

Proof. We begin by proving that (10(i)) implies (10(ii)). Let N be a neighborhood of $0 \in U$ such that $f(x_0 + x) \leq 1$ whenever $x \in N$. Let $K = N^*$ be its polar subset, which is weakly compact by the very definition of the Mackey topology. If p belongs to $S(f^*, x_0, \lambda)$ we deduce from the relation

$$(11) \quad \langle p, x_0 + x \rangle \leq f^*(p) + f(x_0 + x)$$

that

$$(12) \quad \langle p, x \rangle \leq \lambda + 1 \quad \text{whenever } x \in N.$$

Therefore, (12) implies that p belongs to $(\lambda + 1)K$.

Conversely, suppose that the non-empty subset $S = S(f^*, x_0, \lambda)$ is contained in a compact subset of U^* . We first show that

$$(13) \quad f(x_0) = - \inf_{p \in U^*} [f^*(p) - \langle p, x_0 \rangle] = - \inf_{p \in S} [f^*(p) - \langle p, x_0 \rangle]$$

is finite. Since S is bounded and

$$f^*(p) - \langle p, x_0 \rangle \geq \langle p, x - x_0 \rangle - f(x) \quad \text{if } x \in \text{Dom } f$$

it follows that

$$\inf_{p \in S} (f^*(p) - \langle p, x_0 \rangle) \geq \inf_{p \in S} [\langle p, x_0 \rangle - f(x)] > -\infty.$$

Therefore, $x_0 \in \text{Dom}(f)$. Now let $p_0 \in S$ and let M be the closed convex subset defined by

$$(14) \quad \begin{aligned} M &= \{p \in U^* \text{ such that } f^*(p_0 + p) - \langle p, x_0 \rangle \leq f^*(p_0) + c\} \\ &= S(f^*, x_0, \lambda) - p_0 \quad \text{where } c = \lambda + \langle p_0, x_0 \rangle - f^*(p_0). \end{aligned}$$

By assumption, M is contained in a compact subset K .

From Proposition 4 with $x \mapsto f(x)$ is replaced by $p \mapsto f^*(p) - \langle p, x_0 \rangle$, we obtain that

$$(15) \quad \begin{aligned} c\sigma^*(K^*; p - p_0) - c - f(x_0) &\leq c\sigma^*(M^*; p - p_0) - c - f(x_0) \\ &\leq f^*(p) - \langle p, x_0 \rangle \quad \text{whenever } p \in U^*, \end{aligned}$$

since the gauge $\pi(K; p) = \sigma^*(K^*; p)$ of K is smaller than or equal to the gauge $\pi(M; p) = \sigma^*(M^*; p)$ of M and $-f(x_0) = \inf_{p \in F} [f^*(p) - \langle p, x_0 \rangle]$.

Therefore, for any $x \in U$ and $p \in U^*$, we have that

$$(16) \quad \langle p, x_0 + x \rangle - f^*(p) \leq \langle p, x \rangle - c\sigma^*(K^*; p - p_0) + c + f(x_0).$$

Taking the supremum over U^* , we deduce from (16) that

$$(17) \quad f^{**}(x_0 + x) = f(x_0 + x) \leq c\psi\left(K^*; \frac{x}{c}\right) + \langle p_0, x \rangle + c + f(x_0).$$

Therefore, the function $x \mapsto f(x_0 + x) - \langle p_0, x \rangle$ is bounded above on a neighborhood K^* of 0 and is therefore continuous at 0. This implies that f is continuous at x_0 . $\square$

3.4.4. Continuity of support functions

We apply the preceding result to the support function and indicator of a closed convex subset K .

Proposition 6. *Let U and U^* be two paired spaces and let $K \subset U$ be a closed convex subset.*

The lower support function σ_K^b is continuous at $p_0 \in U^$ (supplied with the Mackey topology $\tau(U^*, U)$) if and only if the subsets $\{x \in K \text{ such that } \langle p_0, x \rangle \leq \lambda\}$ are compact (when U is supplied with the weak topology $\sigma(U, U^*)$).*

An element $x_0 \in K$ belongs to the interior of K (for the Mackey topology $\tau(U, U^)$) if and only if $(K - x_0)^*$ is compact (for the weak topology $\sigma(U^*, U)$).*

Proof. To prove the first statement, we apply Proposition 5 to the function $p \mapsto \sigma_K^*(-p) = -\sigma_K^b(p)$. This function is continuous if and only if the sections $\{x \in U \text{ such that } \psi_K(x) - \langle -p, x_0 \rangle \leq \lambda\} = \{x \in K \text{ such that } \langle p_0, x \rangle \leq \lambda\}$ are compact.

To prove the second statement, we apply Proposition 5 to the function ψ_K . This function is continuous at x_0 (i.e. $x_0 \in \text{Int}(K)$) if and only if the sections $\{p \in U^* \text{ such that } \sigma^*(K; p) - \langle p, x_0 \rangle = \sigma^*(K - x_0, p) \leq \lambda\}$ are compact, i.e. if and only if the subsets $\lambda(K - x_0)^*$ are compact. $\square$

3.4.5. Maximum of a convex function: extremal points

Definition 1. Let $X \subset U$ be a convex subset of a vector space U . A point $x \in X$ is said to be an extremal point of X if, whenever elements y and z of X satisfy $x = \lambda y + (1 - \lambda)z$, where $\lambda \in [0, 1]$, then either $\lambda = 0$ or $\lambda = 1$.

We prove next that a convex continuous function achieves its maximum at an extremal point.

Proposition 7. Let U be an Hausdorff locally convex space, X a non-empty compact convex subset of U and f an upper semi-continuous convex function defined on X . Then f achieves its maximum at an extremal point.

Proof. (a) To prove this theorem, we shall use Zorn's lemma applied to the family F of subsets K of X satisfying

$$(18) \quad \begin{aligned} &K \text{ is non-empty, closed and } x \in K, y, z \in X \text{ and} \\ &x = \lambda y + (1 - \lambda)z \text{ implies } y \text{ and } z \text{ belong to } K. \end{aligned}$$

This family F is non-empty because the set X belongs to F . Of course, an element x is extremal if and only if the subset $\{x\}$ belongs to F . The family F ordered by $\supset$ is inductive: Let $C = \{K_i\}_{i \in I}$ be a chain. Then $K = \bigcap_{i \in I} K_i$ is a lower bound of C . The subset K is non-empty and closed because $\{K_i\}$ is a decreasing sequence of non-empty compact subsets. If $x \in K$ can be written $x = \lambda y + (1 - \lambda)z$, then y and z belong to K . Hence K belongs to F and Zorn's lemma implies that

$$(19) \quad \text{any non-empty subset of } F \text{ contains a minimal element.}$$

(b) Recall that

$$M^*(K, f) = \left\{ x \in K \text{ such that } f(x) = \sup_{y \in K} f(y) \right\}.$$

We prove that

(20) If $K \in F$, then $M^*(K, f)$ belongs to F .

Now, $M^*(K, f) \neq \emptyset$ because f achieves its maximum on K , being upper semi-continuous on the non-empty compact set K . Clearly, $M^*(K, f)$ is closed. It remains to check that, if $x \in M^*(K, f)$ is written $x = \lambda y + (1-\lambda)z$, then both y and z belong to $M^*(K, f)$. Since f is convex, we have that

$$\begin{aligned} \sup_{y \in K} f(y) &= f(x) = f(\lambda y + (1-\lambda)z) \leq \lambda f(y) + (1-\lambda)f(z) \\ &\leq (\lambda + (1-\lambda)) \sup_{y \in K} f(y). \end{aligned}$$

This implies that $f(y) = f(z) = \sup_{y \in K} f(y)$. Since K belongs to F , we know that both y and z belong to K . Thus y and z belong to $M^*(K, f)$.

(c) Now, we use properties (19) and (20) to prove that

(21) $K \in F$ is minimal if and only if $K = \{x\}$ contains a single element x (and is therefore an extremal point).

It is obvious that, if $\{x\}$ is extremal, then it is a minimal element of F . Conversely, suppose that a minimal K in F contains two distinct points $x \neq y$. Then there exists a continuous linear form $p \in U^*$ such that $\langle p, x \rangle < \langle p, y \rangle$. But then $M^*(K, p)$ is a proper subset of K which lies in F . This is a contradiction.

(d) Finally, properties (19), (20) and (21) imply the theorem. Given any upper semi-continuous convex function f , the set $M^*(X, f)$ belongs to F by (20) and contains an extremal point x by (21). Such an extremal point x satisfies $f(x) = \sup_{y \in X} f(y)$. $\square$

Proposition 8. *Let X be a compact convex subset of an Hausdorff locally convex space U . Then, for any $p \in U^*$, there exists an extremal point $x \in X$ satisfying*

$$(22) \quad \sigma^*(X; p) = \langle p, x \rangle.$$

Finally, we obtain the fundamental result of Krein and Milman.

Proposition 9 (Krein–Milman). *Let X be a convex compact subset of an Hausdorff locally convex space U . Then X is the closed convex hull of the set of its extremal points.*

Proof. Let Y be the closed convex hull of the set of the extremal points of X . Then $Y \subset X$. If $Y \neq X$, there exists $x \in X$ such that $x \notin Y$. By the Hahn–

Banach separation theorem, there then exists a non-zero continuous linear form $p \in U^*$ such that

$$(23) \quad \sup_{y \in Y} \langle p, y \rangle < \langle p, x \rangle \leq \sup_{x \in X} \langle p, x \rangle = \sigma^*(X, p)$$

This is impossible since $\sigma^*(X, p) = \langle p, y \rangle$ for at least one $y \in Y$ by Proposition 8. $\square$

CHAPTER 4

DIFFERENTIABILITY AND SUBDIFFERENTIABILITY: CHARACTERIZATION OF OPTIMAL SOLUTIONS

The aim of this short chapter is to introduce the various concepts of differentiability in the framework of optimization theory.

We will characterize in Section 4.1 the *minimal set of a convex function f as the subdifferential $\partial f^*(0)$ of its conjugate function f^* at 0*. The *subdifferential* of a function f at x_0 is defined as the (possibly empty) subset

$$\partial f(x_0) = \{p \in U^* \text{ such that } f(x_0) - f(x) \leq \langle p, x_0 - x \rangle \text{ for all } x \in U\}.$$

It is not a stringent concept: we prove that any *convex function f continuous at x_0 has a non-empty compact subdifferential $\partial f(x_0)$* .

We show that when f is convex and lower semi-continuous the correspondences $\partial f: x \rightarrow \partial f(x)$ and $\partial f^*: p \rightarrow \partial f^*(p)$ are inverse. Furthermore, we prove that $x \rightarrow \partial f(x)$ is an *uppersemi-continuous correspondence* on the interior of the domain of f .

In the case of convex functions, the subdifferential $\partial f(x_0)$ is a generalization of the concept of *gradient $Df(x_0)$* . When the gradient exists, it is a continuous, linear functional satisfying

$$\langle Df(x_0), x \rangle = \lim_{\theta \rightarrow 0} \frac{f(x_0 + \theta x) - f(x_0)}{\theta}$$

for all x .

We prove, in fact, that if f is convex and differentiable at x_0 , then $\partial f(x_0) = \{Df(x_0)\}$.

More generally, if a function f is *both differentiable and subdifferentiable* at x_0 , then $\partial f(x_0) = \{Df(x_0)\}$. In this case, we prove that $x_0 \in X$ *minimizes f on X if and only if*

$$\langle Df(x_0), x_0 \rangle \leq \langle Df(x_0), x \rangle \text{ for all } x \in X.$$

This latter minimization problem can be considered as a “linearized minimization problem”, although, as opposed to the linearized extension to the mixed

strategy subsets constructed in Section 1.3.2 and Section 3.1.6, the linearized functional $Df(x_0)$ depends upon the (unknown) minimal solution x_0 .

We also observe that, when f^* is differentiable at p , $\bar{x} = Df^*(p)$ is the unique solution minimizing $x \mapsto f(x) - \langle p, x \rangle$. We may therefore interpret $\bar{x}$ as a "marginal profit" whenever f is regarded as a loss function and f^* as a profit function.

In the next section, we introduce the "intermediate" concept of a function *differentiable from the right*. The derivative from the right of f at x_0 in the direction x is defined by

$$Df(x_0)(x) = \lim_{\theta \rightarrow 0+} \frac{f(x_0 + \theta x) - f(x_0)}{\theta}.$$

The main result states that any convex function continuous at x_0 is differentiable from the right and that $Df(x_0)(x)$ is nothing other than the upper support function of the subdifferential, i.e. $Df(x_0)(x) = \sigma^*(\partial f(x_0); x)$.

The main justification of this concept lies in the following result. Under appropriate assumptions, the pointwise supremum $g(x) = \sup_{p \in P} f(x; p)$ of a set of differentiable functions $f(\cdot; p)$ is differentiable from the right and

$$Dg(x)(y) = \sup_{p \in P(x)} \langle Df(x; p), y \rangle$$

where $P(x) = \{p \in P \text{ such that } g(x) = f(x, p)\}$.

Finally, in the last section, we introduce a very weak concept of differentiability: we shall say that f is locally ε -differentiable at x if there exist $\eta > 0$ and $p \in U^*$ such that

$$f(\bar{x}) \leq f(x) + \langle p, \bar{x} - x \rangle + \varepsilon \|\bar{x} - x\| \quad \text{whenever } \|\bar{x} - x\| \leq \eta.$$

This concept is of interest because a *lower semicontinuous function on a Hilbert space U is locally ε -subdifferentiable on a dense subset of U* .

This is quite a powerful result, which we will use for studying families of perturbed problems

$$\alpha(p) = \inf_{x \in X} f(x, p)$$

as p ranges over a Hilbert space U^* . We shall prove that, if the *perturbations are "smooth"* and the *functions $x \mapsto f(x, p)$ are lower semi-continuous, then a unique minimizing x exists for "almost all" p* (i.e. for every p belonging to a dense denumerable intersection of open subsets). This is obviously of interest, since if a minimization problem $\alpha(p)$ has no unique optimal solution, we may simply choose a small enough perturbation, insignificant for practical purposes, such that the new problem has a unique solution.

4.1. Subdifferentiability

4.1.1. Definitions

In order to characterize the minimal set of a proper function f defined on a topological vector space U , we need the definition of the subgradient of a function f at a point x_0 .

Definition 1. Let f be a proper function defined on a topological vector space U and let $x_0 \in \text{Dom } f$. The “*subdifferential*” of f at x_0 is the (possibly empty) subset $\partial f(x_0)$ of the dual U^* defined by

$$(1) \quad \begin{aligned} \partial f(x_0) = \{p \in U^* \text{ such that } f(x_0) - f(x) \leq \langle p, x_0 - x \rangle \\ \text{for all } x \in U\}. \end{aligned}$$

We call the elements $p \in \partial f(x_0)$ the “*subgradients*” of f at x_0 . We say that f is *subdifferentiable* at x_0 if $\partial f(x_0)$ is non-empty. We define also the “*superdifferential*” $\bar{\partial} f(x_0)$ by

$$(2) \quad \begin{aligned} \bar{\partial} f(x_0) = \{p \in U^* \text{ such that } f(x_0) - f(x) \geq \langle p, x_0 - x \rangle \\ \text{for all } x \in U\}. \\ = -\partial(-f)(x_0). \end{aligned}$$

The following is obviously an equivalent definition for $\partial f(x_0)$.

$$(3) \quad \begin{aligned} p \in \partial f(x_0) \text{ if and only if } x_0 \text{ minimizes} \\ x \mapsto f(x) - \langle p, x \rangle \text{ on } U \end{aligned}$$

The first justification of the concept of subdifferential is its relation with the minimal set of a function.

Proposition 1. Let f be a proper function defined on a topological vector space U . Then

$$(4) \quad x_0 \text{ minimizes } x \mapsto f(x) - \langle p_0, x \rangle \text{ on } U$$

if and only if

$$(5) \quad \begin{cases} \text{(i) } x_0 \in \partial f^*(p_0), \\ \text{(ii) } f(x_0) = f^{**}(x_0). \end{cases}$$

Proof. Suppose that (4) holds. Then the inequalities

$$f(x_0) - \langle p_0, x_0 \rangle \leq f(x) - \langle p_0, x \rangle$$

imply that

$$(6) \quad f^*(p_0) = \langle p_0, x_0 \rangle - f(x_0).$$

But, for all $p \in U^*$,

$$f^*(p) \geq \langle p, x_0 \rangle - f(x_0).$$

We deduce that

$$(7) \quad f^*(p_0) - f^*(p) \leq \langle p_0 - p, x_0 \rangle$$

(i.e. that $x_0 \in \partial f^*(p_0)$). Furthermore, (7) implies that

$$(8) \quad f^{**}(x_0) = \langle p, x_0 \rangle - f^*(p_0)$$

and thus, by (6), $f(x_0) = f^{**}(x_0)$.

Conversely, suppose that (5) holds. Since $x_0 \in \partial f^*(p_0)$,

$$f^*(p_0) - \langle p_0, x_0 \rangle \leq f^*(p) - \langle p, x_0 \rangle$$

and we deduce that

$$f^{**}(x_0) - \langle p_0, x_0 \rangle = -f^*(p_0) = \inf_{x \in U} [f(x) - \langle p, x \rangle].$$

Since $f(x_0) = f^{**}(x_0)$, we have proved that x_0 minimizes $x \mapsto f(x) - \langle p, x \rangle$ on U . $\square$

We can reformulate Proposition 1 as follows.

Proposition 2. *The minimal subset of: $x \mapsto f(x) - \langle p, x \rangle$ is equal to*

$$(9) \quad \{x \in \partial f^*(p) \text{ such that } f(x) = f^{**}(x)\}.$$

We deduce from Theorem 2.4.1 the following consequence.

Proposition 3. *Suppose that f is a proper convex lower semi-continuous function. Then the minimal subset of the perturbed function $x \mapsto f(x) - \langle p, x \rangle$ is equal to $\partial f^*(p)$.*

This result has the alternative formulation

$$(10) \quad p_0 \in \partial f(x_0) \text{ if and only if } x_0 \in \partial f^*(p_0)$$

i.e. the correspondence ∂f^* is the inverse of ∂f .

4.1.2. Examples of subdifferentials

Let X be a non-empty convex subset of a vector space U and let ψ_X be its indicator. Then p belongs to $\partial \psi_X(x)$ if and only if $\psi_X(x) \leq \psi_X(y) + \langle p, x - y \rangle$

for all $y \in U$, i.e. if and only if

$$(11) \quad \begin{cases} \text{(i) } x \in X, \\ \text{(ii) } \langle p, x-y \rangle \geq 0 \text{ for all } y \in X. \end{cases}$$

Definition 2. We shall say that the subdifferential $\partial\psi_X(x)$ (when $x \in X$) is the *normal cone* to X at x .

Proposition 4. Let X be a non-empty closed convex subset of U . For each $p \in U^*$, $\partial\sigma_X^\#(p)$ consists of the elements x (if any) where the linear function $x \mapsto \langle p, x \rangle$ achieves its maximum on X .

Proof. This follows from Proposition 3 with $f = \psi_X$ since $f^* = \sigma_X^\#$. $\square$

Next we characterize the subdifferential of a function $x \mapsto \Phi(\|x\|)$ defined on a normed space.

Proposition 5. Consider the convex function $f(x) = \Phi(\|x\|)$, where $\Phi(t) = \int_0^t \phi(\lambda) d\lambda$ (and ϕ satisfies (2.4.15)).

Then its subdifferential is equal to

$$(12) \quad \partial f(x) = \{p \in U^* \text{ such that } \langle p, x \rangle = \|p\|_* \|x\| \text{ and } \|p\|_* = \phi(\|x\|)\}.$$

Proof. Suppose that $p \in U^*$ satisfies $\langle p, x \rangle = \|p\|_* \|x\|$ and $\|p\|_* = \phi(\|x\|)$. Then, since $\phi(t) = (D\Phi)(t)$, we have that

$$\begin{aligned} \langle p, x \rangle &= \|p\|_* \|x\| \doteq \|x\| (D\Phi)(\|x\|) \\ &= \Phi(\|x\|) + \Phi^*(D\Phi(\|x\|)) \\ &= \Phi(\|x\|) + \Phi^*(\|p\|_*) = f(x) + f^*(p). \end{aligned}$$

This implies that $p \in \partial f(x)$.

Conversely, suppose that $p \in \partial f(x)$. Then

$$(13) \quad \Phi(\|x\|) - \Phi(\|y\|) \leq \langle p, x-y \rangle \quad \text{for all } y \in U.$$

If we take $y = \|x\|(z/\|z\|)$, then $\|y\| = \|x\|$. We deduce from (13) that

$$(14) \quad 0 \leq \langle p, x \rangle - \|x\| \left\langle p, \frac{z}{\|z\|} \right\rangle \quad \text{for all } z \in U.$$

This implies that

$$(15) \quad \|x\| \|p\|_* = \sup_z \|x\| \frac{\langle p, z \rangle}{\|z\|} \leq \langle p, x \rangle$$

and thus that $\langle p, x \rangle = \|p\|_* \|x\|$.

To prove that $\|p\|_* = \phi(\|x\|)$, we take $y = (1+\theta)x$ where $\theta \in]-1, +1[$. Then (13) implies that

$$(16) \quad \theta \langle p, x \rangle \leq \Phi((1+\theta)\|x\|) - \Phi(\|x\|).$$

Therefore, for any $\theta \in]0, 1[$, we deduce that

$$\begin{aligned} \frac{\Phi(\|x\|) - \Phi(\|x\| - \theta\|x\|)}{\theta} &\leq \langle p, x \rangle \\ &\leq \frac{\Phi(\|x\| + \theta\|x\|) - \Phi(\|x\|)}{\theta}. \end{aligned}$$

Letting θ converge to 0, we deduce that

$$(17) \quad D\Phi(x)\|x\| = \phi(\|x\|)\|x\| = \langle p, x \rangle = \|p\|_* \|x\|.$$

Therefore $\|p\|_* = \phi(\|x\|)$. $\square$

Proposition 6. Let $f(x) = \Phi(\|x\|)$.

If $p \in \partial f(x)$ and $q \in \partial f(y)$, then

$$(18) \quad \langle p - q, x - y \rangle \geq (\phi(\|x\|) - \phi(\|y\|))(\|x\| - \|y\|).$$

Proof. By Proposition 5,

$$\begin{aligned} \langle p - q, x - y \rangle &= \langle p, x \rangle + \langle q, y \rangle - \langle q, x \rangle - \langle p, y \rangle \\ &= \|p\|_* \|x\| + \|q\|_* \|y\| - \langle q, x \rangle - \langle p, y \rangle \\ &\geq \|p\|_* \|x\| + \|q\|_* \|y\| - \|q\|_* \|x\| - \|p\|_* \|y\| \\ &= (\|p\|_* - \|q\|_*)(\|x\| - \|y\|) \\ &= (\phi(\|x\|) - \phi(\|y\|))(\|x\| - \|y\|). \quad \square \end{aligned}$$

4.1.3. Subdifferentiability of continuous convex functions

Proposition 7. Suppose that a convex proper function f is continuous at a point x of the interior of its domain. Then $\partial f(x)$ is non-empty and weakly compact.

Proof. Since f is continuous at x , $\mathcal{E}pf$ has a non-empty interior (if $N(x)$ is an open neighborhood contained in $\text{Dom } f$, then $N(x) \times [c, c+1[$ is an open subset contained in $\mathcal{E}p(f)$ (provided that c is an upper bound of f on $N(x)$). Since $\{x, f(x)\}$ does not belong to the interior of the convex set $\mathcal{E}p(f)$, the Hahn-Banach separation theorem implies that there exists a non-zero continuous linear form $\{p, \alpha\} \in U^* \times \mathbb{R}$ such that

$$(19) \quad \forall y \in U, \forall c \geq f(y), \quad \langle p, y \rangle - \alpha c \leq \langle p, x \rangle - \alpha f(x).$$

Taking $y = x$ and $c = f(x) + 1$, we deduce that $\alpha \geq 0$. Actually, $\alpha > 0$ since $\alpha = 0$ implies that $\langle p, y - x \rangle \leq 0$ for all $y \in \text{Dom } f$. This implies the contradiction $p = 0$ because $x \in \text{Int Dom } f$. Taking $c = f(y)$ and dividing both sides of the above inequality by $\alpha > 0$, we deduce that $f^*(p/\alpha) \leq \langle p/\alpha, x \rangle - f(x)$, i.e. that $p/\alpha \in \partial f(x)$.

We next prove that the closed convex subset $\partial f(x)$ is compact in U^* (for the weak topology $\sigma(U^*, U)$). For this purpose, we note that

$$(20) \quad \partial f(x) = \{p \in U^* \text{ such that } f^*(p) - \langle p, x \rangle \leq -f(x_0)\}$$

is a lower section of f^* . Since $f = (f^*)^*$ is continuous at x , Propositions 3.1.3 and 3.1.10 show that $p \mapsto f(p) - \langle p, x \rangle$ is lower semi-compact. Thus $\partial f(x)$ is relatively compact. $\square$

We shall prove (Proposition 13.2.10) that, in Hilbert spaces, the set X of elements $x \in \text{Dom } (f)$ such that $\partial f(x) \neq \emptyset$ is not empty and that its closure is equal to the closure of $\text{Dom } (f)$.

4.1.4. Upper semi-continuity of the subdifferential

Suppose that a convex function f is continuous on an open subset X . Then the correspondence $x \mapsto \partial f(x)$ is an important example of an upper semi-continuous correspondence.

Proposition 8. *Suppose that a convex function f is continuous on an open subset X of a vector space U (for the Mackey topology $\tau(U, U^*)$). Then the correspondence $x \mapsto \partial f(x)$ from U into U^* (supplied with the weak topology $\sigma(U^*, U)$) is upper semi-continuous.*

Proof. We shall prove that the correspondence ∂f maps a neighborhood $x_0 + N$ of x_0 into a compact subset K of U^* and that the restriction of ∂f to $x_0 + N$ has a closed graph. Hence Proposition 2.5.2 implies that the restriction of ∂f to $x_0 + N$ is upper semi-continuous. Since X is open, this implies that ∂f is upper semi-continuous on X .

Since f is continuous at x_0 , there exists a neighborhood $x_0 + 2K^*$ of x_0 such that

$$(21) \quad |f(x_0) - f(y)| \leq \frac{1}{2} \quad \text{for all } y \in x_0 + 2K^*$$

where K is a symmetric convex compact subset of U^* . (Recall that the neighborhoods of U are the polar subsets of the weakly compact subsets of U^* .)

Now, if $x \in x_0 + K^*$, $p \in \partial f(x)$ and $z \in K^*$, then $y = z + x \in x_0 + 2K^*$ and

we obtain

$$(22) \quad -1 \leq f(x) - f(x_0) + f(x_0) - f(x+z) \leq \langle p, -z \rangle$$

by applying (21) twice.

Therefore, for any $x \in x_0 + K^*$ and $p \in \partial f(x)$, $\sup_{z \in K^*} \langle p, z \rangle \leq 1$, i.e. p belongs to K . Thus ∂f maps $x_0 + K^*$ into the weakly compact subset K . Furthermore, the restriction of ∂f to $x_0 + K^*$ has a closed graph. To prove this, suppose that x_μ converges to $\bar{x} \in x_0 + K^*$ and $p_\mu \in \partial f(x_\mu)$ converges to $\bar{p}$ in U^* (for $\sigma(U^*, U)$). We deduce from the inequalities

$$f(x_\mu) - f(y) \leq \langle p_\mu, x_\mu - y \rangle$$

that

$$f(\bar{x}) - f(y) \leq \langle \bar{p}, \bar{x} - y \rangle.$$

Here we have used the fact that $f(x_\mu)$ converges to $f(\bar{x})$, that $\langle p_\mu, y \rangle$ converges to $\langle \bar{p}, y \rangle$ and that $\langle p_\mu, x_\mu \rangle$ converges to $\langle \bar{p}, \bar{x} \rangle$ (because p_μ ranges over the weakly compact subset K and x_μ converges to $\bar{x}$ for the Mackey topology). $\square$

*4.1.5. Characterization of subdifferentiable convex functions

Proposition 9. *Suppose that f is subdifferentiable at x_0 . Then*

$$(23) \quad f(x_0) \leq f(x) + \sigma^*(\partial f(x_0); x_0 - x) \quad \text{for all } x \in U.$$

Proof. When p ranges over $\partial f(x_0)$, we have that

$$\begin{aligned} f(x_0) - f(x) &\leq \langle p, x_0 - x \rangle \\ &\leq \sup_{p \in \partial f(x_0)} \langle p, x_0 - x \rangle = \sigma^*(\partial f(x_0); x_0 - x). \quad \square \end{aligned}$$

We now prove a converse result for which we need “the min-sup theorem” (Theorem 7.1.6) proved in Chapter 7.

Proposition 10. *Let f be a convex proper function and let $x_0 \in \text{Dom } f$. If there exists a weakly compact convex subset $K \subset U^*$ such that*

$$(24) \quad f(x_0) \leq f(x) + \sigma^*(K; x_0 - x) \quad \text{for any } x \in U,$$

then $\partial f(x_0)$ is not empty

Proof. Consider the function

$$(25) \quad \phi(x, p) = f(x) + \langle p, x_0 - x \rangle \quad \text{defined on } X \times K$$

where $X = \text{Dom } f$ is the domain of f . Inequality (24) implies that

$$(26) \quad f(x_0) \leq \inf_{x \in X} \sup_{p \in K} \phi(x, p).$$

Now X is convex and ϕ is convex with respect to x . Also K is convex and compact and ϕ is continuous and concave with respect to p . We deduce the existence of a max-inf $p_0 \in K$, i.e.

$$(27) \quad \begin{aligned} f(x_0) &\leq \inf_{x \in X} \sup_{p \in K} \phi(x, p) = \inf_{x \in X} \phi(x, p_0) \\ &= \inf_{x \in X} (f(x) + \langle p_0, x_0 - x \rangle) \end{aligned}$$

Therefore, p_0 belongs to $\partial f(x_0)$. $\square$

4.2. Differentiability and variational inequalities

4.2.1. Definitions

Subdifferentiability is related to differentiability as follows.

Definition 1. Let f be a proper function mapping a topological vector space U into $]-\infty, +\infty]$.

We shall say that f is *Gâteaux-differentiable* (or, in short, *differentiable*) at $x_0 \in \text{Int Dom } f$ if

$$(1) \quad \begin{cases} \text{(i) } \forall x \in U, \lim_{\theta \rightarrow 0+} (f(x_0 + \theta x) - f(x_0))/\theta = Df(x_0)(x) \text{ exists,} \\ \text{(ii) the functional } x \mapsto Df(x_0)(x) \text{ is linear and continuous.} \end{cases}$$

We shall set in this case

$$(2) \quad Df(x_0)(x) = \langle Df(x_0), x \rangle \quad \text{where } Df(x_0) \in U^*$$

and call $Df(x_0)$ the “*gradient*” or “*derivative*” of f at x_0 .

We shall say that f is differentiable on a subset x if it is differentiable at any point x_0 of x .

The main justification of the concept of differentiability (in the framework of optimization theory) is the *variational principle* which allows us to replace a minimization problem by an equivalent problem in which the loss function is linear.

Proposition 1. *Let us assume that f is differentiable on a convex subset X of a topological vector space U . If $\bar{x} \in X$ minimizes f on X , then*

$$(3) \quad \langle Df(\bar{x}), \bar{x} \rangle = \min_{x \in X} \langle Df(\bar{x}), x \rangle.$$

In particular, if $\bar{x}$ belongs to the interior of X , this condition implies that $Df(\bar{x}) = 0$.

Proof. Since X is convex, $y = \bar{x} + \theta(x - \bar{x}) = (1 - \theta)\bar{x} + \theta x$ belongs to X whenever $x \in X$ and $\theta \in]0, 1[$. Therefore, since $\bar{x}$ minimizes f over X , we obtain

$$(4) \quad \frac{f(\bar{x} + \theta(x - \bar{x})) - f(\bar{x})}{\theta} \geq 0.$$

Taking the limit when $\theta \rightarrow 0+$, we deduce from the differentiability of f at $\bar{x}$ that

$$(5) \quad \langle Df(\bar{x}), x - \bar{x} \rangle \geq 0 \quad \text{for any } x \in X.$$

In particular, if $\bar{x}$ belongs to the strong interior of X , then for any $y \in U$, there exists $\varepsilon > 0$ such that $x = \bar{x} + \varepsilon y$ belongs to X . Thus (5) implies that for any $y \in U$, $\langle Df(\bar{x}), y \rangle \geq 0$. Therefore, $Df(\bar{x}) = 0$. $\square$

4.2.2. Differentiability and subdifferentiability

Conversely, any solution $\bar{x}$ of (3) minimizes f over X provided that f is subdifferentiable at $\bar{x}$.

Proposition 2. *Suppose that f is both differentiable and subdifferentiable at a point $\bar{x}$ of its domain $X = \text{Dom } f$. Then*

$$(6) \quad \partial f(\bar{x}) = \{Df(\bar{x})\}$$

In this case, any solution of

$$(7) \quad \begin{cases} \text{(i) } \bar{x} \in X, \\ \text{(ii) } \forall x \in X, \langle Df(\bar{x}), \bar{x} - x \rangle \leq 0 \end{cases}$$

minimizes f .

Proof. Let $p \in \partial f(\bar{x})$ be any subgradient of f at $\bar{x}$. Then, for any $x \in X$ and any $\theta \in]0, 1[$,

$$(8) \quad \langle p, x \rangle = \frac{1}{\theta} \langle p, \bar{x} + \theta(x - \bar{x}) \rangle \leq \frac{f(\bar{x} + \theta(x - \bar{x})) - f(\bar{x})}{\theta}$$

Taking the limit when θ converges to 0, we deduce that for any $x \in U$,

$$(9) \quad \langle p, x \rangle \leq \langle Df(\bar{x}), x \rangle$$

Therefore, $p = Df(\bar{x})$.

If $\bar{x}$ is a solution of (7), then the fact that $Df(\bar{x})$ is a subgradient implies that $f(\bar{x}) - f(x) \leq \langle Df(\bar{x}), \bar{x} - x \rangle \leq 0$ for any $x \in U$. $\square$

For convex functions, the subdifferentiability hypothesis is redundant.

Proposition 3. *If a convex proper function f is differentiable at a point $\bar{x}$, then it is subdifferentiable and $\partial f(\bar{x}) = \{Df(\bar{x})\}$*

Proof. From the convexity of f , we deduce that, for any $x \in \text{Dom } f$ and $\theta \in]0, 1]$,

$$(10) \quad \frac{f(\bar{x} + \theta(x - \bar{x})) - f(\bar{x})}{\theta} \leq \frac{(1 - \theta)f(\bar{x}) + \theta f(x) - f(\bar{x})}{\theta} \\ = f(x) - f(\bar{x}).$$

Therefore, taking the limit when $\theta \rightarrow 0$, we deduce from the differentiability of f at $\bar{x}$ that

$$(11) \quad \langle Df(\bar{x}), x - \bar{x} \rangle \leq f(x) - f(\bar{x}),$$

i.e. that $Df(\bar{x})$ belongs to the subdifferential of f at $\bar{x}$. $\square$

4.2.3. Legendre transform

Proposition 4. *Suppose that the conjugate function f^* is differentiable at p . Then $\bar{x} = Df^*(p)$ is the unique solution minimizing $x \mapsto f(x) - \langle p, x \rangle$ if and only if*

$$(12) \quad f^{**}[Df^*(p)] = f[Df^*(p)]$$

Remark. Note that the function $p \mapsto \langle p, Df^*(p) \rangle - f[Df^*(p)]$ is called the "Legendre transform" of f .

In particular, when f is a convex function, we obtain the following consequence.

Proposition 5. *Suppose that f is a proper convex lower semi-continuous function. If f^* is differentiable at $p \in U^*$, then $x = Df^*(p)$ is the unique solution which minimizes $x \mapsto f(x) - \langle p, x \rangle$.*

These results motivate the further study of differentiability and/or subdifferentiability of functions and their conjugate functions.

4.2.4. Interpretation: marginal profit

Proposition 5 admits an economic interpretation of the unique minimal solution $\bar{x} = Df^*(p)$ of $x \mapsto f(x) - \langle p, x \rangle$ as the so-called “marginal profit” at p . If $f(x)$ represents the loss or the cost of x , we saw that $f^*(p)$ can be regarded as the maximum profit obtained when price $p \in U^*$ prevails. Hence $Df^*(p)$ is called the “marginal profit”. For any $q \in U^*$,

$$\langle q, Df^*(p) \rangle = \lim_{\theta \rightarrow 0+} \frac{f^*(p + \theta q) - f^*(p)}{\theta}$$

measures the *marginal increase of profit when p moves towards $p + q$* .

More generally, we will interpret the fact that a minimal solution $\bar{x}$ of $f(\cdot) - \langle p, \cdot \rangle$ is a subgradient of the “profit function” f^* at p (Proposition 1.1) by saying that $\bar{x}$ is a “marginal profit”.

4.2.5. Variational inequalities

We consider now the case when $f = g + h$ and

$$(13) \quad \begin{cases} \text{(i) } g \text{ is convex and differentiable on } X, \\ \text{(ii) } h \text{ is convex.} \end{cases}$$

Proposition 6. *Let X be a convex subset of a vector space U . Suppose that (13) holds. Then $\bar{x}$ minimizes $f = g + h$ on X if and only if $\bar{x}$ is a solution of*

$$(14) \quad \begin{cases} \text{(i) } \bar{x} \in X, \\ \text{(ii) } \forall y \in X, \langle Dg(\bar{x}), \bar{x} - y \rangle + h(\bar{x}) - h(y) \leq 0. \end{cases}$$

Proof. Let $\bar{x} \in X$ be an optimal solution and $y \in X$. Then, if $\theta \in]0, 1]$, $x = \bar{x} + \theta(y - \bar{x}) \in X$ and thus,

$$\begin{aligned} g(\bar{x}) + h(\bar{x}) &= f(\bar{x}) \leq f(x) = f(\bar{x} + \theta(y - \bar{x})) \\ &= g(\bar{x} + \theta(y - \bar{x})) + h(\bar{x} + \theta(y - \bar{x})) \\ &\leq g(\bar{x} + \theta(y - \bar{x})) + (1 - \theta)h(\bar{x}) + \theta h(y). \end{aligned}$$

Dividing this inequality by $\theta > 0$, we obtain

$$h(\bar{x}) - h(y) \leq \frac{g(\bar{x} + \theta(y - \bar{x})) - g(\bar{x})}{\theta}.$$

We let θ converge to 0 and obtain (14).

Conversely, since $\{Dg(\bar{x})\} = \partial g(\bar{x})$, we deduce from (14) that, for any $y \in X$,

$$(15) \quad \begin{aligned} f(\bar{x}) - f(y) &= g(\bar{x}) - g(y) + h(\bar{x}) - h(y) \\ &\leq \langle Dg(\bar{x}), \bar{x} - y \rangle + h(\bar{x}) - h(y) \leq 0. \quad \square \end{aligned}$$

These results motivate the following definitions.

Definition 2. Let X be a convex subset of a topological vector space U and let g and h satisfy (13). A solution $\bar{x}$ of

$$(16) \quad \begin{cases} \text{(i) } \bar{x} \in X, \\ \text{(ii) } \forall y \in X, \quad \langle Dg(\bar{x}), \bar{x} - y \rangle + h(\bar{x}) - h(y) \leq 0 \end{cases}$$

is called a *stationary point* of $f = g + h$.

More generally, if S is a map from X into U^* , inequalities of the form

$$(17) \quad \begin{cases} \text{(i) } \bar{x} \in X, \\ \text{(ii) } \forall x \in X, \quad \langle S(\bar{x}), \bar{x} - x \rangle \leq 0 \end{cases}$$

or

$$(18) \quad \begin{cases} \text{(i) } \bar{x} \in X, \\ \text{(ii) } \forall x \in X, \quad \langle S(\bar{x}), \bar{x} - x \rangle + h(\bar{x}) - h(x) \leq 0 \end{cases}$$

will be called “variational inequalities”.

The existence of solutions to such inequalities will be studied in Section 13.2.

4.3. Differentiability from the right

We shall extend slightly the concept of differentiability and show how it is related to the concept of subdifferentiability.

4.3.1. Definition and main inequalities

Definition 1. Let f be a proper function mapping a topological vector space U into $] -\infty, +\infty]$.

We shall say that f is “differentiable from the right” at $x_0 \in \text{Dom } f$ if

$$(1) \quad \forall x \in U, \quad \lim_{\theta \rightarrow 0+} \frac{f(x_0 + \theta x) - f(x_0)}{\theta} = Df(x_0)(x)$$

exists.

The positively homogeneous functional $x \mapsto Df(x_0)(x)$ is called the “*derivative from the right*” of f . We shall say that f is differentiable from the right on a subset M if it is differentiable from the right at any point $x \in M$.

We begin by proving that any subdifferentiable convex function at x_0 is differentiable from the right.

Proposition 1. *Let f be a convex proper function subdifferentiable at x_0 . Then f is differentiable from the right at x_0 . The function $x \mapsto Df(x_0)(x)$ is convex and positively homogeneous and*

$$(2) \quad \sigma^*(\partial f(x_0); x) \leq Df(x_0)(x).$$

Proof. For all $p \in \partial f(x_0)$, $x \in U$ and $\lambda > 0$, we have that

$$(3) \quad \langle p, x \rangle = \frac{1}{\lambda} \langle p, x_0 + \lambda x - x_0 \rangle \leq \frac{f(x_0 + \lambda x) - f(x_0)}{\lambda}.$$

On the other hand, the function $\lambda \mapsto \theta(\lambda) = f(x_0 + \lambda x) - f(x_0)$ is a convex function vanishing at $\lambda = 0$. Therefore, $\lambda \mapsto \frac{\theta(\lambda)}{\lambda}$ is increasing. Thus,

$$(4) \quad \begin{aligned} \langle p, x \rangle &\leq \inf_{\lambda > 0} \frac{f(x_0 + \lambda x) - f(x_0)}{\lambda} = \lim_{\lambda \rightarrow 0+} \frac{f(x_0 + \lambda x) - f(x_0)}{\lambda} \\ &= Df(x_0)(x). \end{aligned}$$

Since this inequality is true for all $p \in \partial f(x_0)$, we obtain inequality (2). The convexity of $x \mapsto Df(x_0)(x)$ is obvious. $\square$

Proposition 2. *Let f be a convex proper function differentiable from the right at x_0 . Then,*

$$(5) \quad Df(x_0)(x - x_0) \leq f(x) - f(x_0)$$

and

$$(6) \quad f(x_0) - f(x) \leq Df(x_0)(x_0 - x).$$

Proof. The first inequality follows from $(f(x_0 + \theta(x - x_0)) - f(x_0))/\theta \leq f(x) - f(x_0)$ by letting θ go to $0+$. To prove the second inequality, we write

$$x_0 = \frac{1}{1+\theta} (x_0 + \theta(x_0 - x)) + \frac{\theta}{1+\theta} x.$$

The convexity of f implies that

$$f(x_0) \leq \frac{1}{1+\theta} [f(x_0 + \theta(x_0 - x)) + \theta f(x)].$$

We deduce that

$$f(x_0) - f(x) \leq \frac{f(x_0 + \theta(x_0 - x)) - f(x_0)}{\theta}$$

from which we obtain the second inequality by letting θ go to $0+$ $\square$.

4.3.2. Derivatives from the right and the support function of the subdifferential

Proposition 3. Let $f: U \rightarrow]-\infty, +\infty]$ be convex and lower semi-continuous. Suppose that $\partial f(x_0)$ is non-empty. Then

$$(7) \quad Df(x_0)^{**}(x) = \sigma^*(\partial f(x_0); x) = \sup_{p \in \partial f(x_0)} \langle p, x \rangle.$$

Proof. Since $\partial f(x_0)$ is a closed convex subset, (7) is equivalent to

$$(8) \quad Df(x_0)^*(p) = \psi(\partial f(x_0); p) \quad \text{for all } p \in U^*.$$

Since

$$\begin{aligned} Df(x_0)(x) &= \inf_{\lambda > 0} \frac{f(x_0 + \lambda x) - f(x_0)}{\lambda}, \\ Df(x_0)^*(p) &= \sup_{x \in U} \left[\langle p, x \rangle - \inf_{\lambda > 0} \frac{f(x_0 + \lambda x) - f(x_0)}{\lambda} \right] \\ &= \sup_{x \in U} \sup_{\lambda > 0} \frac{1}{\lambda} [\langle p, \lambda x \rangle - f(x_0 + \lambda x) + f(x_0)] \\ (9) \quad &= \sup_{\lambda > 0} \frac{1}{\lambda} \left(\sup_{x \in U} [\langle p, \lambda x + x_0 \rangle - f(x_0 + \lambda x)] + \right. \\ &\quad \left. - f(x_0) - \langle p, x_0 \rangle \right) \\ &= \sup_{\lambda > 0} \frac{1}{\lambda} (f^*(p) + f(x_0) - \langle p, x_0 \rangle). \end{aligned}$$

If $p \in \partial f(x_0)$, then $f(p) + f(x_0) - \langle p, x_0 \rangle = 0$ and thus, $Df(x_0)^*(p) = 0$.

If $p \notin \partial f(x_0)$, then $f^*(p) + f(x_0) - \langle p, x_0 \rangle = \theta > 0$ and $Df(x_0)^*(p) = \sup_{\lambda > 0} \theta/\lambda = +\infty$.

Therefore, $Df(x_0)^*(p)$ is the indicator $\psi(\partial f(x_0); \cdot)$ of $\partial f(x_0)$. $\square$

Proposition 4. *Let f be a convex proper function continuous at $x_0 \in \text{Int Dom } f$. Then the map*

$$(10) \quad x \mapsto Df(x_0)(x) = \sigma^*(\partial f(x_0); x) \text{ is continuous on } U.$$

Proof. Since

$$(11) \quad Df(x_0)(x) \leq f(x_0 + x) - f(x_0) = \phi(x)$$

and ϕ is continuous at 0, $Df(x_0)$ is bounded above on a neighborhood of 0. Therefore, $x \mapsto Df(x_0)(x)$ being a convex function bounded from above on a neighborhood, it is continuous (by Proposition (3.3.2)). This implies in particular that

$$(12) \quad Df(x_0)(x) = Df(x_0)^{**}(x) = \sigma^*(\partial f(x_0); x)$$

(see Proposition 3). $\square$

Proposition 5. *Let f be a convex proper function defined on U . If*

$$(13) \quad f \text{ is continuous at } x_0 \text{ and } \partial f(x_0) \text{ contains a unique element,}$$

then f is differentiable at x_0 and $\partial f(x_0) = \{Df(x_0)\}$.

Proof. If f is continuous at x_0 and if $\partial f(x_0) = \{p_0\}$ contains the unique element p_0 , then Proposition 4 implies that $Df(x_0)(x) = \sigma^*(\{p_0\}; x) = \langle p_0, x \rangle$ for any x . Thus $p_0 = Df(x_0)$ is the derivative of f at x_0 . $\square$

*4.3.3. Derivative of a pointwise supremum

Proposition 6. *Consider a family of functions $x \mapsto f(x; p)$ where p ranges over a set P . Suppose that*

$$(14) \quad \left\{ \begin{array}{l} \text{(i) } P \text{ is compact,} \\ \text{(ii) there exists a neighborhood } U \text{ of } x \text{ such that for any } y \in U, \\ \quad p \mapsto f(y; p) \text{ is upper semi-continuous,} \\ \text{(iii) } \forall p \in P, y \mapsto f(y; p) \text{ is convex and differentiable from the} \\ \quad \text{right at } x. \end{array} \right.$$

Consider the function g defined by

$$(15) \quad g(y) = \sup_{p \in P} f(y; p)$$

and the subset

$$(16) \quad P_0 = \{p \in P \text{ such that } g(x) = f(x; p)\}.$$

Then g is differentiable from the right at x and

$$(17) \quad Dg(x)(y) = \sup_{p \in P_0} Df(x; p)(y).$$

Proof. For any $p \in P_0$, we can write

$$(18) \quad \frac{f(x+\theta y; p) - f(x; p)}{\theta} \leq \frac{g(x+\theta y) - g(x)}{\theta}.$$

Letting θ tend to 0, we deduce that

$$(19) \quad \sup_{p \in P_0} Df(x; p)(y) \leq Dg(x)(y).$$

We now prove the converse inequality. Put $a = Dg(x)(y) - \varepsilon$ where $\varepsilon > 0$. We prove that there exists $\bar{p} \in P_0$ such that $a \leq Df(x; \bar{p})(y)$. The function g is convex and so

$$(20) \quad a < Dg(x)(y) = \inf_{\theta > 0} \frac{g(x+\theta y) - g(x)}{\theta}.$$

Therefore, for any $\theta > 0$,

$$(21) \quad B_\theta = \left\{ p \in P \text{ such that } \frac{f(x+\theta y; p) - g(x)}{\theta} \geq a \right\} \neq \emptyset.$$

Let U be the neighborhood of x appearing in assumption 14 (ii). Then there exists $\theta_0 > 0$ such that $x + \theta y \in U$ for any $\theta \in [0, \theta_0]$. Since $p \mapsto f(x + \theta y; p)$ is upper semi-continuous, B_θ is a closed subset of P . Furthermore, $\theta \mapsto B_\theta$ increases with θ . To prove this let $\theta_1 \leq \theta_2$ and $p \in B_{\theta_1}$. The convexity of $x \rightarrow f(x; p)$ implies that

$$\begin{aligned} a &\leq \frac{1}{\theta_1} \left[\left(1 - \frac{\theta_1}{\theta_2}\right) (f(x; p) - g(x)) + \frac{\theta_1}{\theta_2} (f(x + \theta_2 y; p) - g(x)) \right] \\ &\leq \frac{1}{\theta_2} (f(x + \theta_2 y; p) - g(x)) \end{aligned}$$

since $x + \theta_1 y = (1 - \theta_1/\theta_2)x + (\theta_1/\theta_2)(x + \theta_2 y)$ and $f(x; p) - g(x) \leq 0$ for all p by definition of g . Because P is compact, there exists $\bar{p}$ belonging to the non-empty intersection $\bigcap_{0 < \theta \leq \theta_0} B_\theta$. Therefore, for any $\theta \in]0, \theta_0]$, we have $\theta a \leq f(x + \theta y; \bar{p}) - g(x)$. Letting θ tend to 0, we deduce that

$g(x) \leq f(x; \bar{p})$ and thus, that $\bar{p} \in P_0$. Therefore, $g(x) = f(x; \bar{p})$ and we can write

$$(22) \quad a \leq \frac{f(x + \theta y; \bar{p}) - f(x; \bar{p})}{\theta}.$$

This implies that $a \leq Df(x; \bar{p})(y)$. $\square$

*4.4. Local ε -subdifferentiability and perturbed minimization problems

We deal in this section with minimization problems for which the loss function f is lower semi-continuous and bounded below, but not necessarily lower semi-compact. For such problems f does not necessarily achieve its minimum. However, when U is a Banach space, we shall prove the existence of approximate solutions: We show that, for any $\varepsilon > 0$, there exists $\bar{x}$ such that

$$(*) \quad \begin{cases} \text{(i)} & f(\bar{x}) \leq \inf_{x \in X} f(x) + \varepsilon, \\ \text{(ii)} & \forall y \in X, f(\bar{x}) \leq f(y) + \sqrt{\varepsilon} \|\bar{x} - y\|. \end{cases}$$

If X is convex and f is differentiable this implies that $\bar{x}$ is a solution to the variational inequalities.

$$(**) \quad \forall y \in X, \langle Df(\bar{x}), \bar{x} - y \rangle \leq \sqrt{\varepsilon} \|\bar{x} - y\|.$$

In particular, if f is differentiable on its (open) domain X , and bounded from below, we exhibit x_ε such that

$$(**) \quad \begin{cases} \text{(i)} & f(x_\varepsilon) \leq \inf_{x \in X} f(x) + \varepsilon, \\ \text{(ii)} & \|Df(x_\varepsilon)\|_* \leq \sqrt{\varepsilon}. \end{cases}$$

In other words, the equation $Df(x) = 0$, although it need have no solution, always has an "approximate solution" x_ε . We also use (*) to prove that any lower semi-continuous function on a Hilbert space is "locally ε -subdifferentiable" in a dense subset of its domain.

(We say that f is locally ε -subdifferentiable at $\bar{x}$ if there exist $\eta > 0$ and $p \in U^*$ such that $f(\bar{x}) - f(x) \leq \langle p, \bar{x} - x \rangle + \varepsilon \|\bar{x} - x\|$ whenever $\|\bar{x} - x\| \leq \eta$.) This fact is used to study the family of perturbed problems

$$\alpha(p) = \inf_{x \in X} f(x, p)$$

as p ranges over a Hilbert space U^* .

We require only a weak assumption on $x \mapsto f(x, p)$, namely that

$x \mapsto f(x, p)$ is lower semi-continuous and bounded from below for any $p \in U^*$

In addition, however, we need a set of strong assumptions pertaining to the smoothness of the perturbations (for the precise assumptions, see Theorem 3 below).

We then prove that for “almost all” minimization problems $\alpha(p)$ there exists a unique optimal solution, in the sense that the property holds when p ranges over a dense denumerable intersection of open subsets.

As an example, we apply this result to linear perturbations, and prove that any continuous convex function on a Hilbert space is differentiable on a dense denumerable intersection of open subsets.

Finally we note that a best approximant to $x \in U$ by an element of a closed bounded (non convex) subset U exists and is unique for “almost all” x .

4.4.1. Approximate optimal solutions in Banach spaces

Consider a proper function $f: U \rightarrow]-\infty, +\infty]$ which is bounded below. Then, for any $\varepsilon > 0$, there exists $x_\varepsilon \in U$ such that

$$(1) \quad \alpha = \inf_{x \in U} f(x) \leq f(x_\varepsilon) \leq \alpha + \varepsilon.$$

Proposition 1. *Let f be a lower semi-continuous proper function mapping a Banach space U into $]-\infty, +\infty]$. Then, for every x_ε satisfying (1) and every $\lambda > 0$, there exists some $\bar{x} \in U$ such that*

$$(2) \quad \begin{cases} (i) f(\bar{x}) \leq f(x_\varepsilon), \\ (ii) \|x_\varepsilon - \bar{x}\| \leq \lambda, \\ (iii) \forall z \neq \bar{x}, f(\bar{x}) < f(z) + (\varepsilon/\lambda) \|\bar{x} - z\|. \end{cases}$$

The proof of this result is based on a device due to Bishop, Phelps and Browder. Let $\varrho > 0$ be given. We define an ordering on $U \times \mathbf{R}$ by

$$(3) \quad \{y_1, a_1\} \leq \{y_2, a_2\} \text{ if and only if } (a_2 - a_1) + \varrho \|y_2 - y_1\| \leq 0.$$

This relation is easily seen to be reflexive, antisymmetric and transitive. Also, its upper sections $\{\{y, b\} \geq \{x, a\}\}$ are obviously closed. We proceed to show that every closed subset of $U \times \mathbf{R}$ has a maximal element, provided it is “bounded below”.

Lemma 1. (Bishop—Phelps—Browder). *Let S be a closed subset of $U \times \mathbf{R}$ such that*

$$(4) \quad \beta = \inf \{a \text{ such that } \{x, a\} \in S\} > -\infty$$

Then, for any $\{y, b\} \in S$, there exists a maximal element $\{\bar{x}, \bar{a}\} \in S$ larger than $\{y, b\}$.

Proof. We define by induction a sequence $\{x_n, a_n\} \in S$

with $\{x_1, a_1\} = \{y, b\}$. Suppose $\{x_n, a_n\}$ is known. We set

$$(5) \quad \begin{cases} (i) S_n = \{\{x, a\} \text{ such that } \{x, a\} \supseteq \{x_n, a_n\}\}, \\ (ii) \beta_n = \inf \{a \text{ such that } \{x, a\} \in S_n\}. \end{cases}$$

Clearly, $\beta_n \geq \beta$. We now define $\{x_{n+1}, a_{n+1}\}$ to be any point of S_n such that

$$(6) \quad a_n - a_{n+1} \geq \frac{1}{2}(a_n - \beta_n).$$

All the subsets S_n are closed and non-empty and $S_{n+1} \subset S_n$ for every n . Moreover, we deduce from (6) that

$$(7) \quad |a_{n+1} - \beta_{n+1}| \leq \frac{1}{2}|a_n - \beta_n| \leq (1/2^n)|b - \beta|.$$

Hence, for every $\{x, a\} \in S_n$, we obtain from (5(i)) that

$$(8) \quad \begin{cases} (i) |a_{n+1} - a| \leq (1/2^n)|b - \beta|, \\ (ii) \|x_{n+1} - x\| \leq (1/\varrho \cdot 2^n)|b - \beta|. \end{cases}$$

This proves that the diameter of S_n converges to 0 as $n \rightarrow \infty$. Since U is a complete metric space, the sets S_n have a unique element $\{\bar{x}, \bar{a}\}$ in common, i.e.

$$(9) \quad \bigcap_{n=1}^{\infty} S_n = \{\bar{x}, \bar{a}\}.$$

By definition of S_n , we have $\{\bar{x}, \bar{a}\} \supseteq \{x_n, a_n\}$ for all n . In particular, for $n = 1$, this implies that $\{\bar{x}, \bar{a}\} \supseteq \{y, b\}$. Suppose that $\{\bar{x}, \bar{a}\}$ is not maximal. Then there exists $\{x^*, a^*\} \in S$ strictly larger than $\{\bar{x}, \bar{a}\}$. By transitivity, $\{x^*, a^*\}$ is larger than $\{x_n, a_n\}$ for any n . Hence, $\{x^*, a^*\} \in \bigcap_{n=1}^{\infty} S_n$, i.e. $\{x^*, a^*\} = \{\bar{x}, \bar{a}\}$. This proves that $\{\bar{x}, \bar{a}\}$ is indeed maximal. $\square$

Proof of Proposition 1. We take $S = \mathcal{E}p(f)$ to be the epigraph of f . It is closed since f is lower semi-continuous. Take $\varrho = \varepsilon/\lambda$ and $\{y, b\} = \{x_e, f(x_e)\}$. Lemma 1 implies the existence of a maximal element $\{\bar{x}, \bar{a}\}$ in S satisfying $\{\bar{x}, \bar{a}\} \supseteq \{x_e, f(x_e)\}$. Since $\{\bar{x}, \bar{a}\} \in S$, we also have $\{\bar{x}, f(\bar{x})\} \supseteq \{\bar{x}, \bar{a}\}$. But, $\{\bar{x}, \bar{a}\}$ being maximal, this implies that $\bar{a} = f(\bar{x})$. The maximality of $\{\bar{x}, f(\bar{x})\}$ means that

$$(10) \quad (a - f(\bar{x})) + \frac{\varepsilon}{\lambda} \|x - \bar{x}\| > 0 \quad \text{whenever } \{x, a\} \in S$$

unless $x = \bar{x}$ and $a = f(\bar{x})$.

Taking $a = f(x)$ yields (2(iii)). Now, since $\{\bar{x}, \bar{a}\} = \{\bar{x}, f(\bar{x})\} \supseteq \{x_\varepsilon, f(x_\varepsilon)\}$, we have that

$$(11) \quad f(\bar{x}) - f(x_\varepsilon) + \frac{\varepsilon}{\lambda} \|x_\varepsilon - \bar{x}\| \leq 0.$$

Hence, of course, $f(\bar{x}) \leq f(x_\varepsilon)$ (i.e., (2(i))). It follows from (1) that $f(\bar{x}) \geq f(x_\varepsilon) - \varepsilon$. Substituting this result in (11), we obtain that $(\varepsilon/\lambda) \|\bar{x} - x_\varepsilon\| \leq \varepsilon$. But this is (2(ii)) and so the proof is complete. $\square$

4.4.2. The approximate variational principle

We have seen that, if a differentiable function f achieves its minimum at some $\bar{x}$ in the interior of its domain, then $Df(\bar{x}) = 0$. We shall prove that, if a differentiable function is bounded below (although it need not attain its bound) then, for any $\varepsilon > 0$, there exists $x_\varepsilon \in X$ such that $\|Df(x_\varepsilon)\| \leq \varepsilon$ (i.e. its derivative can be made arbitrarily small).

Theorem 1. (Ekeland). *Let X be a convex subset of U and let $f : X \rightarrow \mathbf{R}$ be a differentiable function which is bounded below. Then, for any $\varepsilon > 0$, there exists x_ε such that*

$$(12) \quad f(x_\varepsilon) \leq \alpha + \varepsilon \quad \text{where } \alpha = \inf_{x \in X} f(x).$$

Further, for any $\lambda > 0$, there exists $\bar{x}$ such that

$$(13) \quad \begin{cases} \text{(i)} & f(\bar{x}) \leq f(x_\varepsilon) \leq \alpha + \varepsilon, \\ \text{(ii)} & \|\bar{x} - x_\varepsilon\| \leq \lambda \end{cases}$$

and

$$(14) \quad \forall y \in X, \quad \langle Df(\bar{x}), \bar{x} - y \rangle \leq \frac{\varepsilon}{\lambda} \|\bar{x} - y\|.$$

If we also assume that $\bar{x}$ belongs to the interior of X , inequalities (14) become

$$(15) \quad \|Df(\bar{x})\|_* \leq \frac{\varepsilon}{\lambda}.$$

Proof. By Proposition 1, we know that there exists $\bar{x}$ such that, for any $z \in X$,

$$(16) \quad f(\bar{x}) \leq f(z) + \frac{\varepsilon}{\lambda} \|\bar{x} - z\|.$$

Hence, taking $z = \bar{x} + \theta(y - \bar{x})$, where $y \in X$ and $\theta \in]0, 1[$, we obtain that

$$f(\bar{x}) \leq f(\bar{x} + \theta(y - \bar{x})) + \frac{\varepsilon}{\lambda} \theta \|y - \bar{x}\|.$$

Therefore, dividing by θ and letting θ converge to 0,

$$(17) \quad 0 \leq \langle Df(\bar{x}), y - \bar{x} \rangle + \frac{\varepsilon}{\lambda} \|y - \bar{x}\|.$$

If $\bar{x}$ belongs to the interior of X , we take $y = \bar{x} + \theta u$, with u in the unit ball. Then

$$(18) \quad \langle Df(\bar{x}), u \rangle \leq \frac{\varepsilon}{\lambda} \|u\|,$$

which implies that $\|Df(\bar{x})\|_* \leq \varepsilon/\lambda$. $\square$

Remark. On taking $\lambda = \sqrt{\varepsilon}$, the conclusions (13) and (14) of Theorem 1 become

$$(19) \quad \begin{cases} \text{(i)} & f(\bar{x}) \leq f(x_*) \leq \alpha + \varepsilon, \\ \text{(ii)} & \|\bar{x} - x_*\| \leq \sqrt{\varepsilon}, \\ \text{(iii)} & \langle Df(\bar{x}), \bar{x} - y \rangle \leq \sqrt{\varepsilon} \|\bar{x} - y\|, \forall y \in X. \end{cases}$$

4.4.3. Local ε -subdifferentiability

We shall need to introduce both a stronger concept than differentiability and a weaker concept than subdifferentiability of studying the regularity of perturbed problem. We begin by recalling the concept of a Fréchet-derivative.

Definition 1. Let U be a Banach space. We shall say that a differentiable function f is “Fréchet-differentiable” at $\bar{x}$ if

$$(20) \quad \lim_{\|y\| \rightarrow 0} \frac{|f(\bar{x} + y) - f(\bar{x}) - \langle Df(\bar{x}), y \rangle|}{\|y\|} = 0.$$

Remark. One can give sufficient conditions for a differentiable function to be Fréchet-differentiable. For instance, one can prove that, if f is a differentiable function with open domain $X = \text{Dom } f$ and if $x \mapsto Df(x)$ is continuous from X into the Banach space U^* , then f is Fréchet-differentiable on X .

We now define our weakest concept of differentiability.

Definition 2. We shall say that a function f is “locally ε -subdifferentiable” at $\bar{x}$ if there exist a continuous linear form $p \in U^*$ and a positive number η such that

$$(21) \quad f(\bar{x}) - f(x) \leq \langle \bar{p}, \bar{x} - x \rangle + \varepsilon \|\bar{x} - x\| \quad \text{whenever } \|\bar{x} - x\| \leq \eta.$$

Of course, $\bar{p} \in U^*$ and $\eta > 0$ depend upon ε . Note that if f is locally ε -subdifferentiable at $\bar{x}$, it is locally ε' -subdifferentiable for every $\varepsilon' \geq \varepsilon$. This shows that we can associate with any $\bar{x} \in U$ a number $\varepsilon_0 \in [0, +\infty]$ such that f is locally ε -differentiable at x for $\varepsilon > \varepsilon_0$ but not for $\varepsilon < \varepsilon_0$. The more regular functions have smaller values of ε_0 .

If $\varepsilon = 0$, we shall say that f is "locally subdifferentiable" at $\bar{x}$. With $\varepsilon = 0$ and $\eta = +\infty$, (21) reduces to our familiar concept of subdifferentiability (Definition 4.1.1).

The definition can be stated in another way: f is locally ε -subdifferentiable at $\bar{x}$ if there exists $\bar{p} \in U^*$ such that the function $x \mapsto f(x) - \langle \bar{p}, x \rangle + \varepsilon \|\bar{x} - x\|$ achieves a local minimum at $\bar{x}$.

The concept of local ε -subdifferentiability is so weak that any lower semi-continuous function satisfies it "almost everywhere" (in the sense of the following theorem).

Theorem 2. (Ekeland). *Let f be a proper function mapping a Hilbert space U into $]-\infty, +\infty]$. If f is lower semi-continuous, the set of points where f is locally ε -subdifferentiable is dense in $\text{Dom } f$.*

Proof. Let $x_0 \in \text{Dom } f$, $\delta^* > 0$ and $\varepsilon > 0$ be given. We shall find a point $\bar{x}$ such that $\|\bar{x} - x_0\| \leq \eta$, where f is locally $2\sqrt{\varepsilon}$ -subdifferentiable.

Since f is lower semi-continuous, there exists $\delta \leq \delta^*$ such that

$$f(x_0) - a \leq f(x) \quad \text{whenever } \|x - x_0\| \leq \delta.$$

Consider the function φ defined on U by

$$(22) \quad \varphi(x) = \begin{cases} \exp\left(\frac{1}{\delta^2 - \|x\|^2}\right) & \text{if } \|x\| < \delta, \\ +\infty & \text{if } \|x\| \geq \delta. \end{cases}$$

The domain of φ is an open ball of radius δ which we denote by $B(\delta)$. Furthermore,

$$(23) \quad \varphi \text{ is Fréchet-differentiable on } \text{Dom } \varphi = \dot{B}(\delta).$$

(Its derivative is equal to $(2/(\delta^2 - \|x\|^2)) \varphi(x) Jx$, where J is the duality operator from U into U^* .) We introduce a new function g defined by

$$(24) \quad g(x) = f(x) + \varphi(x - x_0)$$

whose non-empty domain is $\text{Dom } f \cap (x_0 + \dot{B}(\delta))$. It is bounded below by $f(x_0) - a$. From Proposition 1 (with $\lambda = \sqrt{\varepsilon}$), we deduce that there exists

$\bar{x} \in \text{Dom } g \subset x_0 + B(\delta)$ such that

$$(25) \quad \begin{cases} \text{(i) } g(\bar{x}) \leq f(x_0) - a + \varepsilon, \\ \text{(ii) } g(\bar{x}) \leq g(x) + \sqrt{\varepsilon} \|\bar{x} - x\| \text{ for any } x \in \text{Dom } g. \end{cases}$$

Substituting (24) in (25(ii)), we obtain that

$$(26) \quad f(\bar{x}) - f(x) \leq \varphi(x - x_0) - \varphi(\bar{x} - x_0) + \sqrt{\varepsilon} \|\bar{x} - x\|.$$

But, since φ is Fréchet-differentiable, there exists η such that, with $\bar{p} = -D\varphi(\bar{x} - \bar{x}_0)$,

$$(27) \quad |\varphi(\bar{x} - x_0) - \varphi(x - x_0) + \langle \bar{p}, \bar{x} - x \rangle| \leq \sqrt{\varepsilon} \|\bar{x} - x\| \quad \text{whenever} \\ \|\bar{x} - x\| \leq \eta.$$

Hence (26) and (27) imply that f is locally $2\sqrt{\varepsilon}$ differentiable at $\bar{x} \in x_0 + B(\delta)$. This completes the proof. $\square$

Remark. Since we used the assumption that U is a Hilbert space only via (23), *Theorem 2 remains true whenever there exists a function φ satisfying (23).* If a Banach space U has a norm equivalent to a Fréchet-differentiable norm, we can still use the function φ defined by (22).

Remark. When f is convex, we can prove a global version of this theorem.

4.4.4. Perturbation of minimization problems

We consider now a family of minimization problems

$$(28) \quad \alpha(p) = \inf_{x \in X} f(x, p).$$

We shall prove that, for “almost all” $p \in V^*$, (i.e. for every p of a dense denumerable intersection of open subsets), the minimization problem has a *unique* solution $\bar{x}(p)$ to which any minimizing sequence $\{x_n\}$ of $\alpha(p)$ converges. Of course, the necessary assumptions are quite strong, but do not unduly restrict the class of perturbations to which the result applies.

Theorem 3 (Ekeland—Lebourg). *Suppose that*

$$(29) \quad V \text{ is a Hilbert space, } X \text{ is a closed subset of a Banach space}$$

and that

$$(30) \quad \begin{cases} \text{(i) } \forall p \in V^*, x \mapsto f(x, p) \text{ is lower semi-continuous,} \\ \text{(ii) } p \mapsto \alpha(p) \text{ is continuous} \end{cases}$$

Suppose also that

$$(31) \quad \forall x \in X, \quad p \mapsto f(x, p) \text{ is Fréchet-differentiable on } V^*$$

and that

$$(32) \quad \begin{cases} \forall p \in V^*, \forall \varepsilon > 0, \exists \eta > 0 \text{ such that } \|x_1 - x_2\| \leq \varepsilon \\ \text{provided that } f(x_i, p) \leq \alpha(p) + \eta \text{ (} i = 1, 2 \text{) and} \\ \|Df(x_1, p) - Df(x_2, p)\| \leq \eta. \end{cases}$$

Finally, suppose that

$$(33) \quad \begin{cases} \forall p \in V^*, \forall \varepsilon > 0, \exists \eta > 0 \text{ such that} \\ |f(x, q) - f(x, p) - \langle Df(x, p), q - p \rangle| \leq \varepsilon \|p - q\| \\ \text{provided that } \|p - q\| \leq \eta \text{ and } f(x, q) \leq \alpha(p) + \eta \end{cases}$$

and that

$$(34) \quad \begin{cases} \forall p \in V^* \text{ and } \forall \lambda \in \mathbf{R}, \quad |f(p_n, x_n) - f(p, x_n)| \rightarrow 0 \\ \text{provided that } p_n \rightarrow p \text{ and } f(p_n, x_n) \leq \lambda. \end{cases}$$

Then there exists a dense denumerable intersection T of open subsets of V such that, $\forall p \in T$,

$$(35) \quad \begin{cases} \text{(i) the minimization problem } \alpha(p) \text{ has a unique solution } \bar{x}(p), \\ \text{(ii) any minimizing sequence } x_n \text{ converges to } \bar{x}(p). \end{cases}$$

Before proving this theorem, we give an application to the case of perturbation by linear functionals.

Proposition 2. (Asplund). *Let X be a closed bounded subset of a Hilbert space U . Let $f : X \rightarrow \mathbf{R}$ be a lower semi-continuous function which is bounded below and let α be defined on U by*

$$(36) \quad \alpha(p) = \inf_{x \in X} [f(x) - \langle p, x \rangle].$$

Then there exists a dense denumerable intersection T of open subsets such that (35) holds.

Proof. We check the assumptions of Theorem 3. Note that, in this particular case, the functions $p \mapsto f(x) - \langle p, x \rangle$ are k -Lipschitzian on U^* , where $k = \sup_{x \in X} \|x\|$. This is finite because X is bounded. The “equicontinuity assumption” (34) follows. It follows that the function α , which is finite because f is bounded below and X is bounded, is k -Lipschitzian, and hence continuous.

Therefore assumptions (30) are satisfied. If we set $f(x, p) = f(x) - \langle p, x \rangle$, we see that $Df(x, p) = -x$. Thus

$$(37) \quad \|Df(x, p)\|_* = \|x\| \text{ is bounded } \forall x \in X, \forall p \in U^*.$$

We deduce that assumption (32) is satisfied, with $\eta = \varepsilon$, since $\|Df(x_1, p) - Df(x_2, p)\| = \|x_1 - x_2\|$. Finally, assumption (33) is obviously satisfied because we always have that

$$(38) \quad f(x, q) - f(x, p) - \langle Df(x, p), q - p \rangle = 0. \quad \square$$

Remark. Suppose moreover that

$$(39) \quad f \text{ is convex.}$$

Since $\alpha(p) = -f_x^*(p)$, condition (35(i)) amounts to saying that $\partial f_x^*(p)$ has a unique element, namely the derivative of f_x^* at p . In fact, by refining the argument a little, we obtain the following result. This sheds new light on the regularity of convex functions (compare with Proposition 4.3.4).

Proposition 3. *Let U be a Hilbert space. Any continuous convex function defined on U is differentiable on a dense denumerable intersection T of open subsets.*

Proof. We shall prove this result for the conjugate f^* of a convex function f , where f^* is assumed to be continuous. We apply Theorem 3 to the problems

$$\alpha(p) = \inf_{x \in U} [f(x) - \langle p, x \rangle]$$

where $X = \text{Dom } f$ is no longer assumed to be bounded.

Again, all the assumptions of Theorem 3 are trivially satisfied except for the "equicontinuity assumption" (34) which we check directly. Let $\lambda \in \mathbb{R}$ and $\{p_n, x_n\}$ be a sequence such that

$$(40) \quad \begin{cases} \text{(i) } p_n \text{ converges to } p, \\ \text{(ii) } f(x_n) - \langle p_n, x_n \rangle \leq \lambda. \end{cases}$$

We have $f(p_n, x_n) - f(p, x_n) = \langle p - p_n, x_n \rangle$.

On the other hand, for any $q \in U$,

$$(41) \quad \begin{aligned} \langle q, x_n \rangle &= \langle q + p_n, x_n \rangle - \langle p_n, x_n \rangle \\ &\leq f^*(q + p_n) + f(x_n) - \langle p_n, x_n \rangle \leq f^*(q + p_n) + \lambda. \end{aligned}$$

Since p_n converges to p and f^* is continuous at p , there exists $\eta > 0$ and N such that

$$(42) \quad \begin{cases} \text{(i) } f^*(r) \leq f^*(p) + 1 \text{ when } \|r - p\| \leq 2\eta, \\ \text{(ii) } \|p - p_n\| \leq \eta \text{ when } n \geq N. \end{cases}$$

Hence, for any $q \in B(\eta)$ and for any $n \geq N$, $r_n = p_n + q$ belongs to $B(2\eta)$. We deduce from (41) that for any $n \geq N$, $\langle q, x_n \rangle \leq f^*(p) + 1$ when $q \in B(\eta)$. Hence the sequence x_n is bounded. Thus $|f(p_n, x_n) - f(p, x_n)| = |\langle p - p_n, x_n \rangle| \leq \|p - p_n\|_* \|x_n\| \leq k \|p - p_n\|_*$ converges to 0. $\square$

We also mention another application of Theorem 3. This concerns the existence and uniqueness of a best approximant to given points by elements of a closed bounded (non-convex) subset of a Hilbert space.

Proposition 4 (Edelstein). *Let X be a closed bounded subset of a Hilbert space U . There exists a dense denumerable intersection T of open subsets such that, for any $p \in T$, there exists a unique point nearest to p and a unique point farthest from p in X (where a nearest point minimizes the distance to X and a farthest point maximizes the distance to X).*

Proof. We apply Theorem 3 to the function

$$(43) \quad f(x, p) = \pm \frac{1}{2} \|x - p\|^2$$

for which a minimum $\bar{x}$ is either the nearest point to p or the farthest point to p .

The functions given by (43) are Fréchet-differentiable and

$$(44) \quad Df(x, p) = \pm J(x - p)$$

where J is the duality map from U onto U^* .

Hence we have that

$$\begin{aligned} Df(x, p) - Df(y, p) &= \pm J(x - y) \\ f(q, x) - f(p, x) &= \pm \left(\left(q - p, \frac{p + q}{2} - x \right) \right) \\ f(q, x) - f(p, x) - \langle Df(p, x), q - p \rangle &= \pm \|q - p\|^2 \end{aligned}$$

Everything now follows easily from the fact that X is bounded. For instance, the derivative $Df(x, p)$ is locally uniformly bounded, hence the functions $p \mapsto f(x, p)$ are locally uniformly Lipschitzian. Thus f is locally Lipschitzian and therefore continuous. $\square$

4.4.5. Proof of Ekeland–Lebourg's theorem

The proof of Theorem 3 is quite involved. We divide it into four steps.

First step. We prove the following result, which is interesting in itself.

Proposition 5. *Suppose that (31) and (33) hold. If the function α is locally ε -subdifferentiable at $p \in V^*$, then*

$$(45) \quad \begin{aligned} & \|Df(x_1, p) - Df(x_2, p)\| \leq 6\varepsilon \text{ provided that} \\ & f(x_i, p) \leq \alpha(p) + \theta \quad (i = 1, 2) \text{ for some } \theta. \end{aligned}$$

Proof. Since α is locally ε -subdifferentiable at $p \in V^*$, there exists $u \in V = V^{**}$ and $\eta_1 > 0$ such that

$$(46) \quad \alpha(p) \leq \alpha(q) + \langle p - q, u \rangle + \varepsilon \|p - q\|_*,$$

whenever $\|p - q\|_* \leq \eta_1$. Choose $\eta_2 = \min(\eta, \eta_1)$ where η appears in (33). Let $x \in X$ be any point such that

$$(47) \quad f(x, p) \leq \alpha(p) + \varepsilon \eta_2.$$

We write part of (33) as an inequality

$$(48) \quad f(x, q) \leq f(x, p) + \langle Df(x, p), q - p \rangle + \varepsilon \|q - p\|_*.$$

Taking into account (47) and the inequality $\alpha(q) \leq f(x, q)$, we deduce from (48) that

$$(49) \quad \alpha(q) \leq \alpha(p) + \langle Df(x, p), q - p \rangle + 2\varepsilon \eta_2.$$

Now, (46) and (49) imply that

$$(50) \quad \langle u - Df(x, p), q - p \rangle \leq 3\varepsilon \eta_2.$$

Taking the supremum when q ranges over $p + B(\eta_2)$, we deduce from (50) that

$$(51) \quad \|u - Df(x, p)\| \leq 3\varepsilon.$$

If $y \in X$ is another point such that $f(y, p) \leq \alpha(p) + \varepsilon \eta_2$ we also have that

$$(52) \quad \|u - Df(y, p)\| \leq 3\varepsilon.$$

Hence (45) follows from (51) and (52) with $\theta = \theta \eta_2$ $\square$

Second step. We use Proposition 5 and Theorem 2 to prove a further result (Proposition 6). We require the following definition.

$$(53) \quad \begin{aligned} T_\varepsilon = \{ & p \in V^* \text{ such that there exists } \theta > 0 \text{ such that} \\ & \|x_1 - x_2\| \leq \varepsilon \text{ whenever } f(x_i, p) \leq \alpha(p) + \theta \quad (i = 1, 2)\}. \end{aligned}$$

Proposition 6. *Suppose that assumptions (30), (31), (32) and (33) hold. Then, for any $\varepsilon > 0$, the set T_ε is dense in V^* .*

Proof. By assumption (32), there exists η such that $\|x_1 - x_2\| \leq \varepsilon$ whenever $f(x_i, p) \leq \alpha(p) + \eta$ and $\|Df(x_1, p) - Df(x_2, p)\| \leq \eta$. Proposition 5 and Theorem 2 (with $\varepsilon = \eta/6$) imply that the latter property is true as p ranges over a dense subset T_ε of V . $\square$

Third step.

Proposition 7. *Assumption (34) implies that the subsets T_ε are open.*

Proof. Let $p \in T_\varepsilon$. Then there exists $\theta > 0$ such that $\|x_1 - x_2\| \leq \varepsilon$ whenever $f(x_i, p) \leq \alpha(p) + \theta$ ($i = 1, 2$). Taking $\lambda = \alpha(p) + \theta$, we deduce from (34) the existence of $\eta > 0$ such that

$$(54) \quad |f(x, q) - f(x, p)| \leq \frac{1}{3} \theta \quad \text{provided that } \|q - p\| \leq \eta \text{ and } f(x, q) \leq \lambda$$

This clearly implies that

$$(55) \quad \alpha(q) \leq \alpha(p) + \frac{1}{3} \theta \quad \text{when } \|p - q\| \leq \eta.$$

Take any q such that $\|q - p\| \leq \eta$. Proposition 7 will be proved if we can show that q belongs to T_ε . For this purpose, let x_i ($i = 1, 2$) be any two points in X such that

$$(56) \quad f(x_i, q) \leq \alpha(q) + \frac{1}{3} \theta.$$

Taking (55) into account, we deduce that

$$(57) \quad f(x_i, q) \leq \alpha(p) + \frac{2}{3} \theta < \lambda.$$

We can therefore apply (54) and obtain that

$$(58) \quad f(x_i, p) \leq f(x_i, q) + \frac{1}{3} \theta \leq \alpha(p) + \theta \quad (i = 1, 2).$$

Since p belongs to T_ε , $\|x_1 - x_2\| \leq \varepsilon$.

Hence (56) implies that $q \in T_\varepsilon$. $\square$

Fourth step:

Proof of Theorem 3. We define $T \subset V$ as

$$(59) \quad T = \bigcap_{n=1}^{\infty} T_{1/n} = \bigcap_{\varepsilon > 0} T_\varepsilon.$$

Since V^* is a complete metric space and the sets $T_{1/n}$ are dense and open (Propositions 6 and 7), the Baire theorem¹ implies that T is dense.

Let $p \in T$. Consider a minimizing sequence $\{x_n\}$ for $\alpha(p)$, i.e. $\forall n, f(x_n, p) \leq \alpha(p) + 1/n$. Since $p \in T_\varepsilon$ for any $\varepsilon > 0$, we deduce from (53) that

$$(60) \quad \forall \varepsilon > 0, \exists N \text{ such that } \|x_n - x_p\| \leq \varepsilon \text{ when } n, p \geq N.$$

Hence $\{x_n\}$ is a Cauchy sequence. This converges to $\bar{x} \in X$, because X is a closed subset of a Banach space. Since f is lower semi-continuous with respect to x , we have that

$$(61) \quad f(\bar{x}, p) \leq \liminf_n f(x_n, p) = \alpha(p).$$

It follows that $\bar{x}$ is an optimal solution of problem $\alpha(p)$. This optimal solution is unique. If there were another solution $\bar{y}$, we would have $f(\bar{x}, p) = f(\bar{y}, p) = \alpha(p)$. Since $p \in T_\varepsilon$ for all $\varepsilon > 0$, this would imply that $\|\bar{x} - \bar{y}\| \leq \varepsilon$ for every $\varepsilon > 0$, and thus that $\bar{x} = \bar{y}$. $\square$

¹ See note in Section 3.3.3.

CHAPTER 5

INTRODUCTION TO DUALITY THEORY

This chapter is devoted to an introduction to duality theory. We will continue this study in Chapter 14. In Section 5.1, we state the framework in which duality theory arises. We assume that the strategy set

$$X = \{x \in R \text{ such that, } \forall p \in P, \gamma(x, p) \leq 0\}$$

is a subset of R defined by a family of constraint inequalities $\gamma(x, p) \leq 0$. We assume that

$$\begin{cases} \text{(i) } P \text{ is a convex cone,} \\ \text{(ii) } \forall x, \quad p \rightarrow \gamma(x, p) \text{ is positively homogeneous.} \end{cases}$$

Since minimization problems on R are easier to solve than minimization problems on X , the question arises whether it is possible to replace

$$v = \inf_{x \in X} f(x) \quad (\text{constrained minimization problem})$$

by a minimization problem on R :

$$v = \inf_{x \in R} [f(x) + \gamma(x, \bar{p})] \quad (\text{unconstrained minimization problem})$$

for a convenient $\bar{p} \in P$ (called a *Lagrange multiplier*), by adding to the initial loss function f a (well chosen) constraint $\gamma(\cdot, \bar{p})$ which measures a “cost of violation of the constraint $x \in T$ ”. The problem of finding a Lagrange multiplier is called the *dual problem* of the initial problem. Once the dual problem is solved and a Lagrange multiplier $\bar{p}$ is obtained, it remains to solve the unconstrained minimization problem. We shall check that $\bar{x} \in X$ minimizes f on X if and only if

$$\begin{cases} \text{(i) } \bar{x} \text{ minimizes } x \mapsto f(x) + \gamma(x, \bar{p}) \text{ on } R, \\ \text{(ii) } \gamma(\bar{x}, \bar{p}) = 0. \end{cases}$$

Lagrange multipliers have another fundamental property which emphasizes their importance. Let V^* (the dual of a topological vector space V) be the vector

space containing P . We associate with any $y \in T$ the perturbed subset

$$X(y) = \{x \in R \text{ such that } \gamma(x, p) + \langle p, y \rangle \leq 0 \ \forall \ p \in P\}.$$

Note that $X = X(0)$. Let v be the function defined by

$$v(y) = \inf_{x \in X(y)} f(x)$$

which associates with any perturbation $y \in V$ the minimal loss $v(y)$ on $X(y)$. Then we shall prove that the Lagrange multiplier $\bar{p}$ satisfies

$$\bar{p} \in \partial v(0),$$

i.e. p measures the marginal increase of loss when the strategy set is perturbed by y .

We devote Section 5.2 to the case where the constraints are defined by $\gamma(x, p) = \langle p, Lx \rangle - \sigma_Y^*(p)$, where σ_Y^* denotes the upper support function of a closed convex subset $Y \subset V$ and where $L \in \mathcal{L}(U, V)$ is a linear operator. In other words,

$$X = \{x \in R \text{ such that } Lx \in Y\}.$$

If the loss function f is lower semi-continuous and convex, we shall prove that $\bar{x} \in X$ is an optimal solution and $\bar{p} \in P$ (which is the barrier cone of Y) is a Lagrange multiplier if and only if

$$\bar{x} \in \partial f^*(-L^*\bar{p}) \quad \text{and} \quad \bar{p} \in \partial \psi_Y(L\bar{x})$$

where $\partial \psi_Y(L\bar{x})$ denotes the normal cone to Y at $L\bar{x}$. These relations are called the "extremality relations". They lead to the following *fundamental formula*. This is the multi-valued equation

$$0 \in \partial \sigma_Y^*(\bar{p}) - L \partial f^*(-L^*\bar{p}).$$

of which the Lagrange multipliers are solutions.

Once $\bar{p} \in P$ is obtained, minimal solution can be found from

$$\bar{x} \in \partial f^*(-L^*\bar{p}) \cap L^{-1} \partial \sigma_Y^*(\bar{p}).$$

In the rest of this section we investigate several examples and applications of this formula. For instance, if we assume that

$$f^* \text{ is differentiable on } -L^*P$$

then the minimal solution $\bar{x}$ is equal to $Df^*(-L^*\bar{p})$, where the Lagrange multipliers $\bar{p} \in P$ are solutions of the variational inequalities

$$\langle -LDf^*(-L^*\bar{p}), \bar{p} - p \rangle + \sigma_Y^*(\bar{p}) - \sigma_Y^*(p) \leq 0, \quad \forall p \in P.$$

These variational inequalities become

$$\langle -LDf^*(-L^*\bar{p}) + \omega, \bar{p} - p \rangle \leq 0, \quad \forall p \in P$$

when $Y = w - P^+$ and thus reduce to the equation

$$LDf^*(-L^*\bar{p}) = w$$

when $Y = \{w\}$.

We prove in Section 5.3 two existence theorems for Lagrange multipliers (the Fenchel and Uzawa theorems) in the case where V is a finite dimensional space. Lagrange multipliers exist if f , R and Y are convex and if the constraint qualification assumption

$$0 \in \text{Int}(L(R) - Y)$$

holds.

5.1. Dual problem and Lagrange multipliers

We introduce the Lagrangian of a minimization problem $v = \inf_{x \in X} f(x)$ where

$$X = \{x \in R \text{ such that } \gamma(x, p) \leq 0 \quad \forall p \in P\}.$$

The Lagrangian is defined by

$$l(x, p) = f(x) + \gamma(x, p) \quad \text{when } x \in X, p \in P.$$

We check that

$$f_X(x) = \sup_{p \in P} l(x, p)$$

and we define the Lagrange multipliers $\bar{p}$ by

$$\inf_{x \in R} l(x, \bar{p}) = \inf_{x \in R} \sup_{p \in P} l(x, p).$$

Then we show that $\bar{x} \in X$ is an optimal solution if and only if

$$\begin{cases} \text{(i) } \bar{x} \text{ minimizes } f(x) + \gamma(x, \bar{p}) \text{ on } R, \\ \text{(ii) } \gamma(\bar{x}, \bar{p}) = 0. \end{cases}$$

This being done, we check that

$$\bar{p} \in \partial v(0)$$

where

$$v(y) = \inf_{x \in X(y)} f(x)$$

and

$$X(y) = \{x \in R \text{ such that } \gamma(x, p) + \langle p, y \rangle \leq 0 \text{ for all } p \in P\}.$$

We end this section by interpreting the case where $X = \{x \in R \text{ such that } L(x) \in Y\}$ is the set of x satisfying the constraints $\langle p, L(x) \rangle - \sigma_Y^*(p) \leq 0$. We say that X is represented by the family of constraints.

5.1.1. Lagrangian

In many instances, the strategy set X is a subset of an “unconstrained” set R defined or “represented” by a family of inequality constraints, i.e.

$$(1) \quad X = \{x \in R \text{ such that } \gamma(x, p) \leq 0 \text{ for all } p \in P\}$$

where we *shall assume once and for all* that

$$(2) \quad \begin{cases} \text{(i) } P \text{ is a closed convex cone of a dual topological vector space } V^*, \\ \text{(ii) } \gamma \text{ is defined on } R \times P \text{ and is positively homogeneous with respect to } p \text{ (i.e., } \forall \lambda \geq 0, \gamma(x, \lambda p) = \lambda \gamma(x, p)). \end{cases}$$

The aim of the following approach is to replace an optimization problem on X by an equivalent problem on R (by modifying the loss function) whenever an optimization problem on R is simpler than an optimization problem on X .

The idea is to “measure” the fact that x belongs to X or not by the values of the constraints $\gamma(x, p)$. In other words, by adding to a loss function f defined on R a constraint $\gamma(\cdot, p)$, we “penalize” the fact that a strategy x does not belong to X .

Definition 1. Consider the subset X of R defined by

$$(3) \quad X = \{x \in R \text{ such that } \gamma(x, p) \leq 0 \text{ for all } p \in P\}.$$

Let f be a loss function defined on R . We shall say that the function l defined on $R \times P$ by

$$(4) \quad l(x, p) = f(x) + \gamma(x, p)$$

is the “*Lagrangian*” associated with the minimization problem $\alpha = \inf_{x \in X} f(x)$ and the representation of X by the constraints $\gamma(\cdot, p)$.

Of course, as we shall see later on, there are many ways of defining a subset X by means of inequality constraints (see Section 14.1).

The following proposition gives a precise meaning to the penalization procedure we described above.

Proposition 1. *Suppose that (2) holds. Let $l(x, p)$ be the Lagrangian of an optimization problem. Then*

$$(5) \quad f_X(x) = \sup_{p \in P} l(x, p).$$

Proof. If $x \in X$, then $\gamma(x, p) \leq 0$ for any $p \in P$ and $\gamma(x, 0) = 0$ since $0 \in P$. Therefore,

$$(6) \quad \sup_{p \in P} l(x, p) = f(x) + \sup_{p \in P} \gamma(x, p) = f(x) = f_X(x).$$

Now, if $x \notin X$, there exists at least one element $\bar{p} \in P$ such that $\gamma(x, \bar{p}) = \theta > 0$. Since P is a cone, the elements $\lambda \bar{p}$ belong to P whenever $\lambda > 0$. Because γ is positively homogeneous with respect to p , $\gamma(x, \lambda \bar{p}) = \lambda \theta$. Thus $\sup_{p \in P} l(x, p) \geq \sup_{\lambda \geq 0} l(x, \lambda \bar{p}) = f(x) + \sup_{\lambda \geq 0} \lambda \theta = +\infty = f_X(x)$. $\square$

5.1.2. Lagrange multipliers and dual problem

We deduce from (5) that

$$(7) \quad \inf_{x \in X} f(x) = \inf_{x \in R} \sup_{p \in P} l(x, p).$$

The problem now arises as to whether there exists an element $\bar{p} \in P$ such that

$$(8) \quad \inf_{x \in X} f(x) = \inf_{x \in R} \sup_{p \in P} l(x, p) = \inf_{x \in R} l(x, \bar{p}).$$

If so, the initial minimization problem on X is equivalent to a minimization problem on the unconstrained set R .

Definition 2. We shall say that the elements $\bar{p} \in P$ (if any) satisfying (8) are the “*Lagrange multipliers*” of the problem or the solutions of the “*dual problem*”.

If such a multiplier $\bar{p}$ exists, we obtain the following characterization of a minimal solution $\bar{x} \in X$ of f .

Proposition 2. Suppose that (2) holds. Then $\bar{p} \in P$ is a Lagrange multiplier and $\bar{x} \in X$ minimizes f over X if and only if

$$(9) \quad \begin{cases} (i) \forall x \in X, f(\bar{x}) + \gamma(\bar{x}, \bar{p}) \leq f(x) + \gamma(x, \bar{p}), \\ (ii) \forall p \in P, \gamma(\bar{x}, p) \leq 0 \text{ (i.e. } \bar{x} \in X), \\ (iii) \gamma(\bar{x}, \bar{p}) = 0. \end{cases}$$

Proof. Obviously, (9) implies that $\bar{x} \in X$ (by (9(ii))) and that $\bar{x}$ minimizes f on X because, if $x \in X$, $\gamma(x, \bar{p}) \leq 0$ and

$$(10) \quad f(\bar{x}) = f(\bar{x}) + 0 \leq f(x) + \gamma(x, \bar{p}) \leq f(x) + 0$$

(by (9(iii) and (i))). Inequalities (9) imply also that

$$(8) \quad \inf_{x \in X} f(x) = \inf_{x \in R} l(x, \bar{p}).$$

Conversely, suppose that $\bar{p}$ is a Lagrange multiplier and that $\bar{x}$ minimizes f on X . Since $\bar{x} \in X$, $\gamma(\bar{x}, p) \leq 0$ for all $p \in P$ and, in particular, $\gamma(\bar{x}, \bar{p}) \leq 0$. On the other hand, by (8), $f(\bar{x}) = \inf_{x \in R} l(x, \bar{p}) \leq l(\bar{x}, \bar{p}) = f(\bar{x}) + \gamma(\bar{x}, \bar{p})$. Thus, $\gamma(\bar{x}, \bar{p}) \geq 0$. Therefore $\gamma(\bar{x}, \bar{p}) = 0$. This implies that

$$f(\bar{x}) + \gamma(\bar{x}, \bar{p}) = f(\bar{x}) = \inf_{x \in R} l(x, \bar{p}) = \inf_{x \in R} (f(x) + \gamma(x, \bar{p})). \quad \square$$

Definition 3. We shall say that

$$(11) \quad f^{I*}(p) = -\inf_{x \in R} l(x, p)$$

is the loss function of the dual problem.

We always have

$$(12) \quad \inf_{x \in X} f(x) + \inf_{p \in P} f^{I*}(p) \geq 0.$$

An element $\bar{p} \in P$ is a Lagrange multiplier if and only if

$$(13) \quad \begin{cases} (i) \inf_{x \in X} f(x) + \inf_{p \in P} f^{I*}(p) = 0 \\ (ii) f^{I*}(\bar{p}) = \inf_{p \in P} f^{I*}(p) \end{cases}$$

Remark. If the function f is strictly convex and the constraints $\gamma(\cdot, p)$ are convex for all $p \in P$, then the function $f + \gamma(\cdot, p)$ is also strictly convex. Hence if $\bar{p}$ is a Lagrange multiplier, there exists at most one solution $\bar{x} \in X$ which minimizes f on X . If it exists, it is necessarily the unique solution which

minimizes $x \mapsto f(x) + \gamma(x, \bar{p})$ on R . Hence it necessarily satisfies the condition $\gamma(\bar{x}, \bar{p}) = 0$.

Remark. More generally, the question arises as to whether any solution $\bar{x}$ minimizing $x \mapsto f(x) + \gamma(x, \bar{p})$ is a minimal solution of the initial problem. For this purpose, we replace the minimization problem $v = \inf_{x \in X} f(x)$ by the equivalent minimization problem $v^2 = \inf_{x \in X} [f(x)]^2$. Its Lagrangian is defined by $l(x, p) = [f(x)]^2 + \gamma(x, p)$.

Proposition 3. *Let us assume that there exist a minimal solution of the problem $v^2 = \inf_{x \in X} [f(x)]^2$ and a Lagrange multiplier $\bar{p}$. Then any $\bar{x}$ which minimizes $x \mapsto [f(x)]^2 + \gamma(x, \bar{p})$ on R is a minimal solution.*

Proof. By Proposition 2.1.6, the functions f and $\gamma(\cdot, \bar{p})$ are constant on the minimal set $M^b = \{x \in R \text{ such that } f(x)^2 + \gamma(x, \bar{p}) = v\}$. Thus $f(\bar{x}) = v$ and $\gamma(\bar{x}, \bar{p}) = 0$ for any $\bar{x} \in M^b$. $\square$

The equivalence property (8), which we used to introduce Lagrange multipliers, is not the only fundamental property of Lagrange multipliers.

5.1.3. Marginal interpretation of Lagrange multipliers

We have already studied the perturbed problem

$$-f^*(q) = \inf_{x \in U} [f_X(x) - \langle q, x \rangle]; \quad q \in V^*$$

obtained by adding linear functionals q to the original loss function (see Section 2.4).

When X is represented by inequality constraints $\gamma(x, p) \leq 0$, where P is a subset of the dual space V^* , we perturb X by associating with any $y \in V$ the subsets $X(y)$ defined by

$$(14) \quad X(y) = \{x \in P \text{ such that } \gamma(x, p) + \langle p, y \rangle \leq 0 \quad \forall p \in P\}.$$

We shall study the behavior of the function

$$(15) \quad v(y) = \inf_{x \in X(y)} f(x)$$

and its relations with the Lagrange multipliers.

Proposition 4. *Suppose that there exists a Lagrange multiplier $p_0 \in P$ of the problem*

$$(16) \quad v(y_0) = \inf_{x \in X(y_0)} f(x)$$

for the Lagrangian $f(x) + \gamma(x, p) + \langle p, y_0 \rangle$. Then

$$(17) \quad p_0 \text{ belongs to the subdifferential } \partial v(y_0).$$

Proof. Since $v(y) = \inf_{x \in R} \sup_{p \in P} [f(x) + \gamma(x, p) + \langle p, y \rangle]$ and $v(y_0) = \inf_{x \in U} [f(x) + \gamma(x, p_0) + \langle p_0, y_0 \rangle]$ by assumption, we obtain the following inequalities.

$$\begin{aligned} v(y_0) &= \inf_{x \in R} [f(x) + \gamma(x, p_0) + \langle p_0, y_0 \rangle] \\ &= \inf_{x \in R} [f(x) + \gamma(x, p_0) + \langle p_0, y \rangle] + \langle p_0, y_0 - y \rangle \\ &\leq \inf_{x \in R} \sup_{p \in P} [f(x) + \gamma(x, p) + \langle p, y \rangle] + \langle p_0, y_0 - y \rangle \\ &= v(y) + \langle p_0, y_0 - y \rangle. \end{aligned}$$

Thus $p_0 \in \partial v(y_0)$. $\square$

Remark. We shall prove the converse statement when $p \mapsto \gamma(x, p)$ is concave and upper semi-continuous (see Section 14.1.6). In this case, we shall prove that p_0 is a Lagrange multiplier of the problem $v(y_0)$ if and only if $p_0 \in \partial v(y_0)$ and $v(y_0) = v^{**}(y_0)$. This result is analogous to Proposition 4.1.1. More generally, we shall associate with families of optimization problems

$$(18) \quad v(y) = \inf_{x \in U} \Delta(x, y)$$

(where Δ is a proper function defined on the product $U \times V$ of two vector spaces) a “generalized” Lagrangian defined by

$$(19) \quad l(x, p) = \inf_{y \in V} [\Delta(x, y) - \langle p, y \rangle].$$

An analogous duality theory can be devised in this case.

5.1.4. Example

Consider the framework described in 1.1.1, where the set of feasible strategies is constructed from a subset of available resources.

We assume that

$$(20) \quad \text{The space } V \text{ of resources is a topological vector space}$$

and we regard its dual V^* as the space of “resource prices” p (associating with any resource $y \in V$ its value $\langle p, y \rangle \in \mathbb{R}$). We define a closed convex set Y of available resources by

$$(21) \quad Y = \{y \in V \text{ such that } \langle p, y \rangle \leq r(p) \forall p \in P\}$$

where

$$(22) \quad \left\{ \begin{array}{l} \text{(i) } P \text{ is a (closed convex) cone of } V^*, \text{ regarded as the cone of} \\ \text{feasible prices} \\ \text{(ii) } r : P \rightarrow \mathbf{R} \text{ is a positively homogeneous function associating} \\ \text{with any feasible price } p \text{ a maximum profit } r(p). \end{array} \right.$$

In other words, the set Y of available resources is described by “budgetary constraints”, i.e. as the set of resources $y \in Y$ whose values do not exceed the maximum profit for any feasible price.

Let R be the unconstrained strategy set and $L : R \rightarrow V$ be the resource operator. Then the strategy set X defined by

$$(23) \quad X = \{x \in R \text{ such that } L(x) \in Y\}$$

is represented by the constraints $\gamma(x, p) = \langle p, L(x) \rangle - r(p)$ since statement (23) is equivalent to

$$(24) \quad X = \{x \in R \text{ such that } \forall p \in P, \langle p, L(x) \rangle - r(p) \leq 0\}.$$

We regard the value $\langle p, L(x) \rangle$ of the resource $L(x)$ needed to implement the strategy x , as the “cost” of the strategy x when the price p prevails.

Hence, we interpret $\langle p, L(x) \rangle - r(p)$ as the value of the loss occurred by implementing the strategy x under the price p . This loss is positive whenever the cost of x is larger than the maximum profit allowed. It is quite natural to add this loss penalizing the fact that x requires a non-available resources $L(x)$ to the initial loss function of an optimization problem, i.e. to introduce the functions

$$(25) \quad x \mapsto l(x, p) = f(x) + \langle p, L(x) \rangle - r(p)$$

as new possible loss functions, which depend upon the prevailing price. Proposition 2 implies the following result.

Proposition 5. *Suppose that (22) and (24) hold. Then $\bar{p}$ is a Lagrange multiplier of the Lagrangian $l(x, p) = f(x) + \langle p, L(x) \rangle - r(p)$ and $\bar{x}$ minimizes f on X if and only if*

$$(26) \quad \left\{ \begin{array}{l} \text{(i) } \bar{x} \text{ minimizes } x \mapsto f(x) + \langle \bar{p}, L(x) \rangle \text{ on } X, \\ \text{(ii) } \forall p \in P, \langle p, L(\bar{x}) \rangle \leq r(p), \\ \text{(iii) } \langle \bar{p}, L(\bar{x}) \rangle = r(\bar{p}). \end{array} \right.$$

In other words, if the price $\bar{p}$ is a Lagrange multiplier (when such a multiplier exists), then an element $\bar{x} \in X$ minimizes the loss function on X if and only if it

minimizes on R the sum of the loss function f and the cost $\langle \bar{p}, L(\cdot) \rangle$ under the price $\bar{p}$ and the cost $\langle \bar{p}, L(\bar{x}) \rangle$ is equal to the maximum cost $r(\bar{p})$.

The subsets

$$(27) \quad X(y) = \{x \in R \text{ such that } L(x) + y \in Y\}$$

are represented by the constraints

$$(28) \quad \gamma(x, p) + \langle p, y \rangle = \langle p, L(x) + y \rangle - r(p).$$

This is because $L(x) + y$ belongs to Y if and only if

$$\langle p, L(x) + y \rangle - r(p) \leq 0 \quad \text{for all } p \in P.$$

Regard $Y - y$ as the set of available resources obtained by removing a new resource $y \in V$. Then

$$(29) \quad v(y) = \inf_{L(x) \in Y - y} f(x)$$

describes the behavior of the minimal loss with respect to perturbations of the set of available resources Y obtained by removing resources. Proposition 4 implies that the price $\bar{p}$ measures the marginal increase of loss when a new resource y is taken from the set Y of available resources (see Section 4.2.1).

5.2. Case of linear constraints: extremality relations

We devote this section to the case where

$$X = \{x \in R \text{ such that } Lx \in Y\}$$

and L is a linear operator. Let ψ_Y denote the indicator of Y . Since we can write

$$v = \inf_{x \in X} f(x) = \inf_{x \in R} [f(x) + \psi_Y(Lx)]$$

we are led to a more general study of minimization problems of the form

$$v = \inf_{x \in X} F(x, Lx).$$

where F maps $R \times Y$ into $\mathbb{R}$. We define a Lagrange multiplier p as the solution of the problem

$$\begin{cases} \text{(i)} & v^* = \inf_{p \in P} F^*(-L^*p, p) = F^*(-L^*\bar{p}, \bar{p}), \\ \text{(ii)} & v^* + v = 0. \end{cases}$$

We prove that when F is convex and lower semi-continuous, $\bar{p}$ is a Lagrange multiplier and $\bar{x}$ is an optimal solution if and only if the "extremality relation" $\{-L^*\bar{p}, \bar{p}\} \in \partial F(\bar{x}, L\bar{x})$ holds.

In the case where $F(x, y) = f(x) + g(y)$, the extremality condition can be written

$$\begin{cases} \text{(i)} \quad \bar{x} \in \partial f(-L^*\bar{p}) \cap L^{-1} \partial g^*(\bar{p}), \\ \text{(ii)} \quad 0 \in \partial g^*(\bar{p}) - L \partial f^*(-L^*\bar{p}). \end{cases}$$

This formula leads, first of all, to a solution of the dual problem ($0 \in \partial g^*(\bar{p}) - L \partial f^*(-L^*\bar{p})$) and secondly, to an optimal solution $\bar{x}$. If f^* is differentiable on $-L^*P$, the Lagrange multipliers are solutions of the variational inequalities

$$\langle -LDf^*(-L^*\bar{p}), \bar{p} - p \rangle + g^*(\bar{p}) - g^*(p) \leq 0 \quad \text{for all } p \in P.$$

Hence $\bar{x} = Df^*(-L^*\bar{p})$.

We specify the particular form of the above variational inequalities in the following special cases:

$$g^*(p) = \sigma_W^*(p), \quad g^*(p) = \sigma_{W-P^+}^*(p) \quad \text{and} \quad g^*(p) = \sigma_W^*(p) = \langle p, w \rangle.$$

Since the decomposition principle implies that

$$v = \inf_{x \in X} [f(x) + g(Lx)] = \inf_{y \in Y} [Lf(y) + g(y)]$$

we check that these two equivalent minimization problems have the same dual problem.

We also emphasize the so-called "decentralization" principle. When $U = \prod_{i=1}^n U^i$, $R = \prod_{i=1}^n R^i$ and $f(x) = \sum_{i=1}^n f_i(x^i)$, $Lx = \sum_{i=1}^n L_i x^i$, the extremality relations of the minimization problem

$$v = \inf_{x \in X} \left[\sum_{i=1}^n f_i(x^i) + g\left(\sum_{i=1}^n L_i x^i\right) \right]$$

can be written

$$\begin{cases} \text{(i)} \quad i = 1, \dots, n, \bar{x}^i \in \partial f_i^* L_i^*(-\bar{p}), \\ \text{(ii)} \quad \sum_{i=1}^n L_i \bar{x}^i \in \partial g^*(\bar{p}), \end{cases}$$

i.e. the relations "decentralize". The rest of this section deals with examples of conjugate functions.

5.2.1. Generalized minimization problem

Let

$$(1) \quad \begin{cases} \text{(i)} \quad U \text{ and } V \text{ be topological vector spaces,} \\ \text{(ii)} \quad L \in \mathcal{L}(U, V) \text{ be a linear continuous operator,} \\ \text{(iii)} \quad Y \subset V \text{ be a closed convex subset of } V, \\ \text{(iv)} \quad R \subset U \text{ be a convex subset of } U. \end{cases}$$

We assume that the strategy set X is defined by

$$(2) \quad X = \{x \in R \text{ such that } Lx \in Y\}.$$

Since $Y = \{y \in V \text{ such that } \langle p, y \rangle \leq \sigma_Y^*(p) \text{ for all } p \in V^*\}$, we deduce that X is represented by the constraints $\gamma(x, p)$ defined on $R \times V^*$ by

$$(3) \quad \gamma(x, p) = \langle p, Lx \rangle - \sigma_Y^*(p).$$

Recall that $\sigma_Y^* = \psi_Y^*$ is the conjugate function of the indicator of Y . Hence, if $R = \text{Dom } f$ is the domain of a proper function $f: U \rightarrow]-\infty, +\infty]$, we can write

$$(4) \quad v = \inf_{x \in X} f(x) = \inf_{x \in R} [f(x) + \psi_Y^*[Lx]].$$

The loss function of the dual problem can be written

$$\begin{aligned} f^{I*}(p) &= - \inf_{x \in U} [l(x, p)] \\ &= \sup_{x \in U} [\langle -L^*p, x \rangle - f(x) + \psi_Y^*(p)] \\ &= f^*(-L^*p) + \psi_Y^*(p). \end{aligned}$$

With a view to other applications, it is worthwhile to study more general problems of the type

$$(5) \quad v = \inf_{x \in X} F(x, Lx)$$

where F is a function defined on $R \times Y$. We shall say that $p \rightarrow F^*(-L^*p, p)$ *the loss function of the dual problem*, where F^* is the conjugate function of F defined by

$$(6) \quad F^*(q, p) = \sup_{x \in R} \sup_{y \in Y} [\langle q, x \rangle + \langle p, y \rangle - F(x, y)].$$

Its minimal value is denoted by

$$(7) \quad v^* = \inf_{p \in V^*} F^*(-L^*p, p).$$

We notice that

$$(8) \quad v + v^* \geq 0$$

since, for all $x \in R$ and $p \in V^*$, we always have that

$$(9) \quad 0 = \langle -L^*p, x \rangle + \langle p, Lx \rangle \leq F(x, Lx) + F^*(-L^*p, p).$$

Definition 1. We shall say that $\bar{p}$ is a “*Lagrange multiplier*” of the minimization problem (14) or a “*solution of its dual problem*” if

$$(10) \quad \begin{cases} \text{(i) } v + v^* = 0, \\ \text{(ii) } v^* = F^*(-L^*\bar{p}, \bar{p}). \end{cases}$$

This definition is consistent with the previous one because, if we take $F(x, y) = f(x) + \psi_Y(y)$, then

$$F^*(q, p) = f^*(q) + \sigma_Y^*(p).$$

Hence

$$(11) \quad \begin{aligned} v^* &= \inf_p F^*(-L^*p, p) = \inf_p [f^*(-L^*p) + \psi_Y^*(p)] \\ &= \inf_p f^{I*}(p). \end{aligned}$$

An intermediate case occurs when

$$(12) \quad F(x, y) = f(x) + g(y), \quad R = \text{Dom } f, \quad Y = \text{Dom } g.$$

Since $F^*(q, p) = f^*(q) + g^*(p)$, we obtain that

$$(13) \quad v = \inf_{x \in X} [f(x) + g(Lx)], \quad v^* = \inf_{p \in V^*} [f^*(-L^*p) + g^*(p)].$$

In this case, we can say that $\bar{p}$ is a *Lagrange multiplier* if and only if

$$(14) \quad v = \inf_x [f(x) + \langle \bar{p}, Lx \rangle] - g^*(\bar{p})$$

since we can write

$$(15) \quad v = -v^* = -f^*(-L^*\bar{p}) - g^*(\bar{p}) = -\sup_x [\langle -L^*\bar{p}, x \rangle - f(x)] - g^*(\bar{p}).$$

5.2.2. Extremality relations

Proposition 1. Suppose that F is a lower semi-continuous convex function whose domain is $R \times Y$. Then $\bar{p}$ is a *Lagrange multiplier* and $\bar{x}$ minimizes $x \rightarrow F(x, Lx)$ on X if and only if

$$(16) \quad \{-L^*\bar{p}, \bar{p}\} \in \partial F(\bar{x}, L\bar{x}).$$

If $F(x, y) = f(x) + g(y)$ where f and g are lower semi-continuous convex functions, this means that either

$$(17) \quad -L^*\bar{p} \in \partial f(\bar{x}) \quad \text{and} \quad \bar{p} \in \partial g(L\bar{x})$$

Remark. We can interpret the above result as follows. To solve the minimization problem, we begin by looking for a solution $\bar{p}$ of the multivalued equation

$$(23) \quad 0 \in \partial g^*(\bar{p}) - L \partial f^*(-L^*\bar{p}).$$

Then, we look for a solution $\bar{x}$ of

$$(24) \quad \bar{x} \in \partial f^*L^*(-\bar{p}) \cap L^{-1} \partial g^*(\bar{p}).$$

Proof of Proposition 2. Let $\bar{x}$ and $\bar{p}$ satisfy extremality relations (18). Then $L\bar{x} \in \partial g^*(\bar{p})$ and $L\bar{x} \in L \partial f^*L^*(-\bar{p})$. Thus $0 \in \partial g^*(\bar{p}) - L \partial f^*L^*(-\bar{p})$.

Conversely, let $\bar{p}$ be a solution of $0 \in \partial g^*(\bar{p}) - L \partial f^*L^*(-\bar{p})$. There exists $\bar{y} \in \partial g^*(\bar{p}) \cap L \partial f^*L^*(-\bar{p})$. Therefore, there exists $\bar{x} \in \partial f^*L^*(-\bar{p})$ such that $\bar{y} = L\bar{x} \in \partial g^*(\bar{p})$. Thus the extremality relations (18) are satisfied. $\square$

Proposition 3. Suppose that

$$(25) \quad f^* \text{ is differentiable on } -L^*P \text{ where } P = \text{Dom } g^*.$$

Then the correspondence ∂f^* is a map Df^* (see Proposition 4.2.3). Thus relations (22) can be written

$$(26) \quad \begin{cases} \text{(i) } \bar{x} = Df^*L^*(-\bar{p}), \\ \text{(ii) } \bar{p} \in P = \text{Dom } g^*, \\ \text{(iii) } \forall p \in P, \quad \langle (-L)Df^*(-L^*\bar{p}), \bar{p} - p \rangle + g^*(\bar{p}) - g^*(p) \leq 0. \end{cases}$$

Proof. Statements (26(i) and (iii)) amount to saying that

$$(27) \quad LDf^*L^*(-\bar{p}) \in \partial g^*(\bar{p}),$$

i.e. that $0 \in \partial g^*(\bar{p}) - LDf^*L^*(-\bar{p})$. $\square$

Remark. The following lemma provides a sufficient condition for assumption (25).

Lemma 1. Suppose that

$$(28) \quad \begin{cases} \text{(i) } f \text{ is strictly convex,} \\ \text{(ii) } \forall p \in P, \text{ the function } x \mapsto f(x) + \langle p, Lx \rangle \text{ is lower semi-compact and semi-continuous.} \end{cases}$$

Then the property

$$(25) \quad f^* \text{ is differentiable on } -L^*P$$

is satisfied.

Proof. Assumption (28(ii)) implies that for any $q = -L^*p \in -L^*P$ and for any $\lambda \in \mathbb{R}$ the subsets $\{x \in U \text{ such that } f(x) - \langle q, x \rangle \leq \lambda\}$ are (weakly) compact. Proposition 3.4.5 implies that f^* is continuous on $-L^*P$ and thus, subdifferentiable at any point of $-L^*P$. Since f is strictly convex, the subdifferentials $\partial f^*(-L^*p)$ contain a unique element. Proposition 4.3.5 then implies that this unique element is nothing other than the gradient $Df^*(-L^*p)$. $\square$

*5.2.4. Minimization problem under linear constraints

We consider the case when

$$(29) \quad X = \{x \in R \text{ such that } Lx - w \in -P^+\}$$

where

$$(30) \quad \begin{cases} \text{(i) } R \text{ is the domain of a lower semi-continuous convex proper function } f, \\ \text{(ii) } P \text{ is a closed convex cone of } V^*, P^+ \text{ is its polar cone and } w \text{ belongs to } V, \\ \text{(iii) } L \in \mathcal{L}(U, V). \end{cases}$$

Proposition 4. Suppose that (28) and (29) hold and that

$$(25) \quad f^* \text{ is differentiable on } -L^*P^+.$$

Then $\bar{x} \in X$ minimizes f on X and $\bar{p}$ is a Lagrange multiplier if and only if

$$(31) \quad \begin{cases} \text{(i) } \bar{x} = Df^*L^*(-\bar{p}), \\ \text{(ii) } \bar{p} \in P, \\ \text{(iii) } \forall p \in P, \quad \langle (-L)Df^*(-L^*\bar{p}) + w, \bar{p} - p \rangle \leq 0. \end{cases}$$

If $P = V^*$ (i.e. $P^+ = \{0\}$ or $X = \{x \in R \text{ such that } Lx = w\}$), these relations become

$$(32) \quad \begin{cases} \text{(i) } \bar{x} = Df^*L^*(-\bar{p}). \\ \text{(ii) } LDf^*L^*(-\bar{p}) = w. \end{cases}$$

*5.2.5. Minimization of a quadratic functional under linear constraints

Suppose that

$$(33) \quad U \text{ is a Hilbert space and that } f(x) = \frac{1}{2} \|x - u\|^2.$$

Let $J \in \mathcal{L}(U, U^*)$ be the associated duality map from U onto U^* (see Section 2.2.5).

Proposition 5. Let g be a lower semi-continuous convex proper function and $L \in (U, V)$. Then $\bar{x}$ minimizes $x \mapsto \frac{1}{2} \|x - u\|^2 + g(Lx)$ and $\bar{p}$ is a Lagrange multiplier if and only if

$$(34) \quad \begin{cases} \text{(i)} \bar{x} = u - J^{-1}L^*\bar{p}, \\ \text{(ii)} \bar{p} \in P = \text{Dom } g^*, \\ \text{(iii)} \forall p \in P, \quad \langle (LJ^{-1}L^*)\bar{p} - Lu, \bar{p} - p \rangle + g^*(\bar{p}) - g(p) \leq 0. \end{cases}$$

Proof. From the fact that $f(x) = \frac{1}{2} \|x - u\|^2$ we deduce that $Df(x) = J(x - u)$, $Df^*(p) = u + J^{-1}p$ and that $LDf^*L^*(-p) = Lu - LJ^{-1}L^*p$. Thus eq. (34) follow from eqs. (26). $\square$

We shall now generalize Theorem 2.3.1 and obtain an explicit formula for the minimal solution on subspaces.

Suppose now that

$$X = \{x \in U \text{ such that } Lx - w \in -P^+\}$$

Proposition 6. An element $\bar{x} \in X$ minimizes $x \mapsto \frac{1}{2} \|x - u\|^2$ on X if and only if

$$(35) \quad \begin{cases} \text{(i)} \bar{x} = u - J^{-1}L^*\bar{p}, \\ \text{(ii)} \bar{p} \in P, \\ \text{(iii)} \langle (LJ^{-1}L^*)\bar{p} - (Lu - w), \bar{p} - p \rangle \leq 0, \quad p \in P. \end{cases}$$

When L is surjective, $\bar{p}$ is the orthogonal projection on P (for the scalar product $((p, q))_{V^*} = ((L^*p, L^*q))_{U^*}$) of the Lagrange multiplier $\bar{p}_0 = (LJ^{-1}L^*)^{-1}(Lu - w)$ of the minimization problem with equality constraints (case where $P^+ = \{0\}$).

Proof. If $g(y) = \psi(w - P^+; y)$, then

$$g^*(p) = \begin{cases} \langle p, w \rangle & \text{if } p \in P, \\ +\infty & \text{if } p \notin P. \end{cases}$$

Therefore, (35(iii)) follows from (34(iii)). If $P^+ = \{0\}$, the Lagrange multiplier $\bar{p}_0$ satisfies the equation $(LJ^{-1}L^*)\bar{p}_0 = Lu - w$. When L is surjective, we deduce that $\bar{p}_0 = (LJ^{-1}L^*)^{-1}(Lu - w)$. We can write

$$(36) \quad \langle (LJ^{-1}L^*)(\bar{p} - \bar{p}_0), \bar{p} - p \rangle = ((\bar{p} - \bar{p}_0, \bar{p} - p))_{V^*} \leq 0, \quad \forall p \in P. \quad \square$$

*5.2.6. Minimization problem under linear equality constraints

By taking $g = \psi_{\{w\}}$ to be the indicator of the point w , we obtain

$$(37) \quad v = (Lf)(w) \quad \text{and} \quad v^* = -(f^*L^*)^*(w)$$

since $g^*(p) = \langle p, w \rangle$.

Proposition 7. Consider the case where $g = \psi_{\{w\}}$. Then $\bar{p}$ is a Lagrange multiplier of the minimization problem $v = (Lf)(w)$ if and only if

$$(38) \quad \begin{cases} \text{(i)} (Lf)(w) = (f^*L^*)^*(w) = (Lf)^{**}(w), \\ \text{(ii)} -\bar{p} \in \partial(Lf)(w). \end{cases}$$

If we assume furthermore that f is convex and lower semi-continuous, then $\bar{x}$ minimizes f under the constraint $Lx = w$, if and only if

$$(39) \quad \bar{x} \in \partial f^*L^*[L\partial f^*L^*]^{-1}(w).$$

Proof. We have that $\bar{p}$ is a Lagrange multiplier if and only if

$$\begin{aligned} v = (Lf)(w) &= -v^* = (f^*L^*)^*(w) \\ &= \sup_p [\langle -p, w \rangle - f^*L^*(-p)] = \langle -\bar{p}, w \rangle - f^*L^*(-\bar{p}). \end{aligned}$$

This amounts to saying that $\bar{p} \in -\partial(Lf)(w)$ because $f^*L^* = (Lf)^*$ (see Proposition 2.4.4) and we can write

$$(40) \quad (Lf)(w) = \langle -\bar{p}, w \rangle - (Lf)^*(-\bar{p}).$$

Since $g^*(p) = \langle p, w \rangle$, the extremality relations can be written

$$(41) \quad \bar{x} \in \partial f^*L^*(-\bar{p}) \quad \text{and} \quad L\partial f^*L^*(-\bar{p}) = w.$$

This is the same as

$$(42) \quad \bar{x} \in \partial f^*L^*[L\partial f^*L^*]^{-1}(w). \quad \square$$

Remark. This formula is an extension of a formula we proved in the case of the quadratic functional $f(x) = \frac{1}{2} \|x - u\|^2$ (see Theorem 2.3.1).

5.2.7. Duality and the decomposition principle

Consider the case when $F(x, y) = f(x) + g(y)$. Recall that

$$(43) \quad v = \inf_{x \in R} [f(x) + g(Lx)] = \inf_{y \in Y} [Lf(y) + g(y)].$$

It is clear that these two equivalent problems have the same dual problem.

In the first case, we take $F(x, y) = f(x) + g(y)$. Then the loss function of its dual problem is

$$p \mapsto f^*(L^*(-p)) + g^*(p).$$

In the second case, we take $U = V$, $L = 1$ and $G(x, y) = (Lf)(x) + g(y)$. Then the loss function of its dual problem is

$$p \mapsto (Lf)^*(-p) + g^*(p) = f^*L^*(-p) + g^*(p)$$

since $(Lf)^* = f^*L^*$.

The extremality relations for the second problem are given by

$$(44) \quad \bar{p} \in \partial g(\bar{y}) \quad \text{and} \quad -\bar{p} \in \partial(Lf)(\bar{y}).$$

5.2.8. The decentralization principle

Consider the case when

$$(45) \quad \begin{cases} U = \prod_{i=1}^n U^i, \\ f(x) = \sum_{i=1}^n f_i(x^i) \quad \text{where } f_i : U^i \rightarrow]-\infty, +\infty], \\ Lx = \sum_{i=1}^n L_i x^i \quad \text{where } L_i \in \mathcal{L}(U^i, V). \end{cases}$$

The domain R of f is the product $R = \prod_{i=1}^n R^i$ of the domains R^i of the functions f_i .

Proposition 8. Let $\bar{p} \in V^*$ be a Lagrange multiplier of the minimization problem

$$(46) \quad v = \inf_{x \in X} \left[\sum_{i=1}^n f_i(x^i) + g\left(\sum_{i=1}^n L_i x^i\right) \right].$$

This amounts to writing

$$(47) \quad \begin{aligned} v = -v^* &= -\sum_{i=1}^n f_i^*(-L_i^* \bar{p}) - g^*(\bar{p}) \\ &= \sum_{i=1}^n \inf_{x^i \in R^i} [f_i(x^i) + \langle L_i^* \bar{p}, x^i \rangle] - g^*(\bar{p}). \end{aligned}$$

In this case, $\bar{x} = \{\bar{x}^1, \dots, \bar{x}^n\}$ is a solution of the minimization problem (46) if and only if

$$(48) \quad \begin{cases} \text{(i)} \quad \forall i = 1, \dots, n, \bar{x}^i \in \partial f_i^* L_i^*(-\bar{p}), \\ \text{(ii)} \quad \sum_{i=1}^n L_i \bar{x}^i \in \partial g^*(\bar{p}). \end{cases}$$

Proof. This is left as an exercise. $\square$

Interpretation. We interpret the functions f_i as loss functions of n players $i = 1, \dots, n$. Extremality relations (48(i)) show that knowledge of the value of a Lagrange multiplier $\bar{p}$ (if one exists) allows one to deduce that the component

$\bar{x}^i$ of an optimal solution $\bar{x} = \{\bar{x}^1, \dots, \bar{x}^n\}$ lies in the subdifferential $\partial f_i^* L_i^*(-\bar{p})$ of the loss function of player i .

In particular, whenever f_i^* are differentiable on $-L_i^*P$ (where $P = \text{Dom } g^*$), optimal solutions $\bar{x}^i$ and the Lagrange multiplier $\bar{p}$ are solutions of the equations

$$(49) \quad \begin{cases} \text{(i) } \forall i = 1, \dots, n, \quad \bar{x}^i = Df_i^*(-L_i^*\bar{p}) \\ \quad \text{where } \bar{p} \text{ is a solution of} \\ \text{(ii) } \left(\sum_{i=1}^n L_i Df_i^* L_i^* \right) (-\bar{p}) \in \partial g^*(\bar{p}). \end{cases}$$

We interpret relations (48(i)) by saying that $\bar{x} = \{\bar{x}^1, \dots, \bar{x}^n\}$ is obtained by “decentralizing” problem (46) by $\bar{p}$.

5.2.9. Conjugate function of gL

Proposition 9. Consider

$$(50) \quad L \in \mathcal{L}(U, V) \quad \text{and} \quad g : V \mapsto]-\infty, +\infty].$$

Suppose that the following infimum

$$(51) \quad v = -(gL)^*(q) = \inf_{x \in X} [g(Lx) - \langle q, x \rangle]$$

is finite. Then an element $\bar{p} \in V^*$ is a Lagrange multiplier if and only if

$$(52) \quad -v = v^* = (L^*g^*)(q) = g^*(\bar{p}) \quad \text{where } L^*\bar{p} = q.$$

Moreover, if $y \in \partial(L^*g^*)(q)$ and g is convex and lower semi-continuous, we obtain that

$$(53) \quad \partial(gL)(y) = (L^*\partial g^*)(y).$$

Proof. The minimization problem we are considering is the particular case when $f(x) = -\langle q, x \rangle$. Since $f^*(p) = \psi_{\{-q\}}(p)$ is the indicator of $-q$, we deduce that the loss function of the dual problem is defined by $F^*(-L^*p, p) = \psi_{\{-q\}}(-L^*p) + g^*(p) = \psi_{\{q\}}(L^*p) + g^*(p)$. Therefore $v^* = \inf_{L^*p=q} g^*(p) = (L^*g^*)(q)$.

Thus, $\bar{p}$ is a Lagrange multiplier if and only if

$$(54) \quad (gL)^*(q) = g^*(\bar{p}) = \inf_{L^*p=q} g^*(p) \quad \text{where } L^*\bar{p} = q.$$

It is clear that we always have

$$(55) \quad L^*\partial g^*(y) \subset \partial[gL](y).$$

To prove this write $q = L^*p$ where $p \in \partial g(Ly)$. Then

$$\begin{aligned} g(Ly) - g(Lz) &\leq \langle p, Ly - Lz \rangle = \langle L^*p, y - z \rangle \\ &= \langle q, y - z \rangle. \end{aligned}$$

Conversely, suppose that $y \in \partial(L^*g^*)(q)$, i.e. that $q \in \partial(L^*g^*)(y) = \partial[g^{**}L^{**}](y) = \partial(gL)(y)$. We deduce from (54) that

$$(56) \quad \langle q, y \rangle = (gL)(y) + (gL)^*(q) = g[Ly] + g^*(\bar{p})$$

where $L^*\bar{p} = q$. This implies that

$$(57) \quad \langle \bar{p}, Ly \rangle = \langle L^*\bar{p}, y \rangle = \langle q, y \rangle = g[Ly] + g^*(\bar{p})$$

and thus that $\bar{p} \in \partial g[Ly]$. Hence

$$(58) \quad q = L^*\bar{p} \in L^*\partial g[Ly]. \quad \square$$

5.2.10. Conjugate function of $f_1 + f_2$

Proposition 10. *Consider*

$$(59) \quad \text{two proper functions } f_1 \text{ and } f_2 \text{ from } U \text{ into }]-\infty, +\infty]$$

and assume that the following infimum

$$(60) \quad v = -(f_1 + f_2)^*(q) = \inf_{x \in U} [f_1(x) + f_2(x) - \langle q, x \rangle]$$

is finite. Then an element $\bar{p} \in V^$ is a Lagrange multiplier if and only if*

$$(61) \quad -v = v^* = (f_1^* \square f_2^*)(q) = f_1^*(q - \bar{p}) + f_2^*(\bar{p}).$$

Moreover, if f_1 and f_2 are convex and lower semi-continuous and $y \in \partial[f_1^ \square f_2^*](q)$, then*

$$(62) \quad \partial(f_1 + f_2)(y) = \partial f_1(y) + \partial f_2(y).$$

Proof. The minimization problem we are considering is the particular case where $V = U$, $L = 1$, f is defined by $f(x) = f_1(x) - \langle q, x \rangle$ and $g = f_2$. Since $f^*(p) = f_1^*(q + p)$, the loss function of the dual problem is defined by $F^*(-L^*p, p) = f_1^*(q - p) + f_2(p)$. Therefore

$$(63) \quad v^* = \inf_p [f_1^*(q - p) + f_2^*(p)] = [f_1^* \square f_2^*](q).$$

Thus $\bar{p}$ is a Lagrange multiplier if and only if

$$(64) \quad (f_1 + f_2)^*(q) = f_1^*(q - \bar{p}) + f_2^*(\bar{p}) = [f_1^* \square f_2^*](q).$$

It is clear that we always have

$$(65) \quad \partial f_1(y) + \partial f_2(y) \subset \partial(f_1 + f_2)(y).$$

Conversely, suppose that $y \in \partial[f_1^* \square f_2^*](q)$, i.e. that $q \in \partial[f_1^* \square f_2^*]^*(y) = [\partial f_1^{**} + \partial f_2^{**}](y) = \partial(f_1 + f_2)(y)$. We deduce from (64) that $q = (q - \bar{p}) + \bar{p}$ satisfies

$$\begin{aligned} 0 &= \langle q, y \rangle - (f_1^* \square f_2^*)(q) - [(f_1(y) + f_2(y))] \\ &= [\langle q - \bar{p}, y \rangle - f_1^*(q - \bar{p}) - f_1(y)] \\ &\quad + [\langle \bar{p}, y \rangle - f_2^*(\bar{p}) - f_2(y)]. \end{aligned}$$

Since each of these two terms is non-positive, we deduce that both are equal to 0. This implies that $q - \bar{p} \in \partial f_1(y)$ and $\bar{p} \in \partial f_2(y)$, i.e. that $q = q - \bar{p} + \bar{p} \in \partial f_1(y) + \partial f_2(y)$. $\square$

*5.2.11. Minimization of the projection of a function

Proposition 11. Let U_1 and U_2 be two vector spaces and $f: U^1 \times U^2 \rightarrow]-\infty, +\infty]$ be a proper function. An element $\bar{p}^1 \in (U^1)^*$ is a Lagrange multiplier of the problem

$$(66) \quad v = \inf_{x^2 \in U^2} f(0, x^2)$$

if and only if

$$(67) \quad -v = v^* = \inf_{p^1 \in (U^1)^*} f^*(p^1, 0) = f^*(\bar{p}^1, 0).$$

Proof. We take $U = U^1 \times U^2$, $V = U^1$. We define $Lx = -x^1$ and write $F(x, y) = f(x^1, x^2) + \psi_{\{0\}}(y)$. Then

$$(68) \quad v = \inf_{\substack{-x^1=0 \\ x^2 \in U^2}} f(x^1, x^2) = \inf [f(x^1, x^2) + \psi_{\{0\}}(-x^1)].$$

The loss function of the dual problem is defined by $F^*(-L^*p, p) = f^*(-p, 0) + 0$. Thus

$$(69) \quad v^* = \inf_{p^1 \in (U^1)^*} f^*(p^1, 0). \quad \square$$

*5.2.12. Minimization on the diagonal of a product

Proposition 12. Let V be a vector space and $f: V \times V \rightarrow]-\infty, +\infty]$ be a proper function.

An element $\bar{p}$ is a Lagrange multiplier of the problem

$$(70) \quad v = \inf_{x \in V} f(x, x)$$

if and only if

$$(71) \quad -v = v^* = \inf_{p \in V^*} f^*(p, -p) = f^*(\bar{p}, -\bar{p}).$$

Proof. We take

$$U = V \times V, \quad Lx = -x^1 + x^2$$

whenever $x = \{x^1, x^2\} \in U$.

We set $F(x, y) = f(x^1, x^2) + \psi_{\{0\}}(y)$.

Since $Lx = 0$ if and only if $x^1 = x^2$, we can write $v = \inf_{x \in V} f(x, x)$. The loss function of the dual problem is $F^*(-L^*p, p) = f^*(p, -p) + 0$. $\square$

5.3. Existence of Lagrange multipliers in the case of a finite number of constraints

In this section we prove two existence theorems. The first deals with minimization problems of the type

$$v = \inf_{x \in R} F(x, Lx),$$

where F is a *convex* function defined on $R \times Y$, R is a *convex* subset of a vector space U , Y a *convex* subset of $V = \mathbf{R}^m$. The Fenchel theorem states that the constraint qualification assumption

$$0 \in \text{Int}(L(R) - Y)$$

implies the existence of a Lagrange multiplier $\bar{p}$. We also check that the constraint qualification assumption implies the stability of Lagrange multipliers in the sense that, if F depends upon λ , then the Lagrange multipliers $\bar{p}_\lambda$ stay in a compact subset under reasonable assumptions.

We apply this result by proving the subdifferentiability of Lf on $\text{Int}[L \text{ Dom } g]$ and of gL whenever $0 \in \text{Int}[L(U) - \text{Dom } g]$. Finally, we prove the Uzawa theorem which establishes the existence of Lagrange multipliers of

$$v = \inf_{\substack{Ax \leq 0 \\ Bx = 0}} f(x)$$

when f is a convex function, A is a convex operator and B is an affine operator. The required constraint qualification assumption is that

$$\begin{cases} \text{(i) } \exists \bar{x} \text{ such that } A(\bar{x}) \ll 0 \text{ and } B(\bar{x}) = 0 \text{ (Slater condition),} \\ \text{(ii) } 0 \in \text{Int } B(R) \text{ (surjectivity condition).} \end{cases}$$

5.3.1. The Fenchel existence theorem

We shall prove an existence theorem for Lagrange multipliers, in the case when V is a finite dimensional space.

Theorem 1. *Suppose that*

$$(1) \quad V = \mathbf{R}^m \text{ is a finite dimensional subspace}$$

and that

$$(2) \quad F \text{ is a convex function from } R \times Y \text{ into } \mathbf{R} \text{ where } R \subset U \text{ and } Y \subset V \text{ are convex subsets.}$$

If

$$(3) \quad 0 \in \text{Int } (L(R) - Y),$$

then there exists a Lagrange multiplier $\bar{p} \in V^$.*

Remark. Recall that the “constraint qualification” assumption (3) implies that $p \mapsto F^*(-L^*p, p)$ is lower semi-compact on V^* (see Section 3.1.4 and Proposition 3.1.6).

Remark. We do not have to assume that F is lower semi-continuous when we ask only for the existence of a Lagrange multiplier.

Proof. We introduce the following operator ϕ from $R \times Y$ into $\mathbf{R} \times V$ defined by

$$(4) \quad \phi(x, y) = \{F(x, y); L(x) - y\}$$

and the following items

$$(5) \quad \begin{cases} \text{(i) the vector } \{v, 0\} \in \mathbf{R} \times V, \\ \text{(ii) the cone } Q =]0, \infty[\times \{0\} \subset \mathbf{R} \times V. \end{cases}$$

It is clear that the convexity of F and the linearity of L imply that

$$(6) \quad \phi(R \times Y) + Q \text{ is a convex subset of } V$$

(see the proof of Proposition 1.3.10 for instance). Also,

$$(7) \quad \{v, 0\} \notin \phi(R \times Y) + Q$$

[If $\{v, 0\} \in \phi(R \times Y) + Q$, there would exist $x \in R$ and $y \in Y$ such that $Lx - y = 0$ and $v > F(x, y) = F(x, Lx)$. This contradicts the definition of v .] We may therefore use the separation theorem for a point and a convex subset in finite dimensional spaces. There exists a *non-zero* linear form $\{\alpha, p\} \in R \times V^*$ such that

$$(8) \quad \begin{aligned} \alpha v &\leq \langle \{\alpha, p\}, \{v, 0\} \rangle \leq \sigma^b(\phi(R \times Y) + Q; \{\alpha, p\}) \\ &= \sigma^b(\phi(R \times Y); \{\alpha, p\}) + \sigma^b(Q, \{\alpha, p\}). \end{aligned}$$

Since Q is a cone and $\sigma^b(Q; \{\alpha, p\})$ is bounded from below, we deduce from Proposition 1.4.7 that $\{\alpha, p\} \in Q^+$, (i.e. that $\alpha \geq 0$) and that $\sigma^b(Q; \{\alpha, p\}) = 0$. Thus, inequality (8) becomes

$$(9) \quad \begin{aligned} \alpha v &\leq \sigma^b(\phi(R \times Y); \{\alpha, p\}) \\ &= \inf_{x \in R} \inf_{y \in Y} [\alpha F(x, y) + \langle p, Lx - y \rangle]. \end{aligned}$$

We must have $\alpha \neq 0$. Otherwise,

$$0 \leq \inf_{x \in R, y \in Y} \langle p, Lx - y \rangle = \inf_{u \in L(R) - Y} \langle p, u \rangle.$$

Since $0 \in \text{Int}(L(R) - Y)$, this implies that $p = 0$ and thus, that $\{\alpha, p\} = \{0, 0\}$ which is a contradiction.

Therefore, $\alpha > 0$. Dividing both sides of inequality (9) by $\alpha > 0$ and setting $\bar{p} = p/\alpha$, we obtain that

$$\begin{aligned} v &= \inf_{x \in R} \inf_{y \in Y} [F(x, y) - \langle -L^* \bar{p}, x \rangle - \langle \bar{p}, y \rangle] \\ &= -F^*(-L^* \bar{p}, \bar{p}). \end{aligned}$$

Since $F^*(-L^* \bar{p}, \bar{p}) \geq v^* \geq -v$, we deduce that $\bar{p}$ is a Lagrange multiplier. $\square$

Remark. We shall extend this result in the case where V is an infinite dimensional space (see Section 14.1).

*5.3.2. Stability properties

The constraint qualification hypothesis implies with appropriate assumptions that the Lagrange multipliers stay in a fixed compact subset when the loss function F depends upon a parameter.

Proposition 1. Suppose that (1) and (3) hold.

Let Λ and $F: R \times Y \times \Lambda \rightarrow \mathbf{R}$ satisfy

$$(10) \quad \begin{cases} \text{(i) } \Lambda \text{ is a compact subset of a topological vector space } W, \\ \text{(ii) } \lambda \mapsto F(x, y; \lambda) \text{ is upper semi-continuous for all } \{x, y\} \in R \times Y, \\ \text{(iii) } \{x, y\} \mapsto F(x, y; \lambda) \text{ is convex for all } \lambda \in \Lambda, \\ \text{(iv) } \exists d \in W^* \text{ such that, } \forall \lambda \in \Lambda, \langle \lambda, d \rangle \leq \inf_{x \in X} F(x, Lx; \lambda). \end{cases}$$

Then the Lagrange multipliers p_λ of the problems

$$(11) \quad v(\lambda) = \inf_{x \in X} F(x, Lx; \lambda)$$

stay in a compact subset P_0 of P .

Proof. Let $p_\lambda \in V^*$ be a Lagrange multiplier of $v(\lambda)$ where

$$(12) \quad v(\lambda) = -F^*(-L^*p_\lambda, p_\lambda).$$

Such a p_λ exists by Theorem 1. We shall prove that p_λ remains in a bounded (and thus, relatively compact) subset of P , i.e. that, for any $z \in V$, $\sup_{\lambda \in \Lambda} \langle p_\lambda, z \rangle \leq m(z) < +\infty$.

Since $0 \in \text{Int}(L(R) - Y)$, there exist $\varepsilon > 0$, $x \in R$ and $y \in Y$ such that $z = \varepsilon(y - Lx)$. Thus

$$\begin{aligned} \langle p_\lambda, z \rangle &= \varepsilon[\langle -L^*p_\lambda, x \rangle + \langle p_\lambda, y \rangle] \\ &\leq \varepsilon(F^*(-L^*p_\lambda, p_\lambda; \lambda) + F(x, y; \lambda)) \\ &\leq \varepsilon[F(x, y; \lambda) - v(\lambda)] \leq \varepsilon(F(x, y; \lambda) - \langle \lambda, d \rangle) \\ &\leq \sup_{\lambda \in \Lambda} [F(x, y; \lambda) - \langle \lambda, d \rangle] \leq \varepsilon M. \quad \square \end{aligned}$$

*5.3.3. Applications to subdifferentiability

Application: Subdifferentiability of Lf .

Let f be a convex proper function from U into $] -\infty, +\infty]$ and $L \in \mathcal{L}(U, V)$. When V is a finite dimensional space, we know that Lf is continuous (and thus, subdifferentiable) on the interior of its domain $\text{Dom}(Lf) = L \text{Dom } f$. This result can also be obtained by applying the above theorem to the case when $F(x, y) = f(x) + \psi(\{w\}, y)$, since the "constraint qualification" assumption (3) then reduces to $0 \in \text{Int}(L \text{Dom } f - w) = \text{Int } \text{Dom } Lf - w$.

Application: Subdifferentiability of gL .

Proposition 2. Suppose that (1) holds and

$$(13) \quad 0 \in \text{Int}(L(U) - \text{Dom } g).$$

If g is convex and $q \in \text{Dom } (gL)^*$, then there exists $\bar{p}$ such that

$$(14) \quad (gL)^*(q) = (L^*g^*)(q) = g^*(\bar{p}) \quad \text{where } L^*\bar{p} = q.$$

Furthermore, if g is lower semi-continuous, then

$$(15) \quad \partial(gL)(y) = L^* \partial g(Ly)$$

Proof. We apply Theorem 1 to the case where $F(x, y) = f(x) + g(y)$ with $f(x) = -\langle q, x \rangle$ and $R = \text{Dom } f = U$. Then there exists a Lagrange multiplier $\bar{p}$ whenever $(gL)^*(q) < +\infty$. This satisfies the above properties by Proposition 2.9. $\square$

Application: Subdifferentiability of $f_1 + f_2$

Proposition 3. Let f_1 and f_2 be two convex proper functions from V into $]-\infty, +\infty]$. Suppose that (I) holds and that

$$(16) \quad 0 \in \text{Int}(\text{Dom } f_1 - \text{Dom } f_2).$$

If $q \in \text{Dom } (f_1 + f_2)^*$, then there exists $\bar{p} \in U^*$ such that

$$(17) \quad (f_1 + f_2)^*(q) = (f_1^* \square f_2^*)(q) = f_1^*(q - \bar{p}) + f_2^*(\bar{p}).$$

Furthermore, if f_1 and f_2 are lower semi-continuous, then

$$(18) \quad \partial(f_1 + f_2)(y) = \partial f_1(y) + \partial f_2(y).$$

Proof. We apply Theorem 1 to the case when $L = 1$ and $F(x, y) = f(x) + g(y)$ with $f(x) = f_1(x) - \langle q, x \rangle$ and $g(y) = f_2(y)$. Assumption (3) amounts to writing that $0 \in \text{Int}(\text{Dom } f_1 - \text{Dom } f_2)$ since $R = \text{Dom } f = \text{Dom } f_1$ and $Y = \text{Dom } g = \text{Dom } f_2$. Thus there exists a Lagrange multiplier $\bar{p}$ whenever $(f_1 + f_2)^*(q) < +\infty$. This satisfies the above properties by Proposition 2.10. $\square$

5.3.4. Case of nonlinear constraints: The Uzawa existence theorem

Consider

$$(19) \quad \begin{cases} \text{(i) a convex subset } R \text{ of a vector space } U, \\ \text{(ii) } k \text{ convex functions } a_i \text{ defined on } R, \\ \text{(iii) } m - k \text{ affine functions } b_i \text{ defined on } R, \\ \text{(iv) a convex function } f \text{ defined on } R. \end{cases}$$

We associate with these items the subset

$$(20) \quad X = \{x \in R \text{ such that } a_i(x) \leq 0 \text{ for } 1 \leq i \leq k \text{ and } b_j(x) = 0 \text{ for } k+1 \leq j \leq m\}$$

Consider the problem

$$(21) \quad v = \inf_{x \in X} f(x)$$

and the Lagrangian l defined on $R \times \mathbf{R}_+^{k*} + \mathbf{R}^{(m-k)*}$ by

$$(22) \quad l(x, p) = f(x) + \langle p, A(x) \rangle + \langle q, B(x) \rangle$$

where we set

$$(23) \quad A(x) = \{a_1(x), \dots, a_k(x)\}, \quad B(x) = \{b_{k+1}(x), \dots, b_m(x)\}.$$

Proposition 4. *Given the convexity assumptions (19), the "Slater condition"*

$$(24) \quad \begin{aligned} &\exists \tilde{x} \text{ such that } a_i(\tilde{x}) < 0 \quad \forall i = 1, \dots, k \text{ and } b_j(\tilde{x}) = 0, \\ &\forall j = k+1, \dots, m \end{aligned}$$

and the surjectivity condition:

$$(25) \quad 0 \text{ belongs to the interior of } B(R) \text{ in } \mathbf{R}^{(m-k)},$$

it follows that there exists a Lagrange multiplier $\{\bar{p}, \bar{q}\} \in \mathbf{R}_+^{k} \times \mathbf{R}^{(m-k)*}$.*

Proof. This is analogous to the proof of Theorem 1. Consider the operator ϕ mapping X into $\mathbf{R}^k \times \mathbf{R}^{m-k}$ defined by

$$(26) \quad \phi(x) = \{f(x), A(x), B(x)\}.$$

Write

$$(27) \quad \begin{cases} \text{(i)} \quad \theta = \{v, 0, 0\} \in \mathbf{R} \times \mathbf{R}^k \times \mathbf{R}^{m-k}, \\ \text{(ii)} \quad C = \mathbf{R}_+ \times \mathbf{R}_+^k \times \{0\}. \end{cases}$$

Then

$$(28) \quad \theta \notin \phi(R) + C.$$

If θ belongs to $\phi(R) + C$, there would exist $\tilde{x} \in R$ such that

$$\begin{aligned} v &> f(\tilde{x}), \\ 0 &\geq A(\tilde{x}), \\ 0 &= B(\tilde{x}). \end{aligned}$$

This is a contradiction of the definition of v . Convexity assumptions (19) imply that $\phi(R) + C$ is convex (see Proposition 1.3.10). Since $\phi(R) + C$ is contained in a finite dimensional space, we can use the separation theorem. Thus there

exists a non-zero triple $\lambda = \{a, p, q\} \in \mathbf{R} \times \mathbf{R}^{k*} \times \mathbf{R}^{(m-k)*}$ such that

$$(29) \quad av = \langle \lambda, \theta \rangle \leq \inf_{x \in R} \inf_{c \in C} (\langle \lambda, \phi(x) + c \rangle).$$

The fact that C is a cone implies that

$$\lambda = \{a, p, q\} \in C^+ = \mathbf{R}_+ + \mathbf{R}_+^{k*} \times \mathbf{R}_+^{(m-k)*}.$$

The inequality $\langle \lambda, \theta \rangle \leq \inf_{x \in R} \langle \lambda, \phi(x) \rangle$

can then be written

$$(30) \quad av \leq \inf_{x \in X} [af(x) + \langle p, A(x) \rangle + \langle q, B(x) \rangle].$$

If it can be shown that

$$(31) \quad a > 0,$$

then we can divide both sides of inequality (30) by a and set $\bar{p} = p/a$ and $\bar{q} = q/a$. Then

$$(32) \quad v \leq \inf_{x \in X} [f(x) + \langle \bar{p}, A(x) \rangle + \langle \bar{q}, B(x) \rangle]$$

as required. It remains to prove (31). For this purpose, we assume that $a = 0$. The Slater condition (24) then implies that $p = 0$ and condition (25) implies that $q = 0$. But this contradicts the fact that $\lambda = \{a, p, q\}$ is non-zero. [To see this, let $a = 0$ and let $\tilde{x} \in X$ be the element of X appearing in the Slater condition. Then we deduce from (30) that

$$(33) \quad 0 \leq \langle p, A(\tilde{x}) \rangle = \sum_{i=1}^k p^i a_i(\tilde{x}).$$

Since $a_i(\tilde{x}) < 0$ and $p^i \geq 0$, we deduce that $p^i = 0$ for all i , i.e. that $p = 0$. Because $p = 0$, (30) reduces to the inequality

$$(34) \quad 0 \leq \langle q, B(x) \rangle \quad \forall x \in R.$$

But 0 belongs to the interior of $B(R)$ and so this implies that $q = 0$.] $\square$

CHAPTER 6

TWO-PERSON GAMES: AN INTRODUCTION

This chapter is devoted to an account of most of the important concepts of game theory for two-person games.

In Section 6.1, we describe two-person games in “normal” or “strategic” form in terms of the following items: strategy sets X and Y of the two players, a subset $U \subset X \times Y$ of feasible pairs of strategies and a biloss operator $F : U \rightarrow \mathbf{R}^2$ associating with any pair $\{x, y\} \in U$ the biloss $F(x, y) = \{f(x, y), g(x, y)\}$, where $f(x, y)$ is the loss of the first player and $g(x, y)$ the loss of the second.

The influence of a player on the other appears

(a) in the fact that the loss functions f and g depend upon the strategies played by *both* players;

(b) *and* in the fact that $U \subset X \times Y$ is not necessarily the product of two strategy sets: if x is chosen, then the choice of y is required to belong to the set $S(x) = \{y \in Y \text{ such that } \{x, y\} \in U\}$.

We begin by introducing two kinds of “values” of the game. The first is the “shadow minimum” $\alpha = \{\alpha_X, \alpha_Y\} \in \mathbf{R}^2$ defined by

$$\alpha_X = \inf_{\{x, y\} \in U} f(x, y), \quad \alpha_Y = \inf_{\{x, y\} \in U} g(x, y).$$

The definition implies that $F(U) \subset \alpha + \mathbf{R}_+^2$ when both α_X and α_Y are finite. Usually, α does not belong to $F(U)$.

The “*conservative values*” v_X^* and v_Y^* are defined as follows in the case when $U = X \times Y$.

If the first player (named Xavier) has no information whatsoever about the strategy choice of the second player (named Yvette), he calculates the worst possible loss associated with a strategy $x \in X$ by the formula $f^*(x) = \sup_{y \in Y} f(x, y)$. Then $v_X^* = \inf_{x \in X} f^*(x) = \inf_{x \in X} \sup_{y \in Y} f(x, y)$ is called the conservative value for Xavier, and $v_Y^* = \inf_{x \in X} \sup_{y \in Y} g(x, y)$ is called the conservative value for Yvette. If we set $v^* = \{v_X^*, v_Y^*\} \in \mathbf{R}^2$, the subset V of strategies defined by

$$V = \{\{x, y\} \in U \text{ such that } F(x, y) \leq v^*\}$$

is the “useful” part of V , in the sense that Xavier, for example, will “reject” x if $f(x, y) > v^*$ (since he can do better in the worst situation). Notice that $F(V)$ is contained in the square $(\alpha + \mathbf{R}_+^2) \cap (v^* - \mathbf{R}_+^2)$.

We next define the following concepts of “stability” and “equilibrium”.

A pair $\{\bar{x}, \bar{y}\} \in X \times Y$ is *individually stable* if neither Xavier nor Yvette have any incentive to modify their own choice of strategy unless the other player modifies his choice. This means that

$$f(\bar{x}, \bar{y}) = \min_{x \in X} f(x, \bar{y}), \quad g(\bar{x}, \bar{y}) = \min_{y \in Y} g(\bar{x}, y).$$

We say that $\{\bar{x}, \bar{y}\}$ is a “*non-cooperative equilibrium*”. It is *collectively unstable*, if both Xavier and Yvette can find a pair of strategies $\{x, y\} \in U$ such that $f(x, y) < f(\bar{x}, \bar{y})$ and $g(x, y) < g(\bar{x}, \bar{y})$. The *collectively stable* pairs of strategies are called the *Pareto minima*. The ideal situation occurs when there exists a unique pair of strategies $\{\bar{x}, \bar{y}\}$ which is *individually and collectively stable*, i.e. a non-cooperative equilibrium which is Pareto optimal. Unfortunately, this situation is quite exceptional. We are therefore led to let aside the concept of individual stability while retaining that of collective stability (Pareto minima exist under quite weak assumptions). We relax the requirement of individual stability then have to devise selection procedures for Pareto minima. For instance, we require that Pareto minima belong to V , i.e. are rejected neither by Xavier nor by Yvette: the subset of such pairs of strategies is called the *core* of a (two person) game.

In Section 6.2 we examine several examples of games for which some of the above concepts apply and other do not. These examples show that there is no hope of defining a unique “good” concept of solution.

In Section 6.3 and Section 6.4, we examine the two basic economic models. The first illustrates the behavior of a duopoly. In this case, X and Y are production sets, $U = X \times Y$ (i.e. the productions can be chosen independently) but the production costs $f(x, y)$ and $g(x, y)$ depend upon *both* productions. The second deals with the choice of consumptions x and y such that $x + y = w$. Hence the subset U is defined by $U = \{x, y\} \in X \times Y$ such that $x + y = w$. In this model, the loss function f can depend only upon x and g upon y .

Finally, in Section 6.5, we introduce the very important special case of zero-sum games, where $g(x, y) = -f(x, y)$. For such games, any pair of strategies is Pareto minimal and so the concept is not useful. One therefore looks for non-cooperative equilibria. These are nothing other than the “saddle-points” of f .

6.1. Some solution concepts

In this section we describe the main concepts of game theory as they apply to two-person games.

6.1.1. Description of the game

Let Xavier and Yvette be the (first) names of the two players. We describe the two-person game in the “*strategic form*” or “*normal form*” by the following items:

- (1) $\left\{ \begin{array}{l} \text{(i) the strategy set } X \text{ of Xavier,} \\ \text{(ii) the strategy set } Y \text{ of Yvette,} \\ \text{(iii) the loss function } f : X \times Y \rightarrow \mathbf{R} \text{ of Xavier,} \\ \text{(iv) the loss function } g : X \times Y \rightarrow \mathbf{R} \text{ of Yvette.} \end{array} \right.$

We associate with any pair $\{x, y\} \in X \times Y$ of strategies the loss $f(x, y)$ allocated to Xavier and the loss $g(x, y)$ allocated to Yvette. We denote by $F(x, y) = \{f(x, y), g(x, y)\} \in \mathbf{R}^2$ the “biloss” associated with the pair of strategies $\{x, y\}$ and call the map $F : \{x, y\} \in X \times Y \mapsto F(x, y) \in \mathbf{R}^2$ the “biloss operator”.

It can happen that not all pairs of strategies can be implemented, but only those which belong to

- (2) the subset $U \subset X \times Y$ of feasible strategies.

The description of the game in strategic form is then *summarized by the notation* (U, F) *where* $U \subset X \times Y$ *and* F *maps* U *into* $\mathbf{R}^2$.

6.1.2. Shadow minimum

We write

$$(3) \quad \alpha_X = \inf_{\{x, y\} \in U} f(x, y); \quad \alpha_Y = \inf_{\{x, y\} \in U} g(x, y).$$

Definition 1. We shall say that the game is bounded below if both α_X and α_Y are finite. In this case, we shall say that the biloss $\alpha = \{\alpha_X, \alpha_Y\}$ is the “*shadow minimum*” of the game.

In the case where there exists $\{\bar{x}, \bar{y}\} \in U$ such that

$$\alpha_X = f(\bar{x}, \bar{y}); \quad \alpha_Y = g(\bar{x}, \bar{y})$$

such a pair $\{\bar{x}, \bar{y}\}$ yields the minimal loss to *both* Xavier and Yvette, i.e. $F(\bar{x}, \bar{y}) = \alpha$ is the minimum (relative to the appropriate ordering of $\mathbf{R}^2$) of $F(x, y)$

as $\{x, y\}$ ranges over U . Such a pair $\{\bar{x}, \bar{y}\}$ is reasonable as a solution concept but will obviously exist only in exceptional situations. This is why we call the vector α a “shadow minimum” or “virtual minimum”.

We have that

$$F(U) \subset \alpha + \mathbf{R}_+^2$$

Since in most cases the shadow minimum does not belong to $F(U)$, we are led to introduce other solution concepts.

6.1.3. Conservative solutions and values

Consider the case where $U = X \times Y$.

We introduce the functions f^* (f -sharp) and g^* defined by

$$(4) \quad \begin{cases} (i) f^*(x) = \sup_{y \in Y} f(x, y), \\ (ii) g^*(y) = \sup_{x \in X} g(x, y). \end{cases}$$

Since $f(x, y)$ is the loss for Xavier when Xavier chooses x and Yvette y , $f^*(x)$ is the “worst loss” possible for Xavier when he plays the strategy $x \in X$. A consistent policy for Xavier whenever he *has no information* whatsoever about the strategy to be played by Yvette, is to associate with any strategy $x \in X$ the consequent worst loss $f^*(x)$. In the same way, when Yvette has no information about the choice of Xavier, she may associate with any strategy $y \in Y$ the consequent worst loss $g^*(y)$.

Thus, under these behavioral assumptions, Xavier will choose a strategy x^* which minimizes f^* over X and Yvette a strategy y^* which minimizes g^* over Y .

Definition 2. We shall say that a strategy x^* which satisfies

$$(5) \quad f^*(x^*) = v_X^* = \inf_{x \in X} f^*(x)$$

(resp. y^* such that $g^*(y^*) = v_Y^* = \inf_{y \in Y} g^*(y)$) is a “conservative solution” for Xavier (resp. Yvette).

The vector $v^* = \{v_X^*, v_Y^*\}$ is called the “conservative value” of the game.

Conservative values will be used as examples of “threats”, in the following sense. Xavier will *reject* any strategy x yielding a loss $f(x, y)$ larger than $v_X^* = \inf_{x \in X} f^*(x)$, since v_X^* is the loss that he can obtain by unilateral action, whatever the other player does.

Most often these threats are not implemented. Usually, a “conservative solution satisfies neither the “individual stability property” nor the “collective stability property”. We discuss these important properties next.

6.1.4. Non-cooperative equilibrium

For simplicity, we assume that $U = X \times Y$.

Consider a pair $\{x^*, y^*\}$ of conservative strategies for Xavier and Yvette. In some cases, Xavier has a strategy $x \in X$ for which

$$f(x, y^*) < f(x^*, y^*).$$

In other words, $\{x^*, y^*\}$ does not have “*individual stability*” since Xavier, by acting *alone*, is better off by switching from x^* and playing x . We are led to introduce the following concept of “non-cooperative equilibrium” which does satisfy the individual stability property.

Definition 3. We shall say that a pair $\{\bar{x}, \bar{y}\}$ of strategies is a “non-cooperative equilibrium” if

$$(6) \quad \begin{cases} \text{(i)} f(\bar{x}, \bar{y}) = \min_{x \in X} f(x, \bar{y}), \\ \text{(ii)} g(\bar{x}, \bar{y}) = \min_{y \in Y} g(\bar{x}, y). \end{cases}$$

In other words, $\{\bar{x}, \bar{y}\}$ is a non-cooperative equilibrium in the sense that when Yvette implements $\bar{y}$, the optimal strategy for Xavier in respect of the loss function $x \mapsto f(x, \bar{y})$ is $\bar{x}$ and, symmetrically, when Xavier plays $\bar{x}$, the optimal strategy for Yvette in respect of the loss function $y \mapsto g(\bar{x}, y)$ is $\bar{y}$.

Remark. Suppose that there exists a map $\bar{C}$ from Y into X (called the *optimal decision rule* of Xavier) satisfying

$$(7) \quad \forall y \in Y, \quad f(\bar{C}(y), y) = \min_{x \in X} f(x, y)$$

and a map $\bar{D}$ from X into Y (called the *optimal decision rule* of Yvette) satisfying

$$(8) \quad \forall x \in X, \quad g(x, \bar{D}(x)) = \min_{y \in Y} g(x, y).$$

Then any solution $\{\bar{x}, \bar{y}\}$ of the system

$$(9) \quad \bar{C}(\bar{y}) = \bar{x} \quad \text{and} \quad \bar{D}(\bar{x}) = \bar{y}$$

is a non-cooperative equilibrium.

More generally, if we denote by $\bar{\mathcal{E}}$ the correspondence from Y into X defined by

$$(10) \quad \forall y \in Y, \quad \bar{\mathcal{E}}(y) = \left\{ \bar{x} \in X \text{ such that } f(\bar{x}, y) = \min_{x \in X} f(x, y) \right\}$$

and by $\overline{\mathcal{D}}$ the correspondence from X into Y defined by

$$(11) \quad \forall x \in X, \quad \overline{\mathcal{D}}(x) = \{\bar{y} \in Y \text{ such that } g(x, \bar{y}) = \min_{y \in Y} g(x, y)\},$$

then a pair $\{\bar{x}, \bar{y}\}$ is a *non-cooperative equilibrium* if and only if

$$(12) \quad \bar{x} \in \overline{\mathcal{E}}(\bar{y}) \quad \text{and} \quad \bar{y} \in \overline{\mathcal{D}}(\bar{x}),$$

i.e. $\{\bar{x}, \bar{y}\}$ is a *fixed point* of the correspondence $\{x, y\} \mapsto \overline{\mathcal{E}}(y) \times \overline{\mathcal{D}}(x)$.

Remark. We introduce the functions f^b (f -flat) and g^b defined by

$$(13) \quad f^b(y) = \inf_{x \in X} f(x, y); \quad g^b(x) = \inf_{y \in Y} g(x, y).$$

We set

$$(14) \quad F^b(x, y) = \{f^b(y), g^b(x)\}.$$

Note that $\{\bar{x}, \bar{y}\}$ is a *non-cooperative equilibrium* if and only if

$$(15) \quad F(\bar{x}, \bar{y}) = F^b(\bar{x}, \bar{y}).$$

6.1.5. Pareto minimum

Consider a pair $\{x^*, y^*\}$ of conservative strategies for Xavier and Yvette. In some cases, *both* Xavier and Yvette can find a pair of strategies $\{x, y\} \in U$ such that

$$(16) \quad f(x, y) < f(x^*, y^*) \quad \text{and} \quad g(x, y) < g(x^*, y^*).$$

In other words, $\{x^*, y^*\}$ does not have “*collective stability*” since, by *cooperating*, both Xavier and Yvette can find strategies yielding both a smaller loss. We are led to introduce the following concept of “Pareto minimum” which does satisfy the collective stability property.

Definition 4. We shall say that a pair $\{\bar{x}, \bar{y}\} \in U$ of strategies is a “*weak Pareto minimum*” if there is no pair $\{x, y\} \in U$ such that both

$$(17) \quad f(x, y) < f(\bar{x}, \bar{y}) \quad \text{and} \quad g(x, y) < g(\bar{x}, \bar{y}).$$

Remark. A „nice solution concept” is obtained when *both the individual and the collective stability properties hold*. Unhappily, such a requirement is too strong in most instances.

The set of Pareto minima is usually large. The main problem is to find selection procedures for Pareto minima, i.e. to find solution concepts satisfying the collective stability property.

We also notice that, if a pair $\{x, y\}$ is not collectively stable, (in the sense of (17)), then the players will jointly reject $\{x, y\}$. This leads to the concept of the *core* of a game.

6.1.6. Core of a two-person game

For simplicity, we suppose that $U = X \times Y$

Definition 5. We shall say that a pair $\{x, y\}$ of strategies is “rejected” by Xavier if $f(x, y) > v_x^*$, is “rejected” by Yvette if $g(x, y) > v_y^*$ and is “rejected” by both Xavier and Yvette if it is not a weak Pareto minimum.

The “core of the game” is the set of pairs of strategies rejected neither by Xavier, nor by Yvette, nor by both of them.

In other words, the core of the game is the subset of weak Pareto minima $\{\bar{x}, \bar{y}\}$ such that $F(x, y) \leq v^*$.

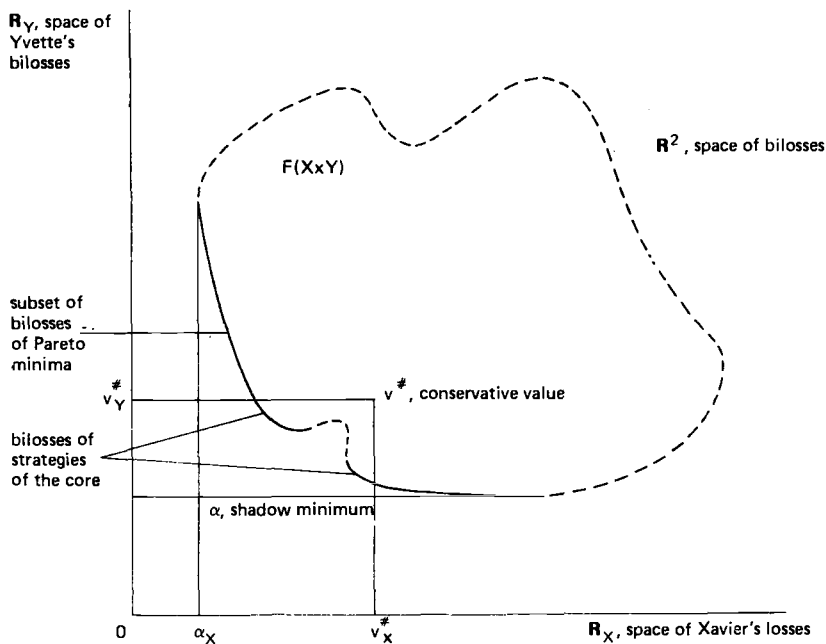


Fig. 1. Image of a game in the space of bilosses

6.1.7. Selection of strategy of the core

The main problem remaining is the selection of a strategy which belongs to the core (whenever it is not empty).

Chapter 10 is devoted to this problem in the case of n person games. We shall only mention here the following selection procedure. We assume that

$$(18) \quad \alpha_X < v_X^* \quad \text{and} \quad \alpha_Y < v_Y^*$$

(instead of $\alpha_X \leq v_X^*$ and $\alpha_Y \leq v_Y^*$) and introduce the following ratios

$$(19) \quad d_X(x, y) = \frac{f(x, y) - \alpha_X}{v_X^* - \alpha_X}; \quad d_Y(x, y) = \frac{g(x, y) - \alpha_Y}{v_Y^* - \alpha_Y}.$$

These measure the distance to the shadow minimum relative to the distance between the conservative value and the shadow minimum.

We assume that both players agree to minimize the worst ratio

$$(20) \quad d(x, y) = \max(d_X(x, y), d_Y(x, y)).$$

We set

$$(21) \quad \bar{d} = \inf_{\substack{x \in X \\ y \in Y}} d(x, y).$$

It is easy to check that *any* bisstrategy $\{\bar{x}, \bar{y}\}$ which minimizes $d(x, y)$ belongs to the core and actually satisfies inequalities

$$(22) \quad f(\bar{x}, \bar{y}) \leq (1 - \bar{d})\alpha_X + \bar{d}v_X^*, \quad g(\bar{x}, \bar{y}) \leq (1 - \bar{d})\alpha_Y + \bar{d}v_Y^*.$$

where at least one equality holds.

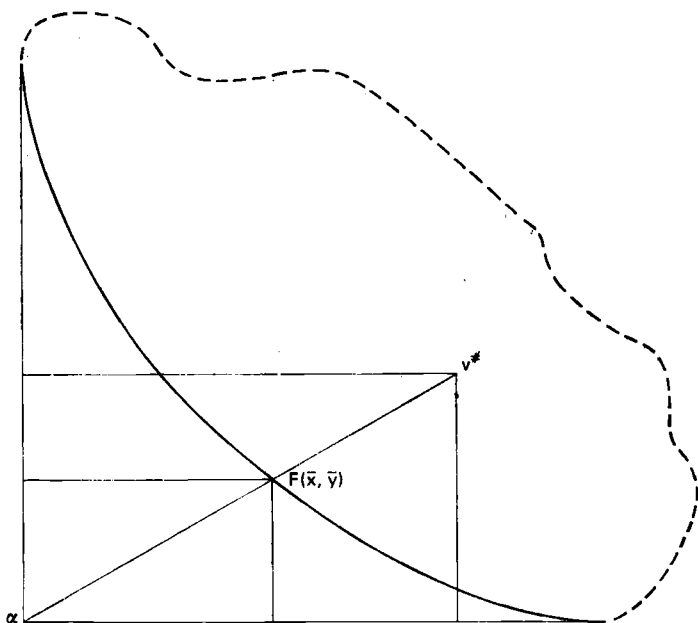


Fig. 2.

*6.2. Examples: some finite games

It is time to illustrate the above concepts with some simple examples with the strategy sets X and Y finite. We begin with an example for which non-coopera-

tive equilibria do not exist but where the core consists of a unique pair of strategies. This pair can be regarded as the “solution concept” for the game. We introduce the “coordination game” as our second example. In one case, we shall describe a situation where the shadow minimum is achieved. In another case, we shall see that this game has non-cooperative equilibria which are Pareto minimal, i.e. which are both *individually and collectively stable*.

In the example of the “prisoner’s dilemma”, we have a situation where the conservative strategies are individually stable (i.e. form a non-cooperative equilibrium), but are not collectively stable. This example shows the need of communication for implementing a Pareto minimal strategy.

In the “game of chicken”, we obtain a pair of strategies which is both a conservative solution, a non-cooperative equilibrium and a Pareto minimum. Finally, the “battle of the sexes” illustrates a case where all pairs of strategies are conservative solutions and where two of them are both individually and collectively stable.

6.2.1. Example

In this first example, non-cooperative equilibria do not exist and the core consists of a unique pair of strategies. For this pair collective stability holds but no strategy pairs are individually stable.

Let Xavier be the first player and Yvette the second player and let $X = \{I, II\}$ and $Y = \{1, 2, 3\}$ be their strategy sets. We represent the biloss operator F defined on $X \times Y$ by the following “bimatrix”, each entry of which is a pair of scalars, the first being the loss to Xavier and the second the loss to Yvette.

	Yvette Xavier	1	2	3
F:	I	(−6, 3)	(3, 0)	(−3, 3)
	II	(3, −2)	(−5, 2)	(4, 7)

irst, we notice that $\alpha_X = -6$ and $\alpha_Y = -2$.

Then we check that

$$f^*(I) = 3, \quad f^*(II) = 4, \quad g^*(1) = 3, \quad g^*(2) = 2 \quad \text{and} \quad g^*(3) = 7$$

and so

$$(1) \quad v_X^* = 3 = f^*(I); \quad v_Y^* = 2 = g^*(2)$$

Therefore

$$(2) \quad \alpha = (-6, -2) \text{ is the shadow minimum}$$

and

(3) $x^* = I$ and $y^* = 2$ form a pair of conservative solutions.

The values of the operator F^b are given in the following “bimatrix”:

F^b :

Xavier \ Yvette	1	2	3
I	$(-6, 0)$	$(-5, 0)$	$(-3, 0)$
II	$(-6, -2)$	$(-5, -2)$	$(-3, -2)$

By comparing the matrices of F and F^b , it is easily seen that *non-cooperative equilibria do not exist*.

In particular, the conservative solutions are not individually stable.

The set of Pareto minima consists of the pairs $\{I, 1\}$, $\{II, 2\}$ and $\{II, 1\}$. The core consists of the pairs $\{II, 2\}$ and $\{II, 1\}$.

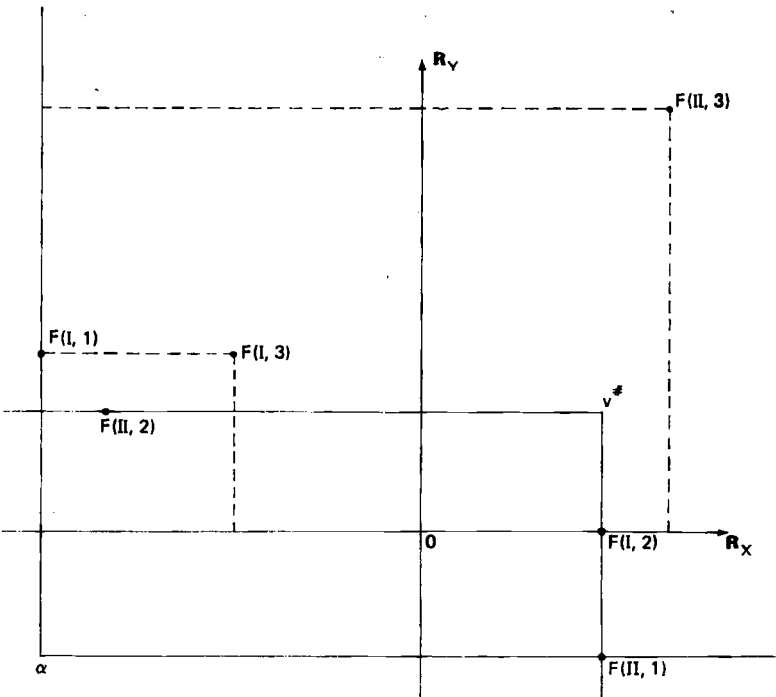


Fig. 3.

6.2.2. Coordination game

Let Xavier and Yvette be the two players and let $X = \{I, II\}$ and $Y = \{1, 2\}$ be their strategy sets. Let F be the biloss operator represented by the following "bimatrix"

F:	Xavier \ Yvette		1	2	$0 < a < b$
	I	(b, b)	$(0, a)$		
	II	$(a, 0)$	(x, x)		

$$0 < a < b$$

It is clear that

$$f^*(I) = b, \quad f^*(II) = \sup(a, x), \quad g^*(1) = b, \quad g^*(2) = \sup(a, x).$$

Hence

$$(4) \quad v^* = \begin{cases} (b, b) = (f^*(I), g^*(1)) & \text{if } b \leq x, \\ (x, x) = (f^*(II), g^*(2)) & \text{if } a \leq x < b, \\ (a, a) = (f^*(II), g^*(2)) & \text{if } x \leq a. \end{cases}$$

On the other hand

$$f^b(1) = a, \quad f^b(2) = \inf(0, x), \quad g^b(I) = a, \quad g^b(II) = \inf(0, x).$$

The operator F^b is represented by the following bimatrix.

F ^b :	Xavier Yvette	1	2
	I	(a, a)	(inf (0, x), a)
	II	(a, inf (0, x))	(inf (0, x), inf (0, x))

Interpretation. Xavier and Yvette are caught in a fire and have to push against a door to escape. But this involves exposing oneself to the flames for the time it takes the door to open.

We denote by I, 1, the strategies: "do not push" and by II, 2, the strategies "push".

If nobody pushes against the door, both stay in the fire and incur a loss of b .

If Xavier pushes against the door (strategy II) and Yvette does not (strategy I), then Yvette escapes first (and loses nothing) and Xavier escapes second incurring a loss of $a < b$.

Let x be the loss incurred by each player when both push against the door together. This is treated as a parameter.

Case 1: $x \leq 0$. In this case, strategy $\{II, 2\}$ yields a gain to both, i.e. a negative loss and it is obvious that it is the optimal strategy. Indeed, in this case, the shadow minimum $\alpha = \{x, x\} = F(II, 2)$ is achieved. Thus $(II, 2)$ is the smallest (in $\mathbb{R}^2$) of the bilosses (and thus, is no longer "shadow").

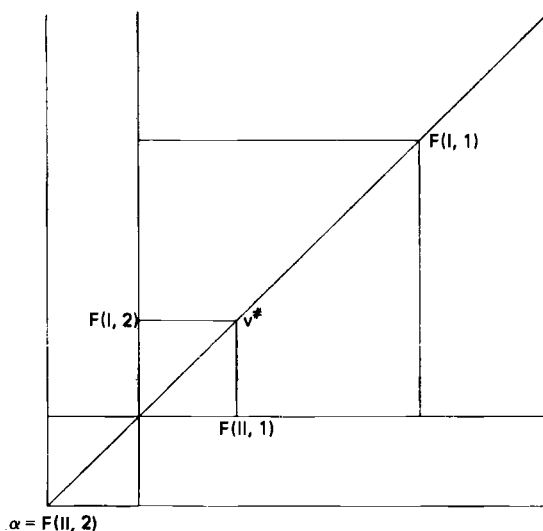


Fig. 4.

Thus $(II, 2)$ is the best strategy.

Case 2: $x > 0$. In this case, bistrategy $(II, 2)$ (i.e. bistrategy push-push) yields a loss $x \geq 0$ to both players (because, for example, they are jamming the door way). The shadow minimum is no longer achieved.

(a) *Non cooperative equilibria.*

Strategies $(II, 1)$ and $(I, 2)$ satisfy the relations $F^b(II, 1) = F(II, 1)$ and $F^b(I, 2) = F(I, 2)$, i.e. the relations

$$(5) \quad \begin{cases} f(II, 1) = a < f(I, 1) = b, \\ g(II, 1) = 0 < g(II, 2) = x \end{cases}$$

and

$$(6) \quad \begin{cases} f(I, 2) = 0 < f(II, 2) = x, \\ g(I, 2) = a < g(I, 1) = b. \end{cases}$$

In other words, if Yvette plays strategy 1 (i.e. does not push), it is optimal for Xavier to play strategy II (i.e. to push) and if Xavier plays strategy II (i.e. pushes), it is optimal for Yvette to play strategy 1 (i.e. not to push).

(The same interpretation holds for $\{I, 2\}$ by exchanging the roles of Xavier and Yvette).

Inequalities (5) and (6) show that bistrategies $\{II, 1\}$ and $\{I, 2\}$ are “individually stable”, i.e. non-cooperative equilibria. Neither player *has an incentive to change his strategy by acting unilaterally*.

There remains the problem of distinguishing between these two non-cooperative equilibria. It is not clear, for example, that Xavier will accept a loss of a and let Yvette escape scot free. Such a choice requires communication or collusion among the players.

(b) *Conservative solutions.*

If Xavier does not push against the door (i.e. plays I), he risks a loss of b if Yvette uses the same strategy. Otherwise, by pushing against the door, the worst possibility is a loss of $\sup(a, x)$.

Hence, a cautious Xavier will naturally choose strategy II (i.e. push) if $x \leq b$ and strategy I (i.e. do not push) if $b \leq x$.

In the same way, a cautious Yvette will choose strategy 2 if $x \leq b$ and strategy 1 if $b \leq x$.

These choices are made independently by each player, *by considering the worst situation. They do not require any communication*.

Naturally, if the game had to be repeated, neither strategy $\{II, 2\}$ nor $\{I, 1\}$ is “individually stable” in the above sense. Again, if $x < a$ and if Xavier plays strategy II (i.e. pushes against the door) and implements his strategy before Yvette, then it is optimal for Yvette to choose strategy 1 (instead of 2).

(c) *Pareto minima and core.*

If $x \geq a$, the conservative solution $\{II, 2\}$ is rejected by both Xavier and Yvette, since it is not collectively stable. *Each* of the players will set a loss (either 0 or a) by replacing the bistrategy $\{II, 2\}$ by either $\{II, 1\}$ and $\{I, 2\}$.

But this reason is no longer valid if $0 \leq x \leq a$, since in this case no multi-strategy yields a smaller biloss than $F(II, 2)$.

(Notice that bistrategies $\{I, 2\}$ and $\{II, 1\}$ share this property in this example).

In other words, the set of Pareto minima coincides with the core, and it is equal to $\{\{I, 2\}, \{II, 1\}, \{II, 2\}\}$ if $x \leq a$, to $\{\{I, 2\}, \{II, 1\}\}$ if $x \geq a$.

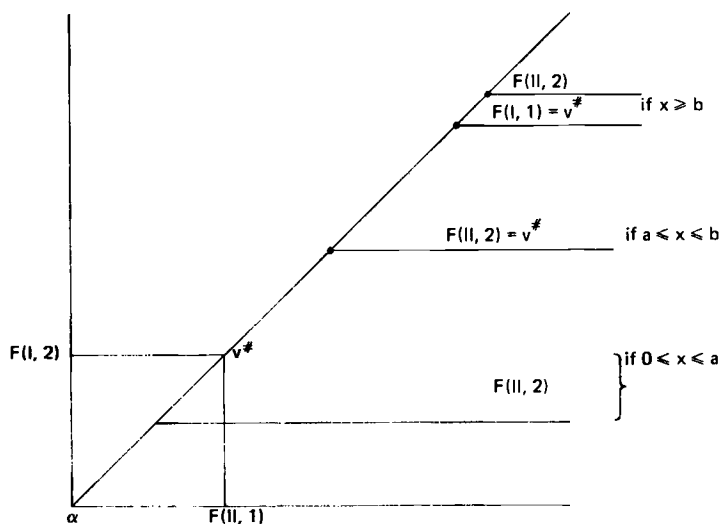


Fig. 5.

6.2.3. Prisoner's dilemma

Let Xavier and Yvette be the players and let $X = \{I, II\}$ and $Y = \{1, 2\}$ be their strategy sets. Let F be the biloss operator described by the bimatrix

		Yvette		
		1	2	
F:	Xavier			
	I	(a, a)	(b, 0)	
	II	(0, b)	(x, x)	

$0 < a < b$

It is clear that

$$f^*(I) = b, \quad f^*(II) = \sup(0, x), \quad g^*(1) = b, \quad g^*(2) = \sup(0, x)$$

and thus, that

$$(7) \quad v^* = \begin{cases} (b, b) = (f^*(I), g^*(1)) & \text{if } b \leq x, \\ (x, x) = (f^*(II), g^*(2)) & \text{if } 0 \leq x \leq b, \\ (0, 0) = (f^*(II), g^*(2)) & \text{if } x \leq 0. \end{cases}$$

On the other hand, we check that

$$f^b(1) = 0, \quad f^b(2) = \inf(b, x), \quad g^b(I) = 0, \quad g^b(II) = \inf(b, x).$$

The operator F^b is described by the following bimatrix

F^b :

	Yvette		
Xavier		1	2
I		(0, 0)	($\inf(b, x)$, 0)
II		(0, $\inf(b, x)$)	($\inf(b, x)$, $\inf(b, x)$)

Interpretation. Both Xavier and Yvette are prisoners, each of whom can either confess or not confess to a crime.

If neither confesses (bistrategy $\{I, 1\}$), they face moderate jail sentences (a years in prison).

If Xavier confesses (strategy II) and Yvette does not, Xavier is set free (0 year in prison) and Yvette faces a hard jail sentence (b years in prison).

The loss x (years of prison) when both players confess (strategy (II, 2)) is treated as a parameter.

Case $x = 0$: This case is obvious, since the shadow minimum $\alpha = \{0, 0\}$ is achieved by the bistrategy (II, 2). Both players confess.

Case $0 < x \leq b$. This means that they face a smaller number of years of prison than b when both confess.

The bistrategy (II, 2) is the only bistrategy which satisfies the relation $F^b(\text{II}, 2) = F(\text{II}, 2)$, i.e.

$$\begin{cases} f(\text{II}, 2) = x < f(\text{I}, 2) = b, \\ g(\text{II}, 2) = x < g(\text{II}, 1) = b. \end{cases}$$

With this bistrategy, neither player has an incentive to make a unilateral switch of strategy. Therefore, the bistrategy is a non-cooperative equilibrium

Moreover, the multistrategy $\{\text{II}, 2\}$ also satisfies the relation $v^* = (f^*(\text{II}), g^*(2))$. In other words, if Xavier does not confess, (strategy I), he risks a years of jail while, by confessing, (strategy II), the worst possibility is $x < b$ years of prison. Hence caution naturally leads Xavier to choose strategy II (and Yvette strategy 2).

Nevertheless if $a \leq x \leq b$, both players can diminish the jail sentence of x years (obtained by playing $\{\text{II}, 2\}$) if neither confesses (strategy $\{I, 1\}$) since it yields only a years of prison. But, if Xavier implements strategy I (does not confess), Yvette can confess (strategy 2), be set free and penalize Xavier with b years of jail!

Since

$$f^b(1) = 0, \quad f^b(2) = b, \quad g^b(I) = 0, \quad g^b(II) = b.$$

The values of F^b are given in the following bimatrix.

F^b :

		Yvette	
		1	2
Xavier	I	(0, 0)	(b, 0)
	II	(0, b)	(b, b)

So, bistrategies $\{I, 2\}$ and $\{II, 1\}$ are non-cooperative equilibria. They are also Pareto minima.

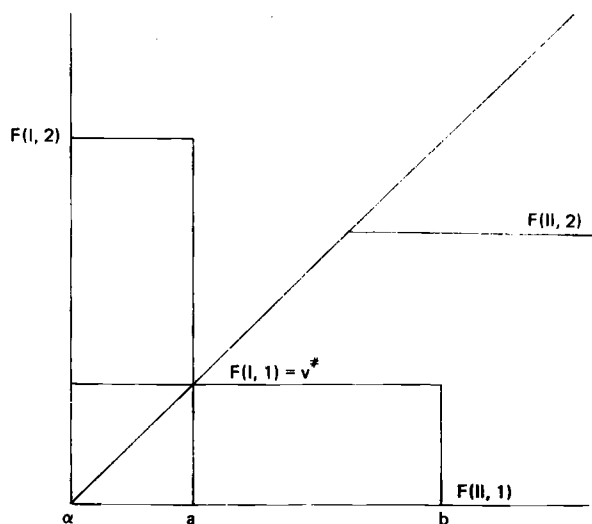


Fig. 7.

Interpretation. Xavier and Yvette drive toward each other at high speed and the first one to swerve “loses”.

We denote by I, 1 the strategies “swerve” and by II, 2 the strategies “don’t swerve”.

If both swerve, they pay an amount of a units. If only one swerves, the winning player has no loss while the losing player pays b . If neither swerves, well, they crash and pay an amount x .

The bistrategies where one player swerves while the other does not are non-cooperative equilibria that are Pareto minima. The bistrategy where both swerve is a conservative solution that is a Pareto minimum.

6.2.5. *The battle of the sexes*

Let Xavier be the first player, Yvette the second and let $X = \{I, II\}$ and $Y = \{1, 2\}$ be their strategy sets. Let F be the biloss operator represented by the following bimatrix

		Yvette		
		Xavier \		
$F:$	I	$(0, a)$	(b, b)	$0 < a < b$
	II	(b, b)	$(a, 0)$	

It is clear that

$$f^*(I) = f^*(II) = g^*(1) = g^*(2) = b.$$

Hence

$$(9) \quad v^* = (b, b)$$

and the four pairs of strategies form conservative solutions!

The values of F^b are summarized in the following bimatrix

		Yvette		
		Xavier \		
$F^b:$	I	$(0, a)$	(a, a)	
	II	$(0, 0)$	$(a, 0)$	

Hence the pairs $\{I, 1\}$ and $\{II, 2\}$ are the non-cooperative equilibria *and* the Pareto minima of the game. Also, they form the core of the game.

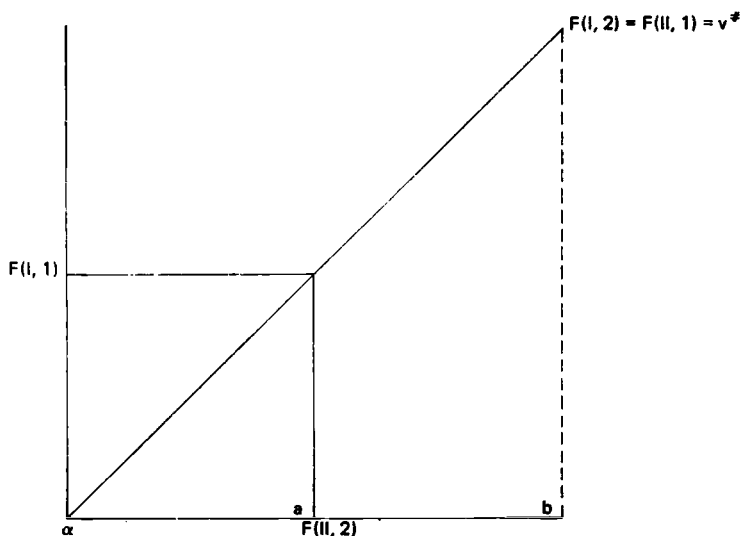


Fig. 8.

Interpretation. Strategies I and 1 represent “attend a football game” and strategies II and 2 represent “go shopping”. Xavier prefers to attend the football game and Yvette prefers to go shopping. But in any case, they prefer to go together.

6.3. Example: Analysis of duopoly

We study in this section the basic example of duopoly in which both players are producers. The loss functions are the cost functions and these depend upon the productions of the two players.

We begin by describing the set of Pareto minima, which yield a profit (negative cost) to both players. The conservative strategies amount to both players producing nothing. We afterwards construct the non-cooperative equilibrium (introduced by Cournot in 1838). This is *not* Pareto minimal.

This non-cooperative equilibrium is constructed as a fixed point of the optimal decision rules. We then compute the productions which result if *one player knows* (or assumes) that *his opponent will use the optimal decision rule*. We will

explain why he can implement a production which yields a smaller loss than the non-cooperative equilibrium to him and a larger loss to his opponent (Stackelberg equilibrium). But if each player guesses (wrongly) that his opponent will use the optimal decision rule, each of them will then implement the so-called *Stackelberg disequilibrium*, which yields to both players a worse loss than the non-cooperative equilibrium.

This example illustrates the danger of *a priori* guesses on the opponent's behavior.

6.3.1. The model of a duopoly

In duopoly, Xavier and Yvette are producers of a given good, assumed to be homogeneous.

Let $x \in \mathbf{R}_+$ and $y \in \mathbf{R}_+$ be the quantities of the good produced by Xavier and Yvette.

We assume that

- (1) the price $p = \alpha - \beta(x+y)$ is an affine function of the total production $x+y$ ($\alpha \geq 0$, $\beta > 0$)

and that the cost functions c and d of each producer are defined by

- (2) $c(x) = \gamma x + \delta$; $c(y) = \gamma y + \delta$, $\gamma > 0$, $\delta \geq 0$

where γ denotes the marginal cost, δ the fixed cost.

The net cost for Xavier is equal to

- (3) $f(x, y) = \gamma x + \delta - px = \beta x \left[x + y + \frac{\gamma - \alpha}{\beta} \right] + \delta$

and the net cost for Yvette is also equal to

- (4) $g(x, y) = \gamma y + \delta - py = \beta y \left[x + y + \frac{\gamma - \alpha}{\beta} \right] + \delta.$

For simplicity, we will assume that $\beta = 1$, $\delta = 0$ and set $\alpha - \gamma = u$. This model of a duopoly is a two-person game whose strategy sets are $X = Y = [0, u]$ and whose biloss operator is defined by

- (5) $F(x, y) = (x(x+y-u), y(x+y-u)).$

This model will be generalized in Section 9.2.5.

6.3.2. The set of Pareto minima

Since $f(x, y) + g(x, y) = (x + y)(x + y - u) = z(z - u)$ (where $z = x + y$ is the total production), we see that the sum of the costs ranges from $-\frac{1}{4}u^2$ to $2u^2$. Thus the subset $F(X \times Y)$ of the biloss space $\mathbb{R}^2$ is the union of the set of vectors $\{a, b\}$ such that $-\frac{1}{4}u^2 \leq a + b \leq 0$, $-\frac{1}{4}u^2 \leq a, b \leq 0$ and of the square $[0, u^2] \times [0, u^2]$.

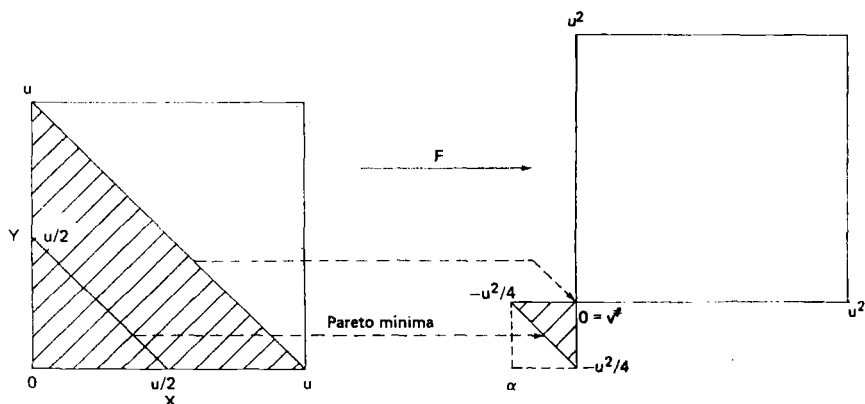


Fig. 9.

Actually, the square $[0, u^2] \times [0, u^2]$ is the image of the subset of $X \times Y$ such that $x + y \geq u$ and the triangle is the image of the subset of $X \times Y$ such that $x + y \leq u$.

We deduce that the set of Pareto minima is the set of bistrategies $\{x, y\}$ such that

$$(6) \quad f(x, y) + g(x, y) = (x + y)(x + y - u) = -\frac{u^2}{4},$$

i.e. such that $x + y = \frac{1}{2}u$. Note that the losses to both players are positive as soon as the total production $x + y$ is larger than u .

6.3.3. Conservative solutions

It is clear that $f^*(x) = \sup_{0 \leq y \leq u} x(x + y - u) = x^2$ achieves its minimum at $x = 0$ and that $g^*(y) = \sup_{0 \leq x \leq u} y(x + y - u) = y^2$ achieves its minimum at $y = 0$.

Thus the *conservative bistrategy* is $\{0, 0\}$, i.e. *both players produce nothing*.

Since $v^* = (0, 0)$, we deduce that the core of the game is equal to the set of Pareto minima.

6.3.4. Non-cooperative equilibria

The concept of non-cooperative equilibria in the case of the duopoly game was introduced by Cournot (in 1838).

Let y be Yvette's production. Then Xavier will implement a production x which minimizes over X his net cost function $x \mapsto x(x+y-u)$. The minimum is achieved at the point

$$(7) \quad x = \bar{C}(y) = \frac{u-y}{2}$$

and is equal to

$$(8) \quad f^b(y) = \inf_x [x(x+y-u)] = -\frac{(u-y)^2}{4}$$

Similarly, Yvette will implement a production

$$(9) \quad y = \bar{D}(x) = \frac{u-x}{2}$$

which minimizes her net loss function when x is produced by Xavier. Thus

$$(10) \quad g^b(x) = \inf_y [y(x+y-u)] = -\frac{(u-x)^2}{4}$$

The non-cooperative equilibrium $\{\bar{x}, \bar{y}\}$ is then the fixed point of the function $\{x, y\} \mapsto \{\bar{C}(y), \bar{D}(x)\}$ i.e. the point

$$(11) \quad \bar{x} = \bar{y} = \frac{u}{3}$$

and yields to each player a loss equal to

$$(12) \quad a_N = b_N = -\frac{u^2}{9}.$$

We notice that the *non-cooperative equilibrium is not Pareto-optimal*. The difference between the total losses is equal to

$$-\frac{2u^2}{9} - \left(-\frac{u^2}{4}\right) = \frac{u^2}{36}.$$

But the total production is greater: $\frac{2}{3}u$ instead of $\frac{1}{2}u$.

6.3.5. Stackelberg equilibria

We assume now that players use “decision rules” $C : Y \rightarrow X$ and $D : X \rightarrow Y$ instead of pure strategies. We then seek “stable solutions” in this framework, i.e. *the fixed points of the map* $\{x, y\} \mapsto \{C(y), D(x)\}$.

Recall that in obtaining a non-cooperative equilibrium in pure strategies, we arrived at unique optimal decision rules $\bar{C}$ and $\bar{D}$.

Assume next that Yvette chooses a fixed decision rule $D : x \mapsto y = D(x)$ and that Xavier knows it.

Then Xavier will choose a strategy x_D which minimizes the loss function $x \mapsto f(x, D(x))$.

For instance, if Xavier knows that Yvette will use the optimal decision rule $\bar{D}$, he will implement the strategy $x_D = \frac{1}{2}u$ which minimizes the function $x \mapsto f(x, \bar{D}(x)) = x(x - u + \frac{1}{2}(u - x)) = \frac{1}{2}x(x - u)$. Yvette will produce $\bar{D}(x_D) = \frac{1}{4}u$. By doing so, the net cost of Xavier is equal to $-\frac{1}{8}u^2$ and that of Yvette is equal to $-\frac{1}{16}u^2$. Xavier is then in a better situation than in the original non-cooperative equilibrium ($-\frac{1}{8}u^2$ instead of $-\frac{1}{9}u^2$) while Yvette is in a worse situation (since she loses $-\frac{1}{16}u^2$ instead of $-\frac{1}{9}u^2$).

The result (x_D, y_D) is called the *Stackelberg equilibrium for Xavier*. In the same way, if we suppose that Yvette knows that Xavier will use the optimal decision rule $\bar{C}$, she will implement the strategy $y_C = \frac{1}{2}u$ and Xavier the strategy $\bar{C}(y_C) = \frac{1}{4}u$.

6.3.6. Stackelberg disequilibrium

We recall that the non-cooperative equilibrium is the fixed point of the map $\{x, y\} \mapsto \{\bar{C}(y), \bar{D}(x)\}$. We call $\bar{C}$ and $\bar{D}$ the “Cournot decision rules”. Since they are affine, their derivatives are constant (equal to $-\frac{1}{2}$). Now, if Yvette uses the Cournot decision rule $\bar{D}$, Xavier will solve the equation

$$(13) \quad \frac{\partial f}{\partial x}(\bar{x}, \bar{D}(\bar{x})) + \bar{D}'(\bar{x}) \frac{\partial f}{\partial y}(\bar{x}, \bar{D}(\bar{x})) = \frac{\partial f}{\partial x}(\bar{x}, \bar{D}(\bar{x})) - \frac{1}{2} \frac{\partial f}{\partial y}(\bar{x}, \bar{D}(\bar{x})) = 0$$

for finding the point $\bar{x}$ which minimizes $x \mapsto f(x, \bar{D}(x))$. This amounts to solving

$$(14) \quad \bar{y} = \bar{D}(\bar{x})$$

and

$$(15) \quad \frac{\partial f}{\partial x}(\bar{x}, \bar{y}) - \frac{1}{2} \frac{\partial f}{\partial y}(\bar{x}, \bar{y}) = 0.$$

The latter equation can be written

$$(16) \quad \bar{x} = C_S(\bar{y}) = \frac{1}{2}(u - \bar{y})$$

where $C_S : y \mapsto \frac{1}{2}(u - y)$ is called the “Stackelberg decision rule”. Thus the Stackelberg equilibrium for Xavier is obtained as the fixed point of the map $\{x, y\} \mapsto \{C_S(y), \bar{D}(x)\}$.

In the same way, the Stackelberg equilibrium for Yvette is obtained as the fixed point of the map $\{x, y\} \mapsto \{\bar{C}(y), D_S(x)\}$ where $D_S : x \mapsto \frac{1}{2}(u - x)$ is the “Stackelberg decision rule” of Yvette.

Now, suppose that the rules of the game neither force Yvette to use the decision rule $\bar{D}$ nor force Xavier to use $\bar{C}$ (and both players are fully aware of these facts).

Then each player can incorrectly believe that the other will use the “naive” optimal decision rule.

In this case, Xavier and Yvette will both use their Stackelberg decision rules. They implement the fixed point $\{x_S, y_S\}$ of the map $\{x, y\} \rightarrow \{C_S(y), D_S(x)\}$, where $x_S = y_S = \frac{2}{5}u$. Their net cost will be equal to $-\frac{2}{25}u^2$, and both of them are in worse situation than in the non-cooperative case.

The bistrategy (x_S, y_S) is called the *Stackelberg disequilibrium*. If we restrict the available “strategies” for each player to either the Cournot decision rule or the Stackelberg decision rule, we obtain a finite game whose bimatrix is given by

Xavier \ Yvette	Cournot	Stackelberg
Cournot	$(-\frac{1}{9}u^2, -\frac{1}{9}u^2)$	$(-\frac{1}{16}u^2, -\frac{1}{8}u^2)$
Stackelberg	$(-\frac{1}{8}u^2, -\frac{1}{16}u^2)$	$-\frac{2}{25}u^2, -\frac{2}{25}u^2$

It is apparent that this game has the same structure as the prisoner’s dilemma game. For both players the Stackelberg decision rules yields a smaller loss than the Cournot decision rule, but both players would be better off if both choose the Cournot decision rule than if both choose the Stackelberg decision rule.

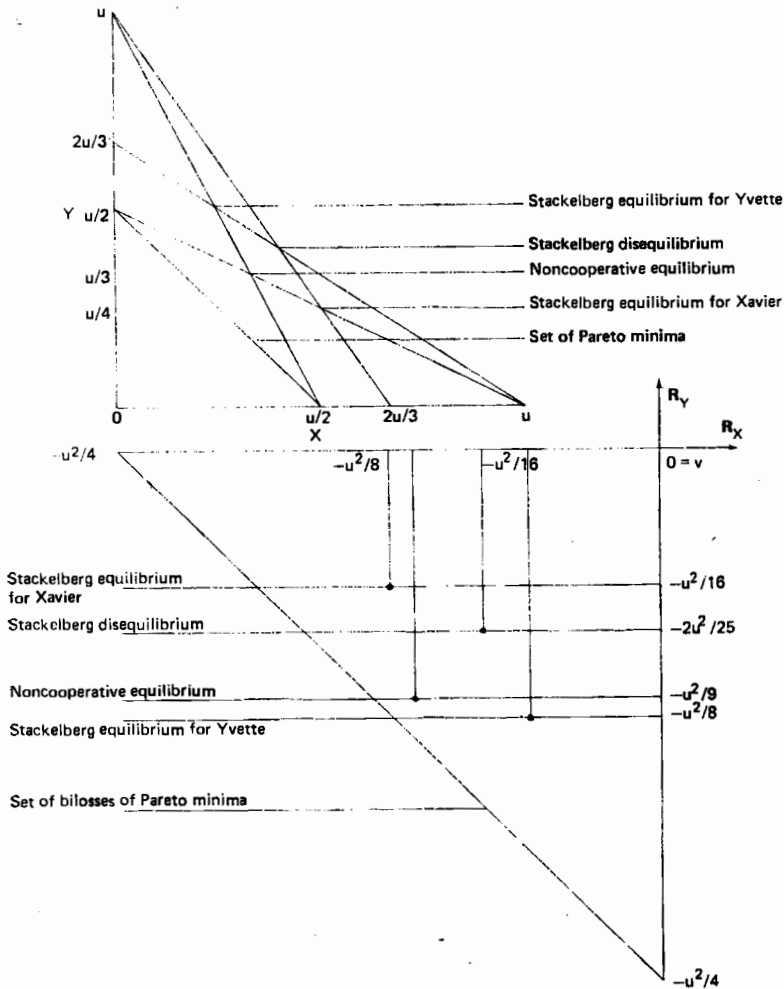


Fig. 10.

6.4. Example: Edgeworth economic game

We consider the basic model of the allocation of a commodity w among two consumers. With the aid of an ingenious graphical representation (the so-called Edgeworth box) due to Edgeworth, we can initiate a study of this model in the case of commodities of two goods. The main feature of this game (as opposed to the duopoly case) is that the set $U = \{x, y\} \in X \times Y$ such that $x + y =$

$= w\}$ is *not a product*, while the loss functions depend only on each player's own consumption.

In such a model the concept of non-cooperative equilibrium makes no sense. We indicate the subset of *Pareto minima* in the Edgeworth box. We then define the *core*. Since unilateral play by either player is impossible our previous definition needs to be extended by assuming that a player rejects an allocation if it yields him a loss larger than that he obtains with his initial endowment (instead of the conservative value).

The core of this game proves to be quite large and the problem arises as to how to choose an allocation in the core. We shall indicate that all Walras equilibria are in the core.

6.4.1. The set of feasible allocations

In this game, Xavier and Yvette are trading two commodities. Their strategy sets are both equal to $X = Y = [0, a] \times [0, b]$, which is the set of commodities $x = \{x_1, x_2\}$ representing x_1 units of a first good and x_2 units of a second.

We assume that Xavier's loss function f depends only on his choice of a commodity x and that Yvette's loss function g depends only on her choice of a commodity y . Thus

$$(1) \quad f: X \rightarrow \mathbf{R}; \quad g: Y \rightarrow \mathbf{R}; \quad F(x, y) = \{f(x), g(y)\} \in \mathbf{R}^2.$$

The new feature is that players cannot use the whole bistrategy set $X \times Y$, but are restricted to a feasible subset $U \subset X \times Y$ representing the possible allocations of commodities that the players can achieve by trading. We assume that

$$(2) \quad \begin{cases} \text{(i) Xavier owns the commodity } w^X = (a, 0), \\ \text{(ii) Yvette owns the commodity } w^Y = (0, b). \end{cases}$$

Then the set U is the subset of pairs $\{x, y\} \in X \times Y$ such that $x + y = w^X + w^Y$, i.e. such that

$$(3) \quad x_1 + y_1 = a, \quad x_2 + y_2 = b.$$

We say that U is the *set of feasible allocations*.

6.4.2. The biloss operator

Consider first the strategy sets X and Y and the loss functions for Xavier and Yvette.

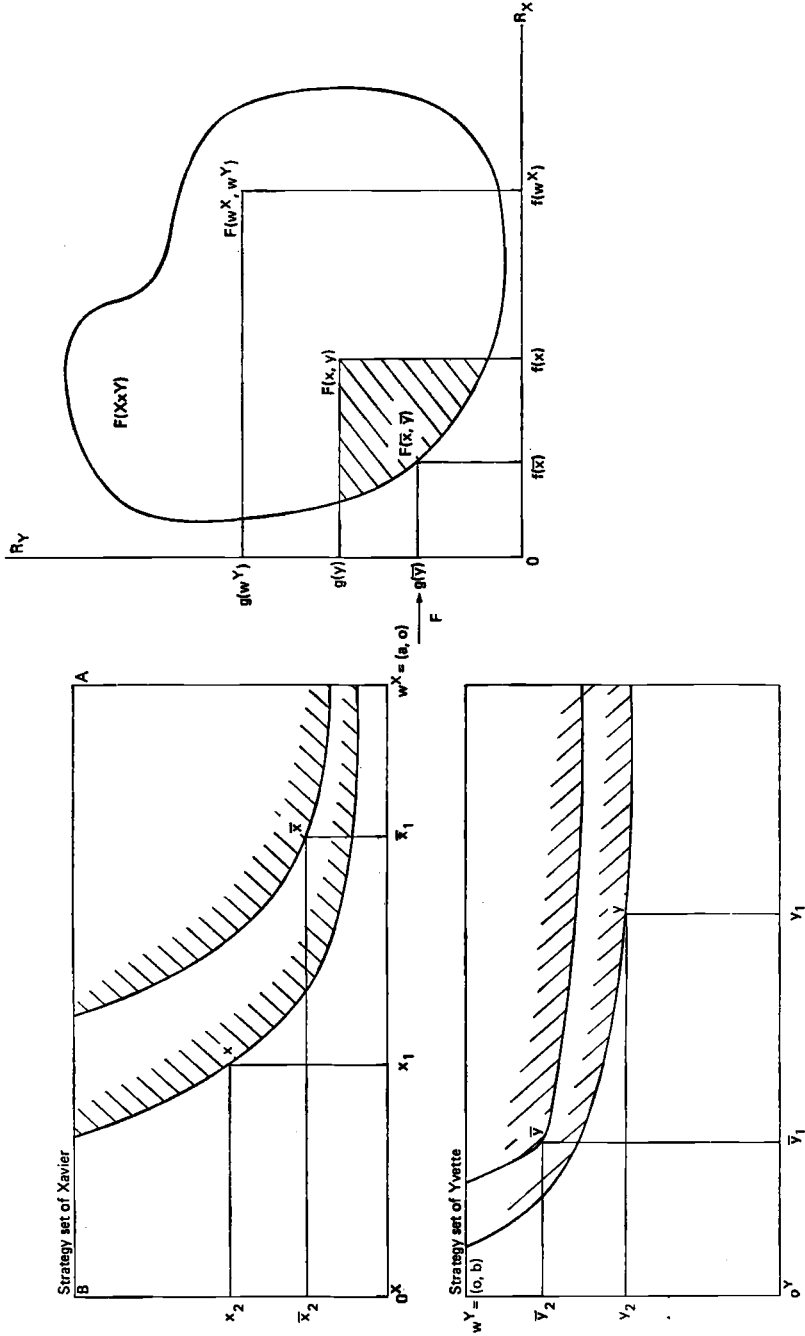


Fig. 11.

The curve passing through x (called an "indifference curve") is the set of points ξ such that $f(\xi) = f(x)$ and the shaded area above represents the set of points ξ such that $f(\xi) < f(x)$. Similarly, the curve passing through y is the set of points η such that $g(\eta) = g(y)$ and the shaded area above represents the set of points η such that $g(\eta) < g(y)$.

We have also drawn indifference curves passing through $\bar{x}$ and $\bar{y}$. In the biloss space $\mathbb{R}^2$, $F(U)$ is indicated. The shaded area is the set of bilosses from $F(U)$ which are strongly smaller than $F(x, y)$.

6.4.3. The Edgeworth box

Edgeworth (1881) gave an ingenious graphical representation of the subset U . We superimpose the rectangle Y on the rectangle X after a rotation which maps 0^Y onto A and w^Y onto w^X .

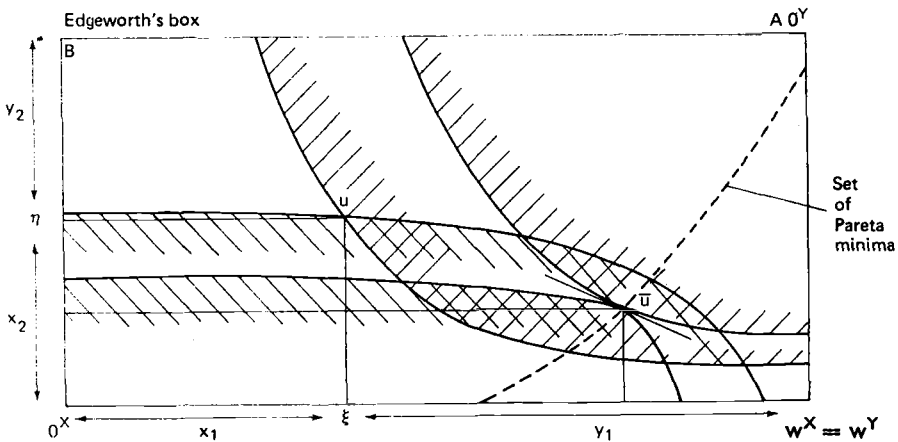


Fig. 12.

A point $u = (\xi, \eta)$ represents a bistrategy $u = \{x, y\}$ of U on the understanding that

$$(4) \quad x_1 = \xi, \quad x_2 = \eta, \quad y_1 = a - \xi, \quad y_2 = b - \eta.$$

We then have that $x + y = w^X + w^Y$ as necessary. This amounts to saying that a given point $u = \{\xi, \eta\}$ of U represents a commodity x for Xavier with respect to the basis $\{\vec{0^X w^X}, \vec{0^X B}\}$ and a commodity y for Yvette with respect to the basis $\{\vec{0^Y B}, \vec{0^Y w^Y}\}$. The diagram is called the "Edgeworth box".

The shaded area of the Edgeworth box represents the set of pairs $\bar{u} = \{\bar{x}, \bar{y}\} \in U$ such that $F(\bar{x}, \bar{y}) \leq F(x, y)$, i.e. the preimage by F of the cone $F(x, y) - \mathbb{R}_+^2$.

6.4.4. Pareto minima

A pair of strategies $\bar{u} = \{\bar{x}, \bar{y}\} \in U$ is a *Pareto minimum* if there is no other strategy $\bar{u} = \{\bar{x}, \bar{y}\}$ such that $f(\bar{x}) < f(\bar{x})$ and $g(\bar{y}) < g(\bar{y})$. This means that the cone $F(\bar{x}, \bar{y}) - \mathbf{R}_+^2$ is disjoint from $F(U)$, i.e. that the shaded areas of $F(U)$ and U are empty. This happens if and only if the indifference curves passing through $\bar{u} = \{\bar{x}, \bar{y}\}$ are tangent.

The set of Pareto minima is represented in the Edgeworth box by the dotted line.

6.4.5. Core

We regard the set of Pareto minima as the set of bistrategies $\bar{u} = \{\bar{x}, \bar{y}\}$ which are not "rejected" by *both* Xavier and Yvette.

We now make the following behavioral assumptions.

Since Xavier owns w^X , he will *reject* any commodity bundle x such that $f(x) > f(w^X) = f(a)$. In other words, he will only trade for commodities above the indifference curve passing through w^X . Similarly, since Yvette owns w^Y , she will *reject* any commodity y such that $g(y) > g(w^Y) = b$. In other words, she will only trade for commodities below the indifference curve passing through w^Y .

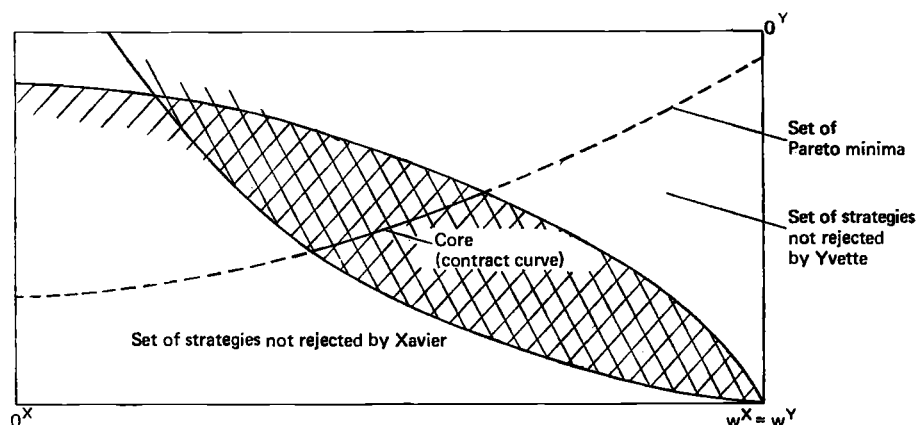


Fig. 13.

By definition, the *core* is the set of bistrategies $\bar{u} = \{\bar{x}, \bar{y}\}$ which are rejected neither by Xavier, nor by Yvette, nor by both Xavier and Yvette acting together.

It is therefore the subset of Pareto minima rejected neither by Xavier nor by Yvette.

It is represented in the Edgeworth box by the part of the dotted curve lying between the indifference curves passing through w^X and w^Y . In this context we call the core the “contract curve”.

The problem arises as to whether it is possible to devise a selection procedure for an allocation in the core. The concept of Walras equilibrium provides such a selection procedure.

6.4.6. Walras equilibria

Let $u = \{x, y\}$ be a bistrategy in the core. Since the indifference curves for Xavier and Yvette passing through u are tangent, they have a common tangent (D) .

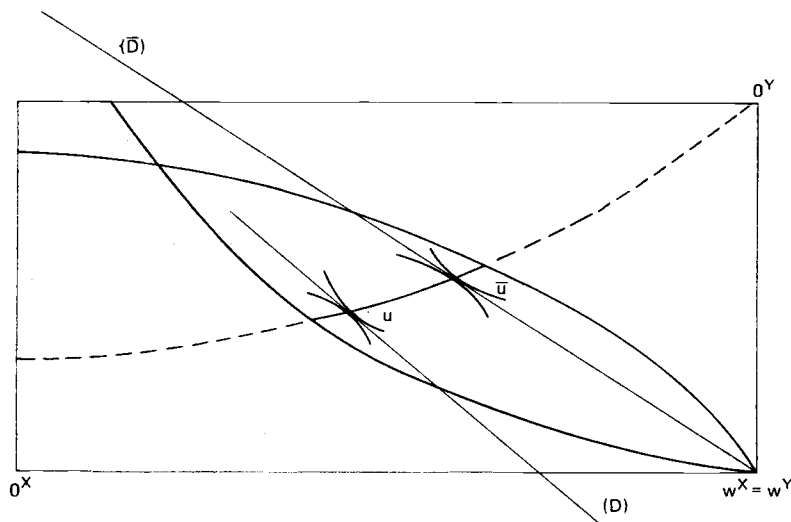


Fig. 14.

Generally, this tangent does not pass through w^X . Assume that $\bar{u} = \{\bar{x}, \bar{y}\}$ is a point of the core such that the common tangent $(\bar{D})$ to the indifference curves passes through $w^X = w^Y$. Consider the equation of the line $(\bar{D})$:

$$(\bar{D}): p_1 \xi + p_2 \eta = p_1 a \quad \text{where } p_1, p_2 \geq 0.$$

We can regard $p = \{p_1, p_2\}$ as a price system, p_1 being the price of a unit of the first good and p_2 the price of a unit of the second good. The half-plane located below $(\bar{D})$ represents Xavier's budget set, i.e. the set of points $x = \{x_1, x_2\}$ such that the value $\langle p, x \rangle = p_1 x_1 + p_2 x_2$ of x is not greater than the value $p_1 a = \langle p, w^X \rangle$ of w^X . The half-plane located above $(\bar{D})$ represents Yvette's budget set,

since it is the subset of points $y = \{y_1, y_2\}$ such that $p_1\xi + p_2\eta = p_1(a - y_1) + p_2(b - y_2) \geq p_1a$, i.e. the subset of points $y = \{y_1, y_2\}$ such that the value $\langle p, y \rangle = p_1y_1 + p_2y_2$ of y is not greater than the value of w^Y .

The fact that the indifference curve passing through $\bar{u}$ is tangent to $(\bar{D})$ at $\bar{u}$ means that $\bar{x}$ minimizes $f(x)$ under the constraints $\langle p, x \rangle \leq \langle p, w^X \rangle$ and that $\bar{y}$ minimizes $g(y)$ under the constraints $\langle p, y \rangle \leq \langle p, w^Y \rangle$.

In summary, such a point $\bar{u} = \{\bar{x}, \bar{y}\}$ has the following properties:

$$(5) \quad \begin{cases} \text{(i)} \quad \bar{x} + \bar{y} = w^X + w^Y, \\ \text{(ii)} \quad \langle p, \bar{x} \rangle = \langle p, w^X \rangle; \langle p, \bar{y} \rangle = \langle p, w^Y \rangle \\ \text{(iii)} \quad \bar{x} \text{ minimizes } f \text{ under the constraint } \langle p, x \rangle \leq \langle p, w^X \rangle \text{ and } \bar{y} \\ \quad \text{minimizes } g \text{ under the constraint} \\ \quad \langle p, y \rangle \leq \langle p, w^Y \rangle. \end{cases}$$

Such a triple $\{\bar{x}, \bar{y}, p\}$ is called a *Walras equilibrium of the economic game*.

6.5. Two-person zero-sum games

We end this chapter by an introduction to the case of “zero-sum games” (also called “duels”), where the loss function $g = -f$ of Yvette is the opposite of the loss function of Xavier. For such games, any pair of strategies is Pareto minimal. Hence this concept is no longer useful. We are left with the concepts of conservative strategies and non-cooperative equilibria.

We notice that the conservative value of Xavier is equal to $v^* = \inf_{x \in X} \sup_{y \in Y} f(x, y)$, whereas the conservative value of Yvette is equal to $-v^b = -\sup_{y \in Y} \inf_{x \in X} f(x, y)$. Note that it is always true that $v^b \leq v^*$.

The interval $[v^b, v^*]$ is called the *duality gap*. If v^* and v^b coincide, the common value $v = v^* = v^b$ is called the “*value of the game*”. The main result is that *a pair $\{\bar{x}, \bar{y}\}$ of conservative strategies is a non-cooperative equilibrium if and only if the game has a value*. In this case, $\{\bar{x}, \bar{y}\}$ is called a “*saddle-point*” of the game. We end this section by studying perturbations of the loss function by linear forms and by displaying examples of games with finite strategy sets for which a saddle point does or does not exist.

6.5.1. Duality gap and value

Let us consider a two-person game with strategy sets X and Y and loss functions f and g .

Definition 1. A two-person zero-sum game is a two-person game in which the sum of the loss functions f and g of Xavier and Yvette is equal to 0.

In other words, Yvette's loss function $g(x, y) = -f(x, y)$ is Xavier's utility function $f(x, y)$. Therefore, Yvette's worst-loss function is

$$(1) \quad g^*(y) = -f^b(y) \quad \text{where } f^b(y) = \inf_{x \in X} f(x, y)$$

and the conservative value v_Y^* is equal to

$$(2) \quad v_Y^* = -v^b \quad \text{where } v^b = \sup_{y \in Y} \inf_{x \in X} f(x, y).$$

We always have that

$$(3) \quad \forall x \in X, \quad \forall y \in Y, \quad f^b(y) \leq f(x, y) \leq f^*(x)$$

and thus

$$(4) \quad v^b = \sup_{y \in Y} \inf_{x \in X} f(x, y) \leq v^* = \inf_{x \in X} \sup_{y \in Y} f(x, y).$$

The interval $[v^b, v^*]$ is called the “*duality gap*”.

In other words, the inequality

$$\forall x \in X, \quad \forall y \in Y, \quad 0 \leq f^*(x) - f^b(y)$$

says that the sum of the two worst-loss functions is always non-negative (i.e. is a true loss). The inequality

$$(5) \quad 0 \leq v^* - v^b = \inf_{\substack{x \in X \\ y \in Y}} [f^*(x) - f^b(y)]$$

follows and may be interpreted as meaning that the sum $v^b - v^*$ of the two minimal worst losses is always non-negative.

Hence, in general, by playing “conservatively” (i.e. as though in the worst situation), the two-players (as a team) obtain a non-negative loss.

We shall distinguish the case where the sum of the minimal worst losses is equal to 0, i.e. where

$$\begin{aligned} v^* &= \text{the minimal worst loss of Xavier} \\ &= v^b = \text{the maximal worst gain of Yvette.} \end{aligned}$$

Definition 2. We shall say that a two-person zero-sum game represented by $f: X \times Y \rightarrow \mathbf{R}$ has a “*saddle-value*” (in short, a “*value*”), if

$$(6) \quad v^* = v^b = v$$

and this common value $v = v^* = v^b$ is called the “*saddle value*” of the game (in short, the value)

6.5.2. *Saddle point*

We introduce also the following definitions.

Definition 3. A strategy $\bar{x} \in X$ which minimizes the worst loss function f^* on X is said to be a “*conservative*” strategy for Xavier:

$$(7) \quad f^*(\bar{x}) = \inf_{x \in X} f^i(x) = v^*.$$

If moreover the game has a value, then $\bar{x}$ is called a “*minisup*” of f , i.e.

$$(8) \quad \sup_y f(\bar{x}, y) = \sup_{y \in Y} \inf_{x \in X} f(x, y) = v.$$

A strategy $y \in Y$ which maximizes $f^b(y)$ is said to be a “*conservative*” strategy of Yvette:

$$(9) \quad f^b(\bar{y}) = \sup_{y \in Y} f^b(y) = v^b.$$

If moreover the game has a value, then $\bar{y}$ is called a “*max-inf*” of f , i.e.

$$(10) \quad \inf_{x \in X} f(x, \bar{y}) = \inf_{x \in X} \sup_{y \in Y} f(x, y) = v$$

A pair $\{\bar{x}, \bar{y}\} \in X \times Y$ is called a “*saddle point*” of f if

$$(11) \quad \sup_{y \in Y} f(\bar{x}, y) = f(\bar{x}, \bar{y}) = \inf_{x \in X} f(x, \bar{y}).$$

We begin by characterizing saddle points.

Proposition 1. A pair $\{\bar{x}, \bar{y}\}$ is a saddle point if and only if $\bar{x}$ is a minisup and $\bar{y}$ is a max-inf

Proof. Assume that $\{\bar{x}, \bar{y}\}$ is a saddle point. Then (11) implies that

$$v^* \leq f^*(\bar{x}) = f(\bar{x}, \bar{y}) = f^b(\bar{y}) \leq v^b$$

and thus $v^* = v^b = v$. Hence $\bar{x}$ minimizes f^* and $\bar{y}$ maximizes f^b . Therefore $\bar{x} \in X$ is a minisup and $\bar{y} \in Y$ is a max-inf.

Conversely, let $\bar{x}$ be a minisup and $\bar{y}$ be a max-inf.

Then the game has a value v . Thus

$$\sup_{y \in Y} f(\bar{x}, y) = f^*(\bar{x}) = v = f^b(\bar{y}) = \inf_{x \in X} f(x, \bar{y}).$$

Taking $y = \bar{y}$ in the supremum of the left-hand side and $x = \bar{x}$ in the infimum of the right-hand side, we obtain

$$f(\bar{x}, \bar{y}) \leq f^*(\bar{x}) = v = f^b(\bar{y}) \leq f(\bar{x}, \bar{y}).$$

Therefore $v = f(\bar{x}, \bar{y})$. Hence $\{\bar{x}, \bar{y}\}$ is a saddle-point. $\square$

Remark. A saddle point $\{\bar{x}, \bar{y}\}$ of f is nothing other than a non-cooperative equilibrium of the two person game described by the biloss operator F defined on $X \times Y$ by $F(x, y) = \{f(x, y), -f(x, y)\}$. In fact (11) reduces to

$$(12) \quad \begin{cases} \text{(i)} f(\bar{x}, \bar{y}) \leq f(x, \bar{y}) \text{ for any } x \in X, \\ \text{(ii)} -f(\bar{x}, \bar{y}) \leq -f(\bar{x}, y) \text{ for any } y \in Y. \end{cases}$$

In other words, in a two-person zero-sum game, a pair $\{\bar{x}, \bar{y}\}$ of conservative strategies is a non-cooperative equilibrium if and only if the game has a value.

In a zero-sum game, the concept of Pareto minimum is not useful, since every bistrategy $\{x, y\}$ is Pareto-minimal.

*6.5.3. Perturbation by linear functions

Suppose that X and Y are convex subsets of vector spaces U and V . We shall perturb the loss functions f by linear forms $\{x, y\} \mapsto -\langle p, x \rangle + \langle q, y \rangle$.

We set

$$(13) \quad v^*(p, q) = \inf_{x \in X} \sup_{y \in Y} [f(x, y) - \langle p, x \rangle + \langle q, y \rangle]$$

and

$$(14) \quad v^b(p, q) = \sup_{y \in Y} \inf_{x \in X} [f(x, y) - \langle p, x \rangle + \langle q, y \rangle].$$

Proposition 2. If y_0 is a max-inf of the loss function $\{x, y\} \mapsto f\{x, y\} - \langle p_0, x \rangle + \langle q_0, y \rangle$, then $y_0 \in \partial v^*(p_0, \cdot)(q_0)$

The converse is true whenever, $\forall x \in X$, the function $y \mapsto f(x, y)$ is concave and upper semi-continuous.

Proof. We shall prove that y_0 is a max-inf if and only if

$$(15) \quad \begin{cases} \text{(i)} y_0 \in \partial v^*(p_0, \cdot)(q_0), \\ \text{(ii)} \inf_{x \in X} [f(x, y_0) - \langle p_0, x \rangle] = \inf_q [v^*(p_0, q) - \langle q, y_0 \rangle] \end{cases}$$

and that the concavity and the upper semi-continuity of f with respect to y implies (15(ii)).

(a) Suppose that y_0 is a max-inf. Then

$$v^*(p_0, q_0) = \inf_{x \in X} [f(x, y_0) - \langle p_0, x \rangle] + \langle q_0, y_0 \rangle$$

and

$$\begin{aligned} v^*(p_0, q) &\geq v^b(p_0, q) = \sup_y \inf_x [f(x, y) - \langle p_0, x \rangle + \langle q, y \rangle] \\ &\geq \inf_x [f(x, y_0) - \langle p_0, x \rangle] + \langle q, y_0 \rangle. \end{aligned}$$

Therefore

$$(16) \quad v^*(p_0, q_0) - v^*(p_0, q) \leq \langle q_0 - q, y_0 \rangle$$

(i.e. $y_0 \in \partial v^*(p_0, \cdot)(q_0)$). Furthermore, we obtain that

$$\begin{aligned} \inf_x [f(x, y_0) - \langle p_0, x \rangle] &= v^*(p_0, q_0) - \langle q_0, y_0 \rangle \\ &= \inf_q [v^*(p_0, q) - \langle q, y_0 \rangle]. \end{aligned}$$

(b) Conversely suppose that y_0 satisfies conditions (15). Since (15(i)) is equivalent to (16), we have that

$$\begin{aligned} v^*(p_0, q_0) - \langle q_0, y_0 \rangle &= \inf_q [v^*(p_0, q) - \langle q, y_0 \rangle] \\ &= \inf_x [f(x, y_0) - \langle p_0, x \rangle] \end{aligned}$$

and thus,

$$v^*(p_0, q_0) = \inf_x [f(x, y_0) - \langle p_0, x \rangle + \langle q_0, y_0 \rangle]$$

(i.e. y_0 is a max-inf).

(c) Suppose that the functions $y \mapsto f(x, y)$ are upper semi-continuous and concave. Write $g(y) = -f(x, y)$. Then

$$\begin{aligned} \inf_q [v^*(p_0, q) - \langle q, y_0 \rangle] &= \\ (17) \quad &= \inf_{x \in X} \inf_q \sup_{y \in Y} [f(x, y) - \langle p_0, x \rangle + \langle q, y - y_0 \rangle]. \end{aligned}$$

For any fixed $x \in X$, we obtain that

$$\begin{aligned} \inf_q \left[\sup_y [f(x, y) + \langle q, y \rangle] - \langle q, y_0 \rangle \right] &= \\ &= \inf_q [g^*(q) - \langle q, y_0 \rangle] = -g^{**}(y_0). \end{aligned}$$

Since g is convex and lower semi-continuous, $-g^{**}(y_0) = f(x_*, y_0)$. Therefore

$$(18) \quad \begin{cases} \inf_{x \in X} \inf_q \sup_y [f(x, y) - \langle p_0, x \rangle + \langle q, y - y_0 \rangle] = \\ = \inf_{x \in X} [f(x, y_0) - \langle p_0, x \rangle]. \end{cases}$$

Therefore (15(ii)) follows from (17) and (18). $\square$

Remark. If $v^*(p, q)$ is replaced by $v^b(p, q)$, the same proof shows that y_0 is a conservative strategy for Yvette if and only if

$$(19) \quad \begin{cases} \text{(i) } y_0 \in \partial v^b(p_0, \cdot)(q_0), \\ \text{(ii) } \inf_{x \in X} [f(x, y_0) - \langle p_0, x \rangle] = \inf_q [v^b(p_0, q) - \langle q, y_0 \rangle]. \end{cases}$$

*6.5.4. Case of finite strategy sets: Matrix games

In the case where the strategy sets

$$(20) \quad X = \{1, \dots, k\} \quad Y = \{1, \dots, l\}$$

are finite, Xavier's loss function $f: X \times Y \rightarrow \mathbf{R}$ is characterized by its values $f(i, j)$ when $1 \leq i \leq k$ and $1 \leq j \leq l$. This is the reason why such games are called "matrix games".

X \ Y		{ 1 2 . . . $\overbrace{j \dots 1}^{\text{Yvette selects columns}}$					
1 ⋮ 2 ⋮ $\vdots$ $\vdots$ $\vdots$ i ⋮ k 1 2 ⋮ i ⋮ k		$f(1, 1)$	$f(1, 2)$	$\dots$	$f(1, j)$	$\dots$	$f(1, l)$
		$f(2, 1)$	$f(2, 2)$	$\dots$	$f(2, j)$	$\dots$	$f(2, l)$
		$\vdots$	$\vdots$	$\dots$	$\vdots$	$\dots$	$\vdots$
		$f(i, 1)$	$f(i, 2)$	$\dots$	$f(i, j)$	$\dots$	$f(i, l)$
		$\vdots$	$\vdots$	$\dots$	$\vdots$	$\dots$	$\vdots$
		$f(k, 1)$	$f(k, 2)$	$\dots$	$f(k, j)$	$\dots$	$f(k, l)$

Xavier selects rows

Once the game is represented by a matrix as above, we identify the i^{th} strategy of Xavier with the i^{th} row of the matrix and the j^{th} strategy of Yvette with the j^{th} column.

In other words, one regards Xavier as selecting rows and Yvette columns. In this case Xavier's loss (or Yvette's utility) is the entry $f(i, j)$ appearing in the corresponding row and column of the matrix.

For Xavier, the worst loss $f^*(i)$ is the maximum loss in the i^{th} row, i.e.

$$(21) \quad f^*(i) = \max_{1 \leq j \leq l} f(i, j)$$

and a conservative strategy is a row which has the smallest worst loss.

$$(22) \quad f^*(\bar{i}) = \min_{1 \leq i \leq k} \max_{1 \leq j \leq l} f(i, j).$$

For Yvette, the situation is symmetric.

The worst gain $f^b(j)$ is the minimum gain in the j^{th} column, i.e.

$$(23) \quad f^b(j) = \min_{1 \leq i \leq k} f(i, j)$$

and a conservative strategy is a column which has the largest worst gain

$$(24) \quad f^b(j) = \max_{1 \leq j \leq l} \min_{1 \leq i \leq k} f(i, j).$$

For instance, consider the game

X \ Y			
	1	2	3
1	-6	2	-3
2	4	-5	-4

The worst losses of Xavier are 2 and 4 respectively and thus, Xavier's conservative strategy is the first row. In this case, Xavier expects Yvette to choose the second column and produce a minimal worst loss of 2. The worst gains for Yvette are respectively -6, -5 and -4 and thus her conservative strategy is the third column. In this case, Yvette expects Xavier to choose the second row and produce a minimal worst loss of 4. The sum of the minimal worst losses is equal to $4 + 2 = 6$. By playing their conservative strategies (first row for Xavier and third column for Yvette), the loss is -3 for Xavier (and +3 for Yvette). Neither player expected such a loss!

Furthermore, if Yvette selects her third column and if we assume that Xavier is informed of this choice, then Xavier would do better selecting his second row (with a loss of -4) instead of the first one. Similarly, if Xavier does choose his conservative strategy, then Yvette would do better playing her second row (with a gain of 2) instead of the third.

It is clear that the conservative strategies $\{1, 3\}$ do not bring satisfaction to the players. The reason is that the game has no value since

$$v^b = -4 < v^* = 2.$$

Now, consider the game

X \ Y	1	2	3
1	-2	-1	-4
2	1	0	-6

Xavier's worst losses being -1 and 1 respectively, $v^* = -1$ and his conservative strategy is given by the first row. In this case, he expects Yvette to choose the second column.

Yvette's worst gains are $-2, -1, -6$. Thus, $v^b = -1$ and her conservative strategy is the second column. In this case, she expects Xavier to choose the first row.

In other words, since the game has a value (equal to -1), the natural expectations of both players are fulfilled and the choices of conservative strategies are consistent. Given that Xavier implements his first row, it is optimal for Yvette to play her second column and conversely, given that Yvette select her second column, it is optimal for Xavier to choose his first row.

Remark. We shall see that the reason why a value does not necessarily exist in matrix game is that the finite strategy sets are "too small". Another way of interpreting this phenomenon is to say that the description of the game is "too poor". Where a saddle-point exists, individual stability can be attained even though the players both choose secretly and independently. Where no saddle-point exists and the game is played only once, the players do not have enough information on the choice of their opponent. But, by "repeating" the game, each player can gather enough information to reach an individually stable situation, even when their choices are made secretly and independently. This is the underlying idea leading to the use of mixed strategies. In this case, we replace the strategy sets $X = \{1, 2, \dots, k\}$ and $Y = \{1, 2, \dots, l\}$ by the mixed strategy sets $\mathcal{M}^k$ and $\mathcal{M}^l$ (see Section 1.3). The loss function f is extended to $\mathcal{M}^k \times \mathcal{M}^l$ by $\tilde{f}$ defined by

$$\tilde{f}(m, n) = \sum_{i=1}^k \sum_{j=1}^l \alpha_i \beta_j f(i, j)$$

where $m = \sum_{i=1}^k \alpha_i \delta(x_i)$ and $n = \sum_{j=1}^l \beta_j \delta(y_j)$.

If $\alpha_i = p_i/p$ where $\sum_{i=1}^k p_i = p$ and $\beta_j = q_j/q$ where $\sum_{j=1}^l q_j = q$, are rational numbers, we interpret m as meaning that Xavier's i th strategy is to be played p_i times out of p ($i = 1, 2, \dots, k$). Similarly, n is interpreted as meaning that Yvette's j th strategy is to be played q_j times out of q ($j = 1, 2, \dots, l$).

CHAPTER 7

TWO-PERSON ZERO-SUM GAMES: EXISTENCE THEOREMS

This chapter deals with the main existence theorems for solutions of two-person zero-sum games. Some of these theorems play a key role not only in this book but in convex and non-convex analysis generally. For instance, many results in *convex analysis* are based on the *minisup theorem*. This states that if a function f satisfies the *topological assumptions*

$$(*) \quad \begin{cases} \text{(i) } \forall y \in Y, x \mapsto f(x, y) \text{ is lower semi-continuous,} \\ \text{(ii) } \exists y_0 \in Y \text{ such that } x \mapsto f(x, y_0) \text{ is lower semi-compact,} \end{cases}$$

together with the *concavity assumption*

$$(**) \quad \forall x \in X, y \mapsto f(x, y) \text{ is concave}$$

and the *convexity assumption*

$$(***) \quad \forall y \in Y, x \mapsto f(x, y) \text{ is convex}$$

then there exists $\bar{x} \in X$ such that

$$\sup_{y \in Y} f(\bar{x}, y) = \sup_{y \in Y} \inf_{x \in X} f(x, y).$$

Many results of *non-convex analysis* and, in particular, various fixed point theorems, can be deduced from the *Ky-Fan theorem*. This states that when $X = Y$, *topological assumptions* (*) and *concavity assumptions* (**) imply the existence of $\bar{x} \in X$ such that

$$\sup_{y \in X} f(\bar{x}, y) \leq \sup_{y \in X} f(y, y).$$

These two results are proved in Section 7.1. In Chapter 13 we shall continue with a more comprehensive study of such results in which the continuity and convexity assumptions are relaxed.

If we assume in addition to the three assumptions above that

$$\begin{cases} \text{(i) } \forall x \in X, y \mapsto f(x, y) \text{ is upper semi-continuous,} \\ \text{(ii) } \exists x_0 \in X \text{ such that } y \mapsto f(x_0, y) \text{ is upper semi-compact,} \end{cases}$$

we may deduce the existence of a saddle point $\{\bar{x}, \bar{y}\}$, i.e.

$$\sup_{y \in Y} f(\bar{x}, y) = f(\bar{x}, \bar{y}) = \inf_{x \in X} f(x, \bar{y}).$$

We have already described examples of finite games which have no saddle-points. But, by embedding $X \times Y$ in the convex set of mixed strategies $\mathcal{M}(X) \times \mathcal{M}(Y)$ and by extending f to $\mathcal{M}(X) \times \mathcal{M}(Y)$ by the function f^A defined by

$$f^A(m, n) = \sum_{i=1}^k \sum_{j=1}^l \alpha^i \beta^j f(x_i, y_j) \quad \text{when } m = \sum_{i=1}^k \alpha^i \delta(x_i) \\ \text{and } n = \sum_{j=1}^l \beta^j \delta(y_j)$$

we obtain the existence of *saddle mixed strategies* since $\mathcal{M}(X)$ and $\mathcal{M}(Y)$ are convex compact subsets and f^A is a bilinear continuous function (Von Neumann theorem).

Hence, even if there is no saddle-point $\{\bar{x}, \bar{y}\} \in X \times Y$ among pure strategies, any (finite) game has *saddle mixed strategies* $\{\bar{m}, \bar{n}\} \in \mathcal{M}(X) \times \mathcal{M}(Y)$. (This result will be extended to compact strategy sets.)

The question arises as to whether this procedure is the only "extension" of games into "playable games", i.e. extended games which always have a saddle point.

This leads us to introduce a concept of game extension due to H. Moulin. The extension is described by the items $\{\mathcal{X}, \mathcal{Y}, i, j, \pi\}$ where i is an injective map from X into $\mathcal{X}$, j is an injective map from Y into $\mathcal{Y}$, and π is a linear operator mapping a bounded loss function on $X \times Y$ onto its *extension* πf on $\mathcal{X} \times \mathcal{Y}$ (which is required to satisfy several reasonable properties).

Besides standard *mixed extension* example (where $\mathcal{X} = \mathcal{M}(X)$, $\mathcal{Y} = \mathcal{M}(Y)$, $\pi f = f^A$), we shall introduce in Section 7.2 a more general class of *extensions without exchange of information*. These have the property that, if the extended game has a value $v(\pi f)$, then this value is the mixed value $v(f^A)$.

Among such extensions, we describe the *sequential extension* where $\mathcal{X} = X^{\mathbb{N}}$ is the set of denumerable sequences of pure strategies, $\mathcal{Y} = Y^{\mathbb{N}}$, and

$$\pi f(\xi, \eta) = \sum_{i,j=1}^{\infty} a_{ij} f(x_i, y_j) \quad \text{whenever } \xi = \{x_i\}_{i \in \mathbb{N}} \text{ and } \eta = \{y_j\}_{j \in \mathbb{N}}$$

The sequence $\{a_{ij}\}$ is assumed to satisfy

$$\forall i, j \in \mathbb{N}, \quad a_{ij} \geq 0 \quad \text{and} \quad \sum_{i,j=1}^{\infty} a_{ij} = 1.$$

We also introduce the concept of extensions *with exchange of information*. The exchange of information is modeled by the use of “decision rules”. Instead of choosing a pure strategy x in X , Xavier is allowed to use a decision rule C in a subset $\mathcal{C}$ of maps from Y into X . Such a decision rule C for Xavier associates with any strategy $y \in Y$ played by Yvette the strategy $x = C(y)$. The rules of the game specify the subset $\mathcal{C}$ of decision rules allocated to Xavier. Also Yvette is allowed to choose a decision rule D in a subset $\mathcal{D}$ of maps from X into Y . Now, if Xavier chooses C in $\mathcal{C}$ and Yvette chooses D in $\mathcal{D}$, the only pure strategies which are *stable* are the fixed points $\{\bar{x}, \bar{y}\}$ of $\{C, D\}$:

$$C(\bar{y}) = \bar{x} \quad \text{and} \quad D(\bar{x}) = \bar{y}.$$

Hence, the exchange of information described by $\mathcal{C} \times \mathcal{D}$ has a sense only if all pairs $\{C, D\} \in \mathcal{C} \times \mathcal{D}$ do have fixed points. If such a fixed point $\{\bar{x}, \bar{y}\}$ is chosen, we assign to Xavier the loss

$$\pi f(C, D) = f(\bar{x}, \bar{y}).$$

Extensions with exchange of information are obtained by taking $\mathcal{X} = \mathcal{C}$, $\mathcal{Y} = \mathcal{D}$ and using the above map π .

Finally, we devote Section 7.3 to the *iterated extensions* which combine sequential extensions and extensions with exchange of information. Let us describe an example. We embed X in the set $\mathcal{C}$ of sequences $C = \{C_i\}$ of non-anticipative decision rules C_i mapping $\prod_{j \leq i-1} Y_j$ into X (where Y_j denotes the strategy set Y used at the j^{th} step).

In the same way, Y is embedded in the set $\mathcal{D}$ of sequences $D = \{D_j\}$ of non-anticipative decision rules D_j mapping $\prod_{i \leq j} X_i$ into Y (where X_i denotes the strategy set X used at the i^{th} step). By choosing $C = \{C_i\}$ and $D = \{D_j\}$, Xavier and Yvette implement the following sequence of pure strategies. At the first step, Xavier plays $x_1 \in X$. Then they play in turns as follows; Yvette plays $y_1 = D_1(x_1)$, Xavier $x_2 = C_2(y_1)$, Yvette $y_2 = D_2(x_1, x_2)$, Xavier $x_3 = C_3(y_1, y_2)$, ..., Yvette $y_p = D_p(x_1, \dots, x_p)$, Xavier $x_{p+1} = C_{p+1}(y_1, \dots, y_p)$, ... and so on.

Then we assign to Xavier the loss $\sum_{i,j=1} a_{ij} f(x_i, y_j)$. This procedure defines the so-called “iterated extension”. The Moulin theorem states that iterated games do have a value.

7.1. The fundamental existence theorems

Let f be Xavier’s loss function, mapping $X \times Y$ into $\mathbf{R}$. Suppose that

- (*)₁ $\left\{ \begin{array}{l} \text{(i) } \forall y \in Y, x \mapsto f(x, y) \text{ is lower semi-continuous,} \\ \text{(ii) } \exists y_0 \in Y \text{ such that } x \mapsto f(x, y_0) \text{ is lower semi-compact.} \end{array} \right.$

Then there exists a *conservative solution* $\bar{x} \in X$, i.e.

$$\sup_{y \in Y} f(x, y) = \inf_{x \in X} \sup_{y \in Y} f(x, y) = v^*.$$

Furthermore, we shall prove that

$$v^* \text{ is equal to } v^\square = \sup_{K \in \mathcal{J}} \inf_{x \in X} \sup_{y \in K} f(x, y)$$

where $\mathcal{J}$ denotes the set of finite subsets of Y . (The inequality $v^\square \leq v^*$ is obvious). We use this result to prove that, under the further assumption,

$$(**)_2 \quad \forall x \in X, \quad y \mapsto f(x, y) \text{ is concave,}$$

we have that

$$v^* = \inf_{C \in \mathcal{C}(Y, X)} \sup_{y \in Y} f(C(y), y)$$

where $\mathcal{C}(Y, X)$ denotes the set of *continuous* decision rules for Xavier. This equality means that, if Yvette is risk-averse (in the sense that $(**)_2$ holds), then Xavier cannot improve his conservative value even if he is informed of the strategy to be played by Yvette. Thus the constant decision rule $\bar{x}$ achieves the minimal worst loss.

In the general case, an “optimal decision rule” $\bar{C}$ is one which satisfies

$$\forall y \in Y, \quad f(\bar{C}(y), y) = \inf_{x \in X} f(x, y).$$

If such an optimal decision rule is continuous, then $\inf_{C \in \mathcal{C}(Y, X)} \sup_{y \in Y} f(C(y), y) = v^*$ and thus $v^* = v^b$. But this latter identity can be obtained under another assumption. If we assume that

$$(**)_1 \quad \forall y \in Y, \quad x \mapsto f(x, y) \text{ is convex}$$

instead of assuming the continuity of $\bar{C}$ we obtain that

$$v^* = \sup_{y \in Y} \inf_{x \in X} f(x, y).$$

We also prove the Lasry theorem, i.e. if

$$(*)_2 \quad \begin{cases} \text{(i) } \forall x \in X, \quad y \mapsto f(x, y) \text{ is upper semi-continuous,} \\ \text{(ii) } \exists x_0 \in X \text{ such that } y \mapsto f(x_0, y) \text{ is upper semi-compact} \end{cases}$$

and $(**)_1$ holds then

$$v^b = \inf_{C \in \mathcal{C}(Y, X)} \sup_{y \in Y} f(C(y), y).$$

Hence assumptions $(*)_1$, $(*)_2$, $(**)_1$, and $(**)_2$ imply the existence of a saddle point. We can also prove the existence of a saddle point when the convexity assumption $(**)_1$ is replaced by the assumption that the correspondence C which associates with any $y \in Y$ the minimal subset $C(y) = \{\bar{x} \in X \text{ such that } f(\bar{x}, y) = \inf_{x \in X} f(x, y)\}$ is lower semi-continuous.

We shall deduce most of the existence theorems of the following chapters from the *Ky-Fan theorem*. This asserts the existence of $\bar{x} \in X$ satisfying $\sup_{y \in Y} f(\bar{x}, y) \leq 0$ provided that $f: X \times X \rightarrow \mathbf{R}$ satisfies assumptions $(*)_1$, $(**)_2$

and

$$\sup_{y \in X} f(y, y) = 0$$

7.1.1. Existence of conservative solutions

Consider the Xavier's loss function $f: X \times Y \rightarrow \mathbf{R}$. A conservative solution exists when the functions $x \mapsto f(x, y)$ are lower semi-continuous and some function $x \mapsto f(x, y_0)$ is lower semi-compact, since in this case the worst loss f^* is lower semi-continuous and lower semi-compact. In fact, we shall prove a stronger result.

We assume that X is a subset of a topological space U and denote by $f_X(x, y)$ the extension of f to U defined by

$$(1) \quad f_X(x, y) = \begin{cases} f(x, y) & \text{if } x \in X, \\ +\infty & \text{if not.} \end{cases}$$

Also, we introduce the following "value"

$$(2) \quad v^\square = \sup_{K \in \mathcal{J}} \inf_{x \in X} \max_{y \in K} f(x, y) \quad (\text{read } v \text{ "natural"})$$

where $\mathcal{J}$ denotes the family of *finite subsets*

$$K = \{y_1, \dots, y_n\} \text{ of } Y.$$

It is clear that

$$(3) \quad v^b \leq v^\square \leq v^*.$$

Because each $y \in Y$ forms a finite subset $K = \{y\}$, we obviously have $v^b = \sup_{\{y\}} \inf_{x \in X} \max_{y \in \{y\}} f(x, y) \leq v^\square$. On the other hand, since $\inf_{x \in X} \sup_{y \in K} f(x, y) \leq v^*$, inequality $v^\square \leq v^*$ holds.

Theorem 1. *Suppose that*

$$(4) \quad \forall y \in Y, \quad x \mapsto f_X(x, y) \text{ is lower semi-continuous on } U$$

(assumption of continuity) and that

$$(5) \quad \exists y_0 \in Y \text{ such that } x \mapsto f(x, y_0) \text{ is lower semi-compact on } X$$

(assumption of compactness). Then there exists a conservative solution $\bar{x}$ and

$$v^* = v^\square.$$

In other words,

$$(6) \quad \sup_{y \in Y} f(\bar{x}, y) = v^\square.$$

Proof. Let $S(y) = \{x \in X \text{ such that } f(x, y) \leq v^\square\}$. Then the set of solutions of (6) is $S = \bigcap_{y \in Y} S(y)$. The subsets $S(y)$ are closed by (4) and $S(y_0)$ is compact by (5). Therefore S is non-empty provided that the finite intersection property holds, i.e. $S_K = S(y_0) \cap \bigcap_{i=1}^n S(y_i) \neq \emptyset$ for all finite subsets $K = \{y_1, \dots, y_n\}$ of Y .

Since $S(y_0)$ is compact and $x \mapsto \max_{i=0, \dots, n} f(x, y_i)$ is lower semi-continuous, there exists $\bar{x}$ which minimizes this function. Therefore

$$(7) \quad \begin{aligned} \max_{i=0, \dots, n} f(\bar{x}, y^i) &= \inf_{x \in X} \max_{i=0, \dots, n} f(x, y^i) \\ &\leq \sup_{K \in \mathcal{J}} \inf_{x \in X} \sup_{y \in K} f(x, y) = v^\square. \end{aligned}$$

This implies that $\bar{x}$ belongs to S_K .

We have proved that S is non-empty, i.e. that there exists a solution $\bar{x}$ satisfying

$$(8) \quad \sup_{y \in Y} f(\bar{x}, y) \leq v^\square.$$

This implies that $v^* \leq \sup_{y \in Y} f(\bar{x}, y) \leq v^\square \leq v^*$,

Thus, (6) holds and $v^\square = v^*$. $\square$

Remark. More generally, we can replace assumption (5) by the weaker assumption

$$(9) \quad \exists y_1, \dots, y_n \in Y \text{ such that } x \mapsto \max_{i=1, \dots, n} f(x, y_i) \text{ is lower semi-compact on } X.$$

Together with assumption (4) this implies that the subset $S_0 = \bigcap_{i=1}^n S(y_i)$ is compact. It is also non-empty. The proof of Theorem 1 with $S(y_0)$ replaced by S_0 then implies the following result.

Proposition 1. *Suppose that assumptions (4) and (9) hold. Then there exists a conservative solution $\bar{x}$ and $v^* = v^\square$.*

This result is useful in the following example.

Example. Consider the case when

$$(10) \quad \begin{cases} \text{(i) } X = \prod_{i=1}^n X^i \text{ where } X^i \subset U^i, \\ \text{(ii) } f(x, y) = \sum_{i=1}^n f_i(x_i, y) \text{ where } f_i \text{ is defined on } X^i \times Y. \end{cases}$$

Instead of assuming that there exists $y_0 \in Y$ such that the function $x \mapsto f(x, y_0)$ is lower semi-compact, we shall assume that $\forall i = 1, \dots, n$ there exists $y_i \in Y$ such that

$$(11) \quad \begin{cases} \text{(a) } f_j^b(y_i) = \inf_{x_j \in X^j} f_j(x_j, y_i) < -\infty, \forall j \neq i, \\ \text{(b) } x_i \mapsto f_i(x_i, y_i) \text{ is lower semi-compact.} \end{cases}$$

Write

$$(12) \quad f_{i, X^i}(x^i, y) = \begin{cases} f_i(x^i, y) & \text{if } x^i \in X^i, \\ +\infty & \text{if } x^i \notin X^i. \end{cases}$$

Proposition 2. *Given (10), the compactness assumption (11) and the continuity assumption*

$$(13) \quad \forall y \in Y, \forall i = 1, \dots, n, \quad x^i \mapsto f_{i, X^i}(x^i, y) \text{ is lower semi-continuous on } U^i,$$

then there exists a conservative solution $\bar{x} = \{\bar{x}^1, \dots, \bar{x}^n\}$ and $v^\square = v^$.*

Proof. We have to show that assumption (9) of Proposition 1 is satisfied.

Taking $K = \{y_1, \dots, y_n\}$, we check that $x \mapsto \max_{1 \leq i \leq n} f(x, y_i)$ is lower semi-compact. If $x = (x_1, \dots, x_n)$ satisfies $\max_{1 \leq i \leq n} f(x, y_i) \leq \lambda$, then, for any i ,

$$f_i(x_i, y_i) \leq \lambda - \sum_{j \neq i} f_j(x_j, y_i) \leq \lambda - \sum_{j \neq i} f_j^b(y_i)$$

Therefore, for any i , x_i belongs to a relatively compact subset. $\square$

Remark. Examples of lower semi-compact functions are given in Section 3.1

Remark. We shall prove that the conclusion (6) of Theorem 1 holds under weaker conditions (see Theorem 13.1.1).

7.1.2. Decision rules

Suppose now that Xavier is *informed* of the choice of Yvette's strategies. Instead of choosing "pure" strategies $x \in X$, Xavier can choose "*Decision rules*" $C : Y \rightarrow X$ mapping Y into X .

The worst loss associated with such a decision rule C is defined by

$$(14) \quad f^*(C) = \sup_{y \in Y} f(C(y), y).$$

Therefore, Xavier will choose from a subset of feasible decision rules a decision rule $\bar{C}$ which minimizes $C \mapsto f^*(C)$.

Our results require that X and Y are topological spaces and that the set of feasible decision rules is the set $\mathcal{C}(Y, X)$ of *continuous decision rules*.

Since any pure strategy $x \in X$ can be identified with the constant (continuous) decision rule $C_x : y \mapsto C_x(y) = x$ it is clear that

$$(15) \quad v_1 = \inf_{C \in \mathcal{C}(Y, X)} f^*(C) \leq \inf_{x \in X} \sup_{y \in Y} f(x, y) = v^*.$$

This means that Xavier's minimal worst loss is *better* when he is informed of Yvette's strategy choice.

However, we shall prove that this information is redundant (in the sense that equality holds in (15)) when the loss function f is concave (or, even, quasi-concave) with respect to y .

Remark. In the case of zero-sum games, the concavity of f with respect to y describes "cautious behavior by Yvette" (see Section 1.3.7).

Before stating and proving the above theorem, we note that, as the topology of Y is strengthened, the set $\mathcal{C}(Y, X)$ of continuous decision rules for Xavier increases. Hence the minimal worst loss $\inf_{C \in \mathcal{C}(Y, X)} f^*(C)$ should decrease. The identity $v^* = \inf_{C \in \mathcal{C}(Y, X)} f^*(C)$ is therefore most powerful when the topology of Y is strongest.

We exploit the comment above by introducing "finite topology" on a convex set Y . This topology is stronger than any vector space topology. Any affine map from Y to a vector space U is continuous provided that both Y and U are equipped with the finite topology.

7.1.3. Finite topology on convex subsets

Suppose that Y is a convex subset of a vector space V . Associate with any finite subset $K = \{y_1, \dots, y_n\}$ of Y the affine map β_K from $\mathcal{M}^n$ (see Section 1.3.4)

into Y defined by

$$(16) \quad \lambda \in \mathcal{M}^n \mapsto \beta_K(\lambda) = \sum_{i=1}^n \lambda^i y_i.$$

Definition 1. The “*finite topology*” defined on a convex subset Y is the strongest topology for which the maps β_K are continuous when K ranges over the family $\mathcal{S}$ of finite subsets of Y .

A map A from Y into a topological space Z is continuous when Y is supplied with the finite topology if and only if the maps

$$(17) \quad A\beta_K: \mathcal{M}^n \mapsto Z \quad \text{are continuous for all } K \in \mathcal{S}.$$

Proposition 3. *The finite topology of Y is stronger than any vector space topology. Any affine operator L from Y into a vector space U is continuous when both Y and U are supplied with the finite topology.*

Proof. Suppose that Z is the vector space V supplied with a vector space topology and that A is the canonical injection from Y into Z . We have to prove that A is continuous. This follows from the fact that, for all $K = \{y_1, \dots, y_n\}$, the map $\lambda \in \mathcal{M}^n \mapsto A\beta_K(\lambda) = \sum_{i=1}^n \lambda^i y_i$ is continuous. We take now $Z = U$ and $A = L$, where L is an affine operator. We have to prove that for any $K = \{y_1, \dots, y_n\}$, the map $L\beta_K: \lambda \in \mathcal{M}^n \mapsto L\beta_K(\lambda) = L(\sum_{i=1}^n \lambda^i y_i) = \sum_{i=1}^n \lambda^i L(y_i) = \beta_{L(K)}(\lambda) \in U$ is continuous. But $L\beta_K = \beta_{L(K)}$ is continuous by the very definition of the finite topology on U . $\square$

7.1.4. Existence of an optimal decision rule

Theorem 2. *Let X be a subset of a topological space U and let Y be a convex subset of a vector space V , supplied with the finite topology.*

Given the continuity and compactness assumptions (4) and (5), and the concavity assumption

$$(18) \quad \forall x \in X, \quad y \mapsto f(y, x) \text{ is quasi-concave,}$$

there exist conservative strategies $\bar{x}$ which, viewed as decision rules, minimize $C \mapsto f^(C)$ on $\mathcal{C}(Y, X)$, i.e.*

$$(19) \quad \sup_{y \in Y} f(\bar{x}, y) = \inf_{C \in \mathcal{C}(Y, X)} \sup_{y \in Y} f(C(y), y) = v^*.$$

Proof. By Theorem 1, the proof reduces to showing that

$$(20) \quad v^\square \leq \inf_{C \in \mathcal{C}(Y, X)} \sup_{y \in Y} f(C(y), y),$$

i.e. for any $K = \{y_1, \dots, y_n\}$ and any $C \in \mathcal{C}(Y, X)$,

$$(21) \quad \inf_{x \in X} \max_{i=1, \dots, n} f(x, y_i) \leq \sup_{y \in Y} f(C(y), y) = c.$$

In fact we shall prove that there exists $\tilde{x} \in X$ such that

$$(22) \quad \max_{i=1, \dots, n} f(\tilde{x}, y_i) \leq c.$$

For this purpose, we introduce the function $\varphi : \mathcal{M}^n \times Y \rightarrow \mathbf{R}$ defined by

$$(23) \quad \varphi(\lambda, y) = f\left(C\left(\sum_{i=1}^n \lambda^i y_i\right), y\right)$$

and the sets

$$(24) \quad F_i = \{\lambda \in \mathcal{M}^n \text{ such that } \varphi(\lambda, y_i) \leq c\}.$$

It is enough to prove that there exists $\tilde{\lambda} \in \bigcap_{i=1}^n F_i$. If so, $\tilde{x} = C(\sum_{i=1}^n \tilde{\lambda}^i y_i)$ clearly satisfies (22).

To prove that the n subsets F_i of $\mathcal{M}^n$ have a non-empty intersection, we use the fundamental *Knaster—Kuratowski—Mazurkiewicz lemma* (see Appendix B).

We have to check that

$$(25) \quad \begin{cases} \text{(i) the } n \text{ subsets } F_i \text{ of } \mathcal{M}^n \text{ are closed,} \\ \text{(ii) } \forall \lambda \in \mathcal{M}^n, \lambda \in \bigcup_{i \in A_\lambda} F_i, \end{cases}$$

where $A_\lambda = \{i \text{ such that } \lambda^i > 0\}$.

The subsets F_i are closed because $\lambda \in \mathcal{M}^n \mapsto C(\sum_{i=1}^n \lambda^i y_i) \in X$ is continuous (for Y is supplied with the finite topology) and $x \mapsto f_\lambda(x, y)$ is lower semi-continuous.

It remains to check that property (25(ii)) holds.

If not, there exists $\lambda_0 \in \mathcal{M}^n$ such that, for any $i \in A_0 = \{i \text{ such that } \lambda_0^i > 0\}$,

$$(26) \quad \varphi(\lambda_0, y_i) = f\left(C\left(\sum_{j=1}^n \lambda_0^j y_j\right), y_i\right) > c.$$

This means that, for any $i \in A_0$, y_i belongs to the open upper section of the function $y \mapsto f(C(\sum_{j=1}^n \lambda_0^j y_j), y)$, which is convex by assumption (18). Therefore $\sum_{j=1}^n \lambda_0^j y_j$ belongs to such a section. Hence $f(C(\sum_{j=1}^n \lambda_0^j y_j), \sum_{j=1}^n \lambda_0^j y_j) > c$. But this contradicts the definition of $c = \sup_{y \in Y} f(C(y), y)$. $\square$

7.1.5. The Ky-Fan inequality

We deduce from Theorem 2 the Ky-Fan inequality. This will play a fundamental role as a tool in proving most of the forthcoming existence theorems.

Theorem 3 (Ky Fan). *Suppose that*

(27) X *is a convex compact subset*

and that the function $f : X \times X \rightarrow \mathbf{R}$ satisfies

$$(28) \quad \begin{cases} \text{(i)} \ \forall y \in X, x \mapsto f(x, y) \text{ is lower semi-continuous,} \\ \text{(ii)} \ \forall x \in X, y \mapsto f(x, y) \text{ is quasi-concave,} \\ \text{(iii)} \ \sup_{y \in X} f(y, y) \leq 0. \end{cases}$$

Then there exists $\bar{x} \in X$ satisfying

$$(29) \quad \sup_{y \in X} f(\bar{x}, y) \leq 0.$$

Proof. Since the identity map I is continuous from X supplied with the finite topology into X (which is compact for a vector space topology), we may deduce from Theorem 2 the existence of $\bar{x} \in X$ satisfying

$$(30) \quad \begin{cases} \sup_{y \in X} f(\bar{x}, y) = \inf_{C \in \mathcal{O}(X, X)} \sup_{y \in X} f(C(y), y) \\ \leq \sup_{y \in X} f(I(y), y) = \sup_{y \in X} f(y, y) \leq 0. \end{cases}$$

Remark. This theorem will be extended in Section 13.2 by replacing the continuity assumption (28(i)) by an assumption of pseudomonotonicity (see Theorem 13.2.1).

7.1.6. The Lasry theorem

Theorem 4 (Lasry). *Suppose that the continuity assumption (4), the compactness assumption (5) and the concavity assumption (18) of Theorem 2 are satisfied. Then*

$$(31) \quad \inf_{C \in \mathcal{O}(Y, X)} \sup_{y \in Y} f(C(y), y) = \sup_{D \in \mathcal{O}(X, Y)} \inf_{x \in X} f(x, D(x)).$$

Proof. We know that $\inf_{C \in \mathcal{O}(X, Y)} \sup_{y \in Y} f(C(y), y) = v^*$ by Theorem 2. Also $\sup_{D \in \mathcal{O}(X, Y)} \inf_{x \in X} f(x, D(x)) \leq v^*$ since $\inf_{x \in X} f(x, D(x)) \leq v^*$ for any map D . It remains to prove that

$$(32) \quad v^* \leq \sup_{D \in \mathcal{O}(X, Y)} \inf_{x \in X} f(x, D(x))$$

Let $\varepsilon > 0$ be fixed.

In the first place, there exists a map $\tilde{D} \in \mathcal{F}(X, Y)$ satisfying

$$(33) \quad \forall u \in X, f^*(u) = \sup_{y \in Y} f(u, y) \leq f(u, \tilde{D}(u)) + \frac{1}{2} \varepsilon.$$

Secondly, since f is lower semi-continuous with respect to x , there exist open neighborhoods $N(u)$ of any $u \in X$ such that

$$(34) \quad \forall x \in N(u), \quad f(u, \tilde{D}(u)) \leq f(x, \tilde{D}(u)) + \frac{1}{2} \varepsilon.$$

We introduce the subset

$$(35) \quad X_0 = \{x \in X \text{ such that } f(x, y^0) \leq v^* + \varepsilon\}$$

which is compact since it is a lower section of the lower semi-compact function $x \mapsto f(x, y^0)$.

Therefore, X_0 is covered by a finite number of open subsets $V_i = N(u_i)$ ($i = 1, \dots, n$). If we set $V_0 = \mathbb{C}_X X_0$ we deduce that

$$(36) \quad X \subset \bigcup_{i=0}^n V_i.$$

Let $\{\beta_i\}_{i=0, \dots, n}$ be a continuous partition of unity subordinate to this finite covering of X . This has the following properties.

If $\beta_0(x) > 0$, then $x \in V_0$. Thus $v^* < f(x, y^0) - \varepsilon < f(x, y^0) + \varepsilon$.

If $\beta_i(x) > 0$, then $x \in N(u_i)$. Thus $v^* \leq f^*(u_i) \leq f(u_i, \tilde{D}(u_i)) + \frac{1}{2} \varepsilon \leq f(x, \tilde{D}(u_i)) + \varepsilon$ by (33) and (34).

In terms of the set

$$(37) \quad S = \{y \in Y \text{ such that } f(x, y) \geq v^* - \varepsilon\}$$

these properties reduce to the following.

$$\begin{aligned} y^0 \in S & \text{ when } \beta_0(x) > 0 \text{ and } \tilde{D}(u_i) \in S \\ & \text{when } \beta_i(x) > 0. \end{aligned}$$

But S is convex since f is assumed to be quasi-concave with respect to y . Since $\beta_i(x) \geq 0$ and $\sum_{i=0}^n \beta_i(x) = 1$, we deduce that

$$(38) \quad D(x) = \beta_0(x)y^0 + \sum_{i=1}^n \beta_i(x) \tilde{D}(u_i) \in S$$

i.e. that

$$(39) \quad \forall x \in X, \quad v^* - \varepsilon \leq f(x, D(x)).$$

Because the functions β_i are continuous, the map $D : x \mapsto D(x)$ is continuous. We have proved that for all $\varepsilon > 0$,

$$(40) \quad v^* - \varepsilon \leq \inf_{x \in X} f(x, D(x)) \leq \sup_{D \in \mathcal{D}(X, Y)} \inf_{x \in X} f(x, D(x)). \quad \square$$

7.1.7. The minisup theorem

If we could find among the optimal decision rules C satisfying

$$(41) \quad f(\bar{C}(y), y) = \inf_{x \in X} f(x, y) \quad \text{for all } y \in Y$$

one which is continuous on Y supplied with the finite topology, we could deduce the existence of $x \in X$ such that

$$(42) \quad v^* = \sup_{y \in Y} f(x, y) \leq \sup_{y \in Y} f(\bar{C}(y), y) = v^b$$

i.e. the existence of a minisup.

In fact this turns out not to be a good approach for proving the existence of a minisup. Instead, we shall proceed on the assumption that f is convex with respect to x and concave with respect to y .

Theorem 5. *Let X and Y be convex subsets of topological vector spaces.*

Suppose that the continuity and compactness assumptions (4) and (5) and the following convexity and concavity assumptions

$$(43) \quad \begin{cases} \text{(i) } \forall x \in X, y \mapsto f(x, y) \text{ is concave,} \\ \text{(ii) } \forall y \in Y, x \mapsto f(x, y) \text{ is convex} \end{cases}$$

hold. Then there exists a minisup $\bar{x}$.

Proof. By theorem 1, we have only to prove that $v^* \leq v^b$ i.e. that for any $K = \{y_1, \dots, y_n\}$,

$$(44) \quad v_K^* = \inf_{x \in X} \max_{i=1, \dots, n} f(x, y_i) \leq v^b.$$

We can write

$$(45) \quad v_K^* = \inf_{x \in X} \sup_{\lambda \in \mathcal{M}^n} \sum_{i=1}^n \lambda^i f(x, y_i)$$

and the concavity of f with respect to y implies that

$$(46) \quad v_K^b = \sup_{\lambda \in \mathcal{M}^n} \inf_{x \in X} \sum_{i=1}^n \lambda^i f(x, y_i) \leq v^b.$$

(Because $\sum_{i=1}^n \lambda^i f(x, y_i) \leq f(x, \sum_{i=1}^n \lambda^i y_i)$ and hence $\inf_{x \in X} \sum_{i=1}^n \lambda^i f(x, y_i) \leq \inf_{x \in X} f(x, \sum_{i=1}^n \lambda^i y_i) \leq v^b$.)

It is therefore enough to prove that

$$(47) \quad v_K^* = \inf_{x \in X} \sup_{i=1, \dots, n} f(x, y_i) \leq \alpha \quad \text{when } \alpha > v_K^b.$$

[Let α converge to v_K^b , in these inequalities. Then $v_K^* \leq v_K^b$. But $v_K^b \leq v^b$ and (44) follows.]

In fact, we prove the following stronger statement.

$$(48) \quad \exists x_\alpha \in X \text{ such that } \sup_{1 \leq i \leq n} f(x_\alpha, y_i) \leq \alpha \quad \text{when } \alpha > v_K^b$$

For this purpose, it is convenient to denote by F the map from X into $\mathbf{R}^n$ defined by

$$(49) \quad F(x) = \{f(x, y_1), \dots, f(x, y_i), \dots, f(x, y_n)\} \in \mathbf{R}^n$$

and to set

$$(50) \quad \begin{cases} \text{(i) } F_+(X) = F(X) + \mathbf{R}_+^n, \\ \text{(ii) } a = \{\alpha, \dots, \alpha\}. \end{cases}$$

It is clear that (48) means that

$$(51) \quad a \in F_+(X) \quad \text{when } \alpha > v_K^b.$$

(This is because $a \in F_+(X)$ is equivalent to the existence of $x_\alpha \in X$ and $c \in \mathbf{R}_+^n$ such that $a = F(x_\alpha) + c$, i.e. the existence of $x_\alpha \in X$ such that $(\forall i, \alpha) \geq F_i(x_\alpha) = f(x_\alpha, y_i)$.)

Now, in order to prove (51), we shall show that

$$(52) \quad \text{if } a \notin F_+(X), \text{ then } \alpha \leq v_K^b.$$

For this purpose we use the separation theorem for a finite dimensional space.

Convexity assumption (43(ii)) implies that $F_+(X)$ is convex (see Proposition 1.3.10). Therefore, if $a \notin F_+(X)$, there exists $\lambda \in \mathbf{R}^{n*}$ ($\lambda \neq 0$) such that

$$(53) \quad \begin{aligned} \langle \lambda, a \rangle &= \alpha \sum_{i=1}^n \lambda^i \leq \inf_{c \in F_+(X)} \langle \lambda, c \rangle \\ &= \sigma^b(F_+(X); \lambda). \end{aligned}$$

This implies that $\lambda \in \mathbf{R}_+^n$ and that $\sigma^b(F_+(X); \lambda) = \sigma^b(F(X); \lambda)$. Therefore, dividing by $\sum_{i=1}^n \lambda^i > 0$ and writing $\bar{\lambda} = \lambda / \sum_{i=1}^n \lambda_i$ we obtain that

$$(54) \quad \alpha \leq \sigma^b(F(X); \bar{\lambda}) = \inf_{x \in X} \sum_{i=1}^n \bar{\lambda}^i f(x_i, y) \leq v_K^b. \quad \square$$

7.1.8. The Nikaido theorem

We now state a corollary of Theorem 5 which will often be useful.

Theorem 6 (Nikaido). *Suppose that*

$$(55) \quad \begin{cases} \text{(i) } X \text{ is a convex compact subset,} \\ \text{(ii) } \forall y \in Y, x \mapsto f(x, y) \text{ is convex and lower semi-continuous} \end{cases}$$

and that

$$(56) \quad \begin{cases} \text{(i) } Y \text{ is a convex subset,} \\ \text{(ii) } \forall x \in X, y \mapsto f(x, y) \text{ is concave.} \end{cases}$$

Then there exists a minisup $\bar{x} \in X$.

7.1.9. Existence of saddle points

Applying the Nikaido theorem to the loss functions f and $-f$ for Xavier and Yvette, we obtain the existence of a saddle-point under convexity assumptions. Theorem 2 and Theorem 4 applied to $-f$ also yield the existence of a saddle-point under assumptions of “quasi-convexity” only.

Theorem 7 (Sion). *Suppose that*

$$(57) \quad \begin{cases} \text{(i) } X \text{ and } Y \text{ are convex compact subsets,} \\ \text{(ii) } \forall y \in Y, x \mapsto f(x, y) \text{ is quasi-convex and lower semi-continuous,} \\ \text{(iii) } \forall x \in X, y \mapsto f(x, y) \text{ is quasi-concave and upper semi-continuous} \end{cases}$$

Then there exists a saddle-point $\{\bar{x}, \bar{y}\}$.

Remark. We can replace the assumption that X and Y are compact by the assumption that $x \mapsto f(x, y_0)$ is lower semi-compact and that $y \mapsto f(x_0, y)$ is upper semi-compact for some $\{x_0, y_0\}$.

*7.1.10. Another existence theorem for saddle points

We can replace the assumption of convexity of f with respect to x (i.e. of cautions behavior by Xavier) in the theorem above by the following regularity assumption.

$$(58) \quad \begin{cases} \text{The minimal decision correspondence } y \mapsto D^b(y) \\ = \{x \in X \text{ which minimizes } f(x, y)\} \text{ is lower semi-continuous.} \end{cases}$$

(Such a regularity assumption may be regarded as describing “regular” behavior by Xavier.)

Theorem 8. *Let Y be a convex subset of a locally convex vector space V . Suppose that*

$$(59) \quad \begin{cases} \text{(i) } \forall x \in X, y \mapsto f(x, y) \text{ is concave,} \\ \text{(ii) } \forall x \in X, y \mapsto f_Y(x, y) \text{ is upper semi-continuous,} \\ \text{(iii) } \exists x^0 \in X \text{ such that } y \mapsto f(x^0, y) \text{ is upper semi-compact} \end{cases}$$

and that X is a topological space for which

$$(60) \quad \begin{cases} \text{(i) } \forall y \in Y, x \mapsto f(x, y) \text{ is lower semi-continuous,} \\ \text{(ii) the minimal decision correspondence } D^b \text{ is lower semi-continuous from } Y \text{ supplied with the finite topology into } X. \end{cases}$$

Then there exists a saddle point.

Proof. Assumptions (59(ii) and (iii)) imply that there exists $\bar{y} \in Y$ satisfying

$$(61) \quad f^b(\bar{y}) = \max_{y \in Y} f^b(y) = \max_{y \in Y} \inf_{x \in X} f(x, y).$$

Let $\bar{x} \in D^b(\bar{y})$ be fixed and let y be any element of Y . Since $(1-\lambda)\bar{y} + \lambda y$ converges to $\bar{y}$ when λ converges to 0, and D^b is lower semi-continuous on Y supplied with the finite topology, there exists $\bar{x}_\lambda \in D^b((1-\lambda)\bar{y} + \lambda y)$ such that $\bar{x}_\lambda$ converges to $\bar{x}$ when λ converges to 0.

We now use the concavity of f with respect to y to obtain the following estimates. If $\lambda \in]0, 1[$,

$$\begin{aligned} (1-\lambda)f^b(\bar{y}) + \lambda f(\bar{x}_\lambda, y) &\leq \\ &\leq (1-\lambda)f(\bar{x}_\lambda, \bar{y}) + \lambda f(\bar{x}_\lambda, y) \\ &\leq f(\bar{x}_\lambda, (1-\lambda)\bar{y} + \lambda y) = f^b((1-\lambda)\bar{y} + \lambda y) \leq f^b(\bar{y}). \end{aligned}$$

Therefore, for any $\lambda > 0$, $f(\bar{x}_\lambda, y) \leq f^b(\bar{y})$.

Since f is lower semi-continuous with respect to x , we deduce that for any $y \in Y$,

$$(62) \quad f(\bar{x}, y) \leq \liminf_{\lambda \rightarrow 0} f(\bar{x}_\lambda, y) \leq f^b(\bar{y}).$$

Hence, $\sup_{y \in Y} f^b(\bar{x}, y) \leq f^b(\bar{y})$. This implies that $\{\bar{x}, \bar{y}\}$ is a saddle point for all $\bar{x} \in D^b(\bar{y})$. $\square$

*7.2. Extension of games without and with exchange of informations

Let X and Y be two strategy sets. An “extension” $\{\mathcal{X}, \mathcal{Y}, i, j, \pi\}$ of a game defined on $X \times Y$ is specified by an injective map i from X into $\mathcal{X}$, an injective map j from Y into $\mathcal{Y}$, a linear operator π mapping bounded loss functions f defined on $X \times Y$ onto bounded functions πf on $\mathcal{X} \times \mathcal{Y}$ satisfying

$$\pi f(i(x), j(y)) = f(x, y)$$

and other requirements. We shall investigate the extensions which are “playable”, i.e. such that, for some class of loss functions f on $X \times Y$, the extended game

π on $\mathcal{X} \times \mathcal{Y}$ has a value. The first example is the “mixed extension” of a finite game. We take $\mathcal{X} = \mathcal{M}(X)$, $\mathcal{Y} = \mathcal{M}(Y)$ to be the convex compact subsets of mixed strategies, $i = \delta_X$ and $j = \delta_Y$ to be Dirac operator and π to be defined by

$$\pi_0 f(m, n) = \langle m \otimes n, f \rangle.$$

Then Theorem 1.7 implies the existence of a saddle pair $\{\bar{m}, \bar{n}\}$ of mixed strategies (Von Neumann’s theorem) for every loss function f .

The second example is given by the “sequential extensions”, in which we take $\mathcal{X} = X^N$ and $\mathcal{Y} = Y^N$ to be the subsets of sequences of strategies, and π to be defined by

$$\pi_a f(\xi, \eta) = \sum_{i,j=1}^{\infty} a_{ij} f(x_i, y_j)$$

where $\forall i, j \in \mathbf{N}$, $a_{ij} \geq 0$ and $\sum_{i,j=1}^{\infty} a_{ij} = 1$. We prove for instance that such an extension is playable when X and Y contain p and q strategies respectively and $a_{ij} = (1/pq) ((p-1)/p)^{i-1} ((q-1)/q)^{j-1}$. Furthermore, the value of the extended game is equal to the value of its mixed extension.

This latter property is shared by all “extensions without exchange of information” (defined in Section 7.2.3).

We also define “extensions with exchange of information” in the following way. We introduce pairs $\mathcal{C} \subset \mathcal{F}(Y, X)$ and $\mathcal{D} \subset \mathcal{F}(X, Y)$ of subsets of decision rules for Xavier and Yvette respectively, which are *consistent* in the sense that, for any $C \in \mathcal{C}$ and $D \in \mathcal{D}$, we can find a fixed point $\{\bar{x}, \bar{y}\}$ of the map $\{x, y\} \rightarrow \{C(y), D(x)\}$. By taking $\mathcal{X} = \mathcal{C}$, $\mathcal{Y} = \mathcal{D}$ and π to be defined by $\pi f(C, D) = f(\bar{x}, \bar{y})$, we obtain the so-called extensions with exchange of information.

For instance, the consistent pair in which $\mathcal{C} = \mathcal{F}(Y, X)$ is the set of *all* decision rules for Xavier (who thus gets all the information) and $\mathcal{D} = Y$ is the set of constant decision rules for Yvette (who has no information whatsoever) defines a *playable extension*. The value of the extended game is Yvette’s conservative value $\sup_{y \in Y} \inf_{x \in X} f(x, y)$ for the initial game. The Lasry theorem implies the same result when we replace $\mathcal{C} = \mathcal{F}(Y, X)$ by the set $\mathcal{C} = \mathcal{C}(Y, X)$ of continuous decision rules, if we assume that f is convex respect to x and upper semi-continuous and semi-compact with respect to y .

We shall devote the next section to the fundamental example of *iterated extensions*.

7.2.1. Definition of extensions of games

Let us consider a two-person zero sum game defined by Xavier’s loss function $f: X \times Y \rightarrow \mathbf{R}$.

Although saddle-points may not exist for the strategy sets X and Y , they may exist for larger strategy sets $\mathcal{X}$ and $\mathcal{Y}$ with the loss functions extended to $\mathcal{X} \times \mathcal{Y}$.

For instance, we can embed X in the set of mixed strategies, the set of sequences of strategies, a subset of decision rules or a subset of sequences of decision rules. It is worth introducing a general definition before studying specific examples. Let $\mathcal{U}(X \times Y)$ and $\mathcal{U}(\mathcal{X} \times \mathcal{Y})$, be the Banach spaces of bounded loss functions defined on $X \times Y$ and $\mathcal{X} \times \mathcal{Y}$ respectively (see Section 3.1.6).

Definition 1. An extension $\{\mathcal{X}, \mathcal{Y}, i, j, \pi\}$ of the games defined on $X \times Y$ is defined by the five following items:

- $$(1) \quad \left\{ \begin{array}{l} \text{(i) and extended strategy set } \mathcal{X} \text{ for Xavier,} \\ \text{(ii) an extended strategy set } \mathcal{Y} \text{ for Yvette,} \\ \text{(iii) an injective map } i \text{ imbedding } X \text{ in } \mathcal{X}, \\ \text{(iv) an injective map } j \text{ imbedding } Y \text{ in } \mathcal{Y}, \\ \text{(v) a linear operator } \pi \text{ mapping loss functions } f \in \mathcal{U}(X \times Y) \\ \text{onto functions } \pi f \in \mathcal{U}(\mathcal{X} \times \mathcal{Y}). \end{array} \right.$$

such that

- $$(2) \quad \left\{ \begin{array}{l} \text{(i) } \pi f(i(x), j(y)) = f(x, y), \\ \text{(ii) } \pi \text{ is positive and } \pi \theta = \bar{\theta} \text{ where } \theta : X \times Y \rightarrow \mathbf{R} \text{ and } \bar{\theta} : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbf{R} \\ \text{denote the constant functions equal to 1,} \\ \text{(iii) } \forall f, \forall x \in X, \sup_{y \in Y} f(x, y) \geq \sup_{\eta \in \mathcal{Y}} \pi f(i(x), \eta), \\ \text{(iv) } \forall f, \forall y \in Y, \inf_{x \in X} f(x, y) \leq \inf_{\xi \in \mathcal{X}} \pi f(\xi, j(y)). \end{array} \right.$$

Remark. In fact condition (2(i)) follows from conditions (2(ii), (iii) and (iv)).

We shall say that π is the “*extension operator*”.

Assumption (2(ii)) means that π is increasing and leaves the constant functions invariant. Assumption (2(iii)) means that if Xavier plays the “pure” strategy $i(x)$ in the extended game, his worst loss in the extended game is better than his worst loss in the initial game (he will not raise objections against the use of extended strategies by Yvette). Similarly, assumption (2(iv)) means that the play of a pure strategy $j(y)$ in the extended game is no worse for Yvette than the play of y in the initial game.

Proposition 1. *The duality gap of the extended game is contained in the duality gap of the initial game, i.e.*

$$(3) \quad \begin{aligned} \sup_{y \in Y} \inf_{x \in X} f(x, y) &\leq \sup_{\eta \in \mathcal{Y}} \inf_{\xi \in \mathcal{X}} \pi f(\xi, \eta) \\ &\leq \inf_{\xi \in \mathcal{X}} \sup_{\eta \in \mathcal{Y}} \pi f(\xi, \eta) \leq \inf_{x \in X} \sup_{y \in Y} f(x, y). \end{aligned}$$

Proof. This follows immediately from (2(i), (iii) and (iv)). $\square$

The most interesting extensions are those such that πf always has a value (even if f does not).

Notice that this value belongs to the duality gap of f .

Definition 2. An extension $\{\mathcal{X}, \mathcal{Y}, i, j, \pi\}$ is said to be “playable” if

$$(4) \quad \forall f \in \mathcal{U}(X \times Y), \quad \sup_{\eta \in \mathcal{Y}} \inf_{\xi \in \mathcal{X}} \pi f(\xi, \eta) = \inf_{\xi \in \mathcal{X}} \sup_{\eta \in \mathcal{Y}} \pi f(\xi, \eta)$$

7.2.2. Mixed extensions

As in Section 1.3, we can enlarge the strategy sets X and Y for Xavier and Yvette by embedding them in the sets $\mathcal{M}(X)$ and $\mathcal{M}(Y)$ of mixed strategies (i.e. discrete probability measures).

We denote by δ or δ_X the Dirac operator from X into $\mathcal{M}(X)$ and by δ or δ_Y the Dirac operator from Y into $\mathcal{M}(Y)$. These will be the injective maps of definition 1.

We extend Xavier’s loss function $f : X \times Y \rightarrow \mathbf{R}$ to a bilinear loss function $\pi_0 f$ mapping $\mathcal{M}(X) \times \mathcal{M}(Y)$ into $\mathbf{R}$ in the following way.

$$(5) \quad \text{Whenever } m = \sum_i \alpha^i \delta(x_i) \text{ and } n = \sum_j \beta^j \delta(y_j), \text{ we define} \\ \pi_0 f(m, n) = \sum_{i,j} \alpha^i \beta^j f(x_i, y_j).$$

It is clear that the items $\{\mathcal{M}(X), \mathcal{M}(Y), \delta_X, \delta_Y, \pi_0\}$ define an extension of the games defined on $X \times Y$. Conditions (2) are satisfied because

$$(6) \quad \begin{cases} \text{(i)} \quad \forall f, \forall x \in X, \sup_{y \in Y} f(x, y) = \sup_{n \in \mathcal{M}(Y)} \pi_0 f(\delta(x), n), \\ \text{(ii)} \quad \forall f, \forall y \in Y, \inf_{x \in X} f(x, y) = \inf_{m \in \mathcal{M}(X)} \pi_0 f(m, \delta(y)). \end{cases}$$

This extension is called the (discrete) mixed extension.

(a) *Case of finite strategy sets.* If X is a finite set of k elements and Y a finite set of l elements, then $\mathcal{M}(X) = \mathcal{M}^k$ and $\mathcal{M}(Y) = \mathcal{M}^l$ are convex compact subsets and $\pi_0 f$ defined by $\pi_0 f(m, n) = \sum_{i=1}^k \sum_{j=1}^l \alpha^i \beta^j f(i, j)$ is a continuous bilinear form.

Thus Theorem 1.7 implies the Von Neumann’s theorem.

Proposition 2 (Von Neumann). *Suppose that X and Y are finite sets.*

Then the mixed extension $\{\mathcal{M}(X), \mathcal{M}(Y), \delta_X, \delta_Y, \pi_0\}$ is playable.

(b) *Case of compact strategy sets.* In the case of (infinite) compact strategy sets X and Y , the sets $\mathcal{M}(X)$ and $\mathcal{M}(Y)$ are convex, but no longer compact. We have seen in Section 3.1.6 that the subsets $\mathcal{M}(X)$ and $\mathcal{M}(Y)$ of Radon prob-

abilities are both convex and compact. The Dirac operators δ_X and δ_Y imbed X and Y in $\mathcal{M}(X)$ and $\mathcal{M}(Y)$ respectively. It remains to define $\pi_0 f$ when $f \in \mathcal{C}(X \times Y)$ is a continuous function on $X \times Y$.

To define $\pi_0 f$, we notice that we can biunivocally associate with $f \in \mathcal{C}(X \times Y)$ continuous linear maps $F \in \mathcal{L}(\mathcal{C}^*(X), \mathcal{C}(Y))$ and $F^* \in \mathcal{L}(\mathcal{C}^*(Y), \mathcal{C}(X))$ defined by

$$(7) \quad \forall m \in \mathcal{C}^*(X), \forall y \in Y, \quad F(m)(y) = \langle m, f(\cdot, y) \rangle$$

and

$$(8) \quad \forall n \in \mathcal{C}^*(Y), \forall x \in X, \quad F^*(n)(x) = \langle n, f(x, \cdot) \rangle.$$

Fubini's theorem implies that F and F^* are transposed operators. Hence we define

$$\pi_0 f(m, n) = \langle F(m), n \rangle = \langle m, F^*(n) \rangle.$$

We denote by $m \otimes n \in \mathcal{C}^*(X \times Y)$ the linear form $f \mapsto \langle m \otimes n, f \rangle = \pi_0 f(m, n)$ defined on $\mathcal{C}(X \times Y)$. It is clear that $\{m, n\} \rightarrow \pi_0 f(m, n)$ is a *separately continuous bilinear form which coincides* with the bilinear form defined by (5) when m and n are discrete measures. Hence the items define an

$$\{\mathcal{M}(X), \mathcal{M}(Y), \delta_X, \delta_Y, \pi_0\}$$

extension of games, also called the *mixed extension*. Thus Theorem 1.7 implies the following result.

Proposition 3. *Suppose that X and Y are compact spaces. Then the mixed extension $\{\mathcal{M}(X), \mathcal{M}(Y), \delta_X, \delta_Y, \pi_0\}$ is playable.*

In other words, these two results imply the existence of a saddle mixed strategy for any two-person zero-sum game whose loss function f is continuous and whose strategy sets are compact.

7.2.3. Extensions without exchange of information

The mixed extension $\{\mathcal{M}(X), \mathcal{M}(Y), \delta_X, \delta_Y, \pi_0\}$ obviously satisfies the following property.

$$(9) \quad \begin{aligned} &\forall g \in \mathcal{C}(X), \forall h \in \mathcal{C}(Y), \quad (\pi_0 gh)(\delta(x), n) = g(x) \pi_Y h(n) \\ &\text{and } (\pi_0 gh)(m, \delta(y)) = (\pi_X g)(m) h(y) \\ &[\text{where } (\pi_X g)(m) = \sum \alpha^i g(x_i) \text{ and } \pi_Y h(n) = \sum \beta^j h(y_j) \\ &\text{when } m = \sum \alpha^i \delta(x_i) \text{ and } n = \sum \beta^j \delta(y_j)]. \end{aligned}$$

We shall distinguish the extensions of games which satisfy an analogous property.

Definition 3. We shall say that $\{\mathcal{X}, \mathcal{Y}, i, j, \pi\}$ is an extension “without exchange of information” if there exist $\pi_X \in \mathcal{L}(\mathcal{U}(X), \mathcal{U}(\mathcal{X}))$ and $\pi_Y \in \mathcal{L}(\mathcal{U}(Y), \mathcal{U}(\mathcal{Y}))$ such that $\forall g \in \mathcal{U}(X), \forall h \in \mathcal{U}(Y), \forall x \in X, \forall y \in Y, \forall \xi \in \mathcal{X}, \forall \eta \in \mathcal{Y}$,

$$(10) \quad \begin{cases} (i) \pi(gh)(i(x), \eta) = g(x)\pi_Y h(\eta), \\ (ii) \pi(gh)(\xi, j(y)) = (\pi_X g)(\xi)h(y). \end{cases}$$

Interpretation. If the initial loss function of the game does not depend on y (i.e., if $f = g\theta_Y$), this means that Yvette is a “dummy” in this game. Then we obtain that $(\pi f)(\xi, j(y)) = \pi_X g(\xi)$, i.e. that Yvette remains a “dummy” in the extended game. In other words, Yvette has no “influence” on the Xavier’s extended loss function. If we agree that any “influence” of one player on the other comes necessarily from an “exchange of information”, we motivate the above terminology.

Proposition 4. Let X and Y be compact strategy sets. Let $\{\mathcal{X}, \mathcal{Y}, \delta_X, \delta_Y, \pi\}$ be an extension without exchange of information. Then, if $v_0(f)$ denotes the value of the mixed extension $\pi_0 f$ of a continuous function f , we obtain:

$$(11) \quad \sup_{\eta \in \mathcal{Y}} \inf_{\xi \in \mathcal{X}} \pi f(\xi, \eta) \leq v_0(f) \leq \inf_{\xi \in \mathcal{X}} \sup_{\eta \in \mathcal{Y}} \pi f(\xi, \eta).$$

Remark. In other words, if an extension without exchange of information is playable, then the value of the extended game πf is the mixed value $v_0(f)$.

Proof. Denote by $\pi^*(\xi, \eta) \in \mathcal{E}^*(X \times Y)$, $\pi_X^*(\xi) \in \mathcal{E}^*(X)$ and $\pi_Y^*(\eta) \in \mathcal{E}^*(Y)$ the Radon measures $f \mapsto \pi f(\xi, \eta)$, $f \mapsto \pi_X f(\xi)$ and $g \mapsto \pi_Y g(\eta)$. Properties (10) imply that

$$\pi^*(i(x), \eta) = \delta(x) \otimes \pi_Y^*(\eta), \pi^*(\xi, j(y)) = \pi_X^*(\xi) \otimes \delta(y).$$

Thus, we obtain the following inequalities

$$\begin{aligned} \inf_{\xi \in \mathcal{X}} \pi f(\xi, \eta) &\leq \inf_{x \in X} \pi f(i(x), \eta) \\ &= \inf_{x \in X} \langle \pi^*(i(x), \eta), f \rangle = \inf_{x \in X} \langle \delta(x) \otimes \pi_Y^*(\eta), f \rangle. \end{aligned}$$

We deduce that

$$\begin{aligned} \inf_{x \in X} \langle \delta(x) \otimes \pi_Y^*(\eta), f \rangle &= \inf_{m \in \overline{\mathcal{M}}(X)} \langle m \otimes \pi_Y^*(\eta), f \rangle \\ &\leq \sup_{n \in \overline{\mathcal{M}}(Y)} \inf_{m \in \overline{\mathcal{M}}(X)} \langle m \otimes n, f \rangle = v_0(f). \end{aligned}$$

Therefore, the above inequalities imply that

$$\sup_{\eta \in \mathcal{U}} \inf_{\xi \in \mathcal{X}} \pi f(\xi, \eta) \leq v_0(f).$$

The other inequality is proved in the same way.

7.2.4. Sequential extensions

Let X and Y be two strategy sets and let $\mathbf{N}$ be the set of integers. We take

$$(12) \quad \begin{cases} \text{(i) } \mathcal{X} = X^{\mathbf{N}} = \text{set of sequences } \xi = \{x_i\}_{i \in \mathbf{N}}, \\ \text{(ii) } \mathcal{U} = Y^{\mathbf{N}} = \text{set of sequences } \eta = \{y_j\}_{j \in \mathbf{N}}. \end{cases}$$

The sets X and Y are canonically embedded in $X^{\mathbf{N}}$ and $Y^{\mathbf{N}}$ by the maps s and t defined by

$$(13) \quad sx = \{x, \dots, x, \dots\}, \quad ty = \{y, \dots, y, \dots\}$$

mapping x and y onto constant sequences.

Let $a = \{a_{ij}\}_{i,j \in \mathbf{N}}$ be a summable sequence on $\mathbf{N} \times \mathbf{N}$ satisfying

$$(14) \quad \forall i, j \in \mathbf{N}, a_{ij} \geq 0, \quad \sum_{i,j=1}^{\infty} a_{ij} = 1.$$

If $f \in \mathcal{U}(X \times Y)$, any sequence $\{f(x_i, y_j)\}$ is bounded.

The convergence of the series

$$(15) \quad \pi_a f(\xi, \eta) = \sum_{i,j=1}^{\infty} a_{ij} f(x_i, y_j)$$

follows. It is obvious that $\{X^{\mathbf{N}}, Y^{\mathbf{N}}, s, t, \pi_a\}$

defines an extension, called the “*sequential extension*” associated with the sequence a .

Taking loss functions of the form gh , we obtain that

$$\begin{aligned} (\pi_a g \cdot h)(s(x), \eta) &= \sum_{i,j=1}^{\infty} a_{ij} g(x) h(y_j) \\ &= g(x) \sum_{j \in \mathbf{N}} a_j^Y h(y_j) \quad \text{where } a_j^Y = \sum_{i \in \mathbf{N}} a_{ij} \\ (\pi_a g \cdot h)(\xi, t(y)) &= \sum_{i,j=1}^{\infty} a_{ij} g(x_i) h(y) \\ &= h(y) \sum_{i \in \mathbf{N}} a_i^X g(x_i) \quad \text{where } a_i^X = \sum_{j \in \mathbf{N}} a_{ij}. \end{aligned}$$

Therefore, if we define π_X and π_Y by

$$\pi_X g(\xi) = \sum_{i=1}^{\infty} a_i^X g(x_i), \quad \pi_Y h(\eta) = \sum_{j=1}^{\infty} a_j^Y h(y_j),$$

we see that *sequential extensions are extensions without exchange of information*.

Interpretation. Sequential extensions may be interpreted in terms of the initial game as follows. Xavier *secretly* chooses a sequence $\xi = \{x_i\}_{i \in \mathbb{N}}$ of "pure" strategies. Yvette *secretly* (and therefore independently) chooses a sequence $\eta = \{y_j\}_{j \in \mathbb{N}}$ of "pure" strategies. The game is then played repeatedly as specified by the sequences of pure strategies chosen by the players. The players evaluate the results of this infinite sequence of plays by discounting over time. In particular, Xavier's extended loss function is the "generalized discounted" sum $\pi_a f(\xi, \eta)$ of the individual losses $f(x_i, y_j)$ at each play.

We give below an example of a playable sequential game in the case of finite strategy sets.

Proposition 5. Suppose that X and Y are finite subsets containing p and q elements respectively. Let $a = \{a_{ij}\}$ be defined by

$$(16) \quad a_{ij} = \frac{1}{pq} \left(\frac{p-1}{p} \right)^{i-1} \left(\frac{q-1}{q} \right)^{j-1}$$

Then the associated sequential game is playable.

Proof. Write $b_i = (1/p) ((p-1)/p)^{i-1}$ and $c_j = (1/q) ((q-1)/q)^{j-1}$.

Then $a_{ij} = b_i c_j$ and we deduce that

$$(17) \quad \pi_a(g \cdot h) = \pi_b(g) \cdot \pi_c(h),$$

where

$$(18) \quad (\pi_b g)(\xi) = \sum_{i=1}^{\infty} b_i g(x_i), \quad (\pi_c h)(\eta) = \sum_{j=1}^{\infty} c_j h(y_j).$$

This implies that

$$(19) \quad \pi_a^*(\xi, \eta) = \pi_b^*(\xi) \otimes \pi_c^*(\eta).$$

Therefore, we can write

$$\begin{aligned} \inf_{\xi \in \mathcal{X}} \sup_{\eta \in \mathcal{Y}} \pi_a f(\xi, \eta) &= \inf_{\xi \in \mathcal{X}} \sup_{\eta \in \mathcal{Y}} \langle \pi_a^*(\xi, \eta), f \rangle \\ &= \inf_{\xi \in \mathcal{X}} \sup_{\eta \in \mathcal{Y}} \langle \pi_b^*(\xi) \otimes \pi_c^*(\eta), f \rangle. \end{aligned}$$

By Proposition 1.3.4, π_b^* is a surjective map from X^N onto $\mathcal{M}(X) = \mathcal{M}^p$ and π_c^* is a surjective map from Y^N onto $\mathcal{M}(Y) = \mathcal{M}^q$. We can therefore write

$$\begin{aligned} \inf_{\xi \in \mathcal{X}} \sup_{\eta \in \mathcal{Y}} \pi_a f(\xi, \eta) &= \inf_{\xi \in X^N} \sup_{\eta \in Y^N} \langle \pi_b^*(\xi) \otimes \pi_c^*(\eta), f \rangle \\ &= \inf_{m \in \mathcal{M}(X)} \sup_{n \in \mathcal{M}(Y)} \langle m \otimes n, f \rangle = v_0(f). \end{aligned}$$

Remark. This result is false when one of the strategy sets is infinite. In fact, one can prove that, when X and Y are infinite, there is no sequence $a = \{a_{ij}\}$ satisfying (14) such that the sequential game is playable.

Remark. When X and Y are both finite, Proposition 5 gives a *deterministic* way of playing the initial game which yields the same value as the mixed extension.

Remark. We can approximate a saddle-point $\{\{\bar{x}_i\}_{i \in N}, \{\bar{y}_j\}_{j \in N}\}$ by the pair of *finite* sequences $\{\bar{x}_i\}_{1 \leq i \leq K}, \{\bar{y}_j\}_{1 \leq j \leq K}$. It is easy to check that the error is estimated by

$$(20) \quad \left| v_0(f) - \sum_{i=1}^K \sum_{j=1}^K a_{ij} f(\bar{x}_i, \bar{y}_j) \right| \leq 2 \|f\| \left[\left(\frac{p-1}{p} \right)^K + \left(\frac{q-1}{q} \right)^K \right].$$

7.2.5. Extensions with exchange of information

Decision rules. We denote by $\mathcal{F}(Y, X)$ (resp. $\mathcal{F}(X, Y)$) the set of maps from Y into X (resp. from X into Y).

We shall interpret a map $C \in \mathcal{F}(Y, X)$ as a “*decision rule*” for *Xavier*.

Such a decision rule can be implemented under the behavioral assumption that *Xavier* is informed of the choice of the strategies y played by *Yvette*.

Xavier's strategy set X can be imbedded in the set $\mathcal{F}(Y, X)$ of decision rules by the map c from X into $\mathcal{F}(Y, X)$ defined by

$$(21) \quad x \in X \mapsto c(x) : y \mapsto c(x)(y) = x.$$

Similarly, we denote by $d : Y \rightarrow \mathcal{F}(Y, X)$ the canonical embedding from Y into the set of decision rules for *Yvette* defined by

$$(22) \quad y \in Y \mapsto d(y) : x \mapsto d(y)(x) = y.$$

These maps are injective.

By embedding X and Y in $\mathcal{F}(Y, X)$ and $\mathcal{F}(X, Y)$ respectively, we identify “pure strategies” x and y with *constant decision rules*.

How can both Xavier and Yvette simultaneously use a decision rule $C \in \mathcal{F}(Y, X)$ and $D \in \mathcal{F}(X, Y)$?

Suppose that there exists a fixed point $\{\tilde{x}, \tilde{y}\}$ of $\{x, y\} \mapsto \{C(y), D(x)\}$, that is a pair $\{\tilde{x}, \tilde{y}\}$ satisfying

$$(23) \quad C(\tilde{y}) = \tilde{x} \quad \text{and} \quad D(\tilde{x}) = \tilde{y}.$$

If Xavier and Yvette implement strategies $\tilde{x}$ and $\tilde{y}$, then these will be consistent with their chosen decision rules if and only if (23) holds. Such a pair $(\tilde{x}, \tilde{y})$ is “stable”: none of the players have an incentive to move from such a fixed point.

We are led to introduce the following definition.

Definition 4. A pair $\{\mathcal{C}, \mathcal{D}\}$ of subsets of decision rules is called “consistent” if

$$(24) \quad \begin{cases} \text{(i) } X \subset \mathcal{C} \subset \mathcal{F}(Y, X), \\ \text{(ii) } Y \subset \mathcal{D} \subset \mathcal{F}(X, Y), \\ \text{(iii) } \forall \{C, D\} \in \mathcal{C} \times \mathcal{D}, \exists \{\tilde{x}, \tilde{y}\} \in X \times Y \text{ such that (23) holds.} \end{cases}$$

If the pair $\{\mathcal{C}, \mathcal{D}\}$ is consistent and if a map associating with $\{C, D\} \in \mathcal{C} \times \mathcal{D}$ a fixed point $\{\tilde{x}, \tilde{y}\} \in X \times Y$ satisfying (23) is selected, we extend a loss function f for Xavier to the loss function $\pi_d f$ defined on $\mathcal{C} \times \mathcal{D}$ by

$$(25) \quad \pi_d f(C, D) = f(\tilde{x}, \tilde{y}).$$

It is clear that

$$(26) \quad \pi_d f(c(x), d(y)) = f(x, y)$$

since $\{\tilde{x}, \tilde{y}\}$ is the unique fixed point of the map

$$\{x, y\} \mapsto \{c(\tilde{x})(y), d(\tilde{y})(x)\} = \{\tilde{x}, \tilde{y}\}$$

It is easy to check that the map $\pi_d : f \mapsto \pi_d f$ is linear, positive and leaves the constant functions invariant. Also, we notice that

$$(27) \quad \begin{cases} \text{(i) } \forall x \in X, \sup_{y \in Y} f(x, y) \geq \sup_{D \in \mathcal{D}} f(x, D(x)), \\ \text{(ii) } \forall y \in Y, \inf_{x \in X} f(x, y) \leq \inf_{C \in \mathcal{C}} f(C(y), y). \end{cases}$$

Therefore, (26) and (27) imply that, if $\{\mathcal{C}, \mathcal{D}\}$ is consistent, then $\{\mathcal{C}, \mathcal{D}, c, d, \pi_d\}$ defines an extension, called an *extension “with exchange of information”*.

Counter-example. The pair $\{\mathcal{F}(Y, X), \mathcal{F}(X, Y)\}$ is *not consistent*, since there are maps $\{x, y\} \mapsto \{C(y), D(x)\}$ which have no fixed points (when X and Y contain more than one element).

In other words, both players cannot have full information about the other player's choice and agree on a "stable" pair of strategies.

Example 1. The pair $\{\mathcal{F}(Y, X), Y\}$ is *consistent*. For any $C \in \mathcal{F}(Y, X)$ and $\bar{y} \in Y$, $\{C(\bar{y}), \bar{y}\}$ is the unique fixed point of the map $\{x, y\} \mapsto \{C(y), y\}$. Xavier's extended loss function is defined by

$$(28) \quad \pi_d f(C, y) = f(C(y), y).$$

The worst loss associated with C is $\sup_{y \in Y} f(C(y), y)$. By associating with any $\varepsilon > 0$ a decision rule C_ε satisfying $f(C_\varepsilon(y), y) \leq \inf_{x \in X} f(x, y) + \varepsilon$, we deduce that Xavier's minimal worst loss is equal to

$$(29) \quad \inf_{C \in \mathcal{F}(Y, X)} \sup_{y \in Y} f(C(y), y) = \sup_{y \in Y} \inf_{x \in X} f(x, y).$$

Hence (27(ii)) and (29) imply that this *extension with exchange of information is playable* and that the value of the extended game is equal to $\sup_{y \in Y} \inf_{x \in X} f(x, y)$, i.e. the conservative value of Yvette in the initial game.

Interpretation. The choice of the pair $\{\mathcal{F}(Y, X), Y\}$ can be interpreted as meaning that Xavier has full information about Yvette's choice of strategies, while Yvette has no information at all about Xavier's choice.

In other words, Yvette first chooses her pure strategy $y \in Y$ and Xavier, knowing this choice, chooses the pure strategy $C(y)$ according to the decision rule C (which is chosen in advance).

Example 2. The Lasry theorem yields another example of an extension with exchange of information. We take $\mathcal{C} = \mathcal{C}(Y, X)$ and $\mathcal{D} = Y$ which obviously form a *consistent pair*. Suppose that

$$(30) \quad \begin{cases} \text{(i) } \forall x \in X, y \mapsto f(x, y) \text{ is upper semi-continuous,} \\ \text{(ii) } \exists x_0 \in X \text{ such that } y \mapsto f(x_0, y) \text{ is upper semi-compact.} \\ \text{(iii) } \forall y \in Y, x \mapsto f(x, y) \text{ is quasi-convex.} \end{cases}$$

Then the Lasry theorem implies that

$$(31) \quad \sup_{y \in Y} \inf_{x \in X} f(x, y) = \inf_{C \in \mathcal{C}(Y, X)} \sup_{y \in Y} f(C(y), y).$$

Hence (27(ii)) and (31) imply that this *extension with exchange of information is playable* when it is restricted to the cone of functions satisfying (30). The value of the extended game is equal to $\sup_{y \in Y} \inf_{x \in X} f(x, y)$, i.e. the conservative value of Yvette in the initial game.

Remark. If we assume that

(32) X and Y are convex compact subsets

then the pair $\{\mathcal{C}(Y, X), \mathcal{C}(X, Y)\}$ where $\mathcal{C}$ denotes the set of continuous functions is *consistent* since the map $\{x, y\} \in X \times Y \rightarrow \{C(y), D(x)\} \in X \times Y$ has a fixed point by the Brouwer theorem. (See Theorem 9.3.5 below.)

*7.3. Iterated games

By combining extensions without and with exchange of information, we can model the concept of extension with *partial exchange of information*. We shall illustrate this concept only with the example of *iterated extensions*. The appropriate scheme of exchange of information *depends upon two subsets S and T of $\mathbb{N}$* . Xavier chooses x_1 . Yvette knows x_1 if the index 1 belongs to T (otherwise she does not know x_1). In the former case, she chooses $y_1 = D_1(x_1)$ according to a decision rule D_1 . Xavier is informed of the choice of y_1 if $1 \in S$, and not informed if $1 \notin S$. In the former case, he implements $x_2 = C_2(y_1)$. Now Yvette is aware of this choice if $2 \in T$ and not aware otherwise ... and so on. Thus S is the set of those "stages" $i \in \mathbb{N}$ such that, before he chooses x_{i+1} , Xavier is informed of the strategies played at stage i . In the same way, T is the set of "stages" $j \in \mathbb{N}$ such that, before she chooses y_j , Yvette knows x_j . Assuming that the players have perfect recall, they choose sequences of non-anticipative decision rules. If $S = T = \mathbb{N}$, the information is "perfect". If $S = R = \emptyset$, there is no exchange of information and we recover the sequential extensions. In any case, once a pair of sequences $\{x_i\}_{i \in \mathbb{N}}$ and $\{y_j\}_{j \in \mathbb{N}}$ is implemented by using sequences of decision rules as above, the extended loss function is defined by $\sum_{i,j=1}^{\infty} a_{ij} f(x_i, y_j)$.

The Moulin theorem states that the length of the duality gap of the iterated game is bounded by

$$2 \|f\| \left(1 - \sum_{\substack{i \in T \\ j \geq i}} a_{ij} - \sum_{\substack{j \in S \\ i \geq j+1}} a_{ij} \right).$$

This bound confirms the intuitive feeling that the "larger" the exchange of information, the smaller the corresponding duality gap.

In particular, iterated games with perfect information are playable.

The proof of the Moulin theorem is quite involved. We begin by assuming that a convenient system of functional equations is solved. We then prove the existence of a solution of such a system by using a fixed point theorem which generalizes the theorem of "successive approximations". With this apparatus, we can then prove the Moulin theorem.

7.3.1. Iterated extensions

An *Iterated* game may be regarded as the “product” of a sequential extension with an extension with pure exchange of information.

To see this, let $a = \{a_{ij}\}_{i,j \in \mathbb{N}}$ be a summable sequence satisfying

$$(1) \quad a_{ij} \geq 0, \quad \sum_{i,j=1}^{\infty} a_{ij} = 1.$$

Let $\{X^{\mathbb{N}}, Y^{\mathbb{N}}, s, t, \pi_s\}$ be the associated sequential extension. In order to construct its extension with exchange of information, we have to introduce consistent *pairs* of subsets of decision rules $\{\mathcal{C}, \mathcal{D}\}$ where $\mathcal{C} \subset \mathcal{F}(Y^{\mathbb{N}}, X^{\mathbb{N}})$ and $\mathcal{D} \subset \mathcal{F}(X^{\mathbb{N}}, Y^{\mathbb{N}})$. The subsets $\mathcal{C}$ and $\mathcal{D}$ which we choose are associated with the subsets $S \subset \mathbb{N}$ and $T \subset \mathbb{N}$ as follows. A decision rule $C = \{C_i\}_{i \in \mathbb{N}}$ belongs to $\mathcal{C}$ if and only if $C\xi = \{C_1\xi, \dots, C_i\xi, \dots\}$, where

$$(2) \quad \forall i, \quad C_i \text{ maps } \prod_{j \in S \cap [0, i-1]} Y_j \text{ into } X.$$

(We denote by Y_j the strategy set Y used at the j th stage). A decision rule $D = \{D_j\}_{j \in \mathbb{N}}$ belongs to $\mathcal{D}$ if and only if $D\eta = \{D_1\eta, \dots, D_j\eta, \dots\}$, where

$$(3) \quad \forall j, \quad D_j \text{ maps } \prod_{i \in T \cap [0, j]} X_i \text{ into } Y.$$

(We denote by X_i the strategy set X used at the i th stage.)

Interpretation. Condition (2) means that at the i th step, Xavier is informed of the choice of the strategies $y_j (j \in S \cap [0, i-1])$ previously played by Yvette and implements the strategy $C_i(y_1, \dots, y_{i-1})$ according to the decision rule C_i .

In the same way, at the j th step, Yvette is informed of the strategies $x_i (i \in T \cap [0, j])$ previously played by Xavier and implements the strategy $D_j(x_1, \dots, x_j)$ according to the decision rule D_j .

Note that at the first step $i = 1$, the set of Xavier's decision rules is the set X of consistent decision rules.

Proposition 1. *The pair $\{\mathcal{C}, \mathcal{D}\}$ of subsets of decision rules defined above is consistent.*

Proof. Let C and D be fixed. We shall prove that there exists a unique fixed point $\xi = \{\bar{x}_i\}_{i \in \mathbb{N}}$ and $\eta = \{\bar{y}_j\}_{j \in \mathbb{N}}$ of the map $\{\xi, \eta\} \rightarrow \{C(\eta), D(\xi)\}$.

Since $C_1 = C_{x_1}$ is a constant decision rule, we take $\bar{x}_1 = x_1$. Since D_1 depends only on x_1 , we take $\bar{y}_1 = D_1(\bar{x}_1)$. Since C_2 depends only on y_1 , we take $\bar{x}_2 = C_2(\bar{y}_1)$, and so on. Then $\bar{x}_n$ is equal to $C_n(\bar{y}_1, \dots, \bar{y}_{n-1})$ and $\bar{y}_n$ is equal to

$D_n(\bar{x}_1, \dots, \bar{x}_n)$. It is clear that the sequences $\bar{\xi}$ and $\bar{\eta}$ form the unique fixed point of $\{\xi, \eta\} \rightarrow \{C(\eta), D(\xi)\}$. $\square$

We denote by i and j the maps from X into $\mathcal{C}$ and from Y into $\mathcal{D}$ defined by

$$(4) \quad \begin{cases} (i) \forall \eta \in Y^{\mathbb{N}}, i(x)(\eta) = \{x, x, \dots, x, \dots\}, \\ (ii) \forall \xi \in X^{\mathbb{N}}, j(y)(\xi) = \{y, y, \dots, y, \dots\}. \end{cases}$$

We can now define the iterated extension.

Definition 1. Let $a = \{a_{ij}\}_{i,j \in \mathbb{N}}$ be a summable sequence satisfying (1) and let $\mathcal{C}$ and $\mathcal{D}$ be the consistent pair of subsets of decision rules associated with subsets S and T of $\mathbb{N}$.

Let $\bar{\pi}_a$ be the map associating with any loss function $g \in \mathcal{U}(X \times Y)$ the loss function

$$(5) \quad \bar{\pi}_a g(C, D) = \sum_{i,j=1}^{\infty} a_{ij} g(\bar{x}_i, \bar{y}_j) = \pi_a g(\bar{\xi}, \bar{\eta})$$

where $\bar{\xi} = \{\bar{x}_i\}_i$ and $\bar{\eta} = \{\bar{y}_j\}_j$ form the fixed point of $\{\xi, \eta\} \rightarrow \{C(\eta), D(\xi)\}$. Then $\{\mathcal{C}, \mathcal{D}, i, j, \bar{\pi}_a\}$ is called the “iterated extension”.

Example. If $S = T = \emptyset$, then $\mathcal{C} \times \mathcal{D}$ can be identified with $X^{\mathbb{N}} \times Y^{\mathbb{N}}$ and we obtain the sequential game (without exchange of information).

Example. If $S = \mathbb{N}$ and $T = \emptyset$, then $\mathcal{C}$ is the set of “non-anticipative” decision rules for Xavier and $\mathcal{D} = Y^{\mathbb{N}}$. In other words, Yvette chooses $y_1, \dots, y_n, \dots$ without information about Xavier’s choices while Xavier chooses x_2 after y_1, x_3 after y_1, y_2 and so on, knowing at each step the strategies implemented by Yvette.

Example. If $S = T = \mathbb{N}$, then the iterated game is called a “game with perfect information”. The subsets $\mathcal{C}$ and $\mathcal{D}$ are the subsets of all “non-anticipative decision rules” for Xavier and Yvette.

Example. We consider the case where $S = T = 2\mathbb{N}$ are subsets of even numbers. This is an example in which each player is only partially informed. Xavier and Yvette choose to keep their odd choices secret and inform their partner only of their even choices.

7.3.2. The Moulin theorem

Theorem 1. *The duality gap of an iterated game associated with S, T and $a = \{a_{ij}\}$ satisfying (1) satisfies the following estimate.*

$$(6) \quad \left| \sup_{D \in \mathcal{D}} \inf_{C \in \mathcal{C}} \bar{\pi}_a g(C, D) - \inf_{C \in \mathcal{C}} \sup_{D \in \mathcal{D}} \bar{\pi}_a g(C, D) \right| \leq 2 \|g\| \left(1 - \sum_{\substack{i \in T \\ j \geq i}} a_{ij} - \sum_{\substack{j \in S \\ i \geq j+1}} a_{ij} \right)$$

where $\|g\| = \sup_{x \in X; y \in Y} |g(x, y)|$.

Note the following fundamental consequence.

Theorem 2. *Iterated games with perfect information are playable.*

Proof. Since $S = T = N$, we have that

$$1 - \sum_{\substack{i \in N \\ j \geq i}} a_{ij} - \sum_{\substack{j \in S \\ i \geq j+1}} a_{ij} = 0. \quad \square$$

We also prove a further consequence.

Theorem 3. *Suppose that X and Y are compact spaces and that the loss function g is continuous. If*

$$(7) \quad 1 - \sum_{\substack{i \in T \\ j \geq i}} a_{ij} - \sum_{\substack{j \in S \\ i \geq j+1}} a_{ij} = 0,$$

then there exists a saddle-point $(\bar{C}, \bar{D}) \in \mathcal{C} \times \mathcal{D}$.

7.3.3. Proof of playability of iterated extensions

If $\xi = (x_1, \dots, x_n, \dots)$ and $\eta = (y_1, \dots, y_n, \dots)$ are sequences of X^N and Y^N respectively, we write $\xi^{(n)} = (x_1, \dots, x_n) \in \prod_{i \in T; i \leq n} X_i$ and by

$$\eta^{(n)} = (y_1, \dots, y_n) \in \prod_{j \in S, j \leq n} Y_j.$$

Suppose for the time being that the following lemma is true.

Lemma 1. *There exist two sequences $\{\lambda_n\}_n$ and $\{\mu_n\}_n$ of non-negative numbers which converge to 0 and satisfy*

$$(8) \quad \lambda_1 \geq \mu_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq \mu_n \geq \lambda_{n+1} \geq \dots \geq 0$$

and two uniformly bounded sequences of functions

$$v_n : \prod_{\substack{i \in T \\ i \leq n-1}} X_i \times \prod_{\substack{j \in S \\ j \leq n-1}} Y_j \rightarrow \mathbf{R}$$

and

$$w_n : \prod_{\substack{i \in T \\ i \leq n}} X_i \times \prod_{\substack{j \in S \\ j \leq n-1}} Y_j \rightarrow \mathbf{R}$$

such that

$$(9) \quad \lambda_n v_n(\xi^{(n-1)}; \eta^{(n-1)}) = \inf_{x_n \in X} \left[\sum_{\substack{j \in S \\ j \leq n-1}} a_{nj} g(x_n, y_j) + \mu_n w_n(\xi^{(n-1)}, x_n; \eta^{(n-1)}) \right]$$

and

$$(10) \quad \mu_n w_n(\xi^{(n)}; \eta^{(n-1)}) = \sup_{y_n \in Y} \left[\sum_{\substack{i \in T \\ i \leq n}} a_{in} g(x_i, y_n) + \lambda_{n+1} v_{n+1}(\xi^{(n)}; \eta^{(n-1)}, y_n) \right]$$

where we set $\xi^{(n)} = (\xi^{(n-1)}, x_n)$; $\eta^{(n)} = (\eta^{(n-1)}, y_n)$.

We shall use these functions v_n and w_n for constructing decision rules. Let $\varepsilon > 0$ and $\varepsilon_n = 2^{-n}\varepsilon$. We can define maps $\Xi_n : \prod_{i \in T; i \leq n-1} X_i \times \prod_{j \in S; j \leq n-1} Y_j \rightarrow X$ and $\Theta_n : \prod_{i \in T; i \leq n} X_i \times \prod_{j \in S; j \leq n-1} Y_j \rightarrow Y$ satisfying

$$(11) \quad \lambda_n v_n(\xi^{(n-1)}; \eta^{(n-1)}) \geq \varepsilon_n + \sum_{\substack{j \in S \\ j \leq n-1}} a_{nj} g(\Xi_n(\xi^{(n-1)}, \eta^{(n-1)}), y_j) + \mu_n w_n(\xi^{(n-1)}, \Xi_n(\xi^{(n-1)}, \eta^{(n-1)}); \eta^{(n-1)}),$$

and

$$(12) \quad \mu_n w_n(\xi^{(n)}, \eta^{(n-1)}) \leq \varepsilon_n + \sum_{\substack{i \in T \\ i \leq n}} a_{in} g(x_i, \Theta_n(\xi^{(n)}, \eta^{(n-1)})) + \lambda_{n+1} v_{n+1}(\xi^{(n)}, \eta^{(n-1)}, \Theta_n(\xi^{(n)}, \eta^{(n-1)})).$$

We define the map $\bar{C} = (\bar{C}_1, \dots, \bar{C}_n, \dots) \in \mathcal{C}$ as follows. Let $\eta = (y_1, \dots, y_n, \dots) \in Y^{\mathbf{N}}$ be any sequence of Yvette's strategies.

For $n = 1$, we define $\bar{C}_1 = \bar{C}_{x1}$ to be the constant decision rule satisfying

$$(13) \quad \lambda_1 v_1 \geq \varepsilon_1 + \mu_1 w_1(x_1)$$

For $n = 2$, we define $\bar{C}_2$ by

$$(14) \quad \bar{C}_2(y_1) = \Xi_1(x_1, y_1).$$

Given $\bar{C}_1, \bar{C}_2, \dots, \bar{C}_{n-1}$, we write $x_j = \bar{C}_j(\eta^{(j-1)})$ for $j \leq n-1$ and

$$\xi^{(n-1)} = (x_1, \dots, x_{n-1}).$$

We then define $\bar{C}_n$ by

$$(15) \quad \bar{C}_n(\eta^{(n-1)}) = \mathcal{E}_n(\xi^{(n-1)}, \eta^{(n-1)}).$$

Thus (11) and (15) imply the following inequalities

$$(16) \quad \begin{aligned} \lambda_n v_n(\xi^{(n-1)}, \eta^{(n-1)}) &\geq \varepsilon_n + \sum_{\substack{j \in S \\ j \leq n-1}} a_{nj} g(\bar{C}_n(\eta^{(n-1)}), y_j) \\ &+ \sum_{\substack{i \in T \\ i \leq n}} a_{in} g(\bar{C}_i(\eta^{(n-1)}), y_n) + \lambda_{n+1} v_{n+1}(\xi^{(n)}, \eta^{(n)}). \end{aligned}$$

Adding these inequalities for $n = 1, \dots, p$, we obtain that

$$(17) \quad \begin{aligned} \lambda_1 v_1 &\geq \sum_{n=1}^p \varepsilon_n + \sum_{\substack{j \in S \\ j \leq p-1 \\ i \geq j+1}} a_{ij} g(\bar{C}_i(\eta), y_j) \\ &+ \sum_{\substack{i \in T \\ i \leq p \\ j \geq i}} a_{ij} g(\bar{C}_i(\eta), y_j) + \lambda_{p+1} v_{p+1}(\xi^{(p)}, \eta^{(p)}). \end{aligned}$$

Since the functions v_p are uniformly bounded and λ_p converges to 0, we deduce that

$$(18) \quad \begin{aligned} \lambda_1 v_1 &\geq \varepsilon + \sum_{\substack{i \in T \\ j \geq i}} a_{ij} g(\bar{C}_i(\eta), y_j) + \sum_{\substack{j \in S \\ i \geq j+1}} a_{ij} g(\bar{C}_i(\eta), y_j) \\ &\geq \varepsilon + \sum_{i,j=1}^{\infty} a_{ij} g(\bar{C}_i(\eta), y_j) - \|g\| \left(1 - \sum_{\substack{j \in T \\ j \geq i}} a_{ij} - \sum_{\substack{j \in S \\ i \geq j+1}} a_{ij} \right) \\ &= \varepsilon + \pi_a g(\bar{C}(\eta), \eta) - \|g\| \alpha, \end{aligned}$$

where $\alpha = 1 - \sum_{i \in T; j \geq i} a_{ij} - \sum_{j \in S; i \geq j+1} a_{ij}$.

Because this inequality is true for any sequence $\eta \in Y^N$, we obtain that

$$(19) \quad \begin{aligned} \lambda_1 v_1 &\geq \varepsilon + \sup_{\eta \in Y^N} \pi_a g(\bar{C}(\eta), \eta) - \|g\| \alpha \\ &= \varepsilon + \sup_{D \in \mathcal{D}} \bar{\pi}_a g(\bar{C}, D) - \|g\| \alpha. \end{aligned}$$

This implies that

$$(20) \quad \lambda_1 v_1 \geq \inf_{C \in \mathcal{C}} \sup_{D \in \mathcal{D}} \bar{\pi}_a g(C, D) - \|g\| \alpha.$$

Similarly, we define decision rules $\bar{D}_n$ by setting

$$(21) \quad \bar{D}_n(\xi^{(n)}) = \Theta_n(\xi^{(n)}, \eta^{(n-1)}),$$

where $\xi = (x_1, \dots, x_n, \dots)$ and $\eta = (y_1, \dots, y_k, \dots)$ with $y_k = \bar{D}_k(\xi^{(k)})$.

We deduce from inequalities (12) that

$$(22) \quad \begin{aligned} & \mu_n w_n(\xi^{(n)}, \eta^{(n-1)}) \\ & \leq \varepsilon_n + \sum_{\substack{i \in T \\ i \leq n}} a_{in} g(x_i, \bar{D}_n(\xi^{(n)})) + \sum_{\substack{j \in S \\ j \leq n}} a_{nj} g(x_{n+1}, \bar{D}_j(\xi^{(n)})) \\ & \quad + \mu_{n+1} w_{n+1}(\xi^{(n+1)}, \eta^{(n)}). \end{aligned}$$

Adding these inequalities, we obtain that

$$(23) \quad \begin{aligned} & \lambda_1 v_1 \leq \mu_1 w_1(\xi_1) \\ & \leq \varepsilon + \sum_{\substack{i \in T \\ j \geq i}} a_{ij} g(x_i, \bar{D}_j(\xi)) + \sum_{\substack{j \in S \\ i \geq j+1}} a_{ij} g(x_i, \bar{D}_j(\varepsilon)) \\ & \leq \varepsilon + \pi_a g(\xi, \bar{D}(\xi)) + \|g\| \alpha. \end{aligned}$$

This implies that

$$(24) \quad \lambda_1 v_1 \leq \sup_{D \in \mathcal{D}} \inf_{C \in \mathcal{C}} \bar{\pi}_a g(C, D) + \|g\| \alpha.$$

Therefore, (20) and (24) imply that

$$(25) \quad \begin{aligned} & \inf_{C \in \mathcal{C}} \sup_{D \in \mathcal{D}} \bar{\pi}_a g(C, D) - \sup_{D \in \mathcal{D}} \inf_{C \in \mathcal{C}} \bar{\pi}_a g(C, D) \\ & \leq 2 \|g\| \alpha. \end{aligned}$$

7.3.4. A system of functional equations

In order to solve systems (9) and (10), we notice that they can be written in the following form

$$(26) \quad \begin{cases} \text{(i)} \quad \lambda_n v_n = \phi_n(w_n), \\ \text{(ii)} \quad \mu_n w_n = \psi_n(v_{n+1}), \end{cases}$$

where the operators ϕ_n and ψ_n are defined by

$$(27) \quad (\phi_n w_n)(\xi^{(n-1)}, \eta^{(n-1)}) = \inf_{x_n \in X} \left[\sum_{\substack{j \in S \\ j \leq n-1}} a_{nj} g(x_n, y_j) + \mu_n w_n(\xi^{(n-1)}, x_n; \eta^{(n-1)}) \right]$$

and

$$(28) \quad (\psi_n v_{n+1})(\xi^{(n)}, \eta^{(n-1)}) = \sup_{y_n \in Y} \left[\sum_{\substack{i \in T \\ i \leq n}} a_{in} g(x_i, y_n) + \lambda_{n+1} v_{n+1}(\xi^{(n)}, \eta^{(n-1)}, y_n) \right]$$

Note that

$$(29) \quad \|\phi_n w_n\| \leq \sum_{\substack{j \in S \\ j \leq n-1}} a_{nj} \|g\| + \mu_n \|w_n\|.$$

and

$$(30) \quad \|\psi_n v_{n+1}\| \leq \sum_{\substack{i \in T \\ i \leq n}} a_{in} \|g\| + \lambda_{n+1} \|v_{n+1}\|.$$

We deduce that, if $\|w_n\| \leq \|g\|$ and $\|v_{n+1}\| \leq \|g\|$, then

$$(31) \quad \begin{cases} \text{(i)} \|\phi_n w_n\| \leq \lambda_n \|g\|. \\ \text{(ii)} \|\psi_n v_{n+1}\| \leq \mu_n \|g\|. \end{cases}$$

provided that the sequences $\{\lambda_n\}$ and $\{\mu_n\}$ satisfy

$$(32) \quad \begin{cases} \text{(i)} \lambda_n = \mu_n + \sum_{j \in S; j \leq n-1} a_{nj}, \quad n \geq 2, \\ \text{(ii)} \mu_n = \lambda_{n+1} + \sum_{i \in T; i \leq n} a_{in}, \quad n \geq 1. \end{cases}$$

For $n = 1$, we set

$$(33) \quad \lambda_1 = \sum_{\substack{i \in T \\ j \leq i}} a_{ij} + \sum_{\substack{i \geq j+1 \\ j \in S}} a_{ij}.$$

It is obvious that

$$(34) \quad \lambda_1 \geq \mu_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq \mu_n \geq \lambda_{n+1} \geq \dots$$

since the coefficients a_{ij} are non-negative. Furthermore, λ_n and μ_n are non-negative. To prove this, add equations (32(i) and (ii)). We obtain that

$$\lambda_n = \lambda_{n+1} + \sum_{\substack{j \in S \\ j \leq n-1}} a_{nj} + \sum_{\substack{i \in T \\ i \leq n}} a_{in}.$$

Hence, adding these equations for $n = 1, \dots, p$,

$$\lambda_{p+1} + \sum_{\substack{j \in S \\ j \leq n-1 \\ n \leq p}} a_{nj} + \sum_{\substack{i \in T \\ i \leq n \\ n \leq p}} a_{in} = \lambda_1 = \sum_{\substack{i \in T \\ i \geq j}} a_{ij} + \sum_{\substack{j \in S \\ j \geq i-1}} a_{ij}.$$

Therefore, $\lambda_{p+1} \geq 0$ and $\mu_p \geq \lambda_{p+1} \geq 0$ for all p . Furthermore, both $\{\lambda_p\}$ and $\{\mu_p\}$ converge to 0. Suppose that one of the λ_n 's (or one of the μ_n 's) is equal to 0. Assume for instance that

$$\lambda_1 \neq 0, \dots, \lambda_{n_0} \neq 0, \quad \mu_{n_0} = 0, \quad \lambda_{n_0+1} = 0.$$

Then we can solve the system (9), (10).

We make an arbitrary choice for the functions w_n for $n \geq n_0$ and v_n for $n \geq n_0 + 1$, since, in this case, eqs. (9) and (10) can be written $0 = 0$. This follows from the fact that, if $\lambda_n = 0$, then eqs. (32(i)) and (32(ii)) imply that $\sum_{j \in S; j \leq n-1} a_{nj} = 0$ and $\sum_{i \in T; i \leq n} a_{in} = 0$. Hence $a_{nj} = 0$ when $j \in S$ and $j \leq n-1$; $a_{in} = 0$ when $i \in T$ and $i \leq n$.

For $n = n_0$, eq. (9) shows that

$$v_{n_0}(\xi^{n_0-1}, \eta^{n_0-1}) = \frac{1}{\lambda_{n_0}} \inf_{x_{n_0} \in X} \sum_{\substack{j \in S \\ j \leq n_0-1}} a_{n_0 j} g(x_{n_0}, y_j)$$

is well-defined.

We then define $w_{n_0-1} = (1/\mu_{n_0-1}) \psi_{n_0}(v_{n_0})$, $v_{n_0-1} = (1/\lambda_{n_0-1}) \phi_n(w_{n_0-1})$, and so on.

Consider now the case when $\lambda_n > 0$ and $\mu_n > 0$ for all n . The system can be written

$$(35) \quad \begin{cases} \text{(i)} & v_n = \phi_n^*(w_n) = \frac{1}{\lambda_n} \phi_n(w_n), \\ \text{(ii)} & w_n = \psi_n^*(v_{n+1}) = \frac{1}{\mu_n} \psi_n(v_{n+1}). \end{cases}$$

The maps ϕ_n^* and ψ_n^* satisfy

$$(36) \quad \begin{cases} \text{(i)} & \|\phi_n^*(w_n^1) - \phi_n^*(w_n^2)\| \leq \frac{\mu_n}{\lambda_n} \|w_n^1 - w_n^2\|, \\ \text{(ii)} & \|\psi_n^*(v_{n+1}^1) - \psi_n^*(v_{n+1}^2)\| \leq \frac{\lambda_{n+1}}{\mu_n} \|v_{n+1}^1 - v_{n+1}^2\|. \end{cases}$$

Let us set

$$(37) \quad U_{2p} = \text{the subset of functions } w_p : \prod_{i \in T; i \leq p} X_i \times \prod_{j \in S; j \leq p-1} Y_j \rightarrow \mathbf{R} \text{ such that } \|w_p\| \leq \|g\|,$$

$$(38) \quad U_{2p-1} = \text{the subset of functions } v_p : \prod_{i \in T; i \leq p-1} X \times \prod_{j \in S; j \leq p-1} Y_j \rightarrow \mathbf{R} \text{ such that } \|v_p\| \leq \|g\|.$$

Then inequalities (31(i) and (ii)) imply that

$$(39) \quad \begin{cases} \text{(i)} & h_{2p} = \psi_p^* \text{ maps } U_{2p+1} \text{ into } U_{2p}, \\ \text{(ii)} & h_{2p-1} = \phi_p^* \text{ maps } U_{2p} \text{ into } U_{2p-1}. \end{cases}$$

Inequalities (36(i) and (ii)) show that the maps h_n satisfy

$$(40) \quad \|h_n(u_n^1) - h_n(u_n^2)\| \leq \alpha_n \|u_n^1 - u_n^2\|,$$

where

$$(41) \quad \alpha_{2p} = \frac{\lambda_{p+1}}{\mu_p}; \quad \alpha_{2p-1} = \frac{\mu_p}{\lambda_p}.$$

We have that $\prod_{i=1}^{2p} \alpha_i = \lambda_{p+1}$ and $\prod_{i=1}^{2p+1} \alpha_i = \mu_{p+1}$. Hence $\prod_{i=1}^n \alpha_i$ converges to 0 because the sequences $\{\lambda_n\}$ and $\{\mu_n\}$ converge to 0. Now, any sequence $\{u_n\}$ of elements $u_n \in U_n$ satisfying

$$(42) \quad u_n = h_n(u_{n+1})$$

defines a solution of the system (9), (10) by setting

$$v_p = u_{2p-1} \text{ and } w_p = u_{2p}.$$

Such a sequence $\{u_n\}$ exists as shown by the following lemma.

7.3.5. A lemma on successive approximations

Lemma 2. Let $\{U_n\}$ be a sequence of complete uniformly bounded metric spaces and let $\{h_n\}$ be a sequence of maps $h_n: U_{n+1} \rightarrow U_n$ satisfying

$$(43) \quad \begin{cases} \text{(i) } d(h_n(x), h_n(y)) \leq \alpha_n d(x, y) \quad \forall x, y \in U_{n+1}, \\ \text{(ii) } \prod_{i=1}^{\infty} \alpha_i = 0. \end{cases}$$

Then there exists a unique sequence $\{\bar{x}_n\}_n$ satisfying

$$(44) \quad \begin{cases} \text{(i) } \bar{x}_n \in U_n, \\ \text{(ii) } \bar{x}_n = h_n(\bar{x}_{n+1}). \end{cases}$$

Proof. If $n \geq p$, we write

$$(45) \quad K_n^p = \text{closure } (h_p \circ h_{p+1} \circ \dots \circ h_n[U_{n+1}]) \subset U_p.$$

Let δ be a common upper bound of the diameters of the subsets U_n . Then we can check that

$$(46) \quad \text{diameter } (K_n^p) \leq \left(\prod_{i=p}^n \alpha_i \right) \delta.$$

Thus $\{K_n^p\}_{n \geq p}$ is a decreasing sequence of closed subsets of U_p whose diameters converge to 0. Since U_p is complete, there exists $\bar{x}_p \in U_p$ such that

$$(47) \quad \bigcap_{n \geq p} K_n^p = \{\bar{x}_p\}.$$

Therefore, since $\bar{x}_{p+1} \in K_n^{p+1}$ whenever $n \geq p+1$,

$$(48) \quad \forall n \geq p+1, \quad h_p(\bar{x}_{p+1}) \in h_p(K_n^{p+1}) \subset K_n^p.$$

Hence $h_p(\bar{x}_{p+1})$ belongs to the intersection of the decreasing sequence $\{K_n^p\}_{n \geq p+1}$, which is equal to $\{\bar{x}_p\}$.

Thus $h_p(\bar{x}_{p+1}) = \bar{x}_p$ for all p , i.e. there exists a sequence $\{\bar{x}_p\}$ satisfying (44). It is unique because, if $\{\bar{y}_p\}$ is another sequence satisfying (44), then if $p < q$, $d(\bar{x}_p, \bar{y}_q) \leq (\prod_{l=p}^{q-1} \alpha_l) d(\bar{x}_q, \bar{y}_q) \leq (\prod_{l=p}^{q-1} \alpha_l) \delta$. Letting q tend to ∞ , we deduce that $d(\bar{x}_p, \bar{y}_p) = 0$ and thus that $\bar{x}_p = \bar{y}_p$. $\square$

7.3.6. Proof of existence of saddle decision rules

Proof of Theorem 3. Note that, if X and Y are compact and g is continuous, then we can take $\varepsilon = 0$ in inequalities (11) and (12). Since $\alpha = 0$, inequalities (19) and (23) with $\varepsilon = 0$ imply that

$$(49) \quad \sup_{D \in \mathcal{D}} \pi_{ag}(\bar{C}, D) \leq \lambda_1 v_1 \leq \inf_{C \in \mathcal{C}} \pi_{ag}(C, \bar{D}).$$

These inequalities show that the pair $\{\bar{C}, \bar{D}\}$ constructed above is a saddle-point of the game. $\square$

CHAPTER 8

THE FUNDAMENTAL ECONOMIC MODEL: WALRAS EQUILIBRIA

This chapter is devoted to the description of a fundamental economic model. We define Walras and pre-Walras equilibria and prove several existence theorems (using Ky-Fan Theorem 7.1.3).

Briefly, the problem is this. There are n consumers (labelled $i = 1, 2, \dots, n$) who have to choose $x^i \in R^i \subset \mathbf{R}^I$ such that the sum $\sum_{i=1}^n x^i$ of their consumptions belongs to a given subset $Y \subset \mathbf{R}^I$ of available commodities. They use a selection mechanism called "demand correspondence" which associates with any price $p \in \mathbf{R}^{I*}$ and income $r \in \mathbf{R}$ a subset $D_i(p, r)$ of the budget set $B_i(p, r) = \{x \in \mathbf{R}^I \text{ such that } \langle p, x \rangle \leq r\}$.

When the total income $r(p) = \sup_{y \in Y} \langle p, y \rangle$ is appropriated by allocating to each consumer i an income $r_i(p)$ such that $\sum_{i=1}^n r_i(p) = r(p)$ for all p , the problem arises as to whether it is possible to find a price $\bar{p}$ and consumptions $\bar{x}^i$ such that

$$\begin{cases} \text{(i) } \bar{x}^i \in D_i(\bar{p}, r_i(\bar{p})), & i = 1, 2, \dots, n. \\ \text{(ii) } \sum_{i=1}^n \bar{x}^i \in Y. \end{cases}$$

When these conditions are satisfied, we shall say that $\bar{p}$ and $\bar{x}^i$ ($i = 1, 2, \dots, n$) constitute a *Walras pre-equilibrium*.

We shall prove an existence theorem in Section 8.2. The main assumption is that the demand correspondences are upper semi-continuous. In Section 8.3 we study the case when the demand correspondences are determined by the minimization of a loss function $x \mapsto f(x, p)$ (which can depend upon the price p). Thus

$$D_i(p, r) = \{x \in B_i(p, r) \text{ such that } f_i(x, p) = \min_{y \in B_i(p, r)} f_i(y, p)\}.$$

Finally, in Section 8.4, we introduce explicitly m producers (labelled $j = 1, 2, \dots, m$) into the model. These are described by their cost functions $g_j: \mathbf{R}^I \rightarrow \mathbf{R}$. An existence theorem for a suitably defined Walras equilibrium is proved. These existence theorems are deduced from a general result about the surjectivity of correspondences which follows directly from the Ky-Fan theorem. The result concerns an upper semi-continuous correspondence S mapping elements

p of a convex compact subset $P \subset U^*$ onto convex compact subsets $S(p) \subset U$ which satisfy the general "Walras law"

$$\forall p \in P, \quad \sigma^b(S(p), p) = \inf_{y \in S(p)} \langle p, y \rangle \leq 0.$$

The conclusion is that the problem of finding

$$\bar{x} \in S(\bar{p}) \cap -P^+$$

admits a solution $\{\bar{x}, \bar{p}\}$.

8.1. Description of the model

In this section we define the model and the Walras and pre-Walras equilibria. We describe several examples of appropriations of the subset of available commodities. In the case of quadratic demand functions, we prove the existence of a Walras equilibrium of an exchange economy.

8.1.1. The subset of available commodities

Consider the commodity space $\mathbf{R}^I$. We begin the description of an economy with

- (1) the subset $Y \subset \mathbf{R}^I$ of available commodities.

Consider a set $N = \{1, \dots, n\}$ of n consumers (households) denoted by $i = 1, \dots, n$. With each consumer $i \in N$ we associate

- (2) its "consumption subset" $R^i \subset \mathbf{R}^I$, which is assumed *once and for all* to be convex, closed and bounded from below.

The basic problem is to allocate the available commodities $y \in Y$ among the consumers. An "allocation" among consumers in N is an element $x = \{x^1, \dots, x^n\}$ belonging to $R^N = \prod_{i=1}^n R^i$ such that the corresponding aggregate consumption $\sum_{i=1}^n x^i$ is an available commodity. We denote by $X(N) \subset R^N$ the *subset of allocations* defined by

- (3) $X(N) = \{x \in R \text{ such that } \sum_{i \in N} x^i \in Y\}.$

The upper support function of Y will be denoted by

- (4) $r(p) = \sigma^*(Y; p) = \sup_{y \in Y} \langle p, y \rangle.$

We interpret $r(p)$ as the “profit function” of the economy, where $p \in \mathbf{R}^{l*}$ is understood as a price (which associates with any $y \in Y$ its value $\langle p, y \rangle \in \mathbf{R}$).

We denote the *barrier cone* of Y by $P(Y^*) = \{p \in \mathbf{R}^{l*} \text{ such that } r(p) < +\infty\}$ (see Definition 1.4.1). In most examples, we assume the existence of

- (5) a non-zero vector $\xi \in \text{Int}(-P_\infty(Y))$ which we call the numéraire.

In such a case, the cone $P(Y^*)$ is spanned by the subset

- (6) $P = \{p \in P(Y^*) \text{ such that } \langle p, \xi \rangle = 1\}$ of normalized prices.

Let us recall that in this case, Theorem 1.4.3 implies that

- (7) P is a convex compact subset of $\mathbf{R}^{l*}$

and that

- (8) $P^+ = -P_\infty(Y)$ is the negative of the recession cone.

Recall that r is *lower semi-continuous, convex and positively homogeneous*. The property that

- (9) Y is closed and convex and satisfies $Y = Y + P_\infty(Y)$

implies that Y is characterized by its maximum profit function, i.e.

- (10) $Y = \{y \in \mathbf{R}^l \text{ such that } \langle p, y \rangle \leq r(p) \text{ for all } p \in P\}.$

If

- (11) $0 \in Y,$

then r is a non-negative function on P .

8.1.2. Appropriation of the economy

In order to devise a decentralized mechanism which allocates an available commodity $y \in Y$, we shall assume that the subset Y of available commodities is “appropriated”. This means that consumers $i \in N$ are endowed with subsets $Y(i)$ of commodities such that

- (12) $Y = \sum_{i \in N} Y(i) \text{ where } Y(i) \subset \mathbf{R}^l$

In other words, any available commodity $y = \sum_{i \in N} y^i \in Y$ is assumed to be appropriated in the sense that each consumer $i \in N$ is entitled to an available commodity $y^i \in Y(i)$.

We shall say that $Y(i)$ is the initial *endowment of available commodities*. Such an appropriation of Y allows the profit function to be shared among the consumers, i.e.

$$(13) \quad \forall p \in P, \quad r(p) = \sum_{i=1}^n r_i(p)$$

where

$$(14) \quad r_i(p) = \sigma^*(Y(i); p) = \sup_{y \in Y(i)} \langle p, y \rangle$$

denotes the maximum profit that consumer i obtains from his initial endowment $Y(i)$. This profit $r_i(p)$ is used as income. If we define the *budget set* $B_i(p, r)$ of consumer i as

$$(15) \quad B_i(p, r) = \{x \in R^I \text{ such that } \langle p, x \rangle \leq r\}$$

the budget set of player i defined by the appropriation $Y = \sum_{i=1}^n Y(i)$ is

$$B_i(p, r_i(p)).$$

8.1.3. Demand correspondences

Once the the set of available commodities has been appropriated, each consumer knows his budget set $B_i(p, r_i(p))$ when the price p prevails.

We shall assume that each consumer i devises a selection procedure which enables him to consume a good in a subset $D_i(p, r)$ of his budget set $B_i(p, r)$.

Definition 1. We shall say that a correspondence $D_i : P \times \mathbf{R} \rightarrow \mathbf{R}^I$ is a “*demand correspondence*” if

$$(16) \quad \forall p \in P, \forall r \in \mathbf{R}, \quad D_i(p, r) \subset B_i(p, r).$$

(We shall construct examples of demand correspondences later (see Section 8.3 for instance).) An economy is therefore defined by the items

$$(17) \quad \{R^I, Y(i), D_i\}_{i \in N}.$$

We can now state the basic problem. If we assume that

$$(18) \quad \begin{cases} \text{(i) } Y = \sum_{i=1}^n Y(i) \text{ is appropriated,} \\ \text{(ii) each consumer } i \text{ chooses a commodity according to his demand correspondence } D_i, \end{cases}$$

is it possible to find a price $\bar{p}$ such that there exist commodities $\bar{x}^i$ satisfying

$$(19) \quad \begin{cases} \text{(i) } \forall i \in N, \quad \bar{x}^i \in D_i(\bar{p}, r_i(\bar{p})), \\ \text{(ii) } \sum_{i \in N} \bar{x}^i \in Y \end{cases}$$

8.1.4. *Walras equilibrium*

Definition 2. We shall say that a pair $\{\bar{x}, \bar{p}\} = \{\bar{x}^1, \dots, \bar{x}^n, \bar{p}\} \in R^N \times P$ satisfying (19) is a Walras pre-equilibrium. We call $\bar{x} = \{\bar{x}^1, \dots, \bar{x}^n\}$ a Walras allocation and $\bar{p}$ a Walras price. We shall say that $\{\bar{x}, \bar{p}\}$ is a Walras equilibrium if, in addition,

$$(20) \quad \forall i \in N, \quad \langle \bar{p}, \bar{x}_i \rangle = r_i(\bar{p}).$$

8.1.5. *Examples of subsets of available commodities and of appropriations*

Example 1. The simplest example is when

$$(21) \quad Y = w - R_+^l$$

is the set of commodities less than or equal to an initial holding w . If each consumer i is assumed to hold an initial endowment w^i of goods such that $w = \sum_{i=1}^n w^i$, then the appropriation of Y is defined by

$$(22) \quad Y = \sum_{i \in N} Y(i) \quad \text{where } Y(i) = w^i - R_+^l.$$

The functions $r(p)$ and $r_i(p)$ are equal to

$$(23) \quad r_i(p) = \langle p, w^i \rangle, \quad r(p) = \langle p, w \rangle.$$

Example 2. Let us consider m firms j , described by their production subsets Z^j . If w is the initial holding, then

$$(24) \quad Y = w + \sum_{j=1}^m Z^j - R_+^l.$$

The usual way to appropriate Y is to assume that

$$(25) \quad w = \sum_{i=1}^n w^i$$

is the sum of initial endowments and that each consumer i owns a fraction $\theta_j^i \in [0, 1]$ of firm j and *shares* to that extent in the production. The system $\theta = \{\theta_j^i\}$ of shares satisfies obviously

$$(26) \quad \forall j = 1, \dots, m, \quad \sum_{i \in N} \theta_j^i = 1.$$

In other words, $Y = w + \sum_{j=1}^m Z^j - R_+^l$ is appropriated in the following way

$$(27) \quad Y = \sum_{i \in N} Y(i); \quad Y(i) = w^i + \sum_{j=1}^m \theta_j^i Z^j - R_+^l.$$

The profit functions $r(p)$ and $r_i(p)$ are defined by

$$(28) \quad \begin{cases} \text{(i)} & r_i(p) = \langle p, w^i \rangle + \sum_{j=1}^m \theta_j^i \sigma^*(Z^j; p), \\ \text{(ii)} & r(p) = \langle p, w \rangle + \sum_{j=1}^m \sigma^*(Z^j; p). \end{cases}$$

The income of consumer i derives from two sources, i.e. the sale of his initial endowment w^i and the shares $\theta_j^i \sigma^*(Z^j, p)$ held in the profits of firms j .

Proposition 1. *Let $\{\bar{x}, \bar{p}\}$ be a Walras equilibrium and let $\bar{y} = \sum_{i \in N} \bar{x}^i$ be the "aggregate demand vector". Then for any allocation $\{\bar{z}^1, \dots, \bar{z}^n\}$ of the production satisfying*

$$(29) \quad \sum_{i \in N} \bar{x}^i \leq w + \sum_{j=1}^m \bar{z}^j$$

we have that

$$(30) \quad \forall j = 1, \dots, m, \quad \langle \bar{p}, \bar{z}^j \rangle = \max_{z \in Z^j} \langle \bar{p}, z \rangle$$

i.e. $\bar{z}^j$ maximizes the profit over the production set Z^j when the Walras price $\bar{p}$ prevails.

Proof. Since $\bar{y} = \sum_{i \in N} \bar{x}^i$ belongs to Y , there exist $\bar{z}^j \in Z^j$ for all $j = 1, \dots, m$ such that (29) holds. Hence

$$\begin{aligned} r(\bar{p}) &= \sum_{i \in N} r_i(\bar{p}) = \sum_{i \in N} \langle \bar{p}, \bar{x}^i \rangle = \langle \bar{p}, \bar{y} \rangle \\ &\leq \langle \bar{p}, w \rangle + \sum_{j=1}^m \langle \bar{p}, \bar{z}^j \rangle \leq r(\bar{p}). \end{aligned}$$

Therefore, since $r(\bar{p}) = \langle \bar{p}, w \rangle + \sum_{j=1}^m \sigma^*(Z^j, \bar{p})$, we obtain that

$$\sum_{j=1}^m (\langle \bar{p}, \bar{z}^j \rangle - \sigma^*(Z^j; \bar{p})) = 0.$$

But $\langle \bar{p}, \bar{z}^j \rangle - \sigma^*(Z^j; \bar{p}) \leq 0$ for all j . We deduce that $\langle \bar{p}, \bar{z}^j \rangle = \sigma^*(Z^j; \bar{p})$ for all j . $\square$

*8.1.6. Example: Quadratic demand functions

We take $R^i = R^I$ for all consumers i .

Consider the case where the demand correspondences $D_i(p, r)$ are the "quadratic demand functions" D_i defined by

$$(31) \quad D_i(p, r) = u^i - \frac{\langle p, u^i \rangle - r}{\langle p, J^{-1}p \rangle} J^{-1}p \quad \text{if } p \neq \emptyset$$

where

$$(32) \quad \begin{cases} \text{(i) } u_i \in \mathbf{R}^I \text{ is a consumption objective,} \\ \text{(ii) } J \in \mathcal{L}(\mathbf{R}^I, \mathbf{R}^{I*}) \text{ is a self-adjoint positive definite linear operator.} \end{cases}$$

(Recall that $D_i(p, r)$ minimizes the distance $\|x - u^i\|$ to the objective u^i subject to the budget constraint $\langle p, x \rangle = r$. See Proposition 2.3.1.)

Definition (31) satisfies the requirements for a demand function since

$$(33) \quad \forall p \neq 0, \quad \langle p, D_i(p, r) \rangle = r.$$

Consider the case of an exchange economy, where

$$(34) \quad Y(i) = \{w^i\}; \quad Y = \{w\} \quad \text{with } w = \sum_{i=1}^n w^i.$$

We regard

$$(35) \quad \begin{cases} \text{(i) } v^i = u^i - w^i \text{ as the "net demand" of consumer } i, \\ \text{(ii) } v = \sum_{i \in N} v^i = \sum_{i \in N} u^i - \sum_{i \in N} w^i \text{ as the "aggregate net demand".} \end{cases}$$

Proposition 2. Suppose that the aggregate net demand $v \neq 0$, then there exists a Walras equilibrium $\langle \bar{x}, \bar{p} \rangle$ satisfying

$$(36) \quad \begin{cases} \text{(i) } \forall i, \bar{x}^i = D_i(\bar{p}, \langle \bar{p}, w^i \rangle), \\ \text{(ii) } \sum_{i \in N} \bar{x}^i = w, \end{cases}$$

where $\bar{x} = \{\bar{x}^1, \dots, \bar{x}^n\}$ is unique and $\bar{p}$ is unique up to the multiplication by a scalar $\alpha \neq 0$. The quantities $\bar{x}$ and $\bar{p}$ are given by

$$(37) \quad \begin{aligned} \text{(i) } \bar{x}^i &= u^i - (\langle Jv, v^i \rangle / \langle Jv, v \rangle) v, \\ \text{(ii) } \bar{p} &= \alpha Jv, \quad \alpha \neq 0. \end{aligned}$$

Proof. Since $w = \sum_{i \in N} D_i(\bar{p}, \langle \bar{p}, w^i \rangle) = \sum_{i \in N} u^i - (\langle \bar{p}, v \rangle / \langle \bar{p}, J^{-1} \bar{p} \rangle) J^{-1} \bar{p}$, we have that $\bar{p} = \alpha Jv$. Thus $\bar{x}^i = D_i(\alpha Jv, \alpha \langle Jv, w^i \rangle) = D^i(Jv, \langle Jv, w^i \rangle) = u^i - (\langle Jv, v^i \rangle / \langle Jv, v \rangle) v$. $\square$

Since $((x, y)) = \langle Jx, y \rangle$ is a scalar product, we deduce from (37) that, for any player i ,

$$(38) \quad \frac{\|\bar{x}^i - u^i\|}{\|w^i - u^i\|} = \frac{|((v, v^i))|}{\|v\| \|v^i\|} = |\cos(v, v^i)|$$

where the Hilbertian cosine is defined by

$$\cos(x, y) = \frac{((x, y))}{\|x\| \|y\|} \in [-1, +1].$$

If the entries of the matrix J are denoted by a^{hk} ($1 \leq h, k \leq l$), then the component $\bar{p}_h$ of $\bar{p}$ is equal to

$$(39) \quad \bar{p}_h = \alpha \sum_{k=1}^l a^{hk} v_k.$$

This implies the "law of supply and demand": We have that $d\bar{p}_h/dv_h = \alpha a^{hh} > 0$ since the entries a^{hh} of J are positive. Hence $\bar{p}_h$ increases with respect to the aggregate net demand v_h and the aggregate objective of consumption $u_h = \sum_{j=1}^J u_h^j$, while it decreases with respect to the aggregate endowment (supply) $w_h = \sum_{i \in N} w_h^i$. If $\xi \neq 0$, $\xi \in \mathbf{R}^l$ plays the role of a "numéraire", i.e. the price is normalized by the condition $\langle p, \xi \rangle = 1$. Then the Walras price $\bar{p}$ is unique and define:

$$(40) \quad \bar{p} = \frac{1}{\langle Jv, \xi \rangle} Jv. \quad \square$$

8.2. Existence of a Walras equilibrium

In this section we prove the existence of a Walras equilibrium when the demand correspondences are upper semi-continuous.

For this purpose we require an abstract theorem on the surjectivity of correspondences.

8.2.1. Existence of a Walras pre-equilibrium

Theorem 1. Suppose that the consumption sets R^i satisfy

- (1) R^i is closed, convex and bounded below
that the subset Y of available commodities satisfies
- (2) Y is closed, convex and $-\mathbf{R}_+^l$ is its recession cone,
and
- (3) $\exists w \in Y$ such that $(Y - w) \cap \mathbf{R}_+^l = \{0\}$

Let $\xi \in \mathbf{R}_+^l$ be a numéraire and $P = \{p \in \mathbf{R}_+^l \text{ such that } \langle p, \xi \rangle = 1\}$. Suppose that the demand correspondences D^i and the income function r_i satisfy

- (4) $\forall i \in N, p \in P \rightarrow D_i(p, r_i(p))$ is upper-semi-continuous
with non-empty closed convex images.

Then there exists a Walras pre-equilibrium $\{\bar{x}, \bar{p}\} \in \mathbf{R}_+^{l*} \times P$.

Remark. Assumption (3) is, by Proposition 3.2.7, a sufficient condition for the map

$$(5) \quad \{x^1, \dots, x^n, y\} \in R^N \times Y \mapsto \sum_{i \in N} x^i - y \in \mathbf{R}^I \text{ to be proper}$$

We shall actually prove that Theorem 1 holds when assumption (3) is replaced by assumption (5).

Proof. It is clear that $\bar{p}$ is a Walras price if and only if $0 \in S(\bar{p})$, where

$$(6) \quad S(p) = \sum_{i \in N} D_i(p, r_i(p)) - Y.$$

Since the subsets $D_i(p, r_i(p))$ and Y are closed, we deduce from (5) that $S(p)$ is closed. It is convex (being the sum of convex subsets) and satisfies $S(p) = S(p) + \mathbf{R}_+^I$ since $Y = Y - \mathbf{R}_+^I$. Thus property (5) implies that:

$$(7) \quad \forall p \in P, \quad S(p) = S(p) + \mathbf{R}_+^I \text{ is closed and convex.}$$

Furthermore, the upper semi-continuity of the correspondence $p \mapsto D_i(p, r_i(p))$ implies that the functions $p \mapsto \sigma^*(D_i(p, r_i(p)), q)$ are upper semi-continuous (see Proposition 2.5.1).

Therefore,

$$(8) \quad \begin{aligned} & (i) \quad \forall q \in \mathbf{R}^{I*}, p \mapsto \sigma^*(S(p); q) \text{ is upper semi-continuous,} \\ & (ii) \quad S(p) \text{ is non-empty closed, convex and satisfies } S(p) = S(p) + \mathbf{R}_+^I. \end{aligned}$$

Let $p \in P$ and z be any point of $S(p)$.

Since $z = \sum_{i \in N} x^i - y$ where $x^i \in D_i(p, r_i(p))$ and $y \in Y$, we deduce that $\langle p, z \rangle \leq \sum_{i \in N} r_i(p) - \langle p, y \rangle \leq r(p) - \langle p, y \rangle$. Then

$$\inf_{z \in S(p)} \langle p, z \rangle \leq r(p) - \sup_{y \in Y} \langle p, y \rangle = 0.$$

Therefore, we have proved that

$$(9) \quad \forall p \in P, \quad \sigma^b(S(p), p) \leq 0.$$

Finally, since $\xi \in \mathbf{R}_+^I$ and $-\mathbf{R}_+^I$ is the recession cone of Y , Theorem 1.4.3 implies that

$$(10) \quad P = \{p \in \mathbf{R}_+^{I*} \text{ such that } \langle p, \xi \rangle = 1\} \text{ is convex and compact.}$$

The existence of $\bar{p} \in P$ such that $0 \in S(\bar{p})$ follows from the following abstract existence theorem.

8.2.2. Surjectivity of correspondences: the Debreu–Gale–Nikaido theorem

Theorem 2. *Let U and U^* be two paired spaces. Suppose that*

$$(11) \quad P \text{ is a convex compact subset of } U^* \text{ which does not contain } 0$$

and $P^+ \subset U$ is its polar cone.

Let $S : P \rightarrow U$ be a correspondence satisfying

$$(12) \quad \begin{cases} \text{(i)} \quad \forall q \in U^*, p \mapsto \sigma^*(S(p); q) \text{ is upper semi-continuous}^1, \\ \text{(ii)} \quad \forall p \in P, S(p) + P^+ \text{ is non-empty, closed and convex,} \\ \text{(iii)} \quad \forall p \in P, \sigma^b(S(p); p) = \inf_{y \in S(p)} \langle p, y \rangle \leq 0. \end{cases}$$

Then there exists $\bar{p} \in P$ such that

$$(13) \quad S(\bar{p}) \cap -P^+ \neq \emptyset \text{ (i.e., } 0 \in S(\bar{p}) + P^+).$$

Proof. We introduce the function φ defined on $P \times P$ by

$$(14) \quad \varphi(p, q) = \sigma^b(S(p), q).$$

This function is concave with respect to q , lower semi-continuous with respect to p (since the correspondence $p \mapsto S(p)$ is upper hemi-continuous) and satisfies $\varphi(p, p) \leq 0$ for all $p \in P$. Since P is convex and compact, the Ky-Fan theorem implies the existence of $\bar{p} \in P$ such that

$$(15) \quad \forall q \in P, \quad \varphi(\bar{p}, q) = \sigma^b(S(\bar{p}), q) \leq 0$$

(see Theorem 6.1.3).

This implies that $0 \in \overline{\text{co}}(S(\bar{p}) + P^+) = S(\bar{p}) + P^+$ (since $S(\bar{p}) + P^+$ is closed and convex). This means that $S(\bar{p}) \cap -P^+ \neq \emptyset$. $\square$

***Remark.** We can relax the assumption that P is compact and assume the following property instead.

$$(16) \quad \begin{array}{l} \text{There exists } q_0 \in P \text{ such that, for any } \lambda, \text{ the subset of elements} \\ p \in P \text{ such that } S(p) \text{ intersects the half-space } H(q_0, \lambda) = \{z \in U \\ \text{such that } \langle q_0, z \rangle \leq \lambda\} \text{ is relatively compact.} \end{array}$$

Proposition 1. *Suppose that*

$$(17) \quad P \text{ is closed convex subset of } U^*$$

and that $S : P \rightarrow U$ is a correspondence satisfying assumption (12) of Theorem 2 and property (16). Then there exists a solution $\bar{p}$ of (13).

¹ We recall that such a correspondence S satisfying (12(i)) is said to be “upper hemi-continuous” (see Definition 2.5.2.).

Proof. Property (16) amounts to saying that the function $p \mapsto \varphi(p, q^0) = \inf_{z \in S(p)} \langle q^0, z \rangle$ is lower semi-compact, since the subset of elements $p \in P$ such that $S(p) \cap H(q^0, \lambda) \neq \emptyset$ is just the subset $\{p \in P \text{ such that } \sigma^b(S(p), q^0) = \varphi(p, q^0) \leq \lambda\}$.

Furthermore, φ is lower semi-continuous with respect to p and concave with respect to q . Therefore, Theorem 6.1.2 implies that there exists $\bar{p} \in P$ such that

$$\begin{aligned} \sup_q \varphi(\bar{p}, q) &\leq \inf_{D \in \mathcal{D}(P, P)} \sup_{q \in P} \varphi(D(q), q) \\ &\leq \sup_q \varphi(q, q) = \sup_q \sigma^b(S(q), q) \leq 0. \end{aligned}$$

This implies that $0 \in S(\bar{p}) + P^+$. $\square$

8.3. Demand correspondences defined by loss functions

In this section we investigate the case when the demand correspondences are defined by

$$D_i(p, r) = \{x \in B_i(p, r) \text{ such that } f_i(x, p) = \min_{y \in B_i(p, r)} f_i(y, p)\}.$$

To prove the upper semi-continuity of such a demand correspondence, we need the compactness of the consumption sets R^i . Therefore, we have to “compactify” the economy. For this purpose we introduce an assumption implying that the map $\{x^1, \dots, x^n, y\} \mapsto \sum_{i=1}^n x^i - y$ is *proper*. Hence we construct a new economy in which the consumption subsets R^i are compact and the assumptions of the existence Theorem 2.1 hold. We check that these two economies are equivalent in the sense that their subsets of equilibria coincide.

8.3.1. Statement of the existence theorem

Consider the demand correspondences D_i obtained by assuming that consumer i chooses a commodity in the budget set B_i by achieving his minimal loss. Specifically, we assume that each consumer chooses a commodity according to a loss function

$$(1) \quad f_i : R^i \times P \mapsto \mathbf{R}$$

which can be indexed by the price prevailing in the economy.

Then we define D_i by

$$(2) \quad D_i(p, r) = \{x \in B_i(p, r) \text{ such that } f_i(x, p) = \min_{y \in B_i(p, r)} f_i(y, p)\}.$$

In such a situation, we shall summarize the features describing the economy by

$$(3) \quad \{R^i, Y(i), f_i\}_{i \in N}$$

since demand correspondences D_i depend upon loss functions f_i . We begin by stating an existence theorem for a Walras equilibrium.

Theorem 1. *Suppose that*

$$(4) \quad \text{the consumption sets } R^i \text{ are closed convex and bounded below}$$

that

$$(5) \quad \text{the initial endowments } Y(i) \text{ are closed, convex and } -R^i_+ \text{ is their recession cone.}$$

and that

$$(6) \quad \forall i, \quad 0 \in R^i - \text{Int } Y(i).$$

Suppose also that

$$(7) \quad \text{the map } \{x, y\} \in R^N \times Y \mapsto \sum_{i \in N} x^i - y \in R^I \text{ is proper}$$

and that the loss functions f_i satisfy

$$(8) \quad \begin{cases} \text{(i) } \forall p \in P, x^i \mapsto f_i(x^i, p) \text{ is convex} \\ \text{(ii) } f_i \text{ is continuous on } R^I \times P. \end{cases}$$

Suppose furthermore that the loss functions f^i satisfy the non-satiation property

$$(9) \quad \begin{cases} \forall p \in P \text{ and } \forall x^i \in R^i \text{ there exists } y^i \in R^i \text{ such that } f_i(y^i, p) < \\ < f_i(x^i, p). \end{cases}$$

Then there exists a Walras equilibrium $\{\bar{x}, \bar{p}\}$.

The proof of this theorem will follow from several preliminary results. To apply Theorem 2.1, we first need to investigate under what conditions the demand correspondences D_i are upper semi-continuous. We will see that this requires compactness properties (i.e. that R^i be compact and that $Y(i) = Y_0(i) - R^i_+$ where $Y_0(i)$ is compact). We therefore need to reduce the initial economy to an equivalent "compact" economy (in the sense that the sets of Walras equilibria coincide). We begin by studying the budget and demand correspondences.

8.3.2. Upper semi-continuity of the demand correspondence

Proposition 1. Suppose that

$$(10) \quad \begin{cases} \text{(i) the income function } r_i \text{ is continuous on } P, \\ \text{(ii) } \forall p \in P, \exists \tilde{x}^i \in R^i \text{ such that } \langle p, \tilde{x}^i \rangle < r^i(p). \end{cases}$$

Then the budget correspondence $p \mapsto B(p, r_i(p))$ has non-empty closed convex images and is closed and lower semi-continuous.

If we assume furthermore that

$$(11) \quad \begin{cases} \text{(i) } R^i \text{ is compact,} \\ \text{(ii) the loss function } f_i : R^i \times P \mapsto \mathbf{R} \text{ is continuous,} \end{cases}$$

then the demand correspondence $p \mapsto D_i(p, r_i(p))$ is upper semi-continuous.

Proof. (a) The budget subsets $B_i(p, r_i(p))$ are non-empty by (10(ii)). The graph of the correspondence $p \mapsto B_i(p, r_i(p))$ is closed since it is the subset of pairs $\{p, x\}$ satisfying $\langle p, x \rangle - r_i(p) \leq 0$ and $p \mapsto r_i(p)$ is upper semi-continuous by assumption (10(i)).

(b) The budget correspondence is lower semi-continuous because, if $p_0 \in P$, $x_0 \in B_i(p_0, r_i(p_0))$ are fixed and $N(x_0)$ is a neighborhood of x_0 , we can construct $x_\theta \in N(x_0)$ and a neighborhood $N(p_0)$ of p_0 such that $x_\theta \in B_i(p, r_i(p))$ for all $p \in N(p_0)$.

For this construction, let $\tilde{x}^i \in R^i$ be defined by assumption (10(ii)). Then

$$(12) \quad \langle p_0, \tilde{x}^i \rangle - r^i(p_0) = -c_0 < 0.$$

Also, there exists $\theta > 0$ such that $x_\theta = \theta \tilde{x}^i + (1 - \theta)x_0$ belongs to $N(x_0)$. Since $\langle p_0, x_0 \rangle - r_i(p_0) \leq 0$, we deduce that $\langle p_0, x_\theta \rangle - r_i(p_0) \leq -c_0\theta$.

Take $\varepsilon = \frac{1}{4}c_0\theta$. Since r_i is lower semi-continuous, there exists a neighborhood $N(p_0)$ of p_0 such that, for, all $p \in N(p_0)$,

$$\langle p, x \rangle - r_i(p) \leq \langle p_0, x_\theta \rangle - r_i(p_0) + \varepsilon + \langle p - p_0, x_\theta \rangle \leq -\frac{1}{2}c_0\theta < 0.$$

This implies that x_θ belongs to $B(p, r_i(p))$ whenever $p \in N(p_0)$.

(c) If we assume furthermore that R^i is compact, then $B_i(p, r_i(p))$ stays in the compact set R^i and thus, is upper semi-continuous by Proposition 2.5.3. Thus it is continuous. Since the loss function f^i is continuous, we deduce from Theorem 2.5.3 that the demand correspondence is upper semi-continuous. $\square$

Note that the income function $r_i = \sigma^*(Y(i); \cdot)$ is continuous on P whenever $Y(i) = Y_0(i) - \mathbf{R}_+^I$ and $Y_0(i)$ is compact. We therefore obtain the following existence result.

Proposition 2. *Suppose that*

$$(13) \quad \begin{cases} \text{(i) the consumption subsets } R^i \text{ are compact and convex,} \\ \text{(ii) } Y(i) = Y_0(i) - \mathbf{R}_+^l \text{ where } Y_0(i) \text{ are compact,} \end{cases}$$

$$(14) \quad \text{the loss functions } f_i : R^i \times P \rightarrow \mathbf{R} \text{ are continuous and convex with respect to } x^i,$$

and

$$(15) \quad \forall i, \quad 0 \in R^i - \text{Int } Y(i).$$

Then there exists a Walras pre-equilibrium.

Proof. This is a straightforward consequence of Theorem 2.1 and Proposition 1, since (15) implies that there exists $\tilde{x}^i \in R^i$ such that $\langle p, \tilde{x}^i \rangle < r_i(p)$ for all $p \in P$. $\square$

We shall say that an economy $\{R^i, Y(i), f_i\}$ satisfying (13) is a *compact economy*. The question arises as to whether any economy is equivalent to a compact economy in the sense that the subsets of Walras equilibria coincide.

8.3.3. Compactification of an economy

Suppose that

$$(7) \quad \text{the map } \{x^1, \dots, x^n, y\} \in \prod_{i=1}^n R^i \times Y \mapsto \sum_{i=1}^n x^i - y \in \mathbf{R}^l \text{ is proper.}$$

Then the subset

$$(16) \quad X(N) = \left\{ x \in \prod_{i=1}^n R^i \text{ such that } \sum_{i=1}^n x^i \in Y \right\}$$

is compact. Therefore, the projections $\pi_i X(N)$ of $X(N)$ onto the factor spaces R^i are compact.

If B is a closed ball of positive radius, then $\pi_i X(N) + B$ is also compact. We set

$$(17) \quad \begin{cases} \tilde{R}^i = R^i \cap (\pi_i X(N) + B); & \tilde{Y}_0(i) = Y(i) \cap (\pi_i X(N) + B); \\ \tilde{Y}(i) = \tilde{Y}_0(i) - \mathbf{R}_+^l; & \tilde{Y} = \sum_{i \in N} \tilde{Y}(i) = Y \cap \sum_{i \in N} (\pi_i X(N) + B). \end{cases}$$

Definition 1. Suppose that (7) holds. Then we shall say that $\{\tilde{R}^i, \tilde{Y}(i), f_i\}_{i \in N}$ is the “*compactified economy*” of $\{R^i, Y(i), f_i\}_{i \in N}$.

We denote the “*compactified budget sets*” by

$$(18) \quad \tilde{B}(p, r) = \{x \in \tilde{R}^i \text{ such that } \langle p, x \rangle \leq r\}$$

and the “compactified demand subsets” by

$$(19) \quad \tilde{D}_i(p, r) = \{x \in \tilde{B}_i(p, r) \text{ such that } f_i(x, p) = \min_{y \in \tilde{B}_i(p, r)} f_i(y, p)\}.$$

The “compactified income functions”

$$(20) \quad \tilde{r}_i(p) = \sigma^*(\tilde{Y}(i); p) \leq r_i(p)$$

are continuous since $\tilde{Y}_0(i)$ is compact for all $i \in N$.

It is clear that any Walras pre-equilibrium $\{\bar{x}, \bar{p}\} \in \prod_{i=1}^n R^i \times P$ of the economy $\{X^i, Y(i), f_i\}_{i \in N}$ is a Walras pre-equilibrium of the compactified economy.

Proposition 3. Suppose that (7) holds and that

$$(21) \quad \text{the subsets } R^i, Y(i) \text{ and the loss functions } x^i \mapsto f_i(x^i, p) \text{ are convex.}$$

Then any Walras equilibrium of the compactified economy $\{\tilde{R}_i, \tilde{Y}(i), f_i\}_{i \in N}$ is a Walras equilibrium of the initial economy $\{R_i, Y(i), f_i\}_{i \in N}$.

Proof. Let $\{\bar{x}^1, \dots, \bar{x}^n, \bar{p}\}$ be a Walras equilibrium of the compactified economy. Then $\bar{x}^i \in \tilde{R}^i \subset R^i$ for all i and $\bar{y} = \sum_{i=1}^n \bar{x}^i \in \tilde{Y} \subset Y$. Thus $\bar{x} \in X(N)$. This implies that

$$(22) \quad \forall i \in N, \quad \bar{x}^i \in \pi_i X(N).$$

Furthermore, we know that $\langle \bar{p}, \bar{y} \rangle = \tilde{r}(\bar{p})$.

(a) We prove that $\langle \bar{p}, \bar{y} \rangle = r(\bar{p})$. If this is false, there exists $y \in Y$ such that $\langle \bar{p}, y \rangle > \langle \bar{p}, \bar{y} \rangle$. Let $y_\theta = (1 - \theta)\bar{y} + \theta y$. Since Y is convex and $\bar{y} \in Y$, it follows that $y_\theta \in Y$. Since $y = \sum_{i \in N} y^i$ where $y^i \in Y(i)$ and $\bar{x}^i \in \pi_i X(N)$, $\bar{x}^i + \theta(y^i - \bar{x}^i) \in Y(i) \cap (\pi_i X(N) + B) = \tilde{Y}_0(i)$ for θ small enough. Hence $y_\theta = \sum_{i \in N} (\bar{x}^i + \theta(y^i - \bar{x}^i)) \in \tilde{Y}$. On the other hand, $\langle \bar{p}, y_\theta \rangle = (1 - \theta)\langle \bar{p}, \bar{y} \rangle + \theta \langle \bar{p}, y \rangle > \langle \bar{p}, \bar{y} \rangle$. This is a contradiction of the fact that $\langle \bar{p}, \bar{y} \rangle$ maximizes $\langle \bar{p}, \cdot \rangle$ over $\tilde{Y}$ (i.e. that $\langle \bar{p}, \bar{y} \rangle = \tilde{r}(\bar{p})$).

(b) We deduce that $\forall i \in N, r_i(\bar{p}) = \tilde{r}_i(\bar{p})$. Since $\langle \bar{p}, \bar{x}^i \rangle \leq \tilde{r}_i(\bar{p})$ and $\langle \bar{p}, \bar{y} \rangle = \tilde{r}(\bar{p}) = r(\bar{p}) = \sum_{i=1}^n r_i(\bar{p})$, we obtain that

$$\sum_{i=1}^n (\tilde{r}_i(\bar{p}) - r_i(\bar{p})) = 0.$$

But $\tilde{r}_i(p) \leq r_i(p)$ for all i . This implies that

$$\tilde{r}_i(\bar{p}) = r_i(\bar{p}) \quad \text{for all } i.$$

(c) Therefore $\bar{x}^i \in \bar{D}^i(\bar{p}, r_i(\bar{p})) = \bar{D}_i(\bar{p}, r_i(\bar{p}))$. It remains to prove that $\bar{x}^i$ belongs to $D_i(\bar{p}, r_i(\bar{p}))$.

To prove this, suppose that $\bar{x}^i \notin D_i(\bar{p}, r_i(\bar{p}))$.

Then there exists $x^i \in D_i(\bar{p}, r_i(\bar{p}))$ such that

$$(23) \quad f_i(x^i, \bar{p}) < f_i(\bar{x}^i, \bar{p}).$$

Therefore, for all $\theta \in]0, 1[$, $x_\theta^i = (1 - \theta)\bar{x}^i + \theta x^i$ satisfies the budget constraint $\langle \bar{p}, x_\theta^i \rangle \leq r_i(\bar{p})$.

For θ small enough, x_θ^i belongs to $\pi_i X(N) + B$ because $\bar{x}^i \in \pi_i X(N)$. Since x^i and $\bar{x}^i$ belong to the convex set R^i , x_θ^i also belongs to R^i . Therefore $x_\theta^i \in \bar{B}_i(\bar{p}, r_i(\bar{p}))$ for θ small enough.

But, since f_i is convex with respect to x^i , we deduce the following contradiction

$$(24) \quad f_i(x_\theta^i, \bar{p}) \leq (1 - \theta)f_i(\bar{x}^i, \bar{p}) + \theta f_i(x^i, \bar{p}) < f_i(\bar{x}^i, \bar{p}). \quad \square$$

Proposition 4. *Suppose that (7) holds. If the property*

$$(25) \quad \forall i \in N, \quad 0 \in R^i - \text{Int } Y(i)$$

holds in the original economy, then it also holds in the compactified one, i.e.

$$(26) \quad \forall i \in N, \quad 0 \in \bar{R}^i - \text{Int } \bar{Y}(i).$$

Proof. Property (25) means that, $\forall i$, there exists $\bar{x}^i \in R^i \cap \text{Int } Y(i) \subset R^i \cap Y(i)$. Thus $\sum_{i=1}^n \bar{x}^i \in Y$, i.e. $\bar{x} = (\bar{x}^1, \dots, \bar{x}^n) \in X(N)$. This implies that $\bar{x}^i \in \pi_i X(N)$. Hence $\bar{x}^i \in \bar{R}^i$. On the other hand, $\bar{x}^i + B(\eta) \subset Y(i)$ where $B(\eta)$ is a ball of sufficiently small radius η . Since $\bar{x}^i + B(\eta) \subset \pi_i X(N) + B$ if η is small enough, then $\bar{x}^i + B(\eta) \subset \bar{Y}(i)$, i.e. $\bar{x}^i$ belongs to the interior of $\bar{Y}(i)$.

8.3.4. Proof of the existence of a Walras equilibrium

Assumptions (4), (5), (8(i)) and (7) and Proposition 3 imply that we can replace the initial economy $\{R^i, Y(i), f_i\}$ by the compactified economy $\{\bar{R}^i, \bar{Y}(i), f_i\}$. Assumptions (6) and (8) and Proposition 2 imply the existence of a Walras pre-equilibrium of the compactified economy $\{\bar{R}^i, \bar{Y}(i), f_i\}$.

Assumptions (8) and (9) imply that $\bar{x}^i \in D_i(\bar{x}, r_i(\bar{p}))$ satisfies $\langle \bar{p}, \bar{x}^i \rangle = r_i(\bar{p})$ by Proposition 2.1.8.

Hence there exists a Walras equilibrium of the compactified economy, which is a Walras equilibrium of the initial economy by Proposition 3.

*8.4. Economies with producers

In this section we introduce m producers (labelled $j = 1, 2, \dots, m$) who are described by proper lower semi-continuous convex cost functions g_j defined on $\mathbf{R}^l$. The (maximum) profit functions are defined by

$$g^*(p) = \sup_{z \in \mathbf{R}^l} [\langle p, z \rangle - g_j(z)].$$

Hence the income $r_i(p) = \langle p, w^i \rangle + \sum_{j=1}^m \theta_j^i g_j^*(p)$ of i is the sum of the value of his endowment w^i and his shares $\theta_j^i g_j^*(p)$ in the profits of firms j . We shall prove the existence of $\{\bar{x}^1, \dots, \bar{x}^n, \bar{z}^1, \dots, \bar{z}^m, \bar{p}\}$ such that $\bar{x}^i \in D_i(\bar{p}, r_i(\bar{p}))$ for all i , $\bar{z}^j$ maximizes the profit $\langle \bar{p}, z \rangle - g_j(z)$ of the j^{th} firm for all j and $\sum_{i=1}^n \bar{x}^i \leq w + \sum_{j=1}^m \bar{z}^j$.

For this purpose, we begin by "compactifying" the economy and apply the abstract Theorem 2.2 to the equivalent compactified economy.

8.4.1. Description of the model

We now introduce into the model m firms (or producers) j described by their cost functions

$$(1) \quad g_j : \mathbf{R}^l \mapsto]-\infty, +\infty]$$

whose domains $Z^j = \{z \in \mathbf{R}^l \text{ such that } g^j(z) < \infty\}$ are assumed to be non-empty.

For any $p \in \mathbf{R}^{l*}$, we regard

$$(2) \quad g_j^*(p) = \sup_{z \in Z^j} [\langle p, z \rangle - g_j(z)]$$

as the profit function of producer j .

In this case, if each consumer i is endowed with a vector w^i , the income function r_i of consumer i is defined by

$$(3) \quad r_i(p) = \langle p, w^i \rangle + \sum_{j=1}^m \theta_j^i g_j^*(p),$$

i.e. $r_i(p)$ is the sum of the value of his initial endowment w^i and his shares $\theta_j^i g_j^*(p)$ in the profits of firms j .

We summarize the description of an economy with producers in the notation

$$\{R^i, f_i, w^i, \{\theta_j^i\}, g_j\}_{i,j}.$$

Definition 1. We shall say a triple $\{\bar{x}, \bar{z}, \bar{p}\} \in \prod_{i=1}^n R^i \times \prod_{j=1}^n Z^j \times R^{l*}$ is a *Walras pre-equilibrium* if

$$(4) \quad \begin{cases} (i) \forall i = 1, \dots, n, \bar{x}^i \in D_i(\bar{p}, r_i(\bar{p})), \\ (ii) \forall j = 1, \dots, n, \bar{z}^j \text{ maximizes the profit } \langle \bar{p}, \cdot \rangle - g_j(\cdot). \\ (iii) \sum_{i=1}^n \bar{x}^i \leq \sum \bar{z}^j + w. \end{cases}$$

We shall say that the triple is a *Walras equilibrium* if, in addition,

$$(5) \quad \forall i, \quad \langle \bar{p}, \bar{x}^i \rangle = \langle \bar{p}, w^i \rangle + \sum_{j=1}^m \theta_j^i g_j^*(\bar{p}).$$

Remark. The usual models deal with the case where the cost function is

$$(6) \quad g_j = \psi_{Z_j}, \quad \text{the indicator of } Z^j$$

and thus, where

$$(7) \quad g_j^*(p) = \sigma^*(Z^j; p).$$

Otherwise, g_j can be regarded as special fixed costs, anti-pollution taxes, etc.

8.4.2. Statement of the existence theorem

Theorem 1. Suppose that

(8) the consumption sets R^i are closed, convex and bounded from below, and that the map

$$(9) \quad \{x, z, u\} \in \prod_{i=1}^n R^i \times \prod_{j=1}^m Z^j \times R^l_+ \mapsto \sum_{i=1}^n x^i - \sum_{j=1}^m z^j + u \in R^l$$

is proper.

Suppose also that the cost functions g_j satisfy

$$(10) \quad \begin{cases} (i) g_j \text{ is a convex lower semi-continuous function defined on the closed set } Z^j, \\ (ii) 0 = g_j(0) = \min_{z \in Z^j} g_j(z). \end{cases}$$

that the endowments w^i satisfy

$$(11) \quad \forall i, \quad \text{there exists } \tilde{x}^i \in R^i \text{ such that } \tilde{x}^i \ll w^i$$

and that the loss functions f_i satisfy

$$(12) \quad \begin{cases} (i) \forall p \in P, x^i \mapsto f_i(x^i, p) \text{ is convex,} \\ (ii) f_i \text{ is continuous on } R^i \times P. \end{cases}$$

Assume also that the loss functions f_i satisfy the non-satiation property

$$(13) \quad \forall p \in P \text{ and } \forall x^i \in R^i, \text{ there exists } y^i \in R^i \text{ such that } f_i(y^i, p) < f_i(x^i, p).$$

Then there exists a Walras equilibrium $\{\bar{x}, \bar{z}, \bar{p}\}$.

Remark. Recall that the compactness assumption (9) is satisfied if

$$(14) \quad \text{the production sets } Z^j \text{ are bounded above}$$

or, if

$$(15) \quad \begin{cases} \text{(i) } \forall j = 1, \dots, m, Z^j \text{ is closed and } 0 \in Z^j, \\ \text{(ii) } Z = \sum_{j=1}^m Z^j \text{ is closed and convex,} \\ \text{(iii) } Z \cap -Z = \{0\} \text{ and } Z \cap R_+^l = \{0\}. \end{cases}$$

(See Proposition 4.2.9.)

Remark. Assumption (10) means that producing nothing is costless ($g_j(0) = 0$) and that all costs $g_j(z)$ are non-negative (i.e. are true costs).

Assumption (10) is satisfied in the case where there are no costs, i.e. in the case where

$$(16) \quad g_j(z) = \psi(Z_j; z), \quad g_j^*(p) = \sigma^*(Z^j; p).$$

Remark. Before proving Theorem 1, we state several preliminary results. We construct a compactified economy and show that it has a Walras equilibrium which is also a Walras equilibrium of the initial economy.

8.4.3. Compactification

Suppose that the map

$$(9) \quad \{x, z, v\} \in \prod_{i=1}^n R^i \times \prod_{j=1}^m Z^j \times R_+^l \mapsto \sum_{i=1}^n x^i - \sum_{j=1}^m y^j + v \in R^l$$

is proper.

Then the subset

$$(17) \quad U_w = \left\{ \{x, z\} \in \prod_{i=1}^n R^i \times \prod_{j=1}^m Z^j \text{ such that } \sum_{i=1}^n x^i - \sum_{j=1}^m z^j \leq w \right\}$$

is compact. Therefore we can replace the initial consumption sets R^i and production sets Z^j by the projections $\pi_i U_w$ and $\pi_j U_w$ onto R^i and Z^j respectively. If B is a closed ball of positive radius, then $\pi_i U_w + B$ and $\pi_j U_w + B$ are also compact.

We set

$$(18) \quad \tilde{R}^i = R^i \cap (\pi_i U_w + B), \quad \tilde{Z}^j = Z^j \cap (\pi_j U_w + B).$$

We replace the cost and profit functions g_j of firms j by the “compactified cost and profit functions”

$$(19) \quad \begin{cases} \text{(i) } \tilde{g}_j(z) = g_j(z) + \psi(\tilde{Z}_j^j; z), \\ \text{(ii) } \tilde{g}_j^*(p) = \sup_{z \in \tilde{Z}_j^j} [\langle p, z \rangle - g_j(z)]. \end{cases}$$

Definition 2. Suppose that (9) holds. We shall say that

$$\{\tilde{R}^i, f_i, w^i, \{\theta_j^i\}, \tilde{g}_j\}_{i,j} \text{ is the compactified economy of } \{R^i, f_i, w^i, \{\theta_j^i\}, g_j\}_{i,j}.$$

The compactified income functions r_i are defined by

$$(20) \quad r_i(p) = \langle p, w^i \rangle + \sum_{j=1}^m \theta_j^i \tilde{g}_j^*(p).$$

Consider the compactified budget sets and demand correspondences defined by

$$(21) \quad \tilde{B}(p, r) = \{x \in \tilde{R}^i \text{ such that } \langle p, x \rangle \leq r\}$$

and

$$(22) \quad \tilde{D}(p, r) = \{x \in \tilde{B}(p, r) \text{ such that } f_i(x, p) = \min_{y \in \tilde{B}(p, r)} f_i(y, p)\}.$$

It is clear that any Walras pre-equilibrium of the initial economy is a Walras pre-equilibrium of the compactified economy.

Proposition 1. Suppose that (9) and (13) hold and that

$$(23) \quad \begin{cases} \text{the subsets } R^i, Z^j, \text{ the cost functions } g_j \text{ and the loss functions} \\ x^i \rightarrow f_i(x^i, p) \text{ are convex} \end{cases}$$

Then any Walras equilibrium of the compactified economy is a Walras equilibrium of the initial economy.

Proof. Let $\{\bar{x}, \bar{z}, \bar{p}\}$ be a Walras equilibrium. Since

$$\sum_{i=1}^n \bar{x}^i - \sum_{j=1}^n \bar{z}^j \leq w = \sum_{i=1}^n w^i,$$

then

$$(24) \quad \bar{x}^i \in \pi_i U_w, \quad \bar{z}^j \in \pi_j U_w.$$

We prove that $\bar{z}^j$ maximizes the profit $\langle \bar{p}, \cdot \rangle - g_j(\cdot)$ over Z^j (instead of $\tilde{Z}_0^j$). If not, there exists $z^j \in Z^j$ such that

$$(25) \quad \langle \bar{p}, \bar{z}^j \rangle - g_j(\bar{z}^j) < \langle \bar{p}, z^j \rangle - g_j(z^j).$$

For any $\theta \in]0, 1]$, $z_\theta^j = (1-\theta)\bar{z}^j + \theta z^j$ belongs to Z^j . For θ small enough, $z_\theta^j = \bar{z}^j + \theta(z^j - \bar{z}^j)$ belongs to $\pi_j U_w + B$ since $\bar{z}^j \in \pi_j U_w$. Therefore z_θ^j belongs to $\tilde{Z}_0^j$ for θ small enough. The convexity of g_j implies that

$$(26) \quad \langle \bar{p}, \bar{z}^j \rangle - g_j(\bar{z}^j) < \langle \bar{p}, z_\theta^j \rangle - g_j(z_\theta^j).$$

This is a contradiction of the fact that $\bar{z}^j$ maximizes the profit over $\tilde{Z}^j$. This implies in particular that for all j , $\tilde{g}_j^*(\bar{p}) = g_j^*(\bar{p})$ and thus, that $r_i(\bar{p}) = \bar{r}_i(\bar{p})$ for all $i = 1, \dots, n$. We deduce that $\bar{x}^i \in D_i(\bar{p}, \bar{r}(\bar{p}))$ as in Proposition 2.3. $\square$

Proposition 2. Suppose that (9) holds. Then the compactified cost functions are continuous. If we suppose that $\inf_{z \in Z^j} g_j(z) \geq 0$ for all $j = 1, \dots, m$, then inequalities

$$(27) \quad \forall p \in \mathbb{R}^{I^*}, \quad \tilde{g}_j^*(p) \leq D\tilde{g}_j^*(p)(p)$$

hold for all j .

Proof. First, observe that

$$\tilde{g}_j^*(p) = \sup_{z \in \tilde{Z}_0^j} (\langle p, z \rangle - g_j(z)) \leq \sup_{z \in \tilde{Z}_0^j} \langle p, z \rangle \leq \sigma^*(\tilde{Z}_0^j; p)$$

since $g_j(z) \geq 0$ for all z . Since $\tilde{Z}_0^j$ is compact, $\sigma^*(\tilde{Z}_0^j; \cdot)$ is a continuous function. Therefore, $\tilde{g}_j^*$ being a convex function dominated by a continuous function, is a continuous function. Hence it is differentiable from the right and satisfies $\tilde{g}_j^*(p) \leq D\tilde{g}_j^*(p)(p) + \tilde{g}_j^*(0)$ (see Proposition 4.3.2). But $\tilde{g}_j^*(0) \leq g_j^*(0) = -g_j(0) = 0$. Thus (27) is satisfied. $\square$

Proposition 3. Suppose that

$$(28) \quad \begin{cases} \text{(i)} \quad \forall i, \text{ there exists } \bar{x}^i \in R^i \text{ such that } \bar{x}^i \ll w^i, \\ \text{(ii)} \quad \forall j, g_j(0) = 0. \end{cases}$$

Then

$$(29) \quad \forall j = 1, \dots, m, \forall p \in P, \quad \tilde{g}_j^*(p) \geq 0,$$

and

$$(30) \quad \forall i = j, \dots, n, \forall p \in P, \quad \langle p, \bar{x}^i \rangle < \bar{r}(p).$$

Proof. Since $\sum_{i=1}^n \tilde{x}^i - \sum_{j=1}^m 0 \leq w$, we have that, for all $j = 1, \dots, m$, $0 \in \pi_j U_w \subset \tilde{Z}_0^j$. Therefore, $\tilde{g}_j^*(p) \geq \langle p, 0 \rangle - g_j(0)$ for all p . Since $\tilde{x}^i \ll w^i$, we deduce that, for all $p \in P$,

$$\langle p, \tilde{x}^i \rangle < \langle p, w^i \rangle \leq \langle p, w^i \rangle + \sum_{j=1}^m \theta_j^i \tilde{g}_j^*(p) = r^i(p).$$

8.4.4. Proof of the existence of a Walras equilibrium

Proposition 4. Suppose that (9) and (28) hold. Then there exists a Walras pre-equilibrium of the compactified economy.

Proof. Indeed, $\bar{p}$ is a Walras pre-equilibrium price if and only if

$$(31) \quad \left(\sum_{i=1}^n \tilde{D}_i(\bar{p}; r_i(\bar{p})) - \sum_{j=1}^m \partial \tilde{g}_j^*(\bar{p}) - w \right) \cap -P^+ \neq 0.$$

Let S be the correspondence defined on the convex compact subset P by

$$(32) \quad S(p) = \sum_{i=1}^n \tilde{D}_i(p, r_i(p)) - \sum_{j=1}^m \partial \tilde{g}_j^*(p) - w.$$

The subsets $S(p)$ are non-empty, convex and compact.

Since the compactified profit functions $\tilde{g}_j^*$ are continuous, the compactified income functions r_i are also continuous. But there exists $\tilde{x}^i$ such that $\langle p, \tilde{x}^i \rangle < r_i(p)$ for all p and $\tilde{R}^i$ is compact. Hence the compactified demand correspondences are upper semi-continuous (see Proposition 2.1). Since the functions $\tilde{g}_j^*$ are continuous, the correspondences $\partial \tilde{g}_j^*$ are upper semi-continuous (see Proposition 4.1.8). Thus the correspondence S is upper semi-continuous.

Finally,

$$\begin{aligned} \sigma^b(S(p), p) &= \sum_{i=1}^n \sigma^b(\tilde{D}_i(p, r_i(p)), p) - \sum_{j=1}^m \sigma^*(\partial \tilde{g}_j^*(p), p) - \langle p, w \rangle \\ &\leq \sum_{i=1}^n r_i(p) - \sum_{j=1}^m \tilde{g}_j^*(p) - \langle p, w \rangle = 0 \end{aligned}$$

since $\sigma^*(\partial \tilde{g}_j^*(p); p) = D\tilde{g}_j^*(p)(p) \geq \tilde{g}_j^*(p)$ and $\sum_{i=1}^n r_i(p) = \langle p, w \rangle + \sum_{j=1}^m \tilde{g}_j^*(p)$. Hence Theorem 2.2 implies the existence of a price $\bar{p} \in P$ such that (31) holds, i.e. a Walras pre-equilibrium price $\bar{p}$. $\square$

Proof of Theorem 1. By Proposition 4, there exists a Walras pre-equilibrium of the compactified economy. It is actually a Walras equilibrium thanks to the non-satiation property (13) (see Proposition 2.5.8). Thus Proposition 1 implies that it is also a Walras equilibrium of the initial economy.

CHAPTER 9

NON-COOPERATIVE n -PERSON GAMES

This chapter is devoted to the study of non-cooperative n -person games with and without constraints and their applications to the non-cooperative Walras equilibria of an economic model with externalities. In passing, we prove the fundamental Brouwer and Kakutani fixed-point theorems (by means of the Ky-Fan theorem) and some selection theorems for fixed points. We begin by describing a game $\{X(N), F\}$ in strategic form, where $X(N) \subset \prod_{i=1}^n X^i$ is a subset of multistrategies and $F: X(N) \rightarrow \mathbf{R}^n$ is a multiloss operator. After definitions of the shadow minimum, the conservative values and multistrategies, we introduce the concept of a non-cooperative equilibrium $x \in X(N)$. For any player $i \in N$, this satisfies

$$f_i(x) = \inf_{\substack{y \in X(N) \\ \pi^i y = \pi^i x}} f_i(y)$$

where f_i is the i th component of F and π^i is the projector from $\prod_{i=1}^n X^i$ onto $\prod_{j \neq i} X^j$. It “models” the individual stability of a non-cooperative equilibrium (i.e. no player can obtain a smaller loss by switching choices under the assumption that the remaining players make no change in their strategies). We prove the Nash theorem for the existence of a non-cooperative equilibrium very simply by means of the Ky-Fan theorem.

We do not need the Ky-Fan theorem when the loss functions are quadratic. In this case, we are even able to exhibit explicit formulas when $X(N)$ is defined by linear equality constraints. We use these to obtain the Walras–Cournot equilibrium (thus generalizing the analysis of duopoly of Section 6.3). We then assume that the choice of the i th player is constrained by those of the remaining players. Given $\pi^i x \in \prod_{j \neq i} X^j$, player i is constrained to choose his strategy in a set $S_i(\pi^i x)$, where S_i is a correspondence mapping $\prod_{j \neq i} X^j$ into X^i . Hence we generalize the concept of non-cooperative equilibrium by requiring that, for all players i , $x^i \in S_i(\pi^i x)$ and that $f_i(x) = \min_{y^i \in S_i(\pi^i x); \pi^i y = \pi^i x} f_i(y)$. We notice that this problem amounts to finding x satisfying for some function φ :

$$\begin{cases} \text{(i) } x \in S(x) \\ \text{(ii) } \sup_{y \in S(x)} \varphi(x, y) \leq 0 \end{cases}$$

i.e. a selection of a fixed point x of a correspondence S mapping X into itself by a function φ mapping $X \times X$ into $\mathbb{R}$.

In particular, on taking $\varphi = 0$, this problem reduces to finding a fixed point of S . Hence we shall prove the Brouwer and Kakutani theorems. We end this section by applying the existence theorem for a non-cooperative equilibrium to find a non-cooperative Walras equilibrium of the fundamental economic model where the loss functions f_i depend not only upon the choice of the i th consumer's own consumption x^i , but also on the consumptions of the remaining consumers (which are sometimes called "externalities").

9.1. Existence of a non-cooperative equilibrium

We begin by describing n -person games in strategic form (or normal form) and by defining the conservative values and multistrategies. We then introduce non-cooperative equilibria and prove the Nash theorem. This asserts that, if the multistrategy subset $X(N) \subset \prod_{i=1}^n X^i$ is convex and compact and the loss functions f_i are continuous and convex with respect to the i th projection x^i of x , then there exists a non-cooperative equilibrium. We also study the stability properties with respect to perturbations of the loss functions by linear forms. Finally, we characterize the non-cooperative equilibria as solutions of variational inequalities when the loss functions are convex and differentiable with respect to the i th projection x^i of x .

9.1.1. Games described in strategic form

We denote by $N = \{1, 2, \dots, n\}$ the set of n players. We shall describe a n -person game in the *strategic* (or *normal*) form by strategy sets and loss functions for each player.

Specifically, we associate with each player i

- (1) a strategy set X^i

and denote by

- (2) $X^N = \prod_{i=1}^n X^i$ the set of multistrategies $x = \{x^1, \dots, x^n\}$.

When needed, we write $x^N = \{x^{N,1}, \dots, x^{N,n}\} \in X^N$ instead of using the notation $x = \{x^1, \dots, x^n\}$ in order to avoid confusion between elements of X^N and elements of a given factor X^i of the product.

The context will make clear whether the notation x denotes a multistrategy or a strategy.

We begin the description of the game by specifying

- (3) a multistrategy set $X(N) \subset X^N$ of feasible multistrategies x .

Next, the behavior of the i th player will be described by a loss function $f_i: X(N) \rightarrow \mathbf{R}$ associating with any multistrategy x a real number $f_i(x)$ measuring his loss. He prefers multistrategies yielding smaller losses.

It is useful to regard

- (4) $\mathbf{R}^n = \mathbf{R}^N = \prod_{i \in N} \mathbf{R}$ as the *multiloss space*.

The behavior of the n players is then described by the multiloss operator $F: X(N) \rightarrow \mathbf{R}^n$ defined by

- (5) $x \in X(N) \mapsto F(x) = \{f_1(x), \dots, f_n(x)\} \in \mathbf{R}^n$.

Definition 1. An n -person game $\{X(N), F\}$ is described in the strategic form by a multiloss operator F mapping a multistrategy set $X(N) \subset X^N$ into the multiloss space $\mathbf{R}^n$.

We shall meet two important particular cases:

- (a) when $X(N) = X^N$ is a product of strategy sets;
- (b) when the loss functions $f_i: X^i \rightarrow \mathbf{R}$ depend only upon strategies implemented by player i .

Definition 2. We shall say that the game is “*bounded below*” if, for any i , $\alpha_i = \inf_{x \in X(N)} f_i(x) > -\infty$. In this case, we shall say that the multiloss $\alpha = \{\alpha_1, \dots, \alpha_n\}$ is the “*shadow minimum*” of the game.

Note that a game is bounded below if and only if

- (6) $F(X) \subset \alpha + \mathbf{R}_+^n$.

If $\alpha = F(\bar{x})$ belongs to $F(X)$, the multistrategy $\bar{x}$ achieves the minimum of the loss function for *each* player i . In this case, $\bar{x}$ is the natural solution concept.

This is seldom the case and we have to investigate other solution concepts.

9.1.2. Conservative values and multistrategies

We shall denote by $i^c = N - i$ the coalition adverse to player i . From the point of view of player i , the set X^N of multistrategies is split as follows:

- (7) $X^N = X^i \times X^{i^c}$ where $X^{i^c} = \prod_{j \neq i} X^j$.

If π^i and π^j denote the projectors from X^N onto X^i and X^j , we set $x^i = \pi^i x$ and $x^j = \pi^j x$. Player i will use the decomposition $x = \{x^i, x^j\} \in X^i \times X^j$ for a multistrategy $x \in X^N$, i.e. he will distinguish the component $x^i \in X^i$ on which he can act alone from the components $x^j \in X^j$ over which he has no control.

We regard

$$(8) \quad f_i^*(x^i) = \sup_{\substack{x \in X(N) \\ \pi^i x = x^i}} f_i(x)$$

as the *worst loss* achieved by player i when he implements $x^i \in X^i$.

Suppose that *each player i has no information* about the choice of the other players. If cautious, each player i will choose a strategy x^{i*} which minimizes the worst loss function f_i over X^i .

Definition 3. We shall say that the vector v^* of components v_i^*

$$(9) \quad v_i^* = \inf_{x^i \in X^i} f_i^*(x^i)$$

is the “*conservative value*” of the game and that any strategy x^* such that

$$(10) \quad \forall i \in N, \quad f_i^*(x^{*i}) = v_i^*$$

is a “*conservative multistrategy*” (or a “*minimax multistrategy*”).

Remark. In fact, we use the conservative value as a “threat” functional, the understanding being that player i will *reject* any multistrategy x yielding a loss $f_i(x)$ larger than v_i^* .

The set of multistrategies accepted by all players (i.e. rejected by no players) is the set of $x \in X(N)$ such that $F(x) \leq v^*$.

Proposition 1. Suppose that each strategy set X^i is a subset of a topological space U^i . If the extended loss functions $f_{i, X(N)}$ are lower semi-continuous and lower semi-compact, then there exists a conservative multistrategy x^* .

Proof. This is left as an exercise. $\square$

9.1.3. Non-cooperative equilibria

To define conservative strategies, we assumed that each player can select a specific strategy x^i independently of the remaining players. In doing so, he cannot take advantage of the choice x^j of the complementary coalition x^j .

We shall consider another type of behavior in which any player i may vary his strategy as a function of the complementary coalition's choice.

It is as though a player i announces his intention to play, forces the complementary coalition $\{i\}$ to move first, and then responds.

With such a rule, a multistrategy will be in "equilibrium" if no player i can obtain a lower loss by making an alternative choice under the assumption that the remaining players make no change in their strategies.

In other words, given the complementary coalition's choice x^i , player i responds by playing a strategy x^i which minimizes $y^i \rightarrow f_i(y^i, x^i)$, i.e.

$$(11) \quad f_i(x^i, x^i) = f_i^b(x) = \inf_{\substack{y \in X(N) \\ \pi^i y = x^i}} f_i(y).$$

Therefore, a multistrategy x will be an equilibrium if eqs. (11) hold for any player i .

Definition 4. We shall say that a multistrategy $x \in X(N)$ is a "non-cooperative equilibrium" (or a "Nash-equilibrium") if

$$(12) \quad \forall i = 1, \dots, n, \quad f_i(x) = \min_{\substack{y \in X(N) \\ \pi^i y = x^i}} f_i(y).$$

In other words, non-cooperative equilibria are the solutions of the equation

$$(13) \quad F(x) - F^b(x) = 0$$

where $F^b: X(N) \rightarrow \mathbf{R}^n$ associates with any x the multiloss

$$F^b(x) = \{f_i^b(x), \dots, f_n^b(x)\}.$$

Note that $F(x) - F^b(x) \in \mathbf{R}_+^l$ for all x .

9.1.4. The Nash theorem

We introduce the function $\varphi: X(N) \times X(N) \rightarrow \mathbf{R}$ defined by

$$(14) \quad \varphi(x, y) = \sum_{i=1}^n [f_i(x) - f_i(y^i, x^i)].$$

Proposition 2. Suppose that $x \in X(N)$ satisfies

$$(15) \quad \sup_{y \in X(N)} \varphi(x, y) \leq 0.$$

Then x is a non-cooperative equilibrium. The converse is true when $X(N)$ is a product of strategy sets.

Proof. Let $x \in X(N)$ be a solution of (15) and let $y \in X(N)$ satisfy $\pi^j y = x^j$. Then $y^j = x^j$ for all $j \neq i$ and $\pi^i y = y^i$. Thus we can write the inequality $\varphi(x, y) \leq 0$ as

$$\begin{aligned} \sum_{j \neq i} (f_j(x) - f_j(x^i, x^j)) + (f_i(x) - f_i(y^i, x^j)) \\ = f_i(x) - f_i(y^i, x^j) = f_i(x) - f_i(y) \leq 0 \end{aligned}$$

Therefore $f_i(x) = \inf_{y \in X(N); \pi^j y = x^j} f_i(y) = f_i^b(x)$.

This being true for each i , x is a non-cooperative equilibrium.

Conversely, if $X(N)$ is a product of strategy sets, we obtain $\varphi(x, y) \leq 0$ by adding the inequalities $f_i(x) - f_i(y^i, x^j) \leq 0$ (which hold by the very definition of a non-cooperative equilibrium). $\square$

Proposition 2 suggests the use of the Ky-Fan theorem to prove the existence of a non-cooperative equilibrium.

Theorem 1 (Nash). *Suppose that the multistrategy set*

$$(16) \quad X(N) \text{ is a convex compact subset}$$

and that, for each player i , the loss function

$$(17) \quad f_i \text{ is continuous and } f_i(\cdot, x^j) \text{ is convex for all } x^j \in X^j.$$

Then there exists a non-cooperative equilibrium.

Proof. We check that the assumptions of the Ky-Fan theorem 7.1.3 are satisfied. In the first place, $X(N)$ is compact and convex. Secondly, φ is lower semi-continuous with respect to x , concave with respect to y and satisfies $\varphi(y, y) = \sum_{i=1}^n (f_i(y) - f_i(y)) = 0$. Thus there exists $x \in X(N)$ satisfying $\sup_{y \in X(N)} \varphi(x, y) \leq 0$. Thus x is a non-cooperative equilibrium by Proposition 2. $\square$

*9.1.5. Stability

Suppose that X^i is a subset of a topological vector space U^i . We introduce the "conjugate" functions defined by

$$(18) \quad \forall p_i \in U^{i*}, \quad f_i^*(p_i; x^j) = \sup_{\pi^j y = x^j} [\langle p_i, y^i \rangle - f_i(y^i, x^j)]$$

which are lower semi-continuous convex functions with respect to p_i .

Proposition 3. *If $x \in X(N)$ is a non-cooperative equilibrium, then*

$$(19) \quad \forall i \in N, \quad x^i \in \partial f_i(0; x^j).$$

The converse is true when $X(N) = X^N$ and the extended functions $y^i \in U^i \mapsto f_i(x^i(y^i, x^j))$ are convex and lower semi-continuous.

Proof. If $x \in X(N)$ is a non-cooperative equilibrium, then

$$\begin{aligned} f_i^*(0; x^j) &= -f_i(x^i, x^j) = -\langle q_i, x^i \rangle + \langle q_i, x^i \rangle - f_i(x) \\ &\leq \langle 0 - q_i, x^i \rangle + f_i^*(q_i; x^j) \end{aligned}$$

i.e. $x^i \in \partial f_i^*(0, x^j)$. Conversely, we use the fact that $x^i \in \partial f_i^*(0; x^j)$ if and only if x^i minimizes the function $y^i \mapsto f_i(y^i, x^j)$ over X^i (see Proposition 4.1.3). $\square$

*9.1.6. Associated variational inequalities

Suppose that

$$(20) \quad \forall x^j \in X^j, \text{ the functions } y^i \mapsto f_i(y^i, x^j) \text{ are convex and differentiable.}$$

Then any solution $x \in X(N)$ of (15) is a solution of the variational inequalities

$$(21) \quad \langle S(x), x - y \rangle = \sum_{i=1}^n \langle D_i f_i(x), x^i - y^i \rangle \leq 0$$

where we denote the derivative of f_i at $x = \{x^i, x^j\}$ in the direction $\{y^i, 0\}$ by

$$(22) \quad \langle D_i f_i(x), y^i \rangle = \lim_{\theta \rightarrow 0} \frac{f_i(x^i + \theta y^i, x^j) - f_i(x^i, x^j)}{\theta}$$

and define the map S from X^N into U^{N*} by

$$(23) \quad S(x) = \{D_1 f_1(x), \dots, D_n f_n(x)\}$$

(see Section 4.2). Therefore, Proposition 4.2.2 implies the following result.

Proposition 4. *Suppose that (20) holds. Any solution $x \in X(N)$ of the variational inequalities (21) is a non-cooperative equilibrium. The converse is true if $X(N)$ is a product of strategy sets.*

We will use this proposition for proving the existence of a non-cooperative equilibrium in the case of quadratic loss functions.

*9.2. Case of quadratic loss functions; application to Walras–Cournot equilibria

We do not have to use the Ky-Fan theorem to prove the existence of a non-cooperative equilibrium when the loss functions

$$f_i(x) = \frac{1}{2} \left\| x^i + \sum_{j \neq i} M_j^i x^j - u^i \right\|_i^2$$

are quadratic ($\|\cdot\|_i$ denotes the norm of a Hilbert space U^i , $u^i \in U^i$ are objectives, $M_j^i \in \mathcal{L}(U^j, U^i)$). A non-cooperative equilibrium exists if we assume that the operators M_j^i satisfy

$$\sum_{j,k=1}^n ((M_j^k x^j, x^k)) \geq c \sum_{j=1}^m \|x^j\|^2$$

$$\text{for all } x \in \prod_{j=1}^m U^j \quad (\text{where } c > 0)$$

and that

$$X(N) \text{ is a closed convex subset of } \prod_{i=1}^n U^i.$$

For this purpose, we use the Lions–Stampacchia theorem which states that a solution $x \in X$ of the variational inequalities

$$\langle Gx - p, x - y \rangle \leq 0 \quad \text{for all } y \in X$$

exists if $G \in \mathcal{L}(V, V^*)$ is V -elliptic and $X \subset V$ is a closed convex subset.

We even obtain explicit formulas when $X(N)$ is an affine subspace defined by linear equality constraints, i.e.

$$X(N) = \left\{ x \in \prod_{i=1}^n U^i \text{ such that } Lx = \sum_{i=1}^n L_i x^i = w \right\}.$$

In this case, we can write $x = M^{-1}(u - J^{-1}L^*p)$ where

$$p = (LMJ^{-1}L^*)^{-1}(LM^{-1}u - w) \left[\text{we set } Mx = \left\{ \sum_{j=1}^n M_j^i x^j \right\}_i \text{ and} \right.$$

$$\left. Jx = \{J_i x^i\}_i, J \text{ being the duality operator of } \prod_{i=1}^n U^i \right].$$

We apply this result to compute the Walras–Cournot equilibrium of an economy with n consumers and m producers. We assume that consumers choose their consumptions using their demand correspondences (which are assumed to be quadratic) and that producers choose their production non-cooperatively.

9.2.1. Non-cooperative games with quadratic loss functions

Suppose that the strategy sets

- (1) U^i are Hilbert spaces for the scalar products
 $((x^i, y^i))_i = \langle J_i x^i, y^i \rangle$ where $J_i \in \mathcal{L}(U^i, U^{i*})$.

We consider the case when the loss functions f_i are quadratic loss functions defined by

$$(2) \quad f_i(x) = f_i(x^i, x^j) = \frac{1}{2} \left\| x^i + \sum_{j \neq i} M_j^i x^j - u^i \right\|_i^2$$

where

- (3) $\forall j \neq k, \quad M_j^k \in \mathcal{L}(U^j, U^k)$ (we let M_i^i be the identity).

Interpretation. By choosing x^j , the adverse coalition transforms the original objective u^i into a perturbed objective $u^i - \sum_{j \neq i} M_j^i x^j$. The aim of each player i is to get as close as possible to this perturbed objective. $\square$

Denote by

- (4) $M = \{M_j^k\}_{j,k} \in \mathcal{L}(U^N, U^N)$

the "matrix" of operators M_j^k .

We shall make the following assumption about M . There exists a constant $c > 0$ such that for any $x \in U^N$

$$(5) \quad \sum_{j,k=1}^n ((M_j^k x^j, x^k))_k \geq c \sum_{i=1}^n \|x^i\|_i^2$$

which is a kind of "consistency" for the perturbation operators M_j^k .

Suppose now that the multistrategy set satisfies

- (6) $X(N)$ is a closed convex subset of U^N .

Theorem 1. Suppose that assumptions (5) and (6) hold. Then there exists a unique non-cooperative equilibrium $x \in X(N)$ which is defined by the following variational inequalities: For all $y \in X(N)$

$$(7) \quad \sum_{j,k=1}^n ((M_j^k x^j - u^k, x^k - y^k))_k \leq 0.$$

Proof. Inequalities (7) can be written

$$(8) \quad \langle S(x), x - y \rangle = \sum_{k=1}^n \langle S(x)_k, x^k - y^k \rangle \leq 0$$

where

$$(9) \quad S(x)_k = \sum_{j=1}^n J_k M_j^k x^j - J_k u^k = D_k f_k(x)$$

is the derivative of f_k at x in the direction $\{y_k, 0\}$. $\square$

We set

$$(10) \quad \begin{cases} \text{(i)} & V = U^N, \|x\| = (\sum_{i=1}^n \|x^i\|_i^2)^{1/2} \\ \text{(ii)} & Jx = \{J_1 x_0^1, \dots, J_n x_0^n\} \in V^*, \\ \text{(iii)} & G(x) = \{\sum_{j=1}^n J_k M_j^k x^j\}_{k=1, \dots, n} = JMx \in V^*, \\ \text{(iv)} & K = X(N). \end{cases}$$

Assumption (5) amounts to saying that G is V -elliptic, in the sense that

$$(11) \quad \forall x \in V \quad \langle G(x), x \rangle \geq c \|x\|^2 \quad \text{where } c > 0.$$

9.2.2. Existence of solutions of variational inequalities

Theorem 1 follows from the following

Theorem 2. (Lions–Stampacchia). *Let $G \in \mathcal{L}(V, V^*)$ be a V -elliptic operator and let K be a closed convex subset of V . Then there exists a unique solution $x = G_K^{-1}(p) \in K$ of the variational inequalities*

$$(12) \quad \forall y \in K, \quad \langle G(x) - p, x - y \rangle \leq 0.$$

The map $G_K^{-1} : V^* \rightarrow K$ satisfies

$$(13) \quad \|G_K^{-1}(p) - G_K^{-1}(q)\| \leq \frac{1}{c} \|p - q\|_*.$$

Proof. Suppose that x is a solution of (12) and that y is a solution of (12) with p is replaced by q . Adding the inequalities $\langle G(x) - p, x - y \rangle \leq 0$ and $\langle G(y) - q, y - x \rangle \leq 0$, we deduce from the V -ellipticity of G that

$$(14) \quad \begin{aligned} c \|x - y\|^2 &\leq \langle G(x) - G(y), x - y \rangle \leq \\ &\leq \langle p - q, x - y \rangle \leq \|p - q\|_* \|x - y\|. \end{aligned}$$

On taking $p = q$, this implies that there exists at most one solution $x = G_K^{-1}(p)$ of (12). If $y = G_K^{-1}(q)$ exists, we have proved inequality (13). We now prove the existence of $y = G_K^{-1}(q)$ by writing (12) in the form

$$\forall y \in K, \quad \langle (x - x + \lambda J^{-1}(Gx - p), x - y) \rangle \leq 0$$

i.e. $x = t(x - \lambda J^{-1}(Gx - p))$, where t denotes the projector of best approximation onto K .

We consider the iteration

$$(15) \quad x_{n+1} = t(x_n - \lambda J^{-1}(Gx_n - p)); \quad x_0 \in K.$$

Since the projector t satisfies $\|tx - ty\| \leq \|x - y\|$, we obtain

$$(16) \quad \begin{aligned} \|x_{n+1} - x_n\| &\leq \|x_n - x_{n-1} - \lambda J^{-1}G(x_n - x_{n-1})\| \\ &\leq \|(1 - \lambda J^{-1}G)\| \cdot \|x_n - x_{n-1}\|. \end{aligned}$$

We estimate $\|(1 - \lambda J^{-1}G)\|$ in the following way.

$$(17) \quad \begin{aligned} \|(1 - \lambda J^{-1}G)x\|^2 &= \|x\|^2 + \lambda^2 \|Gx\|_*^2 - 2\lambda \langle Gx, x \rangle \\ &\leq (1 + \lambda^2 \|G\|^2 - 2\lambda c) \|x\|^2 \end{aligned}$$

since $\langle Gx, x \rangle \geq c \|x\|^2$. Taking $\lambda = c/\|G\|^2$, we deduce that $\theta = \|(1 - \lambda J^{-1}G)\| \leq \sqrt{1 - c/\|G\|^2} < 1$.

Thus

$$(18) \quad \|x_{n+1} - x_n\| \leq \theta \|x_n - x_{n-1}\|$$

and the (Cauchy) sequence converges to an element x of the Hilbert space U . Since $x \mapsto t(x - \lambda J^{-1}(Gx - p))$ is continuous, we deduce that the limit x satisfies $x = t(x - \lambda J^{-1}(Gx - p))$, i.e. that $x = G_K^{-1}(p)$. $\square$

Remark. We can replace the assumption of V -ellipticity by the weaker assumption of K -ellipticity

$$(19) \quad \forall x, y \in K, \quad \langle Gx - Gy, x - y \rangle \geq c \|x - y\|^2 \quad \text{where } c > 0.$$

Proposition 1. (Lax-Milgram). *Any V -elliptic operator $G \in \mathcal{L}(V, V^*)$ is an isomorphism. Its inverse $G^{-1} \in \mathcal{L}(V^*, V)$ is also V -elliptic.*

Proof. Take $K = V$ in the above theorem and note that $G^{-1} = G_V^{-1}$ is nothing other than the inverse of G since the variational inequalities reduce to $\langle Gx - p, z \rangle = 0$ for any $z \in V$, i.e. $Gx - p = 0$. It is clear that $G^{-1} \in \mathcal{L}(V^*, V)$ is V -elliptic since we can write

$$\begin{aligned} \langle p, G^{-1}p \rangle &= \langle Gx, x \rangle \geq c \|x\|^2 \geq (c/\|G\|^2) \|p\|^2 \\ (\text{because } \|p\|_* &= \|Gx\|_* \leq \|G\| \cdot \|x\|). \quad \square \end{aligned}$$

9.2.3. Examples

Proposition 2. *Consider the case when $X(N) = U^N$. If assumption (5) holds, then the unique non-cooperative equilibrium is equal to $M^{-1}u = G^{-1}Ju$.*

Proof. The non-cooperative equilibrium x is the solution of $Gx = Ju$, i.e. of $JMx = Ju$ since $G = JM$, where $J \in \mathcal{L}(U^N, U^{N*})$ is the duality mapping defined by $Jx = \{J_1x^1, \dots, J_nx^n\}$. $\square$

Now, consider the case when

(20) $X(N) = X^N$ is the product of n closed convex subsets X^i .

Denote by $t_i = t_{X^i}$ the projector of best approximation from U^i onto X^i .

Proposition 3. *Suppose that assumptions (5) and (20) hold.*

Then there exists a unique non-cooperative equilibrium $x \in X(N)$ which is defined by the following equations

$$(21) \quad \forall i \in N, \quad x^i = t_i \left(u^i - \sum_{j \neq i} M_j^i x^j \right).$$

Proof. This is left as an exercise. $\square$

Remark. The “natural” iteration defined by $x_{n+1}^i = t_i(u^i - \sum_{j \neq i} M_j^i x_n^j)$ for solving (21) does not necessarily converge.

Theorem 2 shows that if $\lambda = c/\|G\|^2$, the iteration defined by

$$(22) \quad x_{n+1}^i = t_i \left(x_n^i + \lambda \left(u^i - \sum_{j=1}^n M_j^i x_n^j \right) \right)$$

converges to the non-cooperative equilibrium.

In other words, at the $(n+1)$ th step, player i minimizes over X^i its distance to the objective $u^i - \lambda \sum_{j \neq i} M_j^i x_n^j + (1-\lambda)(x_n^i - u^i)$.

9.2.4. Multistrategy sets defined by linear constraints

We introduce n continuous linear operators $L_j \in \mathcal{L}(U^j, V)$ such that the operator

$$(23) \quad L = \sum_{j=1}^n L_j \in \mathcal{L}(U^N, V) \text{ is surjective.}$$

We define the multistrategy set

$$(24) \quad X(N) = \{x \in U^N \text{ such that } Lx = \sum_{j=1}^n L_j x^j = w\}.$$

Interpretation. The space V is regarded as a resource space, $w \in V$ as a scarce resource, $L_j x^j$ as the resource needed by player j to implement x^j . Then $X(N)$ is the set of multistrategies such that the sum Lx of resources $L_j x^j$ needed to implement them is equal to w .

Then a non-cooperative equilibrium x is defined by

$$(25) \quad \begin{cases} \text{(i)} \sum_{j=1}^n L_j x^j = w, \\ \text{(ii)} \forall i \in N, x^i \text{ minimizes } \|y^i - (u^i - \sum_{j \neq i} M^j x^j)\|_i \text{ under the constraint } L_i y^i = w - \sum_{j \neq i} L_j x^j. \end{cases}$$

Theorem 3. Suppose that (5) and (25) hold. Then there exists a non-cooperative equilibrium defined by

$$(26) \quad \begin{cases} \text{(i)} x = M^{-1}(u - J^{-1}L^*p), \\ \text{(ii)} \text{ where } p = (LM^{-1}J^{-1}L^*)^{-1}(LM^{-1}u - w). \end{cases}$$

Proof. This is analogous to the proof of Theorem 2.3.1.

Note that $y \in X(N)$ if and only if $y = x - z$ where $z \in \text{Ker } L$. Then inequalities (7) imply that

$$(27) \quad \langle Gx - Ju, z \rangle = 0 \quad \text{for any } z \in \text{Ker } L,$$

i.e. $Gx - Ju \in (\text{Ker } L)^\perp = \text{Im } L^*$. Thus there exists $p \in V^*$ such that

$$(28) \quad Gx - Ju = -L^*p,$$

i.e. such that $x = G^{-1}Ju - G^{-1}Lp = M^{-1}u - G^{-1}Lp$ since $G^{-1}J = M^{-1}$.

Applying L to both sides of this equation, we obtain that $Lx = w = LM^{-1}u - (LG^{-1}L^*)p$.

Since G is U^N -elliptic, G^{-1} is also U^{N*} -elliptic. On the other hand, since L is surjective, L^* is an isomorphism from V^* onto its closed range in U^{N*} . Thus $LG^{-1}L^*$ is V^* -elliptic since $\langle LG^{-1}L^*q, q \rangle = \langle G^{-1}L^*q, L^*q \rangle \geq c\|L^*q\|_{U^N}^2 \geq c'\|q\|_{V^*}^2$.

Therefore $LG^{-1}L^*$ is an isomorphism from V^* onto V (by Proposition 1). We can write

$$\begin{cases} p = (LG^{-1}L^*)^{-1}(LM^{-1}u - w) = (LM^{-1}J^{-1}L^*)^{-1}(LM^{-1}u - w), \\ x = M^{-1}u - G^{-1}L^*p = M^{-1}u - M^{-1}J^{-1}L^*p. \end{cases}$$

Example. Consider the case when

$$(29) \quad V = \mathbb{R}, \quad Lx = \sum_{j=1}^n \langle p_j, x^j \rangle \quad \text{where } p_j \in U^{j*}, \quad w = r \in \mathbb{R}.$$

A non-cooperative equilibrium is defined by

$$(30) \quad \begin{cases} \text{(i)} \sum_{j=1}^n \langle p_j, x^j \rangle = w, \\ \text{(ii)} \forall i \in N, x^i \text{ minimizes } \|y^i - (u^i - \sum_{j \neq i} M_j^i x^j)\| \\ \text{under the constraint } \langle p_i, x^i \rangle = r - \sum_{j \neq i} \langle p_j, x^j \rangle. \end{cases}$$

The solution x is defined by

$$(31) \quad x = M^{-1} \left(u - \frac{\langle p, M^{-1}u \rangle - r}{\langle p, M^{-1}J^{-1}p \rangle} J^{-1}p \right).$$

9.2.5. Walras-Cournot equilibria

Consider an economy with n consumers and m producers. The behavioral assumption of the model is that consumers choose their consumptions according to the Walrasian model (by using their demand function) and that *producers choose their production in a non-cooperative way*.

For simplicity, we use a model with quadratic demand functions (Section 8.1.6) and with production processes described by linear production operators.

Specifically, we assume that the behavior of consumers i is described by their

$$(32) \quad \begin{cases} \text{(i)} \text{ consumption sets } R^i = \mathbf{R}^l, \\ \text{(ii)} \text{ demand functions } D_i \text{ defined by } D_i(p, r) = \\ \quad = u^i - (\langle p, u^i \rangle - r) / (\langle p, J^{-1}p \rangle) J^{-1}p, \\ \text{(iii)} \text{ initial endowment } w^i \in \mathbf{R}^l, \\ \text{(iv)} \text{ system } \theta = \{\theta_j^i\} \text{ of shares } (\forall j, \sum_{i=1}^n \theta_j^i = 1). \end{cases}$$

Assume that each producer j produces an output $z^j \in \mathbf{R}^l$. If the price p prevails, consumer i will choose

$$x^i = D_i \left(p, \langle p, w^i \rangle + \sum_{j=1}^n \theta_j^i \langle p, z^j \rangle \right).$$

By Proposition 8.1.2, we obtain the equality

$$(33) \quad \sum_{i=1}^n x^i = \sum_{i=1}^n w^i + \sum_{j=1}^m z^j$$

if and only if p is equal to

$$(34) \quad p(z) = \theta J \left(\sum_{i=1}^n (u^i - w^i) - \sum_{j=1}^m z^j \right).$$

Normalize the price system by setting $\theta = 1$ and denote by $v = \sum_{i=1}^n (u^i - w^i)$ the aggregate net demand when there is no production. Then

$$(35) \quad p(z) = J \left(v - \sum_{j=1}^m z^j \right).$$

We now describe the behavior of producers j . If each producer j produces z^j , then the *profit* for j is

$$g(z) = \langle p(z), z^j \rangle = - \left[\|z^j\|^2 - \left((z^j, v - \sum_{k \neq j} z^k) \right) \right].$$

The loss function $f_j = -g_j$ is of quadratic type, i.e.

$$(36) \quad f_j(z) = \left\| z^j - \frac{1}{2} \left(v - \sum_{k \neq j} z^k \right) \right\|^2 - \frac{1}{4} \left\| v - \sum_{k \neq j} z^k \right\|^2.$$

On the other hand, we assume that the input space is equal to $V = \mathbb{R}^k$, that the production process of each producer j is described by

$$(37) \quad \text{a production operator } L_j \in \mathcal{L}(\mathbb{R}^l, \mathbb{R}^k)$$

and that

$$(38) \quad y^0 \in \mathbb{R}^k \text{ is the scarce resource.}$$

We assume that producers will choose a non-cooperative equilibrium (i.e. Cournot equilibrium) defined by

$$(39) \quad \begin{cases} \text{(i) } \sum_{j=1}^m L_j \bar{z}^j = y^0, \\ \text{(ii) } \forall j, \bar{z}^j \text{ maximizes its profit } z^j \rightarrow \langle p(\bar{z}^1, \dots, z^j, \dots, \bar{z}^m), z^j \rangle \\ \text{under the constraint } L_j z^j = y^0 - \sum_{k \neq j} L_k \bar{z}^k. \end{cases}$$

Proposition 4. Suppose that

$$(40) \quad L = \sum_{j=1}^m L_j \in \mathcal{L}(\mathbb{R}^l, \mathbb{R}^k) \text{ is surjective.}$$

Then there exists a unique Walras-Cournot equilibrium.

Specifically, if we set

$$(41) \quad q = \left[(m+1) \sum_{k=1}^m L_k J^{-1} L_k^* - \sum_{j,k=1}^m L_j J^{-1} L_k^* \right]^{-1} \left[\left(\sum_{k=1}^m L_k \right) v - (m+1) y^0 \right]$$

the production z^k are equal to

$$(42) \quad z^k = \frac{1}{m+1} v - J^{-1} \left[L_k^* - \frac{1}{m+1} \left(\sum_{j=1}^m L_j^* \right) \right] q$$

the equilibrium price is equal to

$$(43) \quad p = \frac{1}{m+1} \left[J v + \left(\sum_{k=1}^m L_k^* \right) q \right]$$

and the consumptions x^i are equal to

$$(44) \quad x^i = u^i - \frac{\left\langle p, v^i - \sum_{j=1}^m \theta_j^i z^j \right\rangle}{\langle p, J^{-1}p \rangle} J^{-1}p.$$

Proof. The matrix $M = \{M_j^k\}$ of operators $M_j^k \in \mathcal{L}(\mathbf{R}^l, \mathbf{R}^l)$ is defined by

$$(45) \quad M_j^j = \mathbf{1} \text{ (= identity)}, \quad M_j^k = \frac{1}{2} \mathbf{1} \quad \text{if } j \neq k.$$

Assumption (5) is satisfied since we can write

$$\sum_{j,k=1}^m ((M_j^k z^j, z^k)) = \frac{1}{2} \sum_{j=1}^m \|z^j\|^2 + \frac{1}{2} \left(\left(\sum_{j=1}^m z^j, \sum_{j=1}^m z^j \right) \right) \geq \frac{1}{2} \sum_{j=1}^m \|z^j\|^2.$$

The inverse M^{-1} of M is defined by the matrix $\{N_j^k\}_{j,k}$ of operators $N_j^k \in \mathcal{L}(\mathbf{R}^l, \mathbf{R}^l)$ defined by

$$(46) \quad N_i^i = \frac{2m}{m+1} \mathbf{1}, \quad N_j^k = \frac{-2}{m+1} \mathbf{1} \quad \text{if } i \neq k.$$

Since the objectives x_0^j are equal to $\frac{1}{2}v$, we have that

$$(47) \quad (M^{-1}x_0)^j = v - \frac{m}{m+1} v = \frac{1}{m+1} v.$$

The matrix $LM^{-1}J^{-1}L^*$ is defined by

$$(48) \quad LM^{-1}J^{-1}L^* = 2 \left[\sum_{k=1}^m L_k J^{-1} L_k^* - \frac{1}{m+1} \sum_{j,k=1}^m L_j J^{-1} L_k^* \right].$$

Theorem 3 implies that, if we set

$$(49) \quad q = \left[(m+1) \sum_{k=1}^m L_k J^{-1} L_k^* - \sum_{j,k=1}^m L_j J^{-1} L_k^* \right]^{-1} \left(\left(\sum_{k=1}^m L_k \right) v - (m+1) p^0 \right)$$

then we can write

$$(50) \quad z^k = \frac{1}{m+1} v - J^{-1} L_k^* q - \frac{1}{m+1} \left(\sum_{i=1}^m J^{-1} L_i^* \right) q.$$

The equilibrium price can then be written

$$(51) \quad p = J \left(v - \sum_{k=1}^m z^k \right) = \frac{1}{m+1} \left(Jv + \left(\sum_{k=1}^m L_k^* \right) q \right).$$

Therefore, the consumption x^i is equal to

$$(52) \quad x^i = u^i - \frac{\langle p, v^i \rangle - \sum_{j=1}^m \theta_j^i \langle p, z^j \rangle}{\langle p, J^{-1}p \rangle} J^{-1}p.$$

9.3. Constrained non-cooperative games and fixed point theorems

This section deals with the fundamental Brouwer and Kakutani theorems and with the problem of selecting a fixed point of a correspondence. Curiously enough, such results in non-convex analysis are nothing other than existence theorems for equilibria of non-cooperative games with constraints. In the framework of game theory, the problem amounts to each player finding $x^i \in S_i(x^i)$ such that $f_i(x) = \min_{y^i \in S_i(x^i)} f_i(y^i, x^i)$, where S_i is a correspondence mapping each multistrategy $x^i \in X^i$ of the complementary coalition onto a constrained subset $S_i(x^i)$ of strategies of the i th player. We shall notice that this amounts to solving the following problem:

$$(*) \quad x \in S(x)$$

(where S is a correspondence from X into X) and

$$(**) \quad \alpha(x) = \sup_{y \in S(x)} \varphi(x, y) \leq 0,$$

where φ is a function mapping $X \times X$ into $\mathbf{R}$.

If $\varphi = 0$, then the problem reduces to $(*)$ alone and a fixed point x of S exists under the assumptions of the Kakutani theorem. If S is the constant correspondence defined by $S(x) = X$, then the problem reduces to $(**)$ alone and the Ky-Fan theorem implies the existence of a solution.

We shall prove the existence of a solution of $(*)$ and $(**)$ under the assumptions of the Kakutani and Ky-Fan theorems and a consistency assumption between φ and S .

We continue the study of fixed points of a correspondence in Section 15.1.

In the rest of the section we state and prove a theorem on the existence of an equilibrium of a constrained game.

9.3.1. Selection of a fixed-point

Theorem 1. *Suppose that*

$$(1) \quad X \text{ is a convex compact subset.}$$

Suppose that a function $\varphi : X \times X \rightarrow \mathbb{R}$ satisfies

$$(2) \quad \begin{cases} \text{(i)} \quad \forall y, x \mapsto \varphi(x, y) \text{ is lower semi-continuous,} \\ \text{(ii)} \quad \forall x, y \mapsto \varphi(x, y) \text{ is concave,} \\ \text{(iii)} \quad \sup_{y \in X} \varphi(y, y) \leq 0. \end{cases}$$

and that a correspondence $S : X \rightarrow X$ satisfies

$$(3) \quad \begin{array}{l} S \text{ is upper hemi-continuous with non-empty closed convex images} \\ \text{(see Definition 2.5.2).} \end{array}$$

Suppose also that the function φ and the correspondence S are consistent in the sense that

$$(4) \quad \text{the function } x \mapsto \alpha(x) = \sup_{y \in S(x)} \varphi(x, y) \text{ is lower semi-continuous.}$$

Then there exists $x \in X$ satisfying

$$(5) \quad \begin{cases} \text{(i)} \quad x \in S(x), \\ \text{(ii)} \quad \sup_{y \in S(x)} \varphi(x, y) \leq 0. \end{cases}$$

Remark. We recall that upper semi-continuous correspondences are upper hemi-continuous (see Proposition 2.5.1).

Remark. Recall that (5(i)) amounts to saying that x is a “fixed-point” of the correspondence S . Hence any solution of (5) can be regarded as a selection of a fixed point of S .

Recall that Theorem 2.5.2 implies that assumption (4) is satisfied if we assume that

$$(6) \quad \begin{cases} \text{(i)} \quad \varphi : X \times X \rightarrow \mathbb{R} \text{ is lower semi-continuous,} \\ \text{(ii)} \quad S : X \rightarrow X \text{ is lower semi-continuous.} \end{cases}$$

Hence we can state the following theorem.

Theorem 2. Suppose that assumptions (1), (2), (3) and (6) are satisfied. Then there exists a solution x of (5).

Proof of Theorem 1. We have to prove that

$$(7) \quad \exists x \in X \text{ such that } x \in S(x) \text{ and } \alpha(x) = \sup_{y \in S(x)} \varphi(x, y) \leq 0.$$

Suppose that this conclusion is false. Then, for any $x \in X$, either $x \notin S(x)$ or $\alpha(x) > 0$. But $x \notin S(x)$ means that there exists $p \in U^*$ such that $\langle p, x \rangle - \sigma^*(S(x), p) > 0$.

In other words, if

$$(8) \quad \begin{cases} \text{(i) } V_0 = \{x \in X \text{ such that } \alpha(x) > 0\}, \\ \text{(ii) } V_p = \{x \in X \text{ such that } \langle p, x \rangle - \sigma^*(S(x), p) > 0\}, \end{cases}$$

then

$$(9) \quad X \subset V_0 \cup \bigcup_{p \in U^*} V_p$$

when the conclusion of the theorem is false.

But, V_0 is open (by (4)) and the V_p 's are open because the correspondence is upper hemi-continuous (see Definition 2.5.2). Since X is compact (by (1)), there exist n elements $p_i \in U^*$ such that

$$(10) \quad X \subset V_0 \cup \bigcup_{i=1}^n V_{p_i}.$$

We now introduce a continuous partition of unity $\{\beta_0, \beta_1, \dots, \beta_n\}$ subordinate to this covering.

We consider the following function a defined on $X \times X$ by

$$(11) \quad a(x, y) = \beta_0(x) \varphi(x, y) + \sum_{i=1}^n \beta_i(x) \langle p_i, x - y \rangle.$$

We can apply the Ky-Fan theorem (see Theorem 7.1.3) to deduce that there exists $\bar{x} \in X$ such that

$$(12) \quad \forall y \in X, \quad a(\bar{x}, y) \leq 0.$$

To see this, note that $a(y, y) \leq 0$ since $\varphi(y, y) \leq 0$ by assumption (2(iii)). The functions $y \mapsto a(x, y)$ are concave because the functions $y \mapsto \varphi(x, y)$ are concave by assumption (2(iii)) and the functions $x \mapsto a(x, y)$ are lower semi-continuous because the functions $x \mapsto \varphi(x, y)$ are lower semi-continuous and the functions β_i ($i = 1, \dots, n$) are continuous.

A contradiction to (12) is obtained by proving that there exists an element $\bar{y} \in X$ such that

$$(13) \quad a(\bar{x}, \bar{y}) > 0.$$

We take

$$(14) \quad \begin{aligned} &\bar{y} \in S(\bar{x}) \text{ if } \alpha(\bar{x}) \leq 0 \text{ and } \bar{y} \in S(\bar{x}) \text{ satisfying} \\ &\varphi(\bar{x}, \bar{y}) \geq \alpha(\bar{x}) - \varepsilon \text{ where } \varepsilon = \frac{1}{2} \alpha(\bar{x}) \text{ if } \alpha(\bar{x}) > 0. \end{aligned}$$

Since $\bar{x} \in X$, either $\beta_0(\bar{x}) > 0$ or $\beta_i(\bar{x}) > 0$ for at least one index i .

We prove that

$$(15) \quad \begin{cases} \text{(i) } \beta_0(\bar{x}) > 0 \text{ implies that } \varphi(\bar{x}, \bar{y}) > 0, \\ \text{(ii) } \beta_i(\bar{x}) > 0 \text{ implies that } \langle p_i, \bar{x} - \bar{y} \rangle > 0 \end{cases}$$

from which it follows immediately that $\alpha(\bar{x}, \bar{y}) > 0$.

Suppose that $\beta_0(\bar{x}) > 0$. This implies that $\bar{x} \in V_0$ and thus, that $\alpha(\bar{x}) > 0$. Therefore, by (14),

$$\varphi(\bar{x}, \bar{y}) \geq \alpha(\bar{x}) - \frac{1}{2} \alpha(\bar{x}) = \frac{1}{2} \alpha(\bar{x}) > 0.$$

Suppose that $\beta_i(\bar{x}) > 0$. This implies that $\bar{x} \in V_{p_i}$, i.e. that

$$\langle p_i, \bar{x} \rangle > \sigma^*(S(\bar{x}), p_i) \geq \langle p_i, \bar{y} \rangle$$

(since $\bar{y} \in S(\bar{x})$). Thus $\langle p_i, \bar{x} - \bar{y} \rangle > 0$.

Remark. We shall extend this theorem in Chapter 15 (see Theorems 15.2.1 and 15.2.2) by relaxing the continuity assumption (2(i)).

9.3.2. Equilibria of constrained non-cooperative games

Associate with the multistrategy set $X(N) \subset X^N$ the correspondences S^i mapping X^j into X^i which assign to the choice x^j of strategies of players $j \neq i$ the subset

$$(16) \quad S_i(x^j) = \{y^i \in X^i \text{ such that } \{y^i, x^j\} \in X(N)\}$$

of feasible strategies for the i th player.

We can restate the definition of a non-cooperative equilibrium in the following way:

$$(17) \quad \begin{cases} \text{(i) } \forall i \in N, x_i \in S_i(x^j), \\ \text{(ii) } \forall i \in N, f_i(x^i, x^j) = \min_{y^i \in S_i(x^j)} f_i(y^i, x^j). \end{cases}$$

In this section, we shall study the existence of a solution x of problem (17) when the correspondences S_i are no longer defined by a multistrategy set $X(N)$.

Consider n correspondences $S_i : X^i \rightarrow X^i$.

Definition 1. A "constrained n -person game" $\{S, F\}$ is described by n correspondences $S_i : X^i \rightarrow X^i$ and a multiloss operator $F : X^N \rightarrow \mathbb{R}^N$.

A multistrategy $x \in X^N$ is said to be a non-cooperative equilibrium if

$$(18) \quad \begin{cases} \text{(i) } \forall i \in N, x^i \in S_i(x^j), \\ \text{(ii) } \forall i \in N, f_i(x^i, x^j) = \min_{y^i \in S_i(x^j)} f_i(y^i, x^j). \end{cases}$$

Write

$$(19) \quad \varphi(x, y) = \sum_{i=1}^n (f_i(x^i, x^i) - f_i(y^i, x^i))$$

and

$$(20) \quad S(x) = \prod_{i=1}^n S_i(x^i).$$

Proposition 1. *A multistrategy $x \in X^N$ is a non-cooperative equilibrium if and only if*

$$(21) \quad \begin{cases} \text{(i) } x \in S(x), \\ \text{(ii) } \sup_{y \in S(x)} \varphi(x, y) \leq 0. \end{cases}$$

Proof. This is analogous to the proof of Proposition 1.2. $\square$

We are now able to prove the following existence theorem.

Theorem 3 (Arrow–Debreu–Nash). *Suppose that*

$$(22) \quad \text{the strategy sets } X^i \text{ are convex and compact}$$

and that the loss functions satisfy

$$(23) \quad f_i \text{ is continuous and } f_i(\cdot, x^i) \text{ is convex for all } x^i \in X^i.$$

Suppose also that the correspondences S_i satisfy

$$(24) \quad S_i \text{ is a continuous correspondence from } X^i \text{ into } X^i \text{ with non-empty closed convex images.}$$

Then there exists a non-cooperative equilibrium.

Proof. It is clear that if we set $X = X^N$ and take φ and S to be as defined by (19) and (20), assumptions (22), (23) and (24) of Theorem 3 imply assumptions (1), (2), (3) and (6) of Theorem 2. Thus there exists a solution $x \in X^N$ of (21), which is a non-cooperative equilibrium by Proposition 1. $\square$

9.3.3. Fixed-point theorems

Note that, if we take for S the constant correspondence $S(x) = X$, assumptions (3) and (4) of Theorem 1 are satisfied. We therefore recover the Ky-Fan theorem.

If we take $\varphi = 0$, assumptions (2) and (4) of Theorem 1 are satisfied. We then obtain the Kakutani theorem.

Theorem 4 (Kakutani). *Suppose that*

- (1) X is a convex compact subset

and that a correspondence $S : X \longrightarrow X$ satisfies

- (25) S is upper hemi-continuous with non-empty closed convex images.

Then there exists a fixed point $\bar{x}$ of S .

In the case where S is a function, we obtain the Brouwer theorem generalized by Schauder in the infinite dimensional case.

Theorem 5 (Brouwer–Schauder). *Suppose that*

- (1) X is a convex compact subset

and that

- (26) f is a continuous map from X into itself.

Then there exists a fixed point x of f , i.e. a solution of $x = f(x)$.

***Remark.** We have deduced the Brouwer fixed point theorem from the Ky-Fan theorem. *Actually they are equivalent. If we assume that (10) and (11) hold and that any continuous map D from X into X has a fixed point, then there exists $\bar{x} \in X$ such that $\sup \varphi(\bar{x}, y) \leq 0$.*

The proof is as follows. By the Lasry theorem (Theorem 7.1.4), there exists $\bar{x} \in X$ such that $\sup_{y \in Y} \varphi(\bar{x}, y) \leq \sup_{D \in \mathcal{D}(X, X)} \inf_{x \in X} \varphi(x, D(x))$. Now, since any continuous map D from X into X has a fixed point $\tilde{x}$ by assumption, then, $\inf_{x \in X} \varphi(x, D(x)) \leq \varphi(\tilde{x}, D(\tilde{x})) \leq \varphi(\tilde{x}, \tilde{x}) \leq 0$. Hence $\sup_{y \in Y} \varphi(\bar{x}, y) \leq 0$. $\square$

In other words, we can either, as in our development, assume Ky-Fan theorem (via the Knaster–Kuratowski–Mazurkiewicz lemma) and deduce the Brouwer theorem or assume the Brouwer theorem and deduce the Ky-Fan theorem. Note that Knaster, Kuratowski and Mazurkiewicz proved the Brouwer theorem directly using their lemma (see Appendix B).

Remark. We will give another proof of the Kakutani fixed point theorem in Section 15.1, as well as some other fixed point theorems (see Theorem 15.1.3).

9.4. Non-cooperative Walras equilibria

We devote this section to the fundamental economic model in which the loss functions f_i of the n consumers depend not only upon their own consumptions, but also on the consumptions of the other consumers.

We are led to define a non-cooperative Walras pre-equilibrium $\{\bar{x}^1, \dots, \bar{x}^n, \bar{p}\}$ by the following requirements

$$\begin{cases} \text{(i)} \sum_{i=1}^n \bar{x}^i \in Y, \\ \text{(ii)} \forall i \in N, \bar{x}^i \in B_i(\bar{p}, r_i(\bar{p})), \\ \text{(iii)} \forall i \in N, f_i(\bar{x}^i, \bar{x}^j; \bar{p}) = \min_{y^i \in B_i(\bar{p}, r_i(\bar{p}))} f_i(y^i, \bar{x}^j; \bar{p}). \end{cases}$$

We shall prove the existence of such an equilibrium under assumptions analogous to the ones which imply the existence of a Walras equilibrium. To prove such a result, we construct an $(n+1)$ -player non-cooperative game in which the $(n+1)$ th-player is the "market". Its role is to choose a price p which minimizes $p \mapsto r(p) - \langle p, \sum_{i=1}^n \bar{x}^i \rangle$. We prove that any non-cooperative equilibrium of such a game is a non-cooperative Walras equilibrium of the economy. Then we check that the assumptions of the Arrow-Debreu-Nash theorem are satisfied.

In the last part of this section, we introduce producers in the model and so extend the former results.

9.4.1. Description of the model

We consider in this section an economy

$$\{R^i, Y(i), f_i\}_{i \in N}$$

where the loss functions $f_i : R^N \times P \mapsto \mathbb{R}$ depend upon the allocation $x = \{x^1, \dots, x^n\}$ instead of on the consumption x^i of the i th player only.

Consider the budget correspondences

$$B_i(p, r_i(p)) = \{x \in R^i \text{ such that } \langle p, x^i \rangle \leq r_i(p)\}$$

where

$$r_i(p) = \sup_{y \in Y(i)} \langle p, y \rangle.$$

Definition 1. We shall say that $\{\bar{x}, \bar{p}\}$ is a *non-cooperative Walras pre-equilibrium* if

$$\begin{cases} \text{(i)} \sum_{i \in N} \bar{x}^i \in Y, \\ \text{(ii)} \forall i \in N, \bar{x}^i \in B_i(\bar{p}, r_i(\bar{p})), \\ \text{(iii)} \forall i \in N, f_i(\bar{x}^i, \bar{x}^j; \bar{p}) = \min_{y^i \in B_i(\bar{p}, r_i(\bar{p}))} f_i(y^i, \bar{x}^j; \bar{p}). \end{cases}$$

We shall say that it is a *non-cooperative Walras equilibrium* if also

$$\forall i \in N \quad \langle \bar{p}, \bar{x}^i \rangle = r_i(\bar{p}).$$

Theorem 8.2.1 on the existence of a Walras pre-equilibrium can be generalized in the following way.

Theorem 1. *Suppose that*

- (1) *the consumption sets R^i are closed, convex and bounded below.*

Suppose that

- (2) *the initial endowments $Y(i)$ are closed, convex and $-\mathbf{R}_+^l$ is their recession cone*

and

- (3) $\forall i, \quad 0 \in X^i - \text{Int } Y(i).$

Suppose further that the map

- (4) $\{x, y\} \in R^N \times Y \mapsto \sum_{i \in N} x^i - y \in \mathbf{R}^l$

is proper and that the loss functions $f_i: R^N \times P \rightarrow \mathbf{R}$ satisfy

- (5) $\begin{cases} \text{(i)} \quad \forall p \in P = \mathcal{M}^l, \forall x^i \in R^i, x^i \mapsto f_i(x^i, x^i, p) \text{ is convex,} \\ \text{(ii)} \quad f_i \text{ is continuous on } R^N \times P. \end{cases}$

Finally, we require the non-satiation property

- (6) $\forall i \in N, \forall p \in P, \forall x \in R^N, \text{ there exists } y^i \in R^i \text{ such that}$
 $f_i(y^i, x^i; p) < f_i(x^i, x^i, p).$

Given all these assumptions, there exists a non-cooperative Walras equilibrium.

9.4.2. Existence of a non-cooperative Walras equilibrium: the Arrow-Debreu theorem

To prove this theorem, we shall construct an $(n+1)$ -person constrained non-cooperative game in which player 0 is called the “market” (or the “invisible hand”) and where the n other players are just the n consumers.

Consider the set

- (7) $X(N) = \{x \in R^N \text{ such that } \sum_{i \in N} x^i \in Y\}$

of allocations and its projections $\pi_i X(N)$ onto the factor spaces R^i . Let B denote any closed ball of positive radius.

We introduce the following strategy sets

$$(8) \quad \begin{cases} X^0 = P = M^1 & \text{for } i = 0 \\ X^i = R^i \cap (\pi_i X(N) + B) & \text{for } i = 1, \dots, n. \end{cases}$$

We consider the subsets

$$(9) \quad \tilde{Y}_0(i) = Y(i) \cap (\pi_i X(N) + B); \quad \tilde{Y}(i) = \tilde{Y}_0(i) - R_+^i$$

The compactified income functions are defined by

$$(10) \quad \tilde{r}_i(p) = \sigma^*(\tilde{Y}(i); p), \quad \tilde{r}(p) = \sum_{i=1}^n \tilde{r}_i(p)$$

and the compactified budget correspondences by

$$(11) \quad \tilde{B}_i(p, \tilde{r}_i(p)) = \{x \in X^i \text{ such that } \langle p, x^i \rangle \leq \tilde{r}_i(p)\}.$$

We define the correspondences $S_i : X^i \rightarrow X_i$ by

$$(12) \quad \begin{cases} (i) S_0(x) = X^0 = P & \text{for } i = 0, \\ (ii) S_i(x^i; p) = \tilde{B}_i(p, \tilde{r}_i(p)) & \text{for } i = 1, \dots, n. \end{cases}$$

Finally, the loss functions of the $(n+1)$ -person game are defined by

$$(13) \quad \begin{cases} (i) f_0(x, p) = \tilde{r}(p) - \langle p, \sum_{i \in N} x^i \rangle \\ (ii) f_i(x, p) \text{ are the initial loss functions.} \end{cases}$$

To prove Theorem 1, we have to check that, any non-cooperative equilibrium of the constrained non-cooperative game is a non-cooperative Walras pre-equilibrium and that there exists such a non-cooperative equilibrium.

Proposition 1. *Suppose that the subsets R^i and $Y(i)$ are closed and convex, that the loss functions are convex with respect to x^i and satisfy (6).*

Any non-cooperative equilibrium of the constrained $(n+1)$ -person game defined by (8), (12) and (13) is a non-cooperative Walras equilibrium.

Conversely, any non-cooperative Walras equilibrium is a non-cooperative equilibrium of the $(n+1)$ -person game.

Proof. Let $\{\bar{p}, \bar{x}\} \in X^0 \times \prod_{i=1}^n X^i$ be a non-cooperative equilibrium.

Since $\bar{x}^i \in \tilde{B}^i(\bar{p}, \tilde{r}_i(\bar{p}))$ and $\bar{x}^i$ minimizes $f_i(\cdot, \bar{x}^i, \bar{p})$ on $\tilde{B}(\bar{p}, \tilde{r}_i(\bar{p}))$, we deduce from (6) and Proposition 2.1.8 that

$$(14) \quad \langle \bar{p}, \bar{x}^i \rangle = \tilde{r}_i(\bar{p}) \leq r_i(\bar{p}).$$

Summing these inequalities, we obtain that

$$\begin{aligned} 0 &\leq f(\bar{p}) - \langle \bar{p}, \sum \bar{x}^i \rangle = f_0(\bar{x}, \bar{p}) \\ &\leq f_0(p, \bar{x}) = f(p) - \langle p, \sum \bar{x}^i \rangle \quad \text{for all } p \in X^0 = P. \end{aligned}$$

This implies that

$$(15) \quad \sum_{i \in N} x^i \in \tilde{Y} = \sum_{i \in N} \tilde{Y}(i) \subset Y.$$

Hence $\bar{x} \in X(N)$ and thus

$$(16) \quad \forall i \in N, \quad \bar{x}^i \in \pi_i X(N).$$

We prove that $r_i(\bar{p}) = f_i(\bar{p})$ for all $i \in N$ as in the first two parts of the proof of Proposition 8.3.3.

Now, since $\{\bar{p}, \bar{x}\}$ is a non-cooperative equilibrium of the constrained $(n+1)$ -person game, we deduce that, for all $i \in N$,

$$(17) \quad f_i(\bar{x}^i, \bar{x}^i, \bar{p}) = \min_{y^i \in B_i(\bar{p}, r_i(\bar{p}))} f_i(y^i, \bar{x}^i, \bar{p}).$$

Since $\tilde{B}_i(\bar{p}, f_i(\bar{p})) = \tilde{B}_i(\bar{p}, r_i(\bar{p}))$, we deduce that

$$(18) \quad f_i(\bar{x}^i, \bar{x}^i, \bar{p}) = \min_{y^i \in B_i(\bar{p}, r_i(\bar{p}))} f_i(y^i, \bar{x}^i, \bar{p})$$

as in the proof of the third part of Proposition 8.3.3.

The proof of the converse statement is obvious. $\square$

Proposition 2. *Suppose that the assumptions of Theorem 1 hold.*

Then there exists a non-cooperative equilibrium of the constrained $(n+1)$ -person game defined by (8), (12) and (13).

Proof. We shall check that the assumptions of Theorem 3.3 are satisfied.

(a) The strategy set X^0 is convex and compact. The strategy sets $X^i = R^i \cap (\pi_i X(N) + B)$ are also compact since the map $\{x, y\} \mapsto \sum^n x^i - y$ is proper on $R^N \times Y$ (in this case $X(N)$ is compact, hence $\pi_i X(N)$ is compact and thus, $\pi_i X(N) + B$ is also compact).

They are obviously convex.

(b) When $i \in N$, the loss function f_i is continuous and convex with respect to x^i by assumption. The function f_0 is obviously convex with respect to p and continuous, since the compactified income functions f_i are continuous (for they are the support functions of the compact sets $\tilde{Y}_0(i)$).

(c) The correspondences S_i are continuous with non-empty convex compact images. This is obvious for $i = 0$. Let $i \in N$, $S_i(x^i, p) = \tilde{B}_i(p, r_i(p))$. Since the assumptions of Proposition 8.3.2 are satisfied, the result also follows for $i = 1, 2, \dots, n$. Thus there exists a non-cooperative equilibrium. $\square$

*9.4.3. Non-cooperative Walras equilibria of economies with producers

We introduce m producers described by their production sets $Z^j \subset \mathbf{R}^l$.

We consider an economy $\{R^i, f^i, w^i, \{\theta_j^i\}, z^j\}_{i,j}$ in which we assume that the loss functions

$$f_i : \prod_{i=1}^n R^i \times \prod_{j=1}^m Z^j \times P \rightarrow \mathbf{R}$$

depend upon $\{x, z, p\}$ instead of x^i and p . We introduce

$$r_i : P \times \prod_{j=1}^m Z^j \rightarrow \mathbf{R}$$

defined by

$$(19) \quad r_i(p, z) = \langle p, w^i \rangle + \sum_{j=1}^m \theta_j^i \langle p, z^j \rangle.$$

Definition 2. We shall say that $\{\bar{x}, \bar{z}, \bar{p}\} \in \prod_{i=1}^n R^i \times \prod_{j=1}^m Z^j \times P$ is a “non-cooperative Walras pre-equilibrium” if

$$(20) \quad \begin{cases} \text{(i)} \sum_{i=1}^n \bar{x}^i \leq \sum_{i=1}^n w^i + \sum_{j=1}^m \bar{z}^j, \\ \text{(ii)} \forall i \in N, \bar{x}^i \in B_i(\bar{p}, r_i(\bar{p}, \bar{z})) \text{ and} \\ \quad f_i(x^i, \bar{x}^i, \bar{z}, \bar{p}) = \min_{y^i \in B_i(\bar{p}, r_i(\bar{p}, \bar{z}))} f_i(y^i, \bar{x}^i, \bar{z}, \bar{p}), \\ \text{(iii)} \forall j \in M, \bar{z}^j \text{ maximizes the profit } \langle \bar{p}, \cdot \rangle \text{ on } Z^j. \end{cases}$$

We shall say that it is a “non-cooperative Walras equilibrium” if also

$$(21) \quad \forall i \in N, \langle \bar{p}, \bar{x}^i \rangle = r_i(\bar{p}, \bar{z}).$$

Theorem 2. Suppose that

(22) the consumption sets R^i are closed, convex and bounded below and that the map

$$(23) \quad \{x, z, u\} \in \prod_{i=1}^n R^i \times \prod_{j=1}^m Z^j \times \mathbf{R}_+^l \mapsto \sum_{i=1}^n x^i - \sum_{j=1}^m z^j + u \in \mathbf{R}^l$$

is proper.

Suppose also that the production sets Z^j satisfy

$$(24) \quad Z^j \text{ is closed and convex, contains } 0$$

and that the endowments w^i satisfy

$$(25) \quad \forall i \in N, \text{ there exists } \bar{x}^i \in R^i \text{ such that } \bar{x}^i \ll w^i.$$

Finally, suppose that the loss functions satisfy

$$(26) \quad \begin{cases} \text{(i) } \forall p \in P, \forall x \in R^N, \forall z \in Z^M, x^i \mapsto f_i(x^i, x^j, z, p) \text{ is convex,} \\ \text{(ii) } f_i \text{ is continuous on } \prod_{i=1}^n R^i \times \prod_{j=1}^m Z^j \times P, \\ \text{(iii) } f_i \text{ satisfies the non satiation property (6).} \end{cases}$$

Then there exists a non-cooperative Walras equilibrium.

To prove this theorem, we shall construct an $(n+m+1)$ -person constrained non-cooperative game. We shall prove that any non-cooperative equilibrium of this game is a non-cooperative Walras equilibrium and that there exists such a non-cooperative equilibrium.

Consider the subset

$$(27) \quad U_w = \left\{ \{x, z\} \in \prod_{i=1}^n R^i \times \prod_{j=1}^m Z^j \text{ such that } \sum_{i=1}^n x_i - \sum_{j=1}^m z^j \leq w \right\}$$

of allocations and its projections $\pi_i U_w$ and $\pi_j U_w$ onto R^i and Z respectively.

We defined the strategy sets by

$$(28) \quad \begin{cases} X^0 = P = \mathcal{M}^I & \text{if } i = 0, \\ X^i = R^i \cap (\pi_i U_w + B) & \text{if } 1 \leq i \leq n, \\ X^j = Z^j \cap (\overline{\text{co}}(\pi_j U_w \cup \{0\}) + B) & \text{if } n+1 \leq j \leq n+m. \end{cases}$$

Write

$$(29) \quad r_i(p, z) = \langle p, w^i \rangle + \max \left(0, \sum_{j=1}^m \theta_j^i \langle p, z^j \rangle \right)$$

and

$$(30) \quad \tilde{B}_i(p, r_i(p, z)) = \{x \in X^i \text{ such that } \langle p, x \rangle \leq r_i(p, z)\}.$$

The correspondences S are defined by

$$(31) \quad \begin{cases} \text{(i) } S_0(x, z, p) = X^0 = \mathcal{M}^I \text{ if } i = 0, \\ \text{(ii) } S_i(x^i, z, p) = \tilde{B}_i(p, r_i(p, z)) \text{ if } 1 \leq i \leq n, \\ \text{(iii) } S_j(x, z^j, p) = X^j \text{ if } n+1 \leq j \leq n+m+1. \end{cases}$$

The loss functions f_i are defined by

$$(32) \quad \begin{cases} \text{(i)} f_0(x, z, p) = \langle p, w + \sum_{j=1}^m z^j - \sum_{i=1}^n x^i \rangle \text{ for } i = 0, \\ \text{(ii)} f_i(x, z, p) \text{ are the original loss functions for } 1 \leq i \leq n, \\ \text{(iii)} f_j(x, z, p) = -\langle p, z^j \rangle \text{ for } n+1 \leq j \leq n+m. \end{cases}$$

Proposition 3. Suppose that the subsets R^i and Z^j are closed and convex and that the functions f_i are convex with respect to x^i and satisfy the non satiation property (6). Then any non-cooperative equilibrium of the $(n+m+1)$ -person game defined by (28), (31) and (32) is a non-cooperative Walras equilibrium.

Proof. Let $\{\bar{p}, \bar{x}, \bar{z}\} \in X^0 \times \prod_{i=1}^n X^i \times \prod_{j=1}^m X^j$ be a non-cooperative equilibrium. We obtain that, for any producer j ,

$$(33) \quad \langle \bar{p}, \bar{z}^j \rangle = \max_{z^j \in X^j} \langle \bar{p}, z^j \rangle \geq 0 \quad (\text{since } 0 \in X^j)$$

and thus, that

$$(34) \quad r_i(\bar{p}, \bar{z}) = \langle \bar{p}, w^i \rangle + \sum_{j=1}^m \theta_j^i \langle \bar{p}, \bar{z}^j \rangle.$$

Since $\bar{x}^i \in \tilde{B}_i(\bar{p}, r_i(\bar{p}))$, we deduce that

$$(35) \quad \left\langle \bar{p}, \sum_{i=1}^n \bar{x}^i \right\rangle \leq \langle \bar{p}, w \rangle + \sum_{j=1}^m \langle \bar{p}, \bar{z}^j \rangle.$$

Therefore

$$\begin{aligned} 0 &\leq \left\langle \bar{p}, w + \sum_{j=1}^m \bar{z}^j - \sum_{i=1}^n \bar{x}^i \right\rangle = f_0(\bar{x}, \bar{z}, \bar{p}) \\ &\leq f_0(\bar{x}, \bar{z}, p) = \left\langle p, w + \sum_{j=1}^m \bar{z}^j - \sum_{i=1}^n \bar{x}^i \right\rangle \quad \text{for all } p \in \mathcal{M}^i \end{aligned}$$

Hence

$$(36) \quad \sum_{i=1}^n \bar{x}^i \leq w + \sum_{j=1}^m \bar{z}^j.$$

Since $\{\bar{p}, \bar{x}, \bar{z}\}$ is a non-cooperative equilibrium, we know that

$$(37) \quad \forall j = 1, \dots, m, \quad \bar{z}^j \text{ maximizes } \langle \bar{p}, \cdot \rangle \text{ over } X^j$$

and that

$$(38) \quad \forall i = 1, \dots, n, \quad \bar{x}^i \in \tilde{B}_i(\bar{p}, r_i(\bar{p}, \bar{z})) \text{ minimizes } x^i \mapsto f_i(x^i, \bar{x}^i, \bar{z}, \bar{p}) \text{ over } \tilde{B}_i(\bar{p}, r_i(\bar{p}, \bar{z})).$$

The proofs of Propositions 8.3.3 and 8.4.1 imply that, in fact,

$$(39) \quad \forall j = 1, \dots, m, \quad \bar{z}^j \text{ maximizes } \langle \bar{p}, \cdot \rangle \text{ over } Z^j$$

and

$$(40) \quad \forall i = 1, \dots, n, \quad \bar{x}^i \text{ minimizes } x^i \mapsto f_i(x^i, \bar{x}^j, \bar{z}, \bar{p}) \text{ over } B_i(\bar{p}, r_i(\bar{p}, \bar{z})).$$

Thus $\{\bar{p}, \bar{x}, \bar{z}\}$ is a non-cooperative Walras equilibrium. $\square$

Proposition 4. *Suppose that the assumptions of Theorem 2 hold.*

Then there exists a non-cooperative equilibrium of the $(n+m+1)$ -person game defined by (28), (31) and (32).

Proof. We shall check that the assumptions of Theorem 3.3 are satisfied.

(a) The strategy sets X^0 , X^i and X^j are clearly convex. The strategy set X^0 is compact and the strategy sets X^i and X^j are also compact since we assumed that the map $\{x, z, u\} \mapsto \sum_{i=1}^n x^i - \sum_{j=1}^m z^j + u$ is proper.

(b) The loss functions f_0 and f_j are clearly continuous and the loss functions f_i are continuous by assumption. Obviously, f_0 is convex with respect to p , f_j with respect to z^j and f_i is convex with respect to x_i by assumption.

(c) The correspondences S_0 and S_j are constant.

Since $\langle p, \bar{x}^i \rangle < \langle p, w^i \rangle \Leftarrow r_i(p, z)$ for all $p \in X^0$, $z \in \prod_{j=1}^m Z^j$, we deduce that the correspondences S_i are continuous from Proposition 8.3.1.

Thus there exists a non-cooperative equilibrium. $\square$

CHAPTER 10

MAIN SOLUTION CONCEPTS OF COOPERATIVE GAMES

In this chapter we define cooperative games and introduce the main solution concepts studied in this book except for the concept of the Shapley value (see Chapter 11). We prove here only easy results. The difficult results dealing with the existence of such solution concepts and the equivalences which hold among them will be proved in Chapter 11 in the particular case of games with side-payments and in Chapter 12 in the general case.

Consider a game $\{X(N), F\}$ described in strategic form.

We begin by considering the case when *the players subordinate their interests to those of the whole set N of players* to the extent that they cooperate in excluding all but *collectively stable strategies*, i.e. they exclude all but *weak Pareto multistrategies*. This means that no other multistrategy yields a smaller loss to *each* player. For example, the players may use $\lambda = (\lambda^1, \dots, \lambda^n) \in \mathcal{M}^n$ to “pool” their loss functions f_i , i.e. they play a strategy x_λ which minimizes $x \mapsto \sum_{i=1}^n \lambda^i f_i(\cdot)$ on $X(N)$. Then x_λ is a weak Pareto minimum. The converse is true if we assume that $X(N)$ and F are convex: Any weak Pareto strategy can then be implemented by minimizing a weighted loss function $\sum_{i=1}^n \lambda^i f_i(\cdot)$ for a convenient $\lambda \in \mathcal{M}^n$.

This result also shows that the set of Pareto strategies is “too large”. Since our aim is to devise procedures yielding as small a subset of strategies as possible, the problem of *selecting* Pareto strategies arises.

We have to set aside the concept of a “*strong equilibrium*”, which is *both* a non-cooperative equilibrium and a Pareto minimum i.e. which satisfies both individual and collective stability because such a strong equilibrium exists only in exceptional cases.

We therefore weaken the individual stability requirement by introducing the concept of a “*threat functional*” t_i . This associates with any loss function f_i a value $t_i(f_i)$, which sets a maximum to the amount of loss that player i will accept for himself. We will say that a strategy $\bar{x}$ is an *imputation* if it is a weak Pareto minimum (i.e. it is not rejected by the whole set N of players) and if, for all players i , it satisfies $f_i(\bar{x}) \leq t_i(f_i)$ (i.e. it is not rejected by player i).

An example of an imputation is the “best compromise” $\bar{x}$ defined by

$$\forall i \in N, \quad f_i(\bar{x}) \leq (1 - \bar{d})\alpha_i + \bar{d}t_i(f_i)$$

where,

$$\bar{d} = \inf_{x \in X(N)} \max_{i \in N} \left(\frac{f_i(x) - \alpha_i}{t_i(f_i) - \alpha_i} \right) \in [0, 1]$$

and $\alpha_i = \inf_{x \in X(N)} f_i(x)$.

However, the set of imputations can also be too large.

In order to define a subset of the set of imputations, we assume that the behavior of player i alters when he participates in a coalition A of players (i.e. a subset of the set N of players). We describe the behavior of a coalition A by a pair $\{X(A), F^A\}$ where $X(A)$ represents the multistrategy set of the coalition A and $F^A: X(A) \rightarrow \mathbb{R}^A$ represents the multiloss operator of coalition A . The family $\{X(A), F^A\}_{A \in \mathcal{A}}$ describes a “cooperative game”, which involves the description not only of the behavior of the whole set of players, but also the behavior of each coalition of players. We say that a coalition A *rejects* a multistrategy $\bar{x} \in X(N)$ if it can find a multistrategy x^A in its own multistrategy set $X(A)$ which yields to each player i of A a smaller loss $f_i^A(x^A)$.

The *core* of the cooperative game is the subset of multistrategies $\bar{x} \in X(N)$ which are not rejected by any coalition A .

If we take $t_i(f_i) = \inf_{x^i \in X(\{i\})} f_i^{\{i\}}(x^i)$ as threat functional, we see that the core is contained in the subset of imputations.

When the core is not empty (this happens for a rather large class of games, as we shall see in Chapter 12), it is again too “large”. We can “shrink” the core by asking “more coalitions” to reject strategies. Hence we have to “enlarge the family of coalitions”, i.e. to embed it in the family of the so-called “fuzzy coalitions”. A fuzzy coalition is an element $\tau = \{\tau_1, \dots, \tau_n\} \in [0, 1]^n$, where the i th component $\tau_i \in [0, 1]$ represents the *rate of participation* of the i th player in the fuzzy coalition τ . Clearly, any coalition $A \subset N$ is identified with the fuzzy coalition τ^A , where the rate of participation τ_i^A of player i in A is either 1 when $i \in A$ or 0 when $i \notin A$. If we adequately define the behavior of fuzzy coalitions and, in particular, how a fuzzy coalition rejects a given multistrategy $\bar{x} \in X(N)$, we can introduce the *fuzzy core* of the game. This is the subset of multistrategies $\bar{x} \in X(N)$ which are not rejected by any fuzzy coalition $\tau \in [0, 1]^n$.

Is it possible to identify a useful subset of the fuzzy core?

The answer is positive. We introduce the concept of a canonical cooperative equilibrium $\bar{x} \in X(N)$ for which we require the existence of $\bar{\lambda} \in \mathcal{M}^n$ such that, for any coalition A ,

$$\sum_{i \in A} \bar{\lambda}^i(\bar{x}) \leq \inf_{x^A \in X(A)} \sum_{i \in A} \bar{\lambda}^i f_i^A(x^A).$$

We shall see in Chapter 12 that under quite reasonable assumptions, the fuzzy core and the set of equilibria coincide and that they are not empty.

Throughout this chapter we study the example of the *economic game* constructed from an economy $\{R^i, Y(i), f_i\}_{i \in N}$.

*10.1. Behavior of the whole set of players: Pareto strategies

After defining Pareto strategies, we interpret elements $\lambda \in \mathbf{R}^{n*}$ of the dual of the space of multilosses as “rates of transfer” of side-payments. Then, after noticing that any strategy which minimizes $\sum_{i=1}^n \lambda^i f_i(\cdot)$ is a weak Pareto minimum we prove that the converse statement holds under convexity assumptions. We generalize this theorem in the case of an economic game. Under convenient assumptions, an allocation $\bar{x} = \{\bar{x}^1, \dots, \bar{x}^n\} \in X(N)$ is a weak Pareto minimum if and only if there exist a rate of transfer $\bar{\lambda} \in \mathbf{R}^{n*}$ and a price $\bar{p} \in \mathbf{R}^{n*}$ such that

$$\begin{cases} \text{(i)} \quad \forall i \in N, \bar{\lambda}^i f_i(\bar{x}^i) + \langle \bar{p}, \bar{x}^i \rangle = \min_{y \in R^i} [\bar{\lambda}^i f_i(y) + \langle \bar{p}, y \rangle], \\ \text{(ii)} \quad \sum_{i=1}^n \langle \bar{p}, \bar{x}^i \rangle = \sup_{y \in Y} \langle \bar{p}, y \rangle. \end{cases}$$

10.1.1. Pareto strategies

Although a non-cooperative equilibrium is “individually” stable, it lacks “collective stability”. Indeed, in many instances, the players can find a multi-strategy yielding to each player a smaller loss than a non-cooperative equilibrium.

Thus, by adopting such a multistrategy, *all players* will improve upon a non-cooperative multistrategy. Such a move requires a minimum of cooperation among the players because each player has nothing to lose in making the change. When this small amount of cooperation is assumed, it is natural to set aside any strategy which can be *improved upon by the whole set N of players* and to keep the others. One is then left with the Pareto strategies. These are defined as follows.

Definition 1. Let $\{X(N), F\}$ be a game described in strategic form. We shall say that $x \in X(N)$ is a *weak Pareto strategy* if and only if

- (1) there is no $y \in X(N)$ such that, $\forall i \in N, f_i(y) < f_i(x)$.

In a more compact form, we have the equivalent statement: $x \in X(N)$ is a *weak Pareto strategy* if and only if

- (2) there is no $y \in X(N)$ such that $F(y) \ll F(x)$.

Usually, a somewhat stronger concept is used.

Definition 2. We shall say that $x \in X(N)$ is a (strong) "*Pareto strategy*" if and only if

- (3) there is no $y \in X(N)$ such that $F(y) < F(x)$.

It is clear that *any (strong) Pareto strategy is a weak Pareto strategy.*

Remark. To study Pareto strategies, we do not need to assume that $X(N)$ is a subset of a product $\prod_{i=1}^n X^i$ of strategy sets.

Remark. The concept of Pareto minimum is very often used in modeling "decision problems under several criteria". The set $X(N)$ is regarded as a "decision set", the loss functions f_i as "criteria" to be minimized by the decision maker. A (weak) *Pareto minimum* is usually called a (weak) *efficient decision* in this framework.

Remark. *Preordering on X associated with the multiloss.* A Pareto strategy is nothing other than a maximal element of $X(N)$ for the partial preordering relation on X defined by

- (4) $x \succcurlyeq y$ if and only if $F(x) \leq F(y)$.

The associated indifference relation is given by

- (5) $x \simeq y$ if and only if $F(x) = F(y)$.

We shall set

- (6) $x \succ y$ if and only if $F(x) \ll F(y)$

so that $x \in X(N)$ is a weak Pareto strategy if there is no $y \in X(N)$ such that $y \succ x$.

Remark. The concept of a Pareto minimum depends only upon the partial preordering and not upon its representation by the multiloss operator F from $X(N)$ into $\mathbb{R}^n$.

It is clear that two multiloss operators F and G define the same preordering if and only if there exists a injective strictly increasing map Φ from $F(X(N))$ into $\mathbb{R}^n$ such that $G(x) = \Phi(F(x))$.

Proposition 1. *A strategy $\bar{x} \in X(N)$ is a weak Pareto minimum if and only if $F(\bar{x}) \notin \hat{F}_+(X(N))$.*

Proof. To say that $F(\bar{x})$ belongs to $\hat{F}_+(X(N))$ means that there exists $y \in X(N)$ and $c \in \mathbb{R}_+^n$ such that $F(\bar{x}) = F(y) + c \geq F(y)$, i.e. that $\bar{x}$ is not a Pareto minimum. $\square$

Proposition 2. *The set of weak Pareto minima is $F^{-1}[(F(X(N)) \cap \mathbb{C}\hat{F}_+(X(N)))]$. It is closed whenever F is continuous and $F(X(N))$ is closed.*

10.1.2. Rates of transfer

The problem arises as to whether losses “can be transferred” from one player to another and if so, how this can be done.

The key to this is to use linear functionals $\lambda \in \mathbb{R}^{n*}$ which associate with any multiloss $c \in \mathbb{R}^n$ a number of “units of account” $\langle \lambda, c \rangle$ called a “side-payment”.

The linear functional λ defines “rates of transfer of loss” among the players.

Definition 3. We shall interpret the dual of the multiloss space as the space of “systems of rates of transfer” (in short, “rates of transfer”) of side-payments.

We shall say that two multilosses b and c are “transferable” (under the rate $\lambda \in \mathbb{R}^{n*}$) if and only if $\langle \lambda, b \rangle = \langle \lambda, c \rangle$.

In another words, rates of transfer play the same role vis à vis the multilosses as the prices vis à vis the commodities.

The use of the label “rate of transfer” can be justified in the following way: c_i units of loss of the i th player and c_j units of loss of the j th player are transferable if $\lambda^i c_i = \lambda^j c_j$, i.e. if $c_i = (\lambda^j / \lambda^i) c_j$ (if $\lambda^i \neq 0$) or $c_j = (\lambda^i / \lambda^j) c_i$ (if $\lambda^j \neq 0$). We can say that λ^j / λ^i (resp. λ^i / λ^j) is the rate of transfer of losses from player j to player i (from player i to player j).

We can either use elements of $\mathbb{R}$ supplied with a unit (side-payments) or a given multiloss $c \in \mathbb{R}^n$ (numéraire) for implementing exchanges of multilosses among players.

Usually, we shall choose the unit vector $\mathbf{1} \in \mathbb{R}^n$ as such a numéraire. Then $\langle \lambda, c \rangle / \langle \lambda, \mathbf{1} \rangle = \langle \lambda / \langle \lambda, \mathbf{1} \rangle, c \rangle$ is the number of units $\mathbf{1}$ which can be exchanged with c under the rate λ . Therefore, if we are dealing with non-negative systems of rates of transfer, we shall use rates belonging to the n -simplex $\mathcal{M}^n$ of $\mathbb{R}^{n*}$.

10.1.3. Pareto multipliers

Let $\{X(N), F\}$ be a game defined in strategic form. If a rate of transfer $\lambda \in \mathcal{M}^n$ prevails, the players can “pool” their loss functions and use the collective loss function $x \mapsto \langle \lambda, F(x) \rangle = \sum_{i=1}^n \lambda^i f_i(x)$.

Proposition 3. *Let $\lambda \in \mathcal{M}^n$ and let $x_\lambda \in X(N)$ be a strategy minimizing the function $x \mapsto \langle \lambda, F(x) \rangle$ over $X(N)$. Then x_λ is a weak Pareto strategy.*

Proof. If x_λ is not a weak Pareto strategy, there exists $x \in X(N)$ such that $F(x) \ll F(x_\lambda)$. Since $\lambda \in \mathcal{M}^n$, this implies that $\langle \lambda, F(x) \rangle < \langle \lambda, F(x_\lambda) \rangle$, i.e. that x_λ does not minimize $x \mapsto \langle \lambda, F(x) \rangle$ over $X(N)$. $\square$

The converse is true under convexity assumptions.

Proposition 4. *Suppose that*

$$(7) \quad \begin{cases} \text{(i) the strategy set } X(N) \text{ is convex,} \\ \text{(ii) the loss functions } f_i \text{ are convex.} \end{cases}$$

Then any weak Pareto strategy $\bar{x} \in X(N)$ minimizes over $X(N)$ a collective loss function $\langle \lambda, F(\cdot) \rangle$ for a convenient rate of transfer $\lambda \in \mathcal{M}^n$.

Proof. Write $X = X(N)$. By Proposition 1.3.10, convexity assumptions (7) imply that

$$(8) \quad \mathring{F}_+(X) = F(X) + \mathring{R}_+^n \text{ is convex a subset of } \mathbf{R}^n.$$

Let $\bar{x} \in X$ be a weak Pareto strategy. Hence $1 \cdot F(\bar{x})$ does not belong to the open convex subset $F_+^0(X)$. Therefore, by the separation theorem, we can find $\lambda \in \mathbf{R}^{n*}$, $\lambda \neq 0$, such that

$$(9) \quad \langle \lambda, F(x) \rangle \leq \inf_{c \in \mathring{F}_+(X)} \langle \lambda, c \rangle = \sigma^b(F(X); \lambda) + \sigma^b(\mathbf{R}_+^n; \lambda)$$

This implies that $\sigma^b(\mathbf{R}_+^n; \lambda)$ is bounded below. Hence $\lambda \in \mathbf{R}_+^{n*}$ and $\sigma^b(\mathbf{R}_+^n; \lambda) = 0$. Dividing λ by $\sum_{i=1}^n \lambda^i > 0$, we obtain that

$$(10) \quad \langle \bar{\lambda}, F(\bar{x}) \rangle = \sigma^b(F(X); \bar{\lambda}) = \inf_{x \in X} \langle \bar{\lambda}, F(x) \rangle \quad \text{where } \bar{\lambda} = \frac{\lambda}{\sum_{i=1}^n \lambda^i} \in \mathcal{M}^n.$$

Definition 4. We shall say that a rate of transfer $\lambda \in \mathcal{M}^n$ such that $\langle \lambda, F(\bar{x}) \rangle = \inf_{x \in X} \langle \lambda, F(x) \rangle$ is a “Pareto multiplier” of $\bar{x}$.

Proposition 5. *Let λ and μ be Pareto multipliers of weak Pareto minima x and y respectively. Then*

$$(11) \quad \langle \lambda - \mu, F(x) - F(y) \rangle = \sum_{i=1}^n (\lambda^i - \mu^i) (f_i(x) - f_i(y)) \leq 0.$$

Proof. We have that

$$\langle \lambda, F(x) \rangle \leq \langle \lambda, F(y) \rangle$$

and

$$\langle \mu, F(y) \rangle \leq \langle \mu, F(x) \rangle.$$

We obtain (11) by summing these two inequalities. $\square$

We denote the lower support function of $F_+(X(N))$ by

$$(12) \quad w(\lambda) = \inf_{x \in X(N)} \langle \lambda, F(x) \rangle = \sigma^b(F_+(X(N)); \lambda).$$

Proposition 6. *A strategy $\bar{x} \in X$ is a weak Pareto minimum associated with a Pareto multiplier $\bar{\lambda}$ if and only if $F(\bar{x})$ belongs to the super-differential of the function w at $\bar{\lambda}$, i.e.*

$$(13) \quad F(\bar{x}) \in \bar{\partial} w(\bar{\lambda}).$$

Proof. Proposition (13) means that

$$\langle \bar{\lambda} - \mu, F(\bar{x}) \rangle \leq w(\bar{\lambda}) - w(\mu).$$

If (13) holds, then $\langle \bar{\lambda}, F(\bar{x}) \rangle = w(\bar{\lambda})$ (by taking $\mu = 0$) and thus, $\bar{x}$ minimizes $\langle \bar{\lambda}, F(\cdot) \rangle$.

Conversely, if $\bar{x}$ minimizes $\langle \bar{\lambda}, F(\cdot) \rangle$, then $\langle \bar{\lambda}, F(\bar{x}) \rangle = w(\bar{\lambda})$. Otherwise, for any $\mu \in \mathbf{R}_+^{n*}$, $\langle \mu, F(x) \rangle \geq w(\mu)$. Thus (13) holds. $\square$

***Remark.** Let $\lambda \in \dot{\mathcal{M}}^n = \mathcal{M}^n \cap \dot{\mathbf{R}}_+^{n*}$ be strongly positive. We can easily check that, if $\bar{x}$ minimizes $\langle \lambda, F(\cdot) \rangle$, then $\bar{x}$ is a (strong) Pareto strategy, i.e. that

$$(14) \quad (F(X) - F(\bar{x})) \cap -\mathbf{R}_\oplus^n = \emptyset \quad \text{where } \mathbf{R}_\oplus^n = \mathbf{R}_+^n \setminus \{0\}.$$

It is even obvious that

$$(15) \quad (F(X) - F(\bar{x}))^{++} \cap -\mathbf{R}_\oplus^n = \emptyset,$$

since the fact that $\bar{x}$ minimizes $\langle \lambda, F(\cdot) \rangle$ amounts to saying that $\lambda \in (F(X) - F(\bar{x}))^+ = (F(X) - F(\bar{x}))^{+++}$. The converse is also true.

Proposition 7. *Let $\bar{x}$ be a strong Pareto strategy. If the closed convex cone spanned by $F(X) - F(\bar{x})$ does not intersect $-\mathbf{R}_\oplus^n$, then there exists a strongly positive Pareto multiplier $\lambda \in \dot{\mathcal{M}}^n$.*

Proof. Suppose that there is no strongly positive Pareto multiplier, i.e. that $(F(X) - F(\bar{x}))^+ \cap \dot{\mathbf{R}}_+^{n*} = \emptyset$. By the separation theorem, we obtain the existence of $c \in \mathbf{R}^n$, $c \neq 0$ such that $c \in (\dot{\mathbf{R}}_+^{n*})^- = -\mathbf{R}_+^n$ and $c \in (F(X) - F(\bar{x}))^{++}$, i.e. such that $c \in (F(X) - F(\bar{x}))^{++} \cap -\mathbf{R}_\oplus^n$. This is a contradiction of the assumption. $\square$

We can devise sufficient conditions for a strong Pareto strategy to have a strongly positive Pareto multiplier. For instance, we have the following result.

Proposition 8. Let x belong to $X(N)$. Suppose that, for any non-empty subset $A \neq N$ of players, we can find $y \in X(N)$ such that

$$(16) \quad f_i(y) < f_i(x) \quad \forall i \in A \quad \text{and} \quad f_j(y) \geq f_j(x) \quad \forall j \notin A.$$

If x minimizes $\langle \lambda, F(\cdot) \rangle$ over $X(N)$, then λ belongs to $\mathcal{M}^n$.

Proof. If $\lambda \in \mathcal{M}^n - \mathcal{M}^n$, take $A = \{i \in N \text{ such that } \lambda^i > 0\}$ and $y \in X(N)$ satisfying (16). We obtain the following contradiction:

$$\langle \lambda, F(x) \rangle = \sum_{i \in A} \lambda^i f_i(x) > \sum_{i \in A} \lambda^i f_i(y) = \langle \lambda, F(y) \rangle.$$

10.1.4. Pareto allocations

Consider the set of allocations defined by

$$(17) \quad X(N) = \left\{ x \in R^N \text{ such that } \sum_{i \in N} L_i x_i \in Y \right\}$$

where

$$(18) \quad \begin{cases} \text{(i) } R^i \text{ is subset of a vector space } U^i, \\ \text{(ii) } Y \text{ is a subset of } V = R^k, \\ \text{(iii) } L_i \text{ is a map from } U^i \text{ into } V. \end{cases}$$

Consider also

$$(19) \quad n \text{ loss functions } f_i : R^i \longrightarrow R$$

and the profit function r defined by

$$(20) \quad \forall p \in R^{k*}, \quad r(p) = \sup_{y \in Y} \langle p, y \rangle = \sigma^*(Y; p).$$

Denote by P the barrier cone of Y , i.e. the domain of r .

Proposition 9. Let $\bar{\lambda} \in \mathcal{M}^n$ and $\bar{p} \in P$ be fixed. If $\bar{x} = \{\bar{x}^1, \dots, \bar{x}^n\} \in R^N$ satisfies the following properties

$$(21) \quad \begin{cases} \text{(i) } \forall i \in N, \bar{\lambda}^i f_i(\bar{x}^i) + \langle \bar{p}, L_i \bar{x}^i \rangle = \inf_{y \in R^i} [\bar{\lambda}^i f_i(y) + \langle \bar{p}, L_i y \rangle], \\ \text{(ii) } \sum_{i \in N} \langle \bar{p}, L_i \bar{x}^i \rangle = r(\bar{p}), \end{cases}$$

then $\bar{x}$ is a weak Pareto allocation.

Proof. If $\bar{x}$ is not a Pareto multistrategy, there exists $y \in X(N)$ such that $f_i(y_i) < f_i(\bar{x}^i)$ for all i . Thus

$$(22) \quad \sum_{i \in N} \bar{\lambda}^i f_i(y^i) < \sum_{i \in N} \bar{\lambda}^i f_i(\bar{x}^i).$$

Since $y \in X(N)$, $\langle \bar{p}, \sum_{i \in N} L_i y^i \rangle \leq r(\bar{p})$. Using (21(ii)), we deduce that

$$(23) \quad \sum_{i \in N} (\bar{\lambda}^i f_i(y^i) + \langle \bar{p}, L_i y^i \rangle) < \sum_{i \in N} (\bar{\lambda}^i f_i(\bar{x}^i) + \langle \bar{p}, L_i \bar{x}^i \rangle).$$

Adding inequalities (21(i)), we obtain that

$$(24) \quad \sum_{i \in N} (\bar{\lambda}^i f_i(\bar{x}^i) + \langle \bar{p}, L_i \bar{x}^i \rangle) \leq \sum_{i \in N} (\bar{\lambda}^i f_i(y^i) + \langle \bar{p}, L_i y^i \rangle).$$

This is a contradiction. $\square$

We shall prove the converse statement under convexity assumptions.

Theorem 1. *Suppose that*

$$(25) \quad \begin{cases} \text{(i) The subsets } R^i \text{ and } Y \text{ are convex} \\ \text{(ii) The operators } L_i \text{ from } U_i \text{ into } V = \mathbf{R}^k \text{ are linear,} \\ \text{(iii) The functions } f_i \text{ are convex.} \end{cases}$$

and that

$$(26) \quad 0 \in \text{Int} \left(\sum_{i \in N} L_i(R^i) - Y \right).$$

Let $\bar{x} \in X(N)$ be a weak Pareto allocation. Then there exist $\bar{\lambda} \in \mathcal{M}^n$ and $\bar{p} \in P$ such that properties (21) hold.

Proof. Introduce the map $\phi : R^N \times Y \rightarrow \mathbf{R}^n \times \mathbf{R}^k$ defined by

$$(27) \quad \phi(x, y) = \left\{ F(x), \sum_{i \in N} L_i x^i - y \right\}.$$

Write

$$(28) \quad Q = \mathbf{R}_+^n \times \{0\}.$$

It is clear that $\bar{x} \in X(N)$ is a weak Pareto allocation if and only if

$$(29) \quad \{F(\bar{x}), 0\} \notin \phi(R^N \times Y) + Q.$$

(To prove this, observe that the fact that $\{F(\bar{x}), 0\}$ belongs to $\phi(R^N \times Y) + Q$ means that there exist $x \in R^N$ and $y \in Y$ such that $F(\bar{x}) \gg F(x)$ and $\sum_{i \in N} L_i x^i = y \in Y$ i.e. that $F(x) \gg F(\bar{x})$ where $x \in X(N)$.)

It is also clear that convexity assumptions (25) imply that the subset $\phi(R^N \times Y) + Q$ is convex.

Therefore, the separation theorem in finite dimensional spaces implies the existence of $\{\lambda, p\} \in \mathbf{R}^{n*} \times \mathbf{R}^{k*}$, $\{\lambda, p\} \neq 0$, such that

$$(30) \quad \langle \lambda, F(\bar{x}) \rangle = \inf_{\substack{x \in R^N \\ y \in Y}} \left[\langle \lambda, F(x) \rangle + \left\langle p, \sum_{i \in N} L_i x^i - y \right\rangle \right] + \sigma^b(\mathbf{R}_+^n; \lambda).$$

This implies that $\lambda \in \mathbf{R}_+^{n*}$ and that $\sigma^b(\mathbf{R}_+^n; \lambda) = 0$. On the other hand, assumption (26) implies that $\lambda \neq 0$ [To prove this, assume that $\lambda = 0$. We deduce from (30) that $0 = \inf_{x \in R^N, y \in Y} \langle p, \sum_{i \in N} L_i x^i - y \rangle = \sigma^b(\sum_{i \in N} L_i(R^i) - Y; p)$. Since 0 belongs to the interior of $\sum_{i \in N} L_i(R^i) - Y$, we deduce that $\sigma^b(\sum_{i \in N} L_i(R^i) - Y; p) < 0$ whenever $p \neq 0$. Since $\{\lambda, p\} = \{0, p\} \neq 0$, then $p \neq 0$. Thus λ cannot be equal to 0]. Therefore, $\sum_{i \in N} \lambda^i > 0$. Writing $\bar{\lambda} = \lambda / \sum_{i \in N} \lambda^i \in \mathcal{M}^n$ and $\bar{p} = p / \sum_{i \in N} \lambda^i$, we deduce from (30) that

$$(31) \quad \begin{aligned} \langle \bar{\lambda}, F(\bar{x}) \rangle &= \inf_{x \in R^N} \left[\langle \bar{\lambda}, F(x) \rangle + \left\langle \bar{p}, \sum_{i \in N} L_i x^i \right\rangle \right] - \sup_{y \in Y} \langle \bar{p}, y \rangle \\ &= \inf_{x \in R^N} \left(\langle \bar{\lambda}, F(x) \rangle + \left\langle \bar{p}, \sum_{i \in N} L_i x^i \right\rangle \right) - r(\bar{p}). \end{aligned}$$

This implies in particular that $r(\bar{p}) < \infty$, and thus, that $\bar{p} \in P$. Taking $x = \bar{x}$ in (31), we obtain that $0 \leq \langle \bar{p}, \sum_{i \in N} L_i \bar{x}^i \rangle - r(\bar{p})$. Since $\bar{x} \in X(N)$, we know that $\langle \bar{p}, \sum_{i \in N} L_i \bar{x}^i \rangle \leq r(\bar{p})$. Thus $r(\bar{p}) = \sum_{i \in N} \langle \bar{p}, L_i \bar{x}^i \rangle$. Inequality (31) can be rewritten

$$\begin{aligned} \langle \bar{\lambda}, F(\bar{x}) \rangle + \sum_{i \in N} \langle \bar{p}, L_i \bar{x}^i \rangle &= \\ &= \inf_{x \in R^N} \left[\langle \bar{\lambda}, F(x) \rangle + \sum_{i \in N} \langle \bar{p}, L_i x^i \rangle \right]. \end{aligned}$$

Thus property (21) holds. $\square$

Remark. In the case when $n = 1$, we obtain Theorem 5.3.1 which asserts the existence of a Lagrange multiplier. We shall say that $\bar{\lambda} \in \mathcal{M}^n$ is the *Pareto multiplier* and $\bar{p} \in P$ the *Lagrange multiplier* associated with the weak Pareto allocation $\bar{x} \in X(N)$.

Remark. The following definition is often used.

Definition 5. We shall say that $\{\bar{x}, \bar{p}\}$ is a “*pseudo-equilibrium*” if the following properties hold for all i :

$$(32) \quad f_i(x^i) \leq f_i(\bar{x}^i) \Rightarrow \langle \bar{p}, L_i x^i \rangle \geq \langle \bar{p}, L_i \bar{x}^i \rangle.$$

Then the above theorem implies the following result.

Proposition 10. *Suppose that (25) and (26) hold. We can associate with any weak Pareto allocation $\bar{x} \in X(N)$ an element $\bar{p} \in P$ such that $\{\bar{x}, \bar{p}\}$ is a pseudo-equilibrium.*

Proof. We deduce from (21) that

$$\langle \bar{p}, L_i \bar{x}^i \rangle - \langle \bar{p}, L_i x^i \rangle \leq \lambda^i (f_i(x^i) - f_i(\bar{x}^i))$$

from which we deduce statements (32). $\square$

10.2. Selection of Pareto strategies and imputations

The concept of Pareto optimality is useful in dismissing strategies which can be improved upon by the whole set of players acting as a unit under the behavioral assumption that it is good to make someone better off provided no one else is made worse off.

This cooperation requirement among players is too weak. In general, the set of Pareto minima may be rather "large". This is quite a drawback for a solution concept.

We therefore have to find selection procedures enabling us to discriminate among Pareto minima.

We shall see that the concepts of imputations, core and cooperative equilibria are just such selection procedures. Before defining them, we shall begin by surveying examples of a simple kind of selection procedures, namely, procedures using "selection functions" $s: \mathbf{R}_+^n \rightarrow \mathbf{R}$ which are increasing. We select Pareto strategies by minimizing $s[F_T(x)]$ on $X(N)$, where the multiloss operator F_T is deduced from F by the formulas

$$f_{t_i}(x) = \frac{f_i(x) - \alpha(f_i)}{t_i(f_i) - \alpha(f_i)}, \quad F_T(x) = \{f_{t_i}(x)\}_{i \in N}$$

with

$$\begin{cases} \text{(i)} & \alpha(f_i) = \inf_{x \in X(N)} f_i(x), \\ \text{(ii)} & t_i(f_i) > \alpha(f_i) \text{ for all } i \in N. \end{cases}$$

In particular, by taking $s(c) = \sup_{i \in N} c_i$, we obtain the concept of the "best compromise" $\bar{x}$ defined by: $\forall i \in N, f_i(\bar{x}) \leq (1 - \bar{d}) \alpha(f_i) + \bar{d} t_i(f_i)$ where $\bar{d} = \inf_{x \in X(N)} \max_{i \in N} f_{t_i}(x) \in [0, 1]$. We require that the functionals $t_i: f_i \rightarrow t_i(f_i)$ satisfy $t_i(af + b) = at_i(f) + b$ whenever $a \in \mathbf{R}_+, b \in \mathbf{R}$. They are interpreted as *threat functionals*, in the sense that they set a maximum $t_i(f_i)$ to the amount of loss that player i accepts.

We see that a "best compromise" $\bar{x}$ is a weak Pareto strategy such that, for any $i \in N$, $f_i(\bar{x}) \leq t_i(f_i)$. This is a particular example of an *imputation*.

We end this section by using selection functions to choose particular imputations. We show that the Nash bargaining solution is an example of such a selection procedure in the case of two players.

10.2.1. Normalized games

We shall require that a selection procedure does not depend upon positive affine transformations of the loss functions. For this purpose, we introduce functionals $t : \mathcal{S}(X(N)) \rightarrow \mathbf{R}$ satisfying the following property:

$$(1) \quad \forall \alpha > 0, \forall \beta \in \mathbf{R}, \quad t(\alpha f(\cdot) + \beta) = \alpha t(f) + \beta.$$

For instance, the following functionals defined by

$$(2) \quad t(f) = \sum_{n=1}^K \lambda^n f(\bar{x}_n),$$

$$(3) \quad t(f) = \inf_{x \in X(N)} f(x) \quad \text{or} \quad t(f) = \sup_{x \in X(N)} f(x),$$

$$(4) \quad t(f) = \inf_{x' \in X^I} \sup_{x'' \in X^I} f_i(x', x'')$$

satisfy the above property.

Definition 1. Let $\{X(N), F\}$ be a game bounded below and let $T = (t_1, \dots, t_n)$ be n functionals satisfying (1) and

$$(5) \quad \begin{cases} \text{(i)} \quad \forall i \in N, t_i(f_i) > \alpha(f_i) = \inf_{x \in X(N)} f_i(x), \\ \text{(ii)} \quad \exists x \in X(N) \text{ such that } f_i(x) \leq t_i(f_i) \text{ for all } i \in N. \end{cases}$$

We shall say that the game $\{X(N), F_T\}$ defined by the loss functions

$$(6) \quad f_{t_i}(x) = \frac{f_i(x) - \alpha(f_i)}{t_i(f_i) - \alpha(f_i)}$$

is the T -normalized (or, *normalized*) game.

We shall say that $s : \mathbf{R}_+^n \rightarrow \mathbf{R}$ is a "*selection function*" (resp. "*strong selection function*") if

$$(7) \quad \forall c, d \in \mathbf{R}_+^n \quad \text{such that } c \gg d, \text{ then } s(c) > s(d) \text{ [resp. } c > d \text{ implies that } s(c) > s(d)\text{].}$$

Remarks. Notice that the sets of Pareto multistrategies of the games $\{X(N), F\}$ and $\{X(N), F_T\}$ are the same and that the shadow minimum of the normalized game $\{X(N), F_T\}$ is 0. Also

$$(8) \quad \forall i \in N, \quad t_i(f_i) = 1$$

Notice further that, if

$$(9) \quad \forall i \in N, \quad g_i(x) = \alpha_i f_i(x) + \beta_i, \quad \alpha_i > 0, \beta_i \in \mathbf{R},$$

then $F_T = G_T$.

In other words, if we say that two games $\{X(N), F\}$ and $\{X(N), G\}$ are *strategically equivalent* if (9) holds, the normalized game is an equivalence class of mutually strategically equivalent games.

10.2.2. Pareto strategies obtained by using selection functions

Proposition 1. Let T be a set of n functionals satisfying (I) and (5) and let s be a selection function. Then any $\bar{x} \in X(N)$ which minimizes the function $x \rightarrow s(F_T(x))$ over $X(N)$ is a weak Pareto strategy.

Proof. We have that $F_T(X) \subset \mathbf{R}_+^n$ since 0 is the shadow minimum of $\{X(N), F_T\}$. Thus the function $x \rightarrow s(F_T(x))$ is defined on $X(N)$. If $\bar{x}$ is not a weak Pareto minimum, there exists $x \in X(N)$ such that $F_T(x) \ll F_T(\bar{x})$

Therefore $s(F_T(x)) < s(F_T(\bar{x}))$ by (7) which is a contradiction. $\square$

Remark. If s is a strong selection function, $\bar{x}$ is a strong Pareto minimum.

Remark. Functions $\langle \lambda, \cdot \rangle$ are selection functions if $\lambda \in \mathcal{M}^n$ and strong selection functions if $\lambda \in \hat{\mathcal{M}}^n$. Proposition 1 generalizes Proposition 1.3.

Proposition 2. Suppose that

$$(10) \quad \begin{cases} \text{(i) } X(N) \text{ is a convex subset,} \\ \text{(ii) the loss functions } f_i \text{ are convex} \end{cases}$$

and that

$$(11) \quad \text{the selection function } s \text{ is non-decreasing and differentiable.}$$

If $\bar{x}$ minimizes $s[F_T(\cdot)]$, then the rate of transfer λ of components $\bar{\lambda}^i = D_{\lambda} s(F_T(\bar{x})) / (t_i(f_i) - \alpha(f_i))$ is a Pareto multiplier of $\bar{x}$.

Proof. Assumptions (10) imply that $F_T(X(N)) + \mathbf{R}_+^n$ is convex by Proposition 1.3.10.

Then $F_T(\bar{x}) + \theta(F_T(x) - F_T(\bar{x}))$ is larger than or equal to $F_T(y)$ for a suitable y when $\theta \in [0, 1]$. Thus $s(F_T(\bar{x})) \leq s(F_T(y)) \leq s(F_T(\bar{x}) + \theta(F_T(x) - F_T(\bar{x})))$.

This implies that

$$(12) \quad \langle Ds(F_T(\bar{x})), F_T(x) - F_T(\bar{x}) \rangle = \lim_{\theta \rightarrow 0} \frac{s(F_T(\bar{x}) + \theta(F_T(x) - F_T(\bar{x}))) - s(F_T(\bar{x}))}{\theta} \geq 0$$

for all $x \in X(N)$.

Since s is not decreasing, $Ds(F_T(\bar{x})) \in \mathbf{R}_+^{n*}$.

Therefore, $\bar{\lambda}^i = D_i s(F_T(\bar{x})) / (t_i(f_i) - \alpha(f_i))$ is non-negative. Thus (12) implies that $\bar{x}$ minimizes $\langle \bar{\lambda}, F(\cdot) \rangle$ where $\bar{\lambda} = \{\bar{\lambda}^1, \dots, \bar{\lambda}^n\}$. $\square$

We give sufficient conditions of existence in the following proposition.

Proposition 3. Suppose that $X(N)$ is a subset of a topological space U^N and

$$(13) \quad \begin{cases} \text{(i) the extended loss functions } f_{i, X(N)} : U^N \mapsto]-\infty, +\infty] \\ \text{are lower-semi-continuous,} \\ \text{(ii) a loss function } f_i \text{ is lower semi-compact.} \end{cases}$$

Suppose also that

$$(14) \quad \text{the selection function } s \text{ is continuous and its lower sections are bounded above.}$$

Then there exists $\bar{x} \in X(N)$ which minimizes $s(F_T(\cdot))$.

Proof. We define g by $g(x) = s(F_T(x))$. Then g is clearly lower semi-continuous. It is also lower semi-compact because, if $g(x) \leq \lambda$, then $F_T(x) \leq c$ since the lower section of s is bounded above. Therefore, since f_i is lower semi-compact, the inequality

$$f_{i, X(N)}(x) = \frac{f_i(x) - \alpha(f_i)}{t_i(f_i) - \alpha(f_i)} \leq c_i$$

implies that x belongs to a relatively compact subset.

Therefore g achieves its minimum. $\square$

*10.2.3. Closest strategy to the shadow minimum

Simple examples of strong selection functions are the functions $s_p : c \mapsto s_p(c) = (\sum_{i \in N} |c_i|^p)$ where $1 \leq p < +\infty$ (for $p = 1$, we have the linear selection studied in Section 10.1.3).

In this case, $\bar{x} \in X(N)$ minimizes the function $x \mapsto s_p(F_T(x))$ over $X(N)$ if and only if $F(\bar{x})$ is the projection of the shadow minimum onto $F_T(X(N))$ for the norm

$$\|c\|_p = \left(\sum_{i \in N} \frac{|c_i|^p}{|t_i(f_i) - \alpha_i|^p} \right)^{1/p}$$

where we set $\alpha_i = \alpha(f_i)$.

The simplest case is that when $p = 2$.

Proposition 4. *Suppose that assumption (13) of Proposition 3 hold and that*

$$(15) \quad \begin{cases} \text{(i) } X(N) \text{ is convex,} \\ \text{(ii) the loss functions } f_i \text{ are convex.} \end{cases}$$

Then there exists a (strong) Pareto strategy $\bar{x}$ whose multiloss $F_T(\bar{x})$ is closest to the shadow minimum. Its Pareto multiplier $\bar{\lambda} \in \mathbb{R}_+^n$ is defined by

$$(16) \quad \bar{\lambda}_i = \frac{f_i(\bar{x}) - \alpha_i}{|t_i(f_i) - \alpha_i|^2}.$$

Proof. It is clear that assumptions (13) of Proposition 3 imply the existence of $\bar{x}$ minimizing

$$x \mapsto \sum_{i=1}^n \frac{|f_i(x) - \alpha_i|^2}{|t_i(f_i) - \alpha_i|^2}$$

Since $F_T(X(N)) + \mathbb{R}_+^n$ is convex, by (15), we deduce that for all $x \in x(N)$

$$(17) \quad \sum_{i \in N} \frac{(f_i(\bar{x}) - \alpha_i)}{|t_i(f_i) - \alpha_i|^2} (f_i(\bar{x}) - f_i(x)) = \langle \bar{\lambda}, F(\bar{x}) - F(x) \rangle \leq 0.$$

10.2.4. The best compromise

The function s_∞ defined by $s_\infty(c) = \|c\|_\infty = \sup_{i \in N} |c_i|$ is clearly a selection function.

Proposition 5. *Let $\bar{d} = \inf_{x \in X(N)} s_\infty(F_T(x))$. Then $\bar{x} \in X(N)$ achieves the minimum of $s_\infty(F_T(\cdot))$ if and only if*

$$(18) \quad \forall i \in N, \quad f_i(\bar{x}) \leq (1 - \bar{d})\alpha(f_i) + \bar{d}t_i(f_i)$$

where the equality holds for at least one i . Further, $\bar{x}$ is a weak Pareto strategy.

Proof. We know that $\bar{x}$ is a weak Pareto minimum by Proposition 1. Since

$$\bar{d} = \max_{i \in N} \frac{f_i(\bar{x}) - \alpha(f_i)}{t_i(f_i) - \alpha(f_i)},$$

we deduce (18).

Conversely, if $\bar{x}$ satisfies (18), it is clear that $s_\infty(F_T(\bar{x})) \leq \bar{d}$.

Definition 2. Any strategy $\bar{x}$ satisfying (18) is called a best compromise. It is both as close as possible to the shadow minimum and as far as possible from the vector $\{t_i(f_i)\}_{i \in N}$.

10.2.5. Existence of Pareto strategies

We can use this proposition to prove the existence of Pareto multistrategies.

Proposition 6. Suppose that $X(N)$ is a subset of a topological space U^N and that

$$(19) \quad \begin{cases} \text{(i)} \quad \forall i \in N, \quad f_i|_{X(N)} : U^N \mapsto]-\infty, +\infty] \text{ is lower semi-continuous} \\ \text{(ii)} \quad \exists i \in N \text{ such that } f_i \text{ is lower semi-compact} \end{cases}$$

Then, for any $x_0 \in X(N)$ such that $F(x_0) \gg \alpha$, there exists a weak Pareto strategy $\bar{x} \in X(N)$ such that $F(\bar{x}) \leq F(x_0)$.

Proof. We take the functionals defined by $t_i(f_i) = f_i(x_0)$. Then the function

$$x \mapsto s_\infty(F_T(x)) = \sup_{i \in N} \frac{f_i(x) - \alpha_i(f)}{f_i(x_0) - \alpha_i(f)}$$

is lower semi-continuous and lower semi-compact. Therefore there exists $\bar{x}$ which achieves its minimum. By Proposition 5, $\bar{x}$ is a weak Pareto strategy satisfying $f_i(\bar{x}) \leq (1 - \bar{d}) \alpha(f_i) + \bar{d} f_i(x_0) \leq f_i(x_0)$ since $\bar{d} = \inf_{x \in X(N)} s_\infty(F_T(x)) \in [0, 1]$.

*10.2.6. Interpretation: threat functionals

We can interpret the functionals $t_i : f_i \mapsto t_i(f_i)$ as threat functionals in the following sense. If the n players are reluctant to come to an agreement for choosing a Pareto strategy, each player can threaten to implement his "threat" value $t_i(f_i)$. Once these threat functionals are chosen, the normalized loss functions take their value between their minimum equal to 0 and their threat value equal to 1.

Therefore, a *selection procedure takes threat-functionals into account if it yields to each player a loss not larger than $t_i(f_i)$* .

This is the case when we use the selection function s_∞ and we choose a *best compromise*. The choice by players of a given multistrategy x_0 can define a "threat functional" by setting $t_i(f_i) = f_i(x_0)$. It is agreed that x_0 is the worst strategy which must be implemented if the players do not come to an agreement to play a strategy yielding to each of them a smaller loss.

In other cases, we can take for the threat multistrategy either a conservative strategy or a non-cooperative equilibrium (if they exist). Such a choice describes the following two-stage procedure. Players begin by playing as individuals. They may choose a conservative strategy or, if it is possible, arrive at a non-cooperative equilibrium.

Secondly, they use their individually chosen strategies as "threat strategies" and look for a Pareto minimum better for all the players than the result obtained when the players implement their "threat strategies".

Another example of a threat functional is given by the functional

$$v^* : f_i \longrightarrow v^*(f_i) = \inf_{x^i \in X^i} \sup_{x^{-i} \in X^{-i}} f(x^i, x^{-i})$$

which describes the minimum loss that players i can obtain by unilateral actions, whatever the other players do. We can interpret v^* as a particular example of an "implicit" *threat functional*, in the sense that $v^*(f_i)$ sets a maximum to the amount of loss that player i will accept for himself.

*10.2.7. Imputations: the Nash bargaining solution

The above comments motivate the following definition.

Definition 3. Consider a game $\{X(N), F\}$ and n threat functionals t_i satisfying (1) and (5). We shall say that $\bar{x} \in X(N)$ is an "*imputation*" if

$$(20) \quad \begin{cases} \text{(i) } \bar{x} \text{ is a weak Pareto multistrategy,} \\ \text{(ii) } \forall i \in N, \quad f_i(\bar{x}) \leq t_i(f_i). \end{cases}$$

Property (20(ii)) is often called "*individual rationality*". We can give a systematic way of selecting an imputation.

Let us introduce the subset

$$(21) \quad X_T(N) = \{x \in X(N) \text{ such that } f_i(x) \leq t_i(f_i) \forall i \in N\}$$

and a selection function s .

We set $1 = \{1, \dots, 1\} \in \mathbb{R}^n$, which is the *threat vector* of the normalized game.

Proposition 7. Consider n threat functionals t_i satisfying (1) and (5) and a selection function s . Any $\bar{x} \in X_T(N)$ which maximizes the function $x \mapsto s(1 - F_T(x))$ is an imputation.

Proof. The proof is left as an exercise. $\square$

Example. The best compromise. The best compromise is obviously a selection of an imputation.

Example. Nash bargaining selection. Let us consider the case of $n = 2$ players. We take for selection function the function s defined by $s(c) = c_1 c_2$. Then

$$g(x) = s(1 - F_T(x)) = \frac{(f_1(x) - t_1(f_1))(f_2(x) - t_2(f_2))}{(t_1(f_1) - \alpha(f_1))(t_2(f_2) - \alpha(f_2))}.$$

The solution $\bar{x} \in X_T(\{1, 2\})$ which maximizes g over $X_T(\{1, 2\})$ was introduced by Nash and is called the solution to the “Nash bargaining problem”.

If we assume that

- (22) the strategy set X is convex,
 the loss functions f_1 and f_2 are convex,

we deduce that the rate of transfer $\bar{\lambda} = (\bar{\lambda}_1, \bar{\lambda}_2)$ defined by

$$(23) \quad \bar{\lambda}_1 = t_2(f_2) - f_2(\bar{x}), \quad \bar{\lambda}_2 = t_1(f_1) - f_1(\bar{x})$$

is a *Pareto multiplier* of the imputation $\bar{x}$.

Remark. The question arises as to how to choose a “threat strategy” $x = \{x^1, x^2\}$ (i. e. $t_i(f_i) = f_i(x)$ for $i = 1, 2$). The “threat strategies” can be obtained as the non-cooperative solution of a new game, whose loss functions are the losses assigned to each player by the Nash bargaining solution obtained from this threat. Other methods can be devised.

10.3. Behavior of coalitions of players: the core

We can regard a weak Pareto multistrategy $\bar{x} \in X(N)$ as a multistrategy which is not “rejected” by the whole set of players provided it is understood that x is “rejected by N ” (or “blocked”) only if there is another multistrategy $y \in X(N)$ such that $f_i(y) < f_i(x)$ for all $i \in N$.

If we introduce threat functionals $t_i (i \in N)$ for which there exists at least one $x \in X(N)$ such that $f_i(x) \leq t_i(f_i)$ for all $i \in N$, we can regard an imputation as a multistrategy which is neither rejected by the whole set N of players nor by the individual players i . It is to be understood that $x \in X(N)$ is rejected by player i only if $f_i(x) > t_i(f_i)$. In order to extend this selection procedure for Pareto multistrategies, we shall broaden the description of the game by including assumptions about the behavior of a family of coalitions A . We define the multistrategies "rejected" by coalitions A . The multistrategies not rejected by any coalition constitute the "core" of the game.

10.3.1. Coalitions

We shall regard a non-empty subset $A \subset N = \{1, \dots, n\}$ as a *coalition* of players. We denote the "adverse coalition" by

$$(1) \quad \bar{A} = N - A.$$

If coalition A forms, it "splits" the multistrategy set x^N in two parts

$$(2) \quad X^N = X^A \times X^{\bar{A}} \quad \text{where } X^A = \prod_{i \in A} X^i \quad \text{and} \quad X^{\bar{A}} = \prod_{j \notin A} X_j.$$

X^A (resp. $X^{\bar{A}}$) is the set of multistrategies "controlled" by A (resp. $\bar{A}$). We denote the projector from X^N onto X^A by

$$(3) \quad \pi^A : X^N \rightarrow X^A$$

and write

$$(4) \quad x = \{\pi^A x, \pi^{\bar{A}} x\} = \{x^A, x^{\bar{A}}\} \in X^A \times X^{\bar{A}} = X^N$$

whenever coalition A is involved.

We shall *identify* any coalition A with its characteristic function $\tau^A : N \rightarrow \{0, 1\}$ associating with any player i his *rate of participation* τ_i^A defined by $\tau_i^A = 1$ if i participates in coalition A and by $\tau_i^A = 0$ if i does not participate in A .

We shall denote by τ^A the projector of $\mathbb{R}^n$ defined by

$$(5) \quad (\tau^A \cdot c)_i = \tau_i^A c_i = \begin{cases} c_i & \text{if } i \in A, \\ 0 & \text{if } i \notin A. \end{cases}$$

We notice that τ^N is the identity and that $\tau^\emptyset = 0$. We set

$$(6) \quad \mathbf{R}^A = \tau^A \cdot \mathbf{R}, \quad \mathbf{R}_+^A = \tau^A \cdot \mathbf{R}_+, \quad \dot{\mathbf{R}}_+^A = \tau^A \cdot \dot{\mathbf{R}}_+$$

and also

$$(7) \quad \mathbf{R}^N = \tau^N \cdot \mathbf{R}^n = \mathbf{R}^n, \quad \mathbf{R}_+^N = \mathbf{R}_+^n, \quad \dot{\mathbf{R}}_+^N = \dot{\mathbf{R}}_+^n, \quad \mathbf{R}^\emptyset = \{0\}$$

By a *family* $\mathcal{A}$ of *coalitions*, we mean a family of non-empty subsets of N such that

$$(8) \quad \underline{\mathcal{A}} \subset \mathcal{A} \subset \bar{\mathcal{A}}$$

where

$$(9) \quad \underline{\mathcal{A}} = \{N, \{1\}, \dots, \{n\}\} \text{ is the minimal family of the } (n+1) \text{ coalitions } N \text{ (whole set of players) and } \{i\} \text{ (individual players)}$$

and where

$$(10) \quad \bar{\mathcal{A}} \text{ is the family of the } 2^n - 1 \text{ non-empty subsets of } N.$$

In this section, we shall assume that a family $\mathcal{A}$ is given once and for all.

10.3.2. Cooperative game described in strategic form and its core

Definition 1. Let $\mathcal{A}$ be a family of coalitions. A *cooperative game* $\{X(A), F^A\}_{A \in \mathcal{A}}$ is described in strategic form by associating with each coalition $A \in \mathcal{A}$

$$(11) \quad \begin{cases} \text{(i) a multistrategy subset } X(A) \subset X^A, \\ \text{(ii) a multiloss operator } F^A : X(A) \rightarrow \mathbf{R}^A \\ \text{defined by } F^A(x) = \{f_i^A(x)\}_{i \in A}. \end{cases}$$

We shall say that a coalition $A \in \mathcal{A}$ “*rejects*” (or “*blocks*”, “*can improve upon*”) a multistrategy $x \in X(N)$ if there exists a multistrategy x^A of the multistrategy set $X(A)$ controlled by coalition A yielding to each player $i \in A$ of A a loss $f_i^A(x^A)$ smaller than the loss $f_i(x)$, i.e.

$$(12) \quad \exists x^A \in X(A) \text{ such that, } \forall i \in A, \quad f_i^A(x^A) < f_i(x).$$

We shall say that the “*core of the game*” is the set $\mathcal{C}(\{X(A), F^A\}_{A \in \mathcal{A}})$ satisfying,

$$(13) \quad \begin{cases} \text{(i) } x \in X(N), \\ \text{(ii) } \forall A \in \mathcal{A}, \quad x \text{ is not rejected by } A, \end{cases}$$

i. e. the core is the set of multistrategies $x \in X(N)$ which are not rejected by any coalition $A \in \mathcal{A}$.

Remark. We notice that if $\mathcal{A}$ is the family $\underline{\mathcal{A}}$ of the $(n+1)$ coalitions $N, \{1\}, \dots, \{n\}$, then the core is nothing other than the set of imputations when the threat functionals t_i are defined by $t_i(f_i) = \inf_{x^i \in X(\{i\})} f_i^{(i)}(x^i)$.

Remark. The core of the game is contained in the subset of imputations, which is itself contained in the subset of (weak) Pareto multistrategies. Actually, the larger the family $\mathcal{A}$ of coalitions which is allowed to form, the smaller will be the associated core $\mathcal{C}(\{X(A), F^A\}_{A \in \mathcal{A}})$.

10.3.3. The multiloss operator F^{A*} of the coalition A

We shall give a canonical way of constructing a multiloss operator for the coalition A .

As a member of a given coalition A , a player $i \in A$ will modify his loss function, by taking into account the fact that he is cooperating with other players $j \in A$ and not cooperating (in fact, anti-cooperating) with players of the adverse coalition $\hat{A}$.

Therefore, if cautious, players i of A will assume that players $j \in \hat{A}$ will choose a multistrategy $\pi^{\hat{A}}x$ which maximizes their loss function. Therefore, they associate with a multistrategy $x^A \in X(A)$ the worst losses

$$(14) \quad f_i^{A*}(x^A) = \sup_{\substack{y \in X(N) \\ \pi^{\hat{A}}y = x^A}} f_i(y) = \sup_{\{x^A, x^{\hat{A}}\} \in X(N)} f_i(x^A, x^{\hat{A}}).$$

In other words, we can say that f_i^{A*} is the loss function for player i as a member of the coalition A in opposition to the adverse coalition $\hat{A}$.

Notice that when $f_i : X^I \rightarrow \mathbb{R}$ depends only upon the strategies of player i , $f_i^{A*}(x^I) = f_i(x^I)$ when $i \in A$.

Consider the game $\{x^A, F^{A*}\}_{A \in \mathcal{A}}$. We can reformulate the definition of the core in this case.

A multistrategy $\bar{x} \in X(N)$ belongs to the core if and only if for all coalitions $A \in \mathcal{A}$, for all $\varepsilon > 0$ and for all multistrategies $x^A \in X(A)$ feasible for A , there exists at least one player $i \in A$ and a multistrategy $x^{\hat{A}}$ of the complementary coalition $\hat{A}$ such that

$$(15) \quad f_i(x) \leq f_i^{A*}(x^A) \leq f_i(x^A, x^{\hat{A}}) + \varepsilon.$$

10.3.4. Examples of multistrategy sets $X(A)$

An important class of examples consists of those in which we have to allocate scarce resources among n players. We define

$$(16) \quad \begin{cases} \text{(i) subsets } Y(A) \subset V \text{ of available resources to the coalition } A, \\ \text{(ii) } n \text{ resource operators } L_i \in \mathcal{L}(U^I, V). \end{cases}$$

The subsets $X(A)$ are defined by

$$(17) \quad X(A) = \{x^A \in R^A \text{ such that } \sum_{i \in A} L_i(x^A, \cdot) \in Y(A)\}.$$

In other words, $X(A)$ is the subset of multistrategies $x^A = \{x^{A,i}\}_{i \in A}$ such that the sum of the resources needed to implement them is available to the coalition A .

10.3.5. Economic games and core of an economy

Consider an economy $\{R^i, Y(i), f_i\}_{i \in N}$ where R^i is the consumption set for i , $Y(i) \subset R^i$ is his initial endowment in available resources and f_i is his loss function (see Section 8.1).

We associate with this economy the following game $\{X(A), F^A\}_{A \in \mathcal{A}}$.

We assume that members $i \in A$ of coalition A place at coalition A 's disposal the sum

$$(18) \quad Y(A) = \sum_{i \in A} Y(i)$$

of their endowments in resources.

Therefore, the subset $X(A) \subset R^A$ of allocations of coalition A is defined by

$$(19) \quad X(A) = \{x^A \in R^A \text{ such that } \sum_{i \in A} x^{A,i} \in Y(A)\}.$$

We thus define F^A by

$$(20) \quad f_i^A(x) = f_i(x^i) \quad \text{whenever } i \in A.$$

Definition 2. We shall say that the game $\{X(A), F^A\}_{A \in \mathcal{A}}$ defined by (19) and (20) is the economic game associated with the economy $\{R^i, Y(i), f_i\}_{i \in N}$. Then a coalition A *rejects* an allocation $x \in x(N)$ if it can find an allocation $x^A \in X(A)$ yielding to each member $i \in A$ a loss $f_i(x^{A,i})$ smaller than the loss $f_i(x)$. We shall say that the core $\mathcal{C}(\{X(A), F^A\})$ of the game is the “core of the economy $\{R^i, Y(i), f_i\}_{i \in N}$ ”.

10.3.6. Cooperative game described in characteristic form and its core

Let $\{X(A), F^A\}_{A \in \mathcal{A}}$ be a game described in strategic form. Denote by

$$(21) \quad J(A) = F^A(X(A)) \text{ the set of feasible multilosses of coalition } A.$$

We shall say that

$$(22) \quad \begin{cases} \text{(i) } J_+(A) = J(A) + \mathbf{R}_+^A \text{ is the set of “admissible multilosses” of} \\ \text{of the coalition } A, \\ \text{(ii) } \hat{J}_+(A) = J(A) + \hat{\mathbf{R}}_+^A \text{ is the set of multilosses “rejected” by the} \\ \text{coalition } A. \end{cases}$$

We can prove the following proposition

Proposition 1. *A coalition A “rejects” the multistrategy $x \in X(N)$ if and only if the multiloss $c = F(x)$ satisfies*

$$(23) \quad \tau^A \cdot c \in \dot{J}_+(A).$$

A multistrategy $x \in X(N)$ belongs to the core of the game if and only if the multiloss $c = F(x)$ satisfies

$$(24) \quad \begin{cases} \text{(i) } c \in J(N), \\ \text{(ii) } \forall A \in \mathcal{A}, \tau^A \cdot c \notin \dot{J}_+(A). \end{cases}$$

Proof. This is left as an exercise. $\square$

Since the concept of the core depends upon the strategy subsets $X(A)$ only via their images $J(A)$ by F^A , we can get by with “poorer” description of the game.

Definition 3. We shall say that a correspondence $(\mathcal{A}, J)$ mapping coalitions $A \in \mathcal{A}$ into subsets $J(A) \subset \mathbb{R}^A$ of multilosses of the coalition A “describes a cooperative game in characteristic form”.

The subset $J_+(A) = J(A) + \mathbb{R}_+^A$ is called the *subset of admissible multilosses of the coalition A* and the subset $\dot{J}_+(A) = J(A) + \dot{\mathbb{R}}_+^A$ is called the *subset of multilosses “rejected” or “blocked” by the coalition A* .

We shall say that the subset $\mathcal{C}_0(\mathcal{A}, J)$ of multilosses satisfying

$$(25) \quad \begin{cases} \text{(i) } c \in J(N), \\ \text{(ii) } \forall A \in \mathcal{A}, \tau^A \cdot c \notin \dot{J}_+(A) \end{cases}$$

is the “*feasible core*” of the game $(\mathcal{A}, J)$ and that the subset $\mathcal{C}(\mathcal{A}, J)$ of multilosses c satisfying

$$(26) \quad \begin{cases} \text{(i) } c \in J_+(N), \\ \text{(ii) } \forall A \in \mathcal{A}, \tau^A \cdot c \notin \dot{J}_+(A) \end{cases}$$

is the “*admissible core*”.

Remark. Since $\mathcal{C}_0(\mathcal{A}, J)$ is obviously contained in $\mathcal{C}(\mathcal{A}, J)$, the non-emptiness of $\mathcal{C}_0(\mathcal{A}, J)$ implies the non-emptiness of $\mathcal{C}(\mathcal{A}, J)$. The converse is also true.

Proposition 2. *The two following statements are equivalent*

$$(27) \quad \begin{cases} \text{(a) the feasible core } \mathcal{C}_0(\mathcal{A}, J) \text{ is non-empty,} \\ \text{(b) the admissible core } \mathcal{C}(\mathcal{A}, J) \text{ is non-empty.} \end{cases}$$

Proof. We have actually to prove that, if $c \in J_+(N)$ and $\forall A \in \mathcal{A}, \tau^A \cdot c \notin J_+(A)$, then there exists $d \in J(N)$ such that, $\forall A \in \mathcal{A}, \tau^A \cdot d \notin J_+(A)$. There exist $d \in J(N)$ and $b \in \mathbb{R}_+^N$ such that $c = d + b$. We have to check that d is rejected by no coalition. If A rejects d , then $\tau^A \cdot d$ belongs to $J_+(A)$ and thus, since $\tau^A \cdot c = \tau^A d + \tau^A \cdot b \geq \tau^A d$, c is rejected by A . But this is impossible. $\square$

Remark. In view of the above result, since the definition of the admissible core is more symmetric, we shall use the *admissible core* instead of the feasible core in most instances.

10.4. Behavior of fuzzy coalitions: the fuzzy core

By extending the minimal family $\underline{\mathcal{A}} = \{N, \{1\}, \dots, \{n\}\}$ of coalitions to a larger family $\mathcal{A}$, we “shrank” the set of imputations to the core of the game.

In many instances, the core remains too large a set. Since we would like to have solution concepts yielding a set of solutions as small as possible, the question arises as to whether it is possible to “shrink” the core again by “enlarging” the set of “coalitions”.

This is done by embedding the set of coalitions (identified with $\{0, 1\}^n$) into the subset $\mathcal{C} = [0, 1]^n$ of “fuzzy coalitions”.

We define a fuzzy game $(\mathcal{C}, J)$ as a correspondence mapping any fuzzy coalition τ onto its subset $J(\tau)$ of multilosses. Its core is the set of multilosses which are not rejected by any fuzzy coalition. It is obviously contained in the core of the restriction of $(\mathcal{C}, J)$ to the set of usual coalitions. We end this section by constructing fuzzy economic games and by associating canonically a fuzzy game with a game $\{X(A), F^A\}_{A \in \mathcal{A}}$ described in strategic form.

10.4.1. Fuzzy coalitions

Recall that we identified a coalition A with its “characteristic function” $\tau^A \in \{0, 1\}^n$ defined by its rates of participation

$$\tau_i^A = \begin{cases} 1 & \text{if } i \in A, \\ 0 & \text{if } i \notin A. \end{cases}$$

In other words, a player i either participates wholly in the coalition A or does not participate in it at all.

A little thought shows that it is too strong a restriction to allow only such “crude” coalitions to form. A “finer” analysis should allow “fuzzy coalitions” to form, i.e. coalitions in which a player i can participate with a “rate of participation” $\tau_i \in [0, 1]$ (instead of $\{0, 1\}$).

Definition 1. We shall say that any $\tau \in [0, 1]^n$ is a "fuzzy coalition" and that its i th component τ_i is the rate of participation of player i in this fuzzy coalition. A subset $\mathcal{C}$ of $[0, 1]^n$ which contains $\underline{\mathcal{A}}$ is called a family of fuzzy coalitions.

Remark. The concepts of fuzzy coalitions $\tau \in [0, 1]^n$ and of rates of transfer $\lambda \in \mathcal{M}^n$ are different. We recall that a rate of transfer λ is a rule for sharing a "side payment" among players i participating in coalition A .

Naturally λ also enables a fuzzy coalition τ to share a given side payment w among players by means of the rule

$$(1) \quad \{c \in \mathbb{R}^N \text{ such that } \sum_{i=1}^n \lambda^i \tau_i c_i = \langle \lambda, \tau \cdot c \rangle = w\}.$$

Remark. From now on we shall associate with $\tau = \{\tau_1, \dots, \tau_n\} \in [0, 1]^n$ the map τ from $\mathbb{R}^N$ into $\mathbb{R}^N$ defined by

$$(2) \quad (\tau \cdot c)_i = \tau_i c_i$$

We shall denote the support of the fuzzy coalition τ by

$$(3) \quad A_\tau = \{i \in N \text{ such that } \tau_i > 0\},$$

i.e. A_τ is the set of active players in the fuzzy coalition. We shall set

$$(4) \quad \begin{aligned} R^\tau &= \tau \cdot R^n = R^{A_\tau}; & R_+^\tau &= \tau \cdot R_+^n = R_+^{A_\tau}; \\ \dot{R}_+^\tau &= \tau \cdot \dot{R}_+^n = \dot{R}_+^{A_\tau} \end{aligned}$$

10.4.2. Extension of a family of coalitions

Let $\mathcal{A}$ be a family of coalitions $A \subset N$, identified with the subset of $\{0, 1\}^n \subset \mathbb{R}^n$ of their characteristic functions τ^A . With $\mathcal{A}$ we shall associate its convex hull $\mathcal{C} = \text{co } \mathcal{A}$ which consists of the fuzzy coalitions

$$(5) \quad \tau = \sum_{A \in \mathcal{A}} m(A) \tau^A$$

where $\{m(A)\}_{A \in \mathcal{A}}$ ranges over the set of probabilities $\mathcal{M}(\mathcal{A})$.

The rates of participation of such coalitions are equal to

$$(6) \quad \forall i \in N, \quad \tau_i = \sum_{\substack{A \in \mathcal{A} \\ i \in A}} m(A).$$

Such a formula speaks for itself. If $m(A)$ denotes the probability of coalition A forming, the associated rate of participation of player i is the sum of the probabilities of the formation of coalitions A to which i belongs.

For instance, we can check that

$$(7) \quad \underline{\mathcal{C}} = \text{co}(\underline{\mathcal{A}}) \text{ is the set of fuzzy coalitions } \tau \text{ satisfying } \sum_{i \in N} \tau_i \geq 1 \text{ and, } \forall i \in N, \tau_i \geq (\sum_{j \in N} \tau_j - 1)/(n-1).$$

It is clear that $\tau = m_0 \tau^N + \sum_{i=1}^n m_i \tau^{(i)}$ satisfies these properties. Conversely, suppose that $\tau \in [0, 1]^n$ satisfies the properties. We set

$$m_0 = (\sum_{i=1}^n \tau_i - 1)/(n-1) \geq 0 \text{ and } m_i = \tau_i - m_0 \geq 0.$$

Since $\tau_i = m_0 + m_i$, we can write that $\tau = m_0 \tau^N + \sum_{i=1}^n m_i \tau^{(i)}$. Finally,

$$m_0 + \sum_{i=1}^n m_i = m_0 + \sum_{i=1}^n \tau_i - nm_0 = 1.$$

Thus $\tau \in \text{co}(\underline{\mathcal{A}}) = \underline{\mathcal{C}}$. $\square$

In the same way, if $\bar{\mathcal{A}}$ denotes the family of *all non-empty* coalitions, we can check that

$$(8) \quad \bar{\mathcal{C}} = \text{co } \bar{\mathcal{A}} = \{\tau \in [0, 1]^n \text{ such that } \sum_{i \in N} \tau_i \geq 1\}$$

and that

$$(9) \quad \bar{\mathcal{A}} = \{0, 1\}^n - \{0\} \text{ is the set of extremal points of } \bar{\mathcal{C}}.$$

*10.4.3. Debreu-Scarf coalitions

We shall say that a fuzzy coalition $\tau \in \bar{\mathcal{C}}$ is a “*Debreu-Scarf*” (fuzzy) coalition if its rates of participations $\tau_i = p_i/q \in \mathbb{Q}$ are rational. This implies that $q \geq p_i \forall i = 1, \dots, n$.

Debreu-Scarf fuzzy coalitions are usually interpreted in the following way. We regard i as a “type of player”, p_i as the number of players of type i and q total number of players of each type. Then the rate of participation of “type” i is the proportion of players of type i involved.

Recall that any fuzzy coalition can be approximated by a Debreu-Scarf coalition. If $\bar{\mathcal{J}} = \bar{\mathcal{C}} \cap \mathbb{Q}^n$ is the set of Debreu-Scarf coalitions, we have that

$$(10) \quad \bar{\mathcal{J}} \text{ is dense in } \bar{\mathcal{C}}.$$

*10.4.4. Fuzzy coalitions on a continuum of players

Suppose that the set of players N is no longer finite, but infinite and is in fact a set Ω of players homeomorphic to $[0, 1]$ (Ω is said to be a “*continuum of players*”). We denote by $\mathcal{A}$ the σ -algebra of Borel subsets $A \subset \Omega$, interpreted

as the “family of coalitions” of players. It is identified with the subset

$$(11) \quad \mathcal{A} = \{\tau^A \in L^1(\Omega) \text{ such that } \tau^A(\omega) \in \{0, 1\} \text{ almost everywhere}\}.$$

Now, we can regard the subset

$$(12) \quad \mathcal{T} = \{\tau \in L^1(\Omega) \text{ such that } \tau(\omega) \in [0, 1] \text{ almost everywhere}\}$$

as the family of “fuzzy coalitions” of players $\omega \in \Omega$.

Let us supply $L^1(\Omega)$ with the weak topology $\sigma(L^1(\Omega), L^\infty(\Omega))$.

We can prove the following result.

Proposition 1 (Castaing). *The family $\mathcal{A}$ of coalitions is dense in the family $\mathcal{T}$ of fuzzy coalitions, which is a convex compact subset of $L^1(\Omega)$.*

Proof. Since $\mathcal{T}$ is clearly a bounded subset of $L^\infty(\Omega)$, it is relatively compact in $L^\infty(\Omega)$ supplied with the weak topology $\sigma(L^\infty(\Omega), L^1(\Omega))$. Since $L^\infty(\Omega)$ is contained in $L^1(\Omega)$ (for Ω is compact) and since $\sigma(L^\infty(\Omega), L^1(\Omega))$ is stronger than $\sigma(L^1(\Omega), L^\infty(\Omega))$, then $\mathcal{T}$ is relatively compact in $L^1(\Omega)$ supplied with the weak topology. On the other hand, $\mathcal{T}$ is clearly convex and closed in $L^1(\Omega)$ supplied with the initial topology. Hence $\mathcal{T}$ is closed in $L^1(\Omega)$ supplied with the weak topology.

Therefore, $\mathcal{T}$ is compact. We prove that $\mathcal{A}$ is dense in $\mathcal{T}$, i.e., for any semi-norm $\mathbf{P}_K(\cdot) = \sup_{i=1, \dots, n} |\langle f_i, \cdot \rangle|$ ($K = \{f_1, \dots, f_n\} \subset L^\infty(\Omega)$) of the weak topology and for any $\varepsilon > 0$, we can find $A \in \mathcal{A}$ such that

$$(13) \quad \mathbf{P}_K(\tau - \tau^A) = \sup_{i=1, \dots, n} \left| \int_{\Omega} f_i(\omega) \tau_i(\omega) d\omega - \int_A f_i(\omega) d\omega \right| \leq \varepsilon.$$

For this purpose, consider the vector measure F associating with any $A \in \mathcal{A}$ the vector $F(A) = \{\int_A f_i(\omega) d\omega\}_i$ of $\mathbf{R}^n$. By the Lyapunov convexity theorem (see Appendix C) we know that $F(\mathcal{A})$ is a convex subset of $\mathbf{R}^n$, which is equal to $F(\mathcal{T})$. Hence, we can associate with any τ a coalition $A \in \mathcal{A}$ such that $\mathbf{P}_K(\tau - \tau^A) = 0 \leq \varepsilon$. $\square$

Remark. The fact that the family of Debreu–Scarf coalitions of a finite set N of players is *dense* in the family of fuzzy coalitions (or the fact that the family of coalitions on a continuum of players is *dense* in the family of fuzzy coalitions) shows that under convenient continuity assumptions, properties which are true for the Debreu–Scarf coalitions (or the coalitions on a continuum of players) remain true for fuzzy coalitions.

10.4.5. Fuzzy games described in characteristic form

Definition 2. Let $\mathcal{T} \subset [0, 1]^n$ be a family of fuzzy coalitions. We shall say that $(\mathcal{T}, J)$ describes a fuzzy game in characteristic form if J is a correspondence associating with any fuzzy coalition $\tau \in \mathcal{T}$ its subset $J(\tau) \subset \mathbb{R}^r$ of multilosses.

We shall say that a fuzzy coalition τ "rejects" a multiloss c if

$$(14) \quad \tau \cdot c \in \dot{J}_+(\tau) (= J(\tau) + \dot{\mathbb{R}}_+^r).$$

The core $\mathcal{C}(\mathcal{T}, J)$ of a fuzzy game is the set of multilosses $c \in J_+(N)$ which are not rejected by any fuzzy coalition $\tau \in \mathcal{T}$, i.e. such that

$$(15) \quad \begin{cases} \text{(i) } c \in J_+(N), \\ \text{(ii) } \forall \tau \in \mathcal{T}, \tau \cdot c \notin \dot{J}_+(\tau). \end{cases}$$

These definitions are the natural extensions of cooperative games and their core. The following statement is obvious.

Proposition 2. The core $\mathcal{C}(\mathcal{T}, J)$ of a fuzzy game is contained in the core $\mathcal{C}(\mathcal{A}, J|_{\mathcal{A}})$ of its restriction to $\mathcal{A} = \mathcal{T} \cap \bar{\mathcal{A}}$.

Proof. This is left as an exercise. $\square$

10.4.6. Characterization of the core of a (fuzzy) game

Let $\mathcal{T}$ be a family of fuzzy (or usual) coalitions and $(\mathcal{T}, J)$ be a fuzzy (or usual) game.

We use the lower support functions $\sigma^b(J(\tau), \lambda) = \inf_{c \in J(\tau)} \langle \lambda, c \rangle$ of the set $J(\tau)$ of multilosses to characterize the core of the game.

For this purpose, we denote by $\mathcal{M}(\tau)$ the set of probability measures on the support A_τ of τ . We introduce the function α defined by

$$(16) \quad \alpha_{\mathcal{T}}(c) = \sup_{\tau \in \mathcal{T}} \inf_{\lambda \in \mathcal{M}(\tau)} [\langle \lambda, \tau \cdot c \rangle - \sigma^b(J(\tau); \lambda)].$$

Theorem 1. Let $c \in J_+(N)$. If $\alpha_{\mathcal{T}}(c) \leq 0$, then c belongs to the core $\mathcal{C}(\mathcal{T}, J)$. Conversely, if

$$(17) \quad \forall \tau \in \mathcal{T}, \quad J(\tau) + \mathbb{R}_+^r \text{ is convex,}$$

then $\alpha_{\mathcal{T}}(c)$ is non-positive when c belongs to the core $\mathcal{C}(\mathcal{T}, J)$.

Proof. Suppose in the first place that $\alpha_{\mathcal{T}}(c) \leq 0$.

Then, for any fuzzy coalition τ , $\inf_{\lambda \in \mathcal{M}(\tau)} [\langle \lambda, \tau \cdot c \rangle - \sigma^b(J(\tau), \lambda)] \leq 0$. Since $\mathcal{M}(\tau)$ is compact and $\lambda \mapsto \langle \lambda, \tau \cdot c \rangle - \sigma^b(J(\tau), \lambda)$ is lower semi-continuous, there exists $\lambda \in \mathcal{M}(\tau)$ which achieves the minimum. Therefore,

$$\forall \tau \in \mathcal{T}, \quad \exists \lambda \in \mathcal{M}(\tau) \text{ such that } \langle \lambda, \tau \cdot c \rangle - \sigma^b(J(\tau); \lambda) \leq 0.$$

This implies that c is not rejected by τ (otherwise there would exist $d \in J(\tau)$ such that $\tau_i c_i > d_i$ for all $i \in A\tau$. Then,

$$\langle \lambda, \tau \cdot c \rangle > \langle \lambda, d \rangle \geq \sigma^b(J(\tau); \lambda)$$

which is a contradiction.)

Conversely, suppose that c belongs to the core $\mathcal{C}(\mathcal{T}, J)$. Since $\tau \cdot c \notin J_+(\tau)$, we deduce from (17) and the separation theorem that there exists $\lambda \in (\mathbb{R}^r)^*$, $\lambda \neq 0$ such that

$$(18) \quad \langle \lambda, \tau \cdot c \rangle \leq \inf_{d \in J_+(\tau)} \langle \lambda, d \rangle = \sigma^b(J_+(\tau); \lambda).$$

Since $\sigma^b(J_+(\tau); \lambda) > -\infty$, this implies that $\lambda \in (\mathbb{R}^r)^*_+$. Dividing both sides of (18) by $\sum_{i \in A\tau} \lambda^i > 0$, we can say that (18) holds with $\lambda \in \mathcal{M}(\tau)$.

Therefore, we have proved that

$$(19) \quad \inf_{\lambda \in \mathcal{M}(A\tau)} [\langle \lambda, \tau \cdot c \rangle - \sigma^b(J(\tau); \lambda)] \leq 0$$

and thus that $\alpha_{\mathcal{T}}(c) \leq 0$.

10.4.7. Fuzzy economic games and fuzzy core of an economy

Consider an economy $\{R^i, Y(i), f_i\}_{i \in N}$.

We assume that members i of a fuzzy coalition τ place at the disposal of fuzzy coalition τ the weighted sum

$$(20) \quad Y(\tau) = \sum_{i=1}^n \tau_i Y(i)$$

of their endowments $Y(i)$ of available commodities (i.e. by participating in τ with a rate of τ_i , player i brings the τ_i th part $\tau_i Y(i)$ of his endowment).

We define the subset $X(\tau) \subset R^{A\tau}$ of allocations of the fuzzy coalition τ by

$$(21) \quad X(\tau) = \{x^\tau \in R^{A\tau} \text{ such that } \sum_{i=1}^n \tau_i x^{\tau, i} \in Y(\tau)\}$$

and the multiloss operator F^τ of τ by

$$(22) \quad f_i^\tau(x) = \tau_i f_i(x^i) \quad \text{whenever } \tau_i > 0.$$

Definition 3. We define the “fuzzy economic game” $(\overline{\mathcal{C}}, J)$ by

$$(23) \quad J(\tau) = F^\tau(X(\tau)) \quad \text{for all } \tau \in \overline{\mathcal{C}}.$$

Therefore, a fuzzy coalition *rejects* an allocation $x \in X(N)$ if it can find $x^\tau \in X(\tau)$ yielding to each player $i \in A_\tau$ (participating in τ) a loss $f_i(x^\tau, i)$ smaller than the loss $f_i(x)$.

Notice that, for any coalition A , we obtain

$$(24) \quad Y(A) = Y(\tau^A), \quad X(A) = X(\tau^A), \quad F^A = F^{\tau^A}.$$

Hence the economic game $(\overline{\mathcal{A}}, J)$ is the restriction of $(\overline{\mathcal{C}}, J)$, since $J(A) = F^A(X(A)) = F^{\tau^A}(X(\tau^A)) = J(\tau^A)$ for all A . Therefore,

$$(25) \quad \text{the core } \mathcal{C}(\overline{\mathcal{C}}, J), \text{ called the “fuzzy core of the economy”, is contained in the core of the economy.}$$

Let $\mathcal{J} = \overline{\mathcal{C}} \cap \mathcal{Q}^n$ be the subset of Debreu–Scarf coalitions. We clearly have that

$$(26) \quad \mathcal{C}(\overline{\mathcal{C}}, J) \subset \mathcal{C}(\mathcal{J}, J) \subset \mathcal{C}(\overline{\mathcal{A}}, J).$$

Since $\mathcal{J}$ is dense in $\overline{\mathcal{C}}$, we obtain that the fuzzy core $\mathcal{C}(\overline{\mathcal{C}}, J)$ is the same as the “Debreu–Scarf core” $\mathcal{C}(\mathcal{J}, J)$ of the economy.

Proposition 3 (Ekeland). *Suppose that*

$$(27) \quad \forall i \in N, \quad \text{the loss function } f_i \text{ is continuous.}$$

Then the fuzzy core and the Debreu–Scarf core of the economy coincide.

Proof. It remains to prove that $\mathcal{C}(\mathcal{J}, J) \subset \mathcal{C}(\overline{\mathcal{C}}, J)$. Suppose that $c = F(x)$ (where $x \in X(N)$) is not rejected by any fuzzy coalition σ of $\mathcal{J}$.

We prove that c belongs to $\mathcal{C}(\overline{\mathcal{C}}, J)$. If not, there exists a fuzzy coalition $\tau \in \overline{\mathcal{C}}$ and $y \in X^{\tau^*}$ such that

$$(28) \quad \begin{cases} \text{(i) } \sum_{i \in A_\tau} \tau_i y^i = \sum_{i \in A_\tau} \tau_i y(i) \in \sum_{i \in A_\tau} \tau_i Y(i) \\ \quad \text{(where } y(i) \in Y(i) \text{ depends on } \tau), \\ \text{(ii) } \forall i \in A_\tau, f_i(y^i) < f_i(x^i). \end{cases}$$

Now, if we set $\varepsilon = \frac{1}{2} \min_{i \in A_\tau} (f_i(x^i) - f_i(y^i)) > 0$, we know that there exists $\eta > 0$ such that $\|y^i - z^i\| \leq \eta$ implies

$$(29) \quad \forall i \in A_\tau, \quad f_i(z^i) \leq f_i(y^i) + \varepsilon < f_i(x^i)$$

It remains to show that we can find such $\sigma \in \bar{\mathcal{F}}$ and $z^i \in X^i$ satisfying $\|y^i - z^i\| \leq \eta$ and

$$(30) \quad \sum_{i \in A_\tau} \sigma_i z^i \in \sum_{i \in A_\tau} \sigma_i Y(i).$$

But since $\bar{\mathcal{F}}$ is dense in $\overline{\mathcal{C}}$, we can find $\sigma \in \bar{\mathcal{F}}$ such that

$$(31) \quad \forall i \in A_\tau, \quad \left\| (\tau_i - \sigma_i) y^i - \frac{1}{\left(\sum_{j \in A_\tau} \sigma_j \right)} \left(\sum_{i \in A_\tau} (\tau_i - \sigma_i) y(i) \right) \right\| \leq \eta.$$

Taking such a fuzzy coalition $\sigma \in \bar{\mathcal{F}}$ and writing

$$(32) \quad z^i = \frac{y^i}{\sigma_i} - \frac{1}{\left(\sum_{j \in A_\tau} \sigma_j \right)} \sum_{i \in A_\tau} (\tau_i - \sigma_i) y(i),$$

we check that

$$(33) \quad \forall i \in A_\tau, \quad \|y^i - z^i\| \leq \eta.$$

Also, by (28(i)) and (32),

$$\begin{aligned} \sum_{i \in A_\tau} \sigma_i z^i &= \sum_{i \in A_\tau} \tau_i y^i - \sum_{i \in A_\tau} \tau_i y(i) + \sum_{i \in A_\tau} \sigma_i y(i) \\ &= \sum_{i \in A_\tau} \sigma_i y(i) \in \sum_{i \in A_\tau} \sigma_i Y(i). \end{aligned}$$

We have therefore proved that the fuzzy coalition $\sigma \in \bar{\mathcal{F}}$ rejects $F(x)$ which is a contradiction.

* Interpretation of Debreu-Scarf economic games

Consider a Debreu-Scarf coalition

$$\sigma = \left\{ \frac{p_1}{q}, \frac{p_2}{q}, \dots, \frac{p_n}{q} \right\}$$

where q is the number of players of each type.

We assume that each player of type i has the same characteristics, i.e. the same consumption set R^i , the same endowment $Y(i)$, the same loss function f_i . We also assume that the allocation of each player of the same type i is the same.

Therefore, we can write the set of commodities available to σ as

$$(34) \quad Y(\sigma) = \frac{1}{q} \sum_{i=1}^n p_i Y(i),$$

i.e. n times the average of the sum of initial endowments brought by the players involved.

The set of allocations $x \in X(\sigma)$ is defined by

$$(35) \quad X(\sigma) = \{x \in R^{A\sigma} \text{ such that } \sum_{i=1}^n p_i x^i \in \sum_{i=1}^n p_i Y(i)\}.$$

We can interpret x^i as the (common) commodity allocated to each player of type i and $p^i x^i$ as the commodity allocated to the i th type of player.

Therefore, an allocation $\bar{x} = \{\bar{x}_1 \dots, \bar{x}_n\} \in X(N)$ (where x^i is the common commodity allocated to each player of type i) is *rejected* by a Debreu-Scarf coalition of p_1 players of type 1, $\dots$, p_n player of type n if this coalition can allocate to each player of type i a commodity x^i such that

$$(36) \quad \begin{cases} \text{(i) } \sum_{i=1}^n p_i x^i \in \sum_{i=1}^n p_i Y(i), \\ \text{(ii) for any } i \text{ such that } p_i > 0, f_i(x^i) < f_i(\bar{x}^i). \end{cases}$$

**10.4.8. Fuzzy games described in strategic form and fuzzy core*

We introduce fuzzy games $\{X(\tau), F^\tau\}$ described in strategic form and, in particular, extend a cooperative game $\{X(A), F^A\}$ described in strategic form to a fuzzy game $\{X(\tau), F^\tau\}$. Since the loss functions f_i^A of players i of the coalition A can depend upon strategies of *all* players i , we are led to use a *multiloss correspondence* $F^\tau: X^\tau \rightarrow \mathbb{R}^\tau$ of the fuzzy coalition τ instead of a multiloss operator. This slight modification does not complicate the situation too much.

We begin by defining

$$(37) \quad \mathcal{U}(\tau) = \{m \in \mathbb{R}_+^{\mathcal{A}} \text{ such that } \sum_{A \in \mathcal{A}} m(A) \tau^A = \tau\}$$

i.e. $\mathcal{U}(\tau)$ is the set of $\{m(A)\}_{A \in \mathcal{A}}$ where

$$(38) \quad \forall A \in \mathcal{A}, m(A) \geq 0 \text{ and } \forall i \in N, \sum_{A \ni i} m(A) = \tau_i.$$

Notice that

$$(39) \quad \text{if } m \in \mathcal{U}(\tau), \text{ then } A \subset A_\tau \text{ whenever } m(A) > 0.$$

Canonical multistrategy sets $X_0(\tau)$ of a fuzzy coalition

We assume that

$$(40) \quad \forall A \in \mathcal{A}, X(A) \text{ is a closed convex subset of}$$



Consider the map

$$(41) \quad \{m, \{x^A\}_A\} \in \mathcal{U} \tau \times \prod X(A) \mapsto x^\tau = \sum_{A \in \mathcal{A}} \frac{m(A)}{\tau} x^A \in U^{A_\tau}$$

where we define $x^\tau = \sum_{A \in \mathcal{A}} (m(A)/\tau) x_A$ by

$$(42) \quad \forall i \in A_\tau, \quad x^{\tau, i} = \sum_{A \ni i} \frac{m(A)x_A^i}{\tau_i}.$$

Definition 4. We define the “canonical multistrategy subset” $X_0(\tau) \subset U^{A_\tau}$ of the fuzzy coalition τ to be the set of multistrategies $x^\tau = \sum_{A \in \mathcal{A}} (m(A)x^A)/\tau$ as m ranges over $\mathcal{M}(\tau)$ and x^A over $X(A)$ for all $A \in \mathcal{A}$.

Construction of the multiloss correspondence F^τ of a fuzzy coalition

Definition 5. The multiloss correspondence F^τ from U^{A_τ} into $\mathbf{R}^\tau$ is defined by

$$(43) \quad F^\tau(x^\tau) = \left\{ \sum_{A \in \mathcal{A}} m(A)F^A \right\}_{\sum_{A \in \mathcal{A}} m(A)x^A/\tau = x^\tau}$$

Definition 6. We shall say that $\{X_0(\tau), F^\tau\}_{\tau \in \mathcal{T}}$ describes the “canonical extension” of the cooperative game $\{X(A), F^A\}_{A \in \mathcal{A}}$ into a fuzzy game described in strategic form. We set

$$(44) \quad \begin{cases} \text{(i) } F_+^\tau(x^\tau) = F^\tau(x^\tau) + \mathbf{R}_+^\tau \\ \text{(ii) } \dot{F}_+^\tau(x^\tau) = F^\tau(x^\tau) + \dot{\mathbf{R}}_+^\tau. \end{cases}$$

Consider now a fuzzy game $\{X(\tau), F^\tau\}_{\tau \in \mathcal{T}}$ described in strategic form, where, for any $\tau \in \mathcal{T}$,

$$(45) \quad \begin{cases} \text{(i) } X(\tau) \subset U^{A_\tau} \text{ is a multistrategy set,} \\ \text{(ii) } F^\tau : X(\tau) \longrightarrow \mathbf{R}^\tau \text{ is a multiloss correspondence.} \end{cases}$$

Definition 7. We shall say that a multistrategy $\bar{x} \in X(N)$ is rejected by a fuzzy coalition τ if there exists $x^\tau \in X(\tau)$ such that $\tau \cdot F(\bar{x}) \in \dot{F}_+^\tau(x^\tau)$.

The core $\mathcal{C}(X(\tau), F^\tau)$ of the fuzzy game is the subset of multistrategies $\bar{x} \in X(N)$ which are not rejected by any fuzzy coalition τ .

The notion of a canonical extension $\{X_0(\tau), F^\tau\}$ leads to the following terminology.

Definition 8. Let $\{X(A), F^A\}$ be a game described in strategic form. We shall say that $\bar{x} \in X(N)$ is “canonically rejected” by a fuzzy coalition $\tau \in \mathcal{T}$ if there exists $x^\tau \in X_0(\tau)$ such that $\tau \cdot F(\bar{x}) \in \dot{F}_+^\tau(x^\tau)$, i.e. if there exist $m \in \mathcal{M}(\tau)$ and $x^A \in X(A)$ such that

$$(46) \quad \forall i \in A_\tau, \quad \tau_i f_i(\bar{x}) > \sum_{i \in A} m(A) f_i^A(x^A).$$

The “canonical fuzzy core” of the game $\{X(A), F^A\}$ is the subset of multistrategies $\bar{x} \in X(N)$ which are not canonically rejected by any fuzzy coalition τ .

The characteristic form $(\mathcal{U}, J)$ of the fuzzy game $\{X(\tau), F^\tau\}$ is obviously defined by

$$(47) \quad J(\tau) = F^\tau(X(\tau)) = \bigcup_{x^\tau \in X_0(\tau)} F^\tau(x^\tau).$$

We next show how convexity, continuity and compactness properties of a game $\{X(A), F^A\}_A$ are carried over to its canonical extension.

Proposition 4. *Suppose that for any $A \in \mathcal{A}$*

$$(48) \quad \begin{cases} \text{(i) } X(A) \text{ is convex and compact,} \\ \text{(ii) } \forall i \in A, f_i^A \text{ is convex, lower semi-continuous.} \end{cases}$$

Let $\{X_0(\tau), F^\tau\}_\tau$ be the canonical extension of $\{X(A), F^A\}_{A \in \mathcal{A}}$. Then, for any fuzzy coalition τ ,

$$(49) \quad \begin{cases} \text{(i) } X_0(\tau) \text{ is convex and compact,} \\ \text{(ii) } \forall x^\tau \in X_0(\tau), F_+^\tau(x^\tau) \text{ is closed and convex,} \\ \text{(iii) } F_+^\tau(X_0(\tau)) \text{ is closed and convex.} \end{cases}$$

and, $\forall \lambda \in \mathcal{M}^n$

$$(50) \quad x^\tau \mapsto f^\tau(x^\tau; \lambda) = \sigma^b(F^\tau(x^\tau), \lambda) \text{ is convex.}$$

Proof. (a) We begin by noticing that $\mathcal{U}(\tau)$ is a convex compact subset of $\mathbf{R}_+^d$.

(b) Therefore, if $x_k^\tau = \sum_{A \in \mathcal{A}} (m^k(A) x_k^A) / \tau \in X_0(\tau)$ for any $k = 1, \dots, K$, we can write any convex combination $x^\tau = \sum_{k=1}^K \alpha_k x_k^\tau$ in the form

$$x^\tau = \sum_{A \in \mathcal{A}} \frac{m(A)}{\tau} \sum_{k=1}^K \frac{\alpha_k m^k(A)}{m(A)} x^A, k \in X(\tau)$$

where $m(A) = \sum_{k=1}^K \alpha_k m^k(A)$. Furthermore $X_0(\tau)$ is compact, being the continuous image of the compact subsets $\mathcal{U}(\tau)$ and $X(A)$.

(c) We prove that $F_+^\tau(x^\tau)$ is convex. It is sufficient to prove that

$$\begin{aligned} d &= \sum_{k=1}^K \alpha_k \left(\sum_{A \in \mathcal{A}} m^k(A) F^A(x_k^A) \right) \\ &= \sum_{A \in \mathcal{A}} m(A) \sum_{k=1}^K \frac{\alpha_k m^k(A)}{m(A)} F^A(x_k^A) \end{aligned}$$

belongs to $F_+^\tau(x^\tau)$, whenever $\sum_{A \in \mathcal{A}} (m^k(A) x_k^A) / \tau = x^\tau$ for all k , where $m(A) = \sum_{k=1}^K \alpha_k m^k(A)$.

Since the functions f_i^A are convex, Proposition 1.3.10 implies that

$$\sum_{k=1}^K \frac{\alpha_k m^k(A)}{m(A)} F^A(x_k^A) \geq F^A\left(\sum_{k=1}^K \frac{\alpha_k m^k(A)}{m(A)} x_k^A\right) = F^A(y^A)$$

where $y^A = \sum_{k=1}^K (\alpha_k m^k(A))/m(A)$ x_k^A belongs to the convex subset $X(A)$. Therefore, since $\sum_{A \in \mathcal{A}} (m(A)/\tau) y^A = x^\tau$, inequality $d \geq \sum_{A \in \mathcal{A}} m(A) F^A(y^A)$ implies that $d \in F_+^\tau(x^\tau)$.

(d) We prove that $F_+^\tau(x^\tau)$ is closed. Since the subsets $\mathcal{N}(\tau)$ and $X(A)$ are compact and the maps

$$\{m, \{x^A\}_A\} \mapsto \sum_{i \in A} m(A) f_i^A(x^A)$$

are lower semi-continuous, Proposition 2.1.4 implies that $F_+^\tau(x^\tau)$ is closed.

(e) A proof analogous to the proof of the convexity of $F_+^\tau(x^\tau)$ shows that $F_+^\tau(X_0(\tau))$ is convex.

(f) $F_+^\tau(X_0(\tau))$ is closed because the subsets $\mathcal{N}(\tau) \times \prod_{A \in \mathcal{A}} X(A)$ are compact and the maps $\{m, \{x^A\}_A\} \mapsto \sum_{i \in A} m(A) f_i^A(x^A)$ are lower semi-continuous (Proposition 2.1.4 implies that $F_+^\tau(X_0(\tau))$ is closed).

(g) Let $x^\tau = \sum_{k=1}^K \alpha_k x_k^\tau$ be a convex combination. For any $\varepsilon > 0$ there exists $m^k \in \mathcal{N}(\tau)$ and $x_k^A \in X(A)$ such that

$$\sum_{A \in \mathcal{A}} \frac{m^k(A) x_k^A}{\tau} = x_k^\tau$$

and

$$\sum_{A \in \mathcal{A}} m^k(A) \sum_{i \in A} \lambda^i f_i^A(x_k^A) \leq f^\tau(x_k^\tau, \lambda) + \varepsilon.$$

Therefore, by setting $m(A) = \sum_{k=1}^K \alpha_k m^k(A)$ and $x^A = \sum_{k=1}^K (\alpha_k m^k(A)/m(A)) x_k^A$, $x^A \in X(A)$, we obtain that

$$f^\tau(x^\tau, \lambda) \leq \sum_{A \in \mathcal{A}} m(A) \sum_{i \in A} \lambda^i f_i^A(x^A) \leq \sum_{k=1}^K \alpha_k f^\tau(x_k^\tau, \lambda) + \varepsilon. \quad \square$$

Remark.

$$(51) \quad F_\tau(X_0(\tau)) = \bigcup_{m \in \mathcal{N}(\tau)} \sum_{A \in \mathcal{A}} m(A) F^A(X(A))$$

because

$$\begin{aligned} F_\tau(X_0(\tau)) &= \bigcup_{x^\tau \in X_0(\tau)} \left[\bigcup_{(\sum_{A \in \mathcal{A}} m(A) x^A / \tau = x^\tau)} \sum_{A \in \mathcal{A}} m(A) F^A(x^A) \right] \\ &= \bigcup_{m \in \mathcal{N}(\tau)} \bigcup_{x^A \in X(A)} \sum_{A \in \mathcal{A}} m(A) F^A(x^A) = \bigcup_{m \in \mathcal{N}(\tau)} \sum_{A \in \mathcal{A}} m(A) F^A(X(A)). \end{aligned}$$

Remark. Consider the case when, for all $i \in N$,

$$(52) \quad \begin{cases} \text{(i) } R^i \text{ is a closed convex subset of a vector space } U^i, \\ \text{(ii) } Y(i) \text{ is a closed convex subset of a vector space } V, \\ \text{(iii) } L_i \in \mathcal{L}(U^i, V) \text{ is a continuous linear operator} \end{cases}$$

and the multistrategy sets $X(A)$ are defined by

$$(53) \quad X(A) = \{x^A \in R^A \text{ such that } \sum_{i \in A} L_i x^{A,i} \in Y(A)\}.$$

Proposition 5. Suppose that (52) and (53) hold. Then the canonical multistrategy set $X_0(\tau)$ of the fuzzy coalition τ is contained in the multistrategy sets $X(\tau)$ defined by

$$(54) \quad X(\tau) = \{x^\tau \in R^{A_\tau} \text{ such that } \sum_{i \in A_\tau} \tau_i L_i x^{\tau,i} \in Y(\tau) = \sum_{i \in A_\tau} \tau_i Y(i)\}.$$

Proof. Any x^τ defined by $x^{\tau,i} = \sum_{i \in A} (m(A)/\tau_i) x^{A,i}$ where $x^A \in X(A)$ satisfies

$$\begin{aligned} \sum_{i=1}^n \tau_i L_i x^{\tau,i} &= \sum_{A \in \mathcal{A}} m(A) \sum_{i \in A} L_i x^{A,i} \in \sum_{A \in \mathcal{A}} m(A) \sum_{i \in A} Y(i) \\ &= \sum_{i=1}^n Y(i) \sum_{A \ni i} m(A) = \sum_{i=1}^n \tau_i Y(i). \quad \square \end{aligned}$$

Remark. Consider the case where $X(A) = X^A$ and

$$(55) \quad \text{the loss functions } f_i^A = f_i \text{ are convex functions from } R^i \text{ into } \mathbf{R}.$$

Then

$$(56) \quad F_+^\tau(x^\tau) = \{\tau_i f_i(x^i)\}_{i \in A_\tau} + \mathbf{R}_+^\tau.$$

To prove this, we observe on the one hand that $\{\tau_i f_i(x^i)\}_{i \in A_\tau}$ belongs to $F^\tau(x^\tau)$ since

$$\tau_i f_i(x^i) = m(\{i\}) f_i\left(\frac{m(\{i\})}{\tau_i} x^i\right)$$

where $m \in \mathcal{M}(\tau)$ is defined by $m(A) = \tau_i$ if $A = \{i\}$ and $m(A) = 0$ otherwise. On the other hand,

$$F_+^\tau(x^\tau) = \{\tau_i f_i(x^i)\}_{i \in A_\tau} + \mathbf{R}_+^\tau$$

since, for all $i \in N$,

$$\sum_{i \in A} m(A) f_i(x^{A,i}) \geq \tau_i f_i\left(\sum_{i \in A} \frac{m(A)}{\tau_i} x^{A,i}\right) \geq \tau_i f_i(x^i)$$

because the loss functions are convex. $\square$

We shall extend this remark to the case when $f_i^A = f_i^{A*}$. See Proposition 11.2.5 below.

10.5. Selection of elements of the core: cooperative equilibrium and nucleolus

Theorem 4.1 characterizes the elements of the core of a game $(\mathcal{T}, J)$ as the subset of $c \in J(N)$ such that

$$(1) \quad \alpha_{\mathcal{T}}(c) = \sup_{\tau \in \mathcal{T}} \inf_{\lambda \in \mathcal{M}(\tau)} [\langle \lambda, \tau \cdot c \rangle - \sigma^b(J(\tau); \lambda)] \leq 0.$$

We are led to look for elements $c \in J(N)$ such that

$$(2) \quad \beta_{\mathcal{T}}(c) = \inf_{\lambda \in \mathcal{M}^n} \sup_{\tau \in \mathcal{T}} [\langle \lambda, \tau \cdot c \rangle - \sigma^b(J(\tau); \lambda)] \leq 0.$$

(called canonical cooperative equilibria) and/or for elements which minimize the function $\alpha_{\mathcal{T}}$, which form the “least-core” of the game.

10.5.1. Canonical cooperative equilibrium

We begin by defining the concept of a canonical cooperative equilibrium.

Definition 1. Let $(\mathcal{T}, J)$ be a fuzzy (or usual) cooperative game. We shall say that a multiloss $c \in J(N)$ is a “*weak canonical cooperative equilibrium*” if there exists a Pareto multiplier $\bar{\lambda} \in \mathcal{M}^n$ such that

$$(3) \quad \forall \tau \in \mathcal{T}, \quad \langle \bar{\lambda}, \tau \cdot c \rangle \leq \inf_{d \in J(\tau)} \langle \bar{\lambda}, d \rangle.$$

We shall say that it is a (strong) “*canonical cooperative equilibrium*” if there exists a Pareto multiplier $\bar{\lambda}$ of c which is strongly positive.

The following statement is obvious.

Proposition 1. Any (strong) canonical cooperative equilibrium of a game $(\mathcal{T}, J)$ belongs to its core $\mathcal{C}(\mathcal{T}, J)$.

Proof. Let $\bar{\lambda} \in \mathcal{M}^n$ be a Pareto multiplier of the canonical cooperative equilibrium c . Then $(\bar{\lambda} / \sum_{i \in A_\tau} \bar{\lambda}_i) \in \mathcal{M}(\tau)$ for any fuzzy coalition τ . Therefore, for any $\tau \in \mathcal{T}$,

$$\begin{aligned} & \inf_{\lambda \in \mathcal{M}(\tau)} [\langle \lambda, \tau \cdot c \rangle - \sigma^b(J(\tau); \lambda)] \leq \\ & \leq \frac{1}{\sum_{i \in A_\tau} \bar{\lambda}_i} [\langle \bar{\lambda}, \tau \cdot c \rangle - \sigma(J(\tau); \bar{\lambda})] \leq 0 \end{aligned}$$

by (3). Hence $\alpha_{\mathcal{T}}(c) \leq 0$ and thus, c belongs to the core $\mathcal{C}(\mathcal{T}, J)$ of $(\mathcal{T}, J)$. $\square$

Example. (Games in strategic form.) Consider a game $\{X(A), F^A\}_{A \in \mathcal{A}}$ described in strategic form. Then $c = F(\bar{x})$ (where $\bar{x} \in X(N)$) is a weak (resp. strong) equilibrium if there exists $\bar{\lambda} \in \mathcal{M}^n$ (resp. $\bar{\lambda} \in \mathcal{M}^n$) such that,

$$(4) \quad \forall A \in \mathcal{A}, \quad \sum_{i \in A} \bar{\lambda}_i f_i(\bar{x}) \leq \inf_{x^A \in X(A)} \sum_{i \in A} \bar{\lambda}_i f_i^A(x^A).$$

We shall say that $\bar{x} \in X(N)$ is a weak (resp. strong) canonical cooperative equilibrium if (4) holds with $\bar{\lambda} \in \mathcal{M}^n$ (resp. $\bar{\lambda} \in \mathcal{M}^n$).

We can prove a stronger result than Proposition 1.

Proposition 2. Any (strong) canonical cooperative equilibrium $\bar{x} \in X(N)$ of a game $\{X(A), F^A\}_{A \in \mathcal{A}}$ belongs to its canonical fuzzy core.

Proof. Suppose that $\bar{x} \in X(N)$ satisfying (4) is canonically rejected by a fuzzy coalition τ . We deduce a contradiction. There exists $x^\tau = \sum_{A \in \mathcal{A}} m(A)x^A/\tau$ such that, for all $i \in A_\tau$, $\tau_i f_i(\bar{x}) > \sum_{A \ni i} m(A) f_i^A(x^A)$. Hence

$$\sum_{i=1}^n \bar{\lambda}_i \tau_i f_i(\bar{x}) > \sum_{i=1}^n \bar{\lambda}_i \sum_{A \ni i} m(A) f_i^A(x^A) = \sum_{A \in \mathcal{A}} m(A) \sum_{i \in A} \bar{\lambda}_i f_i^A(x^A).$$

Therefore, by (4), we obtain that

$$\begin{aligned} \sum_{i=1}^n \bar{\lambda}_i \tau_i f_i(\bar{x}) &> \sum_{A \in \mathcal{A}} m(A) \sum_{i \in A} \bar{\lambda}_i f_i(\bar{x}) = \sum_{i=1}^n \left(\sum_{A \ni i} m(A) \right) \bar{\lambda}_i f_i(\bar{x}) \\ &= \sum_{i=1}^n \bar{\lambda}_i \tau_i f_i(\bar{x}). \end{aligned}$$

This is a contradiction. $\square$

We shall prove the converse statement in Theorem 12.1.2.

Example. (Economic game.) Consider an economy $\{R^i, Y(i)f\}_{i \in N}$. Any Walras allocation $\bar{x} = \{\bar{x}, \dots, \bar{x}^n\}$ belongs to the fuzzy core.

Proposition 3. Any Walras allocation $\bar{x} = \{\bar{x}, \dots, \bar{x}^n\}$ of an economy belongs to its canonical fuzzy core.

Proof. Let $\{\bar{x}, \bar{p}\}$ be a Walras equilibrium. Suppose that a fuzzy coalition τ rejects $\bar{x}$. Then there exists an allocation $x^\tau \in X(\tau)$ such that $f_i(x^{\tau, i}) < f_i(\bar{x}^i)$ for all $i \in A_\tau$. Since $\bar{x}^i$ minimizes $f_i(\cdot)$ over $B_i(\bar{p}, r_i(\bar{p}))$, these strict inequalities imply that $x^{\tau, i} \notin B_i(\bar{p}, r_i(\bar{p}))$, i.e. $\langle \bar{p}, x^{\tau, i} \rangle > r_i(\bar{p})$ for all $i \in A_\tau$. Hence

$\langle \bar{p}, \sum_{i \in A_\tau} \tau_i x^{r,i} \rangle > \sum_{i \in A_\tau} \tau_i r_i(\bar{p})$. This is a contradiction: In fact $x^r \in X(\tau)$, i.e. $\sum_{i \in A_\tau} x^{r,i} \in \sum_{i \in A_\tau} \tau_i Y(i)$. Hence $\langle \bar{p}, \sum_{i \in A_\tau} \tau_i x^{r,i} \rangle \leq \sum_{i \in A_\tau} \tau_i r_i(\bar{p})$.

Remark. We shall see in Chapter 12 that, under additional assumptions, the fuzzy core is (almost) *the set of Walras allocations* (see Theorem 12.3.1).

*10.5.2. Least-core

We now describe a procedure for selecting a Pareto minimum which belongs to the core whenever it is not empty. Consider a game $(\mathcal{A}, J)$. We introduce the lower support functions of the subsets $J(A)$ and put $w(A, \lambda) = \sigma^b(J(A), \lambda)$.

Then Theorem 4.1 implies that, if a multiloss $c \in J(N)$ minimizes over $J(N)$ the function $\alpha = \alpha_{\mathcal{A}}$ defined by

$$(5) \quad \alpha(c) = \sup_{A \in \mathcal{A}} \inf_{\lambda \in \mathcal{M}^A} [\langle \lambda, \tau^A \cdot c \rangle - w(A, \lambda)],$$

then $\bar{c}$ belongs to the core whenever $\alpha(\bar{c}) \leq 0$.

It is quite natural to select from the elements of the core those which minimize $\alpha(\cdot)$.

Furthermore, we can define elements which minimize $\alpha(\cdot)$ even when the core is empty.

Definition 2. The “least-core” is the subset of elements $c \in J(N)$ minimizing over $J(N)$ the function $\alpha = \alpha_{\mathcal{A}}$ defined by (5).

Proposition 4. Suppose that

$$(6) \quad \text{The subset } J(N) \text{ is closed and bounded below}$$

Then the least-core is non-empty and compact. It is contained in the set of Pareto minima. If the core is non-empty and if the subsets $J(A) + \mathbb{R}_+^A$ are convex, then the least-core is contained in the core.

Proof. The function α is lower semi-continuous. To see this observe that, since $\{\lambda, c\} \mapsto \langle \lambda, \tau^A \cdot c \rangle - w(A, \lambda)$ is lower semi-continuous, the function

$$c \mapsto \theta(A, c) = \inf_{\lambda \in \mathcal{M}^A} [\langle \lambda, \tau^A \cdot c \rangle - w(A, \lambda)]$$

is also lower semi-continuous because $\mathcal{M}^A$ is compact (see Theorem 2.5.1). Hence α is lower semi-continuous.

The function α is also lower semi-compact. If $\alpha(c) \leq \eta$, then for any individual coalition i , $\theta_i(c) = c_i - w(i, 1) \leq \eta$.

Therefore, $c \leq w + \eta \mathbf{1}$ where $w = \{w(i, 1)\}_{i \in N}$.

Since c is larger than the shadow minimum, we deduce that c lies in a relatively compact subset. Therefore, there exists an element c which belongs to the least-core.

Let c belong to the least-core. It is Pareto minimal. If not, there exists $\varepsilon > 0$ such that $d = c - \varepsilon \mathbf{1}$ belongs to $J(N)$. In this case, for any $A \in \mathcal{A}$ and $\lambda \in \mathcal{M}(A)$, we obtain $\langle \lambda, \tau^A d \rangle = \langle \lambda, \tau^A c \rangle - \varepsilon$. Hence $\theta(A, d) = \theta(A, c) - \varepsilon$ and $\alpha(d) = \alpha(c) - \varepsilon$. This is a contradiction. Now, if the core is not empty, then $\inf_{d \in J(N)} \alpha(d) \leq 0$ (by Theorem 4.1) and thus, if c belongs to the least-core, $\alpha(c) \leq 0$ and c belongs to the core. $\square$

Remark. By replacing a family $\mathcal{A}$ of usual coalitions by a family $\mathcal{T}$ of fuzzy coalitions and the function α by $\alpha_{\mathcal{T}}$, we can define the concept of “fuzzy least-core” as the minimal set of $\alpha_{\mathcal{T}}$. This is obviously contained in the least-core and in the fuzzy core whenever it is non-empty. Also, we can prove that assumption (6) of Proposition 4 implies the non-emptiness of the fuzzy least-core. $\square$

To investigate further properties of the least-core, we shall make an assumption analogous to assumption (16) of Proposition 10.1.8. Let $J_0(N)$ be the subset of multilosses $c \in J(N)$ satisfying $c_i \leq w(i, 1) + \inf_{d \in J(N)} \alpha(d)$.

(7) For any $c \in J_0(N)$ and any non-empty coalition $A \neq N$, we can find $\varepsilon_0 > 0$ such that, for any $\varepsilon > 0$ less than ε_0 there exists $d \in J(N)$ satisfying $d_i \leq c_i - \varrho \varepsilon$ for any $i \in A$ and $c_j \leq d_j \leq c_j + \varepsilon$ for any $j \in A$ (where $\varrho > 0$).

Proposition 5. Suppose that (7) holds. Let $\mathcal{B}$ be a family of coalitions satisfying

$$(8) \quad B_0 = \bigcup_{B \in \mathcal{B}} B \neq N.$$

Then, for any c belonging to the least-core, we have

$$(9) \quad \alpha(c) = \max_{B \in \mathcal{B}} \theta(B, c).$$

Proof. Let c belong to the least-core and let $\mathcal{C}$ be the family of coalitions A such that $\theta(A, c) = \alpha(c)$. If $\mathcal{C} = \mathcal{A}$, then (9) holds for any family $\mathcal{B}$ of coalitions. If not,

$$b = \max_{A \in \mathcal{C}} \theta(A, c) < \alpha(c).$$

Write $\varepsilon_1 = 1/3(\alpha(c) - b)$ and $\varepsilon = \min(\varepsilon_0, \varepsilon_1)$ where ε_0 appears in assumption (7).

By assumption (7), there exists a multiloss $d \in J(N)$ such that $d_i = c_i - \varrho\varepsilon$ for any $i \in B_0$, $c_j \leq d_j \leq c_j + \varepsilon$ for any $j \notin B_0$, where B_0 is the coalition defined by (8).

If $A \subset B_0$, we have $\theta(A, d) \leq \theta(A, c) - \varrho\varepsilon$ and, if $A \not\subset B_0$, we have $-\varrho\varepsilon \leq \theta(A, d) - \theta(A, c) \leq \varepsilon$.

We now assume that property (9) is false, i.e. $\beta = \max_{B \in \mathcal{B}} \theta(B, c) < \alpha(c)$. Hence, when $\theta(A, c) > \beta$, the coalition A does not belong to $\mathcal{B}$ and thus, is contained in B_0 by (8). Such coalitions A exist and satisfy $\theta(A, d) \leq \theta(A, c) - \varrho\varepsilon \leq \alpha(c) - \varrho\varepsilon$. On the other hand, if $\theta(A, c) \leq \beta$, then either $\theta(A, d) \leq \theta(A, c) - \varrho\varepsilon \leq \alpha(c) - \varrho\varepsilon$ or $\theta(A, d) \leq \theta(A, c) + \varepsilon \leq \beta + \varepsilon \leq b + \varepsilon_1 = \alpha(c) - 2\varepsilon_1$. Hence $\alpha(d) = \max_{A \in \mathcal{A}} \theta(A, d) \leq \alpha(c) - \varepsilon_1 \min(2, \varrho) < \alpha(c)$.

This is impossible since c belongs to the least-core. $\square$

Corollary 1. Suppose that property (7) holds. For any non-empty coalition $C \neq N$, and for any multiloss c of the least core, we have that

$$(10) \quad \alpha(c) = \max_{A \not\subset C} \theta(A, c).$$

In particular, for any player $i \in N$, we have that

$$(11) \quad \alpha(c) = \max_{i \in A} \theta(A, c).$$

Proof. This is left as an exercise. $\square$

*10.5.3. Nucleolus

A drawback in using the least-core as a selection procedure is that it can be large.

We are therefore led to define a finer selection procedure which yields a subset of the least-core called the nucleolus.

Interpret

$$(12) \quad \theta(A, c) = \inf_{\lambda \in \mathcal{M}_A} [\langle \lambda, \tau^A \cdot c \rangle - w(A, \lambda)]$$

as the “complaint of the coalition A ” associated with c . Then the elements of the least-core minimize the “loudest complaint” $\alpha(c) = \sup_{A \in \mathcal{A}} \theta(A, c)$.

Schmeidler suggested a more sophisticated selection method, which

- first, minimizes the loudest complaint (i.e., constructs the least-core)
- second, minimizes the second loudest complaint in the least-core,
- and so on.

To formalize this idea we need to define a "classification map" ω which arranges the components of θ in decreasing order. Having so re-classified the complaints, we then seek the smallest of these with respect to the lexicographic ordering.

Definition 3. Let $|\mathcal{A}|$ be the cardinal of a finite set $\mathcal{A}$. We shall say that the map ω from $\mathbf{R}^{\mathcal{A}}$ into $\mathbf{R}^{|\mathcal{A}|}$ associating with any $\theta = \{\theta(A)\}_{A \in \mathcal{A}} \in \mathbf{R}^{\mathcal{A}}$ the vector $\omega\theta$ defined by

$$(13) \quad \begin{cases} (\omega\theta)_1 = \max_{A \in \mathcal{A}} \theta(A) = \theta(A_1) \\ (\omega\theta)_2 = \max_{A \neq A_1} \theta(A) = \theta(A_2) \\ \vdots \\ (\omega\theta)_i = \max_{A \neq A_1, \dots, A_{i-1}} \theta(A) = \theta(A_i) \end{cases}$$

is a *classification map*.

In other words, the components of $\omega\theta$ are the components of θ , but classified in a decreasing order. It is clear that ω is a *continuous map*.

The map

$$(14) \quad \theta(c) : A \mapsto \theta(A, c)$$

can be identified with the vector in $\mathbf{R}^{\mathcal{A}}$ of complaints associated with the multiloss c . Then $\omega\theta(c)$ is the vector of complaints classified in a decreasing order. We then introduce the lexicographic ordering on $\mathbf{R}^{|\mathcal{A}|}$ defined by

$$(15) \quad x \leq y \Leftrightarrow \exists j \text{ such that } x_i = y_i \quad \forall i \leq j-1, x_j > y_j.$$

This is a total preordering.

Definition 4. We shall say that $\bar{c} \in J(N)$ belongs to the "nucleolus" of the game if the classified vector of complaints $\omega\theta(\bar{c})$ is smallest with respect to the lexicographic ordering.

By the very construction of the nucleolus, it is a subset of the least-core (and thus, is a Pareto minimum).

It is constructed in the following way. We define the least-core, i.e.

$$(16) \quad J_1(N) = \{c_1 \in J(N) \text{ such that } (\omega\theta(c_1))_1 = \inf_{c \in J(N)} (\omega\theta(c))_1\}$$

Secondly, we define

$$(17) \quad J_2(N) = \{c_2 \in J_1(N) \text{ such that } (\omega\theta(c_2))_2 = \inf_{c_1 \in J_1(N)} (\omega\theta(c_1))_2\}$$

and, in the same way,

$$(18) \quad J_k(N) = \{c_k \in J_{k-1}(N) \text{ such that } (\omega\theta(c_k))_k = \min_{c_{k-1} \in J_{k-1}(N)} (\omega\theta(c_{k-1}))_k\}.$$

It is clear that the nucleolus is nothing other than the set $J_{|\mathcal{A}|}(N)$.

Proposition 6. *Suppose that*

(6) $J(N)$ *is closed and bounded below.*

Then the nucleolus is a non-empty compact subset of the least-core.

Proof. The maps $c \mapsto \theta(A, c)$ are lower semi-continuous (see the proof of Proposition 4) and the maps $c \mapsto (\omega\theta(c))_i$ are also clearly lower semi-continuous. We also know that assumption (6) and Proposition 4 imply that the least-core $J_1(N)$ is non-empty and compact. Therefore, the subsets $J_k(N)$ are also non-empty and compact. In particular, the last of these, $J_{|A|}(N)$, which is the nucleolus, is compact and non-empty. $\square$

Remark. We shall prove in the next chapter that the nucleolus of a game with side-payments *consists of a unique element* (see Proposition 11.4.5).

CHAPTER 11

GAMES WITH SIDE-PAYMENTS

In this chapter, we study games and fuzzy games *with side-payments*. We consider conditions under which the core is non-empty and introduce the concept of the *Shapley value* and some related ideas.

We begin by defining a game $(\mathcal{A}, v)$ (resp. a fuzzy game $(\mathcal{T}, v)$) with side-payments to be a game $(\mathcal{A}, J)$ (resp. $(\mathcal{T}, J)$) such that

$$\begin{aligned} \forall A \in \mathcal{A}, \quad J(A) &= \{c \in \mathbf{R}^A \text{ such that } \sum_{i \in A} c_i = v(A)\} \\ (\text{resp. } \forall \tau \in \mathcal{T}, \quad J(\tau) &= \{c \in \mathbf{R}^T \text{ such that } \sum_{i=1}^T c_i = v(\tau)\}). \end{aligned}$$

In other words, the function v , called the characteristic function of the game, is a *loss function* which associates with any coalition A (resp. fuzzy coalition τ) its loss $v(A)$ (resp. $v(\tau)$) measured in (monetary) side-payments. The rules of the game require that the loss $v(A)$ is shared by players $i \in A$ (resp. that the loss $v(\tau)$ is shared by players participating to τ proportionally to their rates of participation).

This implies that

$$J(N) \text{ (or } J(\tau^N)) \text{ is the set of Pareto minima.}$$

The *core* $\mathcal{C}(\mathcal{A}, v)$ is easily expressed. It is the subset of multilosses c satisfying

$$\left\{ \begin{array}{l} \text{(i) } \sum_{i=1}^T c_i = v(N), \\ \text{(ii) } \forall A \in \mathcal{A}, \sum_{i \in A} c_i \leq v(A). \end{array} \right.$$

Hence, an element $c \in \mathcal{C}(\mathcal{A}, v)$ displays rather strong stability properties since, for any coalition A , the loss $\sum_{i \in A} c_i$ it receives is not larger than the initial loss $v(A)$ assigned to A .

For instance, we can associate with a cooperative game $\{X(A), F^A\}_{A \in \mathcal{A}}$ described in strategic form, the game $(\mathcal{A}, v)$ with side-payments defined by

$$\forall A \in \mathcal{A}, \quad v(A) = \inf_{x \in X(A)} \sum_{i \in A} f_i^A(x^A)$$

In this case, a multiloss c belongs to the core if

$$\begin{cases} \text{(i)} \sum_{i=1}^n c_i = \inf_{x \in X(N)} \sum_{i=1}^n f_i(x), \\ \text{(ii)} \forall A \in \mathcal{A}, \sum_{i \in A} c_i \leq \inf_{x^A \in X(A)} \sum_{i \in A} f_i(x^A). \end{cases}$$

In the case of fuzzy games $(\mathcal{T}, v)$ where the characteristic function v is assumed to be positively homogeneous, the core $\mathcal{C}(\mathcal{T}, v)$, which is the set of multilosses $c \in \mathbf{R}^n$ such that

$$\begin{cases} \text{(i)} \sum_{i=1}^n c_i = v(\tau^N), \\ \text{(ii)} \forall \tau \in \mathcal{T}, \sum_{i=1}^n \tau_i c_i \leq v(\tau), \end{cases}$$

is nothing other than the subdifferential of v at τ^N , i.e. $\mathcal{C}(\mathcal{T}, v) = \partial v(\tau^N)$. Hence, if v is also convex, the core of the fuzzy game is non-empty. Note that if v is convex and positively homogeneous, it satisfies inequality

$$v(\tau_1 + \tau_2) \leq v(\tau_1) + v(\tau_2)$$

which means that "l'union fait la force" (unity is strength).

We note also that $\mathcal{C}(\mathcal{T}, v) \subset \mathcal{C}(\mathcal{A}, v|_{\mathcal{A}})$, i.e. that the core of a fuzzy game is contained in the core of its restriction to the family $\mathcal{A}$ of coalitions. The converse problem arises. We can associate with any characteristic function $v \in \mathbf{R}^{\mathcal{A}}$ its convex cover πv defined by

$$\pi v(\tau) = \inf_{m \in \mathcal{A}(\tau)} \sum_{A \in \mathcal{A}} m(A) v(A)$$

where $\mathcal{A}(\tau) = \{m \in \mathbf{R}_+^{\mathcal{A}} \text{ such that } \sum_{A \in \mathcal{A}} m(A) \tau^A = \tau\}$.

The characteristic function πv is clearly convex and positively homogeneous. We shall prove that the core $\mathcal{C}(\mathcal{A}, v) = \mathcal{C}(\mathcal{T}, \pi v) = \partial \pi v(\tau^N)$ is non-empty if and only if $\pi v(\tau^N) = v(N)$. We shall say that a characteristic function v is balanced if $\pi v(\tau^N) = v(N)$. Examples of balanced characteristic functions are given. We mention here only the following result due to Scarf. The characteristic function v defined by

$$v(A) = \inf_{x^A \in X^A} \sum_{i \in A} f_i^*(x^A)$$

is balanced when the functions f_i are convex.

Several objections can be raised against the concept of core. For instance, the fact that the core can be empty for certain types of game is a drawback. Another approach, introduced by Shapley, is to evaluate a game a priori, by assigning to a characteristic function v a multiloss $\varphi_n v \in \mathbf{R}^n$. This is required to obey the following axioms. (1) $\varphi_n v$ is Pareto minimal. (2) $\varphi_n v$ does not depend on

how the players are ordered. (3) The loss assigned to a coalition of any partition is the sum of the losses assigned to each player of the coalition. (4) φ_n is linear. We shall prove that there is a unique sequence of such linear operators φ_n (called fuzzy values) which associate with any positively homogeneous characteristic function differentiable at τ^N the multilosses.

$$\varphi_n v = Dv(\tau^N) \quad (\text{gradient of } v \text{ at } \tau^N).$$

Therefore, $\mathcal{C}(\mathcal{T}, v) = \{\varphi_n v\}$ whenever the characteristic function is positively homogeneous, convex and differentiable at τ^N .

Of course, we can apply the fuzzy value φ_n to the convex cover πv of a characteristic function $v \in \mathbf{R}^{\mathcal{A}}$. This requires first that πv is differentiable at τ^N (this is not always true). Another drawback is that the map $v \mapsto \varphi_n \pi v$ is no longer linear, since π is not a linear operator. This suggests introducing the Cornet linear extension operator ω , associating with any $v \in \mathbf{R}^{\mathcal{A}}$ a positively homogeneous function ωv such that, $\forall A \in \mathcal{A}$, $\omega v(\tau^A) = v(A)$. We shall prove that the map $\psi_n = \varphi_n \omega : v \rightarrow \varphi_n \omega v = D[\omega v](\tau^N)$ is the *Shapley value*. This is the unique linear operator $\psi_n \in \mathcal{L}(\mathbf{R}^{\mathcal{A}}, \mathbf{R}^n)$ mapping v into a Pareto minimal multiloss $\psi_n v$, which does not depend on how the players are ordered and which assigns 0 to each *dummy player* i (in the sense that $v(A \cup \{i\}) = v(A)$ for any coalition A).

11.1. Core of a fuzzy game with side-payments

We define games and fuzzy games with side-payments and describe several examples. Then we prove that the core of a fuzzy game $\mathcal{C}(\mathcal{T}, v)$ is equal to $\partial v(\tau^N)$ when v is positively homogeneous and thus, that it is non-empty whenever v is also convex. We investigate the case where $v(\tau) = \sup_{p \in P} w(\tau, p)$. Under convenient assumptions, the core is the closed convex hull of the gradients $Dw(\tau^N; p)$ as p ranges over the maximal set of $w(\tau^N; \cdot)$. We apply this result to characterize the core of a “market game” in terms of “side-payment competitive equilibria”.

11.1.1. Core of a game with side-payments

Definition 1. Let $\lambda \in \mathring{\mathbf{R}}_+^{n*}$. We shall say that a (fuzzy) game $(\mathcal{T}, w)$ defined by

$$(1) \quad \forall \tau \in \mathcal{T}, \quad J(\tau) = \{c \in \mathbf{R}^{A_\tau} \text{ such that } \langle \lambda, c \rangle = w(\tau)\}$$

is a *game with side-payments* (or with *transferable losses*) and that the function $w : \tau \in \mathcal{T} \mapsto w(\tau)$ is the “characteristic function” of such a game, which associates with each coalition τ its loss $w(\tau)$.

The subset of feasible multilosses of the fuzzy coalition τ is nothing other than the set of multilosses transferable with the amount $w(\tau)$ under the rate of transfer $\lambda \in \mathbf{R}_+^{n*}$.

Such a game describes the case where the players of a fuzzy coalition agree to share a given side-payment allocated to the coalition with respect to the same system of eights λ .

Remark. Even if it entails replacing c by $\lambda c = \{\lambda_i c_i\}_{i \in N}$, we can always take $\lambda = \{1, \dots, 1\}$. In this case, we set $(\mathcal{O}, w)_\lambda = (\mathcal{O}, w)$.

Due to the special form of the sets of admissible multilosses of a game with side payments, a nice description of its core is available.

Proposition 1. Let $\lambda \in \mathbf{R}_+^{n*}$ be fixed. The core $\mathcal{C}(\mathcal{O}, w)$ of a (fuzzy) game with side payments $(\mathcal{O}, w)_\lambda$ is the set of multilosses c satisfying

$$(2) \quad \begin{cases} (i) \langle \lambda, c \rangle = w(N), \\ (ii) \forall \tau \in \mathcal{O}, \langle \lambda, \tau \cdot c \rangle \leq w(\tau). \end{cases}$$

Proof. It is enough to prove that for any $\tau \in \mathcal{O}$

$$(3) \quad J_+(\tau) = \{c \in \mathbf{R}^{A_\tau} \text{ such that } \langle \lambda, c \rangle \geq w(\tau)\}$$

and

$$(4) \quad \hat{J}_+(\tau) = \{c \in \mathbf{R}^{A_\tau} \text{ such that } \langle \lambda, c \rangle > w(\tau)\}.$$

(If (4) is true, to say that “ τ does not reject” c amounts to saying that $\langle \lambda, \tau \cdot c \rangle \leq w(\tau)$. In particular, $\langle \lambda, c \rangle \leq w(N)$ since c is not rejected by N and $\langle \lambda, c \rangle \geq w(N)$ since c belongs to $J_+(N)$.)

Statement (3) is quite obvious.

We prove statement (4). Observe that $c \in \hat{J}_+(\tau)$ if and only if there exists $d \in J_+(\tau)$ and $b \in \mathbf{R}_+^{A_\tau}$ such that $c = d + b$. Therefore, $\langle \lambda, c \rangle > \langle \lambda, d \rangle \geq w(\tau)$.

Conversely, if $s = \langle \lambda, c \rangle - w(\tau) > 0$, the vector b with components $b_i = s/\lambda^i$ where $i \in A_\tau$, belongs to $\mathbf{R}_+^{A_\tau}$ and satisfies $\langle \lambda, b \rangle = s$.

Therefore, the multiloss $d = c - b$ is feasible for the coalition τ since

$$\langle \lambda, d \rangle = \langle \lambda, c \rangle - s = w(\tau).$$

Hence $c \in \hat{J}_+(\tau)$. $\square$

Example. (Games with side-payments associated with cooperative games). Let $\{X(A), F^A\}_{A \in \mathcal{A}}$ be a cooperative game described in strategic form. Let $\lambda \in \mathcal{M}^n$ be fixed.

It is natural for the coalition A to pool their losses. We introduce

$$(5) \quad w(A, \lambda) = \inf_{x^A \in X(A)} \sum_{i \in A} \lambda^i f_i^A(x^A) = \sigma^b(J(A); \lambda)$$

which measures the smallest transferable side payment of the coalition A associated with the rate $\lambda \in \mathcal{M}^n$. We can therefore forget about the underlying multistrategies and just consider a game where coalitions A choose “transferable multilosses” instead of multistrategies or feasible multilosses.

The core $\mathcal{C}(\mathcal{A}, w(\lambda))_\lambda$ of this game with side payments is the set of multilosses satisfying

$$(6) \quad \begin{cases} \text{(i) } \sum_{i \in N} \lambda^i c_i = \inf_{x \in X(N)} \sum \lambda^i f_i(x), \\ \text{(ii) } \forall A \in \mathcal{A}, \sum_{i \in A} \lambda^i c_i \leq \inf_{x \in X(A)} \sum_{i \in A} \lambda^i f_i^A(x). \end{cases}$$

Remark. If there exists $x \in X(N)$ such that $c = F(x)$ belongs to the core $\mathcal{C}(\mathcal{A}, w(\lambda))_\lambda$ of the associated game with side-payments, it is clear that c is a canonical cooperative equilibrium of the game $(\mathcal{A}, J)$ (see Section 10.4.1). We shall use this remark in the next chapter.

11.1.2. Linear games

We shall say that a game with side-payments $(\mathcal{A}, w)$ (resp. a fuzzy game $(\mathcal{T}, w)$) is “linear” (or “inessential”) if

$$(7) \quad \begin{aligned} \forall A \in \mathcal{A}, \quad w(A) &= \sum_{i \in A} w(i) \\ [\text{resp. } \forall \tau \in \mathcal{T}, w(\tau) &= \sum_{i \in N} \tau_i w(i)] \end{aligned}$$

It is clear that the core $\mathcal{C}(\mathcal{T}, w)$ of a linear game contains the unique vector $\{w(i)\}_{i \in N}$.

Example. (Games with side-payments associated with an economy). Consider an economy $\{R^i, Y(i), f_i\}_{i \in N}$ (see 8.3). If a price system $p \in \mathbb{R}^n$ and a rate of transfer $\lambda \in \mathbb{R}_+^{n*}$ prevail, we measure the smallest transferable side-payment of a fuzzy coalition $\tau \in [0, 1]^n$ by

$$(8) \quad w(\tau, \lambda, p) = \sum_{i \in N} \tau_i w(i, \lambda^i, p)$$

where

$$(9) \quad w(i, \lambda^i, p) = \inf_{x^i \in R^i} [\lambda^i f_i(x^i) + \langle p, x^i \rangle - r_i(p)]$$

is the smallest net loss of the i^{th} player.

(Recall that $r_i(p) = \sup_{y \in Y(i)} \langle p, y \rangle$ is the income allocated to the i^{th} player.)

We set

$$(10) \quad w(A; \lambda, p) = w(\tau^A, \lambda, p) = \sum_{i \in A} w(i, \lambda, p).$$

Therefore, the fuzzy game $(\mathcal{C}, w(\lambda, p))$ with side-payments associated with the economy is a linear game. *Its core contains the unique vector $\{w(i, 1, p/\lambda^i)\}_{i \in N}$.*

Remark. If there exists $\bar{x} \in X(N)$ such that $\bar{c} = F(\bar{x})$ belongs to the core $\mathcal{C}(\mathcal{A}, w(\bar{\lambda}, \bar{p})) = \mathcal{C}(\mathcal{C}, w(\bar{\lambda}, \bar{p}))$ of the game with side-payments, we notice that the pair $\{\bar{x}, \bar{p}\}$ is a Walras equilibrium because, writing $\bar{\mu}_i = 1/\bar{\lambda}^i$, we have that

$$f_i(\bar{x}^i) = w(i, 1, \bar{\mu}_i \bar{p}) \leq \inf_{x^i \in B_i(\bar{p}, r_i(\bar{p}))} f_i(x^i).$$

Also, for all $i \in N$,

$$f_i(\bar{x}^i) \leq f_i(\bar{x}^i) + \langle \bar{p}, \bar{x}^i \rangle - r_i(\bar{p}).$$

But, since $\bar{x} \in X(N)$, $\sum_{i=1}^n (\langle \bar{p}, \bar{x}^i \rangle - r_i(\bar{p})) \leq 0$.

Therefore, $\langle \bar{p}, \bar{x}^i \rangle = r_i(\bar{p})$ for all i . In particular, $\bar{x}^i \in B_i(\bar{p}, r_i(\bar{p}))$ minimizes $f_i(\cdot)$ over $B_i(\bar{p}, r_i(\bar{p}))$. This remark will play an important role later.

Remark. Notice that $w(i, \lambda^i, p)$ is the dual utility function of the problem

$$(11) \quad \inf_{x^i \in B_i(\bar{p}, r_i(\bar{p}))} \lambda_i f_i(x^i).$$

11.1.3. Non-emptiness of the core of fuzzy games with side-payments

We take $\lambda = 1$. We consider the class of characteristic functions v defined on $\mathbf{R}_+^n$ which are positively homogeneous. In other words, the characteristic functions v are defined on the subset

$$(12) \quad \{\tau \in [0, 1]^n \text{ such that } \sum_{i \in N} \tau_i = 1\}$$

of fuzzy coalitions. They are extended to $\mathbf{R}_+^n$ by setting

$$(13) \quad \begin{cases} v(\tau) = \left(\sum_{i \in N} \tau_i \right) v\left(\frac{\tau}{\sum_{i \in N} \tau_i} \right) & \text{if } \tau \neq 0. \\ v(0) = v(\tau^0) = 0 & \text{if } \tau = 0. \end{cases}$$

Proposition 2. Suppose that the characteristic function v of the game is positively homogeneous. Then the core of the game is equal to the subdifferential $\partial v(\tau^N)$ of v at τ^N .

Proof. If c belongs to the core, we deduce that for any $\tau \in \mathbf{R}_+^n$

$$\langle \tau, c \rangle - v(\tau) \leq \langle \tau^N, c \rangle - v(\tau^N) = 0$$

and thus, that $c \in \partial v(\tau^N)$. Conversely, if $c \in \partial v(\tau^N)$, we deduce from inequalities $v(\tau^N) - \langle \tau^N, c \rangle \leq v(\tau) - \langle \tau, c \rangle$ with $\tau = 0$ and $\tau = 2\tau^N$ that $v(\tau^N) = \langle \tau^N, c \rangle$. Therefore, $\langle \tau, c \rangle \leq v(\tau)$ for all $\tau \in \mathbf{R}_+^n$. $\square$

In particular, if v is a positively homogeneous function defined on $\mathbf{R}_+^n$, we obtain the following result.

Proposition 3. *Suppose that the characteristic function v satisfies*

$$(14) \quad v \text{ is convex and positively homogeneous from } \mathbf{R}_+^n \text{ into } \mathbf{R}.$$

Then

$$(15) \quad \text{the core of the fuzzy game is non-empty, convex and compact.}$$

Furthermore, if we assume that

$$(16) \quad v \text{ is differentiable at } \tau^N = \{1, \dots, 1\},$$

then the core consists of the single vector $Dv(\tau^N)$.

Proof. Since τ^N belongs to the interior of $\mathbf{R}_+^n$, the convex function v is continuous at τ^N and thus, subdifferentiable at τ^N (Proposition 4.1.7) and the subdifferential $\partial v(\tau^N)$ is convex and compact (Proposition 4.1.7).

Therefore, statement (15) follows from Proposition 2. The core $\partial v(\tau^N)$ contains a unique element if and only if v is differentiable at τ^N and, in this case $\partial v(\tau^N) = \{Dv(\tau^N)\}$ (Proposition 4.2.3). In fact, Proposition 4.3.4 implies that the derivative from the right of v at τ^N , namely

$$(17) \quad Dv(\tau^N)(\tau) = \sup_{c \in \mathcal{C}(\tau, v)} \langle \tau, c \rangle = \sigma^*(\mathcal{C}(\tau, v); c),$$

is the upper support function of the core of the fuzzy game.

Interpretation of positively homogeneous convex characteristic functions

Since any positively homogeneous convex function is subadditive,

$$(18) \quad v(\tau + \sigma) \leq v(\tau) + v(\sigma)$$

we can say that *coalitions gain by joining forces*. In particular, if A and B are disjoint coalitions, $\tau^A + \tau^B = \tau^{A \cup B}$ and we obtain

$$(19) \quad v(\tau^{A \cup B}) \leq v(\tau^A) + v(\tau^B). \quad \square$$

Consider the case when τ is no longer differentiable at τ^N , but is the pointwise supremum of functions differentiable at τ^N .

Proposition 4. *Suppose that the characteristic function v is defined by*

$$(20) \quad \forall \tau \in \mathbf{R}_+^n, \quad v(\tau) = \sup_{p \in P} w(\tau, p)$$

where

$$(21) \quad \begin{cases} \text{(i) } P \text{ is compact,} \\ \text{(ii) } \forall \tau \in \mathbf{R}_+^n, p \mapsto w(\tau, p) \text{ is upper semi-continuous,} \\ \text{(iii) } \forall p \in P, \tau \mapsto w(\tau, p) \text{ is convex, positively homogeneous} \\ \text{and differentiable at } \tau^N. \end{cases}$$

Let $P(N)$ be the subset of P defined by

$$(22) \quad P(N) = \{p \in P \text{ such that } v(\tau^N) = w(\tau^N, p)\}.$$

Then the core is the closed convex hull of the set $\{Dw(\tau^N; p)\}_{p \in P(N)}$ of gradients at τ^N of the functions $w(\cdot, p)$ as p ranges over $P(N)$.

Proof. The convex positively homogeneous function v is finite on $\mathbf{R}_+^n$ (by (21) (i) and (ii)). Therefore,

$$(23) \quad Dv(\tau^N)(\tau) = \sigma^*(\mathcal{C}(\mathcal{T}, v); \tau).$$

On the other hand, Proposition 4.3.6 implies that

$$\begin{aligned} Dv(\tau^N)(\tau) &= \sup_{p \in P(N)} \langle Dw(\tau^N, p), \tau \rangle \\ &= \sigma^*(\overline{\text{co}} [\{Dw(\tau^N, p)\}_{p \in P}]; \tau). \end{aligned}$$

Therefore, $\mathcal{C}(\mathcal{T}, v) = \overline{\text{co}} [\{Dw(\tau^N, p)\}_{p \in P}]$.

11.1.4. Core of fuzzy market games

Associate with an economy $\{R^I, Y(i), f_i\}_{i \in N}$ the “fuzzy market game” defined by the characteristic function

$$(24) \quad v(\tau) = \inf_{x^* \in X(\tau)} \sum_{i \in N} \tau_i f_i(x),$$

where the set $X(\tau)$ of allocations of the fuzzy coalition τ is defined in Section 10.4.7.

Consider the saddle points $\{\bar{x}, \bar{p}\}$ of the Lagrangian

$$(25) \quad l(x, p) = \sum_{i \in N} f_i(x^i) + \left\langle p, \sum_{i \in N} x^i \right\rangle - r(p)$$

of the minimization problem $v(N) = \inf_{x \in X(N)} \sum f_i(x^i)$, i.e. the pairs $\{\bar{x}, \bar{p}\} \in X(N) \times \mathbf{R}_+^{I*}$ satisfying

$$(26) \quad v(N) = l(\bar{x}, \bar{p}) = \min_{x \in R^N} l(x, \bar{p}) \quad \text{and} \quad \left\langle \bar{p}, \sum_{i \in N} \bar{x}^i \right\rangle = r(\bar{p}).$$

We shall say that such a pair is a “*side-payment competitive equilibrium*” of the market game.

Associate with any side-payment competitive equilibrium $\{\bar{x}, \bar{p}\}$ the vector,

$$(27) \quad c(\bar{x}, \bar{p}) = \{f_i(\bar{x}^i) + \langle \bar{p}, \bar{x}^i \rangle - r_i(\bar{p})\}_{i \in N}.$$

Proposition 5. *Suppose that assumptions (4), (5), (6), (7), (8) and (9) of Theorem 8.3.1 for the existence of a Walras equilibrium holds. Then the core of the fuzzy market game is the closed convex hull of the set of vectors $c(\bar{x}, \bar{p})$ as $\{\bar{x}, \bar{p}\}$ ranges over the set of side-payment competitive equilibria.*

Proof. Since the subsets R^i and the loss functions f_i are convex and since $0 \in \text{Int}(R^i - Y(i))$ for all i , we deduce that there exists a Lagrange multiplier $p_\tau \in \mathbf{R}_+^{I*}$ of the minimization problem

$$(28) \quad v(\tau) = \inf_{x_\tau \in R_\tau} \sum_i \tau_i (f_i(x_\tau^i) + \langle p_\tau, x_\tau^i \rangle - r_i(p_\tau)).$$

We prove later that these Lagrange multipliers p_τ remain in a fixed convex compact subset P of $\mathbf{R}_+^{I*}$ (Proposition 12.3.1).

We notice that $P(N)$ is nothing other than the set of Lagrange multipliers of the minimization problem $v(\tau^N)$, i.e. the set of side-payment competitive equilibrium prices.

On the other hand, the characteristic functions $\tau \mapsto w(\tau; p)$ are linear. The core of the associated game consists of the single vector $\{w(i, p)\}_{i \in N}$. Proposition 4 implies that the core of the fuzzy market game is the closed convex hull of the subset of vectors $\{w(i, \bar{p})\}_{i \in N}$ as $\bar{p}$ ranges over $P(N)$. But $w(i, \bar{p}) = f_i(\bar{x}^i) + \langle \bar{p}, \bar{x}^i \rangle - r_i(\bar{p})$ where $\bar{x} \in X(N)$ minimizes $\sum_{i \in N} f_i(x^i)$ over X^N . Hence the proposition is proved. $\square$

11.2. Core of a game with side-payments

Since the core of a fuzzy game is contained in that of its restriction, we can use the existence result, Proposition 1.3, whenever we can in some sense “extend” a game $(\mathcal{A}, v)$ into a fuzzy game $(\mathcal{C}, \pi v)$ where πv is convex and positively homogeneous. This can be done by using the “convex cover operator” π de-

defined by

$$\pi v(\tau) = \inf_{m \in \mathcal{N}(\tau)} \sum_{A \in \mathcal{A}} m(A) v(A),$$

where $\mathcal{N}(\tau) = \{m \in \mathbf{R}_+^{\mathcal{A}} \text{ such that } \sum_{A \in \mathcal{A}} m(A) \tau^A = \tau\}$.

We then prove that $\mathcal{C}(\mathcal{A}, v) = \mathcal{C}(\mathcal{T}, \pi v) = \partial \pi v(\tau^N) \neq \emptyset$ if and only if $v(N) = \pi v(\tau^N)$. (In this case we say that v is “balanced”). On the assumption that the family $\{X(A)\}_{A \in \mathcal{A}}$ of multistrategy sets is “balanced”, i.e.

$$\forall m \in \mathcal{N}(\tau^N), \quad \sum_{A \in \mathcal{A}} m(A) X(A) \subset X(N),$$

we shall prove that the characteristic function v defined by

$$v(A) = \inf_{x^A \in X(A)} \sum_{i \in A} f_i^A(x^A)$$

is balanced whenever the functions f_i are convex.

11.2.1. Convex cover of a game

We have proved that the cores of fuzzy games $(\mathcal{T}, v)$ defined by convex positively homogeneous characteristic functions are non-empty. This implies that the core of their restriction $(\mathcal{A}, v|_{\mathcal{A}})$ to families of usual coalitions is also non-empty. The converse problem arises. Can we in some sense extend a game $(\mathcal{A}, v)$ with side payments into a fuzzy game $(\mathcal{T}, \pi v)$ in such a way that the cores of the two games coincide.

Definition 1. Let $\mathcal{A}$ be a family of coalitions and let τ be a fuzzy coalition. We shall denote by

$$(1) \quad \mathcal{N}(\tau) = \{m \in \mathbf{R}_+^{\mathcal{A}} \text{ such that } \sum_{A \in \mathcal{A}} m(A) \tau^A = \tau\}$$

the subset of non-negative weights $m(A)$ such that

$$(2) \quad \forall i \in N, \quad \tau_i = \sum_{\substack{A \in \mathcal{A} \\ A \ni i}} m(A).$$

We shall say that $\mathcal{N}(\tau^N)$ is the set of “balances”. If $v : A \mapsto \mathbf{R}$ is a characteristic function, we shall say that the function $\pi v : \mathcal{T} \mapsto \mathbf{R}$ defined by

$$(3) \quad \pi v(\tau) = \inf_{m \in \mathcal{N}(\tau)} \sum_{A \in \mathcal{A}} m(A) v(A)$$

is its “convex cover” and that π is the “convex cover operator”.

Remark. Notice that

$$(4) \quad \text{If } m \in \mathcal{N}(\tau), \text{ then } m(A) > 0 \text{ implies that } A \subset A_\tau = \{i \in N \mid \tau_i > 0\}.$$

The following proposition summarizes the obvious properties of the map π .

Proposition 1. *For any v , πv is a positively homogeneous, convex function defined on $\mathbf{R}_+^n$, satisfying*

$$(5) \quad \forall A \in \mathcal{A}, \quad \pi v(\tau^A) \leq v(A).$$

Proof. This is left as an exercise. $\square$

Remark. The map π does not have the interpolation property. Therefore, πv is not a true extension of v .

This fact motivates the following definition.

Definition 2. We shall say that a characteristic function v from $\mathcal{A}$ into $\mathbf{R}$ is “balanced” if

$$(6) \quad v(N) = \pi v(\tau^N) = \inf_{m \in \mathcal{N}(\tau^N)} \sum_{A \in \mathcal{A}} m(A) v(A)$$

and that it is “totally balanced” if

$$(7) \quad \forall A \in \mathcal{A}, \quad v(A) = \pi v(\tau^A).$$

If v is totally balanced, the convex cover πv of v is an extension of v . With these definitions, we can characterize the games having a non-empty core.

11.2.2. Non-emptiness of the core of a balanced game

Theorem 1. *The core $\mathcal{C}(\mathcal{A}, v)$ of a game with side-payments $(\mathcal{A}, v)$ is non-empty if and only if v is balanced. In this case $\mathcal{C}(\mathcal{A}, v) = \mathcal{C}(\mathcal{T}, \pi v)$.*

Proof. We begin by proving that, if there exists $c \in \mathcal{C}(\mathcal{A}, v)$, the function v is balanced. Let $m \in \mathcal{N}(\tau^N)$ be a balance. We obtain that

$$\begin{aligned} v(N) &= \langle \tau^N, c \rangle = \left\langle \sum_{A \in \mathcal{A}} m(A) \tau^A, c \right\rangle \\ &= \sum_{A \in \mathcal{A}} m(A) \langle \tau^A, c \rangle \leq \sum_{A \in \mathcal{A}} m(A) v(A). \end{aligned}$$

Therefore, $v(N) \leq \inf_{m \in \mathcal{N}(\tau^N)} \sum_{A \in \mathcal{A}} m(A) v(A) = \pi v(\tau^N)$.

Conversely, suppose that v is balanced. Then $\mathcal{E}(\mathcal{T}, \pi v) \subset \mathcal{E}(\mathcal{A}, v)$ because, if $c \in \mathcal{E}(\mathcal{T}, \pi v)$, then $\langle \tau^N, c \rangle = \pi v(N) = v(N)$ (since v is balanced) and, for any $A \in \mathcal{A}$, $\langle \tau^A, c \rangle \leq \pi v(\tau^A) \leq v(A)$. Hence $c \in \mathcal{E}(\mathcal{A}, v)$. Now, $\mathcal{E}(\mathcal{A}, v) \neq \emptyset$ since $\mathcal{E}(\mathcal{T}, \pi v)$ is non-empty by Proposition 1.3 (because $\pi v(\cdot)$ is convex and positively homogeneous).

Finally, it remains to prove that, if $c \in \mathcal{E}(\mathcal{A}, v)$, then c belongs to $\mathcal{E}(\mathcal{T}, \pi v)$. We know that $\langle \tau^N, c \rangle = v(N) = \pi v(N)$. Since $\langle \tau^A, c \rangle \leq v(A)$ for all A , we deduce that for any $m \in \mathcal{M}(\tau)$, $\langle \tau, c \rangle = \sum_{A \in \mathcal{A}} m(A) \langle \tau^A, c \rangle \leq \sum_{A \in \mathcal{A}} m(A) v(A)$. Hence $\langle \tau, c \rangle \leq \pi v(\tau)$ for all fuzzy conditions.

11.2.3. Balanced family of multistrategy sets

We now give examples of “balanced” characteristic functions associated with strategic games $\{X(A), F^{A*}\}$. For this purpose, we require the following definition.

Definition 3. We shall say that a family $\{X(A)\}_{A \in \mathcal{A}}$ of multistrategy sets $X(A) \subset X^A$ is “balanced” if

$$(8) \quad \text{for any balance } m \in \mathcal{M}(\tau^N), \quad \sum_{A \in \mathcal{A}} m(A) X(A) \subset X(N).$$

Proposition 2. Suppose that the strategy sets X^i are convex. Then the family $\{X^A\}_{A \in \mathcal{A}}$ is “balanced”.

Proof. Let $m \in \mathcal{M}(\tau^N)$ be a balance and let x^A belong to X^A for all $A \in \mathcal{A}$. Therefore, the i th projection of $x^N = \sum_{A \in \mathcal{A}} m(A) x^A$ is equal to $x^{N,i} = \sum_{A \in \mathcal{A}; i \in A} m(A) x^{A,i}$. Since X^i is convex and $\sum_{A \in \mathcal{A}; i \in A} m(A) = 1$, we deduce that $x^{N,i} \in X^i$. Therefore, x^N belongs to X^N .

Example. Consider the subsets $X(A)$ defined by

$$(9) \quad X(A) = \{x^A \in R^A \text{ such that } \sum_{i \in A} L_i(x^A, i) \in \sum_{i \in A} Y(i)\}$$

where

$$(10) \quad \begin{cases} \text{(i) } Y(i) \subset V \text{ is a convex subset of resources allocated to player } i, \\ \text{(ii) } L_i \in \mathcal{L}(U^i, V) \text{ is a resource operator.} \end{cases}$$

Proposition 3. Suppose that (10) holds. Then the family $\{X(A)\}_{A \in \mathcal{A}}$ of multistrategy sets defined by (9) is balanced.

Proof. We consider multistrategies $x^A \in X(A)$ and check that, for and balance $m \in \mathcal{M}(\tau^N)$,

$$(11) \quad x^N = \sum_{A \in \mathcal{A}} m(A) x^A \quad \text{belongs to } X(N).$$

Since $x^{N,j} = \sum_{A \ni j} m(A) x^{A,j}$, we deduce that

$$\begin{aligned} \sum_{j \in N} L_j(x^{N,j}) &= \sum_{j \in N} \sum_{A \ni j} m(A) L_j(x^{A,j}) = \sum_{A \in \mathcal{A}} m(A) \sum_{j \in A} L_j(x^{A,j}) \\ &\in \sum_{A \in \mathcal{A}} m(A) \sum_{j \in A} Y(j) = \sum_{j \in N} \left(\sum_{A \ni j} m(A) \right) Y(j) = \sum_{j \in N} Y(j) = Y(N). \quad \square \end{aligned}$$

11.2.4. Balanced characteristic functions and convex loss functions

We are now able to state the main result of this chapter.

Theorem 2. (Scarf). *Consider the game with side-payments associated with a cooperative game $\{X(A), F^{A*}\}_{A \in \mathcal{A}}$ by the characteristic function v defined by*

$$(12) \quad v(A) = \inf_{x^A \in X(A)} \sum_{i \in A} f_i^{A*}(x^A)$$

Suppose that

$$(13) \quad \begin{cases} \text{(i) the strategy sets } X^i \text{ are convex,} \\ \text{(ii) the loss functions } f^i \text{ are convex} \end{cases}$$

and that

$$(14) \quad \text{the family } \{X(A)\}_{A \in \mathcal{A}} \text{ is balanced.}$$

Then the core of the game with side-payments is non-empty.

By Theorem 1, we have to prove that the characteristic function v defined by (12) is balanced. This fact will follow from the following lemmas.

Lemma 1. *Let m be a balance. Then for any pair of different players i and j , we have*

$$(15) \quad \sum_{\substack{B \ni j \\ B \not\ni i}} m(B) = \sum_{\substack{A \ni i \\ A \not\ni j}} m(A).$$

Proof.

$$\begin{aligned} 1 &= \sum_{A \ni i} m(A) = \sum_{\substack{A \ni i \\ A \not\ni j}} m(A) + \sum_{\substack{A \ni i \\ A \ni j}} m(A) \\ &= \sum_{B \ni j} m(B) = \sum_{\substack{B \ni j \\ B \not\ni i}} m(B) + \sum_{\substack{B \ni i \\ B \ni j}} m(B). \quad \square \end{aligned}$$

Lemma 2. Suppose that the strategy sets X^i are convex. Let m be a balance and $x^A \in X(A)$. Then the multistrategies $y(i)$ defined by

$$(16) \quad \forall j \neq i, \quad y^j(i) = \sum_{\substack{B \ni j \\ B \ni i}} m(B) x^{B,j} \Bigg/ \sum_{\substack{B \ni i \\ B \ni j}} m(B)$$

belong to $X^i = \prod_{j \neq i} X^j$.

Proof. By (15), we can write

$$y^j(i) = \sum_{\substack{B \ni j \\ B \ni i}} m(B) x^{B,j} \Bigg/ \sum_{\substack{B \ni j \\ B \ni i}} m(B),$$

which is a convex combination of elements $x^{B,j}$ of the convex subset X^j . $\square$

Lemma 3. Let m be a balance and let $\{x^A\}$ be a family of elements $x^A \in X(A)$. Let $y(i)$ be defined by (16).

Then

$$(17) \quad \forall i \in N, \quad \sum_{A \ni i} m(A) \{x^A, \pi^A y(i)\} = \sum_{A \in \mathcal{A}} m(A) x^A.$$

Proof. For any $j \in N, j \neq i$,

$$\begin{aligned} \sum_{A \ni i} m(A) \{x^A, \pi^A y(i)\}^j &= \sum_{\substack{A \ni i \\ A \ni j}} m(A) x^{A,j} + \sum_{\substack{A \ni i \\ A \ni j}} m(A) y^j(i) \\ &= \sum_{\substack{A \ni i \\ A \ni j}} m(A) x^{A,j} + \sum_{\substack{A \ni i \\ A \ni j}} m(A) \frac{\sum_{\substack{B \ni j \\ B \ni i}} m(B) x^{B,j}}{\sum_{\substack{B \ni i \\ B \ni j}} m(B)} \\ &= \sum_{\substack{A \ni i \\ A \ni j}} m(A) x^{A,j} + \sum_{\substack{B \ni j \\ B \ni i}} m(B) x^{B,j} = \sum_{A \ni j} m(A) x^{A,j}. \end{aligned}$$

If $j = i$, then $\sum_{A \ni i} m(A) \{x^A, \pi^A y(i)\}^i = \sum_{A \ni i} m(A) x^{A,i}$. $\square$

Lemma 4. Suppose that convexity assumptions (13) hold. For any balance $m \in \mathcal{M}(\tau^N)$ and for any family $\{x^A\}_{A \in \mathcal{A}}$, we have that

$$(18) \quad \sum_{i \in N} f_i \left(\sum_{A \in \mathcal{A}} m(A) x^A \right) \leq \sum_{A \in \mathcal{A}} m(A) f_i^{A*}(x^A).$$

Proof. Since we can write $\sum_{A \in \mathcal{A}} m(A) x^A = \sum_{A \ni i} m(A) \{x^A, \pi^A y(i)\}$, and since $\sum_{A \ni i} m(A) = 1$, we deduce from the convexity of f_i that

$$\begin{aligned} f_i \left(\sum_{A \in \mathcal{A}} m(A) x^A \right) &= f_i \left(\sum_{A \ni i} m(A) \{x^A, \pi^A y(i)\} \right) \\ &\leq \sum_{A \ni i} m(A) f_i(x^A, \pi^A y(i)) \leq \sum_{A \ni i} m(A) f_i^{A*}(x^A). \end{aligned}$$

Therefore,

$$\begin{aligned} \sum_{i \in N} f_i \left(\sum_{A \in \mathcal{A}} m(A) x^A \right) &\leq \sum_{i \in N} \sum_{A \ni i} m(A) f_i^{A*}(x^A) \\ &= \sum_{A \in \mathcal{A}} m(A) \sum_{i \in A} f_i^{A*}(x^A). \quad \square \end{aligned}$$

Proof of Theorem 2. We can associate with any $\varepsilon > 0$ and any balance $m \in \mathcal{N}(\tau^N)$ multistrategies $x^A \in X(A)$ such that

$$(19) \quad \sum_{i \in A} f_i^{A*}(x^A) \leq v(A) + \frac{\varepsilon}{\sum_{B \in \mathcal{A}} m(B)}.$$

Let $x^N = \sum_{A \in \mathcal{A}} m(A) x^A$. Then x^N belongs to $X(N)$ since the family $\{X(A)\}_{A \in \mathcal{A}}$ is balanced. Lemma 4 implies that

$$\begin{aligned} v(N) &\leq \sum_{i \in N} f_i \left(\sum_{A \in \mathcal{A}} m(A) x^A \right) \leq \sum_{A \in \mathcal{A}} m(A) \sum_{i \in A} f_i^{A*}(x^A) \\ &\leq \sum_{A \in \mathcal{A}} m(A) v(A) + \varepsilon. \end{aligned}$$

Hence $v(N) \leq \sum_{A \in \mathcal{A}} m(A) v(A)$ for all balances m , and thus, $v(N) \leq \pi v(\tau^N)$. $\square$

***Example.** Consider the case when $X(A)$ is defined by (9). If the subsets R^i and $Y(i)$ are convex and if the loss functions f_i are convex, Theorem 2 and Proposition 3 imply that the characteristic function v defined by

$$(20) \quad v(A) = \inf_{x^A \in X(A)} \sum_{i \in A} f_i^{A*}(x^A).$$

is balanced. It is also clear that the characteristic functions $w(\cdot, p)$ defined by

$$(21) \quad w(A; p) = \inf_{x^A \in R^A} \left(\sum_{i \in A} [f_i^{A*}(x^A) + \langle p, L_i x^A \rangle - r_i(p)] \right)$$

are balanced.

Proposition 4. Suppose that

$$(22) \quad \begin{cases} \text{(i) the subsets } R^i \subset U^i \text{ and } Y(i) \subset V \text{ are convex,} \\ \text{(ii) the operators } L_i \in \mathcal{L}(U^i, V) \text{ are linear,} \\ \text{(iii) the loss functions } f_i \text{ are convex.} \end{cases}$$

For any p , the core of the game associated with the characteristic function $w(\cdot, p)$ defined by (21) is non-empty.

Proof. Let $\varepsilon > 0$ and $m \in \mathcal{N}(\tau^N)$ be fixed. There exist multistrategies $x^A \in R^A$ such that

$$(23) \quad \sum_{i \in A} (f_i^A(x^A) + \langle p, L_i x^{A,i} \rangle - r_i(p)) \leq w(A, p) + \frac{\varepsilon}{\sum_{B \in \mathcal{A}} m(B)}.$$

Let $x^N = \sum_{A \in \mathcal{A}} m(A) x^A$ belong to R^N (by Proposition 2).

Since $\sum_{A \in \mathcal{A}} m(A) \sum_{i \in \mathcal{A}} L_i x^{A,i} = \sum_{i \in N} L_i (\sum_{A \ni i} m(A) x^A) = \sum_{i \in N} L_i x^{N,i}$ and since $\sum_{A \in \mathcal{A}} m(A) \sum_{i \in A} r_i(p) = \sum_{i \in N} (\sum_{A \ni i} m(A) r_i(p)) = \sum_{i \in N} r_i(p) = r(p)$,

we deduce from Lemma 4 that

$$\begin{aligned} w(N, p) &\leq \sum_{i \in N} (f_i(x^N) + \langle p, L_i x^{N,i} \rangle - r_i(p)) \leq \\ &\leq \sum_{A \in \mathcal{A}} m(A) \left(\sum_{i \in A} f_i^A(x^A) + \langle p, L_i x^{A,i} \rangle - r_i(p) \right) \leq w(A, p) + \varepsilon. \end{aligned}$$

Hence the characteristic functions $w(\cdot, p)$ are balanced. $\square$

*11.2.5. Further properties of convex functions and balances

We can extend Lemma 3 to positive sequences $m \in \mathcal{N}(\tau)$.

Lemma 5. Let $m \in \mathcal{N}(\tau)$ and $x^A \in X^A$ be given. For any $i \in A_\tau$, we can find $y(i) \in \prod_{j \in A_\tau; j \neq i} X^j$ such that

$$(24) \quad \forall j \in A_\tau, \quad \sum_{A \ni j} \frac{m(A) x^{A,j}}{\tau_j} = \sum_{A \ni i} \frac{m(A)}{\tau_i} \{x^A, \pi^A y(i)\}^j.$$

Proof. This is trivial if $j = i$.

If $j \neq i$, we can write,

$$\sum_{A \ni i} \frac{m(A)}{\tau_i} \{x^A, \pi^A y(i)\}^j = \sum_{A \supset \{i, j\}} \frac{m(A)}{\tau_i} x^{A,j} + \left[\sum_{\substack{A \ni i \\ A \not\ni j}} \frac{m(A)}{\tau_i} \right] y(i)^j.$$

We define $y(i)$ by

$$(25) \quad y(i)^j = \frac{\sum_{B \supset \{i, j\}} \left(\frac{1}{\tau_j} - \frac{1}{\tau_i} \right) m(B) x^{B,j} + \sum_{\substack{B \ni j \\ B \not\ni i}} \frac{m(B)}{\tau_j} x^{B,j}}{\sum_{B \supset \{i, j\}} \left(\frac{1}{\tau_j} - \frac{1}{\tau_i} \right) m(B) + \sum_{\substack{B \ni j \\ B \not\ni i}} \frac{m(B)}{\tau_j}}$$

Since $\tau_i = \sum_{A \supset \{i, j\}} m(A) + \sum_{A \ni i; A \not\ni j} m(A)$ and $\tau_j = \sum_{A \supset \{i, j\}} m(A) + \sum_{A \ni j; A \not\ni i} m(A)$, we deduce that

$$(26) \quad \sum_{\substack{A \ni i \\ A \not\ni j}} \frac{m(A)}{\tau_i} = \sum_{A \supset \{i, j\}} m(A) \left(\frac{1}{\tau_j} - \frac{1}{\tau_i} \right) + \sum_{\substack{A \ni j \\ A \not\ni i}} \frac{m(A)}{\tau_j}.$$

Therefore,

$$(27) \quad \begin{aligned} \sum_{\substack{A \ni i \\ A \not\ni j}} \frac{m(A)}{\tau_i} \{x^A, \pi^A y(i)\} &= \sum_{A \supset \{i, j\}} m(A) x^{A, j} \left(\frac{1}{\tau_i} + \frac{1}{\tau_j} - \frac{1}{\tau_i} \right) + \\ &+ \sum_{\substack{A \ni j \\ A \not\ni i}} \frac{m(A)}{\tau_j} x^{B, j} = \sum_{A \ni j} \frac{m(A) x^{A, j}}{\tau_j}. \quad \square \end{aligned}$$

Consider a game $\{X(A), F^{A*}\}_A$ and its extension $\{X_0(\tau), F^{\tau*}\}_{\tau \in \mathcal{T}}$ (see Definition 10.4.6).

Proposition 5. *Suppose that convexity assumptions (13) hold. Then, for any $x^\tau \in X_0(\tau)$, $\forall \lambda \in \mathbf{R}_+^{A^\tau}$,*

$$(28) \quad \sum_{i=1}^n \lambda^i \tau_i f_i(x^\tau) \leq \sigma^b(F^{\tau*}(x^\tau); \lambda),$$

where $\sigma^b(F^{\tau*}(x^\tau); \lambda)$ is the lower support function of $F^{\tau*}(x^\tau)$

Proof. Write $x^\tau = \sum_{A \in \mathcal{A}} (m(A) x^A) / \tau$, where $m \in \mathcal{M}(\tau)$ and $x^A \in X(A)$ for all $A \in \mathcal{A}$.

By Lemma 5, there exists $y(i)$ such that

$$(29) \quad x^\tau = \sum_{A \ni i} \frac{m(A)}{\tau_i} \{x^A, \pi^A y(i)\}.$$

Hence, the convexity of the loss function f_i implies

$$(30) \quad \begin{aligned} f_i(x^\tau) &= f_i \left(\sum_{A \ni i} \frac{m(A)}{\tau_i} \{x^A, \pi^A y(i)\} \right) \\ &\leq \sum_{A \ni i} \frac{m(A)}{\tau_i} f_i(\{x^A, \pi^A y(i)\}) \leq \sum_{A \ni i} \frac{m(A)}{\tau_i} f_i^{A*}(x^A). \end{aligned}$$

Therefore, for all representations of $x^\tau = \sum_{A \in \mathcal{A}} (m(A) x^A) / \tau$, we have that

$$\begin{aligned} \sum_{i=1}^n \lambda^i \tau_i f_i(x^\tau) &\leq \sum_{i=1}^n \lambda^i \sum_{A \ni i} m(A) f_i^A(x^A) \\ &= \langle \lambda, \sum_{A \in \mathcal{A}} m(A) F^{A*}(x^A) \rangle. \end{aligned}$$

Thus, by taking the infimum, we obtain

$$\sum_{i=1}^n \lambda^i \tau_i f_i(x) \leq \sigma^b(F^{\tau*}(x^{\tau}); \lambda). \quad \square$$

We also mention the following property of balances.

Proposition 6. *Let $c \in \mathbf{R}^n$ and $\tau \in \mathcal{T}$. We can write, for any $m \in \mathcal{M}(\tau)$,*

$$(31) \quad \frac{\sum_{i \in N} \tau_i c_i}{\sum_{i \in N} \tau_i} = \sum_{A \in \mathcal{A}} \frac{m(A) |A|}{\sum_{i \in N} \tau_i} \cdot \left(\frac{1}{|A|} \sum_{i \in A} c_i \right).$$

Proof. Since $\tau_i = \sum_{A \ni i} m(A)$, for all $i \in N$, we deduce that

$$\sum_{i \in N} \tau_i = \sum_{i \in N} \sum_{A \ni i} m(A) = \sum_{A \in \mathcal{A}} \sum_{i \in A} m(A) = \sum_{A \in \mathcal{A}} m(A) |A|.$$

Therefore,

$$\begin{aligned} \sum_{A \in \mathcal{A}} \frac{m(A)}{\sum_{i \in N} \tau_i} \sum_{i \in A} c_i &= \frac{1}{\sum_{i \in N} \tau_i} \sum_{A \in \mathcal{A}} m(A) \sum_{i \in A} c_i \\ &= \frac{1}{\sum_{i \in N} \tau_i} \sum_{i \in N} c_i \sum_{A \ni i} m(A) = \frac{1}{\sum_{i \in N} \tau_i} \tau_i c_i. \end{aligned}$$

Interpretation. For any fuzzy coalition τ and $m \in \mathcal{M}(\tau)$, we can write the barycenter

$$\frac{1}{\sum_{i \in N} \tau_i} \sum_{i \in N} \tau_i c_i$$

as a barycenter of the $|\mathcal{A}|$ barycenters $(1/|A|) \sum_{i \in A} c_i$.

11.3. Values of fuzzy games

We define another solution concept for a game with side-payments different from the core. The concept involves associating a multiloss $c \in V(N)$ satisfying “a priori” axioms with any characteristic function $v: \mathcal{T} \rightarrow \mathbf{R}$. These axioms are apparently quite weak and intuitively reasonable. We shall see that for fuzzy games, the new solution concept and the core coincide when they are both defined.

11.3.1. The diagonal property

Definition 1. We denote by V^n the vector space of functions $v : [0, 1]^n \rightarrow \mathbb{R}$ vanishing at 0 and continuously differentiable on the diagonal $\{t\tau^N\} 0 \leq t \leq 1$ of $[0, 1]^n$.

We supply V^n with the scalar product

$$(1) \quad \begin{aligned} ((v, w))_n &= \int_0^1 \langle Dv(t\tau^N), Dw(t\tau^N) \rangle dt \\ &= \sum_{i=1}^n \int_0^1 \frac{\partial}{\partial \tau_i} v(t, \dots, t) \frac{\partial}{\partial \tau_i} w(t, \dots, t) dt \end{aligned}$$

for which V^n is a *pre-Hilbert space*.

Remark. Notice that if the function $v \in V^n$ is positively homogeneous, then

$$(2) \quad Dv(t\tau^N) = Dv(\tau^N)$$

since

$$\begin{aligned} \langle Dv(t\tau^N), \tau \rangle &= \lim_{\theta \rightarrow 0} \frac{v(t\tau^N + \theta\tau) - v(t\tau^N)}{\theta} = \\ &= \lim_{\theta \rightarrow 0} \frac{v\left(\tau^N + \frac{\theta}{t}\tau\right) - v(\tau^N)}{\frac{\theta}{t}} = \langle Dv(\tau^N), \tau \rangle. \end{aligned}$$

In particular, if v and w are positively homogeneous,

$$(3) \quad ((v, w))_n = \langle Dv(\tau^N), Dw(\tau^N) \rangle.$$

Interpretation. The scalar product $((v, w))_n$ takes into account the following “diagonal property”. For a given collective rate of participation $t \in [0, 1]$, we consider the diagonal fuzzy coalition $t\tau^N = \{t, \dots, t\}$ in which each player participates equally. The i th partial derivative $\partial v(t\tau^N)/\partial \tau_i$ measures the marginal loss imposed on player i when he leaves the diagonal fuzzy coalition $t\tau^N$. In other words, $Dv(t\tau^N)$ can be regarded as the “marginal multiloss” imposed on the players when they want to modify a diagonal coalition. In the case where v is positively homogeneous, we take into account only the marginal multiloss for the whole set of players τ^N .

11.3.2. Sequence of fuzzy values

The aim is to define maps φ_n associating with any characteristic function $v \in V^n$ a multiloss $\varphi_n v \in \mathbb{R}^n$ satisfying the following axioms.

Pareto optimality. This requires that

$$(4) \quad \forall n, \forall v \in V^n, \quad \sum_{i \in N} (\varphi_n v)_i = v(\tau^N).$$

Symmetry axiom. Let θ be any permutation of the set N of n players. We define

$$(5) \quad \begin{cases} (i) (\theta * v)(\tau_1, \dots, \tau_n) = v(\tau_{\theta^{-1}(1)}, \dots, \tau_{\theta^{-1}(n)}), \\ (ii) (\theta * c)_i = c_{\theta(i)} \text{ for all } i \in N. \end{cases}$$

The symmetry axiom requires that

$$(6) \quad \forall n, \forall v \in V^n, \forall \text{ permutation } \theta, \quad \varphi_n(\theta * v) = \theta * \varphi_n v.$$

In other words, the symmetry axiom requires that the value does not depend on "how the players are named". Or, equivalently, that the value does not depend upon the order in which players play.

Atomicity axiom. This requires that the loss imposed on a coalition of players in a game between coalitions is the sum of losses imposed on each player of this coalition in the initial game.

Namely, let $\mathcal{D} = \{A_1, \dots, A_m\}$ be a partition of the set N of players in m non-empty "types" A_j of players.

We assume that each fuzzy coalition $\sigma = \{\sigma_1, \dots, \sigma_m\}$ of types of players induces a fuzzy coalition $\tau = \mathcal{D} \cdot \sigma = \{\tau_1, \dots, \tau_n\}$ of players where the rate of participation τ_i of the i^{th} player is equal to the rate of participation σ_j of the type A_j in which he belongs, i.e.

$$(7) \quad \tau_i = \sigma_j \text{ whenever } i \in A_j.$$

This implies that we can associate with any game $(\mathcal{C}, v)$ of n players and any partition $\mathcal{D}$ in m types of players a m -person game defined by the characteristic function $\mathcal{D} * v$

$$(8) \quad (\mathcal{D} * v)(\sigma_1, \dots, \sigma_m) = v(\mathcal{D} \cdot \sigma) (= v(\tau_1, \dots, \tau_n)) \\ \text{where } \tau_i = \sigma_j \text{ when } i \in A_j.$$

The atomicity axiom states that

$$(9) \quad \forall n, \forall \text{ partition } \mathcal{D} \text{ in } m \text{ types of players, } \forall v \in V^n, \text{ then} \\ \forall j = 1, \dots, m, \quad \varphi_m(\mathcal{D} * v)_j = \sum_{i \in A_j} (\varphi_n v)_i.$$

Definition 2. We shall say that a sequence $\{\varphi_n\}_n$ of maps from V^n into $\mathbf{R}^n$ is a "sequence of fuzzy values" if the maps φ_n are continuous linear operators which satisfy the Pareto optimality, symmetry and atomicity axioms.

11.3.3. Existence and uniqueness of a sequence of fuzzy values

Theorem 1. The sequence of continuous linear operators $\varphi_n \in \mathcal{L}(V^n, \mathbf{R}^n)$ defined by

$$(10) \quad \forall v \in V^n, \quad \varphi_n v = \int_0^1 Dv(t\tau^N) dt \in \mathbf{R}^n$$

is the unique sequence of fuzzy values.

For any linear operator $A \in \mathcal{L}(\mathbf{R}^m, \mathbf{R}^n)$ satisfying $A\tau^M = \tau^N$, we obtain

$$(11) \quad \forall v \in V^n, \quad \varphi_m(v \circ A) = A^* \varphi_n v.$$

If we assume that $v \in V^n$ is convex and positively homogeneous, then the value $\varphi_n v$ is the unique element of the core of the fuzzy game $(\mathcal{T}, v)$.

Interpretation. Formula (10) shows that $\varphi_n v$ is the average marginal multiloss imposed when players leave diagonal fuzzy coalitions.

Proof of existence. It is obvious that φ_n is a continuous linear operator. Since

$$\frac{d}{dt} v(t\tau^N) = \sum_{i=1}^n \frac{\partial}{\partial \tau_i} v(t\tau^N),$$

we obtain by integrating this equality from 0 to 1 that

$$\begin{aligned} \sum_{i=1}^n (\varphi_n \tau)_i &= \sum_{i=1}^n \int_0^1 \frac{\partial}{\partial \tau_i} v(t\tau^N) dt = \int_0^1 \frac{d}{dt} v(t\tau^N) dt = \\ &= v(\tau^N) - v(0) = v(\tau^N). \end{aligned}$$

Hence the Pareto optimality axiom holds. To prove property (11), we recall that $D(v \circ A)(\tau) = A^* Dv(A\tau)$ when $A \in \mathcal{L}(\mathbf{R}^m, \mathbf{R}^n)$. Since $Dv(tA\tau^M) = Dv(t\tau^N)$, we obtain

$$\varphi_m(v \circ A) = \int_0^1 D(v \circ A)(t\tau^M) dt = \int_0^1 A^* Dv(t\tau^N) dt = A^* \varphi_n v.$$

In particular, we may take $m = n$ and associate with any permutation θ of N the matrix $A = (a_{ij}^t)$ where $a_{ij}^t = 1$ if $j = \theta(i)$ and 0 if $j \neq \theta(i)$. We then obtain the symmetry axiom, since $(v \circ A)(\tau) = (\sigma * \theta)(\tau)$ and $A^* c = \theta * c$.

In the same way, by associating with a partition $\mathcal{P}$ of N in m types A_j the matrix $\mathcal{P} \in \mathcal{L}(\mathbf{R}^m, \mathbf{R}^n)$ defined by (8), we have $v \circ \mathcal{P} = \mathcal{P} * v$ and $(\mathcal{P} * c)_j = \sum_{i \in A_j} c_i$. Therefore the atomicity axiom follows from (11) with $A = \mathcal{P}$.

Proof of uniqueness. To prove the uniqueness of the sequence of operators φ_n , we use the fact that, for each n , the set of polynomials (in n variables) is dense in V^n (by the Stone–Weierstrass theorem). Therefore, since φ_n is assumed to be linear and continuous, it is sufficient to prove that any sequence $\{\varphi_n\}_n$ of fuzzy values associates with polynomials $v_k : \tau \rightarrow v_k(\tau) = \tau_1^{k_1} \dots \tau_n^{k_n}$ the same multiloss $\varphi_n v_k$.

For this purpose, we check that the Pareto optimality and symmetry axioms imply that

$$(12) \quad \varphi_n v = \frac{v(\tau^N)}{n} \tau^N \quad \text{whenever } v \text{ is symmetric.}$$

[If v is symmetric, then $\varphi_n v = \varphi_n(\theta * v) = \theta * \varphi_n v$ for any permutation θ of N . Such a vector $\varphi_n v$ satisfies $\varphi_n v = a \tau^N = \{a, \dots, a\}$. By the Pareto optimality axiom, we deduce that $an = \sum_{i=1}^n (\varphi_n v)_i = v(\tau^N)$. Therefore (12) holds.]

We now use the atomicity axiom in the following way. We regard $v_k(\tau) = \tau_1^{k_1} \dots \tau_n^{k_n}$ as $v_k = \mathcal{P} * v_{|k|}$, where $|k| = k_1 + \dots + k_n$, $\mathcal{P}$ denotes the partition of $|k|$ players into k_1 players, k_2 players, $\dots$, k_n players and where $v_{|k|}(\sigma_1, \dots, \sigma_{|k|}) = \sigma_1 \sigma_2 \dots \sigma_{|k|}$.

Since $v_{|k|}$ is symmetric, then $(\varphi_{|k|} v_{|k|})_j = 1/|k|$. Therefore,

$$(13) \quad (\varphi_n v_k)_i = \varphi(\mathcal{P} * v_{|k|})_i = \sum_{j \in A_i} (\varphi_{|k|} v_{|k|})_j = \frac{k_i}{|k|}.$$

We have proved that any sequence $\{\varphi_n\}$ of fuzzy values satisfies

$$(14) \quad (\varphi_n v_k)_i = \frac{k_i}{k_1 + \dots + k_n}.$$

(We can check that formula (10) yields the same result!) This proves the uniqueness.

Proof of equivalence with the core. When $v \in V^n$ and v is positively homogeneous and convex, we have $\partial v(\tau^N) = \{Dv(\tau^N)\}$. But $\partial v(\tau^N) = \mathcal{C}(\mathcal{T}, v)$ by Proposition 1.3 and

$$Dv(\tau^N) = \int_0^1 Dv(\tau^N) dt = \int_0^1 Dv(t\tau^N) dt = \varphi_n v$$

since $v \in V^n$ and is positively homogeneous.

Therefore $\mathcal{C}(\mathcal{T}, v) = \{\varphi_n v\}$.

11.3.4. Relations between core and fuzzy value

We saw that the concepts of core and fuzzy value for fuzzy games coincide when they are both defined. We can even relate these concepts when v is only the supremum of a family of differentiable functions, by reformulating Proposition 1.4 in the following way.

Proposition 1. *Suppose that the characteristic function v is defined by*

$$(15) \quad \forall \tau \in \mathbb{R}_+^n, \quad v(\tau) = \sup_{p \in P} w(\tau, p),$$

where the functions w satisfy assumption (21) of Proposition 1.4. Then the core $\mathcal{C}(\mathcal{T}, v)$ is the closed convex hull of the fuzzy values $\varphi_n w(\cdot, p)$ as p ranges over

$$P(N) = \{p \in P \text{ such that } v(\tau^N) = w(\tau^N, p)\}.$$

*11.3.5. Best approximation property of fuzzy values

We can also characterize fuzzy values by best approximation properties.

Finding a “solution” $c \in \mathbb{R}^n$ to a game $(\mathcal{T}, v)$ amounts to replacing the characteristic function $v : \tau \in \mathcal{T} \mapsto v(\tau)$ by a linear characteristic function $c : \tau \in \mathcal{T} \rightarrow \langle \tau, c \rangle$. The question arises as to whether it is reasonable to propose the best approximation to $v \in V^N$ by linear characteristic functions $c \in \mathbb{R}^n$ as a solution concept.

Proposition 2. *The fuzzy value $\varphi_n v$ of a characteristic function $v \in V^N$ is the best approximation to v by linear characteristic functions.*

Proof. Such a best approximation exists since $((v, w))_n$ induces on $\mathbb{R}^n$ the scalar product $((c, d))_n = \sum_{i=1}^n c_i d_i$.

The best approximation c to v is defined by $((v - c, d)) = 0$, for all $d \in \mathbb{R}^n$. Since c and d are linear functions, this can be written

$$\sum_{i=1}^n \left(\int_0^1 \frac{\partial}{\partial \tau_i} v(t\tau^N) dt - c_i \right) d_i = 0 \quad \text{for all } d \in \mathbb{R}^n.$$

Hence

$$c_i = \int_0^1 \frac{\partial}{\partial \tau_i} v(t\tau^N) dt = (\varphi_n v)_i \quad \text{for all } i. \quad \square$$

Remark. We also mention some other properties satisfied by fuzzy values.

Dummy property. We say that player i is a “dummy” or a “null player” in a fuzzy game $(\mathcal{T}, v)$ if the characteristic function v does not depend on τ_i (i.e. does not depend upon the participation of player i). Then the following “dummy property”

- (16) If i is a dummy player in $(\mathcal{T}, v)$, then $(\varphi_n v)_i = 0$
is satisfied by the fuzzy value.

Positivity property. Consider a non-decreasing characteristic function v . Since the loss becomes greater when a fuzzy coalition increases, it is reasonable that the marginal multiloss is non-negative, i.e. the fuzzy values: satisfies

- (17) If v is non decreasing, then $\varphi_n v \geq 0$.

*11.3.6. Convex values of fuzzy games

The question arises as to whether we can extend the concept of fuzzy value to games whose characteristic function is no longer differentiable on the diagonal, but only differentiable from the right.

For this purpose, instead of replacing a characteristic function $\tau \in \mathcal{T} \mapsto v(\tau)$ by a linear characteristic function, we shall only look for replacements of v by convex positively homogeneous functions $\tilde{\varphi}_n v : \tau \in \mathcal{T} \mapsto (\tilde{\varphi}_n v)(\tau)$. We can prove the following result.

Proposition 3. Let $\tilde{V}^n$ be the cone of characteristic functions v defined on $[0, 1]^n$ vanishing at 0 and differentiable from the right on the diagonal $\{\tau \tau^N\}_{t \in [0, 1]}$:

The maps $\tilde{\varphi}_n$ from $\tilde{V}^n$ into the cone of convex, positively homogeneous functions defined by

$$(18) \quad (\tilde{\varphi}_n v)(\tau) = \int_0^1 Dv(t\tau^N)(\tau) dt$$

are additive, positively homogeneous and satisfy the Pareto optimality axiom

$$(19) \quad (\tilde{\varphi}_n v)(\tau^N) = v(\tau^N).$$

Furthermore, for any operator $A \in \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ satisfying $A\tau^M = \tau^N$, they satisfy

$$(20) \quad \tilde{\varphi}_n(v \circ A) = (\varphi_n v) \circ A.$$

If $v \in \tilde{V}^n$ is positively homogeneous and convex, the function $\tau \mapsto (\tilde{\varphi}_n v)(\tau)$ is the upper support function of the core $\mathcal{C}(\mathcal{T}, v)$.

Proof. This is left as an exercise. $\square$

11.4. Shapley value and nucleolus of games with side-payments

Let $\mathcal{A} = \mathcal{D}(N) - \emptyset$ be the set of all non-empty coalitions. Since we can extend any game $(\mathcal{A}, v)$ into a fuzzy game $(\mathcal{T}, \pi v)$ via the (non-linear) map π defined by (2.3), we can define the *fuzzy value* $\varphi_n \pi v$ of the game $(\mathcal{A}, v)$ whenever the extended function πv is differentiable. This is the first drawback, since there exist functions $v \in \mathbf{R}^{\mathcal{A}}$ such that πv is not differentiable. Secondly, $\varphi_n \pi$ is not linear. The problem arises as to whether it is possible to find *linear* operators satisfying three analogous axioms (Pareto optimality, symmetry and dummy axioms). There exists such a unique operator ψ_n , called the “Shapley value”. It is related to the fuzzy value φ_n by the formula $\psi_n = \varphi_n \omega$ where ω is the Cornet extension operator defined by $(\omega v)(\tau) = \sum_{A \in \mathcal{A}} \alpha_A(v) \left(\prod_{i \in A} \tau_i \right)^{1/|A|}$ where $\alpha_A(v) = \sum_{B \subset A} (-1)^{|A|-|B|} v(B)$. It can also be written $\psi_n = \varphi_n \chi$ where χ is the Owen extension operator defined by

$$(\chi v)(\tau) = \sum_{A \in \mathcal{A}} v(A) \prod_{i \in A} \tau_i \prod_{j \notin A} (1 - \tau_j).$$

We end this section by proving that the nucleolus of a game with side-payments consists of a unique element.

11.4.1. The Shapley value

Let $\psi_n : \mathbf{R}^{\mathcal{A}} \rightarrow \mathbf{R}^n$ be a map associating with any characteristic function $v \in \mathbf{R}^{\mathcal{A}}$ a multiloss $\psi_n v \in \mathbf{R}^n$. We shall say that ψ_n satisfies

the Pareto optimality axiom if

$$(1) \quad \forall v \in \mathbf{R}^{\mathcal{A}}, \quad \sum_{i=1}^n (\psi_n v)_i = v(N),$$

the Symmetry axiom if $\forall v \in \mathbf{R}^{\mathcal{A}}$, for any permutation θ of N ,

$$(2) \quad \psi_n(\theta * v) = \theta * \psi_n v$$

where $\theta * v$ is defined by $(\theta * v)(A) = v[\theta(A)]$ and

the Dummy axiom if

$$(3) \quad \forall v \in \mathbf{R}^{\mathcal{A}}, \quad (\psi_n v)_i = 0 \text{ whenever } i \text{ is a dummy player of } v$$

where we define i to be a “dummy player” of v if

$$(4) \quad \forall A \in \mathcal{A}, \quad v(A \cup \{i\}) = v(A).$$

Definition 1. We shall say that a map ψ_n is a “Shapley value” if it is a linear operator from $\mathbf{R}^{\mathcal{A}}$ into $\mathbf{R}^n$ which satisfies the Pareto optimality, symmetry and dummy axioms.

We shall prove that such an operator ψ_n exists and is unique.

11.4.2. Existence and uniqueness of a Shapley value

Theorem 1. (Shapley). *There exists a unique linear operator $\psi_n \in \mathcal{L}(\mathbf{R}^{\mathcal{A}}, \mathbf{R}^n)$ satisfying the Pareto optimality, symmetry and dummy axioms. This operator can be written $\psi_n = \varphi_n \omega$ where $\omega \in \mathcal{L}(\mathbf{R}^{\mathcal{A}}, V^n)$ is the linear extension operator defined by*

$$(5) \quad \omega v(\tau) = \sum_{A \in \mathcal{A}} \alpha_A(v) \left(\prod_{i \in A} \tau_i \right)^{1/|A|}$$

where $|A| = \text{card}(A)$ denotes the number of elements of A and

$$(6) \quad \alpha_A(v) = \sum_{B \subset A} (-1)^{|A|-|B|} v(B).$$

The following formulas hold

$$(7) \quad \begin{aligned} (\psi_n v)_i &= \sum_{A \ni i} \frac{(|A|-1)!(n-|A|)!}{n!} (v(A) - v(A - \{i\})) \\ &= \frac{1}{n!} \sum_{\theta} (v[A_{\theta}(i)] - v[A_{\theta}^0(i)]) \end{aligned}$$

where θ ranges over the set of permutations and where

$$(8) \quad \begin{cases} \text{(i) } A_{\theta}(i) = \{j \in N \text{ such that } \theta(j) \leq \theta(i)\}, \\ \text{(ii) } A_{\theta}^0(i) = \{j \in N \text{ such that } \theta(j) < \theta(i)\}. \end{cases}$$

Definition 2. We shall say that ψ_n is the “Shapley value” and that ω is the “Cornet extension operator”.

Proof. The proof of the Shapley theorem consists of the five following lemmas.

Lemma 1. The $2^n - 1$ characteristic functions μ_A defined by

$$(9) \quad \mu_A(B) = \begin{cases} 1 & \text{if } B \supset A, \\ 0 & \text{otherwise.} \end{cases}$$

form a basis of the $(2^n - 1)$ dimensional vector space $\mathbf{R}^{\mathcal{A}}$. Any function v can be written

$$(10) \quad v = \sum_{A \in \mathcal{A}} \alpha_A(v) \mu_A \quad \text{where } \alpha_A(v) = \sum_{B \subset A} (-1)^{|A|-|B|} v(B).$$

Proof. The functions μ_A are linearly independent. [If not, there exist coefficients $\alpha_A \neq 0$ such that $\sum \alpha_A \mu_A = 0$. Let A_0 be a coalition with minimum cardinality among the coalitions A such that $\alpha_A \neq 0$. Then we can write

$$(11) \quad \mu_{A_0} = \sum_{A \not\subseteq A_0} \beta_A \mu_A$$

(since if $A \subseteq A_0$, we should have $|A| < |A_0|$). Since $A \not\subseteq A_0$, we have $\mu_A(A_0) = 0$. On the other hand, $\mu_{A_0}(A_0) = +1$. Therefore (11) implies that $1 = 0$. This is impossible.]

We prove (10). It follows from (9) that, for any $C \in \mathcal{A}$,

$$\begin{aligned} \sum_{A \in \mathcal{A}} \alpha_A(v) \mu_A(C) &= \sum_{A \subseteq C} \alpha_A(v) = \sum_{A \subseteq C} \sum_{B \subseteq A} (-1)^{|A|-|B|} v(B) = \\ &= \sum_{B \subseteq C} \left(\sum_{B \subseteq A \subseteq C} (-1)^{|A|-|B|} \right) v(B). \end{aligned}$$

Since there are $\binom{|C|-|B|}{|A|-|B|}$ coalitions A between B and C having a given cardinality $|A|$, we deduce that

$$\begin{aligned} \sum_{B \subseteq A \subseteq C} (-1)^{|A|-|B|} &= \sum_{|B| \leq |A| \leq |C|} \binom{|C|-|B|}{|A|-|B|} (-1)^{|A|-|B|} \\ &= \sum_{0 \leq i \leq |C|-|B|} \binom{|C|-|B|}{i} (-1)^i = (1-1)^{|C|-|B|} \\ &= \begin{cases} 0 & \text{if } |B| < |C|, \\ 1 & \text{if } |B| = |C|. \end{cases} \end{aligned}$$

Therefore, the two above equalities imply that

$$\sum_{A \in \mathcal{A}} \alpha_A(v) \mu_A(C) = v(C). \quad \square$$

Lemma 2. For any coalition $A \in \mathcal{A}$, any "Shapley value" satisfies

$$(12) \quad (\psi_n \mu_A)_i = \begin{cases} \frac{1}{|A|} & \text{if } i \in A, \\ 0 & \text{if } i \notin A. \end{cases}$$

Proof. If $i \notin A$, i is a dummy player of μ_A . (Because, if $B \supset A$, then $B \cup \{i\} \supset A$ and $\mu_A(B) = \mu_A(B \cup \{i\}) = 1$. If $B \not\supset A$, $B \cup \{i\} \not\supset A$ since $i \notin A$. Thus $\mu_A(B) = \mu_A(B \cup \{i\}) = 0$.)

Therefore, the dummy axiom implies that $(\psi_n \mu_A)_i = 0$ whenever $i \notin A$.

Let i and j belong to the coalition A and let θ be the permutation which exchanges i and j and keeps the other players invariant. Then $\theta * \mu_A = \mu_A$ since $\theta(A) = A$ and the symmetry axiom implies that

$$(\psi_n \mu_A)_i = \psi_n(\theta * \mu_A)_i = (\theta * \psi_n \mu_A)_i = (\psi_n \mu_A)_j.$$

Finally, the Pareto optimality axiom implies that

$$\sum_{i \in N} (\psi_n \mu_A)_i = \sum_{i \in A} (\psi_n \mu_A)_i = \mu_A(N) = 1.$$

Since $(\psi_n \mu_A)_i$ does not depend on i , we obtain $(\psi_n \mu_A) = 1/|A|$. Therefore, since ψ_n is assumed to be linear, there exists a unique Shapley value ψ_n defined by

$$\begin{aligned} (\psi_n v)_i &= \sum_{A \in \mathcal{A}} \alpha_A(v) (\psi_n \mu_A)_i = \sum_{A \ni i} \frac{\alpha_A(v)}{|A|} = \\ (13) \quad &= \sum_{A \ni i} \frac{1}{|A|} \sum_{B \subset A} (-1)^{|A|-|B|} v(B). \quad \square \end{aligned}$$

Lemma 3. *The Shapley value ψ_n can be written $\psi_n = \varphi_n \omega$ where ω is defined by (5) and (6).*

Proof. We introduce the functions $\beta_A : \tau \rightarrow \beta_A(\tau) = (\prod_{i \in A} \tau_i)^{1/|A|}$:

$$(14) \quad \varphi_n \omega v = \varphi_n \left(\sum_{A \in \mathcal{A}} \alpha_A(v) \beta_A \right) = \sum_{A \in \mathcal{A}} \alpha_A(v) \varphi_n \beta_A$$

Since β_A is a positively homogeneous (concave) function differentiable at τ^N , we deduce that

$$(15) \quad (\varphi_n \beta_A)_i = \frac{\partial}{\partial \tau_i} \beta_A(\tau^N) = \begin{cases} \frac{1}{|A|} & \text{if } i \in A \\ 0 & \text{if } i \notin A \end{cases} = (\psi_n \mu_A)_i.$$

Therefore, for any player i ,

$$(16) \quad (\varphi_n \omega v)_i = (\psi_n v)_i. \quad \square$$

Lemma 4. *We can write*

$$(17) \quad (\psi_n v)_i = \sum_{A \ni i} \frac{(|A|-1)! (n-|A|)!}{n!} (v(A) - v(A - \{i\})).$$

Proof. We deduce from (13) that

$$\begin{aligned} (\psi_n v)_i &= \sum_{A \ni i} \frac{1}{|A|} \sum_{B \subset A} (-1)^{|A|-|B|} v(B) = \\ &= \sum_{B \in \mathcal{A}} v(B) \left(\sum_{A \supset B \cup \{i\}} \frac{(-1)^{|A|-|B|}}{|A|} \right). \end{aligned}$$

Write

$$(18) \quad \gamma_i(B) = \sum_{A \supset B \cup \{i\}} \frac{(-1)^{|A|-|B|}}{|A|}.$$

If B does not contain i , then $C = B \cup \{i\}$ does contain i . It is easily checked that

$$(19) \quad \begin{aligned} \gamma_i(B) &= \sum_{A \supset C} \frac{(-1)^{|A|-|C|+1}}{|A|} = - \sum_{A \supset C = B \cup \{i\}} \frac{(-1)^{|A|-|C|}}{|A|} \\ &= -\gamma_i(C) = -\gamma_i(B \cup \{i\}). \end{aligned}$$

Therefore, we can write

$$(20) \quad \begin{aligned} (\psi_n v)_i &= \sum_{B \in \mathcal{A}} v(B) \gamma_i(B) = \sum_{B \ni i} v(B) \gamma_i(B) \\ &\quad + \sum_{B \not\ni i} v(B) \gamma_i(B) = \sum_{B \ni i} v(B) (\gamma_i(B) - \gamma_i(B \cup \{i\})) \\ &= \sum_{B \ni i} \gamma_i(B) (v(B) - v(B \setminus \{i\})) \end{aligned}$$

since any B which does not contain i can be written $B \cup \{i\} = C - \{i\}$ where C does contain i and conversely.

We now compute $\gamma_i(B)$ when $i \in B = B \cup \{i\}$.

There are exactly $\binom{|N|-|B|}{|A|-|B|}$ coalitions A between B and N , i.e. containing B . Therefore, we can write

$$\begin{aligned} \gamma_i(B) &= \sum_{B \subset A \subset N} \frac{(-1)^{|A|-|B|}}{|A|} \\ &= \sum_{|B| \leq |A| \leq |N|} (-1)^{|A|-|B|} \binom{|N|-|B|}{|A|-|B|} \int_0^1 t^{|A|-1} dt \\ &= \int_0^1 \sum_{b \leq a \leq n} (-1)^{a-b} \binom{n-b}{a-b} \int_0^1 t^{a-1} dt \\ &= \int_0^1 t^{b-1} \sum_{0 \leq a-b \leq n-b} (-1)^{a-b} \binom{n-b}{a-b} t^{a-b} dt \\ &= \int_0^1 t^{b-1} (1-t)^{n-b} dt. \end{aligned}$$

Hence

$$(21) \quad \gamma_i(B) = \int_0^1 t^{|B|-1} (1-t)^{n-|B|} dt = \frac{(|B|-1)! (n-|B|)!}{n!}. \quad \square$$

Lemma 5. *We can write*

$$(22) \quad (\psi_n v)_i = \frac{1}{n!} \sum_{\theta} (v[A_{\theta}(i)] - v[A_{\theta}^c(i)]).$$

Proof. A and $i \in A$ are fixed, the number of permutation θ such $A = A_{\theta}(i) = \{j \in N \text{ such that } \theta(j) \leq \theta(i)\}$ equals $(|A|-1)!(n-|A|)!$, since the members of $A - \{i\}$ and $N - A$ can be permuted at will. Therefore (22) follows from (17). $\square$

Remark. This formula may be interpreted in terms of the following procedure for allocating a loss to a given player i . We choose an order (i.e. a permutation) θ at random, with equal probability $1/n!$. For a given order θ , the loss of player i is the amount he adds to the loss of the coalition of the preceding player since

$$v(A_{\theta}(i)) - v(A_{\theta}^c(i)) = v(A_{\theta}^c(i) \cup \{i\}) - v(A_{\theta}(i)).$$

Then $(\psi_n v)_i$ is nothing other than the expected loss under this random procedure.

Example. It may be useful to carry out the calculation in a few simple cases. *We begin with a two-person game. We obtain that*

$$(23) \quad \begin{aligned} (\psi_2 v)_1 &= \frac{1! 0!}{2!} [v(1, 2) - v(2)] + \frac{0! 1!}{2!} [v(1) - v(\phi)] \\ &= \frac{1}{2} [v(1, 2) - v(1) - v(2)] \end{aligned}$$

and we deduce the other components by circular permutation. For the three-person game, we obtain

$$(24) \quad \begin{aligned} (\psi_3 v)_1 &= \frac{2! 0!}{3!} [v(1, 2, 3) - v(2, 3)] + \frac{1! 1!}{3!} [v(1, 2) - v(2)] \\ &\quad + \frac{1! 1!}{3!} [v(1, 3) - v(3)] + \frac{0! 2!}{3!} [v(1) - v(\phi)]. \end{aligned}$$

Remark. It is easy to check that if a characteristic function $v \in \mathbf{R}^{\mathcal{A}}$ is symmetric, then its convex cover is also symmetric. If θ is any permutation of N .

$$\begin{aligned}
 (\theta * v)(\tau) &= \pi v(\tau_{\theta} - 1_{(1)}, \dots, \tau_{\theta} - 1_{(n)}) \\
 &= \inf_{\sum_{A \ni i} m(A) = \tau_{\theta} - 1_{(i)}} \sum_{A \in \mathcal{A}} m(A) v(A) \\
 &= \inf_{\sum_{A \ni j} m[\theta(A)] = \tau_j} \sum_{A \in \mathcal{A}} m(A) v(A) \\
 &= \inf_{\sum_{A \ni j} m(A) = \tau_j} \sum_{A \in \mathcal{A}} m[\theta^{-1}(A)] v(A) \\
 &= \inf_{\sum_{A \ni j} m(A) = \tau_j} \sum_{A \in \mathcal{A}} m(A) v[\theta(A)] = \pi(\theta * v)(\tau).
 \end{aligned}$$

***Remark.** *The Owen extension operator.* We can also write $\psi_n = \varphi_n \chi$ where $\chi \in \mathcal{L}(\mathbf{R}^{\mathcal{A}}, V^n)$ is the “Owen extension operator” defined by

$$(25) \quad (\chi v)(\tau) = \sum_{A \in \mathcal{A}} v(A) \prod_{i \in A} \tau_i \prod_{j \notin A} (1 - \tau_j).$$

The function χv is multilinear and interpolates v since

$$(26) \quad (\chi v)(\tau^A) = v(A) \quad \text{for any } A \in \mathcal{A}.$$

If we denote by v_A the function

$$\tau \mapsto v_A(\tau) = \prod_{i \in A} \tau_i \prod_{j \notin A} (1 - \tau_j)$$

we deduce that

$$(\varphi_n v_A)_i = \begin{cases} \int_0^1 t^{|A|-1} (1-t)^{n-|A|} dt & \text{if } i \in A, \\ \int_0^1 t^{|A|} (1-t)^{n-|A|-1} dt & \text{if } i \notin A. \end{cases}$$

In other words, using (21), we find that

$$(\varphi_n v_A)_i = \begin{cases} \gamma_i(A) & \text{if } i \in A, \\ -\gamma_i(A \cup \{i\}) & \text{if } i \notin A. \end{cases}$$

Therefore

$$\begin{aligned}
 (\varphi_n \chi v)_i &= \sum_{A \ni i} \gamma_i(A) v(A) - \sum_{A \not\ni i} v(A) \gamma_i(A \cup \{i\}) \\
 &= \sum_{B \ni i} \gamma_i(B) (v(B) - v(B - \{i\})) = (\psi_n v)_i.
 \end{aligned}$$

***11.4.3. Properties of the Cornet extension operator**

We mention some properties of the extension operator ω .

Proposition 1. *The extension operator ω maps characteristic functions into positively homogeneous fuzzy characteristic functions. It is an interpolation operator, i.e.*

$$(27) \quad \forall \text{ coalition } A, (\omega v)(\tau_A) = v(A).$$

Therefore

$$(28) \quad \mathcal{C}(\mathcal{T}, \omega v) \subset \mathcal{C}(\mathcal{A}, v).$$

Proof. The first statement is obvious. Since

$$\beta_B(\tau^A) = \mu_B(A) = \begin{cases} 1 & \text{if } A \supset B, \\ 0 & \text{otherwise,} \end{cases}$$

we deduce from Lemma 1 that

$$(\omega v)(\tau_A) = \sum_{B \in \mathcal{A}} \alpha(B) \beta_B(\tau^A) = \sum_{B \in \mathcal{A}} \alpha(B) \mu_B(A) = v(A).$$

Since $(\mathcal{A}, v)$ is the restriction to $\mathcal{A}$ of $(\mathcal{T}, \omega v)$, we obtain (28). $\square$

In particular, we deduce the following corollary.

Proposition 2. *Suppose that the characteristic function v satisfies*

$$(29) \quad \forall A \in \mathcal{A} \text{ such that } |A| \geq 2, \quad \alpha_A(v) = \sum_{B \subset A} (-1)^{|A|-|B|} v(B) \leq 0.$$

Then the Shapley value $\psi_n v$ belongs to core $\mathcal{C}(\mathcal{A}, v)$ of the game $(\mathcal{A}, v)$.

Proof. Since μ_A is linear if $|A| = 1$ and concave if $|A| \geq 2$, assumption (29) implies that ωv is convex and positively homogeneous. Hence $\{D\psi_n v\} = \mathcal{C}(\mathcal{T}, \omega v)$ by Proposition 2.3. $\square$

We shall give an example of a game satisfying assumption (29).

Proposition 3 (Cornet). *Let $C \in \mathbb{R}_+^n$ and $p \in \mathbb{N}$, $p \geq 1$. The characteristic function v defined by*

$$(30) \quad v(A) = - \left(\sum_{i \in A} C_i \right)^p \text{ for all } A \in \mathcal{A}$$

satisfies property (29). Its Shapley value belongs to its core.

Proof. We shall prove by induction that

$$(31) \quad \begin{cases} \text{(i) } \alpha_A(v) = - \sum_{p_1 + \dots + p_n = p} C_1^{p_1} \dots C_n^{p_n} (p! / (p_1! p_2! \dots p_n!)) \leq 0 \\ \text{if } |A| \leq p, \\ \text{(ii) } \alpha_A(v) = 0 \text{ for } |A| > p. \end{cases}$$

This result is true when $|A| = 1 \leq p$, since, when $A = \{i\}$, we can write

$$(32) \quad \alpha_A(v) = v(\{i\}) = -C_i^p.$$

Suppose that it is still true for $|A| \leq a-1$. Consider a coalition A satisfying $|A| = a$. Without loss of generality we may take A to be $\{1, \dots, a\}$. Hence, we obtain that

$$(33) \quad \begin{aligned} v(A) &= -(C_1 + \dots + C_a)^p \\ &= - \sum_{B \subset A} \sum_{\substack{p=q_1+\dots+q_b \\ B=\{1, \dots, b\}}} \frac{p!}{q_1! q_2! \dots q_b!} C_1^{q_1} \dots C_b^{q_b}. \end{aligned}$$

If $a \leq p$, the induction hypothesis implies that

$$(34) \quad v(A) = \sum_{\substack{B \subset A \\ |B| < a}} \alpha_B(v) - \sum_{\substack{p=q_1+\dots+q_a \\ B=\{1, \dots, a\}}} \frac{p!}{q_1! \dots q_a!} C_1^{q_1} \dots C_a^{q_a}.$$

On the other hand, we know that $v(A) = \sum_{B \subset A} \alpha_B(v)$. Hence we obtain

$$(35) \quad \alpha_A(v) = - \sum_{\substack{p=q_1+\dots+q_a \\ B=\{1, \dots, a\}}} \frac{p!}{q_1! \dots q_a!} C_1^{q_1} \dots C_a^{q_a}.$$

We have proved (31(i)). Since the components C_i are non-negative we deduce that $\alpha_A(v) \leq 0$. If $a > p$, we can write again that $v(A) = \sum_{B \subset A} \alpha_B(v)$. The induction hypothesis implies that $\alpha_B(v) = 0$ whenever $p < |B| \leq a-1$. Hence

$$(36) \quad v(A) = \sum_{\substack{B \subset A \\ |B| \leq p}} \alpha_B(v) + \alpha_A(v).$$

On the other hand, we deduce from (34) that

$$(37) \quad v(A) = \sum_{\substack{B \subset A \\ |B| \leq p}} \alpha_B(v).$$

Hence $\alpha_A(v) = 0$. $\square$

*11.4.4. Simple games

A game $(\mathcal{A}, v)$ is said to be *simple* if for each coalition, we have either $v(A) = 0$ or $v(A) = 1$. It is a game in which every coalition either wins (value 0) or loses (value 1), with nothing in between. As such, simple games are applicable to political science, as they include voting "games" in elections. In this case, the formula giving the Shapley value becomes especially simple. In fact, the terms $v(A) - v(A - \{i\})$ will always take the value 0 or 1, taking the value 1 whenever A is a losing coalition but $A - \{i\}$ is not.

Hence we have

$$(38) \quad (\varphi_n v)_i = \sum_{A \in \mathcal{A}_i} \frac{(|A| - 1)! (n - |A|)!}{n!}$$

where the summation is taken over the set $\mathcal{A}_i$ of all losing coalitions A such that $A - i$ is winning.

*11.4.5. Nucleolus of games with side-payments

In the case of games with side-payments $(\mathcal{A}, v)$, we can prove that the nucleolus contains only one point. The complaint becomes:

$$(39) \quad \theta(A, c) = \sum_{i \in A} c_i - v(A)$$

and depends *linearly* on c .

Proposition 4. *The nucleolus of a game with side-payments consists of a unique multiloss c .*

Proof. Assume that c and d are two different multilosses in the nucleolus. We shall obtain a contradiction by proving that

$$(40) \quad \omega \theta \left(\frac{c+d}{2} \right) \prec_{\neq} \omega \theta(x) = \omega \theta(d)$$

where $\prec$ denotes the lexicographic ordering.

Consider the first component. Then

$$\begin{aligned} \omega \theta \left(\frac{c+d}{2} \right)_1 &= \max_{A \in \mathcal{A}} \left(\frac{\theta(A, c) + \theta(A, d)}{2} \right) \\ &\leq \frac{1}{2} \max_{A \in \mathcal{A}} \theta(A, c) + \frac{1}{2} \max_{A \in \mathcal{A}} \theta(A, d) \\ &= \frac{1}{2} \omega \theta(c) + \frac{1}{2} \omega \theta(d)_1. \end{aligned}$$

If the inequality is strict, this implies (40). If not, let $A_1 \in \mathcal{A}$ satisfy $\theta(\frac{1}{2}(c+d))_1 = \frac{1}{2}(\theta(A_1, c) + \theta(A_1, d))$. Then we deduce that $\theta(A_1, c) = \max_{A \in \mathcal{A}} \theta(A, c) = \omega\theta(c)_1$ and that $\theta(A_1, d) = \max_{A \in \mathcal{A}} \theta(A, d) = \omega\theta(d)_1$.

Assume that there exist $A_1, \dots, A_{j-1}$ such that, for any $k \leq j-1$,

$$\begin{aligned}
 \omega\theta\left(\frac{c+d}{2}\right)_k &= \max_{A \neq A_1, \dots, A_{j-1}} \theta\left(A, \frac{c+d}{2}\right) \\
 (41) \qquad &= \theta\left(A_k, \frac{c+d}{2}\right) = \frac{1}{2}((\omega\theta(c))_k + (\omega\theta(d))_k) \\
 &= \frac{1}{2}\theta(A_k, c) + \frac{1}{2}\theta(A_k, d).
 \end{aligned}$$

Taking the supremum over the set of coalitions $A \neq A_1, \dots, A_{j-1}$ we obtain that

$$\begin{aligned}
 \omega\theta\left(\frac{c+d}{2}\right)_j &= \max_{A \neq A_1, \dots, A_{j-1}} \left(\frac{\theta(A, c) + \theta(A, d)}{2}\right) \\
 &\leq \frac{1}{2} \max_{A \neq A_1, \dots, A_{j-1}} \theta(A, c) + \frac{1}{2} \max_{A \neq A_1, \dots, A_{j-1}} \theta(A, d) \\
 &= \frac{1}{2} \omega\theta(c)_j + \frac{1}{2} \omega\theta(d)_j.
 \end{aligned}$$

If strict inequality holds, then the contradiction (40) holds. If not, we deduce the existence of a coalition A_j such that (41) holds for all $k \leq j$.

We have proved that either $\omega(\frac{1}{2}(c+d)) < \omega\theta(c) = \omega\theta(d)$ or that for any $A_k (1 \leq k \leq |\mathcal{A}|)$

$$(42) \qquad \theta(A_k, c) = \omega\theta(c)_k = \omega\theta(d)_k = \theta(A_k, d).$$

This implies that $\theta(c) = \theta(d)$, and thus, that $c = d$. $\square$

Symmetry property of the nucleolus

Let $\bar{c}$ be the nucleolus of the game $(\mathcal{A}, v)$.

We associate with any pair $\{j, k\}$ of players, $j \neq k$, the number

$$(43) \qquad s_{kj} = \max_{A \ni k, A \not\ni j} \left\{ \sum_{i \in A} \bar{c}_i - v(A) \right\} = \max_{\substack{A \ni k \\ A \not\ni j}} \theta(A, \bar{c}).$$

We can regard s_{kj} as the maximum complaint that player k can raise against player i . Indeed, to say that $s_{kj} > s_{jk}$ amounts to saying that player k is "stronger" than player j .

We shall prove the following result.

Proposition 5. Let $\bar{c}$ be the nucleolus of the game $(\mathcal{A}, v)$. If $s_{kj} > s_{jk}$, then $\bar{c}_j = v(\{j\})$.

Interpretation. If player k is stronger than player j , then he forces player j to submit to his worst possible loss $v(\{j\})$.

This property can be viewed as a symmetry property of the nucleolus. We have $s_{kj} = s_{jk}$ except in the case where either $\bar{c}_j = v(\{j\})$ or $\bar{c}_k = v(\{k\})$.

Proof of Proposition 5. Assume that there exist two players k and j , $k \neq j$, such that

$$(44) \quad s_{kj} > s_{jk} \quad \text{and} \quad \bar{c}_j < v(\{j\}).$$

Choose δ small enough that

$$(45) \quad s_{kj} - \delta > s_{jk} + \delta \quad \text{and} \quad \bar{c}_j > v(\{j\}) - \delta.$$

Consider the multiloss c defined by

$$(46) \quad \begin{cases} \text{(i)} & c_i = \bar{c}_i \text{ if } i \neq k, j, \\ \text{(ii)} & c_k = \bar{c}_k - \delta, \\ \text{(iii)} & c_j = \bar{c}_j + \delta. \end{cases}$$

We have that

$$(47) \quad \sum_{i \in N} c_i = \sum_{i \in N} \bar{c}_i = v(N); \quad \forall i \in N, \quad c_i \leq v(\{i\}).$$

Also we obtain that

$$\begin{aligned} \sum_{i \in A} c_i - v(A) &= \sum_{i \in A} \bar{c}_i - v(A) && \text{if } A \ni k, A \not\ni j, \\ \sum_{i \in A} c_i - v(A) &= \sum_{i \in A} \bar{c}_i - v(A) && \text{if } A \ni k, A \ni j, \\ \sum_{i \in A} c_i - v(A) &< \sum_{i \in A} \bar{c}_i - v(A) && \text{if } A \ni k, A \not\ni j, \\ \sum_{i \in A} c_i - v(A) &> \sum_{i \in A} \bar{c}_i - v(A) && \text{if } A \ni j, A \not\ni k. \end{aligned}$$

Therefore, in $\mathbf{R}^{\mathcal{A}}$, we obtain $\theta(c)$ from $\theta(\bar{c})$ by keeping the components A such that either A or $\hat{A} = N - A$ contains both j and k , and by modifying only the components which contain either k or j .

Since

$$s_{kj} = \max_{\substack{A \ni k \\ A \not\ni j}} \theta(A, \bar{c}) > s_{jk} = \max_{\substack{A \ni j \\ A \not\ni k}} \theta(A, \bar{c})$$

the largest component which has to be modified is the one such that A contains k and not j . It will be decreased and thus, $\omega\theta(c) < \omega\theta(\bar{c})$.

This is impossible since $\bar{c}$ yields the smallest element for the lexicographic ordering. $\square$

CHAPTER 12

GAMES WITHOUT SIDE-PAYMENTS

We shall prove in this chapter the main theorems dealing with the concepts of the core, the fuzzy core and equilibria of games without side-payments.

We use two fundamental tools for this purpose. The first of these is the “representation” of a game *without* side-payments by a family of games *with* side-payments. The second is the “cover” of a game without side-payments. Such a cover is a fuzzy game without side-payments constructed by “extending in some sense the family of games with side-payments of the representation of the original game into a family of fuzzy games with side-payments.

In other words, each *representation* of a game without side payments defines a cover which is a *fuzzy game* (without side payments).

This allows the definition of the *fuzzy core* of the representation of the game as the subset of *admissible multilosses* of the whole *set* N of players which are *not rejected* by any fuzzy coalition. We may also define an *equilibrium* of the representation as an *admissible multiloss* of the whole *set* N of players which belongs to the core of a game with side-payments of the representation.

The equivalence between these two concepts is proved by applying the minisup theorem.

We then prove that the fuzzy core is non-empty (i.e. that an equilibrium exists) by using the Ky-Fan theorem. In particular, we shall prove that the canonical fuzzy core and the set of canonical equilibria (almost) coincide and that they are non-empty under assumptions slightly stronger than the Scarf theorem for the non-emptiness of the core.

Also, for a convenient representation of the economic game, we shall prove that the fuzzy core of an economy (almost) coincides with the set of Walras allocations.

12.1. Equivalence between the fuzzy core and the set of equilibria

We say that a family $w(\mathcal{M}^n \times P)$ of characteristic functions $w(\lambda, p)$ ($\lambda \in \mathcal{M}^n$, $p \in P$) *represents* a game $(\mathcal{A}, J)$ if for any $A \in \mathcal{A}$, $\sigma^b(J(A); \lambda) = \sup_{p \in P} w(A, \lambda, p)$.

Then, we define a multiloss $c \in J_+(N)$ to be a weak *equilibrium* of such a representation if there exists $\{\bar{\lambda}, \bar{p}\} \in \mathcal{M}^n \times P$ such that

$$\forall A \in \mathcal{A}, \quad \langle \bar{\lambda}, \tau^A \cdot c \rangle \leq w(A, \bar{\lambda}, \bar{p}).$$

It is said to be an “equilibrium” if there exists such a λ which belongs to $\mathcal{M}^n$.

We shall say that a multiloss $c \in J_+(N)$ belongs to the *fuzzy core* of such a representation if, for any fuzzy coalition $\tau \in \mathcal{T}$, there exists $\lambda \in \mathcal{M}(A_\tau)$ and $p \in P$ such that $\langle \lambda, \tau \cdot c \rangle \leq \pi w(\tau, \lambda, p)$, i.e. if $\tau \cdot c \notin \pi_w J(\tau) + \mathbb{R}_+^r$, where

$$\begin{aligned} \pi_w J(\tau) = \{ & c \in \mathbb{R}^n \text{ such that } \langle \lambda, \tau \cdot c \rangle \leq \pi w(\tau, \lambda, p) \\ & \text{for any } \{\lambda, p\} \in \mathcal{M}^n \times P \} \end{aligned}$$

is a subset of multilosses of the fuzzy coalition τ constructed with the representation $w(\mathcal{M}^n \times P)$. This defines a fuzzy game $(\mathcal{T}, \pi_w J)$ which will be called the *cover* of a game $(\mathcal{A}, J)$.

This extension, of course, depends upon the representation $w(\mathcal{M}^n \times P)$, as also do the concepts of the fuzzy core and equilibria. We shall prove that the set of equilibria is contained in the fuzzy core and that the latter is contained in the set of weak equilibria.

12.1.1. Representation of a game

Definition 1. Let $(\mathcal{A}, J)$ (resp. $(\mathcal{T}, J)$) be a game (resp. fuzzy game) without side-payments. Let $w(\mathcal{M}^n \times P)$ be a family of characteristic functions $w(\lambda, p) : A \in \mathcal{A} \rightarrow w(A; \lambda, p)$ (resp. $\tau \in \mathcal{T} \rightarrow w(\tau, \lambda, p)$) of games with side-payments obtained when λ ranges over $\mathcal{M}^n$ and P a set of parameters p .

We shall say that the family $(\mathcal{A}, w(\mathcal{M}^n \times P))$ (resp. $(\mathcal{T}, w(\mathcal{M}^n \times P))$) “represents” the game $(\mathcal{A}, J)$ (resp. $(\mathcal{T}, J)$) if $\forall \lambda \in \mathcal{M}^n$,

$$\begin{aligned} (1) \quad & \forall A \in \mathcal{A}, \quad \sup_{p \in P} w(A, \lambda, p) = \sigma^b(J_+(A); \lambda) = \inf_{c \in J(A)} \langle \lambda, c \rangle \\ & (\text{resp. } \forall \tau \in \mathcal{T}, \quad \sup_{p \in P} w(\tau, \lambda, p) = \sigma^b(J_+(\tau); \lambda) = \inf_{c \in J(\tau)} \langle \lambda, c \rangle). \end{aligned}$$

Remark. If one is looking for minimal requirements, one may replace condition (1) by the weaker condition

$$(2) \quad \begin{cases} \text{(i) } \sup_{p \in P} w(N, \lambda, p) = \sigma^b(J_+(N); \lambda), \\ \text{(ii) } \forall A \in \mathcal{A}, \sup_{p \in P} w(A, \lambda, p) \leq \sigma^b(J_+(A); \lambda). \end{cases}$$

Example. Canonical representation. It is always possible to represent a game $(\mathcal{A}, J)$ of a fuzzy game $(\mathcal{T}, J)$ by the functions

$$w^b(\lambda) : \tau \in \mathcal{T} \rightarrow w(\tau, \lambda) = \sigma^b(J(\tau); \lambda).$$

Definition 2. We shall say that the representation $(\mathcal{A}, w^b(\mathcal{M}^n))$ of a game $(\mathcal{A}, J)$ (resp. $(\mathcal{T}, w^b(\mathcal{M}^n))$ of a fuzzy game $(\mathcal{T}, J)$) is the “canonical representation”.

For instance, the canonical representation of cooperative games $\{X(A), F^A\}_{A \in \mathcal{A}}$ described in strategic form is defined by

$$(3) \quad w^b(A, \lambda) = \inf_{x^A \in X(A)} \sum_{i \in A} \lambda^i f_i^A(x^A)$$

since $J(A) = F^A(X(A))$.

Consider its canonical extension $\{X_0(\tau), F^\tau\}_{\tau \in e}$ described by the multi-strategy sets

$$(4) \quad X_0(A) = \left\{ x^\tau = \sum_{A \in \mathcal{A}} \frac{m(A)}{\tau} x^A \right\}_{m \in \mathcal{U}(\tau); x^A \in X(A)}$$

and by the multiloss correspondences

$$(5) \quad F^\tau(x^\tau) = \left\{ \sum_{A \in \mathcal{A}} m(A) F^A(x^A) \right\}_{\sum_{A \in \mathcal{A}} m(A) x^A / \tau = x^\tau}$$

(see Section 10.4.8).

We shall show that it is represented by the convex cover $w^b(\tau, \lambda)$ of $w^b(A, \lambda)$.

Write $J_0(\tau) = F^\tau(X_0(\tau)) = \{ \sum_{A \in \mathcal{A}} m(A) F^A(x^A) \}_{m \in \mathcal{U}(\tau); x^A \in X(A)}$.

We deduce that $w^b(\tau, \lambda) = \sigma^b(J_0(\tau); \lambda)$ is equal to

$$(6) \quad w^b(\tau, \lambda) = \inf_{x^\tau \in X_0(\tau)} f^\tau(x^\tau; \lambda)$$

where we set

$$(7) \quad \begin{aligned} f^\tau(x^\tau; \lambda) &= \inf_{\sum_{A \in \mathcal{A}} m(A) x^A / \tau = x^\tau} \sum_{i=1}^n \lambda^i \sum_{A \ni i} m(A) f_i^A(x^A) \\ &= \inf_{\sum_{A \in \mathcal{A}} m(A) x^A / \tau = x^\tau} \sum_{A \in \mathcal{A}} m(A) \sum_{i \in A} \lambda^i f_i^A(x^A) \\ &= \sigma^b(F_+^\tau(x^\tau); \lambda). \end{aligned}$$

Therefore, we can also write

$$\begin{aligned} w^b(\tau, \lambda) &= \inf_{x^\tau \in X_0(\tau)} \inf_{\sum_{A \in \mathcal{A}} m(A) x^A / \tau = x^\tau} \sum_{A \in \mathcal{A}} m(A) \sum_{i \in A} \lambda^i f_i^A(x^A) \\ &= \inf_{m \in \mathcal{U}(\tau)} \sum_{A \in \mathcal{A}} m(A) \inf_{x^A \in X(A)} \sum_{i \in A} \lambda^i f_i^A(x^A) \\ &= \inf_{m \in \mathcal{U}(\tau)} \sum_{A \in \mathcal{A}} m(A) w^b(A, \lambda) \end{aligned}$$

where $w^b(A, \lambda)$ is defined by (3). In other words, the functions $w^b(\tau, \lambda)$ and $w^b(A, \lambda)$ are related by

$$(8) \quad w^b(\tau, \lambda) = \pi w^b(\cdot, \lambda) = \inf_{m \in \mathcal{M}(\tau)} \sum_{A \in \mathcal{A}} m(A) w^b(A, \lambda).$$

12.1.2. Equilibrium of a representation

Definition 3. Let $w(\mathcal{M}^n \times P^n)$ be a representation of a game $(\mathcal{A}, J)$ (resp. fuzzy game $(\mathcal{C}, J)$). We shall say that a multilooss $\bar{c}$ is a “weak cooperative equilibrium” of the representation if there exist $\bar{\lambda} \in \mathcal{M}^n$ and $\bar{p} \in P$ such that

$$(9) \quad \begin{cases} (i) \bar{c} \in J_+(N) \\ (ii) \forall A \in \mathcal{A}, \langle \bar{\lambda}, \tau^A \cdot \bar{c} \rangle \leq w(A, \bar{\lambda}, \bar{p}) \\ \quad \text{(resp. } \forall \tau \in \mathcal{C}, \langle \bar{\lambda}, \tau \cdot \bar{c} \rangle \leq w(\tau, \bar{\lambda}, \bar{p})). \end{cases}$$

We shall say that $\bar{c}$ is an “equilibrium” (or a “strong equilibrium”) if there exists $\lambda \in \mathcal{M}^n$ satisfying (9(ii)).

Remark. To say that $\bar{c}$ is a weak cooperative equilibrium of the representation $w(\mathcal{M}^n \times P)$ amounts to saying that

$$(10) \quad \bar{c} \in J(N) \cap \mathcal{C}(\mathcal{A}, w(\bar{\lambda}, \bar{p})),$$

i.e. that $\bar{c}$ is an admissible multilooss of the whole set N of players which belongs to the core of one of the side-payment games defined by a characteristic function $w(\bar{\lambda}, \bar{p})$ of the representation $w(\mathcal{M}^n \times P)$.

Remark. It is clear that any (weak) cooperative equilibrium $\bar{c}$ of the representation $(\mathcal{C}, w(\mathcal{M}^n \times P))$ of a fuzzy game is a (weak) cooperative equilibrium of its restriction $(\mathcal{A}, w(\mathcal{M}^n \times P))$.

Example. Canonical cooperative equilibrium. By using the canonical representation of the game $(\mathcal{A}, J)$, the definition of (weak) cooperative equilibrium of the canonical representation coincides with the definition of canonical cooperative equilibrium introduced in Section 10.5.1.

Proposition 1. Let $(\mathcal{A}, w(\mathcal{M}^n \times P))$ be a representation of a game $(\mathcal{A}, J)$. Then any (weak) cooperative equilibrium of the representation $(\mathcal{A}, w(\mathcal{M}^n \times P))$ is a (weak) canonical cooperative equilibrium.

Proof. Since $w^b(A, \lambda) = \sigma^b(J(A), \lambda) = \sup_{p \in P} w(A, \lambda, p)$, then conditions $\langle \bar{\lambda}, \tau^A \cdot \bar{c} \rangle \leq w(A, \bar{\lambda}, \bar{p})$ imply conditions $\langle \bar{\lambda}, \tau^A \cdot \bar{c} \rangle \leq w^b(A, \bar{\lambda})$. $\square$

12.1.3. Cover associated with a representation

Let $w(\mathcal{M}^n \times P)$ be a representation of a game $(\mathcal{A}, J)$.

We associate with the characteristic functions $A \mapsto w(A; \lambda, p)$ the characteristic functions $\tau \mapsto \pi_w(\tau, \lambda, p) = \inf_{m \in \mathcal{N}(\tau)} \sum_{A \in \mathcal{A}} m(A) w(A, \lambda, p)$ of a fuzzy game. In some sense, we can therefore “extend” the game $(\mathcal{A}, J)$ into a fuzzy game $(\mathcal{T}, \pi_w J)$. The extension depends on the choice of representation of the game $(\mathcal{A}, J)$.

Definition 4. Let $w(\mathcal{M}^n \times P)$ be a representation of a game $(\mathcal{A}, J)$. Its “cover” associated with the representation $w(\mathcal{M}^n \times P)$ is the fuzzy game $(\mathcal{T}, \pi_w J)$ defined by

$$(11) \quad \forall \tau \in \mathcal{T}, \quad \pi_w J(\tau) = \{c \in \mathbb{R}^n \text{ such that } \langle \lambda, c \rangle \geq \pi_w(\tau, \lambda, p) \text{ for any } \lambda \in \mathcal{M}^n, p \in P\}.$$

Notice that

$$(12) \quad \forall \tau \in \mathcal{T}, \quad \pi_w J(\tau) \text{ is a closed convex subset such that } \pi_w J(\tau) = \pi_w J(\tau) + \mathbb{R}_+^r.$$

The lower support function of $\pi_w J(\tau)$ is defined by

$$(13) \quad \sigma^b(\pi_w J(\tau); \lambda) = \sup_{p \in P} \pi_w(\tau, \lambda, p).$$

Also notice that we always have

$$(14) \quad \forall A \in \mathcal{A}, \quad J_+(A) \subset \pi_w J(\tau^A).$$

Example. Canonical cover of a game. By using the canonical representation $w^b(\mathcal{M}^n)$ of a game $(\mathcal{A}, J)$, we obtain its “canonical cover” $(\mathcal{T}, \pi^b J)$ defined by

$$(15) \quad \pi^b J(\tau) = \overline{\text{co}} \left(\bigcup_{m \in \mathcal{N}(\tau)} \sum_{A \in \mathcal{A}} m(A) J_+(A) \right).$$

Indeed, we have, by (13),

$$\begin{aligned} \sigma^b(\pi^b J(\tau); \lambda) &= \inf_{m \in \mathcal{N}(\tau)} \sum_{A \in \mathcal{A}} m(A) \sigma^b(J_+(A); \lambda) \\ &= \inf_{m \in \mathcal{N}(\tau)} \sigma^b \left(\sum_{A \in \mathcal{A}} m(A) J_+(A); \lambda \right) \\ &= \sigma^b \left[\overline{\text{co}} \bigcup_{m \in \mathcal{N}(\tau)} \sum_{A \in \mathcal{A}} m(A) J_+(A); \lambda \right] \end{aligned}$$

by Proposition 1.4.4.

Definition 5. We shall say that the fuzzy game $(\mathcal{T}, \pi^b J)$ defined by (15) is the canonical cover of the game $(\mathcal{A}, J)$.

Proposition 2. Suppose that

$$(16) \quad \forall A \in \mathcal{A}, \quad J_+(A) \text{ is closed, convex and bounded below.}$$

Then, for any fuzzy coalition,

$$(17) \quad \pi^b J(\tau) = \bigcup_{m \in \mathcal{M}(\tau)} \sum_{A \in \mathcal{A}} m(A) J_+(A).$$

Proof. We begin by checking that $K = \bigcup_{m \in \mathcal{M}(\tau)} \sum_{A \in \mathcal{A}} m(A) J_+(A)$ is convex. We can write any $d = \sum_{k=1}^K \alpha_k \sum_{A \in \mathcal{A}} m^k(A) c^{A,k} \in \text{co } K$ in the form $d = \sum_{A \in \mathcal{A}} m(A) d^A$ where $m(A) = \sum_{k=1}^K \alpha_k m^k(A)$ and where $d^A = \sum_{k=1}^K (\alpha_k m^k(A) c^{A,k} / m(A)) \in J_+(A)$ when $c^{A,k} \in J_+(A)$.

Since $\sum_{A \ni i} m(A) = \sum_{k=1}^K \alpha_k \sum_{A \ni i} m^k(A) = \sum_{k=1}^K \alpha_k \tau_i = \tau_i$ when $m^k \in \mathcal{M}(\tau)$, m belongs to $\mathcal{M}(\tau)$. We next prove that K is closed.

Since $\mathcal{M}(\tau)$ is compact and the subsets $J_+(A)$ are bounded below, the map $\{m, \{c^A\}_{A \in \mathcal{A}}\} \in \mathcal{M}(\tau) \times \prod_{A \in \mathcal{A}} J_+(A) \rightarrow \sum_{A \in \mathcal{A}} m(A) c^A$ is proper.

Hence its image $J(\tau)$ is closed. Thus $K = \overline{\text{co}}(K) = \pi^b J(\tau)$. $\square$

Consider a game $\{X(A), F^A\}_{A \in \mathcal{A}}$ described in strategic form and its canonical extension $\{X_0(\tau), F^\tau\}_{\tau \in \mathcal{T}}$ (see Section 10.4.8). These definitions are consistent.

Proposition 3. The canonical cover $(\mathcal{T}, \pi^b J)$ of a game $\{X(A), F^A\}_{A \in \mathcal{A}}$ described in strategic form is defined by

$$(18) \quad \forall \tau \in \mathcal{T}, \quad \pi^b J(\tau) = \overline{\text{co}}(F^\tau(X_0(\tau)) + \mathbf{R}_+^r).$$

In particular, if we assume that for any A ,

$$(19) \quad \begin{cases} \text{(i) } X(A) \text{ is convex and compact,} \\ \text{(ii) } \forall i \in A, f_i^A \text{ is convex and lower semi-continuous,} \end{cases}$$

then

$$(20) \quad \forall \tau \in \mathcal{T}, \quad \pi^b J(\tau) = F^\tau(X_0(\tau)) + \mathbf{R}_+^r.$$

Proof. We proved in Section 10.4.8 (formula (51)) that

$$(21) \quad F^\tau(X_0(\tau)) = \bigcup_{m \in \mathcal{M}(\tau)} \sum_{A \in \mathcal{A}} m(A) F^A(X(A)).$$

Therefore, the first statement follows from (15) and the second statement from Proposition 10.4.4. $\square$

12.1.4. Fuzzy core of a representation

We now introduce the concept of the fuzzy core of a representation $w(\mathcal{M}^n \times P)$ of a game $(\mathcal{A}, J)$.

Definition 6. The fuzzy core $\mathcal{C}_w(\mathcal{A}, J)$ of the representation $w(\mathcal{M}^n \times P)$ of a game $(\mathcal{A}, J)$ is the subset of multilosses c satisfying

$$(22) \quad \begin{cases} \text{(i) } c \in J_+(N), \\ \text{(ii) } \forall \tau \in \mathcal{T}, \tau \cdot c \notin \pi_w J(\tau) + \dot{\mathbf{R}}_+^r. \end{cases}$$

In other words, the fuzzy core $\mathcal{C}_w(\mathcal{A}, J)$ is the subset of available multilosses of the whole set N of players which are not rejected by any fuzzy coalition τ .

Since $J_+(A) + \dot{\mathbf{R}}_+^A \subset \pi_w J(\tau^A) + \dot{\mathbf{R}}_+^A$ for all coalitions $A \in \mathcal{A}$, we deduce the following result.

Proposition 4. Let $w(\mathcal{M}^n \times P)$ be a representation of a game $(\mathcal{A}, J)$. The fuzzy core $\mathcal{C}_w(\mathcal{A}, J)$ is contained in the canonical fuzzy core $\mathcal{C}^b(\mathcal{A}, J)$, which is contained in the core $\mathcal{C}(\mathcal{A}, J)$.

Remark. The fuzzy core $\mathcal{C}_w(\mathcal{A}, J)$ and the core of the fuzzy game $(\mathcal{T}, \pi_w J)$ are related by formula

$$(23) \quad \mathcal{C}_w(\mathcal{A}, J) = \mathcal{C}(\mathcal{T}, \pi_w J) \cap J_+(N).$$

12.1.5. The equivalence theorem

Theorem 1. Let $w(\mathcal{M}^n \times P)$ be a representation of a game $(\mathcal{A}, J)$ without side-payments.

(a) The subset of strong cooperative equilibria of $w(\mathcal{M}^n \times P)$ is contained in the fuzzy core $\mathcal{C}_w(\mathcal{A}, J)$ of $w(\mathcal{M}^n \times P)$.

(b) Conversely, if we assume that

$$(24) \quad \begin{cases} \text{(i) } P \text{ is convex and compact,} \\ \text{(ii) } \forall A \in \mathcal{A}, \{\lambda, p\} \rightarrow w(A, \lambda, p) \text{ is concave and upper semi-continuous} \end{cases}$$

then the fuzzy core $\mathcal{C}_w(\mathcal{A}, J)$ is contained in the subset of weak cooperative equilibria.

Proof. Let $\bar{c} \in J_+(N)$ be a strong cooperative equilibrium, and $\bar{\lambda} \in \mathcal{M}^n$ and $\bar{p} \in P$ the associate multipliers. Since $\langle \bar{\lambda}, \tau^A \cdot \bar{c} \rangle \leq w(A, \bar{\lambda}, \bar{p})$ for any $A \in \mathcal{A}$, we deduce that $\langle \bar{\lambda}, \tau \cdot c \rangle \leq \pi_w J(\tau, \bar{\lambda}, \bar{p}) \leq \sigma^b(\pi_w J(\tau); \bar{\lambda})$ for any fuzzy coalition

$\tau \in \mathcal{T}$. Since $\bar{\lambda} \in \mathcal{M}^n$, then $(\bar{\lambda} / \sum_{i \in A} \bar{\lambda}_i) \in \mathcal{M}(A_\tau)$. Therefore,

$$(25) \quad \alpha_{\mathcal{T}}(c) = \sup_{\tau \in \mathcal{T}} \inf_{\lambda \in \mathcal{M}(A_\tau)} [\langle \lambda, \tau \cdot c \rangle - \sigma^b(\pi_w J(\tau); \lambda)] \leq 0.$$

Hence Theorem 10.4.1 implies that $\bar{c}$ belongs to the core of the fuzzy game $(\mathcal{T}, \pi_w J)$ which is represented by $w(\mathcal{M}^n \times P)$ by construction.

Conversely, suppose that $\bar{c}$ belongs to the fuzzy core $\mathcal{C}_w(\mathcal{A}, J) \subset \mathcal{C}(\mathcal{T}, \pi_w J)$.

Since

$$(26) \quad \forall \tau \in \mathcal{T}, \quad \pi_w J(\tau) + \mathbf{R}_+^r \text{ is convex}$$

Theorem 10.4.1 implies that $\alpha_{\mathcal{T}}(\bar{c}) \leq 0$. We introduce

$$(27) \quad \beta(\bar{c}) = \sup_{\tau \in \mathcal{T}} \inf_{\substack{\lambda \in \mathcal{M}^n \\ p \in P}} [\langle \lambda, \tau \cdot \bar{c} \rangle - \pi_w(\tau, \lambda, p)].$$

We check that

$$\begin{aligned} \beta(\bar{c}) &= \sup_{\tau \in \mathcal{T}} \inf_{\lambda \in \mathcal{M}^n} [\langle \lambda, \tau \cdot c \rangle - \sigma^b(\pi_w J(\tau); \lambda)] \\ &\leq \sup_{\tau \in \mathcal{T}} \inf_{\lambda \in \mathcal{M}(A_\tau)} [\langle \lambda, \tau \cdot c \rangle - \sigma^b(\pi_w J(\tau); \lambda)] \\ &= \alpha_{\mathcal{T}}(\bar{c}) \leq 0. \end{aligned}$$

On the other hand, $\mathcal{T}$ and $\mathcal{M}^n \times P$ are convex compact subsets (by assumption (24(i))) and $\langle \lambda, \tau \cdot \bar{c} \rangle - \pi_w(\tau, \lambda, p)$ is convex lower semi-continuous with respect to $\{\lambda, p\}$ by assumption (24(ii)) and concave with respect to τ (since $\pi_w(\cdot, \lambda, p)$ is convex with respect to τ by Proposition 11.2.1). Then the Nikaido Theorem 7.1.6 implies the existence of $\{\bar{\lambda}, \bar{p}\} \in \mathcal{M}^n \times P$ such that

$$(28) \quad \beta(\bar{c}) = \sup_{\tau \in \mathcal{T}} [\langle \bar{\lambda}, \tau \cdot \bar{c} \rangle - \pi_w(\tau, \bar{\lambda}, \bar{p})] \leq 0.$$

This implies that $\bar{c} \in J(N)$ is a weak equilibrium. $\square$

In particular, the canonical representation of a game obviously satisfies conditions (24). Theorem 1 becomes the following.

Theorem 2. *Let $(\mathcal{A}, J)$ be a game. Any strong canonical cooperative equilibrium belongs to the canonical fuzzy core. Conversely, the canonical fuzzy core is contained in the subset of weak canonical cooperative equilibria.*

The strategic formulation of the above theorem is the following.

Theorem 3. *Let $\{X(A), F^A\}_{A \in \mathcal{A}}$ be a game described in strategic form by multi-strategy sets $X(A)$ and multiloop operators F^A satisfying*

$$(29) \quad \begin{cases} \text{(i) } X(A) \text{ is convex and compact,} \\ \text{(ii) } \forall i \in A, f_i^A \text{ is convex and lower semi-continuous.} \end{cases}$$

Any strong canonical cooperative equilibrium $\bar{x} \in X(N)$ belongs to the core $\mathcal{C}(\{X_0(\tau), F^*\}_{\tau \in \mathcal{T}})$ of its canonical cover. Conversely, the canonical fuzzy core is contained in the subset of weak canonical cooperative equilibria.

We shall now give a sufficient condition for the fuzzy core to be contained in the set of cooperative equilibria. We denote by w^* the multiloss with components

$$(30) \quad w_i^* = \inf_{c \in J(\{i\})} c.$$

It is clear that the core $\mathcal{C}(\mathcal{A}, J)$ of the game is contained in $w^* - R_+^n$.

Proposition 5. Suppose that assumptions (24) of Theorem 1 hold. Suppose that for any $c \in J_+(N) \cap (w^* - R_+^n)$ and for any non-empty coalition $A \in \mathcal{A}_{A \in \mathcal{A}, A \neq N}$ we can find $d \in J(N)$ such that

$$(31) \quad \forall i \in A, \quad d_i < c_i.$$

Then the fuzzy core is equal to the set of (strong) cooperative equilibria.

Proof. If $\bar{c}$ belongs to the fuzzy core, there exist $\bar{\lambda} \in \mathcal{M}^n$ and $\bar{p} \in P$ such that (19) holds. In particular,

$$(32) \quad \begin{cases} \text{(i)} \quad \langle \bar{\lambda}, \bar{c} \rangle = \min_{d \in J(N)} \langle \bar{\lambda}, d \rangle, \\ \text{(ii)} \quad \bar{c} \in J(N) \cap (w^* - R_+^n). \end{cases}$$

If $\bar{\lambda}$ did not belong to $\mathcal{M}^n$, we would obtain a contradiction because we may take $A = \{i \in N \text{ such that } \lambda^i > 0\}$ and $d \in J(N)$ satisfying (31). Then inequality

$$\langle \bar{\lambda}, \bar{c} \rangle = \sum_{i \in A} \bar{\lambda}^i \bar{c}_i > \sum_{i \in A} \bar{\lambda}^i d_i = \langle \bar{\lambda}, d \rangle$$

is a contradiction of (32(i)).

12.2. Non-emptiness of the fuzzy core of a balanced game

We shall prove the fundamental theorem for the non-emptiness of the fuzzy core. This is also an existence theorem for a weak equilibrium by the equivalence theorem.

Roughly speaking, the fuzzy core of a representation $w(\mathcal{M}^n \times P)$ is non-empty whenever the cores $\mathcal{C}(\mathcal{A}, w(\lambda, p))_\lambda$ of the associated games with side-payments are non-empty, i.e. whenever the characteristic functions $A \mapsto w(A,$

λ, p are “balanced”. We also need the upper semi-continuity of the correspondence $\{\lambda, p\} \rightarrow \mathcal{C}(\mathcal{A}, w(\lambda, p))_\lambda$.

Then the non-emptiness of the fuzzy core is proved using the Ky-Fan theorem.

There is nevertheless a difficulty which comes from the fact that $\{\lambda, p\} \rightarrow \mathcal{C}(\mathcal{A}, \lambda, p)$ is not necessarily upper semi-continuous when λ belongs to the boundary of $\mathcal{M}^n$. We overcome this difficulty by an approximation device, which lengthens the proof of this theorem.

We apply the theorem to the canonical representation of a game and obtain an improvement of the Scarf theorem about the non-emptiness of the core.

12.2.1. Statement of theorems of non-emptiness of the fuzzy core

Theorem 1. *Suppose that the game $(\mathcal{A}, J)$ satisfies*

- (1) $J_+(N)$ is convex, closed and bounded below.

We also suppose that the game is represented by a family $w(\mathcal{M}^n \times P)$ of characteristic functions satisfying

- (2)
$$\begin{cases} \text{(i) } P \text{ is convex and compact,} \\ \text{(ii) } \{\lambda, p\} \mapsto w(N, \lambda, p) \text{ is continuous on } \mathcal{M}^n \times P, \\ \text{(iii) } \forall A \in \mathcal{A}, \{\lambda, p\} \mapsto w(A, \lambda, p) \text{ are upper semi-continuous and} \\ \text{concave on } \mathcal{M}^n \times P. \end{cases}$$

Finally, we assume that

- (3) $\forall \{\lambda, p\} \in \mathcal{M}^n \times P, A \mapsto w(A, \lambda, p)$ is “balanced”.

Then the fuzzy core of the representation $w(\mathcal{M}^n \times P)$ is non-empty (and thus, there exists at least a weak cooperative equilibrium of the representation $w(\mathcal{M}^n \times P)$).

Remark. Recall that assumption (3) is equivalent to the assumption

- (4) $\forall \{\lambda, p\} \in \mathcal{M}^n \times P$, the core $\mathcal{C}(\mathcal{A}, w(\lambda, p))_\lambda$ of the associated game with side-payments is non-empty.

(See Theorem 11.2.1).

If we use the canonical representation of a game $(\mathcal{A}, J)$, Theorem 1 becomes the following.

Theorem 2. *Suppose that the game $(\mathcal{A}, J)$ satisfies*

- (1) $J_+(N)$ is convex, closed and bounded below

and that

$$(5) \quad \text{for any balance } m \in \mathcal{M}(\tau^N), \quad \sum_{A \in \mathcal{A}} m(A) J_+(A) \subset J_+(N).$$

Then the canonical fuzzy core is non-empty and thus, there exists $\bar{c} \in J(N)$ and $\bar{\lambda} \in \mathcal{M}^n$ such that

$$(6) \quad \forall A \in \mathcal{A}, \quad \sum_{i \in A} \bar{\lambda}_i \bar{c}_i \leq \inf_{d \in J(A)} \sum_{i \in A} \bar{\lambda}_i d_i.$$

Proof of Theorem 2. Take $P = \emptyset$ and $w(A, \lambda) = \sigma^b(J(A); \lambda)$ which is clearly upper semi-continuous and concave on $\mathbf{R}_+^n$.

Since $J(N)$ is bounded below, $\lambda \rightarrow w(N, \lambda)$ is defined on $\mathbf{R}_+^n$ and thus, continuous on its interior $\mathring{\mathbf{R}}_+^n$. Hence assumptions (2) of Theorem 1 are satisfied. Finally, assumption (5) implies that the characteristic functions $A \rightarrow w(A, \lambda)$ are balanced, since, $\forall m \in \mathcal{M}(\tau^N)$

$$\sigma^b(J_+(N); \lambda) \leq \sum_{A \in \mathcal{A}} m(A) \sigma^b(J_+(A); \lambda),$$

i.e.

$$w(N, \lambda) = \pi w(\tau^N, \lambda).$$

Hence Theorem 2 follows from Theorem 1. $\square$

We now apply Theorem 2 to games $\{X^A, F^{A*}\}$ defined in strategic form.

Theorem 3. Consider a game $\{X^A, F^{A*}\}_{A \in \mathcal{A}}$ described in strategic form by strategy sets $X^A = \prod_{i \in A} X^i$ and by loss functions $f_i^{A*}(x^A) = \sup_{\pi_A y = x^A} f_i(y)$ where

$$(7) \quad \begin{cases} \text{(i)} \quad \forall i \in N, f_i \text{ is convex on } X^N, \\ \text{(ii)} \quad \forall i \in N, f_i \text{ is lower semi-continuous and bounded below on } X^N, \\ \text{(iii)} \quad \exists i_0 \in N \text{ such that } f_{i_0} \text{ is lower semi-compact.} \end{cases}$$

Then the canonical fuzzy core is non-empty and thus, there exists $\bar{x} \in X^N$ and $\bar{\lambda} \in \mathcal{M}^n$ such that,

$$(8) \quad \forall A \in \mathcal{A}, \quad \sum_{i \in A} \bar{\lambda}_i f_i(\bar{x}) \leq \inf_{x^A \in X^A} \sum_{i \in A} \bar{\lambda}_i f_i^{A*}(x^A).$$

Proof. We deduce Theorem 3 from Theorem 2 as follows. Assumptions (7) imply that $J_+(N)$ is convex, bounded below and closed. Also, assumptions (7(i)) and Theorem 11.2.2 imply that the characteristic functions are balanced. Hence the fuzzy core is non-empty and Theorem 1.3 implies that there exists a weak canonical cooperative equilibrium. $\square$

Theorem 3 can be extended to the case of multistrategy sets $X(A) \subset X^A$ which form a balanced family.

Theorem 4. Consider a game $\{X(A), F^{A*}\}_{A \in \mathcal{A}}$. Suppose that hypotheses (7) and

(9) the family $\{X(A)\}_{A \in \mathcal{A}}$ is balanced

hold. Then the canonical fuzzy core is non-empty and thus, there exists a weak canonical cooperative equilibrium.

Proof. This is analogous to the proof of Theorem 3. $\square$

Remark. Theorem 2 implies in particular that the core $\mathcal{C}(\mathcal{A}, J)$ of a game $(\mathcal{A}, J)$ satisfying assumptions (1) and (5) is non-empty. It is a slightly weaker result than the following Scarf theorem, which we will not prove.

Theorem 5 (Sarf). If the game $(\mathcal{A}, J)$ satisfies

(10) $J^+(N)$ is closed and bounded below

and

(11) for any balance $m \in \mathcal{M}(\tau^N)$, $\bigcap_{\{A|m(A)>0\}} (\tau^A)^{-1} J_+(A) \subset J_+(N)$,

then the core, of the game is non-empty.

Notice that assumption (5) implies assumption (11). To see this, let $c \in \bigcap_{\{A|m(A)>0\}} (\tau^A)^{-1} J_+(A)$. For any A such that $m(A) > 0$, there exists $d^A \in J_+(A)$ such that $\tau^A c \geq d^A$. Hence $c \in \sum_{A \in \mathcal{A}} m(A) \tau^A c \geq \sum_{A \in \mathcal{A}} m(A) d^A$.

But $\sum_{A \in \mathcal{A}} m(A) d^A$ belongs to $J_+(N)$ by assumption (5). Hence c belongs to $J_+(N)$.

Also Theorem 3 implies another Scarf theorem, which states that the core of the game $\{X^A, F^{A*}\}_{A \in \mathcal{A}}$ is non-empty (Sarf called the elements of such a core "cooperative equilibria").

Remark on the proof of Theorem 1. There is a small technical difficulty which complicates and lengthens the proof of the theorem. Several assumptions and several properties of the associated side-payment games will fail when λ does not belong to $\mathcal{M}^n$.

We overcome this difficulty by using an approximation device. We replace the set of multipliers $\mathcal{M}^n \times P$ by $\mathcal{M}_\varepsilon^n \times P$ where $\varepsilon < 1/n$ and

(12) $\mathcal{M}_\varepsilon^n = \{\lambda \in \mathcal{M}^n \text{ such that } \lambda_i \geq \varepsilon \text{ for all } i \in N\}$.

We define the polar cone $(\mathcal{M}_\varepsilon^n)^+$ of $\mathcal{M}_\varepsilon^n$, which contains $\mathbf{R}_+^n$ and replace $J_+(N)$ by the larger subset

$$(13) \quad J_+(N) + (\mathcal{M}_\varepsilon^n)^+.$$

We shall prove that, for any $\varepsilon > 0$, there exist $\{\lambda^\varepsilon, p^\varepsilon\} \in \mathcal{M}_\varepsilon^n \times P$ and c^ε satisfying

$$(14) \quad c^\varepsilon \in \mathcal{O}(\mathcal{A}, w(\lambda^\varepsilon, p^\varepsilon))_{\lambda^\varepsilon} \cap (J_+(N) + (\mathcal{M}_\varepsilon^n)^+).$$

This implies that c^ε is rejected by no fuzzy coalition. By letting ε converge to 0, we shall prove that c^ε converges to an element c of the fuzzy core.

We shall prove successively that the subsets $J_+(N) - \mathcal{O}(\mathcal{A}, w(\lambda, p))_{\lambda} + (\mathcal{M}_\varepsilon^n)^+$ are closed and convex and that the correspondence $\{\lambda, p\} \rightarrow \mathcal{O}(\mathcal{A}, w(\lambda, p))$ is upper semi-continuous.

In the third lemma, we prove the existence of c_ε satisfying (14). We deduce finally that c_ε converges to an element c which belongs to the fuzzy core of $(\mathcal{A}, J)$.

*12.2.2. Upper semi-continuity of the associated side-payment games

We begin by proving the following lemma.

Lemma 1. *If $U \subset \mathbf{R}^n$ is closed and bounded below, then the map $\{c, d\} \in U \times (\mathcal{M}_\varepsilon^n)^+ \mapsto c + d \in \mathbf{R}^n$ is proper. In particular, if assumption (I) holds, $\forall \lambda \in \mathcal{M}_\varepsilon^n, \forall p \in P, \forall \varepsilon > 0$,*

$$(15) \quad J_+(N) - \mathcal{O}(\mathcal{A}, w(\lambda, p))_{\lambda} + (\mathcal{M}_\varepsilon^n)^+ \text{ is closed and convex.}$$

Proof. (a) Since $\mathcal{M}_\varepsilon^n$ is the convex hull of the vectors $e_i^{*\varepsilon}$ with components

$$(16) \quad e_i^{*\varepsilon, k} = \begin{cases} \varepsilon & \text{if } k \neq i, \\ 1 - (n-1)\varepsilon & \text{if } k = i, \end{cases}$$

then its positive polar cone $(\mathcal{M}_\varepsilon^n)^+$ is spanned by the n vectors e_i^ε with components

$$(17) \quad e_i^\varepsilon, k = \begin{cases} -\frac{\varepsilon}{1-n\varepsilon} & \text{if } k \neq i, \\ \frac{1-\varepsilon}{1-n\varepsilon} & \text{if } k = i. \end{cases}$$

To see this, one has only to check that

$$\langle e_i^{*\varepsilon}, e_j^\varepsilon \rangle = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j. \end{cases}$$

Therefore, any $p^e \in (\mathcal{M}_e^n)^+$ can be written $p_i^e = \sum_{j=1}^n \lambda_j^e e_j^i$ where $\lambda_j^e \geq 0$ for all $i \in N$.

(b) Now let $U \subset \alpha + \mathbf{R}^n$ be a closed subset. We prove that if $c^m = b^m + p^m \in U + (\mathcal{M}_e^n)^+$ converges to c , we can extract convergent subsequences from $\{b^m\}_m$ and $\{p^m\}_m$. Let $p^m = \sum_{i=1}^n \lambda_i^m e_i^e$. For each $k \in N$, we have

$$\begin{aligned} c_k^m &= b_k^m + \sum_{i=1}^n \lambda_i^m e_{e,k}^i \geq \alpha_k + \sum_{i=1}^n \lambda_i^m e_{e,k}^i \\ &= \alpha_k + \frac{\lambda_k^m}{1 - n\varepsilon} - \frac{\varepsilon}{1 - n\varepsilon} \sum_{i=1}^n \lambda_i^m. \end{aligned}$$

Summing these inequalities, we obtain that

$$(18) \quad \sum_{i=1}^n \lambda_i^m \leq \sum_{k=1}^n (\alpha_k - c_k^m).$$

Since $\lambda_i^m \geq 0$ for all i , we deduce that the sequences λ_i^m are bounded. Thus subsequences of λ_i^m converge to $\lambda_i \geq 0$ for all i . Then a subsequence of p^m converges to $p \in (\mathcal{M}_e^n)^+$ and a subsequence of $b^m = c^m - p^m$ converges to $b \in J_+(N)$.

(c) Let w^* be the multiloss with components $w_i^* = \inf_{c_i \in J(i)} c_i$.

Then

$$(19) \quad \forall \lambda \in \dot{\mathcal{M}}^n, \forall p \in P, \quad \mathcal{Q}((\mathcal{A}, w(\lambda, p))_\lambda) \subset w^* - \mathbf{R}_+^n$$

since, $\forall c \in \mathcal{Q}(\mathcal{A}, w(\lambda, p))_\lambda, \forall i \in N$, we have

$$(20) \quad \lambda^i c_i \leq w(i, \lambda^i, p) \leq \sigma^b(J(i), \lambda^i) = \lambda^i w_i^*.$$

On the other hand, the map $\{c, d, p\} \in (\alpha + \mathbf{R}_+^n) \times (-w^* + \mathbf{R}_+^n) \times \mathcal{M}_e^n \rightarrow c + d + p \in \mathbf{R}^n$ is proper. Therefore, $J_+(N) - \mathcal{Q}(\mathcal{A}, w(\lambda, p))_\lambda + (\mathcal{M}_e^n)^+$ is closed since the subsets $J_+(N) \subset \alpha + \mathbf{R}_+^n$ and $-\mathcal{Q}(\mathcal{A}, w(\lambda, p))_\lambda \subset -w^* + \mathbf{R}_+^n$ are closed. It is also convex since these subsets are convex. $\square$

We shall prove that the assumptions of Theorem 1 imply that the correspondence associating with any $\{\lambda, p\} \in \mathcal{M}_e^n \times P$ the core $\mathcal{Q}(\mathcal{A}, w(\lambda, p))$ of the associated game with side-payments is upper semi-continuous.

Lemma 2. Suppose that

$$(21) \quad \begin{cases} \text{(i) } P \text{ is compact,} \\ \text{(ii) } \{\lambda, p\} \mapsto w(N, \lambda, p) \text{ is continuous on } \mathcal{M}_e^n \times P, \\ \text{(iii) } \forall A \in \mathcal{A}, \{\lambda, p\} \mapsto w(A, \lambda, p) \text{ is upper semi-continuous.} \end{cases}$$

Then the correspondence $\{\lambda, p\} \in \mathcal{M}_e^n \times P \mapsto \mathcal{Q}(\mathcal{A}, w(\lambda, p))_\lambda$ is upper semi-continuous.

Proof. To prove that the correspondence $\{\lambda, p\} \mapsto \mathcal{C}(\mathcal{A}, w(\lambda, p))_\lambda$ is upper semi-continuous on $\mathcal{M}_\varepsilon^n \times P$, we use Proposition 2.5.3 and check that its graph is closed and its values are contained in a fixed compact subset.

The graph of $\mathcal{C}(\mathcal{A}, w(\lambda, p))_\lambda$ is the set of elements $\{c, \lambda, p\} \in \mathbb{R}^n \times \mathcal{M}_\varepsilon^n \times P$ such that

$$(22) \quad \begin{cases} \text{(i)} & \langle \lambda, c \rangle - w(N, \lambda, p) = 0 \\ \text{(ii)} & \forall A \in \mathcal{A}, \langle \lambda, \tau^A \cdot c \rangle - w(A, \lambda, p) \leq 0. \end{cases}$$

Since $\{\lambda, p\} \mapsto w(N, \lambda, p)$ is continuous on $\mathcal{M}_\varepsilon^n \times P$ and the functions $w(A, \lambda, p)$ are upper semi-continuous with respect to $\{\lambda, p\}$, the graph is closed.

We prove that it maps $\mathcal{M}_\varepsilon^n \times P$ into a compact set. Let $(\lambda, p) \in \mathcal{M}_\varepsilon^n \times P$. By (20), $c \leq w^*$ whenever $c \in \mathcal{C}(\mathcal{A}, w(\lambda, p))_\lambda$. Hence, if $\lambda \in \mathcal{M}_\varepsilon^n$,

$$\begin{aligned} \varepsilon \sum_{i=1}^n |c_i - w_i^*| &= \varepsilon \sum_{i=1}^n (w_i^* - c_i) \leq \sum_{i=1}^n \lambda'_i (w_i^* - c_i) \\ &= \langle \lambda, w^* - c \rangle = \langle \lambda, w^* \rangle - w(N, \lambda, p) \\ &\leq \sup_{1 \leq i \leq n} w_i^* - \inf_{\{\lambda, p\} \in \mathcal{M}_\varepsilon^n \times P} w(N, \lambda, p) \leq M_\varepsilon < +\infty, \end{aligned}$$

since $\varepsilon \leq \lambda_i$ for all i , $\langle \lambda, c \rangle = w(N, \lambda, p)$, $\|\lambda\|_1 = 1$ and since $\{\lambda, p\} \mapsto w(N, \lambda, p)$ is continuous on the compact set $\mathcal{M}_\varepsilon^n \times P$. Thus $\sum_{i=1}^n |w_i^* - c_i| \leq \varepsilon^{-1} M_\varepsilon$ and hence, $\mathcal{C}(\mathcal{A}, w(\lambda, p))_\lambda$ is contained in the ball of center c and radius $\varepsilon^{-1} M_\varepsilon$. $\square$

*12.2.3. Existence of approximate cooperative equilibria

The Ky-Fan theorem implies the existence of approximate cooperative equilibria c^ε , i.e. of elements c^ε satisfying (14).

Lemma 3. Suppose that, for any $\varepsilon > 0$, the sets

$$(23) \quad J_+(N) - \mathcal{C}(\mathcal{A}w(\lambda, p))_\lambda - (\mathcal{M}_\varepsilon^n)^+ \text{ are closed and convex when } \lambda \in \mathcal{M}_\varepsilon^n \text{ and } p \in P.$$

Suppose also that

$$(24) \quad \begin{cases} \text{(i)} & P \text{ is compact and convex,} \\ \text{(ii)} & \text{the function } \{\lambda, p\} \mapsto w(N, \lambda, p) \text{ is concave.} \end{cases}$$

Suppose furthermore that

$$(25) \quad \begin{cases} \text{(i)} & \text{the cores } \mathcal{C}(\mathcal{A}, w(\lambda, p))_\lambda \text{ of the associated side-payment games} \\ & \text{are non-empty for any } (\lambda, p) \in \mathcal{M}_\varepsilon^n \times P. \\ \text{(ii)} & \text{the correspondence } \{\lambda, p\} \mapsto \mathcal{C}(\mathcal{A}, w(\lambda, p))_\lambda \text{ is upper semi-} \\ & \text{continuous on } \mathcal{M}_\varepsilon^n \times P. \end{cases}$$

Then there exists $\{\lambda^s, p^s\} \in \mathcal{M}_s^n \times P$ such that

$$(26) \quad \mathcal{C}(\mathcal{A}, w(\lambda^s, p^s))_{\lambda^s} \cap (J_+(N) + (\mathcal{M}_s^n)^+) \neq \emptyset.$$

Proof. Write $\mathcal{C}(\lambda, p) = \mathcal{C}(\mathcal{A}, w(\lambda, p))_{\lambda}$. We have to prove that there exists $(\lambda^s, p^s) \in \mathcal{M}_s^n \times P$ such that (26) holds, i.e. such that

$$(27) \quad 0 \in J_+(N) - \mathcal{C}(\lambda^s, p^s) + (\mathcal{M}_s^n)^+$$

Since this set is closed and convex, this amounts to proving that

$$(28) \quad \sigma^b((J_+(N) - \mathcal{C}(\lambda^s, p^s) + (\mathcal{M}_s^n)^+); \lambda) \leq \langle \lambda, 0 \rangle = 0$$

for any $\lambda \in \mathbb{R}^{n^*}$. But $\sigma^b((J_+(N) - \mathcal{C}(\lambda^s, p^s) + (\mathcal{M}_s^n)^+); \lambda) = -\infty$ when $\lambda \notin \mathcal{M}_s^n$ and, if $\lambda \in \mathcal{M}_s^n$

$$\begin{aligned} \sigma^b((J_+(N) - \mathcal{C}(\lambda^s, p^s) + (\mathcal{M}_s^n)^+); \lambda) &= \\ &= \sigma^b(J_+(N) - \mathcal{C}(\lambda^s, p^s); \lambda) = \sigma^b(J_+(N); \lambda) - \sigma^*(\mathcal{C}(\lambda^s, p^s); \lambda) \\ &= \sup_{p \in P} [w(N, \lambda, p) - \sigma^*(\mathcal{C}(\lambda^s, p^s); \lambda)]. \end{aligned}$$

Therefore, we have to prove that

$$(29) \quad \forall \{\lambda, p\} \in \mathcal{M}_s^n \times P, \quad \varphi(\{\lambda^s, p^s\}, \{\lambda, p\}) \leq 0$$

where we set

$$(30) \quad \varphi(\{\lambda^s, p^s\}, \{\lambda, p\}) = w(N, \lambda, p) - \sigma^*(\mathcal{C}(\lambda^s, p^s), \lambda).$$

For this purpose, we shall apply the Ky-Fan theorem (Theorem 7.1.3).

In the first place, $X = \mathcal{M}_s^n \times P$ is a convex compact subset.

Secondly, since the correspondence $\mathcal{C}$ which associates the core $\mathcal{C}(\lambda^s, p^s)$ with $\{\lambda^s, p^s\} \in \mathcal{M}_s^n \times P$ is upper semi-continuous, then the functions $\{\lambda^s, p^s\} \rightarrow \sigma^*(\mathcal{C}(\lambda^s, p^s); \lambda)$ are upper semi-continuous (see Proposition 2.5.1). Thus, $\varphi(\{\lambda^s, p^s\}, (\lambda, p))$ is lower semi-continuous with respect to $\{\lambda^s, p^s\}$.

Thirdly, $w(N, \lambda, p)$ is concave with respect to $\{\lambda, p\}$ by assumption (24(ii)) and $\lambda \rightarrow \sigma^*(\mathcal{C}(\lambda^s, p^s), \lambda)$ is convex. Thus, $\varphi(\{\lambda^s, p^s\}, \{\lambda, p\})$ is concave with respect to $\{\lambda, p\}$.

Finally, we have to check that

$$(31) \quad \forall \{\lambda, p\}, \quad \varphi(\{\lambda, p\}, \{\lambda, p\}) \leq 0.$$

Since $\sigma^*(\mathcal{C}(\lambda, p); \lambda) = \sup_{c \in \mathcal{C}(\lambda, p)} \langle \lambda, c \rangle$ and $\langle \lambda, c \rangle = w(N, \lambda, p)$ when c belongs to $\mathcal{C}(\lambda, p)$, we obtain that $\varphi(\{\lambda, p\}, \{\lambda, p\}) = w(N, \lambda, p) - \sigma^*(\mathcal{C}(\lambda, p); \lambda) = 0$. Therefore, by the Ky-Fan theorem, there exists a solution $\{\lambda^s, p^s\}$ of (27), i.e. of (26). $\square$

**12.2.4. Proof of the non-emptiness of the core*

We can now prove Theorem 1.

Since the assumptions of Theorem 1 imply the assumptions of Lemmas 1, 2 and 3, we know that there exists a sequence $\{c^e\}_e$ of elements

$$(32) \quad c^e \in (J_+(N) + (\mathcal{M}_e^n)^+) \cap \mathcal{O}(\lambda^e, p^e)$$

where we set $\mathcal{O}(\lambda, p) = \mathcal{O}(\mathcal{A}, w(\lambda, p))_\lambda$.

Let $\varepsilon \leq \bar{\varepsilon} < 1/n$. Since the map $\{c, d, p\} \in \{\alpha + \mathbf{R}_+^n\} \times (-w^* + \mathbf{R}_+^n) \times (\mathcal{M}_e^n)^+ \rightarrow c + d + p \in \mathbf{R}^n$ is proper, we deduce that $(J_+(N) + (\mathcal{M}_e^n)^+ \cap (w^* - \mathbf{R}_+^n))$ is compact.

Therefore c^e stays in a compact subset and we can extract a subsequence which converges to an element $c \in \mathbf{R}^n$.

(a) Since $c^e \in J_+(N) + (\mathcal{M}_e^n)^+ \subset J_+(N) + (\mathcal{M}_e^n)^+$, it can be written $c^e = d^e + p^e$ where $d^e \in J_+(N)$ and $p^e \in (\mathcal{M}_e^n)^+$. By Lemma 1, subsequences of d^e and p^e converges to $d \in J_+(N)$ and $p \in (\mathcal{M}_e^n)^+$ respectively. Actually, since $p^e \in (\mathcal{M}_e^n)^+$, it is easy to check that the limit $p \in \mathbf{R}^+$. Hence $c = d + p \in J_+(N)$.

(b) On the other hand, since $c^e \in \mathcal{O}(\lambda^e, p^e)$, we have

$$(33) \quad \langle \lambda^e, \tau \cdot c^e \rangle \leq \pi w(\tau, \lambda^e, p^e) \quad \text{for any } \tau \in \mathcal{T}.$$

This implies that $\tau \cdot c^e \notin \pi_w J_+(\tau) + \dot{\mathbf{R}}_+^r = \pi_w \dot{J}_+(\tau) = \text{Int}(\pi_w J_+(\tau))$. Therefore, $c^e \in \bigcap_{\tau \in \mathcal{T}} \tau^{-1}(\mathbb{C} \pi_w \dot{J}_+(\tau))$.

Since this set is closed, the limit c belongs to $\bigcap_{\tau \in \mathcal{T}} \tau^{-1}(\mathbb{C} \pi_w \dot{J}_+(\tau))$. In other words, for any coalition τ , $\tau \cdot c \notin \pi_w \dot{J}_+(\tau)$, i.e. c is not rejected by τ . We have proved that c belongs to the fuzzy core.

12.3. Equivalence between the fuzzy core of an economy and the set of Walras allocations

We shall show that the fuzzy economic game is obtained from the economic game by a convenient representation and that the equilibria of this representation are the Walras allocations. We will then deduce from the equivalence theorem that the fuzzy core of an economy and the set of Walras equilibria (almost) coincide.

12.3.1. Representation of economic games

Consider the game $(\mathcal{A}, J)$ and the fuzzy game $(\mathcal{T}, J)$ associated with an economy $\{R^i, Y(i), f_i\}_{i \in N}$ defined by

$$(1) \quad J(\tau) = F^\tau(X(\tau)); \quad F^\tau(x) = \{\tau_i f_i(x^i)\}_{i \in A_\tau}$$

where

$$(2) \quad X(\tau) = \{x \in R^{I^*} \text{ such that } \sum_{i \in I^*} \tau_i x^i \in Y(\tau) = \sum_{i \in N} \tau_i Y(i)\}$$

(see Section 10.4.7).

Since

$$(3) \quad \sigma^b(J(\tau); \lambda) = \inf_{x \in X(\tau)} \sum_{i \in N} \lambda_i \tau_i f_i(x^i)$$

and $X(\tau)$ is represented by the budgetary constraints

$$\left\langle p, \sum_{i \in N} \tau_i x^i \right\rangle - \sum_{i \in N} \tau_i r_i(p) \leq 0$$

as p ranges over R^{I^*} (and where $r_i(p) = \sup_{y \in Y(i)} \langle p, y \rangle$), it is natural to introduce the Lagrangian defined by $\sum_{i \in N} \tau_i (\lambda^i f_i(x^i) + \langle p, x^i \rangle - r_i(p))$ and the utility function of the dual problem defined by

$$(4) \quad w(\tau, \lambda, p) = \sum_{i \in N} \tau_i w(i, \lambda^i, p)$$

where

$$(5) \quad w(i, \lambda^i, p) = \inf_{x^i \in R^i} [\lambda^i f_i(x^i) + \langle p, x^i \rangle - r_i(p)].$$

Therefore duality Proposition 5.3.4 implies that $(\overline{\mathcal{C}}, w(\mathcal{M}^n \times P))$ is a representation of the fuzzy economic game.

Proposition 1. Suppose that, for all $i \in N$,

$$(6) \quad \begin{cases} \text{(i) } R^i \text{ and } f_i \text{ are convex,} \\ \text{(ii) } Y(i) = Y(i) - R_+^i \text{ is convex,} \\ \text{(iii) } 0 \in R^i - \text{Int } Y(i) \end{cases}$$

and that

$$(7) \quad \text{the functions } f_i \text{ are bounded below on } X(N).$$

Then the functions $w(\lambda, p)$ represent the economic fuzzy game. Furthermore, there exists a convex compact subset P of R^{I^*} such that, $\forall \lambda \in \mathcal{M}^n$,

$$(8) \quad \forall \tau \in \overline{\mathcal{C}}, \max_{p \in P} w(\tau, \lambda, p) = \sigma^b(J(\tau); \lambda).$$

Proof. Since the functions f_i are convex and since $0 \in R^i - \text{Int } Y(i) \subset \text{Int } (R^i - Y(i))$ for all i , we deduce that $0 \in \text{Int } (\sum_{i \in N} \tau_i R^i - Y(\tau))$ for all fuzzy coal-

tions τ . Then Proposition 5.3.4 implies the existence of elements $p_{\lambda, \tau} \in \mathbf{R}^{I*}$ such that

$$(9) \quad \sigma^b(J(\tau); \lambda) = w(\tau, \lambda, p_{\lambda, \tau}) = \sup_{p \in \mathbf{R}^{I*}} w(\tau, \lambda, p).$$

Since $\mathbf{R}_+^{I*}$ is the common barrier cone of the subsets $Y(\tau)$ (since $Y(i) = Y(i) - \mathbf{R}_+^I$ for any i), $p_{\lambda, \tau}$ belongs to $\mathbf{R}_+^{I*}$.

It remains to prove that the Lagrange multipliers stay in a bounded set B . In this case, we take $P = \overline{\text{co}}(B)$, which is convex and compact. For this purpose, it is sufficient to prove that, for any $z \in \mathbf{R}^I$, there exists $m(z) < \infty$ such that

$$(10) \quad \sup_{\lambda \in \mathcal{M}^*, \tau \in \mathcal{T}} \langle p_{\lambda, \tau}, z \rangle \leq m(z).$$

We use again the fact that $0 \in \text{Int}(Y(i) - \mathbf{R}_+^I)$.

We can associate with z a scalar $\theta > 0$, elements $x^i \in \mathbf{R}^I$ and $y_i \in Y(i)$ such that $\theta^{-1}z = y^j - x^j$. Thus

$$(11) \quad (\theta^{-1}\Sigma\tau_i)z = y - \sum_{i \in N} \tau_i x^i \quad \text{where } y = \sum_{i \in N} \tau_i y^i \in Y(\tau).$$

Therefore, since $\sum_{i \in N} \tau_i \geq 1$ (for $\tau \in \overline{\mathcal{T}}$), we deduce that

$$\begin{aligned} \langle p_{\lambda, \tau}, z \rangle &\leq \theta \left[\langle p_{\lambda, \tau}, y \rangle - \sum_{i \in N} \tau_i \langle p_{\lambda, \tau}, x^i \rangle \right] \\ &\leq \theta \left[\sum_{i \in N} \tau_i r_i(p_{\lambda, \tau}) - \sum_{i \in N} \tau_i \langle p_{\lambda, \tau}, x^i \rangle \right] \quad (\text{since } y \in Y(\tau)) \\ &\leq \theta \left[\sum_{i \in N} \lambda^i \tau_i f_i(x^i) - w(\tau, \lambda, p_{\lambda, \tau}) \right] \end{aligned}$$

(by definition of $w(\tau, \lambda, p_{\lambda, \tau})$). Hence (9) implies that

$$(12) \quad \langle p_{\lambda, \tau}, z \rangle \leq \theta \left[\sum_{i \in N} \lambda^i \tau_i f_i(x^i) - \sigma^b(J(\tau); \lambda) \right].$$

Since x^i depends only upon z , we deduce that

$$(13) \quad \sum_{i \in N} \lambda^i \tau_i f_i(x^i) \leq \sup_{i \in N} |f_i(x^i)| = m_1(z).$$

Now, we use again the convexity of f_i and the constraint qualification assumption (6(iii)), which implies the existence of $w^i \in \mathbf{R}^I \cap Y(i)$. Indeed if $x^i \in X(\tau)$, then $\sum_{i \in N} \tau_i x^{i,i} + (1 - \tau_i) w^i \in \sum_{i \in N} Y(i) = Y(N)$. Furthermore, $y^j = \tau_j x^{j,j} + (1 - \tau_j) w^j \in \mathbf{R}^I$ for all i . Hence the allocation $y = \{y^1, \dots, y_n\}$ belongs to $X(N)$. Assumption (7) implies that $\forall i \in N, \alpha_i \leq f_i(y^i) \leq \tau_i f_i(x^{i,i}) + (1 - \tau_i) f_i(w^i)$.

Therefore, for all $x^i \in X(\tau)$ and $\lambda \in \mathcal{M}^n$, we have

$$(14) \quad \sum_{i=1}^n \lambda^i (\alpha^i - f_i(w^i)) \leq \sigma^b(J(\tau); \lambda) = \inf_{x^i \in X(\tau)} \sum_{i \in N} \lambda^i \tau_i f_i(x^i, i).$$

Finally, we deduce from (12), (13) and (14) that for any $z \in \mathbb{R}^I$

$$(15) \quad \langle p_{\lambda, \tau}, z \rangle \leq \theta [m_1(z) + \sup_{i \in N} (f_i(w^i) - \alpha^i)].$$

Hence $p_{\lambda, \tau}$ stays in a bounded subset. $\square$

12.3.2. Fuzzy core and Walras allocations

In particular, the characteristic functions

$$(16) \quad A \mapsto w(A, \lambda, p) = w(\tau^A, \lambda, p) = \sum_{i \in A} w(i, \lambda^i, p)$$

represent the economic game. Since they are linear, their convex cover consists of the characteristic functions $\tau \mapsto w(\tau, \lambda, p)$, which represent the fuzzy economic game by Proposition 1.

Therefore, the cover of the economic game by means of the representation defined by (16) is the fuzzy economic game and the fuzzy core of this representation is the fuzzy core of the game defined in Section 10.4.7 by (25).

Now, we check that the cooperative equilibria of the representation defined by (16) are the multilosses of Walras allocations.

Proposition 2. *Let $\bar{c} = F(\bar{x})$ (where $\bar{x} \in X(N)$) be a (weak) cooperative equilibrium of the representation (16) of the economic game. Then, if $\{\bar{\lambda}, \bar{p}\} \in \mathcal{M}^n \times P$ is an associated pair of multipliers, we have, $\forall i \in N$,*

$$(17) \quad \begin{cases} \text{(i)} & \langle \bar{p}, \bar{x}_i \rangle = r_i(\bar{p}), \\ \text{(ii)} & \forall i \in N, \bar{\lambda}^i f_i(\bar{x}^i) = \min_{x^i \in B_i(\bar{p}, r_i(\bar{p}))} \bar{\lambda}^i f_i(x^i). \end{cases}$$

In particular, if $\bar{c} = F(\bar{x})$ is a "strong equilibrium", then $\{\bar{x}, \bar{p}\}$ is a Walras equilibrium.

Proof. For all coalitions $\{i\}$, we obtain that

$$\bar{\lambda}^i \bar{c}_i = \bar{\lambda}^i f_i(\bar{x}^i) \leq w(i, \bar{\lambda}^i, \bar{p}) \leq \bar{\lambda}^i f_i(\bar{x}^i) + \langle \bar{p}, \bar{x} \rangle - r_i(\bar{p}).$$

Then $\langle \bar{p}, \bar{x}^i \rangle - r_i(\bar{p}) \geq 0$ for all $i \in N$. Since $\bar{x} \in X(N)$, we deduce that $\sum_{i \in N} (\langle \bar{p}, \bar{x} \rangle - r_i(\bar{p})) \leq 0$. Therefore (17(i)) holds.

If $x^i \in B_i(\bar{p}, r_i(\bar{p}))$, then $\langle \bar{p}, x^i \rangle - r_i(\bar{p}) \leq 0$ and

$$\bar{\lambda}^i f_i(\bar{x}^i) \leq w(i, \bar{\lambda}, \bar{p}) \leq \bar{\lambda}^i f_i(x^i) + \langle \bar{p}, x^i \rangle - r_i(\bar{p}) \leq \bar{\lambda}^i f_i(x_i).$$

Thus (17(ii)) holds.

If $\bar{\lambda}^i > 0$, we deduce that

$$(18) \quad f_i(\bar{x}^i) = \min_{x^i \in B_i(\bar{p}, r_i(\bar{p}))} f_i(x^i). \quad \square$$

We are led to introduce the following definition.

Definition 1. We shall say that $\{\bar{x}, \bar{p}\} \in X(N) \times P$ is a “*weak Walras equilibrium*” if there exists $\lambda \in \mathcal{M}^n$ such that (17) holds.

12.3.3. The equivalence theorem

The above results imply the following equivalence theorem.

Theorem 1. Any Walras allocation belongs to the fuzzy core of an economy. Conversely, if assumptions (6) and (7) of Proposition 1 hold, the fuzzy core is contained in the set of weak Walras allocations.

Proof. The first statement follows from Proposition 10.5.3. The converse statement is a consequence of Proposition 1 and equivalence Theorem 1.1. $\square$

CHAPTER 13

MINIMAX TYPE INEQUALITIES, MONOTONE CORRESPONDENCES AND γ -CONVEX FUNCTIONS

In this chapter we shall improve the minimax type inequalities of Section 7.1 (which played a fundamental role in game theory). We recall that both the existence theorem for an optimal decision rule (and thus, the Ky-Fan theorem) and the minisup theorem were proved by estimating

$$v \sqcap = \sup_{K \in \mathcal{S}} \inf_{x \in X} \sup_{y \in K} \varphi(x, y)$$

(where $\mathcal{S}$ denotes the family of finite subsets K of Y) and by using the existence theorem for a conservative solution in the particular form

$$(*) \quad \sup_{y \in Y} \varphi(\bar{x}, y) = v \sqcap.$$

The problem arises as to whether it is possible to weaken and/or replace assumptions implying (*).

For this purpose, we devise another proof of (*). We first observe that there *always* exists a generalized sequence $\{x_\mu\}$ of elements x_μ of X such that

$$(**) \quad \forall y \in Y, \quad \limsup_{\mu} \varphi(x_\mu, y) \leq v^+.$$

Then we point out that convenient *compactness assumptions* imply that there exists a sequence x_μ satisfying (**) and

$$(***) \quad x_\mu \text{ stays in a compact set and converges to } \bar{x}.$$

Finally, convenient *continuity assumptions* imply that any sequence $\{x_\mu\}_\mu$ satisfying (**) and (***) converges to solution $\bar{x}$ of (*).

In the first section of this chapter, we prove (*) and analogous inequalities according to the above scheme. We also devote our attention to the relaxation of the compactness assumptions. Recall the compactness assumption of Section 7.1, i.e. that there exists y_0 such that

$$0 \in \text{Int } \text{Dom } \varphi_X^*(\cdot, y_0)$$

where we set $\varphi_X^*(p, y_0) = \sup_{x \in X} (\langle p, x \rangle - \varphi(x, y_0))$.

This is replaced by the much weaker assumption

$$0 \in \text{Int} \bigcup_{y \in Y} \text{Dom } \varphi_X^*(\cdot, y).$$

In the second section, we relax the continuity assumption of Section 7.1 [$\forall y \in Y, x \rightarrow \varphi_X(y, y)$ is lower semi-continuous] in order to generalize the Ky-Fan theorem. We define pseudo-monotone functions $\varphi: X \times X \rightarrow \mathbf{R}$ to be functions such that for any compact generalized sequence $\{x_\mu\}$ converging to $\bar{x}$ and satisfying $\lim_\mu \sup \varphi(x_\mu, \bar{x}) \leq 0$, we obtain

$$\forall y \in X, \quad \varphi(\bar{x}, y) \leq \liminf_\mu \varphi(x_\mu, y).$$

This condition is obviously (much) weaker than the lower semi-continuity of φ with respect to x .

Nevertheless, we will prove a result due to Brézis–Nirenberg–Stampacchia. This asserts that, if $x \rightarrow \varphi(x, y_0)$ is coercive for some y_0 , if the functions $y \rightarrow \varphi(x, y)$ are quasi-concave, if $\sup_{y \in X} \varphi(y, y) = 0$ and if φ is *pseudo-monotone and lower semi-continuous with respect to x for the finite topology*, then there exists $\bar{x} \in X$ such that

$$\sup_{y \in \bar{X}} \varphi(\bar{x}, y) \leq 0.$$

This result generalizes in a useful way the Ky-Fan inequality. Indeed, we apply it to the case where

$$\varphi(x, y) = \langle L(x), x - y \rangle$$

(where L maps a closed convex set X of U into the dual U^*).

We notice that a *completely-continuous* map L (continuous from X supplied with the *weak* topology into U^* supplied with the *strong* topology) implies the lower semi-continuity of φ with respect to x whereas a *monotone* operator L , i.e. satisfying

$$\forall x, y \in X, \quad \langle L(x) - L(y), x - y \rangle \geq 0$$

implies the pseudo-monotonicity of φ . This is important since the gradient of a convex function is monotone, but not completely continuous. Such results imply in particular that *any coercive bounded monotone and finitely continuous operator L from U into U^* is surjective*. We shall extend such results to the case of correspondences and end Section 13.2 with a very short introduction to (maximal) monotone correspondences. In the third section of this chapter, we replace the convexity assumptions by other conditions to prove the estimates

$$(\text{****}) \quad v \square \leq \inf_{C \in \mathcal{C}(Y, X)} \sup_{y \in Y} \varphi(C(y), y)$$

and

$$(\text{*****}) \quad v^{\square} \leq v^b = \sup_{y \in Y} \inf_{x \in X} \varphi(x, y).$$

More precisely, we observe that, in our previous consideration of these inequalities, the convexity assumption was largely made use of via the barycentric inequalities

$$\forall m \in \mathcal{M}(X), \quad f(\beta m) \leq \langle m, f \rangle$$

using only the fact that the barycentric operator β maps $\mathcal{M}(X)$ into X (and using neither the convexity of X nor the explicit definition of βm).

We are led to associate with any map γ from $\mathcal{M}(X)$ into X the cone of γ -convex functions f defined by

$$\forall m \in \mathcal{M}(X), \quad f(\gamma m) \leq \langle m, f \rangle.$$

We prove in particular that if the components f_i of a map F from X into $\mathbf{R}^n$ are γ -convex, then $F(X) + \mathbf{R}_+^n$ is convex. This latter property played a fundamental role in game theory. We will prove the converse result, i.e. if $F(X) + \mathbf{R}_+^n$ is convex, there exists a map γ from $\mathcal{M}(X)$ into X such that the components f_i of F are γ -convex.

This allows us to prove inequality (****) when φ is γ_Y -concave with respect to y (and lower semi-continuous with respect to x) and inequality (*****) when φ is also γ_X -convex with respect to x and γ_Y -concave with respect to y .

Actually, we will prove that there exists a vector space W and a map π from X into W such that $\pi(X)$ is convex and that $f: X \rightarrow \mathbf{R}$ is γ -convex if and only if there exists a convex function g on $\pi(X)$ such that $f = g \circ \pi$.

13.1. Relaxation of compactness assumptions

13.1.1. Existence of a conservative solution

Recall that the two fundamental minimax inequalities (Theorem 7.1.2 for the existence of an optimal decision rule and minisup Theorem 7.1.5) were obtained from Theorem 7.1.1 asserting the existence of $\bar{x} \in X$ satisfying

$$(1) \quad \sup_{y \in Y} \varphi(\bar{x}, y) = v^{\square} = \sup_{K \in \mathcal{B}} \inf_{x \in X} \sup_{y \in K} \varphi(x, y)$$

by estimating this number $v^{\square}$ by $\inf_{C \in \mathcal{C}(X, X)} \sup_{y \in Y} \varphi(C(y), y)$ and $\sup_{y \in Y} \inf_{x \in X} \varphi(x, y)$ respectively.

Therefore, if we want to weaken the assumptions of such results, we need to begin by weakening the assumptions of Theorem 7.1.1. Actually, we will replace the covering $\mathcal{S}$ of Y by finite subsets by any covering $\mathcal{A}$ of Y satisfying

- (2) if K and L belong to $\mathcal{A}$, then $K \cup L$ belongs to $\mathcal{A}$.

We associate with such a covering the scalar

$$(3) \quad v^{\square}(\mathcal{A}) = \sup_{K \in \mathcal{A}} \inf_{x \in X} \sup_{y \in K} \varphi(x, y).$$

Notice that

$$(4) \quad v^{\square} = v^{\square}(\mathcal{S}) \leq v^{\square}(\mathcal{A}) \leq v^{\square}(\mathcal{B}) \leq v^{\square}(\mathcal{U})$$

when $\mathcal{S} \subset \mathcal{A} \subset \mathcal{B}$ and where $\mathcal{U}$ denotes the covering of Y by *all* subsets of Y . Theorem 7.1.1 is generalized as follows.

Theorem 1. *Let X be a subset of a topological space $\mathcal{U}$ and let $\varphi : X \times Y \rightarrow \mathbf{R}$ be a function such that $v^{\square}(\mathcal{A})$ is finite.*

Suppose that

- (5) *there exists $K_0 \in \mathcal{A}$ such that $x \mapsto \sup_{y \in K_0} \varphi(x, y)$ is lower semi-compact on X*

and that,

- (6) *$\forall y \in Y$, the function $x \mapsto \varphi_X(x, y) = \varphi(x, y)$ if $x \in Y$, and $+\infty$ if $x \notin X$ is lower semi-continuous on U .*

Then there exists $\bar{x} \in X$ such that

$$(7) \quad \sup_{y \in Y} \varphi(\bar{x}, y) = v^{\square}(\mathcal{A}).$$

In order to weaken either assumption (5) or assumption (6) (or both), we will replace the proof of Theorem 7.1.1 by a longer one, which use three lemmas.

The first states that there exists a generalized sequence x_{μ} such that

$$(8) \quad \forall y \in Y, \quad \lim_{\mu} \sup \varphi(x_{\mu}, y) \leq v^{\square}(\mathcal{A}).$$

The second lemma shows that the lower semi-compactness of some function $x \mapsto \sup_{y \in K_0} \varphi(x, y)$ implies that there exists a generalized sequence satisfying (8) and

$$(9) \quad x_{\mu} \text{ stays in a compact subset and converges to an element } \bar{x} \in U.$$

The third lemma shows that the lower semi-continuity of the proper functions $x \rightarrow \varphi_X(x, y)$ is a sufficient condition ensuring that

- (10) whatever the generalized sequence $\{x_\mu\}_\mu$ satisfying the properties (8) and (9), the limit $\bar{x}$ of $\{x_\mu\}_\mu$ satisfies $\sup_{y \in Y} \varphi(\bar{x}, y) \leq v^\square(\mathcal{A})$.

Remark. Therefore, any assumption implying (9) (called a “compactness assumption”) and any assumptions implying (10) (called a “continuity assumption”) can replace assumptions (5) and (6) of Theorem 1.

13.1.2. Proof of existence of a conservative solution

We begin by proving the first lemma, which does not need any assumption.

Lemma 1. *There exists a generalized sequence $\{x_\mu\}_\mu$ of elements x_μ of X such that*

$$(11) \quad \forall K \in \mathcal{A}, \quad \limsup_\mu \left[\sup_{y \in K} \varphi(x_\mu, y) \right] \leq v^\square(\mathcal{A}).$$

If $\mathcal{A}$ is spanned by a denumerable covering, there exists a denumerable sequence $\{x_n\}_{n \in \mathbb{N}}$ satisfying (11).

Proof. By the very definition of $v^\square(\mathcal{A})$, inequalities

$$(12) \quad \inf_{x \in X} \sup_{y \in K} \varphi(x, y) \leq v^\square(\mathcal{A})$$

hold for all subsets K belonging to $\mathcal{A}$.

Therefore, we can associate with any $n \in \mathbb{N}$ an element $x_{K, n}$ belonging to X such that

$$(13) \quad \sup_{y \in K} \varphi(x_{K, n}, y) \leq v^\square(\mathcal{A}) + \frac{1}{n}.$$

On the other hand, the set $\mathcal{M} = \mathcal{A} \times \mathbb{N}$ preordered by the relation

$$(14) \quad (K_1, n_1) < (K_2, n_2) \quad \text{if and only if} \quad K_1 \subset K_2 \text{ and } n_1 \leq n_2$$

is “filtered” or “directed” (any pair $\{(K_i, n_i)\}_{i=1,2}$ of $\mathcal{M}$ has an upper bound $(K_1 \cup K_2, \max(n_1, n_2))$). Therefore, the family $(K, n) \in \mathcal{M} \rightarrow x_{K, n} \in X$ is a generalized sequence. By taking $L \subset K$ and $m \geq n$, we deduce from (13) that

$$(15) \quad \sup_{y \in K} \varphi(x_{L, m}, y) \leq \sup_{y \in L} \varphi(x_{L, m}, y) \leq v^\square(\mathcal{A}) + \frac{1}{m} \leq v^\square(\mathcal{A}) + \frac{1}{n}.$$

In other words, if we denote by $\mu = (K, n)$ the indices of $\mathcal{M}$ and by $x_\mu = x_{K,n}$ the elements of the generalized sequence, inequality (15) can be translated as follows.

With any $K \in \mathcal{A}$ and $n \in \mathbb{N}$, we can associate an index $\mu_0(n)$ (namely, $\mu_0(n) = (K, n)$) such that, for any $\nu \geq \mu_0(n)$,

$$\sup_{\nu \geq \mu_0(n)} \sup_{y \in K} \varphi(x_\nu, y) \leq v^\square(\mathcal{A}) + \frac{1}{n}.$$

(Take $\nu = (L, m)$ with $L \subset K$ and $m \geq n$.) Therefore,

$$(16) \quad \limsup_\nu \left(\sup_{y \in K} \varphi(x_\nu, y) \right) = \inf_{\mu \in \mathcal{M}} \sup_{\nu \geq \mu} \left(\sup_{y \in K} \varphi(x_\nu, y) \right) \leq v^\square(\mathcal{A})$$

It is clear that, if $\mathcal{A}$ is a denumerable covering satisfying (2), then the sequence x_μ is a denumerable sequence since $\mathcal{M} = \mathcal{A} \times \mathbb{N}$ is denumerable. $\square$

Lemma 2. *Suppose that the compactness assumption (5) holds. Then there exists a sequence x_μ satisfying (11) and*

$$(17) \quad \{x_\mu\}_\mu \text{ stays in a compact subset and converges to an element } \bar{x} \in U.$$

Proof. Take $K = K_0$, where K_0 is defined in the compactness assumption (5). By Lemma 1, there exists a generalized sequence $\{x_\mu\}_\mu$ of elements x_μ of X such that

$$(18) \quad \limsup_\mu \left(\sup_{y \in K_0} \varphi(x_\mu, y) \right) = \inf_{\mu \in \mathcal{M}} \sup_{\nu \geq \mu} \left(\sup_{y \in K_0} \varphi(x_\nu, y) \right) \leq v^\square(\mathcal{A}).$$

Thus, we can associate with any $n > 0$ an index $\mu_0(n)$ such that

$$(19) \quad \sup_{\nu \geq \mu_0(n)} \sup_{y \in K_0} \varphi(x_\nu, y) \leq v^\square(\mathcal{A}) + \frac{1}{n}.$$

In other words, for all $\nu \geq \mu_0(n)$, the element x_ν belongs to a section of the function $x \mapsto \sup_{y \in K_0} \varphi(x, y)$ and so, by the compactness assumption (5), x_ν lies in a relatively compact subset of U . Therefore, we can extract a generalized subsequence $\{x_{\mu_\nu}\}$ which converges to an element $\bar{x}$ of the closure $\bar{X}$ of X in U . Furthermore,

$$(20) \quad \limsup_\nu \left(\sup_{y \in K} \varphi(x_{\mu_\nu}, y) \right) \leq \limsup_\mu \left(\sup_{y \in K} \varphi(x_\mu, y) \right) \leq v^\square(\mathcal{A}).$$

Hence the sequence $\{x_{\mu_\nu}\}_\nu$ satisfies properties (11) and (17). $\square$

Lemma 3. *Suppose that the continuity assumption (6) holds. Then, for any generalized sequence $\{x_\mu\}$ of elements $x_\mu \in X$ satisfying (11) and (17), its limit $\bar{x}$ satisfies*

$$(21) \quad \sup_{y \in Y} \varphi(\bar{x}, y) = v^\square(\mathcal{A}).$$

Proof. Consider a generalized sequence $\{x_\mu\}$ of elements $x_\mu \in X$ converging to $\bar{x} \in X$ such that, for any $y \in Y$,

$$(22) \quad \limsup_{\mu} \varphi_X(x_\mu, y) = \limsup_{\mu} \varphi(x_\mu, y) \leq v^\square(\mathcal{A})$$

(since $\varphi_X(x, y) = \varphi(x, y)$ whenever $x \in X$).

Since $x \mapsto \varphi_X(x, y)$ is lower semi-continuous, we deduce that

$$(23) \quad \varphi_X(\bar{x}, y) \leq \liminf_{\mu} \varphi_X(x_\mu, y) \leq \limsup_{\mu} \varphi_X(x_\mu, y) \leq v^\square(\mathcal{A}).$$

Since $v^\square(\mathcal{A})$ is finite, this implies that $\bar{x} \in X$ and thus, that $\varphi(\bar{x}, y) = \varphi_X(\bar{x}, y) \leq v^\square(\mathcal{A})$. $\square$

13.1.3. Existence of optimal decision rules and minisup under weaker compactness assumptions

We give an example of compactness assumptions in the case when

$$(24) \quad \begin{aligned} &U = F^* \text{ is the dual of a barreled space } F, \\ &\text{supplied with the weak topology } \sigma(U, F) \end{aligned}$$

or

$$(25) \quad \begin{aligned} &U \text{ and } U^* \text{ are paired spaces, } U \text{ is supplied with the weak topology} \\ &\sigma(U, U^*) \text{ and } U^* \text{ is supplied with the Mackey topology } \tau(U^*, U). \end{aligned}$$

Then, Propositions 3.1.9 and 3.1.10 state that the condition

$$(26) \quad \exists y_0 \in Y \text{ such that } p \mapsto \varphi_X^*(p, y_0) = \sup_{x \in X} [\langle p, x \rangle - \varphi(x, y_0)] \text{ is continuous at } 0$$

implies the compactness assumption

$$(27) \quad \exists y_0 \in Y \text{ such that } x \mapsto \varphi(x, y_0) \text{ is lower semi-compact.}$$

Recall also that condition (26) is equivalent to

$$(28) \quad 0 \in \text{Int Dom } \varphi_X^*(\cdot, y_0)$$

We will relax this assumption and prove that conclusions of Theorems 7.1.2 and 7.1.5 remain valid when we replace (28) by the (much) weaker condition

$$(29) \quad 0 \in \text{Int} \bigcup_{y \in Y} \text{Dom } \varphi_X^*(\cdot, y)$$

Example. In the case where $\bar{x}$ is defined by

$$(30) \quad \varphi(x, y) = \langle L(y), x \rangle + \psi(y), \quad \text{where } L \text{ maps } Y \text{ into } U^*,$$

assumption (28) amounts to writing that

$$(31) \quad \exists y_0 \in Y \text{ such that } L(y_0) \in \text{Int}(X^b)$$

(where X^b is the positive polar subset of X), because

$$\varphi_X^*(p, y) = \sigma^*(X; p - L(y)) - \psi(y).$$

This assumption requires in particular that the interior of X^b is not empty.

But the weaker assumption (29) in this case becomes

$$(32) \quad 0 \text{ belongs to } \text{Int}[L(Y) - X^b],$$

which can have a meaning even when the interior of X^b is empty.

We state and prove the following generalization of Theorems 7.1.2 and 7.1.5.

Theorem 2. *Suppose that*

$$(33) \quad \begin{cases} \text{(i) } Y = \bigcup_{n=0}^{\infty} K_n \text{ is a countable union of an (increasing) sequence} \\ \text{of compact subsets } K_n \subset Y, \\ \text{(ii) } \forall x \in X, y \mapsto \varphi(x, y) \text{ is upper semi-continuous.} \end{cases}$$

We make the following compactness assumption

$$(34) \quad \begin{cases} \text{(i) } X \text{ is a subset of the dual } U = F^* \text{ of a barreled space } F, \\ \text{(ii) } 0 \in \text{Int} \bigcup_{y \in Y} \text{Dom } \varphi_X^*(\cdot, y) \end{cases}$$

the continuity assumption

$$(35) \quad \forall y \in Y, x \mapsto \varphi_X(x, y) \text{ is lower semi-continuous on the space } U = F^* \text{ supplied with the weak } ^* \text{ topology}$$

and the concavity assumption

$$(36) \quad \begin{cases} \text{(i) } Y \text{ is convex,} \\ \text{(ii) } \forall x \in X, y \mapsto \varphi(x, y) \text{ is concave.} \end{cases}$$

Let $\mathcal{C}(Y, X)$ denote the set of continuous decision rules mapping Y into X . Then there exist a solution $\bar{x} \in X$ of

$$(37) \quad \sup_{y \in Y} \varphi(\bar{x}, y) = \inf_{C \in \mathcal{C}(Y, X)} \sup_{y \in Y} \varphi(C(y), y).$$

Furthermore, if we make the convexity assumption

$$(38) \quad \begin{cases} \text{(i) } X \text{ is convex,} \\ \text{(ii) } \forall y \in Y, x \mapsto \varphi(x, y) \text{ is convex,} \end{cases}$$

then there exists a minisup $\bar{x} \in X$ of φ , i.e.

$$(39) \quad \sup_{y \in Y} \varphi(\bar{x}, y) = \sup_{y \in Y} \inf_{x \in X} \varphi(x, y).$$

We give examples where Y can be covered by a countable sequence of compact spaces.

Example 1. Let $V = G^*$ be the dual of the Banach space G . Then $V = \bigcup_{n=1}^{\infty} B_n$, where the balls B_n of radius n are weakly compact. Therefore, any subsets Y of V can be written

$$Y = \bigcup_{n=0}^{\infty} K_n \quad \text{where } K_n = Y \cap B_n.$$

Example 2. Let $Y \subset V = G^*$ be a closed convex cone such that

$$(40) \quad \text{The cone } Y^+ \text{ has a non-empty interior (for the Mackey topology } \tau(G, V)).$$

Therefore, if $p_0 \in \text{Int}(Y^+)$, the subsets

$$(41) \quad Y_n = \{y \in Y \text{ such that } \langle p_0, y \rangle \leq n\}$$

are compact and form a covering of Y . (See Proposition 3.1.5 and 3.1.9.)

Example 3. More generally, let $Y \subset V = G^*$ be a closed convex subset such that

$$(42) \quad \exists p_0 \text{ belonging to the interior of the positive polar subset } Y^b \text{ of } Y \text{ (for the Mackey topology } \tau(G, V)).$$

Then the subsets

$$(43) \quad Y_n = \{y \in Y \text{ such that } \langle p_0, y \rangle \leq n\}.$$

are compact and form a covering of Y . (See Proposition 3.1.5 and 3.1.9.)

Theorem 2 is the combination of

(a) a proposition stating that compactness and continuity assumptions (34) and (35) imply the existence of $\bar{x} \in X$ such that

$$(44) \quad \sup_{y \in Y} \varphi(\bar{x}, y) = \sup_{n \in \mathbb{N}} \inf_{x \in X} \sup_{y \in K_n} \varphi(x, y);$$

(b) a proposition stating that continuity and concavity assumptions (35) and (36) imply the following estimate

$$(45) \quad \sup_{n \in \mathbb{N}} \inf_{x \in X} \sup_{y \in K_n} \varphi(x, y) \leq \inf_{C \in \mathcal{C}(Y, X)} \sup_{y \in Y} \varphi(C(y), y);$$

(c) a proposition stating that concavity and convexity assumptions (36) and (38) imply the following estimate

$$(46) \quad \sup_{n \in \mathbb{N}} \inf_{x \in X} \sup_{y \in K_n} \varphi(x, y) = \sup_{y \in Y} \inf_{x \in X} \varphi(x, y).$$

We begin by proving (44).

Proposition 1. *Suppose that assumptions (33), (34) and (35) of Theorem 5 hold. Then there exists $\bar{x} \in X$ such that*

$$(44) \quad \sup_{y \in Y} \varphi(\bar{x}, y) = \sup_{n \in \mathbb{N}} \inf_{x \in X} \sup_{y \in K_n} \varphi(x, y).$$

Proof. We take for a covering of Y the denumerable covering $\mathcal{A} = \{K_n\}_n$. Lemma 1 implies that there exists a denumerable sequence $\{x_n\}_n$ such that

$$(47) \quad \limsup_n \varphi(x_n, y) \leq v_{\square}(\mathcal{A}) = \sup_{n \in \mathbb{N}} \inf_{x \in X} \sup_{y \in K_n} \varphi(x, y).$$

Assume for a while that there exists a subsequence $\{x_{n_i}\}$ of $\{x_n\}$ which converges to $\bar{x} \in U$. Lemma 3 shows that the lower semi-continuity of the functions $x \mapsto \varphi_x(x, y)$ implies that $\bar{x}$ is a solution of (44).

Hence it remains to prove that the compactness assumption (34) implies that $\{x_n\}$ lies in a relatively weak compact subset. We shall use the fact that any bounded subset of the dual $U = F^*$ of a barreled space F is weakly relatively compact (see Corollary 4 of Appendix A).

To prove the latter statement, we choose any $p \in F$ and check that there exists n_p such that

$$(48) \quad \sup_{n \geq n_p} \langle p, x_n \rangle \leq a(p) < +\infty.$$

(Since the sequence is countable, (48) implies

$$(49) \quad \forall p \in F, \quad \sup_{n \in \mathbb{N}} \langle p, x_n \rangle \leq \tilde{a}(p) < +\infty$$

i.e. that the sequence is bounded.)

By the compactness assumptions (34), we can associate with any $p \in F$ an $\varepsilon_p > 0$ such that $p/\varepsilon_p \in \bigcup_{y \in Y} \text{Dom } \varphi_X^*(\cdot, y)$.

Hence there exists $y_p \in Y$ such that $\varphi_X^*(p/\varepsilon_p, y_p) < +\infty$. Thus, for any $n \in N$, $\langle p, x_n \rangle \leq \varepsilon_p [\varphi(x_n, y_p) + \varphi_X^*(p/\varepsilon_p, y_p)]$. Since $\limsup_n \varphi_n(x_n, y_p) \leq v^\square(\mathcal{A})$, there exists n_p such that $\varphi(x_n, y_n) \leq v^\square(\mathcal{A}) + \varepsilon$ as soon as $n \geq n_p$. Therefore, (48) holds since

$$\forall n \geq n_p, \quad \langle p, x_n \rangle \leq \varepsilon_p \left[(v^\square(\mathcal{A})) + \varepsilon + \varphi_X^*\left(\frac{p}{\varepsilon_p}, y_p\right) \right] = a(p).$$

This completes the proof. $\square$

The second estimate (45) follows from the following.

Proposition 2. *Let Y be a topological space. Suppose that*

(50) *$\mathcal{A}$ is a covering of Y by compact subsets K ,*

that

(51)
$$\begin{cases} \text{(i) } X \text{ is a topological space,} \\ \text{(ii) } \forall y \in Y, x \mapsto \varphi(x, y) \text{ is lower semi-continuous} \end{cases}$$

and that

(52)
$$\begin{cases} \text{(i) } Y \text{ is a convex subset,} \\ \text{(ii) } \forall x \in X, y \mapsto \varphi(x, y) \text{ is quasi-concave on } Y. \end{cases}$$

If $\mathcal{Q}(Y, X)$ denotes the set of continuous decision rules mapping Y into X , then

$$(45) \quad \sup_{K \in \mathcal{A}} \inf_{x \in X} \sup_{y \in K} \varphi(x, y) \leq \inf_{C \in \mathcal{Q}(Y, X)} \sup_{y \in Y} \varphi(C(y), y),$$

Proof. It is sufficient to prove that for any $K \in \mathcal{A}$ and $C \in \mathcal{Q}(Y, X)$, there exists $\bar{x} \in X$ such that

$$(53) \quad \sup_{y \in K} \varphi(\bar{x}, y) \leq \sup_{y \in Y} \varphi(C(y), y).$$

By replacing K by $\overline{\text{co}}(K)$ if necessary we can assume that K is a *convex compact* subset of Y .

We define the function $\hat{\varphi}$ defined on $K \times K$ by

$$(54) \quad \hat{\varphi}(z, y) = \varphi(C(z), y).$$

This function is lower semi-continuous with respect to z (since C is continuous and φ is lower semi-continuous) and quasi-concave with respect to y . Further-

more, K is compact. Thus the Ky-Fan theorem (Theorem 7.1.3) implies the existence of $\bar{z} \in K$ such that

$$\begin{aligned} \sup_{y \in K} \hat{\varphi}(\bar{z}, y) &= \sup_{y \in K} \varphi(C(\bar{z}), y) \\ &\leq \sup_{y \in K} \varphi(C(y), y) \leq \sup_{y \in Y} \varphi(C(y), y). \end{aligned}$$

By setting $\bar{x} = C(\bar{z})$, we deduce (53). $\square$

Finally the third estimate (46) follows from

Proposition 3. *Suppose that*

(55) *$\mathcal{A}$ is a covering of Y by compact subsets of Y ,*

that

(56) *$\forall x \in X, y \mapsto \varphi(x, y)$ is concave and upper semi-continuous*

and that

(57) *$\forall y \in Y, x \mapsto \varphi(x, y)$ is convex.*

Then

$$(46) \quad \sup_{K \in \mathcal{A}} \inf_{x \in X} \sup_{y \in K} \varphi(x, y) = \sup_{y \in Y} \inf_{x \in X} \varphi(x, y).$$

Proof. It is sufficient to prove that for any compact convex subset K of Y ,

$$(58) \quad \inf_{x \in X} \sup_{y \in K} \varphi(x, y) = \sup_{y \in K} \inf_{x \in X} \varphi(x, y).$$

(If (58) is true, we obtain that

$$\begin{aligned} \sup_{K \in \mathcal{A}} \inf_{x \in X} \sup_{y \in K} \varphi(x, y) &= \sup_{K \in \mathcal{A}} \sup_{y \in K} \inf_{x \in X} \varphi(x, y). \\ &= \sup_{y \in Y} \inf_{x \in X} \varphi(x, y). \end{aligned}$$

To prove (58), we use the fact that

(a) K is convex and compact and that, for all $x \in X$, $y \mapsto \varphi(x, y)$ is concave and upper semi-continuous.

(b) X is convex and $\forall y \in Y$, $x \mapsto \varphi(x, y)$ is convex.

Then the Nikaido theorem (Theorem 7.1.8) implies the existence of a max $\inf \bar{y} \in K$ of φ : therefore, (58) holds. $\square$

13.2. Relaxation of continuity assumptions: variational inequalities for monotone correspondences

We shall relax the continuity assumption in the case of functions $\varphi : X \times X \rightarrow \mathbf{R}$ satisfying

$$(*) \quad \begin{cases} \text{(i) } \exists y_0 \in X \text{ such that } x \mapsto \varphi(x, y_0) \text{ is lower semi-compact,} \\ \text{(ii) } \forall y \in X, x \mapsto \varphi(x, y) \text{ is lower semi-continuous for the finite} \\ \quad \text{topology,} \\ \text{(iii) } \forall x \in X, y \mapsto \varphi(x, y) \text{ is concave,} \\ \text{(iv) } \sup_{y \in Y} \varphi(y, y) \leq 0 \end{cases}$$

in proving the existence of $\bar{x}$ satisfying

$$(**) \quad \sup_{y \in X} \varphi(\bar{x}, y) \leq 0.$$

We will prove that assumptions $(*)$ imply the existence of a generalized sequence $\{x_\mu\}$ satisfying

$$(***) \quad \forall y \in X, \quad \limsup_{\mu} \varphi(x_\mu, y) \leq 0$$

and

$$(****) \quad x_\mu \text{ stays in a compact set and converges to } \bar{x}.$$

Among the functions φ having the property that $(**)$ holds whenever $\bar{x}$ is the limit of a generalized sequence satisfying $(***)$ and $(****)$, we single out the *pseudo-monotone functions*, i.e. functions such that $(**)$ holds whenever $\bar{x}$ is the limit of a compact generalized sequence $\{x_\mu\}_\mu$ satisfying only $(****)$ and

$$(***) \quad \limsup \varphi(x_\mu, \bar{x}) \leq 0.$$

(instead of $(****)$ and $(***)$).

Hence *there exists a solution $\bar{x}$ of $(**)$ whenever φ is a pseudo-monotone function satisfying $(*)$.*

We will apply this result in the cases when

$$\varphi(x, y) = \langle L(x), x - y \rangle \quad \text{or} \quad \varphi(x, y) = \sigma^b(S(x), x - y)$$

where L and S are respectively a map and a correspondence from a closed subset X of U into U^* .

We say that L and S are pseudo-monotone if the associated function φ is also pseudo-monotone. We shall prove that completely upper semi-continuous correspondences are pseudo-monotone and that finitely upper semi-continuous monotone correspondences are pseudo-monotone. Moreover, the sum of a pseudo-monotone correspondence and of a finitely upper semi-continuous monotone correspondence remains pseudo-monotone.

The introduction of monotone (and pseudo-monotone) correspondences is motivated by the fact that, in infinite dimensional spaces, the subdifferential of a convex function is not (generally) completely upper semi-continuous, but is monotone.

We then deduce that *there exists a solution $\bar{x} \in X$ to the variational inequalities $\langle L(\bar{x}), \bar{x} - y \rangle \leq 0$ for all $y \in X$ whenever L is a coercive, bounded, finitely continuous and pseudo-monotone operator*. In particular, *such operators are surjective*.

These results will be extended to the case of correspondences. We shall continue the study of monotone correspondences with a short introduction to cyclically monotone and maximal monotone correspondences.

13.2.1. Variational inequalities

We devote this section to the relaxation of the continuity assumption

$$(1) \quad \forall y \in Y, \quad x \mapsto \varphi(x, y) \text{ is lower semi-continuous}$$

in the case where φ is a function defined on the product $X \times X$. The requirement (1) is too strong in many instances. We shall illustrate this point with the problem of finding a solution $\bar{x} \in X$ to the *variational inequalities*.

$$(2) \quad \forall y \in X, \quad \langle L(\bar{x}), \bar{x} - y \rangle \leq 0$$

where

$$(3) \quad \begin{array}{l} X \text{ is a closed convex subset of the dual } U = F^* \text{ of a Banach space} \\ F, \text{ supplied with the weak topology } \sigma(F^*, F) \end{array}$$

and where

$$(4) \quad L \text{ is a map from } X \text{ into } U^* = F$$

(we take $\varphi(x, y) = \langle L(x), x - y \rangle$).

In most instances, variational inequalities arise from optimization problems or, more generally, from non-cooperative games (see Propositions 4.3.2, 4.3.6 and 9.1.4 for instance).

We can extend this problem to the case of correspondences S satisfying

$$(5) \quad S \text{ maps } X \text{ into } U^* \text{ and takes non-empty compact convex values.}$$

We shall say that $\bar{x}$ is a solution of variational inequalities if there exists $\bar{p} \in U^*$ such that

$$(6) \quad \begin{cases} \text{(i) } \bar{x} \in X \text{ and } \bar{p} \in S(\bar{x}), \\ \text{(ii) } \forall y \in X, \langle \bar{p}, \bar{x} - y \rangle \leq 0. \end{cases}$$

We can restate this problem in the following way.

$$(7) \quad \bar{x} \in X \text{ is a solution of } 0 \in S(\bar{x}) + N_X(\bar{x}) \text{ (or } S(\bar{x}) \cap -N_X(\bar{x}) \neq \emptyset).$$

where $N_X(x)$ is the normal cone to X at x (which is equal to 0 if $x \in \text{Int}(X)$)

Lemma 1. Suppose that (5) holds. Then $\bar{x} \in X$ is a solution of the variational inequalities (6) if and only if

$$(8) \quad \sup_{y \in X} \sigma^b(S(\bar{x}); \bar{x} - y) \leq 0.$$

Proof. It is clear that if there exists $\bar{p} \in S(\bar{x})$ satisfying $\langle \bar{p}, \bar{x} - y \rangle \leq 0$ for any $y \in X$, then $\sigma^b(S(\bar{x}); \bar{x} - y) \leq 0$ for any $y \in X$.

Conversely, assume that

$$\sup_{y \in X} \inf_{p \in S(\bar{x})} \langle p, \bar{x} - y \rangle = \sup_{y \in X} \sigma^b(S(\bar{x}), \bar{x} - y) \leq 0.$$

Then, since X is convex, $S(\bar{x})$ is convex and compact and $\langle p, \bar{x} - y \rangle$ is separately continuous, we deduce from the max inf theorem (see Theorem 7.1.5) that there exists $\bar{p} \in S(\bar{x})$ such that

$$\sup_{y \in Y} \langle \bar{p}, \bar{x} - y \rangle = \sup_{y \in Y} \inf_{p \in S(\bar{x})} \langle p, \bar{x} - y \rangle \leq 0. \quad \square$$

Therefore, we have to apply the Ky-Fan theorem (Theorem 7.1.3) to the function φ defined by

$$(9) \quad \varphi(x, y) = \sigma^b(S(x); x - y)$$

in order to solve the variational inequalities (6).

Remark. If X is a cone, any solution $\bar{x}$ of (6) is actually a solution to the problem of finding $\bar{x} \in X$ and $\bar{p} \in X^+$ such that $\bar{p} \in S(\bar{x})$ and $\langle \bar{p}, \bar{x} \rangle = 0$. If $X = U$, $\bar{x}$ is a solution of the multivalued equation $0 \in S(\bar{x})$.

Definition 1. Let S be a correspondence from a subset X of a locally convex space U into its dual U^* . We shall say that

(a) S is *completely upper semi-continuous* if it is upper semi-continuous from X supplied with the weak topology into U^* supplied with the strong topology;

(b) S is *finitely upper semi-continuous* if it is upper semi-continuous from X supplied with the finite topology into U^* supplied with the weak topology;

(c) S is “*coercive*” on X if either X is bounded or

$$(10) \quad \lim_{\substack{\|x\| \rightarrow \infty \\ x \in X}} \frac{\sigma^b(S(x); x)}{\|x\|} = +\infty;$$

(d) S is “*bounded on X* ” if

$$(11) \quad \lim_{\substack{\|x\| \rightarrow \infty \\ x \in X}} \sup_{p \in S(x)} \frac{\|p\|}{\|x\|} < +\infty$$

(e) S is “*monotone*” if for any $x, y \in X$, $p \in S(x)$ and $q \in S(y)$, then $\langle p - q, x - y \rangle \geq 0$.

Lemma 2. Suppose that U is the dual of a Banach space and that

$$(12) \quad L \text{ and } S \text{ are coercive and bounded.}$$

Then, for any $y \in X$, the functions $x \rightarrow \langle L(x), x - y \rangle$ and $x \rightarrow \sigma^b(S(x), x - y)$ are coercive.

Proof. Since S is bounded on X , there exist constants K and M such that $\sigma^*(S(x); y) \leq K\|x\| \|y\|$ when $\|x\| \geq M$. Therefore, we obtain $\varphi(x, y) = \sigma^b(S(x); x - y) \geq \sigma^b(S(x); x) - \sigma^*(S(x); y) \geq \sigma^b(S(x); x) - K\|x\| \|y\|$. Since S is coercive, we deduce from these inequalities that $\varphi(\cdot, y)$ is also coercive for any $y \in Y$. $\square$

13.2.2. Existence of a solution to variational inequalities for completely upper semi-continuous correspondences

Thanks to Lemma 1 and the Ky-Fan inequality, we obtain the following result.

Proposition 1. Suppose that U is a reflexive Banach space and that the map L and the correspondence S from X into U satisfy (12) and

$$(13) \quad L \text{ and } S \text{ are completely upper semi-continuous.}$$

Then there exists a solution $\bar{x} \in X$ to variational inequalities (2) and (6).

Proof. (a) We will use Theorem 7.1.2 (as in proving the Ky-Fan theorem). By setting $\varphi(x, y) = \sigma^b(S(x); x - y)$, we see that φ satisfies

$$\sup_{y \in X} \varphi(y, y) = 0.$$

(b) Since φ is lower semi-continuous with respect to x for the finite topology (which is stronger than any vector space topology) and since the identity is continuous from X into X supplied with the finite topology (see Proposition 7.1.3), we deduce from the proof of Theorem 7.1.2 that

$$v^\square \leq \inf_{C \in \mathcal{C}(X, X)} \sup_{y \in X} \varphi(C(y), y) \leq \sup_{y \in X} \varphi(y, y) \leq 0.$$

(c) Furthermore, the coerciveness and boundedness of S implies that (by Lemma 2)

$$\forall y \in Y, \quad x \mapsto \varphi(x, y) \text{ is coercive.}$$

Therefore, by Proposition 3.1.1, we deduce that

$$(14) \quad \exists y_0 \in Y \text{ such that } x \mapsto \varphi(x, y_0) \text{ is lower semi-compact.}$$

At this stage of the proof, Lemmas 1 and 2 and (1.2) and (14) imply that there exists a generalized sequence $\{x_\mu\}_\mu$ satisfying

$$(15) \quad x_\mu \text{ stays in a compact subset and converges weakly to } \bar{x} \in X$$

and

$$(16) \quad \forall y \in X, \quad \limsup \varphi(x_\mu, y) \leq v^\square \leq 0.$$

(d) It remains to prove the following result, which implies that

$$(17) \quad \forall y \in X, \quad \varphi(\bar{x}, y) \leq \liminf_{\mu} \varphi(x_\mu, y).$$

Proposition 2. *Let S be a completely upper semi-continuous correspondence from a closed subset X of a reflexive Banach space U into U^* . Then, for any bounded generalized sequence $\{x_\mu\}_\mu$ satisfying (15) and (16), its limit $\bar{x}$ has the property that*

$$(18) \quad \forall y \in X, \quad \sigma^b(S(\bar{x}), \bar{x} - y) \leq \lim_{\mu} \sigma^b(S(x_\mu), x_\mu - y).$$

Proof. Let $\{x_\mu\}_\mu$ be a generalized sequence satisfying (15) and let $\bar{x}$ be its limit. We take $p_\mu \in S(x_\mu)$ such that $\langle p_\mu, x_\mu - y \rangle = \sigma^b(S(x_\mu); x_\mu - y)$. Since $\{x_\mu\}$ is contained in a weakly compact subset and since S is completely upper semi-

continuous, the subsets $S(x_\mu)$ stay in a (strong) compact subset of U^* , which is equicontinuous. Hence, we can extract a generalized subsequence (again denoted by) $\{p_\mu\}_\mu$ which converges strongly to $\bar{p} \in U^*$ and a generalized subsequence (again denoted by) $\{x_\mu\}_\mu$ which converges weakly to $\bar{x} \in X$. Therefore $\langle p_\mu, x_\mu - y \rangle$ converges to $\langle \bar{p}, \bar{x} - y \rangle$ and thus, $\sigma^b(S(\bar{x}); \bar{x} - y) \leq \langle \bar{p}, \bar{x} - y \rangle = \lim_\mu \langle p_\mu, x_\mu - y_\mu \rangle = \lim_\mu \sigma^b(S(x_\mu); x_\mu - y)$. $\square$

When U is a finite dimensional space, all vector space topologies coincide and upper semi-continuous correspondences are completely upper semi-continuous. When U is no longer finite-dimensional, the assumption that S is completely upper semi-continuous is much more stringent.

For instance, we proved that the subdifferential ∂f of a differentiable convex function is upper semi-continuous from U supplied with the strong topology into F supplied with the weak topology (see Proposition 4.1.8).

It is not true in general that it remains upper semi-continuous when U is supplied with the weak topology and U^* with the strong topology!

Nevertheless, subdifferentials of convex functions are monotone, i.e.

$$(19) \quad \langle p - q, x - y \rangle \geq 0 \quad \forall p \in \partial f(x), \quad \forall q \in \partial f(y).$$

We obtain (19) by adding the inequalities

$$\begin{aligned} f(x) - f(y) &\leq \langle p, x - y \rangle \quad \text{since } p \in \partial f(x), \\ f(y) - f(x) &\leq \langle q, y - x \rangle \quad \text{since } q \in \partial f(y). \end{aligned}$$

Since optimal solutions of convex functions exist, there is some hope of proving the existence of a solution $\bar{x}$ to variational inequalities if we replace the assumption of complete upper semi-continuity by an assumption of monotonicity.

Indeed, if we look back at the proof of proposition 2, we notice that the complete upper semi-continuity of L implies much more than we need (eq. (18) instead of inequality (17)).

13.2.3. Pseudo-monotone functions: the Brézis–Nirenberg–Stampacchia theorem

We are led to introduce the following concept of pseudo-monotone function.

Definition 2. We shall say a function φ defined on $X \times X$ is “pseudo-monotone” if, for any generalized sequence $\{x_\mu\}$ satisfying

$$(20) \quad \{x_\mu\}_\mu \text{ stays in a compact set and converges to } \bar{x}$$

and

$$(21) \quad \limsup_\mu \varphi(x_\mu, \bar{x}) \leq 0$$

its limit $\bar{x}$ satisfies

$$(22) \quad \forall y \in X, \quad \varphi(\bar{x}, y) \leq \liminf_{\mu} \varphi(x_{\mu}, y).$$

We shall say that an operator L mapping X into U^* is "pseudo-monotone" if the function φ defined by $\varphi(x, y) = \langle L(x), x - y \rangle$ is pseudomonotone.

Proposition 2 shows that *a completely continuous operator is pseudo-monotone*. It is also obvious that *any function φ lower semi-continuous with respect to x is pseudo-monotone*.

We can extend the Ky-Fan theorem in the following way.

Theorem 1 (Brézis–Nirenberg–Stampacchia). *Let us assume that*

$$(23) \quad X \text{ is a closed convex subset,}$$

that

$$(24) \quad \exists y_0 \in X \text{ such that } x \mapsto \varphi(x, y_0) \text{ is lower semi-compact,}$$

and that

$$(25) \quad \begin{cases} \text{(i) } \forall x \in X, y \mapsto \varphi(x, y) \text{ is quasi-concave,} \\ \text{(ii) } \sup_{y \in X} \varphi(y, y) = 0. \end{cases}$$

Let us assume moreover that

$$(26) \quad \begin{cases} \text{(i) } \varphi \text{ is pseudo-monotone,} \\ \text{(ii) } \forall y \in X, x \mapsto \varphi(x, y) \text{ is lower semi-continuous on } X \text{ supplied} \\ \text{with the finite topology.} \end{cases}$$

Then there exists $\bar{x}$ such that $\sup_{y \in X} \varphi(\bar{x}, y) \leq 0$.

Proof. Since φ is quasi-concave with respect to y and lower semi-continuous with respect to x for the finite topology, Theorem 7.1.2 implies that

$$(27) \quad v^{\square} \leq \inf_{C \in \mathcal{C}(X, X)} \sup_{y \in X} \varphi(C(y), y) \leq \sup_{y \in X} \varphi(y, y) \leq 0.$$

[The identity is continuous when X is supplied with the finite topology and $\sup_{y \in X} \varphi(y, y) \leq 0$]

On the other hand, we deduce from Lemmas 1.1 and 1.2 that there exists a generalized sequence $\{x_{\mu}\}$ satisfying

$$(28) \quad \{x_{\mu}\} \text{ stays in a compact set and converges to } \bar{x} \in X$$

and

$$(29) \quad \text{for any finite subset } K \text{ of } X, \quad \limsup_{\mu} \left[\sup_{y \in K} \varphi(x_{\mu}, y) \right] \leq v^{\square} \leq 0.$$

By taking $K = \{\bar{x}\}$, we see that the sequence $\{x_{\mu}\}$ satisfies properties (20) and (21). Thus, since φ is pseudo-monotone, we deduce that, $\forall y \in X$,

$$(30) \quad \varphi(\bar{x}, y) \leq \liminf_{\mu} \varphi(x_{\mu}, y) \leq \limsup_{\mu} \varphi(x_{\mu}, y) \leq 0$$

by (29) with $K = \{y\}$.

***Remark.** By using the pseudo-monotonicity of φ , we do not exploit all the information given by (29). We can introduce a concept of “weak pseudo-monotonicity” by replacing the requirement (21) by the weaker one (29) in Definition 2.

Remark. We point out the following result.

Proposition 3. *If we assume that φ is pseudo-monotone and $f : X \rightarrow \mathbf{R}$ is lower semi-continuous, then the function ψ defined by $\psi(x, y) = \varphi(x, y) + f(x) - f(y)$ is also pseudo-monotone.*

Proof. Let $\{x_{\mu}\}$ be a generalized sequence of $x_{\mu} \in X$ converging to $\bar{x}$ and satisfying

$$(31) \quad \limsup_{\mu} (\varphi(x_{\mu}, \bar{x}) + f(x_{\mu}) - f(\bar{x})) \leq 0.$$

Then

$$\begin{aligned} \limsup_{\mu} \varphi(x_{\mu}, \bar{x}) + \liminf_{\mu} f(x_{\mu}) - f(\bar{x}) &\leq \\ \limsup_{\mu} [\varphi(x_{\mu}, \bar{x}) + f(x_{\mu}) - f(\bar{x})] &\leq 0. \end{aligned}$$

But, since f is lower semi-continuous, $\liminf_{\mu} f(x_{\mu}) \geq f(\bar{x})$. Hence $\limsup_{\mu} \varphi(x_{\mu}, \bar{x}) \leq 0$ and the pseudo-monotonicity of φ imply:

$$(32) \quad \forall y \in X, \quad \varphi(\bar{x}, y) \leq \liminf_{\mu} \varphi(x_{\mu}, y).$$

Therefore,

$$\begin{aligned} \forall y \in X, \quad \varphi(\bar{x}, y) + f(\bar{x}) - f(y) & \\ &\leq \liminf_{\mu} \varphi(x_{\mu}, y) + \liminf_{\mu} f(x_{\mu}) - f(y) \\ &\leq \liminf_{\mu} (\varphi(x_{\mu}, y) + f(x_{\mu}) - f(y)). \end{aligned}$$

Thus we proved that the function ψ defined by $\psi(x, y) = \varphi(x, y) + f(x) - f(y)$ is pseudo-monotone. $\square$

13.2.4. Existence of a solution to variational inequalities for pseudo-monotone correspondences

Theorem 2. *Let U be a reflexive Banach space and*

- (33) *S be a correspondence from X into U^* with non-empty bounded closed convex values*

where X is a closed convex subset of U . Suppose also that

- (34) *S is coercive and bounded on X*

and that

- (35) *S is pseudo-monotone and finitely upper semi-continuous.*

Then there exists a solution $\bar{x} \in X$ to the variational inequalities (6).

Proof. We apply Theorem 1 to the function φ defined by $\varphi(x, y) = \sigma^b(S(x), x - y)$. It is coercive with respect to x by Lemma 2. It is pseudo-monotone by Definition 2 and clearly concave with respect to y . It remains to prove that it is lower semi-continuous with respect to x for the finite topology. This amounts to saying that for any $K = \{x_1, \dots, x_n\}$, the function ψ defined on $\mathcal{M}^n$ by

- (36) $\psi(\lambda) = \sigma^b(S(\beta(\lambda)), \beta(\lambda) - y)$ is lower semi-continuous

where

- (37) $\beta(\lambda) = \sum_{k=1}^n \lambda^k x_k.$

Let $N = \frac{1}{2}\varepsilon(K - y)^*$, which is a neighborhood of 0 for the weak topology of U^* . Since S is finitely upper semi-continuous, there exists $\eta > 0$ such that $\|\lambda - \lambda_0\| \leq \eta$ implies that

- (38) $S(\beta(\lambda)) \subset S(\beta(\lambda_0)) + N$

and that

- (39) $\sup_{p \in S(\beta(\lambda_0))} |\langle p, \beta(\lambda) - \beta(\lambda_0) \rangle| \leq \frac{1}{2} \varepsilon.$

We deduce from inequality (38) that

- (40) $\sigma^b(S(\beta(\lambda)), \beta(\lambda) - y) \geq \sigma^b(S(\beta(\lambda_0)), \beta(\lambda_0) - y) + \sigma^b(S(\beta(\lambda_0)), \beta(\lambda) - \beta(\lambda_0)) + \sigma^b(N, \beta(\lambda) - y)$

But, since $\beta(\lambda) \in \overline{\text{co}} K - y$, we obtain that

- (41) $\sigma^b(N, \beta(\lambda) - y) \geq -\frac{1}{2} \varepsilon.$

Also, (39) implies that

$$(42) \quad \sigma^b(S(\beta(\lambda_0)), \beta(\lambda) - \beta(\lambda_0)) \geq \frac{1}{2} \varepsilon.$$

Therefore, inequalities (40), (41) and (42) imply that

$$\psi(\lambda) \geq \psi(\lambda_0) - \varepsilon \quad \text{when } \|\lambda - \lambda_0\| \leq \eta. \quad \square$$

Taking $X = U$, we obtain the following surjectivity theorem.

Theorem 3. *Suppose that a correspondence S from a reflexive Banach space U into U^* , with non-empty compact convex values, is coercive, bounded, pseudo-monotone and finitely continuous. Then there exists $\bar{x} \in X$ such that $0 \in S(\bar{x})$.*

13.2.5. Pseudo-monotonicity of monotone correspondences

Proposition 4. *Any monotone and finitely upper semi-continuous correspondence is pseudo-monotone.*

The sum of a monotone and finitely upper semi-continuous correspondence and of a pseudo-monotone correspondence is pseudo-monotone.

Proof. (a) We begin by proving that a monotone and finitely upper semi-continuous correspondence is pseudo-monotone. Let $\{x_\mu\}$ be a compact generalized sequence converging to $\bar{x}$ and let $p_\mu \in S(x_\mu)$ satisfy

$$(43) \quad \limsup_{\mu} \langle p_\mu, x_\mu - y \rangle = \limsup_{\mu} \sigma^b(S(x_\mu); x_\mu - y) \leq 0.$$

Let $\bar{p} \in S(\bar{x})$. Since S is monotone, then

$$\langle \bar{p}, x_\mu - \bar{x} \rangle \leq \langle p_\mu, x_\mu - \bar{x} \rangle$$

implies that

$$0 = \lim_{\mu} \langle \bar{p}, x_\mu - \bar{x} \rangle \leq \liminf_{\mu} \langle p_\mu, x_\mu - \bar{x} \rangle$$

and thus, that, by (43),

$$(44) \quad \lim_{\mu} \langle p_\mu, x_\mu - \bar{x} \rangle = 0.$$

We now choose $y \in X$, $z_\theta = \theta y + (1 - \theta)\bar{x} \in X$ ($\theta \in]0, 1[$) and $p_\theta \in S(z_\theta)$.

We write inequality $\langle p_\mu - p_\theta, x_\mu - z_\theta \rangle \geq 0$ in the following form:

$$\begin{aligned} & \langle p_\mu, x_\mu - \bar{x} \rangle - \langle p_\theta, x_\mu - \bar{x} \rangle \\ & - \theta \langle p_\theta, \bar{x} - y \rangle + \theta \langle p_\mu, \bar{x} - y \rangle \geq 0 \end{aligned}$$

Taking the $\liminf$, we deduce from (44) that

$$(45) \quad \langle p_\theta, \bar{x} - y \rangle \leq \liminf_{\mu} \langle p_\mu, \bar{x} - y \rangle.$$

Since the correspondence S is upper semi-continuous for the finite topology, we deduce that p_θ converges to some $p \in S(\bar{x})$ when θ converges to 0. Then

$$\begin{aligned} \sigma^b(S(\bar{x}); \bar{x} - y) &\leq \langle p, \bar{x} - y \rangle \leq \liminf_{\mu} \langle p_\mu, \bar{x} - y \rangle = \\ &\liminf_{\mu} \langle p_\mu, x_\mu - y \rangle = \liminf_{\mu} \sigma^b(S(x_\mu); x_\mu - y) \end{aligned}$$

since, by (44),

$$\begin{aligned} \liminf_{\mu} [\langle p_\mu, \bar{x} - x_\mu \rangle + \langle p_\mu, x_\mu - y \rangle] &= \\ &= \lim_{\mu} [\langle p_\mu, \bar{x} - x_\mu \rangle] + \liminf_{\mu} [\langle p_\mu, x_\mu - y \rangle] \\ &= \liminf_{\mu} [\langle p_\mu, x_\mu - y \rangle]. \end{aligned}$$

(b) To prove the second part of the proposition, suppose that $S = T + R$ where T is monotone and finitely upper semi-continuous, R pseudo-monotone. Let $\{x_\mu\}$ be a compact generalized sequence converging to $\bar{x}$ and satisfying

$$(46) \quad \limsup_{\mu} \langle p_\mu, x_\mu - \bar{x} \rangle = \limsup_{\mu} \sigma^b(S(x_\mu); x_\mu - \bar{x}) \leq 0,$$

where $p_\mu = q_\mu + r_\mu \in S(x_\mu)$, $q_\mu \in T(x_\mu)$ and $r_\mu \in R(x_\mu)$. Since T is monotone, we deduce from $\langle q_\mu - q, x_\mu - \bar{x} \rangle \geq 0$ that

$$\langle r_\mu, x_\mu - \bar{x} \rangle = \langle p_\mu - q_\mu, x_\mu - \bar{x} \rangle \leq \langle p_\mu - q, x_\mu - \bar{x} \rangle$$

where q is any element of $T(\bar{x})$.

Therefore

$$\begin{aligned} \limsup_{\mu} \langle r_\mu, x_\mu - \bar{x} \rangle &\leq \limsup_{\mu} \langle p_\mu, x_\mu - \bar{x} \rangle \\ &\quad - \lim_{\mu} \langle q, x_\mu - \bar{x} \rangle \leq 0. \end{aligned}$$

Since R is pseudo-monotone, we deduce that

$$(47) \quad \forall y \in X, \quad \sigma^b(R(\bar{x}), \bar{x} - y) \leq \liminf_{\mu} \sigma^b(R(x_\mu), x_\mu - y)$$

and, in particular, that $\lim_{\mu} \sigma^b(R(x_\mu), x_\mu - \bar{x}) = 0$. Since T is pseudo-monotone (being monotone and finitely upper semi-continuous) we deduce from

$$\begin{aligned} \limsup_{\mu} \sigma^b(T(x_\mu), x_\mu - \bar{x}) &\leq \limsup_{\mu} \langle q_\mu, x_\mu - \bar{x} \rangle \\ &\leq \limsup_{\mu} \langle p_\mu, x_\mu - \bar{x} \rangle - \lim_{\mu} \sigma^b(R(x_\mu), x_\mu - \bar{x}) \leq 0 \end{aligned}$$

that

$$(48) \quad \sigma^b(T(\bar{x}), \bar{x}-y) \leq \liminf_{\mu} \sigma^b(T(x_{\mu}), x_{\mu}-\bar{x}).$$

Let $\bar{q} \in T(\bar{x})$ satisfy $\langle \bar{q}, \bar{x}-y \rangle = \sigma^b(T(\bar{x}), \bar{x}-y)$ and let $\bar{r} \in R(\bar{x})$ satisfy $\langle \bar{r}, \bar{x}-y \rangle = \sigma^b(R(\bar{x}), \bar{x}-y)$. Let $\bar{p} = \bar{q} + \bar{r} \in S(\bar{x})$. Then

$$\begin{aligned} \sigma^b(S(\bar{x}); \bar{x}-y) &\leq \langle \bar{q} + \bar{r}, \bar{x}-y \rangle \\ &\leq \liminf_{\mu} \langle q_{\mu}, x_{\mu}-\bar{x} \rangle + \liminf_{\mu} \langle r_{\mu}, x_{\mu}-\bar{x} \rangle \\ &\leq \liminf_{\mu} [\langle p_{\mu}, x_{\mu}-\bar{x} \rangle] = \liminf_{\mu} \sigma^b(S(x_{\mu}); x_{\mu}-y). \end{aligned}$$

We thus have proved that S is pseudo-monotone.

13.2.6. Monotone and cyclically monotone correspondences

Definition 3. We shall say that a correspondence S from U into U^* is "cyclically monotone" if

$$(49) \quad \text{for any finite sequence } \{\{x_i, p_i\}\}_{1 \leq i \leq n} \text{ of the graph } G(S), \text{ we have} \\ \langle p_1, x_1-x_2 \rangle + \langle p_2, x_2-x_3 \rangle + \cdots + \langle p_n, x_n-x_1 \rangle \geq 0.$$

Of course, any cyclically monotone correspondence is monotone. In fact, the cyclically monotone correspondences characterize the subdifferentials of convex functions.

Theorem 4. (Rockafellar). *Let S be a correspondence from U into U^* . Then S is cyclically monotone if and only if there exists a lower semi-continuous convex proper function f such that*

$$(50) \quad \text{for all } x \in U, \quad S(x) \subset \partial f(x).$$

Proof. We begin by proving the sufficient condition. For this purpose, it is sufficient to prove that the correspondence ∂f is cyclically monotone, since clearly this implies that any $S \subset \partial f$ is also cyclically monotone. Let $\{\{x_i, p_i\}\}$ be a sequence of n elements of the graph of ∂f ; then

$$\begin{aligned} f(x_1) - f(x_2) &\leq \langle p_1, x_1 - x_2 \rangle \\ f(x_2) - f(x_3) &\leq \langle p_2, x_2 - x_3 \rangle \\ &\vdots \\ f(x_n) - f(x_1) &\leq \langle p_n, x_n - x_1 \rangle. \end{aligned}$$

Adding these inequalities, we obtain that

$$0 \leq \langle p_1, x_1 - x_2 \rangle + \langle p_2, x_2 - x_3 \rangle + \cdots + \langle p_n, x_n - x_1 \rangle.$$

Conversely, we now consider a cyclically monotone correspondence S from U into U^* and construct an appropriate f . If, for any x , $S(x) = \emptyset$, the statement is trivial. If not, fix a pair $\{x_0, p_0\}$ of the graph of S and define f by

$$(51) \quad f(x) = \sup_{\{(x_i, p_i)\} \in \mathcal{S}} (\langle p_n, x - x_n \rangle + \langle p_{n-1}, x_n - x_{n-1} \rangle + \cdots + \langle p_0, x_1 - x_0 \rangle),$$

where $\mathcal{S}$ is the family of finite sequences $\{(x_i, p_i)\}_{1 \leq i \leq n}$ of $G(S)$.

In the first place, since f is a pointwise supremum of continuous affine functionals, it is a lower semi-continuous convex function. Its domain is non-empty since

$$(52) \quad f(x_0) = 0.$$

(Since S is cyclically monotone, $f(x_0)$ is the supremum of non-positive scalars $\langle p_n, x_0 - x_n \rangle + \cdots + \langle p_0, x_1 - x_0 \rangle$, and thus, $f(x_0) \leq 0$. Since $0 = \langle p_0, x_0 - x_0 \rangle + \langle p_0, x_0 - x_0 \rangle$, then $\{x_0, p_0\} \in \mathcal{S}$ achieves this supremum, which is equal to 0.)

It remains to check that $S(x) \subset \partial f(x)$. For this purpose, we take $p \in S(x)$ and show that

$$(53) \quad f(x) - f(y) \leq \langle p, x - y \rangle \quad \text{for all } y \in U.$$

Let $\varepsilon > 0$. By definition (51) of f , there exists a finite sequence $\{x_i, p_i\}_{1 \leq i \leq n}$ of elements of $G(S)$ such that

$$(54) \quad f(x) \leq \langle p_n, x - x_n \rangle + \langle p_{n-1}, x_n - x_{n-1} \rangle + \cdots + \langle p_0, x_1 - x_0 \rangle + \varepsilon.$$

Since $\{x, p\}$ belongs to $G(S)$, we also obtain that

$$(55) \quad \langle p, y - x \rangle + \langle p_n, x - x_n \rangle + \langle p_{n-1}, x_n - x_{n-1} \rangle + \cdots + \langle p_0, x_1 - x_0 \rangle \leq f(y).$$

adding these two inequalities, we obtain that $f(x) - f(y) \leq \langle p, x - y \rangle + \varepsilon$. Letting ε tend to 0, we obtain (53). $\square$

In order to derive the identity $S = \partial f$, we are led to introduce the following definitions.

Definition 4. A correspondence S is said to be *maximal cyclically monotone* (resp. *maximal monotone*) if its graph is not strictly contained in the graph of a cyclically monotone (resp. monotone) correspondence.

With such a definition, Theorem 4 implies the following statement.

Proposition 5. *Let S be a maximal cyclically monotone correspondence. Then there exists a lower semi-continuous convex function f such that $S = \partial f$.*

In the next section, we shall characterize maximal monotone correspondences in Hilbert spaces *only* and mention some of their properties.

13.2.7. Maximal monotone correspondences

In this section, U denotes a Hilbert space identified with its dual U^* . Recall that a correspondence S from $X \subset U$ into U is monotone if and only if its graph $G(S) \subset U \times U$ is monotone in the sense that

$$(56) \quad \text{a subset } G \subset U \times U \text{ is monotone if } \forall \{x, p\}, \{y, q\} \in G, \langle p - q, x - y \rangle \geq 0.$$

We shall use the following characterization of a monotone correspondence in Hilbert space.

Proposition 6. *Let S be a correspondence from X into U . It is monotone if and only if*

$$(57) \quad \forall k > 0, \quad \forall \{x, p\}, \{y, q\} \in G(S), \quad \|x - y + k(p - q)\| \geq \|x - y\|.$$

Proof. Suppose that S is monotone. Then $\|x - y + k(p - q)\|^2 = \|x - y\|^2 + k^2\|p - q\|^2 + 2k\langle p - q, x - y \rangle \geq \|x - y\|^2$ since $\langle p - q, x - y \rangle \geq 0$ for S is monotone.

Conversely, assume (57) and take $\{x, p\}$ and $\{y, q\}$ in $G(S)$. Then $\langle p - q, x - y \rangle = (1/2k)[(\|x - y + k(p - q)\|^2 - \|x - y\|^2) - k^2\|p - q\|^2] \geq -k\|p - q\|^2/2$.

Letting k converge to 0, we deduce that $\langle p - q, x - y \rangle \geq 0$. Thus S is monotone. $\square$

We also note the following consequence.

Proposition 7. *Let S be a monotone correspondence from X into U and let $Y = S(X) \subset U$. Then $(1 + S)^{-1}$ is actually a map satisfying $\|(1 + S)^{-1}(p) - (1 + S)^{-1}(q)\| \leq \|p - q\|$ for all $p, q \in Y$.*

Proof. Let $x \in (1 + S)^{-1}(p)$ and $y \in (1 + S)^{-1}(q)$. Then $p - x \in S(x)$ and $q - y \in S(y)$. Inequality (57), with p and q replaced by $p - x$ and $q - y$, implies that $\|p - q\| = \|x - y + (p - x - q + y)\| \geq \|x - y\|$, i.e. that $\|x - y\| \leq \|p - q\|$. If we take $p = q$, this implies that $(1 + S)^{-1}$ contains a unique point x , i.e. that $(1 + S)^{-1}$ is a map. If we take $p \neq q$, the inequality implies that this map is Lipschitz with constant 1. $\square$

It is clear that the union of an increasing family $\{G_i\}_{i \in I}$ of monotone-subsets of $U \times U$ is monotone. Therefore, by Zorn's axiom, each monotone subset is contained in a maximal monotone subset.

Thus any monotone correspondence can be extended into a maximal monotone correspondence. In fact, we shall use only the following analytic characterization of maximal monotone correspondences:

Proposition 8. *Let S be a monotone correspondence from X into U . It is maximal monotone if and only if*

$$(58) \quad \text{if } \forall \{y, q\} \in G(S), \quad \langle p - q, x - y \rangle \geq 0, \quad \text{then } p \in S(x).$$

Proof. This is left as an exercise. $\square$

We now state the fundamental characterization of maximal monotone correspondences.

Theorem 5 (Minty). *Let S be a monotone correspondence from X into U . Then S is maximal monotone if and only if $1 + S$ maps X onto U . In this case, $(1 + S)^{-1}$ is a map from U into U which is Lipschitz with constant 1.*

Proof. (a) Suppose that $1 + S$ is surjective. We shall prove that (58) holds. For this purpose, take $\{x, p\}$ to satisfy $\langle p - q, x - y \rangle \geq 0$ for all $\{y, q\} \in G(S)$. Since $1 + S$ is surjective, there exist $y_0 \in U$ and $q_0 \in S(y_0)$ such that $p + x = y_0 + q_0$. Hence $p - q_0 = -(x - y_0)$. Thus, $-||x - y_0||^2 \geq 0$. Therefore $x = y_0$ and $p = q_0 \in S(y_0) = S(x)$. Proposition 8 implies that S is maximal monotone.

(b) Suppose that S is maximal monotone. Let $y \in U$. We shall prove that there exist $x \in U$ and $p \in S(x)$ such that $y = x + p$. We write $R(x) = S(x) - y$, which defines a maximal monotone correspondence from X into U . We have to prove that there exists $x \in U$ such that $-x \in R(x)$. By Proposition 8, this amounts to finding $x \in U$ such that,

$$(59) \quad \forall \{y, q\} \in G(R), \quad \langle x + q, x - y \rangle \leq 0.$$

Let y_0 and $q_0 \in R(y_0)$ be fixed. If a solution x of (59) exists, it belongs to the subset

$$(60) \quad \begin{aligned} X_0 &= \{x \in X \text{ such that } \langle x + q_0, x - y_0 \rangle = \\ &= ||x||^2 + \langle x, q_0 - y_0 \rangle - \langle q_0, y_0 \rangle \leq 0\}. \end{aligned}$$

This set is obviously bounded. Thus $C = \overline{\text{co}}(X_0)$ is a weak compact convex subset of U in which we have to find solutions of (59). For this purpose, we introduce the function f defined on $C \times G(R)$ by

$$(61) \quad f(x, \{y, q\}) = \langle x + q, x - y \rangle = ||x||^2 + \langle x, q - y \rangle - \langle q - y, y \rangle$$

It is lower semi-continuous and convex with respect to x (for the weak topology). Theorem 7.1.1 implies that there exists a solution $x \in C$ to the following inequality.

$$(62) \quad \sup_{\{y, q\} \in G(R)} f(x, \{y, q\}) \leq \sup_{K \in \mathcal{J}} \inf_{x \in C} \max_{i=1, \dots, n} f(x, \{y_i, q_i\}) = v \square$$

where $K = \{\{y_1, q_1\}, \dots, \{y_n, q_n\}\}$ ranges over the finite subsets of $G(R)$.

We define the function f_K on $\mathcal{M}^n \times \mathcal{M}^n$ by setting

$$(63) \quad f_K(\lambda, \mu) = \sum_{i=1}^n \mu^i \langle \beta(\lambda) + q_i, \beta(\lambda) - y_i \rangle \quad \text{where } \beta(\lambda) = \sum_{i=1}^n \lambda^i y_i.$$

We have the following estimates.

$$(64) \quad v \square \leq \sup_{K \in \mathcal{J}} \inf_{\lambda \in \mathcal{M}^n} \sup_{\mu \in \mathcal{M}^n} f_K(\lambda, \mu).$$

Now, it is clear that f_K is lower semi-continuous and convex with respect to λ and linear and continuous with respect to μ . By the minimax Theorem 7.1.7, there exists a saddle point $\{\lambda_0, \mu_0\}$. Therefore,

$$(65) \quad \sup_{\mu \in \mathcal{M}^n} f_K(\lambda_0, \mu) = \inf_{\lambda \in \mathcal{M}^n} f_K(\lambda, \mu_0) \leq f_K(\mu_0, \mu_0) \leq \sup_{\mu \in \mathcal{M}^n} f_K(\mu, \mu).$$

Now, it remains to check that

$$(66) \quad f_K(\mu, \mu) \leq 0 \quad \text{for all } \mu \in \mathcal{M}^n.$$

But,

$$\begin{aligned} f_K(\mu, \mu) &= \sum_{i=1}^n \sum_{k=1}^n \mu^i \mu^k \langle \beta(\mu) + q_i, y_k - y_i \rangle \\ &= \sum_{i=1}^n \sum_{k=1}^n \mu^i \mu^k \langle q_i, y_k - y_i \rangle \left(\text{for } \sum_{i, k=1}^n \mu^i \mu^k \langle \beta(\mu), y_k - y_i \rangle = 0 \right). \\ &= \frac{1}{2} \sum_{i, k=1}^n \mu^i \mu^k \langle q_i - q_k, y_k - y_i \rangle \leq 0 \end{aligned}$$

since $\langle q_i - q_k, y_k - y_i \rangle \leq 0$ for all $i, k = 1, \dots, n$ because R is monotone.

Inequalities (62), (64), (65) and (66) imply that (59) holds. Hence

$$-x \in R(x). \quad \square$$

We state some other properties of maximal monotone correspondences.

Proposition 9. *Let S be a maximal monotone correspondence from X into U . Then the images $S(x)$ are closed and convex. If a sequence of elements x_n converges strongly to x and if a sequence of elements $p_n \in S(x_n)$ converges weakly to p , then $p \in S(x)$.*

Proof. By Proposition 8, $S(x)$ is the set of $p \in U$ such that $\langle p - q, x - y \rangle \geq 0$ for all $\{y, q\} \in G(S)$. This implies that $S(x)$ is closed and convex. Let x_n converge strongly to x and $p_n \in S(x_n)$ converge weakly to p . Hence, for any $\{y, q\} \in G(S)$, inequalities $\langle p_n - q, x_n - y \rangle \geq 0$ hold. In the limit, we obtain that, $\langle p - q, x - y \rangle \geq 0$. Then, again by Proposition 8, $p \in S(x)$. $\square$

One can check that a correspondence S is maximal monotone by proving that $1 + S$ is surjective thanks to the Minty theorem.

Example 1. Let b be a maximal monotone map from $\mathbb{R}$ into $\mathbb{R}$. Let Ω be an open subset of $\mathbb{R}^n$ and $U = L^2(\Omega)$. If x is a function from Ω into $\mathbb{R}$, we set $S_b(x) : \omega \mapsto b(x(\omega))$. Let $X = \{x \in L^2(\Omega) \text{ such that } S_b(x) \in L^2(\Omega)\}$. Then S_b is obviously a monotone correspondence from X into $L^2(\Omega)$. Assume also that $0 = b(0)$. Then S_b is maximal monotone.

To see this let $y \in L^2(\Omega)$ be given. Since b is maximal monotone, for almost all $\omega \in \Omega$, there exists $x(\omega) \in \mathbb{R}$ such that $x(\omega) + b(x(\omega)) = y(\omega)$. Since $0 = (1 + b)^{-1}(0)$, we deduce that $|x(\omega)| = |(1 + b)^{-1}(y(\omega)) - (1 + b)^{-1}(0)| \leq |y(\omega) - 0|$. Since $y \in L^2(\Omega)$, then x belongs also to $L^2(\Omega)$. Hence $1 + S_b$ is surjective. $\square$

For instance, the map $x \in L^2(\Omega) \mapsto x^3 \in L^2(\Omega)$ is maximal monotone.

Example 2. Any monotone map S continuous from U supplied with the finite topology into U supplied with the weak topology is maximal monotone.

Proof. Let x and p satisfy $\langle p - S(y), x - y \rangle \leq 0$ for all $y \in U$. By Proposition 8, we have to prove that $p = S(x)$. For this purpose, we take any $z \in U$ and we set $y = \theta z + (1 - \theta)x = x - \theta(x - z)$ where $\theta \in]0, 1[$. We deduce that $\langle p - S(x - \theta(x - z)), x - z \rangle \leq 0$. On letting θ tend to 0, the continuity property of S implies that $\langle p - S(x), x - z \rangle \leq 0$ for all $z \in U$. Hence $p \in S(x)$. $\square$

Example 3. The most important example of a maximal monotone correspondence is the subdifferential of a lower semi-continuous function:

Proposition 10. Let $f : U \mapsto]-\infty, \infty]$ be a proper lower semi-continuous convex function and $X = \{x \in U \text{ such that } \partial f(x) \neq \emptyset\}$ the domain of the subdifferential of f . Then ∂f is a maximal monotone operator. Furthermore, $X = \overline{\text{Dom}(f)}$ and $\text{Int}(X) = \text{Int}(\text{Dom}(f))$.

Proof. (a) To prove that ∂f is maximal monotone, we prove that $1 + \partial f$ is surjective, i.e. if y is given in U , there exists x such that $y - x \in \partial f(x)$. In fact, we shall check that this solution is the minimum of the function $g : z \mapsto f(z) +$

$+\frac{1}{2}\|y-z\|^2$. This function is lower semi-continuous and lower semi-compact (since, if $p \in \text{Dom}(f^*)$, the inequality $g(z) \leq \theta$ implies that $\|y-z\|^2 \leq 4(\theta + f^*(p) + \|p\|^2)$ and thus, that z remains in a weak compact subset of U). Therefore, there exists $x \in U$ which minimizes g . By Proposition 4.2.6, x is a solution to the variational inequalities $f(x) - f(z) + \langle x - y, x - z \rangle \leq 0$. It means that $y - x \in \partial f(x)$, i.e., that $x \in (1 + \partial f)^{-1}(y)$. Hence ∂f is maximal monotone.

(b) Since $X \subset \text{Dom}(f)$, we prove that $\text{Dom}(f) \subset \bar{X}$. Since ∂f is maximal monotone, for any $k > 0$, there exists a solution $x_k \in X$ of the equation

$$(67) \quad y \in x_k + k \partial f(x_k)$$

where y is given in $\text{Dom}(f)$. This implies that

$$(68) \quad f(x_k) - f(x) \leq \left\langle \frac{y - x_k}{k}, x_k - x \right\rangle \quad \text{for all } x \in U.$$

If we take $p \in \text{Dom}(f^*)$ and $x = y$, we obtain the following inequality:

$$\begin{aligned} \|x_k - y\|^2 &\leq kf(y) + kf^*(p) - \langle kp, x_k \rangle \\ &\leq \frac{1}{2} \|x_k - y\|^2 + k(f(y) + f^*(p) - \langle p, y \rangle + \frac{1}{2} \|p\|^2). \end{aligned}$$

Hence x_k converges to y and thus, $\text{Dom}(f) \subset \bar{X} \subset \overline{\text{Dom}(f)}$.

(c) We prove now that any $y \in \text{Int}(\text{Dom}(f))$ belongs to X . For this purpose, consider the solutions x_k of (67) and set $x = y + \theta z$ in (68), where $\|z\| \leq 1$. We obtain that

$$\begin{aligned} f(x_k) - f(y + \theta z) &\leq \left\langle \frac{y - x_k}{k}, x_k - y - \theta z \right\rangle \\ &\leq -\frac{1}{k} \|y - x_k\|^2 - \theta \left\langle \frac{y - x_k}{k}, z \right\rangle. \end{aligned}$$

Therefore, $\sup_{k \geq 0} |\langle (y - x_k)/k, z \rangle|$ is finite. By the Banach–Steinhaus theorem (see Theorem 10 of Appendix A), we deduce that $(y - x_k)/k$ remains in a weak compact subset and that a generalized subsequence converges weakly to an element p . We have seen in (b) that x_k converges to y . Since f is lower semi-continuous, inequalities (68) imply that

$$(69) \quad f(y) - f(x) \leq \langle p, y - x \rangle \quad \text{for all } x \in U.$$

Hence $p \in \partial f(y)$, i.e. y belongs to X . Since $X \subset \text{Dom}(f)$, this shows that the interiors of X and $\text{Dom}(f)$ coincide. $\square$

Maximal monotone correspondences have many other properties (which can be found in the book of Brézis [3] and the paper of Brézis and Haraux for instance). We mention some of these without proof.

Let S be a maximal monotone correspondence from X into U

(a) $\bar{X}$ and $\text{Int}(X)$ are convex.

(b) If $\text{Int}(X) \neq \emptyset$, then, for all $x \in \text{Int}(X)$, S is bounded on some neighborhood of x .

(c) S is surjective if and only if $\forall y \in \bar{S(X)}$, S^{-1} is bounded on some neighborhood of y .

(d) In particular, any coercive maximal monotone correspondence is surjective (compare with Theorem 3).

(e) $\forall k > 0$, the Lipschitz map J_k defined by $J_k(x) = (1 + kS)^{-1}(x)$ satisfies

(70) $J_k(x)$ converges to the projection of x onto $\bar{X}$ when $k \rightarrow 0$.

(f) $\forall k > 0$, the map $S_k = (1/k)(I - J_k)$ is maximal monotone and Lipschitz with constant $1/k$. It satisfies: $S_k(x) \in S(J_k(x))$, $\forall x \in X$. Furthermore, $\forall x \in X$, $S_k(x)$ converges weakly to the projection of 0 onto $S(x)$ (S_k is called the *Yosida approximation* of S).

(g) Let T be a maximal monotone correspondence from Y into U . If either

(71) $\text{Int}(X) \cap Y \neq \emptyset$

or

(72) $\forall k > 0, \forall \{x, p\} \in G(T), \langle p, S_k(x) \rangle \geq 0$,

then $S+T$ is also maximal monotone.

(h) We shall say that a maximal monotone correspondence T from Y into U satisfies the Brézis-Haraux property if $\forall y \in Y, \forall q \in T(Y), \exists c = c(y, q)$ such that, $\forall \{x, p\} \in G(T), \langle p - q, x - y \rangle \geq c$.

It is easy to check that any "tri-monotone correspondence" (satisfying $\langle p - r, x - y \rangle \geq \langle r - q, y - z \rangle$ when $\{x, p\}, \{y, q\}$ and $\{z, r\}$ belong to $G(T)$) and in particular any cyclically monotone correspondence satisfies this property. We can prove that if T is a maximal monotone correspondence from U into U satisfying the Brézis-Haraux property, then

(73)
$$\begin{cases} \text{(i) } \text{Int}((S+T)(X)) = \text{Int}(S(X)+T(U)), \\ \text{(ii) } \overline{(S+T)(X)} = \overline{S(X)+T(U)}. \end{cases}$$

In particular, $S+T$ is surjective whenever S or T is surjective.

Furthermore, if $X = U$, then $I+ST$ is surjective (Brézis and Browder [1, 2]) The latter result enables us to solve non-linear Hammerstein integral equations.

(i) Finally, the fundamental property is the existence and uniqueness of the solution of the multi-valued differential equation $dx/dt \in -S(x(t))$, $x(0) = x_0$, where x_0 is given in X . (See Brézis [3] for instance.)

13.3. Relaxation of convexity assumptions

In most cases, we can replace convexity assumptions by assumptions of γ -convexity: If γ maps the set $\mathcal{M}(X)$ of discrete probabilities into X , we say that $f: X \rightarrow \mathbf{R}$ is γ -convex if

$$\forall m \in \mathcal{M}(X), \quad f(\gamma m) \leq \langle m, f \rangle$$

In other words, we can usually replace the barycentric operator β used in defining ordinary convex functions by any map γ .

We know that $F(X) + \mathbf{R}_+^n$ is convex whenever the n components f_i of F are convex functions. This result remains true if the functions f_i are γ -convex. Moreover, the converse statement holds. If $F(X) + \mathbf{R}_+^n$ is convex, there exists a map γ from $\mathcal{M}(X)$ into X such that the functions f_i are γ -convex.

This is the reason why the minisup theorem remains true when we assume that the function φ is γ_X -convex with respect to x and γ_Y -concave with respect to y for any maps $\gamma_X: \mathcal{M}(X) \rightarrow X$ and $\gamma_Y: \mathcal{M}(Y) \rightarrow Y$. Also, an optimal decision rule continues to exist when we assume only that φ is γ_Y -concave with respect to y .

We can explain this by the fact that *any cone of γ -convex functions is the image of a cone of convex functions by a linear operator $\pi^*: g \rightarrow \pi^*g = g \circ \pi$ where π maps X onto a convex subset $\pi(X)$ of a vector space W .*

In particular, we can construct γ -convex functions by taking products $f = g \circ \pi$ of convex functions g by a given map π from X onto a convex subset $\pi(X)$ of a vector space W .

The most important example is given by the *Lyapunov theorem*. We take X to be a σ -algebra on a set Ω , $W = \mathbf{R}^n$ and $\pi: X \rightarrow \mathbf{R}^n$ to be an *atomless* bounded vector-valued measure. The Lyapunov theorem states that $\pi(X)$ is convex. Hence the set functions $A \in X \rightarrow g(\pi(A))$ are γ -convex whenever g is a convex function defined on $\mathbf{R}^n$. This allows the study of games where “nature” is a player choosing an “event” $A \in X$ in a set Ω of “states of nature”.

We also meet “natural” examples of maps γ from $\mathcal{M}(X)$ into X . For instance, we take X to be the family of the (weakly) compact convex subsets K of a vector space U and the map γ to be

$$\gamma\left(\sum_{i=1}^n \alpha^i \delta(K_i)\right) = \sum_{i=1}^n \alpha^i K_i.$$

If g is a convex lower semi-continuous function on U (in the usual sense), the function f defined on X by

$$f(K) = \inf_{x \in K} g(x)$$

is γ -convex.

13.3.1. Definition of γ -convex functions

Definition 1. Let $\gamma : \mathcal{M}(X) \rightarrow X$ be a correspondence with non-empty values mapping the set $\mathcal{M}(X)$ of discrete probabilities m into subsets $\gamma(m)$ of X .

A function f defined on X is said to be " γ -convex" if and only if

$$(1) \quad \forall m \in \mathcal{M}(X), \forall x \in \gamma(m), \quad f(x) \leq f^A(m) = \langle m, f \rangle.$$

A function f is " γ -affine" if

$$(2) \quad \forall m \in \mathcal{M}(X), \forall x \in \gamma(m), \quad f(x) = f^A(m)$$

and is " γ -concave" if $-f$ is γ -convex.

A subset K of X is said to be " γ -convex" if

$$(3) \quad \gamma(\mathcal{M}(K)) \subset K.$$

By Proposition 1.3.5, convex functions and convex sets are γ -convex functions and convex sets when we take $\gamma = \beta$ to be the barycentric operator.

We will present two other classes of examples in Sections 7 and 8 below.

13.3.2. The fundamental characteristic property of families of γ -convex functions

Let $f : X \times Y \rightarrow \mathbf{R}$ be a function. It is characterized by the map F from X into the space $\mathcal{S}(Y) = \mathbf{R}^Y$ (of functions on Y) defined by

$$(4) \quad \forall x \in X, \quad F(x) : y \rightarrow F(x)(y) = f(x, y).$$

Let $\gamma : \mathcal{M}(X) \rightarrow X$ be a correspondence with non-empty values. If the function f is γ -convex with respect to x , we obtain the following fundamental property.

Proposition 1. Suppose that

$$(5) \quad \forall y \in Y, \quad \text{the function } x \rightarrow f(x, y) \text{ is } \gamma\text{-convex.}$$

Then

$$(6) \quad F(X) + \mathbf{R}_+^Y \text{ is a convex subset of } \mathbf{R}^Y$$

where $\mathbf{R}_+^Y$ denotes the cone of non-negative functions on Y .

Example. If $Y = \{1, \dots, n\}$ is a set of n elements, then $\mathbf{R}^Y = \mathbf{R}^n$, $\mathbf{R}_+^Y = \mathbf{R}^n$ and the map F from X into $\mathbf{R}^n$ is defined by

$$(7) \quad F(x) = \{f(x, 1), \dots, f(x, i), \dots, f(x, n)\} \in \mathbf{R}^n.$$

Proposition 2. *If the n functions $x \rightarrow f(x, i)$ are γ -convex, then*

$$(8) \quad F_+(X) = F(X) + \mathbf{R}_+^n \text{ is convex in } \mathbf{R}^n.$$

This generalizes Proposition 1.3.10, which played a fundamental role.

Proof of Proposition 1. Let

$$(9) \quad g = \sum_{i=1}^n \alpha_i (F(x_i) + h_i) \in \text{co} (F(X) + \mathbf{R}_+^Y)$$

where $x_i \in X$, $h_i \in \mathbf{R}_+^Y$, $\alpha_i \geq 0$ (for $i = 1, \dots, n$) and $\sum_{i=1}^n \alpha_i = 1$. We have to find $x_0 \in X$ and $h_0 \in \mathbf{R}_+^Y$ such that

$$(10) \quad g = F(x_0) + h_0.$$

For this purpose we take

$$(11) \quad x_0 \in \gamma(m) \quad \text{where } m = \sum_{i=1}^n \alpha_i \delta(x_i) \in \mathcal{M}(X)$$

and

$$(12) \quad h_0 = \sum_{i=1}^n \alpha_i h_i + \sum_{i=1}^n \alpha_i F(x_i) - F(x_0).$$

We have that $x_0 \in X$ since γ maps $\mathcal{M}(X)$ into X . The function $\sum_{i=1}^n \alpha_i F(x_i) - F(x_0)$ is positive since,

$$(13) \quad \forall y \in Y, \quad \left(\sum_{i=1}^n \alpha_i F(x_i) - F(x_0) \right) (y) = \sum_{i=1}^n \alpha_i f(x_i, y) - f(x_0, y) \geq 0$$

by the γ -convexity of the function $x \rightarrow f(x, y)$.

Therefore, the functions h being positive, the function h_0 is also positive, i.e. $h_0 \in \mathbf{R}_+^Y$. $\square$

In fact, this property characterizes sets of γ -convex functions on X in the following sense.

Proposition 3. *Let $f : X \times Y \rightarrow \mathbf{R}$ be a function characterized by a map F from X into $\mathbf{R}^Y$ satisfying*

$$(14) \quad F(X) + \mathbf{R}_+^Y \text{ is a convex subset of } \mathbf{R}^Y.$$

Then there exists a correspondence $\gamma : \mathcal{M}(X) \rightarrow X$ with non-empty values such that the functions $x \rightarrow f(x, y)$ are γ -convex.

Proof. Let $m = \sum_{i=1}^n \alpha_i \delta(x_i) \in \mathcal{M}(X)$ be a discrete probability on X . Then (14) implies that $\sum_{i=1}^n \alpha_i F(x_i) = \sum_{i=1}^n \alpha_i (F(x_i) + 0)$ belongs to $F(X) + \mathbf{R}_+^Y$ since this set is convex. Thus there exists at least one $x \in X$ and $h \in \mathbf{R}_+^Y$ such that $\sum_{i=1}^n \alpha_i F(x_i) = F(x) + h$. In other words, for any $y \in Y$, $\sum_{i=1}^n \alpha_i f(x_i, y) = (\sum_{i=1}^n \alpha_i F(x_i))(y) = F(x)(y) + h(y) \geq f(x, y)$. Therefore, if we denote by $\gamma(m)$ the subset of such elements $x \in X$, we see that f is γ -convex with respect to x . $\square$

13.3.3. The minisup theorem for γ_X -convex- γ_Y -concave functions

In the minisup theorem (see Theorem 7.1.5), we can replace the assumptions of convexity of φ with respect to x and concavity of φ with respect to y by assumptions of γ_X -convexity and γ_Y -concavity respectively. Specifically, we consider two correspondences with non-empty values.

$$(15) \quad \gamma_X : \mathcal{M}(X) \longrightarrow X \quad \text{and} \quad \gamma_Y : \mathcal{M}(Y) \longrightarrow Y$$

Theorem 1. Suppose that X is a subset of a topological space U , that

$$(16) \quad \exists y_0 \in Y \text{ such that } x \rightarrow \varphi(x, y_0) \text{ is lower semi-compact}$$

and that

$$(17) \quad \forall y \in Y, \quad x \rightarrow \varphi(x, y) \text{ is lower semi-continuous on } U.$$

Suppose also that

$$(18) \quad \forall y \in Y, \quad x \rightarrow \varphi(x, y) \text{ is } \gamma_X\text{-convex}$$

and

$$(19) \quad \forall x \in X, \quad y \rightarrow \varphi(x, y) \text{ is } \gamma_Y\text{-concave.}$$

Then there exists a minisup $\bar{x}$ of φ .

Proof. Since assumptions (16) and (17) imply that there exists $\bar{x}$ such that $\sup_{y \in Y} \varphi(\bar{x}, y) \leq v^\square$, it remains to prove that

$$(20) \quad v^\square = \sup_{K \in \mathcal{J}} \inf_{x \in X} \sup_{y \in K} \varphi(x, y) \leq v^\flat = \sup_{y \in Y} \inf_{x \in X} \varphi(x, y).$$

The proof of this inequality is analogous to that of Theorem 7.1.5. We have to prove that for any

$$K = \{y_1, \dots, y_n\} \in \mathcal{J},$$

$$(21) \quad v_K^\# = \inf_{x \in X} \sup_{\lambda \in \mathcal{M}^n} \sum_{i=1}^n \lambda^i \varphi(x, y_i) \leq v^b$$

$$[v^\square = \sup_{K \in \mathcal{J}} v_K^\#].$$

Since the n functions $x \mapsto \varphi(x, y_i)$ are γ -convex, then $\phi_+(X) = \phi(X) + \mathbf{R}_+^n$ is convex we set $\phi(x) = \{\varphi(x, y_1), \dots, \varphi(x, y_n)\}$.

This implies that

$$(22) \quad v_K^\# = \sup_{\lambda \in \mathcal{M}^n} \inf_{x \in X} \sum_{i=1}^n \lambda^i \varphi(x, y_i)$$

(by the separation theorem).

Now, since the functions $y \mapsto \varphi(x, y)$ are γ_Y -concave, we deduce that

$$(23) \quad \sup_{\lambda \in \mathcal{M}^n} \inf_{x \in X} \sum_{i=1}^n \lambda^i \varphi(x, y_i) \leq v^b$$

for we associate with any $\lambda \in \mathcal{M}^n$ an element $y_\lambda \in \gamma(\sum_{i=1}^n \lambda^i \delta(y_i))$, and deduce that

$$\inf_{x \in X} \left(\sum_{i=1}^n \lambda^i \varphi(x, y_i) \right) \leq \inf_{x \in X} \varphi(x, y_\lambda) \leq v^b.]$$

Hence (22) and (23) imply (21). $\square$

Remark. We can use this theorem to prove that

$$(24) \quad v^\square(\mathcal{A}) = \sup_{K \in \mathcal{A}} \inf_{x \in X} \sup_{y \in K} \varphi(x, y) = v^b$$

whenever

$$(25) \quad \begin{cases} \text{(i) } \mathcal{A} \text{ is a covering of } Y \text{ by compact subsets of } Y, \\ \text{(ii) } \forall x \in X, y \mapsto \varphi(x, y) \text{ is } \gamma_Y\text{-concave and upper semi-continuous} \\ \text{(iii) } \forall y \in Y, x \mapsto \varphi(x, y) \text{ is } \gamma_X\text{-convex} \end{cases}$$

(see Proposition 1.3).

This allows us to replace the assumptions of convexity and concavity of φ in Theorem 1.2 by assumptions of γ_X -convexity and γ_Y -concavity of φ with respect to x and y respectively.

13.3.4. Existence of optimal decision rules for functions γ_Y -concave with respect to y

We can also replace the assumption of quasi-concavity by an assumption of quasi- γ -concavity in Theorem 7.1.2 concerning the existence of an optimal decision rule.

For this purpose, we need to define a " γ -finite topology" on subsets Y analogous to the finite topology on convex subsets.

Definition 2. Let γ be a map from $\mathcal{M}(Y)$ into Y . The " γ -finite topology" on Y is the strongest topology for which the maps $\lambda \in \mathcal{M}^n \mapsto \gamma_K(\lambda) = \gamma(\sum_{i=1}^n \lambda^i \delta(y_i))$ are continuous when $K = \{y_1, \dots, y_n\}$ ranges over the family of finite subsets.

A map C from Y into a topological space X is continuous when Y is supplied with the γ -finite topology if and only if

$$(26) \quad \forall K \in \mathcal{F}, \quad C\gamma_K \text{ is continuous from } \mathcal{M}^n \text{ into } X.$$

Also, we define a function φ to be quasi- γ -concave with respect to y if and only if

$$(27) \quad \forall x \in X, \forall \alpha \in \mathbf{R}, \quad \text{the subsets } \{y \in Y \text{ such that } \varphi(x, y) \geq \alpha\} \text{ are } \gamma\text{-convex}.$$

We can prove the following theorem.

Theorem 2. Suppose that the compactness and continuity assumptions (16) and (17) hold.

Let γ be a map from $\mathcal{M}(Y)$ into Y and let $\mathcal{C}(Y, X)$ be the set of continuous maps from Y into X when Y is supplied with the γ -finite topology.

If we assume that

$$(28) \quad \forall x \in X, \quad y \rightarrow \varphi(x, y) \text{ is quasi-}\gamma\text{-concave,}$$

then there exists $\bar{x} \in X$ such that

$$(29) \quad \sup_{y \in Y} \varphi(\bar{x}, y) = \inf_{C \in \mathcal{C}(Y, X)} \sup_{y \in Y} \varphi(C(y), y).$$

Proof. This is the same as the proof of Theorem 7.1.2. Since assumptions (16) and (17) imply the existence of $\bar{x}$ such that $\sup_{y \in Y} \varphi(\bar{x}, y) = v^\square$, it is sufficient to prove that we can associate with any $K = \{y_1, \dots, y_n\}$ and any map $C \in \mathcal{C}(Y, X)$ an element $\bar{x} \in X$ such that

$$(30) \quad \max_{i=1, \dots, n} \varphi(\bar{x}, y_i) \leq \sup_{y \in Y} \varphi(C(y), y).$$

For this purpose, we introduce the function $\tilde{\varphi}$ defined on $\mathcal{M}^n \times Y$ by

$$(31) \quad \tilde{\varphi}(\lambda, y) = \varphi(C\gamma_K(\lambda), y) = \varphi\left(C\gamma\left(\sum_{i=1}^n \lambda^i \delta(y_i)\right), y\right).$$

and the n subsets

$$(32) \quad F_i = \{\lambda \in \mathcal{M}^n \text{ such that } \tilde{\varphi}(\lambda, y_i) \leq \sup_{y \in Y} \varphi(C(y), y)\}.$$

These subsets are closed since $C\gamma_K$ is continuous and φ is lower semi-continuous with respect to x .

Since the functions $y \rightarrow \varphi(x, y)$ are quasi- γ -concave, we deduce that $\forall \lambda \in \mathcal{M}^n$, $\lambda \in \bigcap_{i \in A_\lambda} F_i$ where $A_\lambda = \{i \text{ such that } \lambda^i > 0\}$. Hence the assumptions of the Knaster-Kuratowski-Mazurkiewicz lemma are satisfied and there exists $\tilde{\lambda} \in \bigcap_{i=1}^n F_i$.

Thus $\tilde{x} = C\gamma_K(\tilde{\lambda}) = C\gamma(\sum_{i=1}^n \tilde{\lambda}^i \delta(y_i))$ satisfies inequality (30). $\square$

Remark. This theorem implies that

$$(33) \quad v^\square(\mathcal{A}) = \sup_{K \in \mathcal{A}} \inf_{x \in X} \sup_{y \in K} \varphi(x, y) = \inf_{C \in \mathcal{C}(Y, X)} \sup_{y \in Y} \varphi(C(y), y)$$

whenever

$$(34) \quad \begin{cases} \text{(i) } \mathcal{A} \text{ is a covering of } Y \text{ by compact subsets if } Y, \\ \text{(ii) } \forall x \in X, y \mapsto \varphi(x, y) \text{ is quasi-}\gamma\text{-concave (see Proposition 1.2).} \end{cases}$$

(see Proposition 1.2).

We can replace the assumption of concavity with respect to y by an assumption of γ -concavity in Theorem 1.2.

13.3.5. Example: Image of a cone of convex functions by π^*

Let π be a map from a set X into a vector space W such that

$$(35) \quad \text{the image } \pi(X) \text{ is a convex subset of } W.$$

It defines a linear operator π^* from $\mathbf{R}^{\pi(X)}$ into $\mathbf{R}^X$ by

$$(36) \quad \forall x \in X, (\pi^* f)(x) = f[\pi(x)].$$

A question immediately arises. Is the image by π^* of the cone of convex functions on $\pi(X)$ a cone of γ -convex functions on X ? The answer is positive. Let $\pi^A : \mathcal{M}(X) \rightarrow W$ be the linear extension of π defined by

$$(37) \quad \pi^A(m) = \sum_{i=1}^n \lambda^i \pi(x_i) \quad \text{whenever } m = \sum_{i=1}^n \lambda^i \delta(x_i)$$

(see Definition 1.3.4).

Since $\pi(X)$ is convex, $\pi^d(m)$ belongs to $\pi(X)$ whenever $m \in \mathcal{M}(X)$. Therefore, the correspondence γ_π from $\mathcal{M}(X)$ into X defined by

$$(38) \quad \gamma_\pi = \pi^{-1}\pi^d$$

has non-empty values. In other words, if $m \in \mathcal{M}(X)$,

$$\gamma_\pi(m) = \{x \in X \text{ such that } \pi x = \pi^d(m)\}.$$

Proposition 4. *Let $\pi : X \rightarrow W$ satisfy (35). Then the function f is the image π^*g of a convex function g on $\pi(X)$ if and only if f is a γ_π -convex function (where $\gamma_\pi = \pi^{-1}\pi^d$).*

Proof. The first statement is quite obvious. Let $f = \pi^*g = g \circ \pi$ where g is convex. Let $x \in \gamma_\pi(m) = \pi^{-1}\pi^d(m)$ where $m = \sum_{i=1}^n \alpha^i \delta(x_i)$. Then

$$\pi x = \sum \alpha^i \pi(x_i) \text{ and}$$

$$\begin{aligned} f(x) &= g(\pi x) = g\left(\sum_{i=1}^n \alpha^i \pi(x_i)\right) \leq \sum_{i=1}^n \alpha^i g(\pi(x_i)) \\ &= \sum_{i=1}^n \alpha^i f(x_i) = f^d(m). \end{aligned}$$

Conversely, let f be γ_π -convex. First, we notice that

$$(39) \quad \text{if } \pi(x_1) = \pi(x_2), \text{ then } f(x_1) = f(x_2).$$

[We have that $x_1 \in \pi^{-1}\pi x_2 = \gamma_\pi^d(\delta(x_2))$ and $x_2 \in \pi^{-1}\pi x_1 = \gamma_\pi^d(\delta(x_1))$. Thus $f(x_1) \leq f(x_2)$ and $f(x_2) \leq f(x_1)$.]

We denote by $\langle \pi \rangle$ the equivalence relation defined by

$$(40) \quad x_1 \langle \pi \rangle x_2 \Leftrightarrow \pi(x_1) = \pi(x_2).$$

and by

$$(41) \quad \begin{cases} X/\langle \pi \rangle \text{ the factor space and } \theta \text{ the canonical} \\ \text{surjection from } X \text{ onto } X/\langle \pi \rangle. \end{cases}$$

Therefore, any map φ constant on the $\langle \pi \rangle$ -equivalence classes can be written $\varphi = \hat{\varphi} \circ \theta$ where $\hat{\varphi}$ is defined on $X/\langle \pi \rangle$.

In particular, $\pi = \hat{\pi} \circ \theta$ where $\hat{\pi}$ is a bijective map from $X/\langle \pi \rangle$ onto $\pi(X)$ and $f = \hat{f} \circ \theta$ where $\hat{f}$ maps $X/\langle \pi \rangle$ into $\mathbb{R}$.

Since $\hat{\pi}$ is bijective, we can write $\theta = \hat{\pi}^{-1} \circ \pi$ and $f = \hat{f} \circ \theta = \hat{f} \circ \hat{\pi}^{-1} \circ \pi = g \circ \pi$ where $g = \hat{f} \circ \hat{\pi}^{-1}$. It remains to deduce convexity of $g = \hat{f} \circ \hat{\pi}^{-1}$ from the γ_π -convexity of f .

Let $\sum_{i=1}^n \alpha^i \delta(u_i)$ be a discrete probability on $\pi(X)$ and take $x_i \in \pi^{-1}(u_i)$ and

$$x_0 \in \pi^{-1}\left(\sum_{i=1}^n \alpha^i \pi(x_i)\right) = \pi^{-1}\left(\sum_{i=1}^n \alpha^i u_i\right).$$

Therefore, g is convex since

$$\begin{aligned} g\left(\sum_{i=1}^n \alpha^i u_i\right) &= g(\pi x_0) = f(x_0) \\ &\leq \sum_{i=1}^n \alpha^i f(x_i) = \sum_{i=1}^n \alpha^i g(\pi x_i) = \sum_{i=1}^n \alpha^i g(u_i). \quad \square \end{aligned}$$

13.3.6. Relations between convexity and γ -convexity

We now answer the fundamental question: is the cone of γ -convex functions the image by an operator π^* of a cone of convex functions?

Proposition 5. *Let $\gamma : \mathcal{M}(X) \rightarrow X$ be a correspondence with non-empty values. Then there exist a vector space W and a map π from X into W such that*

$$(42) \quad \begin{cases} \text{(i) } \pi(X) \text{ is convex,} \\ \text{(ii) } \gamma = \pi^{-1}\pi^d. \end{cases}$$

Proof. Let $\mathcal{S}^*(X)$ be the space of discrete measures on X and M be the closed vector space spanned by the set N of measures $\delta\gamma m - m$ when m ranges over $\mathcal{S}^*(X)$, i.e.

$$(43) \quad N = \{\delta x - m\}_{x \in \gamma(m), m \in \mathcal{M}(X)}.$$

We introduce

$$(44) \quad \begin{cases} \text{(i) the factor space } W = \mathcal{S}^*(X)/M, \\ \text{(ii) the canonical surjection } \pi^d \text{ from } \mathcal{S}^*(X) \text{ onto } W. \end{cases}$$

Let $\delta : X \rightarrow \mathcal{S}^*(X)$ be the Dirac operator. We define π by

$$(45) \quad \begin{aligned} \pi &= \pi^d \delta \text{ maps } X \text{ into } W \text{ via } \mathcal{S}^*(X) \\ X &\xrightarrow{\delta} \mathcal{S}^*(X) \xrightarrow{\pi^d} \mathcal{S}^*(X)/M = W. \end{aligned}$$

Thus π^d is a linear extension of π , i.e. $\pi^d(\sum \alpha^i \delta(x_i)) = \sum \alpha^i \pi^d(\delta(x_i)) = \sum \alpha^i \pi(x_i)$. On the other hand, since $\delta x - m \in M$ when $x \in \gamma(m)$ and $m \in \mathcal{M}(X)$, we obtain that

$$\pi^d(\delta x - m) = \pi^d \delta x - \pi^d(m) = \pi x - \pi^d m = 0.$$

Therefore

$$(46) \quad x \in \gamma(m) \text{ if and only if } x \in \pi^{-1}\pi^A(m). \quad \square$$

Therefore, Propositions 4 and 5 imply the following theorem.

Theorem 3. *Let γ be a correspondence with non-empty values. Then there exist a vector space W and a map π from X into W satisfying*

$$(47) \quad \begin{cases} \text{(i) } \pi(X) \text{ is convex,} \\ \text{(ii) the cone of } \gamma\text{-convex functions is the image by } \pi^* \text{ of the cone} \\ \text{of convex functions on } \pi(X). \end{cases}$$

We remark also that

$$(48) \quad \text{the vector space } F \text{ of } \gamma\text{-affine functions equals } N^\perp = M^\perp.$$

Therefore, since $\mathcal{S}^*(X)$ is the dual of $\mathcal{S}(X)$ supplied with the topology of pointwise convergence, which coincides with the weak topology $\sigma(\mathcal{S}(X), \mathcal{S}^*(X))$, we deduce that

$$(49) \quad F \text{ is isomorphic with the dual of } W$$

when F is supplied with (the restriction to F of) the topology of pointwise convergence and W is supplied with the factor topology of $\mathcal{S}^*(X)$ supplied with the weak topology.

It is quite obvious that *the weakest topology on X for which π is continuous coincides with the weakest topology for which all the functions of F are continuous.*

We shall denote this topology by $\sigma_\gamma(X, F)$, and call it “*the weak topology on X associated with γ* ”.

It is an Hausdorff topology if and only if *either π is injective or, equivalently, if F “separates the points of X ” in the following sense*

$$(50) \quad \text{If } f(x) = f(y) \text{ for all } f \in F, \text{ then } x = y.$$

Formula (46) shows that π is injective if and only if γ is a map.

Note also that the weak topology $\sigma_\gamma(X, F)$ on X is defined by the semi-distances

$$(51) \quad d_K(x, y) = \sup_{f \in K} |f(x) - f(y)| = P_K(\pi(x) - \pi(y))$$

where K ranges over the family $\mathcal{S}$ of finite subsets of F . Theorem 3 shows that the properties of convex functions can be “carried” into analogous properties

of γ -convex functions. For instance, we can use a vector space of γ -affine functions as a "space of perturbations" of minimization problems of functions on X . We define φ_γ^{**} by

$$(52) \quad \varphi_\gamma^*(f) = \sup_{x \in X} [f(x) - \varphi(x)] \quad (\text{when } f \in F)$$

and

$$(53) \quad \varphi_\gamma^{**}(x) = \sup_{f \in F} [f(x) - \varphi_\gamma^*(f)].$$

The function φ_γ^* is convex and lower semi-continuous on X and the function φ_γ^{**} is γ -convex and lower semi-continuous. As in Theorem 2.4.1, we can characterize γ -convex lower semi-continuous functions.

Proposition 6. *Let F be the vector space of γ -affine functions on X , supplied with the weak topology $\sigma_\gamma(X, F)$. A function φ is γ -convex and lower semi-continuous if and only if*

$$(54) \quad \forall x \in X, \quad \varphi(x) = \varphi_\gamma^{**}(x).$$

Proof. If φ is γ -convex, there exists a convex function ψ defined on $\pi(X)$ such that

$$\forall x \in X, \quad \varphi(x) = \psi(\pi(x)).$$

The function ψ is lower semi-continuous since φ is lower semi-continuous on X supplied with the weakest topology for which π is continuous.

Therefore, if we denote by $\tilde{\psi} = \psi_{\pi(X)}$ the proper lower semi-continuous convex function from W into $]-\infty, +\infty]$, we deduce from Theorem 2.4.1. that

$$\tilde{\psi}(u) = \sup_{f \in F} (\langle f, u \rangle - \psi^*(f))$$

where

$$\begin{aligned} \psi^*(f) &= \sup_{u \in \pi(X)} [\langle f, u \rangle - \psi(u)] \\ &= \sup_{x \in X} [\langle f, \pi(x) \rangle - \varphi(x)] = \varphi_\gamma^*(f). \end{aligned}$$

Therefore

$$\begin{aligned} \varphi(x) &= \psi(\pi x) = \sup_{f \in F} [\langle f, \pi(x) \rangle - \varphi_\gamma^*(f)] \\ &= \sup_{f \in F} [f(x) - \varphi(f)] = \varphi_\gamma^{**}(x). \end{aligned}$$

13.3.7. Example: $\tilde{\beta}$ -convex set functions

Let U and U^* be two paired vector spaces supplied with their weak topologies.

Consider the following strategy set:

(55) X is the set of weakly compact convex subsets K of U .

We define the map $\gamma = \tilde{\beta}$ from $\mathcal{M}(X)$ into X by

$$(56) \quad \tilde{\beta}\left(\sum_{i=1}^n \lambda^i \delta(K_i)\right) = \sum_{i=1}^n \lambda^i K_i \in X$$

(since $\sum_{i=1}^n \lambda^i K_i$ is a weakly compact convex set.)

Therefore, the cone of $\tilde{\beta}$ -convex functions is the cone of functions f defined on X and satisfying

$$(57) \quad f\left(\sum_{i=1}^n \lambda^i K_i\right) \leq \sum_{i=1}^n \lambda^i f(K_i).$$

For instance, if g is a *convex* function defined on U , then the function f defined on X by

$$(58) \quad f(K) = \inf_{x \in K} g(x)$$

is *obviously* $\tilde{\beta}$ -convex.

To prove this we write

$$\begin{aligned} f\left(\sum_{i=1}^n \lambda^i K_i\right) &= \inf_{x_i \in K_i} g\left(\sum_{i=1}^n \lambda^i x_i\right) \\ &\leq \inf_{x_i \in K_i} \sum_{i=1}^n \lambda^i g(x_i) = \sum_{i=1}^n \lambda^i \inf_{x_i \in K_i} g(x_i) \\ &= \sum_{i=1}^n \lambda^i f(K_i). \end{aligned}$$

In particular, for any $p \in U^*$, the functions

$$(59) \quad K \rightarrow \sigma^*(K, p) \quad \text{and} \quad K \mapsto \sum_{i=1}^n \lambda^i \sigma^*(K, p_i)$$

are $\tilde{\beta}$ -affine.

We can give an explicit form for the map π from X into W such that $\tilde{\beta} = \pi^{-1}\pi^A$.

We suppose that

- (60) W is the subspace of $\mathcal{S}(U^*)$ spanned by the support functions $\sigma^b(K; \cdot)$ and $\sigma^*(K; \cdot)$ when K ranges over X .

and π is the map from X into W defined by

- (61) $\forall K \in X, \pi(K) = \sigma^b(K; \cdot) \in \mathcal{S}(U^*)$.

The map π is *injective* since the lower support functions characterize the closed subsets K .

It is clear that $\tilde{\beta} = \pi^{-1}\pi^A$ since

$$\begin{aligned} \pi\tilde{\beta}\left(\sum_{i=1}^n \lambda^i \delta(K_i)\right) &= \pi\left(\sum_{i=1}^n \lambda^i K_i\right) = \\ \sigma^b\left(\sum_{i=1}^n \lambda^i K_i; \cdot\right) &= \sum_{i=1}^n \lambda^i \sigma^b(K_i; \cdot) = \sum_{i=1}^n \lambda^i \pi(K_i) \\ &= \pi^A\left(\sum_{i=1}^n \lambda^i \delta(K_i)\right). \end{aligned}$$

In Section 15.3 we shall study in more detail the weakest topology on X for which π is continuous (when $\mathcal{S}(U^*)$ is supplied with the topology of pointwise convergence). If M is the symmetric convex hull of a finite subset of U^* , we obtain from the minimax theorem that

$$\begin{aligned} \sup_{p \in M} (\sigma^b(K_1; p) - \sigma^b(K_2; p)) &= \\ &= \sup_{p \in M} \inf_{x_1 \in K_1} \sup_{x_2 \in K_2} |\langle p, x_1 - x_2 \rangle| \\ &= \sup_{x_2 \in K_2} \inf_{x_1 \in K_1} \sup_{p \in M} |\langle p, x_1 - x_2 \rangle| \\ &= \sup_{x_2 \in K_2} \inf_{x_1 \in K_1} P_M(x_1 - x_2). \end{aligned}$$

Hence the topology on X is defined by the semi-distances

$$\begin{aligned} (62) \quad d_M(K_1, K_2) &= P_M(\pi(K_1) - \pi(K_2)) \\ &= \max \left[\sup_{x_1 \in K_1} \inf_{x_2 \in K_2} P_M(x_1 - x_2), \sup_{x_2 \in K_2} \inf_{x_1 \in K_1} P_M(x_1 - x_2) \right] \end{aligned}$$

where M ranges over the (symmetric convex hulls of) finite subsets of U^* . It is the weak Hausdorff topology we study in Section 15.3.

Interpretation. If U describes a commodity space, we can represent X as the family of production sets (which may describe firms, for instance).

If g denotes a cost function defined on the commodity space, the function f defined on X by (58) associates with each firm K its minimal cost $f(K) = \inf_{x \in K} g(x)$. Such cost functions are β -convex.

13.3.8. Example: Convex functions of atomless vector measures

We mention a very important example of a map π from a set X into a vector space W such that $\pi(X)$ is convex. We take for the strategy set

$$(63) \quad \text{a } \sigma\text{-algebra } X \text{ on a set } \Omega.$$

This is appropriate whenever Ω describes the set of “states of the nature” and a strategy $A \in X$ represents an “event”. It is also appropriate when Ω describes “a set of players” and $A \in X$ a “coalition of players”.

It is quite natural to construct loss functions on X by means of measures $\mu : X \rightarrow \mathbf{R}$. In particular, we shall devote our attention to loss functions of the following type:

$$(64) \quad \forall A \in X, f(A) = g(\mu_1(A), \dots, \mu_n(A)) = g(\mu(A))$$

where $\mu = \{\mu_1, \dots, \mu_n\}$ is the vector valued measure

$$(65) \quad \mu : A \in X \rightarrow \mu(A) = \{\mu_1(A), \dots, \mu_n(A)\} \in \mathbf{R}^n$$

and g is a convex function defined on the image $\mu(X) \subset \mathbf{R}^n$ of the σ -algebra X by the vector-valued measure μ .

Such loss functions will be γ -convex whenever

$$(66) \quad \mu(X) \text{ is a convex subset of } \mathbf{R}^n.$$

The Lyapunov theorem states that such a property is true whenever the measures μ_i are “atomless”.

Definition 3. A set $A \in X$ is said to be an “atom” of μ_i if and only if

$$(67) \quad \begin{cases} \mu_i(A) \neq 0 \text{ and for any } B \subset A, B \in X, \text{ then either} \\ \mu_i(B) = \mu_i(A) \text{ or } \mu_i(B) = 0. \end{cases}$$

A measure μ_i . (resp. a vector valued measure $\mu = \{\mu_1, \dots, \mu_n\}$) is “atomless” if it has no atoms (resp. if the measures μ_i are atomless).

The explicit statement of Lyapunov’s theorem is given below (see Appendix C).

Theorem 4. (Lyapunov). Let $\mu = \{\mu_1, \dots, \mu_n\}$ be a vector valued measure mapping X into $\mathbf{R}^n$.

If

(68) μ *is bounded and atomless*

then

(69) *the image $\mu(X)$ is a convex compact subset of $\mathbb{R}^n$.*

CHAPTER 14

INTRODUCTION TO CALCULUS OF VARIATIONS AND OPTIMAL CONTROL

We begin this chapter by extending to infinite dimensional spaces the duality theory for optimization problems presented in Chapter 5 for the case of finite dimensional spaces. Using the minimax inequalities of Section 13.1, we prove the existence of a solution $\bar{p} \in V^*$ to the dual problem

$$(*) \quad F^*(-L^*\bar{p}, \bar{p}) = \inf_{p \in V^*} F^*(-L^*p, p) = - \inf_{x \in U} F(x, Lx)$$

where F is a lower semi-continuous proper convex function and $L \in \mathcal{L}(U, V)$.

The second (and independent) section deals with minimization problems with an integral criterion and constraints of the form

$$(**) \quad v = \inf_{x \in L^1(\Omega, U)} \left[\int_{\Omega} \varphi(\omega, x(\omega)) \, d\omega + g\left(\int_{\Omega} A(\omega) x(\omega) \, d\omega\right) \right]$$

where $A(\omega) \in \mathcal{L}(U, \mathbb{R}^m)$. We shall prove the existence of Lagrange multipliers, even when the assumption of the convexity of the functions $\xi \mapsto \varphi(\omega, \xi)$ is dropped.

In the third section, we adapt the results of the first section to calculus of variations problems of the form

$$(***) \quad v = \inf_{\substack{x(0)=\xi_0 \\ x(1)=\xi_1}} \int_0^1 \phi\left(\omega, x(\omega), \frac{d}{d\omega} x(\omega)\right) \, d\omega$$

and optimal control problems

$$(***) \quad v = \inf_{\substack{x(0)=\xi_0, x(1)=\xi_1 \\ \frac{d}{d\omega} x(\omega) + A(\omega) x(\omega) = M(\omega) u(\omega)}} \int_0^1 \phi(\omega, x(\omega), u(\omega)) \, d\omega.$$

This section has to be regarded as a *very short introduction* to the field of calculus of variations and optimal control.

Finally, we devote the fourth (and independent) section to an introduction to the dynamic programming approach to continuous and impulsive control problems (which is the subject of a forthcoming book of Bensoussan and Lions).

We now briefly summarize the main results of this chapter. In the general case, we assume that $R \subset U$, $Y \subset V$ and

F is a convex lower semi-continuous function from $R \times Y$ into $\mathbf{R}$.

We associate with F the function H defined on $U \times V^*$ by

$$H(x, p) = \sup_{y \in Y} [\langle p, y \rangle - F(x, y)].$$

The existence of a Lagrange multiplier will be established either under the Slater condition

$$\exists x_0 \in R \text{ such that } L_0 x_0 \in \text{Int } Y$$

or under the weaker constraint qualification assumption

$$\left\{ \begin{array}{l} \text{(i) } R = \bigcup_{n=1}^{\infty} K_n \text{ is a countable union of compact subsets of } U, \\ \text{(ii) } \forall p \in V^*, x \mapsto H(x, p) \text{ is upper semi-continuous,} \\ \text{(iii) } 0 \in \text{Int } (Y - L(R)). \end{array} \right.$$

Note that the assumption $R = \bigcup_{n=1}^{\infty} K_n$ is always satisfied when U is the dual of a Banach space and in particular, in reflexive Banach spaces and Hilbert spaces. We will point out that the extremality relations

$$\{-L^* \bar{p}, \bar{p}\} \in \partial F(\bar{x}, L\bar{x})$$

are equivalent to the relations

$$L^* \bar{p} \in \bar{\partial}_x H(\bar{x}, \bar{p})$$

and

$$L\bar{x} \in \partial_p H(\bar{x}, \bar{p}).$$

In the second section, we prove the existence of a Lagrange multiplier $\bar{p} \in \mathbf{R}^{m*}$ of $(**)$ in a constructive manner, by approximating to it by problems of the form

$$v = \inf_{x_i \in R^i} \left[\frac{1}{T} \sum_{i=1}^T f_i(x_i) + g\left(\frac{1}{T} \sum_{i=1}^T L_i x_i\right) \right].$$

We study what happens if we drop the assumption of convexity of the functions f_i in the existence theorem for a Lagrange multiplier, and estimate the (positive) length of the consequent duality gap by

$$0 \leq v + v^* \leq \frac{m+1}{T} \max_{i=1, \dots, T} \varrho(f_i),$$

where we denote by $\varrho(f)$ the *modulus of non-convexity* of a function f . This is defined by

$$\varrho(f) = \sup_{m \in \mathcal{M}(X)} [f(\beta m) - f^2(m)]$$

(convex functions are those whose modulus of non-convexity vanishes). This estimate shows that the duality gap shrinks as the number of functions involved in the problem increases and explains why there exists a Lagrange multiplier in the case of a minimization problem with integral criterion and constraints.

We adapt the first section to the case of problems of calculus of variations whose abstract form is

$$v = \inf_{x \in U(L)} F(x, Lx, \gamma x)$$

where γ maps a subspace $U(L)$ of U onto a space $\mathcal{E}$ and L maps $U(L)$ into V . We assume that

$$U_0(L) = \text{Ker } \gamma \text{ is dense in } U.$$

We construct a dual problem which does not use the transpose L^* of L , but the transpose L_0^* of its restriction to $U_0(L)$, defined by

$$\langle L_0^* p, x \rangle = \langle p, Lx \rangle \quad \text{when } p \in V^* = V, x \in U_0(L).$$

We extend this formula when $p \in V(L_0^*) = \{p \in V \text{ such that } L_0^* p \in U = U^*\}$. We prove that there exists a unique operator $\beta^* \in \mathcal{L}(V(L_0^*), \mathcal{E}^*)$ such that the abstract Green formula

$$\langle L^* p, x \rangle = \langle p, Lx \rangle + \langle \beta^* p, \gamma x \rangle \quad \text{when } p \in V(L_0^*), x \in U(L)$$

holds. This leads us to the required dual problem

$$v^* = \inf_{p \in V(L_0^*)} F^*(-L_0^* p, p, \beta^* p).$$

If F is a lower semi-continuous convex function mapping $R \times Y \times C$ into R ($R \subset U, Y \subset V, C \subset \mathcal{E}$), the extremality relation becomes the "Euler-Lagrange system":

$$\{-L_0^* \bar{p}, \bar{p}, \beta^* \bar{p}\} \in \partial F(\bar{x}, L\bar{x}, \gamma \bar{x}).$$

This is equivalent to the Hamiltonian system

$$L_0^* \bar{p} \in \bar{\partial}_x H(\bar{x}, \bar{p}, \beta^* \bar{p}) \quad \text{and} \quad \{L\bar{x}, \gamma \bar{x}\} \in \partial_{p, \pi} H(\bar{x}, \bar{p}, \beta^* \bar{p})$$

where H is the "Hamiltonian", defined on $U \times V \times \mathcal{E}^*$ by

$$H(x, p, \pi) = \sup_{y \in V, \xi \in \mathcal{E}} [\langle p, y \rangle + \langle \pi, \xi \rangle - F(x, y, \xi)].$$

We elaborate an analogous theory in the case of optimal control problems which leads to the famous Pontryagin maximum principle.

We emphasize the fact that attention is restricted to the (very particular) "convex" case and to the (much simpler) abstract formulation of the problems of calculus of variations. This avoids the technical difficulties involved in the explicit computation of the conjugates of the functions $F(x, y) = \int_{\Omega} \phi(\omega, x(\omega), y(\omega)) d\omega$.

Finally, we devote a short section (Section 14.4) to an *introduction* to the "dynamic programming" approach to optimal control stopping time and impulsive control problems. This is a borderline topic in the context of the subject matter of this book, since the methods are more those of partial differential equations than of functional analysis which we chose as our main mathematical method. Above all, the rewarding framework within which to use the dynamic programming approach is the "stochastic" one, while this book deals with a deterministic world.

The main feature of the so-called dynamic programming approach is to determine the "performance function"

$$v(t, \xi) = \inf_{\substack{\dot{x}(s) = \Psi(s, x(s), u(s)) \\ x(t) = \xi}} \left[\int_t^T \varphi(s, x(s), u(s)) ds + h(x(T)) \right]$$

(assumed to be smooth) as the solution of a first order partial differential equation (the Hamilton-Jacobi-Bellman equation). If the performance function is known, we show that we can solve the optimal control problem and even find, at each time t , the optimal control $u(t)$ as a function of the optimal state $x(t)$. We shall also use this approach to solve "impulsive control problems". In this case, the performance function, assumed to be smooth, is a solution to partial differential quasi-variational inequalities (the Bensoussan and Lions quasi-variational inequalities).

14.1. Duality in infinite dimensional spaces

The existence of Lagrange multipliers of the minimization problem

$$v = \inf_{x \in X} F(x, Lx)$$

(where $R \subset U$, $Y \subset V$, $F: R \times Y \rightarrow \mathbb{R}$, $L \in \mathcal{L}(U, V)$) is easily proved when V is a finite dimensional space (see Chapter 5).

We shall extend these results in the case when V is an infinite dimensional space and, for this purpose, apply the minimax theorems to the "Lagrangian" defined on $U \times V^*$ by

$$l(x, p) = \langle p, Lx \rangle - H(x, p)$$

where the "Hamiltonian" H is defined on $U \times V^*$ by

$$H(x, p) = \sup_{y \in Y} [\langle p, y \rangle - F(x, y)].$$

We check that, when F is convex and lower semi-continuous (with respect to y), the Lagrange multipliers are nothing other than the max-inf of the Lagrangian. We recall that $\bar{p}$ is a Lagrange multiplier and $\bar{x}$ a minimal solution if and only if

$$\{-L^*\bar{p}, \bar{p}\} \in \partial F(\bar{x}, L\bar{x})$$

and we prove that an equivalent assertion is that

$$L\bar{x} \in \partial_p H(\bar{x}, \bar{p}) \quad \text{and} \quad L^*\bar{p} \in \bar{\partial}_x H(\bar{x}, \bar{p}).$$

Using the minisup Theorem 7.1.5, we prove that the "Slater condition"

$$\exists x_0 \in R \quad \text{such that } Lx_0 \in \text{Int}(Y)$$

implies the existence of a Lagrange multiplier.

This is quite a stringent requirement, which is not met for instance when Y is the cone of non-negative functions of a Lebesgue space $L^p(\Omega)$ whose interior is empty. But when U and V are reflexive Banach spaces we relax this assumption by replacing it with the constraint qualification condition

$$0 \in \text{Int}(Y - L(R))$$

we used when $V = \mathbf{R}^m$. This is done by applying the minisup Theorem 13.1.2.

In the above discussion no use is made of the actual form of the Lagrangian and so we are led to call a "Lagrangian" of a minimization problem

$$v = \inf_{x \in X} f(x) \quad \text{where } f: U \rightarrow]-\infty, +\infty]$$

any function l defined on $U \times V^*$ satisfying

$$v = \inf_{x \in U} \sup_{p \in V^*} l(x, p).$$

To prove the existence of a Lagrange multiplier (i.e. of a max-inf of a Lagrangian), we introduce the functions Δ defined on $U \times V$ by

$$\Delta(x, y) = \sup_{p \in P} [\langle p, y \rangle + l(x, p)].$$

We prove that either the assumption

$$\exists x_0 \in U \text{ such that } 0 \in \text{Int Dom } \Delta(x_0, \cdot).$$

or, in the case of reflexive Banach spaces, the relaxed assumption

$$0 \in \text{Int } \bigcup_{x \in U} \text{Dom } \Delta(x, \cdot)$$

implies the existence of such a Lagrange multiplier.

What is more, under convexity assumptions, we shall see that defining a duality theory by means of a Lagrangian is equivalent to the use of a family of perturbed minimization problems

$$v(y) = \inf_{x \in X} \Delta(x, y)$$

(since $v = v(0)$).

In this case, we prove that the set of Lagrange multipliers is the subdifferential $\partial v(0)$ of v at 0.

14.1.1. Lagrangian of a minimization problem under linear constraints

Consider a minimization problem of the type

$$(1) \quad v = \inf_{x \in X} F(x, Lx)$$

where

$$(2) \quad \begin{cases} \text{(i) } R \text{ is a convex subset of a vector space } U, \\ \text{(ii) } Y \text{ is a closed convex subset of a topological vector space } V, \\ \text{(iii) } L \in \mathcal{L}(U, V), \\ \text{(iv) } X = \{x \in R \text{ such that } Lx \in Y\}, \\ \text{(v) } F : R \times Y \rightarrow \mathbf{R}. \end{cases}$$

We shall not assume any more that V is a finite-dimensional space. Nevertheless, we shall prove the existence of a Lagrange multiplier $\bar{p}$, i.e. of $\bar{p} \in V^*$ satisfying

$$(3) \quad \begin{cases} \text{(i) } v^* = F^*(-L^*\bar{p}, \bar{p}) = \inf_{p \in V^*} F^*(-L^*p, p), \\ \text{(ii) } v + v^* = 0, \end{cases}$$

where F^* is the conjugate function of F .

For this purpose, we associate with the function F its "Hamiltonian" and with the minimization problem (1) its "Lagrangian".

Definition 1. Let F be a function from $U \times V$ into $]-\infty, +\infty]$. Its “*Hamiltonian*” H is the function defined on $U \times V^*$ by

$$(4) \quad \forall x \in U, \forall p \in V^*, \quad H(x, p) = \sup_{y \in Y} [\langle p, y \rangle - F(x, y)].$$

The “*Lagrangian*” of the minimization problem

$$v = \inf_{x \in X} F(x, Lx)$$

is the function l defined on $U \times V^*$ by

$$(5) \quad l(x, p) = \langle p, Lx \rangle - H(x, p).$$

In other words, for all x , $H(x, \cdot)$ is the conjugate function of the function $y \mapsto F(x, y)$. We point out the following obvious properties.

Proposition 1. The conjugate function F^* of F can be written

$$(6) \quad F^*(q, p) = \sup_{x \in R} [\langle q, x \rangle + H(x, p)].$$

If we assume that

$$(7) \quad \forall x \in X, \quad y \mapsto F(x, y) \text{ is convex and lower semi-continuous}$$

then

$$(8) \quad F(x, y) = \sup_{p \in P} [\langle p, y \rangle - H(x, p)].$$

Proof. This is left as an exercise. $\square$

Example. If $F(x, y) = f(x) + g(y)$, where $f: R \rightarrow \mathbf{R}$ and $g: Y \rightarrow \mathbf{R}$, we deduce that

$$(9) \quad H(x, p) = g^*(p) - f(x)$$

and that

$$(10) \quad l(x, p) = f(x) + \langle p, Lx \rangle - g^*(p).$$

These definitions are consistent with the definition of the Lagrangian of a minimization problem $v = \inf_{Lx \in Y} f(x)$ by taking $g = \psi_Y$ to be the indicator of Y .

The following proposition motivates the introduction of the Lagrangian and relates it to the concept of a dual problem.

Proposition 2. *We have*

$$(11) \quad \sup_{p \in V^*} \inf_{x \in R} l(x, p) = -v^*.$$

If we assume furthermore that (7) holds, then

$$(12) \quad \inf_{x \in R} \sup_{p \in V^*} l(x, p) = v.$$

In this case the Lagrange multipliers are the max-inf of the Lagrangian.

Proof. We have that

$$\begin{aligned} \inf_{x \in R} l(x, p) &= \inf_{x \in R} [\langle p, Lx \rangle - H(x, p)] \\ &= -\sup_{x \in R} [\langle -L^*p, x \rangle + H(x, p)] = -F^*(-L^*p, p) \end{aligned}$$

using (6). Hence

$$\sup_{p \in V} \inf_{x \in R} l(x, p) = -\inf_{p \in V^*} F^*(-L^*p, p) = -v^*.$$

In the same way, we obtain that

$$\sup_p l(x, p) = \sup_p [\langle p, Lx \rangle - H(x, p)] = F(x, Lx)$$

by (8) (since assumption (7) holds). Hence (12) holds.

But, to say that $\bar{p}$ is a Lagrange multiplier amounts to saying that p is a max-inf of l , i.e.

$$(13) \quad \inf_{x \in R} l(x, \bar{p}) = \inf_{x \in R} \sup_{p \in P} l(x, p) = v = -v^*. \quad \square$$

We mention the following property of the Lagrangian.

Proposition 3. *The function $p \rightarrow H(x, p)$ is convex and lower semi-continuous. If we assume that*

$$(14) \quad F \text{ is a convex function on } R \times Y,$$

then $x \mapsto H(x, p)$ is concave.

Proof. This is left as an exercise. $\square$

We also need the conjugate function of the Lagrangian with respect to p .

Proposition 4. *Suppose that (7) holds. Then the conjugate function $\Delta(x, \cdot)$ of $p \mapsto -l(x, p)$ is defined by*

$$(15) \quad \Delta(x, y) = F(x, Lx + y).$$

Its domain is equal to

$$(16) \quad \text{Dom } \Delta(x, \cdot) = Y - Lx \quad \text{when } x \in R.$$

Proof. We check that

$$\begin{aligned} \Delta(x, y) &= \sup_{p \in V^*} [\langle p, y \rangle + l(x, p)] \\ &= \sup_{p \in V^*} [\langle p, y + Lx \rangle - H(x, p)] \\ &= F(x, Lx + y) \end{aligned}$$

by (8). Since the domain of F is $R \times Y$, it is clear that the domain of $\Delta(x, \cdot)$ is the set of $y \in V$ such that $Lx + y \in Y$.

14.1.2. Extremality relations

We have proved (Proposition 5.2.1) that $\bar{p}$ is a Lagrange multiplier and $\bar{x}$ a minimal solution if and only if

$$(17) \quad \{-L^*\bar{p}, \bar{p}\} \in \partial F(\bar{x}, L\bar{x}).$$

Proposition 5. Suppose that (7) holds. Then the extremality relations (17) are equivalent to

$$(18) \quad L\bar{x} \in \partial_p H(\bar{x}, \bar{p}) \quad \text{and} \quad L^*\bar{p} \in \bar{\partial}_x H(\bar{x}, \bar{p}).$$

Proof. In the first place, (17) amount to

$$(19) \quad F(\bar{x}, L\bar{x}) - F(x, y) \leq \langle -L^*\bar{p}, \bar{x} - x \rangle + \langle \bar{p}, L\bar{x} - y \rangle.$$

By taking $x = \bar{x}$, this implies that

$$\begin{aligned} \sup_y [\langle \bar{p}, y \rangle - F(\bar{x}, y)] &= \langle \bar{p}, L\bar{x} \rangle - F(\bar{x}, L\bar{x}) \\ &= H(\bar{x}, \bar{p}). \end{aligned}$$

Hence, the very definition of H implies that

$$(20) \quad H(\bar{x}, \bar{p}) - H(\bar{x}, p) \leq \langle \bar{p} - p, L\bar{x} \rangle.$$

On the other hand, we can write (19) in the form

$$\begin{aligned} \langle \bar{p}, y \rangle - F(x, y) &\leq \langle L^*\bar{p}, x \rangle - F(\bar{x}, L\bar{x}) \\ &= \langle L^*\bar{p}, x - \bar{x} \rangle + H(\bar{x}, \bar{p}). \end{aligned}$$

Taking the supremum over Y , we obtain that

$$H(x, \bar{p}) - H(\bar{x}, \bar{p}) \leq \langle -L^*\bar{p}, \bar{x} - x \rangle.$$

This means that $L^*\bar{p} \in \bar{\partial}_x H(\bar{x}, \bar{p})$.

Conversely, let $L\bar{x} \in \partial_p H(\bar{x}, \bar{p})$ and $L^*\bar{p} \in \bar{\partial}_x H(\bar{x}, \bar{p})$. Since $F(x, y) = \sup_p [\langle p, y \rangle - H(x, p)]$ by (8) and since

$$\langle \bar{p}, L\bar{x} \rangle = F(\bar{x}, L\bar{x}) + H(\bar{x}, \bar{p})$$

we deduce that

$$F(\bar{x}, L\bar{x}) - F(x, y) + \langle \bar{p}, y - L\bar{x} \rangle \leq H(x, \bar{p}) - H(\bar{x}, \bar{p}).$$

Since $L^*\bar{p} \in \partial_x H(\bar{x}, \bar{p})$, we obtain

$$F(\bar{x}, L\bar{x}) - F(x, y) - \langle \bar{p}, L\bar{x} - y \rangle \leq \langle -L^*\bar{p}, \bar{x} - x \rangle.$$

Hence $\{-L^*\bar{p}, \bar{p}\} \in \partial F(\bar{x}, L\bar{x})$. $\square$

14.1.3. Existence of a Lagrange multiplier under the Slater condition

In this section, we shall replace the assumption that V is a finite-dimensional space by the assumption that

- (21) V is an infinite dimensional space supplied with the Mackey topology $\tau(V, V^*)$.

Recall that the Mackey topology $\tau(V, V^*)$ coincides with the initial topology when V is a barrelled space (see Corollary 3 of Appendix A).

Theorem 1. *Suppose that*

- (22) $\begin{cases} \text{(i) } F \text{ is a convex function from } R \times Y \text{ into } R, \\ \text{(ii) } \forall x \in R, y \rightarrow F(x, y) \text{ is lower semi-continuous} \end{cases}$

and that the following "Slater condition" holds:

- (23) $\exists x_0 \in R$ such that $Lx_0 \in \text{Int } Y$ (for $\tau(V, V^*)$).

Then there exists a Lagrange multiplier $\bar{p} \in V^*$

Proof. Write $P = \{p \in V^* \text{ such that, } \forall x \in R, l(x, p) > -\infty\}$. We have to prove that there exists a minisup $\bar{p}$ of the function $\phi(p, x) = -l(p, x)$ defined on $P \times R$ (by Proposition 2). For this purpose, we apply the minisup theorem (Theorem 7.1.5). It is clear that, $\forall x \in X, p \mapsto \phi(p, x)$ is convex and lower

semi-continuous and that, $\forall p \in P, x \mapsto \phi(p, x)$ is concave. It *remains* to prove that there exists x_0 such that $p \mapsto \phi(p, x_0)$ is lower semi-compact.

By assumption (23) and Proposition 4, we see that $0 \in \text{Int Dom } \Delta(x_0, \cdot)$. Since $\Delta(x_0, \cdot)$ is the conjugate function of $\phi(\cdot, x_0)$, we deduce from Proposition 3.1.10 that the function $\phi(\cdot, x_0)$ is lower semi-compact.

Example. In the case where $F(x, y) = f(x) + g(y)$, the constraint qualification assumption (23) is implied by

$$(24) \quad \exists x_0 \in R \quad \text{such that } g \text{ is continuous at } Lx_0$$

(and actually is equivalent to (23) whenever g is convex and lower semi-continuous).

Remark. Assumption (23) is very restrictive. For instance, let us assume that Y is the cone of non-negative functions of a space of functions on Ω . If we take either

$$(25) \quad V = \mathcal{U}(\Omega) \text{ is the space of bounded functions on } \Omega,$$

$$(26) \quad V = \mathcal{C}(\Omega) \text{ is the space of continuous functions on a compact set } \Omega,$$

or

$$(27) \quad V = L^\infty(\Omega) \text{ is the space of (classes) of bounded measurable functions on } \Omega,$$

is the space of (classes) of bounded measurable functions on Ω , it is easy to prove that the constant function $1 : \omega \mapsto 1(\omega) = 1$ belongs to the interior of the cone of non-negative functions $Y = \mathcal{U}_+(\Omega)$ (resp. $Y = \mathcal{C}_+(\Omega)$, $Y = L^\infty_+(\Omega)$).

This property is false if we take

$$(28) \quad V = L^p(\Omega), \quad Y = L^q_+(\Omega) \quad \text{for } p > 1.$$

We prove this fact in the case when $\Omega =]-1, +1]$ supplied with the Lebesgue measure. To say that $L^p_+(\Omega)$ has a *non-empty interior* amounts to saying that its positive polar cone $L^q_+(\Omega)$ ($1/p + 1/q = 1$) is spanned by a *non-compact* convex subset (for the weak topology) [see Proposition 1.5.5]. It is clear that $L^q_+(\Omega)$ is spanned by the set

$$S = \left\{ x \in L^q_+(\Omega) \text{ such that } \int_{-1}^{+1} x(\omega) d\omega = 1 \right\}$$

To prove that S is not compact, we associate with a function $x \in S$ (with compact support) the functions $\omega \rightarrow nx(n\omega)$ which belong to S (since $\int_{-1}^{+1} nx(n\omega) d\omega = 1$). They *do not* converge in $L^q(\Omega)$ (since $nx(n, \cdot)$ converges to the Dirac measure at 0!).

Thus, when Y is a cone of non-negative functions, the use of Theorem 1 requires the choice of one of the spaces $\mathcal{U}(\Omega)$, $\mathcal{Q}(\Omega)$ or $L^\infty(\Omega)$. This is quite a drawback, since these spaces are *not reflexive*. Furthermore, the characterization of their duals is not always possible. [We know that $\mathcal{Q}^*(\Omega)$ is the space of Radon measures on the compact set Ω . The space $L^\infty(\Omega)^*$ is the bidual of $L^1(\Omega)$. The characterization of (at least) one supplement of $L^1(\Omega)$ is given by Ioffe–Levin–Valadier: *it is the vector subspace of continuous linear functionals $p \in L^\infty(\Omega)^*$ such that there exists a decreasing sequence of Borel subsets A_n satisfying $\bigcap_{n=1}^\infty A_n = \emptyset$ and, $\forall x \in L^\infty(\Omega)$, $\langle p, \chi_{\Omega-A_n} - x \rangle = 0$ (where χ_A is the characteristic function of A). In other words, this means that p is supported by every A_n .*]

14.1.4. Relaxation of the Slater condition

Furthermore, the constraint qualification assumption (23) is much stronger than the “constraint qualification assumption”

$$(29) \quad 0 \in \text{Int } (Y - L(R))$$

we used in Theorem 5.3.1 when V is a finite-dimensional space. The question arises as to whether it is possible to replace (23) by (29). We show that the answer is positive by using the minisup Theorem 13.1.2. We assume now that

$$(30) \quad V = G^* \text{ is the dual of a barreled space } G.$$

Theorem 2. *Suppose that (30) holds and that*

$$(31) \quad \begin{cases} \text{(i) } F \text{ is a convex function from } R \times Y \text{ into } \mathbf{R}, \\ \text{(ii) } \forall x \in R, y \rightarrow F(x, y) \text{ is weakly lower semi-continuous,} \\ \text{(iii) } \forall p \in P, x \rightarrow H(x, p) \text{ is upper semi-continuous on } U, \end{cases}$$

and that

$$(32) \quad \begin{cases} \text{(i) } R = \bigcup_{n=1}^\infty K_n \text{ is a countable union of compact subsets of } U, \\ \text{(ii) } 0 \in \text{Int } (Y - L(R)). \end{cases}$$

Then there exists a Lagrange multiplier $\bar{p} \in V^*$.

Proof. Since a Lagrange multiplier is a minisup $\bar{p}$ of the function ϕ defined by $\phi(p, x) = -l(x, p)$, we apply Theorem 13.1.2. The assumptions (31) imply that ϕ is lower semi-continuous and convex with respect to p , upper semi-continuous and concave with respect to x . It remains to check that the compactness assumption of Theorem 13.1.2 is satisfied. Since $\Delta(x, \cdot)$ is the conjugate function of $\phi(\cdot, x)$ this compactness assumption is

$$(33) \quad 0 \in \text{Int} \bigcup_{x \in R} \text{Dom } \Delta(x, \cdot).$$

Since $\text{Dom } \Delta(x, \cdot) = Y - Lx$, property (33) is nothing other than assumption (32(ii)).

Remark. Recall $R = \bigcup_{n=1}^{\infty} K_n$ whenever

$$(34) \quad \text{Int } R^b = \emptyset \text{ for the Mackey topology } \tau(U^*, U)$$

(see Example 3 of Section 13.1.3).

This property also holds for *any closed subset* R of the dual $U = F^*$ of a Banach space.

We point out the following consequence (among other possible corollaries of Theorem 2).

Theorem 3. *Suppose that*

$$(35) \quad U \text{ and } V \text{ are reflexive Banach space,}$$

that

$$(36) \quad f: U \rightarrow]-\infty, +\infty] \text{ and } g: V \rightarrow]-\infty, +\infty] \text{ are two lower semi-continuous convex functions}$$

and that

$$(37) \quad 0 \in \text{Int} (\text{Dom } g - L \text{Dom } f).$$

Then there exists a Lagrange multiplier $\bar{p}$ of the minimization problem

$$v = \inf_{x \in X} [f(x) + g(Lx)].$$

Remark. In the case of reflexive Banach spaces (and, in particular, in the case of Hilbert spaces), we do not have to assume the restrictive constraint qualification assumption (24).

In particular, for minimization problem under constraints, Theorem 3 can be restated as follows.

Theorem 4. *Suppose that*

(38) *U and V are reflexive Banach spaces,*

and that

(39)
$$\begin{cases} \text{(i) } f \text{ is a convex lower semi-continuous function defined on } R \subset U, \\ \text{(ii) } Y \text{ is a closed convex subset of } V, \\ \text{(iii) } L \in \mathcal{L}(U, V). \end{cases}$$

Then, for any $w \in V$ satisfying

(40) $w \in \text{Int}(Y - L(R)),$

there exists a Lagrange multiplier $\bar{p}$ of the minimization problem

$$v = \inf_{\substack{x \in R \\ Lx \in Y - w}} f(x).$$

This result allows us to use Hilbert spaces $L^2(\Omega)$ as constraint spaces. For instance, consider a continuous linear operator $L \in \mathcal{L}(U, L^2(\Omega))$ and a cone R of U such that

(41) L maps R onto $L_+^2(\Omega)$.

Theorem 4 obviously implies the following result

$$[\text{since } L(R) - L_+^2(\Omega) = L_+^2(\Omega) - L_+^2(\Omega) = L^2(\Omega)].$$

Proposition 6. *Let U be a Hilbert space, R a closed convex cone of U and $L \in \mathcal{L}(U, L^2(\Omega))$ satisfying (41).*

Let X be the subset of R defined by

(42) $X = \{x \in R \text{ such that } (Lx)(\omega) \leq w(\omega) \text{ for almost all } \omega \in \Omega\}$

where $w \in L^2$ is given.

Let f be a convex lower semi-continuous convex function defined on R . Then there exists a Lagrange multiplier $\bar{p} \in L_+^2(\Omega)$ of the minimization problem

$$v = \inf_{x \in X} f(x).$$

14.1.5. Generalized Lagrangian of a minimization problem

In the above existence theorems, we used *only* the fact that we could represent the function f defined by $f(x) = F(x, Lx)$ as the pointwise supremum of the Lagrangian:

(43)
$$f(x) = \sup_p l(x, p) \quad \text{whenever } x \in X$$

(see property (8) of Proposition 2).

This representation also holds in the case of the Lagrangian of the optimization problems constructed in Section 5.1 (see Proposition 5.1.1).

Therefore, we can construct a “duality theory” whenever such a representation holds.

Definition 2. Let $f: U \rightarrow]-\infty, +\infty]$ be a function defined on U whose domain is X . Let

$$(44) \quad v = \inf_{x \in X} f(x).$$

We shall say that a function l defined on $R \times P$ where $R \subset U$ and $P \subset V^*$ is a *Lagrangian of the minimization problem (44)* if

$$(45) \quad v = \inf_{x \in R} \sup_{p \in P} l(x, p).$$

We shall say that $p \in P$ is “*Lagrange multiplier*” if $\bar{p}$ is a *max-inf of the Lagrangian*, i.e. if

$$(46) \quad v = \inf_{x \in R} l(x, \bar{p}).$$

We shall say that the function f^{I*} defined by

$$(47) \quad f^{I*}(p) = - \inf_{x \in R} l(x, p)$$

is the *loss function of the dual problem*. We will set

$$(48) \quad v^* = \inf_{p \in P} f^{I*}(p) = - \sup_{p \in P} \inf_{x \in R} l(x, p).$$

In this framework, the existence of a Lagrange multiplier can be obtained by applying the minisup Theorems 7.1.5 and 13.1.2 to the function ϕ defined on $P \times R$ by $\phi(p, x) = -l(x, p)$.

For this purpose, we introduce the conjugate functions $\Delta(x, \cdot)$ of the functions $\phi(\cdot, x) = -l(x, \cdot)$ defined on $P \times R$ by

$$(49) \quad \Delta(x, y) = \sup_{p \in P} [\langle p, y \rangle + l(x, p)].$$

We mention the following properties of Δ .

Proposition 7. *If the Lagrangian satisfies*

$$(50) \quad \forall p \in P, \quad x \mapsto l(x, p) \text{ is convex (resp. lower semi-continuous),}$$

then the function Δ satisfies

$$(51) \quad \Delta : R \times V \rightarrow]-\infty, +\infty] \text{ is convex (resp. lower semi-continuous)}$$

Proof. This is left as an exercise. $\square$

Consider the function v defined by

$$(52) \quad v(y) = \inf_{x \in R} \Delta(x, y).$$

Then properties (46) and (49) show that

$$(53) \quad v = v(0),$$

since $\Delta(x, 0) = \sup_{p \in P} l(x, p)$.

We can regard the minimization problem (52) as a "perturbed" problem by elements $y \in V$. Proposition 5.1.4 can be generalized.

Proposition 8. Suppose that there exists a Lagrange multiplier $\bar{p}$. Then $\bar{p}$ belongs to the subdifferential of v at 0, i.e.

$$(54) \quad \bar{p} \in \partial v(0).$$

Proof. For any $y \in V$ we have

$$v + \langle \bar{p}, y \rangle = \inf_{x \in R} l(x, \bar{p}) + \langle \bar{p}, y \rangle.$$

Since $v = v(0)$, we can write this inequality in the form $v(0) - v(y) \leq \langle \bar{p}, 0 - y \rangle$ which means that $\bar{p} \in \partial v(0)$. $\square$

Remark. We shall prove the converse statement in Theorem 6 below.

Theorem 5. Suppose that the Lagrangian satisfies

$$(55) \quad \begin{cases} \text{(i)} \ \forall x \in R, p \mapsto l(x, p) \text{ is concave and upper semi-continuous,} \\ \text{(ii)} \ \forall p \in P, x \mapsto l(x, p) \text{ is convex.} \end{cases}$$

If we assume that either

$$(56) \quad \exists x_0 \in R \text{ such that } 0 \in \text{Int Dom } \Delta(x_0, \cdot)$$

or that

$$(57) \quad \begin{cases} \text{(i)} \ R = \bigcup_{n=1}^{\infty} K_n \text{ is a countable union of compact subsets } K_n \text{ of } U, \\ \text{(ii)} \ \forall p \in P, x \mapsto l(x, p) \text{ is lower semi-continuous,} \\ \text{(iii)} \ 0 \in \text{Int } \bigcup_{x \in X} \text{Dom } \Delta(x, \cdot) = \text{Int Dom } v, \\ \text{(iv)} \ V = G^* \text{ is the dual of a barreled space } G, \end{cases}$$

then there exists a Lagrange multiplier $\bar{p}$.

Proof. This is a restatement of the minisup Theorems 7.1.5 and 13.1.2 applied to

$$\varphi(p, x) = -l(x, p). \quad \square$$

Example 1. Consider the Lagrangian $l(x, p) = f(x) + \langle p, L(x) \rangle - \sigma^*(Y; p)$ of the minimization problem

$$(58) \quad v = \inf_{L(x) \in Y} f(x).$$

Since $l(x, p) + \langle p, y \rangle = f(x) + \langle p, L(x) \rangle - \sigma^*(Y - y; p)$, we see that

$$(59) \quad v(y) = \inf_{Lx \in Y - y} f(x).$$

Example 2. We can also represent a closed convex subset containing 0 by its gauge $\pi(Y; \cdot)$ and associate with the minimization problem (58) the Lagrangian defined on $R \times R_+$ by

$$(60) \quad \tilde{l}(x, p) = f(x) + p(\pi(Y; L(x)) - 1)$$

since we clearly have that

$$\sup_{p \geq 0} \tilde{l}(x, p) = \begin{cases} f(x) & \text{if } L(x) \in Y, \\ +\infty & \text{if not.} \end{cases}$$

Since $l(x, p) + py = f(x) + p(\pi(Y; L(x)) - (1 - y))$ we see that

$$\Delta(x, y) = \begin{cases} f(x) & \text{if } L(x) \in (1 - y)Y, \\ +\infty & \text{if not.} \end{cases}$$

Hence the perturbed problem is defined by

$$(61) \quad \tilde{v}(y) = \inf_{L(x) \in (1-y)Y} f(x)$$

where $y \in R$.

Example 3 (Rockafellar). We associated the Lagrangian $l(x, p)$ defined on $R \times R_+^n$ by $l(x, p) = f(x) + \langle p, L(x) \rangle$ with the problem $v = \inf_{L(x) \leq 0} f(x)$ (where L maps R into R^n). The reason was that $\psi(-R_+^n; y) = \sup_{p \geq 0} \langle p, y \rangle$. The question arises as to whether it is possible to represent the indicator $\psi(-R_+^n; \cdot)$ of $-R_+^n$ as the pointwise supremum of a function $\mu(y, p)$, the domain of which is $R^n \times R^{n*}$ [instead of $R^n \times R_+^{n*}$]. In this case, we can associate with the above problem the new Lagrangian $\tilde{l}$ defined on $R \times R^{n*}$ (instead of $R \times R_+^{n*}$) by

$$(62) \quad \tilde{l}(x, p) = f(x) + \mu(Lx, p).$$

The following function μ defined by

$$(63) \quad \mu(y, p) = \sum_{i=1}^n \mu_r(y_i, p_i)$$

where

$$(64) \quad \mu_r(y_i, p_i) = \begin{cases} -\frac{p_i^2}{4r} & \text{if } p_i \leq -2ry_i, \\ p_i y_i + r y_i^2 & \text{if } p_i \geq -2ry_i \end{cases}$$

satisfies the above property. Indeed

$$(65) \quad \begin{cases} \text{(i) if } y_i \leq 0, \sup_{p_i \in \mathbb{R}} \mu_r(y_i, p_i) = 0, \\ \text{(ii) if } y_i < 0, \sup_{p_i \in \mathbb{R}} \mu_r(y_i, p_i) = +\infty \end{cases}$$

as can be seen by examining the graphs of the function $p \rightarrow \mu_r(y, p)$ when $y \leq 0$ and $y > 0$.

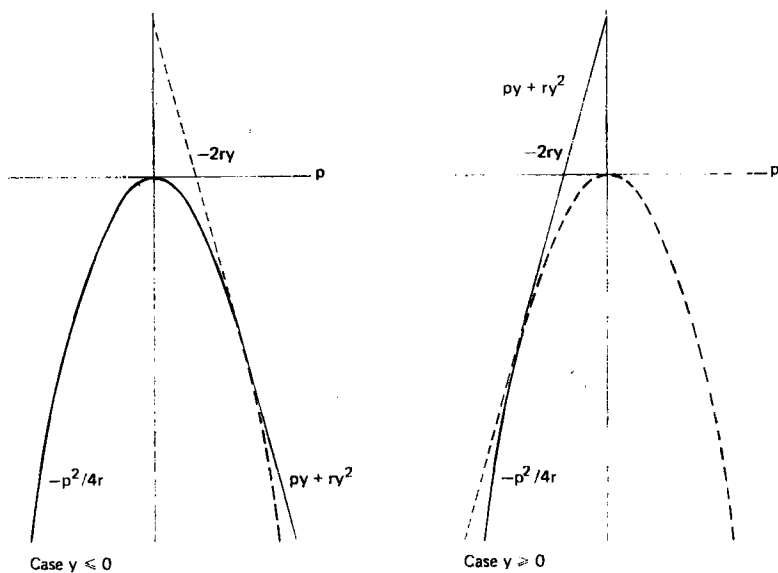


Fig. 15.

Therefore

$$(66) \quad \psi(-\mathbf{R}_+^n; y) = \sup_{p \in \mathbb{R}^n} \mu_r(y, p).$$

Remark. Any kind of function μ whose graph has the same shape as the graph of μ_r satisfies the above property.

14.1.6. Characterization of a Lagrangian by perturbations of the minimization problem

Consider now a family of loss functions $x \mapsto \Delta(x, y)$ where y ranges over a topological vector space V , which are perturbations of the loss function f defined by

$$(67) \quad f(x) = \Delta(x, 0).$$

Consider the minimal values

$$(68) \quad v(y) = \inf_{x \in U} \Delta(x, y) \quad \text{and} \quad v = v(0) = \inf_{x \in U} f(x).$$

We shall show that under mild conditions, we can associate with Δ a Lagrangian $l_\Delta(x, p)$ such that its set of Lagrange multipliers is nothing other than the subdifferential of v at 0.

We define l_Δ on $U \times V^*$ by

$$(69) \quad l_\Delta(x, p) = \inf_{y \in V} [\Delta(x, y) - \langle p, y \rangle].$$

Proposition 9. *If we assume that*

$$(70) \quad \forall x \in U, \quad y \mapsto \Delta(x, y) \text{ is convex and lower semi-continuous,}$$

then l_Δ is a Lagrangian, i.e.

$$(71) \quad f(x) = \sup_{p \in V^*} l_\Delta(x, p).$$

If we assume also that

$$(72) \quad \Delta : U \times V \rightarrow]-\infty, +\infty] \text{ is a convex function,}$$

then l_Δ satisfies

$$(73) \quad \begin{cases} \text{(i) } \forall x \in U, p \mapsto l_\Delta(x, p) \text{ is concave and upper semi-continuous,} \\ \text{(ii) } \forall p \in V^*, x \mapsto l_\Delta(x, p) \text{ is convex.} \end{cases}$$

Proof. Fix $x \in U$ and set $g(y) = \Delta(x, y)$. Therefore $l_\Delta(x, p) = -\sup_p [\langle p, y \rangle - g(y)] = -g^*(p)$ is concave and upper semi-continuous with respect to p . If g is convex and lower semi-continuous, we obtain that

$$\begin{aligned} \Delta(x, y) = g(y) &= g^{**}(y) = \sup_{p \in V^*} [\langle p, y \rangle - g^*(p)] \\ &= \sup_{p \in V^*} [\langle p, y \rangle + l_\Delta(x, p)]. \end{aligned}$$

Finally, it is clear that the convexity of Δ with respect to both x and y implies the convexity of the Lagrangian l with respect to x . $\square$

The formula

$$(74) \quad \Delta(x, y) = \sup_{p \in V^*} [\langle p, y \rangle + l_A(x, p)]$$

shows that the perturbed problems associated with the Lagrangian are nothing other than the initial perturbed problems. In the same way, the Lagrangian associated with perturbed problems is nothing other than the initial Lagrangian.

Proposition 10. *Suppose that a Lagrangian l satisfies*

$$(75) \quad \forall x \in R, \quad p \rightarrow l(x, p) \text{ is concave and upper semi-continuous.}$$

Then $l = l_A$ and the conjugate function of $y \rightarrow v(y)$ satisfies

$$(76) \quad v^*(p) = -\inf_x l(x, p).$$

Proof. Consider the perturbed problem

$$(77) \quad \Delta(x, y) = \sup_p [\langle p, y \rangle + l(x, p)].$$

Write $\theta(p) = -l(x, p)$. Hence θ is a convex lower semi-continuous function. Therefore $\Delta(x, y) = \theta^*(y)$ and

$$\begin{aligned} l(x, p) &= -\theta^{**}(p) = -\sup_y [\langle p, y \rangle - \Delta(x, y)] \\ &= \inf_y [\Delta(x, y) - \langle p, y \rangle] = l_A(x, p). \end{aligned}$$

Now we deduce that

$$\begin{aligned} v^*(p) &= -\inf_y [v(y) - \langle p, y \rangle] = -\inf_x \inf_y [\Delta(x, y) - \langle p, y \rangle] = \\ &= -\inf_x l(x, p). \quad \square \end{aligned}$$

Theorem 6. *Suppose that the Lagrangian satisfies (75). Then $\bar{p}$ is a Lagrange multiplier if and only if*

$$(78) \quad \begin{cases} \text{(i) } v(0) = v^{**}(0), \\ \text{(ii) } \bar{p} \in \partial v(0). \end{cases}$$

Proof. We have already proved that any Lagrange multiplier $\bar{p}$ belongs to $\partial v(0)$ (Proposition 8). Furthermore, $\sup_p \inf_x l(x, p) = \inf_x \sup_p l(x, p)$. But

$$(79) \quad v(0) = \inf_x \sup_p l(x, p)$$

by the very definition of the Lagrangian and

$$(80) \quad v^{**}(0) = \sup_p [\langle p, 0 \rangle - v^*(p)] = \sup_p \inf_x l(x, p)$$

by formula (76). Hence the *Lagrangian has a value if and only if* $v(0) = v^{**}(0)$. In this case, if $\bar{p} \in \partial v(0)$, $\bar{p}$ minimizes $v^*(p) = -\inf_x l(x, p)$, i.e. $\bar{p}$ is a Lagrange multiplier [since $\inf_x l(x, \bar{p}) = \sup_p \inf_x l(x, p) = \inf_x \sup_p l(x, p) = \inf_x f(x)$]. $\square$

Remark. We can also associate with a Lagrangian $l(x, p)$ a family of perturbed dual problems

$$(81) \quad \Gamma(p, q) = \inf_x [l(p, x) + \langle q, x \rangle]$$

where

$$(82) \quad \Gamma(p, 0) = f^{I*}(p)$$

is the loss function of the dual problem. A symmetric study is then possible.

14.2. Duality in the case of non-convex integral criterion and constraints

We shall prove in this section the Aumann-Perles theorem, i.e. there exists a Lagrange multiplier $\bar{p} \in \mathbb{R}^{m*}$ of the following minimization problem with integral criterion and constraints:

$$(*) \quad v = \inf_{x \in L^1(\Omega, U)} \left[\int_{\Omega} \varphi(\omega, x(\omega)) \, d\omega + g \left(\int_{\Omega} A(\omega) x(\omega) \, d\omega \right) \right]$$

whenever we assume (essentially) that the constraint qualification assumption $0 \in \text{Int} \left(\text{Dom } g - \left(\int_{\Omega} A(\omega) \, d\omega \right) U \right)$ is satisfied, that $A(\omega) \in \mathcal{L}(U, \mathbb{R}^m)$ for almost all $\omega \in \Omega$ and that g is a lower semi-continuous convex function. But, *we do not assume that the functions $\xi \rightarrow \varphi(\omega, \xi)$ are convex.*

We shall prove this theorem in a constructive way, by approximating it by minimization problems of the form

$$v = \inf_{x_i \in R_i} \frac{1}{T} \left[\sum_{i=1}^T f_i(x_i) + g \left(\frac{1}{T} \sum_{i=1}^T L_i x_i \right) \right],$$

where g is a convex function defined on $\mathbb{R}^n$, R_i is a convex subset of U , and where $L_i \in \mathcal{L}(U, \mathbb{R}^m)$. Suppose that the constraint qualification assumption holds, i.e.

$$0 \in \text{Int} \left(\text{Dom } g - \frac{1}{T} \sum_{i=1}^T L_i(R_i) \right).$$

If we do not assume that the functions f_i are convex, the length $v+v^*$ of the duality gap is positive, where we set

$$(**) \quad v^* = \inf_{p \in \mathbf{R}^{m*}} \left[\frac{1}{T} \sum_{i=1}^T f_i^*(-L_i^* p) + g^*(p) \right]$$

We will estimate the duality gap by measuring the lack of convexity of a function f in terms of the *modulus of non-convexity*

$$\varrho(f) = \sup_{m \in \mathcal{M}(X)} [f(\beta m) - f^A(m)].$$

(We see that convex functions are those whose modulus of non-convexity $\varrho(f)$ is zero.) Roughly speaking, the less convex the behaviour of a function, the larger is its modulus $\varrho(f)$.

We will then prove the fundamental estimate

$$0 \leq v + v^* \leq \frac{m+1}{T} \max_{0 \leq i \leq T} \varrho(f_i).$$

This estimate shows that the duality gap “shrinks” as the number of functions increases. This explains why the Aumann–Perles holds.

The proof uses a fundamental result due to Shapley and Folkman, which states that, in some sense, the sum of a large number of sets is approximatively convex.

14.2.1. Modulus of non-convexity of a function

Consider

$$(1) \quad \left\{ \begin{array}{l} \text{(i) a convex subset } X \text{ of a vector space } U, \\ \text{(ii) the subset } \mathcal{M}(X) \text{ of discrete probabilities} \\ \quad m = \sum_{i=1}^n \alpha^i \delta(x_i) \text{ on } X, \\ \text{(iii) the barycentric operator } \beta : \mathcal{M}(X) \rightarrow X \text{ associating with any} \\ \quad m = \sum_{i=1}^n \alpha^i \delta(x_i) \text{ the barycenter } \beta m = \sum_{i=1}^n \alpha^i x_i. \end{array} \right.$$

We shall introduce a modulus of non-convexity $\varrho(f)$ of a function f defined on X which “measures” the non-convexity of f .

Definition 1. Let $f: X \rightarrow \mathbf{R}$ be a function defined on X . We shall say that

$$(2) \quad \varrho(f) = \sup_{m \in \mathcal{M}(X)} [f(\beta m) - \langle m, f \rangle]$$

is the “*modulus of non-convexity*” of f .

It is clear that a function f is convex if and only if its modulus of non-convexity $\varrho(f)$ is equal to zero.

Remark. We can relax the assumption (1) and associate with any

(3) map γ from $\mathcal{M}(X)$ into X

the modulus of non- γ -convexity ϱ_γ defined by

$$(4) \quad \varrho_\gamma(f) = \sup_{m \in \mathcal{M}(X)} (f(\gamma m) - f^d(m)), \text{ where } f^d(m) = \langle m, f \rangle.$$

We notice that a γ -convex function is a function such that $\varrho_\gamma(f) = 0$.

For a given set X , we can use γ as parameter. For instance, we can associate a map γ with a family of functions.

Proposition 1. Consider a family $\mathcal{F}$ of functions f satisfying

(5) f is lower semi-continuous and lower semi-compact.

Then there exists an optimal map $\gamma_{\mathcal{F}}: \mathcal{M}(X) \rightarrow X$ associated with $\mathcal{F}$. It is defined by

(6) $\gamma_{\mathcal{F}}(m) \in X$ minimizes on X the function, $x \mapsto \sup_{f \in \mathcal{F}} (f(x) - f^d(m))$.

Proof. The function $x \mapsto \sup_{f \in \mathcal{F}} (f(x) - f^d(m))$ is lower semi-continuous and lower semi-compact. Hence, there exists a minimum $\gamma_{\mathcal{F}}(m)$, i.e.

$$\sup_{f \in \mathcal{F}} (f(\gamma_{\mathcal{F}}(m)) - f^d(m)) = \inf_{x \in X} \sup_{f \in \mathcal{F}} (f(x) - f^d(m)).$$

Therefore,

$$\begin{aligned} \sup_{f \in \mathcal{F}} \varrho_{\gamma_{\mathcal{F}}}(f) &= \sup_{f \in \mathcal{F}} \sup_{m \in \mathcal{M}(X)} (f(\gamma_{\mathcal{F}}(m)) - f^d(m)) \\ &= \sup_{m \in \mathcal{M}(X)} \inf_{x \in X} \sup_{f \in \mathcal{F}} ((f(x) - f^d(m))). \quad \square \end{aligned}$$

For the sake of simplicity, we shall assume that (1) holds. The extensions to the case of γ -convex functions are left as exercises.

Proposition 2. Let $F(x, y) = f(x) + g(y)$ be the sum of two functions $f: X \rightarrow \mathbf{R}$ and $g: Y \rightarrow \mathbf{R}$. Then

$$(7) \quad \varrho(F) \leq \varrho(f) + \varrho(g).$$

Proof. For any $\varepsilon > 0$, there exists $m = \sum_{i=1}^n \alpha^i \delta(x_i, y_i)$ such that

$$\begin{aligned} \varrho(F) &\leq F(x, y) - \sum_{i=1}^n \alpha^i F(x_i, y_i) + \varepsilon \\ &= f(x) - \sum_{i=1}^n \alpha^i f(x_i) + g(y) - \sum_{i=1}^n \alpha^i g(y_i) + \varepsilon \\ &\leq \varrho(f) + \varrho(g) + \varepsilon \end{aligned}$$

where we set $x = \sum_{i=1}^n \alpha^i x_i$ and $y = \sum_{i=1}^n \alpha^i y_i$. $\square$

14.2.2. Estimate of the duality gap

Consider a minimization problem

$$(8) \quad v = \inf_{x \in X} F(x, Lx)$$

where

$$(9) \quad \begin{cases} \text{(i) } R \text{ is a convex subset of a vector space } U, \\ \text{(ii) } Y \text{ is a closed convex subset of } V = \mathbf{R}^m, \\ \text{(iii) } L \text{ is a linear operator from } U \text{ into } V, \\ \text{(iv) } X = \{x \in R \text{ such that } Lx \in Y\}, \\ \text{(v) } F : R \times Y \rightarrow \mathbf{R}. \end{cases}$$

The dual minimization problem is

$$(10) \quad v^* = \inf_{p \in \mathbf{R}^m} F^*(-L^*p, p).$$

Recall that $-v^* \leq v$. We want to estimate the length $v + v^*$ of the duality gap.

Theorem 1. Suppose that (9) holds and that

$$(11) \quad 0 \in \text{Int}(L(R) - Y).$$

Then there exists $\bar{p} \in \mathbf{R}^m$ such that

$$(12) \quad 0 \leq v + F^*(-L^*\bar{p}, \bar{p}) \leq \varrho(F).$$

Remark. If we also assume that F is convex, then $\varrho(F) = 0$. We obtain the existence Theorem 5.3.1 for a Lagrange multiplier.

Proof. This is analogous to the proof of Theorem 5.3.1. Consider the operator $\phi : R \times Y \rightarrow \mathbf{R} \times \mathbf{R}^m$ defined by

$$\phi(x, y) = (F(x, y), L(x) - y).$$

We set

$$(13) \quad w = \inf_{m \in \mathcal{M}_L(R \times Y)} \sum \alpha^i F(x_i, y_i) \text{ where } \mathcal{M}_L(R \times Y) \text{ is the set of discrete probabilities } m = \sum_{i=1}^k \alpha^i \delta(x_i, y_i) \text{ such that } \sum_{i=1}^k \alpha^i (L(x_i) - y_i) = 0.$$

Consider

$$(14) \quad \begin{cases} \text{(i) the vector } \theta = (1, 0) \in \mathbf{R} \times \mathbf{R}^m \\ \text{(ii) the cone } Q =]0, \infty[\times \{0\} \subset \mathbf{R} \times \mathbf{R}^m. \end{cases}$$

We shall divide the proof into the following three steps

Step a : $w\theta \notin \text{co}(\Phi(R \times Y) + Q)$.

Step b: there exists $\bar{p}$ such that

$$w \leq -F^*(-L^*\bar{p}, \bar{p}).$$

Step c: $v \leq w + \varrho(F)$.

**Step a:* If $w\theta$ belongs to $\text{co}(\Phi(R \times Y) + Q)$, there exists $m = \sum_{i=1}^n \alpha^i \delta(x_i, y_i) \in \mathcal{M}(R \times Y)$ satisfying

$$w > \sum_{i=1}^n \alpha^i F(x_i, y_i)$$

and

$$0 = \sum_{i=1}^n \alpha^i (y_i - L(x_i)) = \sum_{i=1}^n \alpha^i y_i - L\left(\sum_{i=1}^n \alpha^i x_i\right).$$

The latter equality means that $m \in \mathcal{M}_L(R \times Y)$. Therefore, we obtain a contradiction of the definition of w .

Step b: Since $\mathbf{R}^m$ is a finite-dimensional space, we can apply the separation theorem to separate $w\theta$ from the convex subset $\text{co}(\Phi(R \times Y) + Q)$. There exists a non-zero continuous linear form $\{\alpha, p\} \in (\mathbf{R} \times \mathbf{R}^m)^*$ such that

$$(15) \quad \langle \{\alpha, p\}, w\theta \rangle = \alpha w \leq \inf_{\substack{x \in R \\ y \in Y \\ c > 0}} [\alpha F(x, y) + \langle p, Lx - y \rangle + \alpha c].$$

We first deduce that $\inf_{c > 0} \alpha c$ is finite, and thus, that $\alpha \geq 0$ and $\inf_{c > 0} \alpha c = 0$.

Secondly, we deduce from the constraint qualification assumption (11) that $\alpha > 0$ (see the proof of Theorem 5.3.1).

Finally, by setting $\bar{p} = p/\alpha$, we deduce from inequality (15) that

$$(16) \quad w \leq \inf_{\substack{x \in R \\ y \in Y}} [F(x, y) + \langle p, Lx \rangle - \langle p, y \rangle] = -F^*(-L^*\bar{p}, \bar{p}).$$

Step c: By the very definition of w , we can associate with any $\varepsilon > 0$ a discrete probability $m = \sum_{i=1}^n \alpha^i \delta(x_i, y_i)$ satisfying

$$(17) \quad \begin{cases} \text{(i)} \sum_{i=1}^n \alpha^i F(x_i, y_i) \leq w + \varepsilon, \\ \text{(ii)} \sum_{i=1}^n \alpha^i (L(x_i) - y_i) = 0. \end{cases}$$

If we set

$$(18) \quad x = \sum_{i=1}^n \alpha^i x_i, \quad y = \sum_{i=1}^n \alpha^i y_i$$

we may deduce from (17(ii)) that $y = Lx$ and from (17(i)) that

$$(19) \quad F(x, y) + \sum_{i=1}^n \alpha^i F(x_i, y_i) - F(x, y) \leq w + \varepsilon.$$

This implies that

$$(20) \quad F(x, Lx) - \varrho(F) \leq F(x, y) + \sum_{i=1}^n \alpha^i F(x_i, y_i) - F(x, y) \leq w + \varepsilon.$$

Hence, we have proved that

$$(21) \quad v = \inf_{x \in X} F(x, Lx) \leq w + \varrho(F).$$

14.2.3. The Shapley-Folkman theorem

We now proceed to state the fundamental result of Shapley and Folkman, which shows that, in some sense, the vector sum of a large number of sets is approximately convex.

Theorem 2 (Shapley-Folkman). *Consider T subsets K_t ($1 \leq t \leq T$) of $\mathbb{R}^m$. If*

$$(22) \quad x \in \text{co} \left(\sum_{1 \leq t \leq T} K_t \right)$$

there exists a subset S of at most m indices t such that

$$(23) \quad x \in \sum_{t \in S} K_t + \sum_{t \notin S} \text{co } K_t.$$

Proof. First step: We begin by replacing the subsets K_t by finite subsets $K_t(x) \subset K_t$. Since $x \in \text{co}(\sum_{t=1}^T K_t)$, there exist n real numbers $\lambda^j \in]0, 1[$ such that $\sum_{1 \leq j \leq n} \lambda^j = 1$ and points x_t^j ($1 \leq j \leq n$, $1 \leq t \leq T$) such that

$$x = \sum_{j=1}^n \lambda^j \left(\sum_{1 \leq t \leq T} x_t^j \right) = \sum_{1 \leq t \leq T} \left(\sum_{j=1}^n \lambda^j x_t^j \right).$$

We put

$$K_i(x) = \{x_i^1, \dots, x_i^j, \dots, x_i^n\},$$

and write

$$x_i = \sum_{1 \leq j \leq n} \lambda_j x_i^j \in \text{co}(K_i(x)).$$

Second step: Denote by H the set of decompositions $\bar{x} = \{x_t\}_{1 \leq t \leq T}$ of the sum $x = \sum_{t=1}^T x_t$ of elements $x_t \in \text{co} K_t(x)$. If we set $\sigma \bar{x} = \sum_{t=1}^T x_t$, this subset H can be written

$$(24) \quad H = \prod_{1 \leq t \leq T} \text{co} K_t(x) \cap \sigma^{-1}(x).$$

It is a convex compact subset. Hence there exists at least one extremal point $\bar{x} \in H$. We shall prove that all but m of the components x_t of such an extremal point $\bar{x}$ belong to $K_t(x)$. To see this, assume that the subset $S \subset \{1, \dots, T\}$ of indices t such that x_t does not belong to $K_t(x)$ contains at least $(m+1)$ elements. We shall prove that this implies that $\bar{x}$ is not extremal, i.e. a contradiction. Since $K_t(x)$ is finite, we can associate with any x_t a direction $y_t \in \mathbb{R}^m$ and $\varepsilon_t > 0$ such that

$$(25) \quad x_t + \theta y_t \in \text{co} K_t(x) \quad \text{for any } \theta \text{ such that } |\theta| \leq \varepsilon_t.$$

(See note at the end of the proof if necessary.)

Let $\varepsilon = \min_{1 \leq t \leq T} \varepsilon_t > 0$. Since S contains at least $m+1$ elements we can find scalars α_t (with $\alpha_t \neq 0$ for at least one t) such that

$$(26) \quad |\alpha_t| \leq 1, \quad \sum_{t \in S} \alpha_t y_t = 0.$$

Thus, for any $t \in S$,

$$(27) \quad \begin{cases} \text{(i) } u_t = x_t + \varepsilon \alpha_t y_t \text{ belongs to } \text{co} K_t(x), \\ \text{(ii) } v_t = x_t - \varepsilon \alpha_t y_t \text{ belongs to } \text{co} K_t(x). \end{cases}$$

We write $u_t = v_t = x_t$ whenever $t \notin S$.

Thus $\bar{u} = \{u_t\}_{1 \leq t \leq T}$ and $\bar{v} = \{v_t\}_{1 \leq t \leq T}$ both belong to $\prod_{1 \leq t \leq T} \text{co} K_t(x)$ and satisfy

$$\begin{aligned} \sigma \bar{u} &= \sigma \bar{x} + \varepsilon \sum \alpha_t y_t = \sigma \bar{x} = x, \\ \sigma \bar{v} &= \sigma \bar{x} - \varepsilon \sum \alpha_t y_t = \sigma \bar{x} = x. \end{aligned}$$

In other words, $\bar{u}$ and $\bar{v}$ belong to H . Since $\alpha \neq 0$ for at least one t , $u \neq v$.

Furthermore,

$$\vec{x} = \frac{1}{2}(\vec{u} + \vec{v}).$$

Hence $\vec{x}$ is not extremal in H . We thus have proved that the set S has at most m elements. $\square$

Note: Proof of (25). Since x_t does belong to $\text{co } K_t(x)$ and does *not* belong to $K_t(x)$, there exist at least two positive scalars $\lambda_t^i \in]0, 1[$ such that

$$x_t = \sum_i \lambda_t^i x_t^i = \lambda_t^{i_0} x_t^{i_0} + (1 - \lambda_t^{i_0}) z_t^0$$

where

$$z_t^0 = \sum_{i \neq i_0} \frac{\lambda_t^i}{1 - \lambda_t^{i_0}} x_t^i \in \text{co } K_t(x).$$

We set $y_t = x_t^{i_0} - z_t^0$ and $\varepsilon_t = \min(\lambda_t^{i_0}, 1 - \lambda_t^{i_0})$.

Therefore, if $|\theta| \leq \varepsilon_t$, then $\lambda_t^{i_0} + \theta \in]0, 1[$ and

$$x_t + \theta y_t = (\lambda_t^{i_0} + \theta) x_t^{i_0} + (1 - (\lambda_t^{i_0} + \theta)) z_t^0 \in \text{co } K_t(x). \quad \square$$

14.2.4. Sharp estimate of the duality gap

Suppose that

$$(28) \quad \begin{cases} \text{(i) } R = \prod_{1 \leq t \leq T} R_t \text{ where } R_t \text{ is a subset of a vector space } U_t, \\ \text{(ii) } f(x) = (1/T) \sum_{t=1}^T f_t(x_t) \text{ where } f_t : R_t \rightarrow \mathbf{R}, \\ \text{(iii) } L(x) = (1/T) \sum_{t=1}^T L_t(x_t) \text{ where } L_t \in \mathcal{L}(U_t, \mathbf{R}^m) \\ \text{is a linear operator from } U_t \text{ into } \mathbf{R}^m. \end{cases}$$

Proposition 2 shows that

$$\varrho(f) \leq \frac{1}{T} \sum_{t=1}^T \varrho(f_t) \leq \max_{1 \leq t \leq T} \varrho(f_t).$$

Using the Shapley-Folkman theorem, we shall prove that we can use in the estimate (12) of the duality gap $((m+1)/T) \max_{1 \leq t \leq T} \varrho(f_t)$ rather than $\max_{1 \leq t \leq T} \varrho(f_t)$, where m is the dimension of the space $\mathbf{R}^m$.

Theorem 3. Suppose that

$$(29) \quad \begin{cases} \text{(i) } g \text{ is a convex function from } Y \text{ into } \mathbf{R}, \\ \text{(ii) the subsets } R_t \subset U_t \text{ are convex,} \\ \text{(iii) the maps } L_t \text{ are linear.} \end{cases}$$

If we suppose that

$$(30) \quad 0 \in \text{Int} \left(Y - \frac{1}{T} \sum_{i=1}^T L_i(R_i) \right),$$

then there exists $\bar{p} \in \mathbf{R}^{m*}$ such that

$$(31) \quad \begin{aligned} v &= \inf_{x_i \in R_i} \left(\frac{1}{T} \sum_{i=1}^T f_i(x_i) + g \left(\frac{1}{T} \sum_{i=1}^T L_i x_i \right) \right) \\ &\leq - \left[\frac{1}{T} \sum_{i=1}^T f_i^*(-L_i^* \bar{p}) + g^*(\bar{p}) \right] + \frac{m+1}{T} \max_{1 \leq i \leq T} \varrho(f_i) \\ &\leq -v^* + \frac{m+1}{T} \max_{1 \leq i \leq T} \varrho(f_i). \end{aligned}$$

Proof. Consider the proof of Theorem 1 with

$$(32) \quad F(x, y) = \frac{1}{T} \sum_{i=1}^T f_i(x_i) + g(y).$$

The first and second steps of this proof show that there exists $\bar{p} \in \mathbf{R}^{m*}$ such that

$$(33) \quad w \leq -F^*(-L^* \bar{p}, \bar{p}).$$

Since

$$\begin{aligned} F^*(q, p) &= \sup_{\substack{x_i \in R_i \\ y \in Y}} \left[\sum_{i=1}^T \langle q_i, x_i \rangle + \langle q, y \rangle - \frac{1}{T} \sum_{i=1}^T f_i(x_i) - g(y) \right] \\ &= \frac{1}{T} \sum_{i=1}^T \sup_{x_i \in R_i} (\langle Tq_i, x_i \rangle - f_i(x_i)) + \sup_{y \in Y} (\langle p, y \rangle - g(y)) \\ &= \frac{1}{T} \sum_{i=1}^T f_i^*(Tq_i) + g^*(p) \end{aligned}$$

and

$$L^* p = \left\{ \frac{1}{T} L_i^* p \right\}_{1 \leq i \leq T} \in \prod_{1 \leq i \leq T} U_i^*,$$

we deduce that

$$(34) \quad w \leq - \left[\frac{1}{T} \sum_{i=1}^T f_i^*(-L_i^* \bar{p}) + g^*(\bar{p}) \right].$$

The third step of the proof of Theorem 1 is modified as follows. We deduce from (13) (i.e. from the definition of w) that, for any $\varepsilon > 0$,

$$(35) \quad (w + \varepsilon)\theta \in \text{co} (\Phi(R \times Y) + ([0, \infty[\times \{0\})).$$

Since

$$\begin{aligned}\Phi(x, y) &= \left\{ \frac{1}{T} \sum_{i=1}^T f_i(x_i) + g(y), \frac{1}{T} \sum_{i=1}^T L_i x_i - y \right\} \\ &= \sum_{i=1}^T \left\{ \frac{1}{T} f_i(x_i), \frac{1}{T} L_i x_i \right\} + \{g(y), -y\},\end{aligned}$$

we can write

$$(36) \quad \text{co } \Phi(R \times Y) + ([0, \infty[\times \{0\}) = G + \text{co } \sum_{1 \leq i \leq T} E_i,$$

where

$$(37) \quad E_i = \left\{ \left\{ \frac{1}{T} f_i(x_i), \frac{1}{T} L_i x_i \right\} \right\}_{x_i \in R_i}$$

and where

$$(38) \quad G = \{g(y), -y\}_{y \in Y} + ([0, \infty[\times \{0\}).$$

Since g is convex, G is a convex subset of $R \times R^m$. Therefore, we deduce from (35) and (36) that there exist $y \in Y$ and $c \geq 0$ such that

$$(39) \quad (w + \varepsilon)\theta - \{g(y) + c, -y\} \in \text{co } \sum_{1 \leq i \leq T} E_i.$$

But, by the Shapley-Folkman theorem (see Theorem 2), there exists a set S of at most $(m+1)$ indices such that

$$(40) \quad (w + \varepsilon)\theta - \{g(y) + c, -y\} \in \sum_{i \notin S} E_i + \sum_{i \in S} \text{co } (E_i).$$

Therefore, we can find $x_i \in R_i$ for $i \notin S$ and $m_i \in \mathcal{M}(R_i)$ for $i \in S$ such that

$$\begin{aligned}(w + \varepsilon)\theta - \{g(y) + c, -y\} &= \sum_{i \notin S} \left\{ \frac{1}{T} f_i(x_i), \frac{1}{T} L_i x_i \right\} \\ &\quad + \sum_{i \in S} \left\{ \frac{1}{T} f_i^A(m_i), \frac{1}{T} L_i^A m_i \right\}.\end{aligned}$$

Since L_i is linear, $L_i^A m_i = L_i \beta_i m_i$. By setting $x_i = \beta_i m_i$ whenever $i \in S$, we deduce that

$$\begin{aligned}(w + \varepsilon)\theta - \{g(y) + c, -y\} &= \sum_{1 \leq i \leq T} \left\{ \frac{1}{T} f_i(x_i), \frac{1}{T} L_i x_i \right\} \\ &\quad - \sum_{i \in S} \left\{ \frac{1}{T} (f_i(\beta_i m_i) - f_i^A(m_i)), 0 \right\}.\end{aligned}$$

Therefore, we obtain that

$$(41) \quad \begin{aligned} w + \varepsilon &\geq \frac{1}{T} \sum_{1 \leq i \leq T} f_i(x_i) + g(y) - \frac{1}{T} \sum_{i \in S} (f_i(m_i) - f_i^d(m_i)) \\ &\geq \frac{1}{T} \sum_{1 \leq i \leq T} f_i(x_i) + g(y) - \frac{1}{T} \sum_{i \in S} \varrho(f_i) \end{aligned}$$

and

$$(42) \quad y = \frac{1}{T} \sum_{1 \leq i \leq T} L_i x_i.$$

These two statements together imply that

$$(43) \quad \begin{aligned} w + \varepsilon &\geq \frac{1}{T} \sum_{1 \leq i \leq T} f_i(x_i) + g\left(\frac{1}{T} \sum_{1 \leq i \leq T} L_i x_i\right) - \frac{1}{T} \sum_{i \in S} \varrho(f_i) \\ &\geq v - \frac{m+1}{T} \max_{1 \leq i \leq T} \varrho(f_i). \end{aligned}$$

Hence, the conclusion of Theorem 3 follows from inequalities (34) and (43).

14.2.5. Applications

Let $f: R \rightarrow \mathbf{R}$ be a function defined on a convex subset R of U .

Associate with an integer T the function

$$(44) \quad f_T(x) = \inf_{(1/T) \sum_{i=1}^T x_i = x} \frac{1}{T} \sum_{i=1}^T f(x_i).$$

It is clear that

$$(45) \quad \text{for any } x \in R, \quad f^{**}(x) \leq f_T(x) \leq f(x).$$

Proposition 3. Suppose that

$$(46) \quad \begin{cases} \text{(i) } g \text{ is a convex function from } Y \text{ into } \mathbf{R}, \\ \text{(ii) } R \text{ is convex,} \\ \text{(iii) } L \in \mathcal{L}(U, \mathbf{R}^m). \end{cases}$$

If we suppose that

$$(47) \quad 0 \in \text{Int}(L(R) - Y),$$

then there exists $\bar{p} \in \mathbf{R}^{m*}$ such that

$$(48) \quad \begin{aligned} v_T &= \inf_{x \in R} (f_T(x) + g(Lx)) \\ &\leq -(f^*(-L^*\bar{p}) + g^*(\bar{p})) + \frac{m+1}{T} \varrho(f) \\ &\leq -v^* + \frac{m+1}{T} \varrho(f). \end{aligned}$$

Proof. This result follows from Theorem 3 with $R_t = R$, $f_t = f$ and $L_t = L$ for all t .

Therefore, there exists $\bar{p} \in \mathbf{R}^{m*}$ such that

$$\begin{aligned}
 v &= \inf_{t_i \in R} \left(\frac{1}{T} \sum_{1 \leq i \leq T} f(x_i) + g \left(\frac{1}{T} \sum_{1 \leq i \leq T} x_i \right) \right) \\
 &= \inf_{x \in R} \inf_{(1/T) \sum_{1 \leq i \leq T} x_i = x} \left[\frac{1}{T} \sum_{1 \leq i \leq T} f(x_i) + g(Lx) \right] \\
 &= \inf_{x \in R} (f_T(x) + g(Lx)) \\
 &\leq - \left(\frac{1}{T} \sum_{1 \leq i \leq T} f^*(-L^* \bar{p}) + g^*(\bar{p}) \right) + \frac{m+1}{T} \varrho(f) \\
 &= -(f^*(-L^* \bar{p}) + g^*(\bar{p})) + \frac{m+1}{T} \varrho(f) \leq -v^* + \frac{m+1}{T} \varrho(f). \quad \square
 \end{aligned}$$

Let $f: R \rightarrow \mathbf{R}$ be a function defined on a convex subset $R \subset U$ and $L \in \mathcal{L}(U, \mathbf{R}^m)$ be a linear operator.

We shall associate with any T the function

$$(49) \quad (Lf)_T(y) = \inf_{(1/T) \sum_{i=1}^T Lx_i = y} \frac{1}{T} \sum_{i=1}^T f(x_i)$$

It is clear that

$$(50) \quad (Lf)_T(y) \leq (Lf)(y) = \inf_{Lx=y} f(x)$$

Proposition 4. *If we suppose that*

$$(51) \quad y \in \text{Int } L(R),$$

then there exists $\bar{p} \in \mathbf{R}^{m}$ such that*

$$(52) \quad (Lf)_T(y) - (f^* L^*)^*(y) = (Lf)_T(y) - (Lf)^{**}(y) \leq \frac{m+1}{T} \varrho(f).$$

Proof. We apply Proposition 3 with $g = \psi(Y; \cdot)$. $\square$

Consider T functions $h_t: R_t \rightarrow \mathbf{R}$, where R_t is a convex subset of the space $\mathbf{R}^n$. Let $\square_{1 \leq i \leq T} h_t$ be the inf-convolution of these functions, defined by

$$(53) \quad \square_{1 \leq i \leq T} h_t(y) = \inf_{\sum_{1 \leq i \leq T} x_i = y} \sum_{1 \leq i \leq T} h_t(x_i)$$

(see Definition 1.2.2).

Proposition 5. *Suppose that*

(54) *the subsets R_i are convex*

and that

$$(55) \quad y \in \text{Int} \left(\sum_{1 \leq i \leq T} R_i \right).$$

Then

$$(56) \quad \left[\bigcap_{1 \leq i \leq T} h_i(y) \right] - \left[\sum_{1 \leq i \leq T} h_i^* \right]^*(y) \leq (m+1) \max_{1 \leq i \leq T} \varrho(h_i).$$

Proof. We apply the above proposition with $f_i = Th_i$ and $L = T\mathbf{1}$. $\square$

14.2.6. Extremality relations

The extremality relations still hold with an error which can be estimated with the modulus of non-convexity.

Proposition 6. *Suppose that the assumptions of Theorem 3 are satisfied and that g is lower semi-continuous. Let $\bar{p} \in \mathbf{R}^{m*}$ satisfy inequality (31).*

If there exists $\hat{x}_i \in R_i$ satisfying

$$(57) \quad \begin{cases} \text{(i)} \ (1/T) \sum_{i=1}^T L_i \hat{x}_i \in \partial g^*(\bar{p}), \\ \text{(ii)} \ \hat{x}_i \in \partial f_i^*(-L_i^* \bar{p}), \\ \text{(iii)} \ f_i(\hat{x}_i) = f_i^{**}(\hat{x}_i), \end{cases}$$

then

$$(58) \quad v - \left[\frac{1}{T} \sum_{i=1}^T f_i(\hat{x}_i) + g \left(\frac{1}{T} \sum_{i=1}^T L_i \hat{x}_i \right) \right] \leq \frac{m+1}{T} \max_{1 \leq i \leq T} \varrho(f_i).$$

Remark. This proposition is analogous to Proposition 5.2.8 concerning the decentralization principle. It shows that *this principle still holds with an error which decreases as the number of functions increases.*

Proof. Condition (57(i)) means that

$$g \left(\frac{1}{T} \sum_{i=1}^T L_i \hat{x}_i \right) = \left\langle \bar{p}, \frac{1}{T} \sum_{i=1}^T L_i \hat{x}_i \right\rangle - g^*(\bar{p})$$

and (57(ii) and (iii)) mean that $\hat{x}_i$ minimizes $x \mapsto f_i(x) + \langle p, L_i x \rangle$ (see Proposition 4.1.1), i.e.

$$f_i(\hat{x}_i) + \langle \bar{p}, L_i \hat{x}_i \rangle = \min_x [f_i(x) + \langle L_i^* \bar{p}, x \rangle] = -f_i^*(-L_i^* \bar{p}).$$

Therefore,

$$\begin{aligned} & \frac{1}{T} \sum_{1 \leq i \leq T} f_i(\hat{x}_i) + g\left(\frac{1}{T} \sum_{1 \leq i \leq T} L_i \hat{x}_i\right) \\ &= - \left[\frac{1}{T} \sum_{1 \leq i \leq T} f_i^*(-L_i^* \bar{p}) + g^*(\bar{p}) \right] \\ &\geq v - \frac{m+1}{T} \max_{1 \leq i \leq T} \varrho(f_i). \quad \square \end{aligned}$$

***Remark.** We can investigate the properties of a minimum $x = x_i$ of the minimization problem. For this purpose, recall that

$$(59) \quad \partial_a f(x) = \{p \in U^* \text{ such that } f(x) + f^*(p) - \langle p, x \rangle \leq a\}$$

is the a -subdifferential of f at x .

Proposition 7. Suppose that the assumptions of Theorem 3 are satisfied. Let $\bar{p} \in \mathbb{R}^{m*}$ satisfy inequality (31). If $\bar{x} = \{\bar{x}_i\}_{1 \leq i \leq T}$ minimizes

$$\frac{1}{T} \sum_{1 \leq i \leq T} f_i(x_i) + g\left(\frac{1}{T} \sum_{1 \leq i \leq T} L_i x_i\right),$$

then

$$(60) \quad \begin{cases} \text{(i)} \quad \bar{p} \in \partial_{((m+1)/T)\varrho} g(1/T \sum_{i=1}^T L_i \bar{x}_i), \\ \text{(ii)} \quad -L_i \bar{p} \in \partial_{(m+1)\varrho} f_i(\bar{x}_i) \end{cases}$$

where we set

$$(61) \quad \varrho = \max_{1 \leq i \leq T} \varrho(f_i).$$

Proof. We can write inequality (31) in the form

$$\begin{aligned} & \sum_{1 \leq i \leq T} \frac{1}{T} (f_i(\bar{x}_i) + f_i^*(-L_i^* \bar{p}) - \langle -L_i^* \bar{p}, \bar{x}_i \rangle) \\ &+ g\left(\frac{1}{T} \sum_{1 \leq i \leq T} L_i \bar{x}_i\right) + g^*(\bar{p}) - \langle \bar{p}, (1/T) \sum_{1 \leq i \leq T} L_i \bar{x}_i \rangle \\ &\leq \frac{m+1}{T} \varrho. \end{aligned}$$

Since all the terms of this sum are non-negative, we deduce that each of the m is less than or equal to $((m+1)/T)\varrho$, i.e. that statements (60) are satisfied. $\square$

14.2.7. The Aumann-Perles duality theorem

Let Ω be a bounded open subset Ω of $\mathbf{R}^n$ and let K be a convex subset of some Banach space F .

We introduce the following notation.

(62) $U = L^1(\Omega, F)$ is the Banach space of (classes of) measurable functions x from Ω into F with integrable norm $\|x(\omega)\|$.

(63) $R = L^1(\Omega, K)$ is the subset of functions $x \in L^1(\Omega, F)$ such that $x(\omega) \in K$ for almost all $\omega \in \Omega$.

Let

(64) $g : \mathbf{R}^m \rightarrow]-\infty, +\infty]$ be a convex lower semi-continuous function whose domain is denoted by Y .

We consider a minimization problem of the form

$$(65) \quad v = \inf_{x \in R} [f(x) + g(Lx)]$$

where the function f and the linear operator L are “integral” in the sense that

$$(66) \quad \begin{cases} \text{(i) } f(x) = \int_{\Omega} \varphi(\omega, x(\omega)) \, d\omega, \\ \text{(ii) } L(x) = \int_{\Omega} A(\omega) x(\omega) \, d\omega. \end{cases}$$

We shall assume that φ is a function from $\Omega \times K$ into $\mathbf{R}$ satisfying

$$(67) \quad \begin{cases} \text{(i) for almost every } \omega \in \Omega, \xi \in K \rightarrow \varphi(\omega, \xi) \text{ is continuous,} \\ \text{(ii) for every } \xi \in K, \omega \in \Omega \rightarrow \varphi(\omega, \xi) \text{ is measurable,} \\ \text{(iii) } |\varphi(\omega, \xi)| \leq C(\omega) + M \|\xi\| \text{ where } C \in L^1(\Omega), M > 0, \end{cases}$$

and that A is a map from $\Omega \times F$ into $\mathbf{R}^m$ satisfying

$$(68) \quad \begin{cases} \text{(i) for almost every } \omega \in \Omega, A(\omega) \in \mathcal{L}(F, \mathbf{R}^m), \\ \text{(ii) } \omega \rightarrow A(\omega) \text{ is a measurable function from } \Omega \text{ into } \mathcal{L}(F, \mathbf{R}^m), \\ \text{(iii) } \omega \rightarrow \|A(\omega)\| \text{ belongs to } L^\infty(\Omega). \end{cases}$$

It follows from the Lebesgue convergence theorem that

$$(69) \quad \begin{cases} \text{(i) } f : R = L^1(\Omega, K) \rightarrow \mathbf{R} \text{ is continuous,} \\ \text{(ii) } L \in \mathcal{L}(L^1(\Omega, F), \mathbf{R}^m). \end{cases}$$

Remark. A function φ satisfying (67(i) and (ii)) is called a “Carathéodory function”.

In order to define the dual problem of the minimization problem

$$(65) \quad v = \inf_{x \in \mathbf{R}} \left[\int_{\Omega} \varphi(\omega, x(\omega)) \, d\omega + g \left(\int_{\Omega} A(\omega) x(\omega) \, d\omega \right) \right].$$

we introduce the conjugate function f^* defined on $L^\infty(\Omega, F^*)$ by

$$(70) \quad f^*(q) = \sup_{x \in \mathbf{R}} \left[\int_{\Omega} [\langle q(\omega), x(\omega) \rangle - \varphi(\omega, x(\omega))] \, d\omega \right],$$

and the transpose L^* of L defined on $\mathbf{R}^{m*}$ by

$$(71) \quad L^*p : \omega \rightarrow A^*(\omega)p.$$

We define the dual problem

$$(72) \quad v^* = \inf_{p \in \mathbf{R}^{m*}} [f^*(-L^*p) + g^*(p)].$$

Remark. Note that the measurable selection theorem implies that

$$(73) \quad f^*(q) = \int_{\Omega} \varphi^*(\omega, q(\omega)) \, d\omega$$

where, for almost every $\omega \in \Omega$, $\varphi^*(\omega, p) = \sup_{\xi \in K} [\langle p, \xi \rangle - \varphi(\omega, \xi)]$ (see for instance in the book of Ekeland and Temam [1974, Theorem VIII-1-2, p. 220 and Proposition IX-2-1, p. 251]).

In this case, we can write

$$(74) \quad v^* = \inf_{p \in \mathbf{R}^{m*}} \left[\int_{\Omega} \varphi^*(\omega, -A^*(\omega)p) \, d\omega + g^*(p) \right].$$

We will not use this result. $\square$

Denote by $L_0 \in \mathcal{L}(F, \mathbf{R}^m)$, the operator defined by

$$(75) \quad \forall \xi \in F, \quad L_0(\xi) = \left(\int_{\Omega} A(\omega) \, d\omega \right) \xi.$$

Theorem 4 (Aumann-Perles). Suppose that hypotheses (64), (67), (68) and the constraint qualification assumption

$$(76) \quad 0 \in \text{Int}(Y - L_0(K))$$

hold. Suppose also that the moduli of non-convexity are bounded, i.e.

$$(77) \quad \varrho = \sup_{\omega \in \Omega} \varrho(\varphi(\omega, \cdot)) < +\infty.$$

Then there exists a Lagrange multiplier $\bar{p} \in \mathbf{R}^{m*}$, i.e.

$$(78) \quad v = -v^* = -(f^*(-L^*\bar{p}) + g^*(\bar{p}))$$

14.2.8. The approximation procedure

Let h be some small parameter, and let $\mathcal{O}_h = \{\Omega_h^i\}_i$ be a partition of Ω into a finite number of subsets Ω_h^i , such that

$$(79) \quad |\mathcal{O}_h| = \max_i \text{meas}(\Omega_h^i) \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

Denote by $L_h(\Omega, K)$ the space of step functions from Ω into K , constant over each member of $\mathcal{O}_h$. We approximate the minimization problem (3) by

$$(80) \quad \begin{aligned} v_h &= \inf_{x_h \in L_h(\Omega, K)} \left\{ \int_{\Omega} \varphi(\omega, x_h(\omega)) \, d\omega + g \left(\int_{\Omega} A(\omega, x_h(\omega)) \, d\omega \right) \right\} \\ &= \inf_{\{\xi_h^i\}} \left\{ \sum_i \int_{\Omega_h^i} \varphi(\omega, \xi_h^i) \, d\omega + g \left(\sum_i \int_{\Omega_h^i} A(\omega) \xi_h^i \, d\omega \right) \right\} \end{aligned}$$

Its dual problem can be written

$$(81) \quad v_h^* = \inf_{p \in \mathbb{R}^{m*}} [f_h^*(-L^*p) + g^*(p)]$$

where

$$(82) \quad \begin{aligned} f_h^*(-L^*p) &= \sup_{x_h \in L_h(\Omega, K)} \left[\int_{\Omega} [\langle -A^*(\omega)p, x_h(\omega) \rangle - \varphi(\omega, x_h(\omega))] \, d\omega \right] \\ &= \sum_i \sup_{\xi \in K} \int_{\Omega_h^i} (\langle -A^*(\omega)p, \xi \rangle - \varphi(\omega, \xi)) \, d\omega \end{aligned}$$

Proposition 8. Under the assumptions of Theorem 4, for any h there exists $\bar{p}_h \in \mathbb{R}^m$ such that

$$(83) \quad -v_h^* \leq v_h \leq -[f_h^*(-L^*\bar{p}_h) + g^*(\bar{p}_h)] + M |\mathcal{O}_h|,$$

where M is a constant independent of h .

Proof. We apply Theorem 3 with $R_i = K$,

$$f_i(\xi) = T \int_{\Omega_h^i} \varphi(\omega, \xi) \, d\omega \quad \text{and} \quad L_i = T \int_{\Omega_h^i} A(\xi) \, d\omega$$

(where T denotes the number of elements of the partition $\mathcal{O}_h$). The modulus of non-convexity of f_i is estimated by

$$\begin{aligned} \varrho(f_i) &= T \sup_{m \in \mathcal{M}(K)} \left[\int_{\Omega_h^i} \varphi(\omega, \beta m) \, d\omega - \left\langle m, \int_{\Omega_h^i} \varphi(\omega, \cdot) \, d\omega \right\rangle \right] \\ &\leq T \int_{\Omega_h^i} \varrho \, d\omega \leq T \varrho |\mathcal{O}_h|. \end{aligned}$$

Finally, the constraint qualification is satisfied. Hence Theorem 3 implies the existence of $\bar{p}_h$ such that (83) is satisfied with $M = (m+1)\varrho$. $\square$

Theorem 4 is a consequence of the following more precise proposition.

Proposition 9. *Under the assumptions of Theorem 4, there exists a subsequence of the family of $\bar{p}_h$ satisfying (83) which converges to a Lagrange multiplier $\bar{p} \in \mathbf{R}^{m*}$ (i.e. which satisfies (78)). Furthermore, we have the estimate*

$$(84) \quad v = -[f^*(-L^*\bar{p}) + g^*(\bar{p})] \leq -f_h^*(-L^*\bar{p}_h) - g^*(\bar{p}_h) + M|\mathcal{C}_h|.$$

Proof. We clearly have the inequalities

$$(85) \quad -v^* \leq v \leq v_h \leq -[f_h^*(-L^*\bar{p}_h) + g^*(\bar{p})] + M|\mathcal{C}_h|.$$

It therefore remains to prove that

(a) the family $\bar{p}_h$ lies in a relatively compact subset and thus, that a suitable subsequence converges to $\bar{p}$ in $\mathbf{R}^{m*}$.

(b) for any $\varepsilon > 0$, $\exists h_0$ such that, $\forall h \leq h_0$,

$$(86) \quad f^*(-L^*\bar{p}) + g^*(\bar{p}) \leq f_h^*(-L^*\bar{p}_h) + g^*(\bar{p}_h) + \varepsilon.$$

It is clear that (85) and (86) imply

$$-v^* \leq v \leq v_h \leq -[f^*(-L^*\bar{p}) + g^*(\bar{p})] + M|\mathcal{C}_h| + \varepsilon.$$

Letting h and ε go to 0, we deduce that

$$(78) \quad v = -v^* = -[f^*(-L^*\bar{p}) + g^*(\bar{p})]$$

i.e. $\bar{p}$ is a Lagrange multiplier. Hence estimate (84) follows from (85) and (78).

Proof of (a). Since $0 \in \text{Int}(Y - L_0(K))$ and the elements ξ can be identified with step functions of $L_h(\Omega, K)$ for all h , we can write any $z \in \mathbf{R}^m$ in the form $z = \varepsilon \left(y - \int_{\Omega} A(\omega) \xi \, d\omega \right)$ where $\varepsilon > 0$, $y \in Y$ and $\xi \in K$.

Hence

$$\begin{aligned} \left\langle \bar{p}_h, \frac{z}{\varepsilon} \right\rangle &= \left\langle \bar{p}_h, y - \int_{\Omega} A(\omega) \cdot \xi \, d\omega \right\rangle \\ &\leq g^*(\bar{p}) + g(y) + f_h^*(-L^*\bar{p}_h) + f(\xi) \\ &\leq f(\xi) + g(y) - v_h \leq f(\xi) + g(y) - v. \end{aligned}$$

Thus the family $\{\bar{p}_h\}$ is bounded, i.e. relatively compact. A suitable subsequence (again denoted by) $\bar{p}_h$ converges to an element $\bar{p} \in \mathbf{R}^{m*}$.

Proof of (b). Let ε be fixed. There exists $\bar{x} \in L^1(\Omega, K)$ such that

$$f^*(-L^*\bar{p}) \leq \int_{\Omega} \langle -A^*(\omega)\bar{p}, \bar{x}(\omega) \rangle - \varphi(\omega, \bar{x}(\omega)) \, d\omega + \frac{1}{4}\varepsilon$$

by the very definition of f^* . Also, there exist $\eta > 0$ and $\delta > 0$ such that

$$(87) \quad \begin{cases} \text{(i)} \left| \left\langle q, \int_{\Omega} A(\omega) y(\omega) d\omega \right\rangle - \left\langle \bar{p}, \int_{\Omega} A(\omega) x(\omega) d\omega \right\rangle \right| \leq \frac{1}{4} \varepsilon, \\ \text{(ii)} g^*(\bar{p}) \leq g(q) + \frac{1}{4} \varepsilon, \\ \text{(iii)} \int_{\Omega} [\varphi(\omega, y(\omega)) - \varphi(\omega, \bar{x}(\omega))] d\omega \leq \frac{1}{4} \varepsilon, \end{cases}$$

whenever $\|\bar{p} - q\|_{R^{m*}} \leq \eta$ and $\|\bar{x} - y\|_U \leq \delta$. [This follows from the fact that $\{q, y\} \rightarrow \left\langle q, \int_{\Omega} A(\omega) y(\omega) d\omega \right\rangle$ is a continuous function on $R^{m*} \times L^1(\Omega, F)$, $q \rightarrow g^*(q)$ is lower semi-continuous and f is continuous on $U = L^1(\Omega, F)$.]

On the other hand, we can find h_1 such that, for any $h \leq h_1$, $\|\bar{p} - \bar{p}_h\| \leq \eta$. Since the space of step functions is dense in $L^1(\Omega, K)$, we can find h_2 such that, for any $h \leq h_2$, there exists $x_h \in L_h(\Omega, K)$ satisfying $\|\bar{x} - x_h\| \leq \delta$ (we take $x_h = x_{h_2}$). Then, for all $h \leq h_0 = \min(h_1, h_2)$, we deduce from (87) with $q = \bar{p}_h$ and $y = x_h$ that

$$\begin{aligned} v^* &\leq f^*(-L^*\bar{p}) + g^*(\bar{p}) \\ &\leq \int_{\Omega} (\langle \bar{p}, A(\omega) \bar{x}(\omega) \rangle - \varphi(\omega, \bar{x}(\omega))) d\omega + g_*(\bar{p}) + \frac{1}{4} \varepsilon \\ &\leq \int_{\Omega} (\langle \bar{p}_h, A(\omega) x_h(\omega) \rangle - \varphi(\omega, x_h(\omega))) d\omega + g_*(\bar{p}_h) \\ &\quad + \int_{\Omega} (\langle \bar{p}, A(\omega) \bar{x}(\omega) \rangle - \langle \bar{p}_h, A(\omega) x_h(\omega) \rangle) d\omega \\ &\quad + \int_{\Omega} (\varphi(\omega, x_h(\omega)) - \varphi(\omega, \bar{x}(\omega))) d\omega \\ &\quad + g^*(\bar{p}) - g_*(\bar{p}_h) + \frac{1}{4} \varepsilon \\ &\leq f_h^*(-L^*\bar{p}_h) + g^*(\bar{p}_h) + \frac{1}{4} \varepsilon. \end{aligned}$$

14.3. Duality in calculus of variations

Consider the following example of calculus of variations (problem of Bolza).

$$v = \inf_{\substack{x(0) = \xi_0 \\ x(1) = \xi_1}} \int_0^1 \phi(\omega, x(\omega), \dot{x}(\omega)) d\omega.$$

where $\dot{x}$ denotes the derivative of x and ϕ maps $]0, 1[\times R^n \times R^n$ into $] -\infty, +\infty]$. We leave until later a specification of the precise assumptions on ϕ and the description of the (infinite-dimensional) space of functions in which we look for a solution of the minimization problem. We shall prove that the dual problem can be written

$$v^* = \inf_{p(\cdot)} \left[\int_0^1 \phi^*(\omega, \dot{p}(\omega), p(\omega)) d\omega + \langle p(0), \xi_0 \rangle - \langle p(1), \xi_1 \rangle \right].$$

We restrict our study to the convex case, where for almost all $\omega \in]0, 1[$, the function $\{u, v\} \rightarrow \phi(\omega, u, v)$ is convex. We shall prove that $\bar{p}(\cdot)$ is a Lagrange multiplier and $\bar{x}(\cdot)$ is a solution of the minimization problem *if and only* if the following extremality relation holds. For almost all $\omega \in]0, 1[$,

$$(*) \quad \{\dot{\bar{p}}(\omega), \bar{p}(\omega)\} \in \partial\phi(\omega, \bar{x}(\omega), \dot{\bar{x}}(\omega)).$$

If the function ϕ is assumed to be differentiable with respect to $\{u, v\}$, this relation becomes

$$\dot{\bar{p}}(\omega) = D_u\phi(\omega, \bar{x}(\omega), \dot{\bar{x}}(\omega)); \quad \bar{p}(\omega) = D_v\phi(\omega, \bar{x}(\omega), \dot{\bar{x}}(\omega)).$$

We recognize the Euler-Lagrange equation in the classical setting of the calculus of variations.

Introduce the conjugate function ψ of the function ϕ with respect to v , defined by

$$\psi(\omega, u, r) = \sup_v [\langle r, v \rangle - \phi(\omega, u, v)]$$

(which coincides with the Legendre transform of ϕ with respect to v whenever it exists).

We shall see that the extremality relations can be rewritten in the form: for almost all ω ,

$$(**) \quad \begin{cases} \text{(i)} & -\dot{\bar{p}}(\omega) \in \bar{\partial}_u\psi(\omega, \bar{x}(\omega), \bar{p}(\omega)), \\ \text{(ii)} & \dot{\bar{x}}(\omega) \in \partial_r\psi(\omega, \bar{x}(\omega), \bar{p}(\omega)). \end{cases}$$

where $\bar{\partial}_u$ and ∂_r denote the super-differential with respect to u and the sub-differential with respect to r of the function ψ . If we assume that ψ is differentiable with respect to $\{u, r\}$, this system becomes

$$\begin{cases} \text{(i)} & -\dot{\bar{p}}(\omega) = D_u\psi(\omega, \bar{x}(\omega), \bar{p}(\omega)), \\ \text{(ii)} & \dot{\bar{x}}(\omega) = D_r\psi(\omega, \bar{x}(\omega), \bar{p}(\omega)). \end{cases}$$

We recognize the Hamiltonian system in the classical setting of the calculus of variations.

Now since we have devised a duality theory in infinite-dimensional spaces and since extremality relations $(*)$, $(**)$ look like the extremality relations

$$(***) \quad \{-L^*\bar{p}, \bar{p}\} \in \partial F(\bar{x}, L\bar{x})$$

and

$$(***) \quad L\bar{x} \in \bar{\partial}_p H(\bar{x}, \bar{p}), \quad L^*\bar{p} \in \bar{\partial}_x H(\bar{x}, \bar{p}),$$

it is strongly tempting to deduce the above results from the general theory by setting

$$F(x, y) = \int_{\Omega} \phi(\omega, x(\omega), y(\omega)) d\omega$$

and $Lx = Dx = \dot{x}$. This is not as simple as it may appear, since we have to relate the transpose D^* of the differential operator D with $-D$ in order to link (*) and (**), (**) and (****) respectively. It is time now to become more specific.

Under suitable (and natural) assumptions on ϕ , we can assume that the function F maps $U \times V$ into $]-\infty, +\infty]$ where

$U = V = L^2(0, 1)$ is the space of (classes of)
square integrable functions from $]0, 1[$ in $\mathbb{R}^n$.

Then the minimization problem has a meaning if we take x in the "Sobolev space"

$$H^1(0, 1) = \{x \in L^2(0, 1) \text{ such that } \dot{x} = Dx \in L^2(0, 1)\}.$$

This definition requires the derivative Dx to be regarded as the derivative in the *distributional sense*. Supplied with the graph norm, the space $H^1(0, 1)$ is a Hilbert space, dense in $L^2(0, 1)$. The important fact underlying all the theory is the following. The operator γ associating with any $x \in H^1(0, 1)$ its "trace" $\gamma x = \{x(0), x(1)\} \in \mathbb{R}^n \times \mathbb{R}^n$ satisfies the following properties:

- (*****) $\begin{cases} \text{(i) } \gamma \text{ maps } H^1(0, 1) \text{ onto } \mathbb{R}^n \times \mathbb{R}^n, \\ \text{(ii) Ker } \gamma \text{ is dense in } L^2(0, 1), \end{cases}$

since one can prove that $\text{Ker } \gamma = H_0^1(0, 1)$ is the closure in $H^1(0, 1)$ of the space $\mathcal{D}(0, 1)$ of infinitely differentiable functions with compact support in $]0, 1[$. The latter statement implies that the *transpose* $(D_0)^*$ of the restriction D_0 to $H_0^1(0, 1)$ of D is equal to $-D$ by the very definition of the derivative in the distributional sense, i.e.

$$\begin{aligned} \forall p \in L^2(0, 1), \forall x \in H_0^1(0, 1), \quad & \int_0^1 \langle p(\omega), Dx(\omega) \rangle d\omega \\ &= - \int_0^1 \langle Dp(\omega), x(\omega) \rangle d\omega. \end{aligned}$$

Now the formula for integration by parts relates the transpose D^* of D with the transpose $D_0^* = -D$ of D_0 , i.e.

$$\begin{aligned} \forall p \in H^1(0, 1), \forall x \in H^1(0, 1). \\ & \int_0^1 \langle p(\omega), Dx(\omega) \rangle d\omega \\ &= - \int_0^1 \langle Dp(\omega), x(\omega) \rangle d\omega + \langle p(1), x(1) \rangle - \langle p(0), x(0) \rangle. \end{aligned}$$

This formula can be proved directly (approximating by the usual integration by parts formula) or in an abstract way using only the trace property (*****). We choose the latter approach and prove in fact an abstract "Green formula" which is quite versatile in that it allows various differential operators L to be used. We describe the results briefly. Assume that $U(L)$, U and $\mathcal{E}$ are three Hilbert spaces such that

$$\begin{cases} \text{(i)} & U(L) \subset U, \\ \text{(ii)} & \gamma \in L(U(L), \mathcal{E}) \text{ is surjective,} \\ \text{(iii)} & U_0(L) = \text{Ker } \gamma \text{ is dense in } U. \end{cases}$$

Let $L \in \mathcal{L}(U(L), V)$, where V is another Hilbert space. Assume that $U^* = U$ and $V^* = V$ for simplicity and that we "know" the transpose $L_0^* \in \mathcal{L}(V, U_0(L)^*)$ of the restriction L_0 of L to $U_0(L)$. We define the domain of L_0^* by

$$V(L_0^*) = \{p \in V \text{ such that } L_0^*p \in U\}.$$

We shall prove that there exists a *unique* operator $\beta^* \in \mathcal{L}(V(L_0^*), \mathcal{E}^*)$ such that the abstract Green formula

$$\forall p \in V(L_0^*), \forall x \in U(L), \quad \langle L_0^*p, x \rangle - \langle p, Lx \rangle = \langle \beta^*p, \gamma x \rangle$$

holds.

We then modify the results of Section 14.1 as follows. We associate with a minimization problem

$$v = \inf_{x \in U(L)} F(x, Lx, \gamma x)$$

the dual problem

$$v^* = \inf_{p \in V(L_0^*)} F^*(-L_0^*p, p, \beta^*p).$$

We check that the extremality relations can be written

$$\{-L_0^*\bar{p}, \bar{p}, \beta^*\bar{p}\} \in \partial F(\bar{x}, L\bar{x}, \gamma\bar{x})$$

or, equivalently,

$$\begin{cases} \text{(i)} & \{L\bar{x}, \gamma\bar{x}\} \in \partial_{p, \pi} H(\bar{x}, \bar{p}, \beta^*p), \\ \text{(ii)} & L_0^*\bar{p} \in \partial_x H(\bar{x}, \bar{p}, \beta^*\bar{p}). \end{cases}$$

where H is the Hamiltonian defined by

$$H(x, p, \pi) = \sup_{y, \xi} [\langle p, y \rangle + \langle \pi, \xi \rangle - F(x, y, \xi)].$$

We deduce the existence of a Lagrange multiplier under the constraint qualification assumption that $0 \in \text{Int Dom } v$ where the perturbation function v is defined by

$$v(y, \xi) = \inf_{x \in U(L)} F(x, Lx + y, \gamma x + \xi).$$

We end this section with an analogous study in the framework of optimal control problems.

14.3.1. The Green formula

Consider three Hilbert spaces $U, V, \mathcal{E}$ together with a linear operator L mapping its domain $U(L) \subset U$ into V and a linear operator γ mapping $U(L)$ onto $\mathcal{E}$. We require,

- (1) for the norm $\|x\| = (\|x\|_U^2 + \|Lx\|_V^2)^{1/2}$, $U(L)$ is a Hilbert space and
- (2) $\begin{cases} \text{(i) } \gamma \in \mathcal{L}(U(L), \mathcal{E}) \text{ is surjective,} \\ \text{(ii) } U_0(L) = \text{Ker } \gamma \text{ is dense in } U. \end{cases}$

Remark. Property (2(ii)) implies that $U(L)$ is dense in U . Property (1) amounts to saying that the unbounded operator $(U(L), L)$ is "closed".

We shall say that assumption (2) is the "trace property".

Notation. For any map M defined on $U(L)$, we shall set

- (3) $M_0 = M|_{U_0(L)}$ to be the restriction of M to $U_0(L)$.

For the sake of simplicity, we shall assume that U and V are identified with their respective duals

- (4) $U = U^*$ and $V = V^*$.

Let i and i_0 be the canonical injections from $U(L)$ and $U_0(L)$ into U respectively. Since they are injective (resp. have a dense image) their transposes $i^* \in \mathcal{L}(U, U(L)^*)$ and $i_0^* \in \mathcal{L}(U, U_0(L)^*)$ have a dense image (resp. are injective).

We assume that

- (5) the transpose i_0^* is identified with the canonical injection from U into $U_0(L)^*$

in such a way that

- (6) $\begin{cases} \text{(i) } U_0(L) \subset U \subset U_0(L)^*, \\ \text{(ii) each space is dense in the larger spaces.} \end{cases}$

With these identifications, we have one alternative. If M is any continuous linear operator mapping $U(L)$ into W , we may either "use" the transpose $(M_0)^* \in \mathcal{L}(W^*, U_0(L)^*)$ of the restriction M_0 to $U_0(L)$ or we may "use" the transpose $M^* \in \mathcal{L}(W^*, U(L)^*)$ of M . Because of the identification (6), we will use the transpose $(M_0)^*$ of M_0 in preference to M . In order to emphasize this point, we introduce the following definition.

Definition 1. Suppose that (1), (2), (4) and (5) hold. We shall say that the transpose $(M_0)^*$ of the restriction of M to $U_0(L)$ is the "formal adjoint" of M . The formal adjoint $L_0^* \in \mathcal{L}(V, U_0(L)^*)$ is defined by

$$(7) \quad \forall p \in V, \forall x \in U_0(L), \quad \langle L_0^* p, x \rangle = \langle p, Lx \rangle.$$

We define the domain of L_0^* by

$$(8) \quad V(L_0^*) = \{p \in V \text{ such that } L_0^* p \in U\}.$$

(This has a meaning since U is a dense subspace of $U_0(L)^*$.) We endow $V(L_0^*)$ with the graph norm

$$(9) \quad \|p\| = (\|p\|_V^2 + \|L_0^* p\|_U^2)^{\frac{1}{2}}.$$

Proposition 1. The domain $V(L_0^*)$ is a Hilbert space.

Proof. If p_n is a Cauchy sequence of $V(L_0^*)$ for the graph norm, then p_n is a Cauchy sequence of V and $L_0^* p_n$ a Cauchy sequence of U . Then p_n converges to p in V and $L_0^* p_n$ to q in U . Since $\langle L_0^* p_n, x \rangle = \langle p_n, Lx \rangle$ for any $x \in U_0(L)$, we deduce that $\langle q, x \rangle = \langle p, Lx \rangle = \langle L_0^* p, x \rangle$ for any $x \in U_0(L)$. Thus $L_0^* p = q \in U$. Hence $p \in V(L_0^*)$ and p_n converges to p . $\square$

Theorem 1. Suppose that (1), (2), (4) and (5) hold. Then there exists a unique operator $\beta^* \in \mathcal{L}(V(L_0^*), E^*)$ such that

$$(10) \quad \forall p \in V(L_0^*), \forall x \in U(L), \quad \langle L_0^* p, x \rangle - \langle p, Lx \rangle = \langle \beta^* p, \gamma x \rangle.$$

Proof. We begin by proving that $L^* - i^* L_0^*$ maps $V(L_0^*)$ into $U_0(L)^\perp$ where $i^* \in \mathcal{L}(U, U(L)^*)$ is the transpose of the canonical injection i from $U(L)$ into U . If $p \in V(L_0^*)$ and $x \in U_0(L)$, we obtain that

$$\langle L^* p - i^* L_0^* p, x \rangle = \langle p, Lx \rangle - \langle p, i L_0 x \rangle = \langle p, Lx - Lx \rangle = 0.$$

Since γ is a surjective operator from $U(L)$ onto E , it has a (continuous linear) right inverse σ . Hence $\sigma\gamma$ is a projector whose kernel is $U_0(L)$ and $\gamma^* \sigma^*$ is a

projector whose range is $U_0(L)^\perp = (\text{Ker } \sigma\gamma)^\perp$. We write

$$(11) \quad \beta^* = -\sigma^*(L^* - i^*L_0^*) \in \mathcal{L}(V(L_0^*), \mathcal{E}^*).$$

Since $L^*p - i^*L_0^*p$ belongs to $U_0(L)^\perp$, we can write

$$(12) \quad L^*p - i^*L_0^*p = \gamma^*\sigma^*(L^*p - i^*L_0^*p) = -\gamma^*\beta^*p.$$

Therefore, by applying $L^*p - i^*L_0^*p + \gamma^*\beta^*p = 0$ to $x \in U(L)$, we obtain formula (10). The operator β^* is unique. Indeed, if formula (10) holds, it can be written

$$(13) \quad L^*p - i^*L_0^*p = -\gamma^*\beta^*p \quad \text{for all } p \in V(L_0^*).$$

If σ is (any) right inverse of γ , we obtain, by applying σ^* to both sides of (13), that

$$\beta^*p = -\sigma^*(L^*p - i^*L_0^*p).$$

Definition 2. We shall say that formula (12) is an *(abstract) Green formula*.

14.3.2. Abstract problem of calculus of variations

Now consider a function

$$(14) \quad F : U \times V \times \mathcal{E} \rightarrow]-\infty, +\infty]$$

and its conjugate function F^* defined on $U \times V \times \mathcal{E}^*$ by

$$(15) \quad F^*(p, q, \pi) = \sup_{\substack{x \in U \\ y \in V \\ \xi \in \mathcal{E}}} [\langle q, x \rangle + \langle p, y \rangle + \langle \pi, \xi \rangle - F(x, y, \xi)].$$

Assume once and for all that (1), (2), (4) and (5) hold.

We introduce the minimization problems

$$(16) \quad v = \inf_{x \in U(L)} F(x, Lx, \gamma x)$$

and

$$(17) \quad v^* = \inf_{p \in V(L_0^*)} F^*(-L_0^*p, p, \beta^*p).$$

These two minimization problems are related by the formula

$$(18) \quad 0 \leq v + v^*.$$

To see this, observe that, since $x \in U(L)$ and $p \in V(L_0^*)$, the Green formula implies that

$$\begin{aligned} 0 &= \langle -L_0^* p, x \rangle + \langle p, Lx \rangle + \langle \beta^* p, \gamma x \rangle \\ &\leq F(x, Lx, \gamma x) + F^*(-L_0^* p, p, \beta^* p). \end{aligned}$$

Taking the infimum when x ranges over $U(L)$ and p over $V(L^*)$, we obtain (18).

Definition 3. We shall say that $\bar{p} \in V(L_0^*)$ is a *Lagrange multiplier* if and only if

$$(19) \quad \begin{cases} \text{(i)} & 0 = v + v^*, \\ \text{(ii)} & v^* = F^*(-L_0^* \bar{p}, \bar{p}, \beta^* \bar{p}). \end{cases}$$

Proposition 2. Suppose that F is a lower semi-continuous convex function. Then $\bar{p}$ is a Lagrange multiplier and $\bar{x}$ minimizes

$$x \rightarrow F(x, Lx, \gamma x) \text{ on } U(L)$$

if and only if

$$(20) \quad \{-L_0^* \bar{p}, \bar{p}, \beta^* \bar{p}\} \in \partial F(\bar{x}, L\bar{x}, \gamma \bar{x}).$$

If $F(x, y, \xi) = f(x) + g(y) + h(\xi)$ where f , g , and h are lower semi-continuous proper convex functions defined on U , V and E respectively, this relation becomes

$$(21) \quad \begin{cases} \text{(i)} & -L_0^* \bar{p} \in \partial f(\bar{x}) \text{ (or } \bar{x} \in \partial f^*(-L_0^* \bar{p})), \\ \text{(ii)} & \bar{p} \in \partial g(L\bar{x}) \text{ (or } L\bar{x} \in \partial g^*(\bar{p})), \\ \text{(iii)} & \beta^* \bar{p} \in \partial h(\gamma \bar{x}) \text{ (or } \gamma \bar{x} \in \partial h^*(\beta^* \bar{p})). \end{cases}$$

Proof. We have that $\{L_0^* \bar{p}, \bar{p}, \beta^* \bar{p}\}$ belongs to $\partial F(\bar{x}, L\bar{x}, \gamma \bar{x})$ if and only if

$$\begin{aligned} 0 &= \langle -L_0^* \bar{p}, \bar{x} \rangle + \langle \bar{p}, L\bar{x} \rangle + \langle \beta^* \bar{p}, \gamma \bar{x} \rangle \\ &= [F(\bar{x}, L\bar{x}, \gamma \bar{x}) + F^*(-L_0^* \bar{p}, \bar{p}, \beta^* \bar{p})]. \end{aligned}$$

Extremality relations (21) obviously follow from (20). $\square$

Remark. Denote by A and A^* the "boundary-value" operators

$$(22) \quad \begin{cases} \text{(i)} & A : x \in U(L) \mapsto Ax = \{Lx, \gamma x\} \in V \times E, \\ \text{(ii)} & A^* : p \in V(L_0^*) \mapsto A^* p = \{-L_0^* p, \beta^* p\} \in U \times E^*. \end{cases}$$

[Despite the notation used, A^* is not the transpose of A !]

Since relations (20) imply that

$$A^* \bar{p} \in \partial_x F(\bar{x}, L\bar{x}, \gamma \bar{x}); \quad p \in \partial_y F(\bar{x}, L\bar{x}, \gamma \bar{x}),$$

we see that $\bar{x}$ satisfies the equation

$$0 \in \Lambda^* \partial_y F(\bar{x}, L\bar{x}, \gamma\bar{x}) - \partial_{x,\xi} F(\bar{x}, L\bar{p}, \gamma\bar{x}).$$

Remark. When $F(x, y, z) = f(x) + g(y) + h(z)$, we can write the extremality relations (21) in the form

$$(23) \quad \begin{cases} \text{(i) } \bar{x} \in \partial f(-L_0^* \bar{p}) \cap \Lambda^{-1}[\partial g^*(\bar{p}) \times \partial h^*(\beta^* \bar{p})], \\ \text{(ii) } 0 \in \partial g^*(\bar{p}) \times \partial h^*(\beta^* \bar{p}) - \Lambda \partial f^*(-L_0^* \bar{p}). \end{cases}$$

In particular, if we assume that

$$f \text{ is differentiable on } -L_0^* \text{ Dom } g^*$$

we can write that

$$\bar{x} = Df^*(L_0^*(-\bar{p}))$$

where $\bar{p}$ is the solution of the variational inequalities

$$(24) \quad \begin{cases} \text{(i) } \langle LDf^* L_0^*(-\bar{p}), \bar{p} - p \rangle + g^*(\bar{p}) - g^*(p) \leq 0, \forall p \in V, \\ \text{(ii) } \langle \gamma Df^* L_0^*(-\bar{p}), \beta^* \bar{p} - \pi \rangle + h^*(\beta^* \bar{p}) - h^*(\pi) \leq 0, \forall \pi \in \mathcal{E}^* \end{cases}$$

Proof. Let $\bar{x} \in U(L)$ and $\bar{p} \in V(L_0^*)$ satisfy relations (21). Then $\Lambda \bar{x} = \{L\bar{x}, \gamma\bar{x}\} \in \partial g^*(\bar{p}) \times \partial h^*(\beta^* \bar{p})$ and $\Lambda \bar{x} \in \Lambda \partial f^*(-L_0^* \bar{p})$. Hence $0 = \Lambda \bar{x} - \Lambda \bar{x}$ belongs to $\partial g^*(\bar{p}) \times \partial h^*(\beta^* \bar{p}) - \Lambda \partial f^*(-L_0^* \bar{p})$.

Conversely, let $\bar{p}$ be a solution of (23(ii)).

There exists $\{\bar{y}, \bar{\xi}\} \in \partial g^*(\bar{p}) \times \partial h^*(\beta^* \bar{p})$ and $\bar{x} \in \partial f^*(-L_0^* \bar{p})$ such that $\bar{y} = L\bar{x}$ and $\bar{\xi} = \gamma\bar{x}$. Hence $-L_0^* \bar{p} \in \partial f(\bar{x})$, $L\bar{x} \in \partial g^*(\bar{p})$ and $\gamma\bar{x} \in \partial h^*(\beta^* \bar{p})$.

Finally, variational inequalities (24) make explicit the formulas $L\bar{x} \in \partial g^*(\bar{p})$ and $\gamma\bar{x} \in \partial h^*(\beta^* \bar{p})$ when $\bar{x} = Df^*(-L_0^* \bar{p})$.

14.3.3. The Hamiltonian system

Associate with the minimization problem (16), the Hamiltonian H mapping $U \times V \times \mathcal{E}^*$ into $]-\infty, +\infty]$. This is defined by

$$(25) \quad H(x, p, \pi) = \sup_{\substack{y \in V \\ \xi \in \mathcal{E}}} [\langle p, y \rangle + \langle \pi, \xi \rangle - F(x, y, \xi)].$$

Then, if the functions $\{y, \xi\} \rightarrow F(x, y, \xi)$ are convex and lower semi-continuous for every x , we verify that

$$(26) \quad F(x, y, \xi) = \sup_{\substack{p \in V \\ \pi \in \mathcal{E}^*}} [\langle p, y \rangle + \langle \pi, \xi \rangle - H(x, p, \pi)].$$

Furthermore, we also have the relation

$$(27) \quad F^*(q, p, \pi) = \sup_{x \in U} [\langle q, x \rangle + H(x, p, \pi)].$$

We now introduce the function l mapping $U(L) \times V \times \Xi^*$ into $]-\infty, +\infty]$ defined by

$$(28) \quad l(x, p, \pi) = \langle p, Lx \rangle + \langle \pi, \gamma x \rangle - H(x, p, \pi).$$

Proposition 3. *The function l defined on $U(L) \times V \times \Xi^*$ is a Lagrangian of the minimization problem $v = \inf_{x \in U(L)} F(x, Lx, \gamma x)$, i.e.*

$$(29) \quad F(x, Lx, \gamma x) = \sup_{\substack{p \in V \\ \pi \in \Xi^*}} l(x, p, \pi).$$

Proof. This follows immediately from relations (26). $\square$

Proposition 4. *Let F be a lower semi-continuous convex proper function and H its Hamiltonian. Then the extremality conditions*

$$(20) \quad \{-L_0^* \bar{p}, \bar{p}, \beta^* \bar{p}\} \in \partial F(\bar{x}, L\bar{x}, \gamma \bar{x})$$

are equivalent to the relations

$$(30) \quad \begin{cases} \text{(i)} \quad \Lambda \bar{x} = \{L\bar{x}, \gamma \bar{x}\} \in \partial_{p, \pi} H(\bar{x}, \bar{p}, \beta^* \bar{p}), \\ \text{(ii)} \quad L_0^* \bar{p} \in \bar{\partial}_x H(\bar{x}, \bar{p}, \beta^* \bar{p}). \end{cases}$$

Proof. This is analogous to the proof of Proposition 14.1.3. Suppose that $\{-L_0^* \bar{p}, \bar{p}, \beta^* \bar{p}\} \in \partial F(\bar{x}, L\bar{x}, \gamma \bar{x})$. Then,

$$\begin{aligned} F(\bar{x}, L\bar{x}, \gamma \bar{x}) - F(x, y, \xi) &\leq \langle -L_0^* \bar{p}, \bar{x} - x \rangle + \langle \bar{p}, L\bar{x} - y \rangle + \\ &\quad + \langle \beta^* \bar{p}, \gamma \bar{x} - \xi \rangle. \end{aligned}$$

Taking $x = \bar{x}$, we deduce that $\Lambda \bar{x} = \{L\bar{x}, \gamma \bar{x}\}$ belongs to $\partial_{p, \pi} H(\bar{x}, \bar{p}, \beta^* \bar{p})$, since $H(\bar{x}, \cdot, \cdot)$ is the conjugate function of $F(\bar{x}, \cdot, \cdot)$. This amounts to writing that $\langle \bar{p}, L\bar{x} \rangle + \langle \beta^* \bar{p}, \gamma \bar{x} \rangle = F(\bar{x}, L\bar{x}, \gamma \bar{x}) + H(\bar{x}, \bar{p}, \beta^* \bar{p})$. Hence we can write

$$\langle \bar{p}, y \rangle + \langle \beta^* \bar{p}, \xi \rangle - F(x, y, \xi) \leq \langle L_0^* \bar{p}, x - \bar{x} \rangle + H(\bar{x}, \bar{p}, \beta^* \bar{p}).$$

Taking the supremum over $\{y, \xi\}$, we deduce that $L_0^* \bar{p} \in \bar{\partial}_x H(\bar{x}, \bar{p}, \beta^* \bar{p})$.

Conversely, suppose that (30) holds.

Since $\Lambda \bar{x} \in \partial_{p, \pi} H(\bar{x}, \bar{p}, \beta^* \bar{p})$, we have that

$$(31) \quad \langle \bar{p}, L\bar{x} \rangle + \langle \beta^* \bar{p}, \gamma \bar{x} \rangle = F(\bar{x}, L\bar{x}, \gamma \bar{x}) + H(\bar{x}, \bar{p}, \beta^* \bar{p}).$$

By definition of $H(x, \bar{p}, \beta^* \bar{p})$, we have that

$$(32) \quad \langle \bar{p}, y \rangle + \langle \beta^* \bar{p}, \xi \rangle \leq F(x, y, \xi) + H(x, \bar{p}, \beta^* \bar{p}).$$

Since $L^* \bar{p} \in \bar{\partial}_x H(\bar{x}, \bar{p}, \beta^* \bar{p})$, it follows that

$$(33) \quad H(x, \bar{p}, \beta^* \bar{p}) - H(\bar{x}, \bar{p}, \beta^* \bar{p}) \leq \langle L^* \bar{p}, x - \bar{x} \rangle.$$

We deduce from the three inequalities (31), (32) and (33) that

$$\begin{aligned} F(\bar{x}, L\bar{x}, \gamma\bar{x}) - F(x, y, \xi) &\leq \langle -L^* \bar{p}, \bar{x} - x \rangle + \langle \bar{p}, L\bar{x} - y \rangle + \\ &\quad + \langle \beta^* \bar{p}, \gamma\bar{x} - \xi \rangle, \end{aligned}$$

which means that $\{-L^* \bar{p}, \bar{p}, \gamma^* \bar{p}\} \in \partial F(\bar{x}, L\bar{x}, g\bar{x})$.

Remark. In the case when

$$(34) \quad F(x, y, \xi) = F_0(x, y) + h(\xi),$$

the Hamiltonian can be written

$$(35) \quad H(x, p, \pi) = H_0(x, p) - h^*(\xi).$$

The system (20) can be written

$$(36) \quad \begin{cases} \text{(i)} \quad \{-L^* \bar{p}, \bar{p}\} \in \partial F_0(\bar{x}, L\bar{x}), \\ \text{(ii)} \quad \beta^* \bar{p} \in \partial h(\gamma\bar{x}). \end{cases}$$

The conditions (20(i)) are called the *Euler-Lagrange conditions* and conditions (20(ii)) the *transversality conditions*. The system (30) can be written

$$(37) \quad \begin{cases} \text{(i)} \quad L\bar{x} \in \partial_p H_0(x, p) \quad \text{and} \quad L^* \bar{p} \in \bar{\partial}_x H_0(\bar{x}, \bar{p}), \\ \text{(ii)} \quad \gamma\bar{x} \in \partial g^*(\beta^* \bar{p}). \end{cases}$$

Conditions (37(i)) are called *Hamiltonian conditions*.

14.3.4. Lagrangian of a problem of calculus of variations

We have to make explicit the dual problem associated with this Lagrangian and check that it is the same as (17).

Proposition 5. Suppose that (1), (2), (4), (5) and

$$(38) \quad \forall q, p, \pi, \quad F^*(q, p, \pi) = \sup_{x \in U(L)} [\langle q, x \rangle + H(x, p, \pi)]$$

hold. Suppose also that

$$(39) \quad \exists \tilde{x}, \tilde{y}, \tilde{\xi} \quad \text{such that } x \rightarrow F(\tilde{x}, \tilde{y}, \tilde{\xi}) \text{ is continuous at } \tilde{x} \text{ (on } U).$$

Then

$$-\inf_{x \in U(L)} l(x, p, \pi) = \begin{cases} F^*(-L_0^*p, p, \beta^*p) & \text{if } p \in V(L_0^*) \text{ and } \pi = \beta^*p, \\ +\infty & \text{otherwise.} \end{cases}$$

Proof. We can write

$$-\inf_{x \in U(L)} l(x, p, \pi) = \sup_{x \in U(L)} [H(x, p, \pi) - \langle p, Lx \rangle - \langle \pi, \gamma x \rangle].$$

We show that this quantity is infinite if $p \notin V(L_0^*)$, i.e. if $-L_0^*p \notin U$. This can be written

$$(40) \quad \| -L_0^*p \|_U = \sup_{\substack{x \in U_0(L) \\ \|x\|_U=1}} |\langle -L_0^*p, x \rangle| = +\infty.$$

Therefore, there exists a sequence $x_n \in U_0(L)$, such that $\|x_n\|_U = 1$ and $-\langle L^*p, x_n \rangle \rightarrow +\infty$. Even if it means replacing $F(x, y, \xi)$ by $F(x - \tilde{x}, y, \xi)$ and x_n by λx_n , we can assume that there exists $\{\tilde{y}, \tilde{\xi}\}$ such that $F(\cdot, \tilde{y}, \tilde{\xi})$ is bounded above on the unit ball of U . Hence, since $x_n \in U_0(L)$,

$$\begin{aligned} & \sup_{x \in U(L)} [H(x, p, \pi) - \langle p, Lx \rangle - \langle \pi, \gamma x \rangle] \\ & \geq \sup_n [H(x_n, p, \pi) - \langle L_0^*p, x_n \rangle] \\ & \geq \sup_n [\langle p, \tilde{y} \rangle + \langle \pi, \tilde{\xi} \rangle - F(x_n, \tilde{y}, \tilde{\xi}) - \langle L_0^*p, x_n \rangle] \\ & \geq a + \sup_n [\langle -L_0^*p, x_n \rangle] = +\infty \end{aligned}$$

where $a = \langle p, \tilde{y} \rangle + \langle \pi, \tilde{\xi} \rangle - \sup_{\|x\| \leq 1} F(x, \tilde{y}, \tilde{\xi})$.

Suppose that $p \in V(L_0^*)$ and that $\beta^*p \neq \pi$. Since $p \in V(L_0^*)$, we can apply the Green formula. Hence $-l(x, p, \pi) = H(x, p, \pi) - \langle L_0^*p, x \rangle + \langle \beta^*p - \pi, \gamma x \rangle$.

Since γ maps $U(L)$ onto E , we deduce that

$$\sup_{x \in U(L)} -l(x, p, \pi) = +\infty \quad \text{whenever } \pi^*p \neq \pi.$$

Suppose that $p \in V(L_0^*)$ and $\beta^*p = \pi$. Then

$$\begin{aligned} \sup_{x \in U(L)} -l(x, p, \pi) &= \sup_{x \in U(L)} [\langle -L_0^*p, x \rangle + H(x, p, \beta^*p)] \\ &= F^*(-L_0^*p, p, \beta^*p) \quad \text{by assumption (38). } \square \end{aligned}$$

14.3.5. Existence of a Lagrange multiplier

Consider the function v defined on $V \times E$ by

$$(41) \quad v(y, \xi) = \inf_{x \in U(L)} F(x, Lx + y, \gamma x + \xi).$$

We assume that

(42) the domain of F is $R \times Z$ where $R \subset U$, $Z \subset V \times E$.

If A is the operator defined by $Ax = \{Lx, \gamma x\}$, then

(43) $Z - A(R)$ is the domain of v .

It is clear that

$$(44) \quad v(y, \xi) = \sup_{p \in V, \pi \in E^*} [\langle p, y \rangle + \langle \pi, \xi \rangle + l(x, p, \pi)]$$

since $\langle p, y \rangle + \langle \pi, \xi \rangle + l(x, p, \pi) = \langle p, Lx + y \rangle + \langle \pi, \gamma x + \xi \rangle - H(x, p, \pi)$.

Proposition 6. Suppose that (38), (39) and (42) hold. Then, if p is any Lagrange multiplier, we have

$$(45) \quad \bar{p} \in \partial_p v(y, \xi), \quad \beta^* \bar{p} \in \partial_\xi v(y, \xi).$$

Proof. By Theorem 1.6, we know that $\partial v(y, \xi)$ is the set of Lagrange multipliers. By Proposition 5 we know that if $\{p, \pi\}$ is a Lagrange multiplier, then $\pi = \beta^* p$. Hence (45) holds. $\square$

Theorem 2. Suppose that

(46) F is a convex lower semi-continuous function on $U \times V \times E$, whose domain is $R \times Z$ where $R \subset U$ and $Z \subset V \times E$

and that

(47) $\forall \{p, \pi\}, \quad x \rightarrow H(x, p, \pi)$ is upper semi-continuous on U .

If

$$(48) \quad 0 \in \text{Int}(Z - A(R)),$$

then there exists a Lagrange multiplier.

Proof. This is analogous to the proof of Theorem 1.2. We apply the minisup theorem Theorem 13.1.2 to the function $\phi(p, \pi; x) = -l(x, p, \pi)$, whose conjugate function with respect to $\{p, \pi\}$ is $\{y, \xi\} \rightarrow F(x, Lx + y, \gamma x + \xi)$. $\square$

14.3.6. Example: The Dirichlet variational problem

We now consider the most famous classical example of the calculus of variations. This is the same as Bolza's problem but set in n dimensions. Let Ω be a smooth bounded open subset of $\mathbb{R}^n$ with boundary Γ .

(a) The operator $L = \text{grad}$ and $L_0^* = -\text{div}$

The space

$$(49) \quad H^1(\Omega) = \{x \in L^2(\Omega) \text{ such that } D_i x \in L^2(\Omega) \text{ for } i = 1, \dots, n\}$$

is called the Sobolev space. The derivative $D_i x = \partial x(\omega) / \partial \omega_i$ is to be understood as a derivative in the distributional sense, i.e. it is defined by

$$(50) \quad \int_{\Omega} (D_i x)(\omega) y(\omega) d\omega = - \int_{\Omega} x(\omega) D_i y(\omega) d\omega$$

for all infinitely differentiable functions y with compact support in Ω . With the graph norm

$$\|x\| = \left(|x|^2 + \sum_{i=1}^n |D_i x|^2 \right)^{\frac{1}{2}} \quad \text{where } |y| = \left(\int_{\Omega} |y(\omega)|^2 d\omega \right)^{\frac{1}{2}}$$

$H^1(\Omega)$ is a Hilbert space.

If we set

$$(51) \quad U = L^2(\Omega), \quad V = L^2(\Omega)^n,$$

we see that

$$(52) \quad H^1(\Omega) \text{ is the domain of the operator } L = \text{grad}$$

defined by

$$Lx = \text{grad } x = \{D_1 x, \dots, D_n x\}.$$

One can prove the following fundamental trace theorem.

If Ω is smooth enough, there exists a Hilbert space $\mathcal{E}$ of functions ξ defined on Γ (called $H^{\frac{1}{2}}(\Gamma)$) such that the trace operator γ defined by $\gamma x = x|_{\Gamma}$ satisfies

$$(53) \quad \begin{cases} \text{(i) } \gamma \in \mathcal{L}(H^1(\Omega), H^{\frac{1}{2}}(\Gamma)) \text{ is surjective,} \\ \text{(ii) } H_0^1(\Omega) = \text{Ker } \gamma \text{ is dense in } L^2(\Omega) \end{cases}$$

and

$$(54) \quad H_0^1(\Omega) \text{ is the closure in } H^1(\Omega) \text{ of the space of infinitely differentiable functions with support in } \Omega.$$

Hence, we deduce from (50) that the transpose L_0^* of the restriction of grad to $H_0^1(\Omega)$ is defined by

$$(55) \quad L_0^* p = -\text{div } p = - \sum_{i=1}^n D_i p_i.$$

Let

$$(56) \quad D^1(\Omega) = \{p \in L^2(\Omega)^n \text{ such that } \operatorname{div} p \in L^2(\Omega)\}$$

be the domain of div , supplied with the graph norm.

Theorem 1 implies the existence of a unique operator $\beta^* \in \mathcal{L}(D^1(\Omega), H^{-\frac{1}{2}}(\Gamma))$, where we set $H^{\frac{1}{2}}(\Gamma)^* = H^{-\frac{1}{2}}(\Gamma)$, such that

$$(57) \quad \forall p \in D^1(\Omega), \forall x \in H^1(\Omega), \quad \int_{\Omega} (-\operatorname{div} p)x - \int_{\Omega} \langle p, \operatorname{grad} x \rangle = \int_{\Gamma} x \tilde{\beta}^* p.$$

Comparing this with the classical formula, we can set

$$(58) \quad \beta^* p = -\langle p, \tilde{n} \rangle \quad \text{where } \tilde{n} \text{ denotes the normal to } \Gamma.$$

[For $n = 1$ and $\Omega =]0, 1[$, we have $D^1(\Omega) = H^1(\Omega)$,

$$(59) \quad \mathcal{E} = \mathbf{R}^2, \quad \mathcal{E}^* = \mathbf{R}^2, \quad \gamma x = \{x(0), x(1)\}, \quad \beta^* p = \{p(0), -p(1)\}$$

and (57) is nothing other than the formula for integration by parts.]

(b) *The Dirichlet problem*

Let suppose that

$$(60) \quad F_0 : L^2(\Omega) \times L^2(\Omega)^n \rightarrow]-\infty, +\infty] \text{ is a lower semi-continuous, convex function}$$

whose conjugate function is denoted by F_0^* . If $\xi \in H^{\frac{1}{2}}(\Gamma)$ is given, we define the Dirichlet variational problem to be

$$(61) \quad v = \inf_{\substack{x \in H^1(\Omega) \\ \gamma x = \xi}} F_0(x, \operatorname{grad} x) = \inf_{x \in H^1(\Omega)} [F_0(x, \operatorname{grad} x) + \psi(\xi; \gamma x)]$$

and its dual problem to be

$$(62) \quad v^* = \inf_{p \in D^1(\Omega)} \left[F_0^*(\operatorname{div} p, p) - \int_{\Gamma} \langle p, \tilde{n} \rangle \xi \right]$$

which is a Neumann variational problem.

The Euler-Lagrange system can be written

$$(63) \quad \{\operatorname{div} \bar{p}, \bar{p}\} \in \partial F_0(\bar{x}, \operatorname{grad} \bar{x}); \quad \gamma \bar{x} = \xi.$$

Example. In the case where

$$(64) \quad F_0(x, y) = \frac{1}{2} \left(|x|^2 + \sum_{i=1}^n |y_i|^2 \right),$$

this system becomes

$$\operatorname{div} \bar{p} = \bar{x} \quad \text{and} \quad \bar{p} = \operatorname{grad} \bar{x},$$

i.e.

$$\bar{x} = \operatorname{div} \operatorname{grad} \bar{x} = \Delta x.$$

The Euler–Lagrange system is then

$$(65) \quad \begin{cases} -\Delta \bar{x} + \bar{x} = 0, \\ \gamma \bar{x} = \xi. \end{cases}$$

The minimum $\bar{x}$ is the solution of the Dirichlet problem for the Laplacian.

Let H_0 be the Hamiltonian defined by

$$(66) \quad H_0(x, p) = \sup_{y \in L^2(\Omega)^n} \left(\sum_{i=1}^n \int_{\Omega} p_i y_i \, d\omega - F_0(x, y) \right).$$

Then the Hamiltonian system can be written

$$(67) \quad \begin{cases} \text{(i)} \quad \operatorname{grad} \bar{x} \in \partial_p H_0(\bar{x}, \bar{p}), \\ \text{(ii)} \quad -\operatorname{div} \bar{p} \in \partial_x H_0(\bar{x}, \bar{p}), \\ \text{(iii)} \quad \gamma \bar{x} = \xi. \end{cases}$$

Remark. Assumptions (38) and (39) are, of course, satisfied when the functions $x \rightarrow F_0(x, y)$ are continuous with respect to x . We refer to the book of Ekeland and Temam and to a note of Berliocchi and Lasry for minimal assumptions. In the case when $n = 1$, the comprehensive study of the Bolza problem is due to Rockafellar.

Remark. Note that we can deduce from a measurable selection theorem that if

$$(68) \quad F_0(x, y) = \int_{\Omega} \phi(\omega, x(\omega), y(\omega)) \, d\omega$$

where ϕ is a Carathéodory function on $\Omega \times \mathbf{R}^n \times \mathbf{R}^n$, we can write

$$(69) \quad F_0^*(p, q) = \int_{\Omega} \phi^*(\omega, p(\omega), q(\omega)) \, d\omega$$

and

$$(70) \quad H_0(x, p) = \int_{\Omega} \psi(\omega, x(\omega), q(\omega)) \, d\omega$$

where, for almost all $\omega \in \Omega$,

$$\phi^*(\omega, r, s) = \sup_{\substack{u \in \mathbb{R} \\ v \in \mathbb{R}^n}} [r \cdot u + \langle s, v \rangle - \phi(\omega, u, v)]$$

and

$$\psi(\omega; x, s) = \sup_{v \in \mathbb{R}^n} [\langle s, v \rangle - \phi(\omega, u, v)].$$

[See the book of Ekeland and Temam (1974, Theorems VIII-2, p. 220 and Proposition IX-2-1, p. 251).]

Since

$$\begin{aligned} \langle q, x \rangle + \langle p, y \rangle - F_0(x, y) - F_0(p, q) &= \\ &= \int_{\Omega} [\langle q(\omega), x(\omega) \rangle + \langle p(\omega), y(\omega) \rangle - \phi(\omega, x(\omega), y(\omega)) \\ &\quad - \phi^*(\omega, q(\omega), p(\omega))] d\omega = 0 \end{aligned}$$

if and only if the integrand is equal to 0 (since the integrand is always non-positive), we deduce that $\{q, p\} \in F(x, y)$ if and only if, for almost all ω , $\{q(\omega), p(\omega)\} \in \partial\phi(\omega, x(\omega), y(\omega))$. This allows us to write the global relations (63) and (67) locally, i.e. for almost $\omega \in \Omega$,

$$(63 \text{ bis}) \quad \{\operatorname{div} \bar{p}(\omega), \bar{p}(\omega)\} \in \partial\phi(\omega, \bar{x}(\omega), \operatorname{grad} \bar{x}(\omega))$$

and

$$(67 \text{ bis}) \quad \begin{cases} \text{(i) } \operatorname{grad} \bar{x}(\omega) \in \partial_s \psi(\omega, \bar{x}(\omega), \bar{p}(\omega)), \\ \text{(ii) } -\operatorname{div} \bar{p}(\omega) \in \partial_u \psi(\omega, \bar{x}(\omega), \bar{p}(\omega)). \end{cases}$$

14.3.7. The maximum principle for optimal control problems

It is possible to regard an optimal control problem as a particular case of a problem in the calculus of variations. However, we prefer to outline an analogous study in the framework of Control Theory. Consider

$$(71) \quad \begin{cases} \text{a Hilbert space } W \text{ (of "controls")}, \\ \text{a continuous linear operator } M \in \mathcal{L}(W, V) \end{cases}$$

and the constraint

$$Lx = Mu \quad \text{where } x \in U(L), \quad u \in W$$

relating a "control" u to a "state" x of the system. Let

$$(72) \quad F : U \times W \times \mathcal{E} \rightarrow]-\infty, +\infty] \text{ be a lower semi-continuous, convex function.}$$

We consider the "optimal control" problem

$$(73) \quad v = \inf_{Lx = Mu} F(x, u, \gamma x).$$

In order to construct the dual problem, we introduce the "Hamiltonian" H defined by

$$(74) \quad H(x, r, \pi) = \sup_{\substack{u \in W \\ \xi \in \Xi}} [\langle r, u \rangle + \langle \pi, \xi \rangle - F(x, u, \xi)].$$

Then, we can write

$$(75) \quad F(x, u, \xi) = \sup_{\substack{r \in W^* \\ \pi \in \Xi^*}} [\langle r, u \rangle + \langle \pi, \xi \rangle - H(x, r, \pi)].$$

We construct the Lagrangian of the problem defined by

$$(76) \quad l(x, u; p, r, \pi) = \langle p, Lx \rangle + \langle r - M^*p, u \rangle + \langle \pi, \gamma x \rangle - H(x, r, \pi).$$

It is clearly a Lagrangian since

$$(77) \quad \sup_{p, r, \pi} l(x, u, p, r, \pi) = \begin{cases} F(x, u, \gamma x) & \text{if } Lx = Mu, \\ +\infty & \text{if } Lx \neq Mu. \end{cases}$$

Proposition 7. Suppose that (75) holds, that

$$(78) \quad \forall q, r, \pi, F^*(q, r, \pi) = \sup_{x \in U(L)} [\langle q, x \rangle + H(x, r, \pi)]$$

and

$$(79) \quad \exists \tilde{x}, \tilde{u}, \tilde{\xi} \text{ such that } x \rightarrow F(x, \tilde{u}, \tilde{\xi}) \text{ is continuous on } U \text{ at } \tilde{x}.$$

Then the dual problem is defined by

$$(80) \quad \inf_{\substack{x \in U(L) \\ u \in W}} l(x, u, p, r, \pi) = \begin{cases} F^*(-L_0^*p, M^*p, \beta^*p) & \text{if } p \in V(L_0^*), \\ r = M^*p, \pi = \beta^*p, \\ +\infty & \text{otherwise.} \end{cases}$$

Proof. This is analogous to the proof of Proposition 5 and is left as an exercise. $\square$

The above Lagrangian is associated with the following perturbations:

$$(81) \quad v(y, k, \xi) = \inf_{Lx = Mu - y} F(x, u + k, \gamma x + \xi).$$

Proposition 8. A Lagrange multiplier $\bar{p}$ exists whenever $0 \in \text{Int Dom } v$ and $x \rightarrow H(x, r, \pi)$ is upper semi-continuous on U .

Let $\bar{p}$ be a Lagrange multiplier. It is easy to check that $\{\bar{x}, \bar{u}\}$ is an optimal solution if and only if

$$(82) \quad \begin{cases} \text{(i)} & L\bar{x} = M\bar{u}, \\ \text{(ii)} & \{-L_0^*\bar{p}, M^*\bar{p}, \beta^*\bar{p}\} \in \partial F(\bar{x}, \bar{u}, \gamma\bar{x}). \end{cases}$$

By a proof analogous to the proof of Proposition 4, we deduce that conditions (82(ii)) are equivalent to

$$(83) \quad \begin{cases} \text{(i)} & L_0^*\bar{p} \in \bar{\partial}_x H(\bar{x}, M^*\bar{p}, \beta^*\bar{p}), \\ \text{(ii)} & \{\bar{u}, \gamma\bar{x}\} \in \partial_{r, \xi} H(\bar{x}, M^*\bar{p}, \beta^*\bar{p}). \end{cases}$$

In the case where

$$(84) \quad F(x, u, \xi) = F_0(x, u) + h(\xi)$$

and where $H_0(x, r) = \sup_u [\langle r, u \rangle - F_0(x, u)]$, these relations can be written

$$(85) \quad \begin{cases} \text{(i)} & L_0^*\bar{p} \in \bar{\partial}_x H_0(\bar{x}, M^*\bar{p}), \\ \text{(ii)} & \bar{u} \in \partial_r H_0(\bar{x}, M^*\bar{p}), \\ \text{(iii)} & \beta^*\bar{p} \in \partial h(\gamma\bar{x}). \end{cases}$$

Relation (85(i)) is called the “adjoint equation”, relation (85(ii)) is the famous “maximum principle” and relations (85(iii)) are called “transversality conditions”.

Notice that (85(ii)) can be reformulated as follows:

$$(86) \quad \bar{u} \text{ maximizes } \langle \bar{p}, Mu \rangle - F_0(\bar{x}, u).$$

Example. Consider the case when $\Omega =]t, T[$,

$$(87) \quad U = L^2(\Omega, \mathbb{R}^n), \quad W = L^2(\Omega, \mathbb{R}^p), \quad E = \mathbb{R}^n \times \mathbb{R}^n$$

and

$$(88) \quad \begin{aligned} U(L) &= H^1(\Omega), \quad Lx(\omega) = Dx(\omega) - A(\omega)x(\omega), \\ (Mu)(\omega) &= M(\omega)u(\omega), \quad \gamma x = \{x(t), x(T)\}, \quad \beta^*p = \{-p(t), p(T)\} \end{aligned}$$

where

$$(89) \quad A(\cdot) \in L^\infty(\Omega; \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)); \quad M(\cdot) \in L^\infty(\Omega; \mathcal{L}(\mathbb{R}^p, \mathbb{R}^n)).$$

Let $F_0: L^2(\Omega, \mathbb{R}^n) \times L^2(\Omega, \mathbb{R}^p) \rightarrow]-\infty, +\infty]$ be a convex lower semi-continuous proper function and C be a subset of $\mathbb{R}^n \times \mathbb{R}^n$.

We consider the optimal control problem

$$(90) \quad \begin{aligned} v &= \inf F_0(x, u) \text{ under the constraint} \\ Dx(\omega) - A(\omega)x(\omega) &= M(\omega)u(\omega) \text{ almost everywhere and} \\ \{x(t), x(T)\} &\in C. \end{aligned}$$

Under the assumptions (78) and (79), the dual problem is

$$(91) \quad v^* = \inf_{p \in H^1(\Omega)} [F^*(Dp - A^*p; M^*p) + \sigma^*(C; \{p(t), -p(T)\})].$$

The Hamiltonian system becomes:

$$(92) \quad \begin{cases} \text{(i)} & -D\bar{p} - A^*\bar{p} \in \bar{\partial}_x H_0(\bar{x}, M^*\bar{p}), \\ \text{(ii)} & \bar{u} \in \partial_r H_0(\bar{x}, M^*\bar{p}), \\ \text{(iii)} & \{\bar{p}(t), -\bar{p}(T)\} \text{ is normal to } C \text{ at } \{\bar{x}(t), \bar{x}(T)\}. \end{cases}$$

The perturbation function is equal to

$$(93) \quad v(y, k, \xi_t, \xi_T) = \inf_{\substack{Dx(\omega) - A(\omega)x(\omega) \\ = M(\omega)u(\omega) + M(\omega)k(\omega) + y(\omega) \\ x(\omega) \in C - \{\xi_t, \xi_T\}}} F_0(x, u).$$

Hence, we deduce that

$$(94) \quad \begin{cases} \text{(i)} & \bar{p}(\cdot) \in \partial_y v(0, 0, 0, 0), \\ \text{(ii)} & M^*(\cdot)\bar{p}(\cdot) \in \partial_k(0, 0, 0, 0) \\ \text{(iii)} & \bar{p}(t) \in \partial_{\xi_t} v(0, 0, 0, 0), \\ \text{(iv)} & -\bar{p}(T) \in \partial_{\xi_T} v(0, 0, 0, 0). \end{cases}$$

In particular, if we associate with any ξ the optimal control problem

$$(95) \quad v(t, \xi) = \inf_{\substack{Dx(\omega) - A(\omega)x(\omega) = M(\omega)u(\omega), \\ \omega \in]t, T[, \quad T[a, e], \\ \{x(t), x(T)\} \in C - \{\xi, 0\}}} F_0(x, u)$$

the solutions $\bar{p}(\xi, \cdot)$ of its dual problem satisfy the condition

$$(96) \quad \bar{p}(\xi, t) \in -\partial_{\xi} v(t, \xi).$$

Consider the case where there is no state constraint, i.e. where

$$(97) \quad \begin{cases} \text{(i)} & F_0 : U \times K \rightarrow \mathbf{R}, \text{ where } K \subset W, K \text{ is convex, } \neq \emptyset \\ \text{(ii)} & C = C_0 \times \mathbf{R}^n \text{ where } C_0 \subset \mathbf{R}^n, C_0 \text{ is convex, } \neq \emptyset \end{cases}$$

Then, if we assume (89), (97) and (72), (78), and (79) (with F_0 replacing F) we may deduce the existence of a function $\bar{p}$ satisfying (92). [If we set $R = U(L) \times K$, $Y = \{0 \times C_0\}$ and $\Lambda(x, u) = \{Dx - Ax - Mu, x(0)\}$, the constraint qualification $0 \in \text{Int}(Y - \Lambda(R))$ is satisfied since clearly, Λ maps $U(L) \times K$ onto $L^2(0, T; \mathbf{R}^n) \times \mathbf{R}^n$.]

Example (the quadratic case). Suppose that F_0 and h are quadratic functionals:

$$(98) \quad \begin{cases} \text{(i)} & F_0(x, u) = \frac{1}{2} \int_t^T [j(s, x(s) - x_0(s))^2 + k(s, u(s) - u_0(s))^2] ds, \\ \text{(ii)} & h(\xi) = \frac{1}{2} g(\xi - \xi_T)^2 \end{cases}$$

where we set

$$(99) \quad \begin{cases} \text{(i)} & j(s, \xi)^2 = \langle J(s)\xi, \xi \rangle, \quad g(\xi)^2 = \langle G\xi, \xi \rangle, \\ \text{(ii)} & k(s, \eta)^2 = \langle K(s)\eta, \eta \rangle \end{cases}$$

where

$$(100) \quad J(s) \in \mathcal{L}(\mathbf{R}^n, \mathbf{R}^n), \quad G \in \mathcal{L}(\mathbf{R}^n, \mathbf{R}^n) \text{ and } K(s) \in \mathcal{L}(\mathbf{R}^p, \mathbf{R}^p) \text{ are self-transpose, positive semidefinite matrices such that the functions } s \rightarrow J(s) \text{ and } s \rightarrow K(s) \text{ belong to } L^\infty(\Omega, \mathcal{L}(\mathbf{R}^n, \mathbf{R}^n)) \text{ and } L^\infty(\Omega, \mathcal{L}(\mathbf{R}^p, \mathbf{R}^p)) \text{ respectively.}$$

Consider the differential equations describing the state of the system:

$$(101) \quad \begin{cases} \text{(i)} & \dot{x}(s) = A(s)x(s) + M(s)u(s), \\ \text{(ii)} & x(t) = \xi. \end{cases}$$

The adjoint system (92(i) and (iii)) becomes

$$(102) \quad \begin{cases} \text{(i)} & -\dot{p}(s) = A^*(s)p(s) - J(s)(x(s) - x_0(s)), \\ \text{(ii)} & p(T) = -G(x(T) - \xi_T), \end{cases}$$

since the Hamiltonian can be written

$$\begin{aligned} H(x, r, \pi) = & -\frac{1}{2} \int_t^T j(s, x(s) - x_0(s))^2 ds \\ & + \frac{1}{2} \int_t^T [k_*(s, r(s))^2 + \langle r(s), u_0(s) \rangle] ds + \frac{1}{2} g_*(\pi)^2 + \langle \pi, \xi_T \rangle \end{aligned}$$

where we set

$$(103) \quad k_*(s, r)^2 = \langle K(s)^{-1}r, r \rangle, \quad g_*(\pi)^2 = \langle G^{-1}\pi, \pi \rangle.$$

The maximum principle states that u maximizes the function $u \mapsto \langle p, Mu \rangle - F_0(x, u)$ over K , i.e.

$$(104) \quad u(s) = u_0(s) + K(s)^{-1} M^*(s) p(s).$$

On writing $u(s)$ in the system (101), the control problem reduces to solving the two-point boundary value problem for the system

$$(105) \quad \begin{cases} \text{(i)} & \dot{x}(s) = A(s)x(s) + N(s)p(s) + M(s)u_0(s), \\ \text{(ii)} & -\dot{p}(s) = A^*(s)p(s) - J(s)(x(s) - x_0(s)), \\ \text{(iii)} & x(t) = \xi, p(T) = -G(x(T) - \xi_T) \end{cases}$$

where we set

$$(106) \quad N(s) = M(s)K(s)^{-1}M^*(s).$$

14.4. Optimal control and impulsive control problems

Consider the optimal control problem

$$v(t, \xi) = \inf_{\substack{u(\cdot) \\ \dot{x}(s) = \varphi(s, x(s), u(s)) \\ x(t) = \xi \\ \{u(s)\} \in K}} \left[\int_t^T \varphi(s, x(s), u(s)) \, ds + h(x(T)) \right].$$

The approach of dynamic programming is to take the initial time t and the initial state ξ of the system as parameters and to show that the “optimal performance function” $\{t, \xi\} \rightarrow v(t, \xi)$ is a solution of a first order partial differential equation

$$\begin{cases} \frac{\partial v}{\partial t}(t, \xi) = \max_{u \in K} \left[- \left\langle \frac{\partial v}{\partial \xi}(t, \xi), \varphi(t, \xi, u) \right\rangle - \varphi(t, \xi, u) \right], \\ v(T, \xi) = h(\xi) \end{cases}$$

by assuming a lot of smoothness. This equation is the Hamilton–Jacobi–Bellman equation.

We shall prove that knowledge of the performance function allows one to “solve” the optimal control problem in the following way. We shall assume that for any fixed $\{t, \xi\}$, there is a unique control $u = D^*(t, \xi)$ achieving the maximum of the function

$$u \rightarrow \left\langle - \frac{\partial v}{\partial \xi}(t, \xi), \varphi(t, \xi, u) \right\rangle - \varphi(t, \xi, u)$$

over K . This decision rule D^* is called a “feed-back” map, which yields the “optimal closed loop control” $\bar{u}(t)$ at each time t via the formula

$$\bar{u}(t) = D^*(t, \bar{x}(t))$$

where $\bar{x}(\cdot)$ is the optimal state. We shall also prove the following fundamental principle of optimality.

An optimal pair $\{\bar{x}(\cdot), \bar{u}(\cdot)\}$ has the property that, whatever the initial state ξ at time t is, the remaining states $\bar{x}(s)$ must constitute an optimal state when $s \geq \tau$ with regard to the state $\bar{x}(\tau)$ resulting from the first initial state. We will give the Hamilton–Jacobi–Bellman equation explicitly in the case of quadratic control problems. In this case, the performance function $v(t, \xi)$ is quadratic with respect to ξ : $v(t, \xi) = \frac{1}{2} \langle P(t)\xi, \xi \rangle + \langle r(t), \xi \rangle + \sigma(t)$. We show that $P(\cdot)$, $r(\cdot)$ and $\sigma(\cdot)$ are solutions of a non-linear system of differential equations called the “Riccati system”. Knowledge of $P(\cdot)$ and $r(\cdot)$ enables one to construct the “closed loop control”.

In summary, we have two methods for solving optimal control problems. The maximum principle is particularly well-suited when the performance function is not smooth (which can happen) and leads to a two-point boundary-value problem for a system of two differential equations. The principle of optimality assumes some smoothness and leads to a non-linear partial differential equation.

Next, we use the dynamic programming approach to solve optimal stopping time problems of the type

$$v(t, \xi) = \inf_{\substack{t \leq \tau \leq T \\ \dot{x}(s) = \psi(s, x(s)) \\ x(t) = \xi}} \left[\int_t^{\tau} \varphi(s, x(s)) ds + h(\tau, x(\tau)) \right].$$

We shall prove that the performance function v , assumed to be smooth, is a solution to the Bensoussan–Lions variational inequalities

$$\begin{cases} \text{(i)} \quad \partial v(t, \xi) / \partial t + \langle \partial v(t, \xi) / \partial \xi, \psi(t, \xi) \rangle + \varphi(t, \xi) \geq 0, \\ \text{(ii)} \quad v(t, \xi) - h(t, \xi) \leq 0, \\ \text{(iii)} \quad [v(t, \xi) - h(t, \xi)] [\partial v(t, \xi) / \partial t + \langle \partial v(t, \xi) / \partial \xi, \psi(t, \xi) \rangle + \varphi(t, \xi)] = 0 \end{cases}$$

and

$$v(T, \xi) = h(T, \xi)$$

We use the performance function v to construct the optimal stopping time $\bar{\tau}$ as follows. We define the “stopping set” $\mathcal{S}$ by

$$\mathcal{S} = \{(t, \xi) \in [0, T] \times \mathbb{R}^n \text{ such that } v(t, \xi) = h(t, \xi)\}.$$

Then $\bar{\tau}$ is the smallest of the elements $\tau \in]t, T[$ such that $\{\tau, \xi\} \in \mathcal{S}$. We disregard in this section all technical difficulties which arise from the lack of smoothness of the functions involved in the problems. A book of Bensoussan and Lions (1978) will study thoroughly such problems from both the optimal control and the partial differential equations view points.

14.4.1. The Hamilton–Jacobi–Bellman equation for a control problem

Consider an optimal control problem denoted by

$$(1) \quad v(t, \xi) = \inf \left[\int_t^T \varphi(s, x(s), u(s)) ds + h(x(T)) \right]$$

under the constraint

$$(2) \quad \begin{cases} \dot{x}(s) = \psi(s, x(s), u(s)) & \text{for } s \in]t, T[, \quad \dot{x} = dx/ds, \\ x(t) = \xi \end{cases}$$

and

$$(3) \quad u(s) \in K \quad \text{for } s \in]t, T[.$$

We assume that

$$(4) \quad K \subset \mathbf{R}^p \text{ is a convex compact subset}$$

and that

$$(5) \quad \begin{cases} \varphi:]0, T[\times \mathbf{R}^n \times K \rightarrow \mathbf{R} \\ \psi:]0, T[\times \mathbf{R}^n \times K \rightarrow \mathbf{R}^n \end{cases}$$

are at least twice differentiable on a neighborhood of $]0, T[\times \mathbf{R}^n \times K$. (We do not attempt a treatment in terms of minimal assumptions in this introductory presentation.)

The main assumption we make is that

$$(6) \quad \begin{array}{l} \text{the performance function } v \text{ defined on } [0, T] \times \mathbf{R}^n \\ \text{by (1) is continuously differentiable.} \end{array}$$

This assumption is by no means necessary, but allows a quick development of the theory. We shall prove that the performance function is the solution of the Hamilton–Jacobi–Bellman equation.

Proposition 1. *Suppose that (6) holds. Then the performance function is a solution of the first order partial differential equation*

$$(7) \quad \begin{cases} \text{(i) } \partial v(t, \xi) / \partial t = \max_{u \in K} [\langle -\partial v(t, \xi) / \partial \xi, \psi(t, \xi, u) \rangle - \varphi(t, \xi, u)], \\ \text{(ii) } v(T, \xi) = h(\xi). \end{cases}$$

Remark. Consider the case when

$$(8) \quad \psi(t, \xi, u) = M(t)u + A(t)\xi.$$

Then

$$(9) \quad \begin{aligned} \max_{u \in K} \left[\left\langle -\frac{\partial v}{\partial \xi}(t, \xi), \psi(t, \xi, u) \right\rangle - \varphi(t, u) \right] \\ = - \left\langle \frac{\partial v}{\partial \xi}(t, \xi), A(t)\xi \right\rangle \\ + \max_{u \in K} \left[\left\langle -M(t)^* \frac{\partial v}{\partial \xi}(t, \xi), u \right\rangle - \varphi(t, \xi, u) \right]. \end{aligned}$$

The last term of the right-hand side inequality is just the Hamiltonian of ψ .

Proof. We begin by proving the terminal condition (7(ii)). If we take $t = T$, then

$$v(T, \xi) = \inf_{x(T)=\xi} \left[\int_T^T \varphi(T, x(T), u(T)) \, dT + h(x(T)) \right] = h(\xi).$$

We now prove that v satisfies the partial differential equation. Let $t < T$. We introduce Δt (which will tend to 0) and split the control function u into the form

$$(10) \quad u(s) = u_1(s) + u_2(s)$$

where

$$(11) \quad \begin{cases} u_1(s) = \begin{cases} u(s) & \text{if } s \in]t, t + \Delta t[, \\ 0 & \text{if } s \notin [t + \Delta t, T], \end{cases} \\ u_2(s) = \begin{cases} 0 & \text{if } s \in]t, t + \Delta t[, \\ u(s) & \text{if } s \notin [t + \Delta t, T]. \end{cases} \end{cases}$$

Hence the solution x of the differential eq. (2) can be split in the following way : $x(\cdot) = x_{u_1}(\cdot)$ is solution of

$$(12) \quad \begin{cases} \dot{x}(s) = \varphi(s, x(s), u_1(s)) & \text{if } s \in]t, t + \Delta t[, \\ x(t) = \xi \end{cases}$$

and $x(\cdot) = x_{u_1}(\cdot; u_2)$ is solution of

$$(13) \quad \begin{cases} \dot{x}(s) = \varphi(s, x(s), u_2(s)) & s \in]t + \Delta t, T[, \\ x(t + \Delta t) = x_{u_1}(t + \Delta t). \end{cases}$$

Consider

$$(14) \quad v(t + \Delta t, x_{u_1}(t + \Delta t)) = \inf_{u_2(\cdot)} \int_{t + \Delta t}^T \varphi(s, x_{u_1}(s; u_2), u_2(s)) \, ds + h(x(T, u_2)).$$

Then we can write

$$(15) \quad v(t, \xi) = \inf_{u_1(\cdot)} \left[\int_t^{t + \Delta t} \varphi(s, x_{u_1}(s), u_1(s)) \, ds + v(t + \Delta t, x_{u_1}(t + \Delta t)) \right],$$

[since

$$\begin{aligned} \int_t^T \varphi(s, x(s), u(s)) \, ds + h(x(T)) &= \\ &= \int_t^{t + \Delta t} \varphi(s, x_{u_1}(s), u_1(s)) \, ds \\ &\quad + \int_{t + \Delta t}^T \varphi(s, x_{u_1}(s; u_2), u_2(s)) \, ds + h(x_{u_1}(T, u_2)). \end{aligned}$$

In computing the infimum over $u = u_1 + u_2$, we take first the infimum over u_2 when u_1 is fixed and then the infimum over u_1 .]

We now use the Taylor series expansion formula

$$(16) \quad \begin{aligned} x_{u_1}(t + \Delta t) &= x_{u_1}(t) + \Delta t \dot{x}_{u_1}(t) + o(\Delta t) \\ &= \xi + \Delta t \left(\dot{x}_{u_1}(t) + \frac{o(\Delta t)}{\Delta t} \right) = \xi + \Delta \xi \end{aligned}$$

where we set

$$\Delta \xi = \Delta t \left(\dot{x}_{u_1}(t) + \frac{o(\Delta t)}{\Delta t} \right).$$

Also, we obtain

$$(17) \quad \begin{aligned} v(t + \Delta t, \xi + \Delta \xi) &= v(t, \xi) + \left\langle \frac{\partial v}{\partial \xi}(t, \xi), \Delta \xi \right\rangle + \frac{\partial v}{\partial t}(t, \xi) \Delta t + o(\Delta t, \Delta \xi) \\ &= v(t, \xi) + \left[\left\langle \frac{\partial v}{\partial \xi}(t, \xi), \dot{x}_{u_1}(t) \right\rangle + \frac{\partial v}{\partial t}(t, \xi) \right] \Delta t \\ &\quad + o(\Delta t). \end{aligned}$$

Furthermore, we can replace $\dot{x}_{u_1}(t)$ by $\psi(t, x_{u_1}(t), u_1(t)) = \psi(t, \xi, u_1(t))$. Also, we can write

$$(18) \quad \begin{aligned} \int_t^{t+\Delta t} \varphi(s, x_{u_1}(s), u_1(s)) \, ds &= \varphi(t, x_{u_1}(t), u_1(t)) \Delta t + o(\Delta t) \\ &= \varphi(t, \xi, u_1(t)) \Delta t + o(\Delta t). \end{aligned}$$

Hence, we obtain that

$$\begin{aligned} &\int_t^{t+\Delta t} \varphi(s, x_{u_1}(s), u_1(s)) \, ds + v(t + \Delta t, x_{u_1}(t + \Delta t)) \\ &= v(t, \xi) + \Delta t \left[\varphi(t, \xi, u_1(t)) + \left\langle \frac{\partial v}{\partial \xi}(t, \xi), \psi(t, \xi, u_1(t)) \right\rangle \right. \\ &\quad \left. + \frac{\partial v}{\partial t}(t, \xi) \right] + o(\Delta t). \end{aligned}$$

Therefore, eq. (15) becomes

$$\begin{aligned} \frac{\partial v}{\partial t}(t, \xi) &= \max_{u_1(t) \in K} \left[\left\langle -\frac{\partial v}{\partial \xi}(t, \xi), \psi(t, \xi, u_1(t)) \right\rangle - \varphi(t, \xi, u_1(t)) \right] \\ &\quad + \frac{o(\Delta t)}{\Delta t}. \end{aligned}$$

Hence, by letting Δt tend to 0, we can check that v is a solution of the partial differential eq. (7).

14.4.2. Construction of the closed loop control

Assume now that we know the performance function $v(t, \xi)$. We shall assume that

- (19) there exists a unique control $D^*(t, \xi) \in K$ which achieves the maximum over K of the function $u \mapsto [\langle -\partial v / \partial \xi(t, \xi), \psi(t, \xi, u) \rangle - \varphi(t, \xi, u)]$, which depends smoothly upon t and ξ .

This decision rule is called a “*feed back map*” or a “*closed loop control*”. This gives the optimal control at each t as shown in the following proposition.

Proposition 2. Suppose that (6) and (19) hold. Then, for each t , the optimal control $\bar{u}(t)$ and the optimal state $\bar{x}(t)$ are related by

$$(20) \quad \bar{u}(t) = D^*(t, \bar{x}(t)).$$

Proof. Consider the solution $z(\cdot)$ of the differential equation

$$(21) \quad \begin{cases} \dot{z}(s) = \psi(s, z(s), D^*(s, z(s))), \\ z(t) = \xi. \end{cases}$$

We differentiate the function $s \mapsto v(s, z(s))$ on $]t, T[$ and obtain

$$(22) \quad \begin{aligned} \frac{d}{ds} v(s, z(s)) &= \frac{\partial}{\partial s} v(s, z(s)) + \left\langle \frac{\partial v}{\partial \xi}(s, v(s)), \dot{z}(s) \right\rangle \\ &= \frac{\partial}{\partial s} v(s, z(s)) + \left\langle \frac{\partial v}{\partial \xi}(s, z(s)), \psi(s, z(s), D^*(s, z(s))) \right\rangle. \end{aligned}$$

But, by the very definition of D^* ,

$$\begin{aligned} \left\langle \frac{\partial v}{\partial \xi}(s, z(s)), \psi(s, z(s), D^*(s, z(s))) \right\rangle &= \\ &= -\varphi(s, z(s), D^*(s, z(s))) - \\ &\quad - \max_u \left[\left\langle -\frac{\partial v}{\partial \xi}(s, z(s), \psi(s, z(s), u)) \right\rangle - \varphi(s, z(s), u) \right] \\ &= -\varphi(s, z(s), D^*(s, z(s))) - \frac{\partial v}{\partial t}(s, z(s)). \end{aligned}$$

Therefore, we deduce that

$$(23) \quad \frac{d}{ds} v(s, z(s)) = -\varphi(s, z(s), D^*(s, z(s))).$$

Hence, by integrating from t to T ,

$$\begin{aligned} v(T, z(T)) - v(t, z(t)) &= h(z(T)) - v(t, \xi) \\ &= - \int_t^T \varphi(s, z(s), D^*(s, z(s))) \, ds, \end{aligned}$$

i.e.

$$(24) \quad v(t, \xi) = \int_t^T \varphi(s, z(s), D^*(s, z(s))) \, ds + h(z(T)).$$

This means that the state $z(\cdot)$ which is the solution of the differential equation for the control $u(\cdot)$ defined by $u(t) = D^*(t, z(t))$, achieves the minimum. Hence it is the optimal state and u is the optimal control.

14.4.3. The principle of optimality

We now proceed to prove the principle of optimality.

Proposition 3. *Let $\bar{x}$ be an optimal state associated with an optimal control $\bar{u}$ of the minimization problem $v(t, \xi)$. Then the restrictions $\bar{x}_\tau$ and $\bar{u}_\tau$ to $]t, \tau[$ (where $t < \tau < T$) are respectively an optimal state and an optimal control of the problem $v(\bar{x}(\tau), \tau)$.*

Proof. We have, by definition, that

$$(25) \quad v(t, \xi) = \int_t^T \varphi(s, \bar{x}(s), \bar{u}(s)) \, ds + h(\bar{x}(T)).$$

If the restriction $\{\bar{x}_\tau, \bar{u}_\tau\}$ is not an optimal pair of the problem $v(\bar{x}(\tau), \tau)$, there exists a pair $\{\hat{x}, \hat{u}\}$ such that

$$\begin{aligned} (26) \quad & \int_t^T \varphi(s, \hat{x}(s), \hat{u}(s)) \, ds + h(\hat{x}(T)) \\ & < \int_t^T \varphi(s, \bar{x}(s), \bar{u}(s)) \, ds + h(\bar{x}(T)). \end{aligned}$$

We shall exhibit a contradiction by proving that the pair $\{\hat{\bar{x}}, \hat{\bar{u}}\}$ defined by

$$(27) \quad \{\hat{\bar{x}}(s), \hat{\bar{u}}(s)\} = \begin{cases} \{\bar{x}(s), \bar{u}(s)\} & \text{if } s \in]t, \tau[, \\ \{\hat{x}(s), \hat{u}(s)\} & \text{if } s \notin]t, \tau[\end{cases}$$

yields a loss smaller than $v(t, \xi)$. Indeed, we have that

$$\begin{aligned} v(t, \xi) &\leq \int_t^T \varphi(s, \bar{x}(s), \bar{u}(s)) \, ds + \int_t^T \varphi(s, \hat{x}(s), \hat{u}(s)) \, ds + h(\hat{x}(T)) \\ &< \int_t^T \varphi(s, \bar{x}(s), \bar{u}(s)) \, ds + \int_t^T \varphi(s, \bar{u}(s), \bar{u}(s)) \, ds + h(\bar{x}(T)) \\ &= v(t, \xi). \quad \square \end{aligned}$$

Proposition 4. Suppose also that v is twice differentiable. Suppose that (6) and (19) hold. Then the function $-\partial v(\cdot, \xi)/\partial \xi$ is the solution of the adjoint system

$$(28) \quad \begin{cases} \text{(i)} \, d\bar{p}/dt = -\langle D\psi(t, \bar{x}(t), \bar{u}(t)), \bar{p}(t) \rangle + D\varphi(t, \bar{x}(t), \bar{u}(t)), \\ \text{(ii)} \, \bar{p}(T) = -Dh(\bar{x}(T)), \end{cases}$$

when $D = \partial/\partial \xi$ denotes the gradient with respect to ξ .

Proof. Let $\bar{x}(\cdot)$ be the optimal state and let $\bar{u}(t) = D^*(t, \bar{x}(t))$ be the optimal control.

Write $\bar{p}(t) = -\partial v(t, \bar{x}(t))/\partial \xi$. It is clear that $\bar{p}(T) = Dh(\bar{x}(T))$ since $v(T, \bar{x}(T)) = -h(\bar{x}(T))$.

We now compute the derivative of the i th component $\bar{p}_i(\cdot)$ of $\bar{p}(\cdot)$. Then

$$\begin{aligned} (29) \quad \frac{d\bar{p}_i}{dt} &= -\frac{\partial^2}{\partial \xi_i \partial t} v(t, \bar{x}(t)) - \sum_{j=1}^n \frac{\partial^2}{\partial \xi_i \partial \xi_j} v(t, \bar{x}(t)) \cdot \frac{d\bar{x}_j}{dt} \\ &= -\frac{\partial^2}{\partial \xi_i \partial t} v(t, \bar{x}(t)) - \sum_{j=1}^n \frac{\partial^2}{\partial \xi_i \partial \xi_j} v(t, \bar{x}(t)) \psi_j(t, \bar{x}(t), \bar{u}(t)). \end{aligned}$$

Now compute $\partial^2 v(t, \xi)/\partial \xi_i \partial \xi_j$ by using the Hamilton–Jacobi–Bellman equation. Since $D^*(t, \xi)$ is the unique solution which achieves the maximum in (19), we deduce that

$$\begin{aligned} (30) \quad \frac{\partial^2}{\partial \xi_i \partial t} v(t, \xi) &= -\sum_{j=1}^n \frac{\partial^2}{\partial \xi_i \partial \xi_j} v(t, \xi) \cdot \psi_j(t, \xi, D^*(t, \xi)) \\ &\quad - \sum_{j=1}^n \frac{\partial v}{\partial \xi_j} \cdot \frac{\partial \psi_j}{\partial \xi_i}(t, \xi, D^*(t, \xi)) - \frac{\partial}{\partial \xi_i} \varphi(t, \xi, D^*(t, \xi)). \end{aligned}$$

On setting $\xi = \bar{x}(t)$ and $\bar{u}(t) = D^*(t, \bar{x}(t))$, eqs. (29) and (30) yield

$$(31) \quad \frac{d\bar{p}_i(t)}{dt} = -\sum_{j=1}^n \bar{p}_j(t) \frac{\partial \psi_j}{\partial \xi_i}(t, \bar{x}(t), \bar{u}(t)) + \frac{\partial}{\partial \xi_i} \varphi(t, \bar{x}(t), \bar{u}(t))$$

which is the adjoint equation.

14.4.4. The quadratic case: Riccati equations

We continue the study of the quadratic control problem initiated in Section 14.2.7.

Consider the following norms

$$(32) \quad j(s, \xi) = \langle J(s)\xi, \xi \rangle^{\frac{1}{2}}, \quad g(\xi) = \langle G\xi, \xi \rangle^{\frac{1}{2}}, \quad k(s, \eta) = \langle K(s)\eta, \eta \rangle^{\frac{1}{2}}$$

associated with the duality operators

$$(33) \quad J(s) \in \mathcal{L}(\mathbf{R}^n, \mathbf{R}^{n*}), \quad G \in \mathcal{L}(\mathbf{R}^n, \mathbf{R}^{n*}), \quad K(s) \in \mathcal{L}(\mathbf{R}^p, \mathbf{R}^{p*}).$$

We introduce linear operators

$$(34) \quad A(s) \in \mathcal{L}(\mathbf{R}^n, \mathbf{R}^n), \quad M(s) \in \mathcal{L}(\mathbf{R}^p, \mathbf{R}^n)$$

and set

$$(35) \quad N(s) = M(s)K(s)^{-1}M^*(s).$$

We shall assume that the functions $J(\cdot)$, $K(\cdot)$, $A(\cdot)$, $M(\cdot)$, $N(\cdot)$ are sufficiently smooth. We set

$$(36) \quad \begin{cases} \text{(i)} \quad \varphi(s, x, u) = \frac{1}{2}j(s, x - x_0(s))^2 + \frac{1}{2}k(s, u - u_0(s))^2, \\ \text{(ii)} \quad h(\xi) = \frac{1}{2}g(\xi - \xi_T)^2, \\ \text{(iii)} \quad \psi(s, x, u) = A(s)\xi + M(s)u \end{cases}$$

and study the quadratic control problem

$$(37) \quad \begin{cases} v(t, \xi) = \inf \left[\int_t^T \{j(s, x(s) - x_0(s))^2 + k(s, u(s) - u_0(s))^2\} ds \right. \\ \quad \left. + g(x(T) - \xi_T)^2 \right], \\ \dot{x}(s) = A(s)x(s) + M(s)u(s), \\ x(t) = \xi \end{cases}$$

without constraints on the control (i.e. $u(\cdot)$ ranges over $L^2(t, T; \mathbf{R}^p)$). It is clear that, for each $t, \xi \mapsto v(t, \xi)$ is a quadratic function, which can be written

$$(38) \quad v(t, \xi) = \frac{1}{2} \langle P(t)\xi, \xi \rangle + \langle r(t), \xi \rangle + \sigma(t)$$

where, for each t ,

$$(39) \quad \begin{cases} \text{(i)} \quad P(t) \in \mathcal{L}(\mathbf{R}^n, \mathbf{R}^{n*}) \text{ is a symmetric matrix,} \\ \text{(ii)} \quad r(t) \in \mathbf{R}^{n*}, \\ \text{(iii)} \quad \sigma(t) \in \mathbf{R}. \end{cases}$$

Proposition 5. *The Hamilton–Jacobi–Bellman equation can be written*

$$(40) \quad \begin{cases} \frac{\partial v}{\partial t}(t, \xi) = \frac{1}{2} \left\langle N(t) \frac{\partial v}{\partial \xi}(t, \xi), \frac{\partial v}{\partial \xi}(t, \xi) \right\rangle \\ - \left\langle \frac{\partial v}{\partial \xi}(t, \xi), A(t)\xi + M(t)u_0(t) \right\rangle - \frac{1}{2} j(t, \xi - x_0(t))^2 \end{cases}$$

with the terminal condition

$$(41) \quad v(T, \xi) = \frac{1}{2} g(\xi - \xi_T)^2.$$

The performance function v can be written in the form (38) where $P(\cdot)$, $r(\cdot)$ and $(\cdot)$ are solutions of the “Riccati system of equations”

$$(42) \quad \begin{cases} \text{(i)} \quad \dot{P}(t) = P(t)N(t)P(t) - A^*(t)P(t)P(t) - P(t)A(t) - J(t), \\ \text{(ii)} \quad \dot{r}(t) = P(t)N(t)r(t) + J(t)x_0(t) - A^*(t)r(t) - P(t)M(t)u_0(t), \\ \text{(iii)} \quad \dot{\sigma}(t) = \langle N(t)r(t), r(t) \rangle - \frac{1}{2} j(t, x_0(t))^2 - \langle r(t), M(t)u_0(t) \rangle \end{cases}$$

with the terminal conditions

$$(43) \quad P(T) = G, \quad r(T) = -G\xi_T, \quad \sigma(T) = \frac{1}{2} g(\xi_T)^2.$$

Then, if $P(\cdot)$, $r(\cdot)$ are solved, the optimal co-state function $\bar{p}(\cdot)$ is given by

$$(44) \quad -\bar{p}(t) = P(t)\bar{x}(t) + r(t)$$

and the optimal “closed loop” control is defined by

$$(45) \quad \bar{u}(t) = u_0(t) - K(t)^{-1} M^*(t) [P(t)\bar{x}(t) + r(t)].$$

The optimal state is the solution of the differential equation

$$(46) \quad \begin{cases} \text{(i)} \quad \dot{\bar{x}}(s) = A(s)\bar{x}(s) - N(s)P(s)\bar{x}(s) - N(s)r(s) + M(s)u_0(s), \\ \text{(ii)} \quad \bar{x}(t) = \xi. \end{cases}$$

Proof. In the case of quadratic control problems, we have that

$$\begin{aligned} \max_{u \in \mathbb{R}^n} \left[\left\langle -\frac{\partial v}{\partial \xi}, \psi(t, \xi, u) \right\rangle - \varphi(t, \xi, u) \right] &= \\ &= - \left\langle \frac{\partial v}{\partial \xi}, A(t)\xi \right\rangle - \frac{1}{2} j(s, \xi - x_0(s))^2 \\ &\quad + \max_{u \in \mathbb{R}^n} \left[\left\langle -\frac{\partial v}{\partial \xi}, M(t)u \right\rangle - \frac{1}{2} k(t, u - u_0(t))^2 \right] \\ &= \left\langle -\frac{\partial v}{\partial \xi}, A(t)\xi \right\rangle - \frac{1}{2} j(t, \xi - x_0(t))^2 \\ &\quad + \frac{1}{2} k_* \left(t, -M^*(t) \frac{\partial v}{\partial \xi} \right)^2 - \left\langle \frac{\partial v}{\partial \xi}, M(t)u_0(t) \right\rangle \\ &= \frac{1}{2} k_* \left(t, M^*(t) \frac{\partial v}{\partial \xi} \right)^2 - \left\langle \frac{\partial v}{\partial \xi}, A(t)\xi + M(t)u_0(t) \right\rangle \\ &\quad - \frac{1}{2} j(t, \xi - x_0(t))^2. \end{aligned}$$

But

$$\begin{aligned} k_*\left(t, M^*(t) \frac{\partial v}{\partial \xi}\right)^2 &= \left\langle K(t) M^*(t) \frac{\partial v}{\partial \xi}, M^*(t) \frac{\partial v}{\partial \xi} \right\rangle \\ &= \left\langle N(t) \frac{\partial v}{\partial \xi}, \frac{\partial v}{\partial \xi} \right\rangle. \end{aligned}$$

Thus we have proved (40).

Let us write $v(t, \xi) = \frac{1}{2} \langle P(t), \xi, \xi \rangle + \langle r(t), \xi \rangle + \sigma(t)$ where, for each t , $P(t) \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$ is a symmetric operator. Since

$$(47) \quad \frac{\partial v}{\partial t}(t, \xi) = \frac{1}{2} \langle \dot{P}(t) \xi, \xi \rangle + \langle \dot{r}(t), \xi \rangle + \dot{\sigma}(t)$$

and since

$$(48) \quad \frac{\partial v}{\partial \xi}(t, \xi) = P(t) \xi + r(t),$$

the Hamilton–Jacobi–Bellman equation can be written

$$\begin{aligned} &\frac{1}{2} \langle \dot{P}(t) \xi, \xi \rangle + \langle \dot{r}(t), \xi \rangle + \dot{\sigma}(t) \\ &= \frac{1}{2} \langle N(t) (P(t) \xi + r(t)), P(t) \xi + r(t) \rangle \\ &\quad - \langle P(t) \xi + r(t), A(t) \xi + M(t) u_0(t) \rangle - \frac{1}{2} \langle J(t) \xi, \xi \rangle \\ &\quad + \langle J(t) x_0(t), \xi \rangle - \frac{1}{2} j(t, x_0(t))^2 \\ &= \frac{1}{2} \langle P(t) N(t) P(t) \xi, \xi \rangle - \langle P(t) \xi, A(t) \xi \rangle - \frac{1}{2} \langle J(t) \xi, \xi \rangle \\ &\quad + \langle P(t) N(t) r(t) - A^*(t) r(t) - P(t) M(t) u_0(t) + J(t) x_0(t), \xi \rangle \\ &\quad + \frac{1}{2} \langle N(t) r(t), r(t) \rangle - \langle r(t), M(t) u_0(t) \rangle - \frac{1}{2} j(t, x_0(t))^2. \end{aligned}$$

We notice that, since the matrix P is symmetric, we can write

$$(49) \quad \langle P(t) \xi, A(t) \xi \rangle = \frac{1}{2} \langle A^*(t) P(t) \xi, \xi \rangle + \frac{1}{2} \langle P(t) A(t) \xi, \xi \rangle.$$

Hence we have, for any $\xi \in \mathbb{R}^n$,

$$\begin{aligned} &\frac{1}{2} \langle (\dot{P}(t) - [P(t) N(t) P(t) - P(t) A(t) - A^*(t) P(t) - J(t)]) \xi, \xi \rangle \\ &\quad + \langle \dot{r}(t) - [P(t) N(t) r(t) - A^*(t) r(t) - P(t) M(t) u_0(t) \\ &\quad + J(t) x_0(t)], \xi \rangle \\ &\quad + \dot{\sigma}(t) - [\frac{1}{2} \langle N(t) r(t), r(t) \rangle - \langle r(t), M(t) u_0(t) \rangle - \frac{1}{2} j(t, x_0(t))^2] \\ &= 0. \end{aligned}$$

This implies that “Riccati” eq. (42) hold.

Since for all $\xi \in \mathbf{R}^n$ the terminal condition can be written

$$\begin{aligned} \frac{1}{2} \langle P(T)\xi, \xi \rangle + \langle r(T), \xi \rangle + \sigma(T) &= v(T, \xi) \\ &= \frac{1}{2} \langle G\xi, \xi \rangle - \langle G\xi_T, \xi \rangle + \frac{1}{2} \langle G\xi_T, \xi_T \rangle, \end{aligned}$$

we obtain the terminal conditions (43).

Let $P(\cdot)$ and $r(\cdot)$ be solutions of the Riccati equation. It is clear that $P(t)$ is symmetric for all t . Let $\bar{p}(\cdot)$, $\bar{x}(\cdot)$ be defined by (44) and (46). Then

$$\begin{aligned} -\frac{\partial \bar{p}}{\partial t}(t) &= P(t) \bar{x}(t) + \dot{r}(t) + P(t) \dot{x}(t) \\ &= -P(t) N(t) \bar{p}(t) + A^*(t) \bar{p}(t) - P(t) A(t) \bar{x}(t) + P(t) \dot{x}(t) \\ &\quad - J(t) (\bar{x}(t) - x_0(t)) - P(t) M(t) u_0(t) \\ &= A^*(t) \bar{p}(t) - J(t) (\bar{x}(t) - x_0(t)) \\ &\quad + P(t) [\dot{x}(t) - (A(t) \bar{x}(t) + M(t) u_0(t) + N(t) \bar{p}(t))] \\ &= A^*(t) \bar{p}(t) - J(t) (\bar{x}(t) - x_0(t)) \\ &\quad + P(t) [\dot{x}(t) - (A(t) \bar{x}(t) + M(t) u_0(t) - N(t) (P(t) \bar{x}(t) \\ &\quad + r(t))]. \end{aligned}$$

Hence, since $\bar{x}(\cdot)$ is the solution of (46), we see that $\bar{p}(\cdot)$ is the solution of the adjoint system

$$\begin{aligned} -\frac{\partial \bar{p}}{\partial t}(t) &= A^*(t) \bar{p}(t) - J(t) (\bar{x}(t) - x_0(t)), \\ -\bar{p}(T) &= P(T) \bar{x}(T) + r(T) = G(\bar{x}(T) - \xi_T). \end{aligned}$$

This proves that $\bar{x}(\cdot)$ is the optimal solution and that the feed-back map is given by

$$\bar{u}(t) = u_0(t) - K(t)^{-1} M^*(t) [P(t) \bar{x}(t) + r(t)],$$

since the function $u \mapsto \langle -\partial v / \partial \xi, M(t)u \rangle - \frac{1}{2}k(t, u - u_0(t))^2$ achieves its maximum at

$$(50) \quad D^*(t, \xi) = u_0(t) - K(t)^{-1} M^*(t) \frac{\partial v}{\partial \xi}(t, \xi).$$

14.4.5. The Bensoussan–Lions variational inequalities of a stopping time problem

We consider an optimal “stopping time” problem

$$(51) \quad v(t, \xi) = \inf_{t \leq \tau \leq T} \left[\int_t^\tau \varphi(s, x(s)) \, ds + h(\tau, x(\tau)) \right],$$

where $x(\cdot)$ is the solution of the differential equation

$$(52) \quad \begin{cases} \text{(i)} \quad \dot{x}(s) = \varphi(s, x(s)) & \text{for } s \in]t, T[, \\ \text{(ii)} \quad x(t) = \xi. \end{cases}$$

where $\varphi :]0, T[\times \mathbf{R}^n \rightarrow \mathbf{R}$ and $\psi :]0, T[\times \mathbf{R}^n \rightarrow \mathbf{R}^n$ are *smooth functions* which satisfy the requirements implying existence and uniqueness of a smooth solution. We shall say that the function v defined by (51) is the *performance function of the optimal stopping time problem*.

We shall assume that

$$(53) \quad \text{the performance function } v \text{ is continuously differentiable.}$$

[Actually, it is sufficient to assume that v is Lipschitz].

Remark. Since the function $\tau \rightarrow \int_t^\tau \varphi(s, x(s)) ds + h(\tau, x(\tau))$ is continuous and τ ranges over a compact set, there exists an optimal stopping time $\bar{\tau}$.

Interpretation. We can use such a model to determine the optimal time to sell a commodity ξ available at time t , whose state evolves according to (2), when $\varphi(s, x(s))$ is the maintenance cost at time s and $h(s, \xi)$ is the value of ξ at time s .

Proposition 6 (Bensoussan–Lions). *Suppose that (53) holds. Then the performance function v is a solution to the variational inequalities: for any $\{t, \xi\} \in (0, T] \times \mathbf{R}^n$,*

$$(54) \quad \begin{cases} \text{(i)} \quad \partial v(t, \xi) / \partial t + \langle \partial v(t, \xi) / \partial \xi, \psi(t, \xi) \rangle + \varphi(t, \xi) \geq 0, \\ \text{(ii)} \quad v(t, \xi) - h(t, \xi) \leq 0, \\ \text{(iii)} \quad [v(t, \xi) - h(t, \xi)] [\partial v(t, \xi) / \partial t + \langle \partial v(t, \xi) / \partial \xi, \psi(t, \xi) \rangle + \varphi(t, \xi)] = 0 \end{cases}$$

and the terminal condition

$$(55) \quad v(T, \xi) = h(T, \xi).$$

Proof. The terminal condition $v(T, \xi) = h(T, \xi)$ follows from the definition by taking $t = \tau = T$. Now, by taking $\tau = t$, we deduce that

$$(56) \quad v(t, \xi) \leq \int_t^t \varphi(t, x(t)) dt + h(t, x(t)) = h(t, x(t)).$$

Hence (54(ii)) is satisfied. We now take Δt (which will converge to 0) and $\tau \geq t + \Delta t$. We write

$$(57) \quad \begin{aligned} v(t, \xi) &\leq \int_t^\tau \varphi(s, x(s)) ds + h(\tau, x(\tau)) \\ &= \int_t^{t+\Delta t} \varphi(s, x(s)) ds + \int_{t+\Delta t}^\tau \varphi(s, x(s)) ds + h(\tau, x(\tau)) \end{aligned}$$

Taking the infimum over $\tau \in [t + \Delta t, T]$, we deduce that

$$(58) \quad v(t, \xi) \leq \int_t^{t+\Delta t} \varphi(s, x(s)) \, ds + v(t + \Delta t, x(t + \Delta t)).$$

Using the Taylor formula, we can write $x(t + \Delta t) = \xi + \Delta \xi$ where $\Delta \xi = \Delta t \cdot \partial x / \partial t + o(\Delta t)$.

Then

$$\begin{aligned} v(t, \xi) &\leq v(t, \xi) + \Delta t \cdot \left[\varphi(t, \xi) + \frac{\partial v}{\partial t}(t, \xi) \right] + \left\langle \frac{\partial v}{\partial \xi}(t, \xi), \Delta \xi \right\rangle \\ &\quad + o(\Delta t, \Delta \xi) \leq v(t, \xi) + \\ &\quad + \Delta t \cdot \left[\varphi(t, \xi) + \frac{\partial v}{\partial t}(t, \xi) + \left\langle \frac{\partial v}{\partial \xi}, \psi(t, \beta) \right\rangle \right] + o(\Delta t). \end{aligned}$$

Dividing by Δt and letting Δt tend to 0, we deduce that (54(i)) holds. It remains to check that (54(iii)) holds. This amounts to proving that

$$(59) \quad \frac{\partial v}{\partial t}(t, \xi) + \left\langle \frac{\partial v}{\partial \xi}(t, \xi), \psi(t, \xi) \right\rangle + \varphi(t, \xi) = 0$$

whenever

$$(60) \quad v(t, \xi) < h(t, \xi).$$

Let t be a point satisfying (60) and $\bar{\tau}$ be the optimal stopping strategy. Then $\bar{\tau} > t$ (If not, we would have $v(\bar{\tau}, \xi) = v(t, \xi) = h(\bar{\tau}, \xi)$, which is impossible).

Let Δt be so small that $t + \Delta t < \bar{\tau}$.

Since

$$\begin{aligned} v(t, \xi) &= \int_t^{t+\Delta t} \varphi(s, x(s)) \, ds + \int_{t+\Delta t}^{\bar{\tau}} \varphi(s, x(s)) \, ds + h(\bar{\tau}, x(\bar{\tau})) \\ &\geq \int_t^{t+\Delta t} \varphi(s, x(s)) \, ds + v(t + \Delta t, x(t + \Delta t)), \end{aligned}$$

we deduce from the Taylor formula applied to $x(t + \Delta t)$ and $v(t + \Delta t, x(t + \Delta t))$ that

$$v(t, \xi) \geq v(t, \xi) + \Delta t \left[\varphi(t, \xi) + \frac{\partial v}{\partial t}(t, \xi) + \left\langle \frac{\partial v}{\partial \xi}(t, \xi), \psi(t, \xi) \right\rangle \right] + o(\Delta t)$$

as in the proof of Proposition 1.

Dividing by Δt and letting Δt converge to 0, we deduce that

$$(61) \quad 0 \geq \varphi(t, \xi) + \frac{\partial v}{\partial t}(t, \xi) + \left\langle \frac{\partial v}{\partial \xi}(t, \xi), \psi(t, \xi) \right\rangle.$$

Hence (59) follows from (61) and (54(i)).

14.4.6. Construction of the optimal stopping time

Knowledge of the performance function v allows us to find the optimal stopping time. We introduce the following "continuation subset $\mathcal{C}$ " of $[0, T] \times \mathbb{R}^n$ defined by

$$(62) \quad \mathcal{C} = \{ \{t, \xi\} \in [0, T] \times \mathbb{R}^n \text{ such that } v(t, \xi) < h(t, \xi) \}$$

and the "stopping subset $\mathcal{S}$ " defined by

$$(63) \quad \mathcal{S} = \{ \{t, \xi\} \in [0, T] \times \mathbb{R}^n \text{ such that } v(t, \xi) = h(t, \xi) \}.$$

Proposition 7. Suppose that (56) holds. If $\xi \in \mathbb{R}^n$ is given, the optimal stopping time $\bar{\tau}$ is the smallest of the elements $\tau \in [t, T]$ such that $\{\tau, \xi\}$ belongs to the stopping set $\mathcal{S}$.

Proof. Let $\xi \in \mathbb{R}^n$ be fixed and let $\bar{\tau}$ be the smallest of the elements $\tau \in [t, T]$ such that $\{\tau, \xi\}$ belongs to the stopping set $\mathcal{S}$. This set is non-empty since $\{T, \xi\}$ belongs to $\mathcal{S}$ by (52). Consider the function $s \mapsto v(s, x(s))$. Its derivative is equal to

$$\begin{aligned} \frac{d}{ds} v(s, x(s)) &= \frac{\partial v}{\partial s}(s, x(s)) + \left\langle \frac{\partial v}{\partial \xi}(s, x(s)), \frac{dx(s)}{ds} \right\rangle \\ (64) \quad &= \frac{\partial v}{\partial s}(s, x(s)) + \left\langle \frac{\partial v}{\partial \xi}(s, x(s)), \varphi(s, x(s)) \right\rangle \\ &= -\varphi(s, x(s)) \quad \text{whenever } s \in]t, \bar{\tau}[\end{aligned}$$

[since in this case, $\{s, \xi\}$ belongs to $\mathcal{C}$, (i.e. $v(s, x(s)) < h(s, x(s))$ and thus, equality (59) holds].

Hence, by integrating this equality from t to $\bar{\tau}$, we deduce that

$$\begin{aligned} h(\bar{\tau}, x(\bar{\tau})) - v(t, \xi) &= v(\bar{\tau}, x(\bar{\tau})) - v(t, \xi) \\ &= - \int_t^{\bar{\tau}} \varphi(s, x(s)) \, ds, \end{aligned}$$

i.e. $\bar{\tau}$ achieves the minimum. $\square$

14.4.7. The Bensoussan-Lions quasi-variational inequalities of an impulsive control problem

We study an "impulsive control problem"

$$(65) \quad v(t, \xi) = \inf_{u \in \mathcal{M}_t} \left[\int_t^T \varphi(s, x(s)) \, ds + N_t(u) \right]$$

where u ranges over the set $\mathcal{M}_t$ of discrete vector measures

$$(66) \quad u = \sum_{i=1}^m \xi_i \delta(\theta_i) \quad \text{where } \theta_i \in [t, T], \xi_i \in \mathbf{R}_+^n, N_t(u) = m$$

on the interval $[t, T]$ and where $x(\cdot)$ is the solution of the following systems of differential equations:

$$(67) \quad \begin{cases} \text{if } s \in [t, \theta_1[, & \dot{x}_0(s) = \psi(s, x_0(s)); & x_0(t) = \xi, \\ \text{if } s \in [\theta_1, \theta_2[, & \dot{x}_1(s) = \psi(s, x_1(s)); & x_1(\theta_1) = x_0(\theta_1) + \xi_1, \\ \text{if } s \in [\theta_i, \theta_{i+1}[, & \dot{x}_i(s) = \psi(s, x_i(s)); & x_i(\theta_i) = x_{i-1}(\theta_i) + \xi_i, \\ \text{if } s \in [\theta_m, T[, & \dot{x}_m(s) = \psi(s, x_m(s)); & x_m(\theta_m) = x_{m-1}(\theta_{m-1}) + \xi_m \end{cases}$$

We assume that $\psi :]0, T[\times \mathbf{R}^n \rightarrow \mathbf{R}$ and $\psi :]0, T[\times \mathbf{R}^n \rightarrow \mathbf{R}^n$ are *smooth functions* which satisfy the requirements for the existence and uniqueness of a *smooth solution*.

Interpretation. The typical example of such an impulsive control problem arises in inventory management. We regard $\mathbf{R}^n$ as the space of "inventories" ξ . We denote by $x(t)$ the state of the inventory at time t . An "impulsive control" $u = \sum_{i=1}^m \xi_i \delta(\theta_i)$ denotes a "supplying policy". At time θ_i , the firm orders $\xi_i \in \mathbf{R}_+^n$. The evolution of the state $x(\cdot)$ is described by the system of differential eqs. (67). We regard $\int_{\tau_1}^{\tau_2} \varphi(s, x(s)) ds$ as the maintenance cost of the inventory $x(\cdot)$ from τ_1 to τ_2 . We add to the maintenance cost $\int_t^T \varphi(s, x(s)) ds$ the sum $N_t(u) = m$ of the fixed costs (assumed to be equal to 1) arising at each order of a commodity. The aim of the problem is to find the optimal times θ_i at which to place the orders as well as the optimal quantities ξ_i to order.

Many other management problems (production maintenance, marketing, finance, etc.), can be "modeled" by impulsive control problems. $\square$

We shall require that the performance function $v(\cdot, \cdot)$ defined by (65) is smooth. For instance, we shall assume that

$$(68) \quad \text{the performance function is continuously differentiable.}$$

We shall prove that v is a solution of quasi-variational inequalities. For this purpose, we define the operator R associating with a function $v(\cdot, \cdot)$ the function $Rv(\cdot, \cdot)$ defined by

$$(69) \quad Rv(t, \xi) = 1 + \inf_{\eta \in \mathbf{R}_+^n} v(t, \xi + \eta).$$

We introduce the correspondence S defined by

$$(70) \quad Sv = \{w(\cdot, \cdot) \text{ such that } w \leq Rv\}.$$

Proposition 8 (Bensoussan–Lions). *Suppose that the regularity assumption (68) holds.*

Then the performance function $v(\cdot, \cdot)$ is a solution to the following quasi-variational inequalities

$$(71) \quad v \in S(v)$$

and

$$(72) \quad \begin{cases} \text{(i)} \quad \partial v(t, \xi)/\partial t + \langle \partial v(t, \xi)/\partial \xi, \psi(t, \xi) \rangle + \varphi(t, \xi) \geq 0, \\ \text{(ii)} \quad [\partial v(t, \xi)/\partial t + \langle \partial v(t, \xi)/\partial \xi, \psi(t, \xi) \rangle + \varphi(t, \xi)] (v(t, \xi) - Rv(t, \xi)) = 0 \end{cases}$$

and the terminal condition

$$(73) \quad v(T, \xi) = 0.$$

The quasi-variational inequalities (71), (72) are called the “Bensoussan–Lions” quasi-variational inequalities.

Proof. The terminal condition $v(T, \xi) = 0$ is obvious since $N_T(u) = 0$ and $\int_T^T \varphi(s, x(s)) ds = 0$. Consider now $t < T$. We use controls of the form

$$(74) \quad u = \eta \delta(t) + w \quad \text{where } w = \sum_{i=2}^n \xi_i \delta(\theta_i) \in \mathcal{M}_t, \quad \theta_i \in]t, T].$$

Then, since $\theta_1 = t$, the state $x(\cdot)$ satisfies

$$(75) \quad \dot{x}(s) = \psi(s, x(s)), \quad x(\theta_1) = x(t) + \eta = \xi + \eta.$$

Therefore, since $N_t(u) = 1 + N_t(w)$, we deduce that

$$(76) \quad \begin{aligned} v(t, \xi) &\leq 1 + \inf_{w \in \mathcal{M}_t} \left[\int_t^T \varphi(s, x(s)) ds + N_t(w) \right] \\ &= 1 + v(t, \xi + \eta). \end{aligned}$$

Taking the infimum as η ranges over $\mathbf{R}_+^n$, we obtain that

$$(77) \quad v(t, \xi) \leq 1 + \inf_{\eta \in \mathbf{R}_+^n} v(t, \xi + \eta) = Rv(t, \xi),$$

i.e. that $v \in S(v)$.

We now check that (72(i)) holds. For this purpose, we take controls $u = \sum_{i=1}^n \xi_i \delta(\theta_i)$ where $t < \theta_1 < \theta_2 < \dots < \theta_n < T$ and Δt such that $t + \Delta t < \theta_1$.

Then we can write

$$(78) \quad v(t, \xi) \leq \int_t^{t+\Delta t} \varphi(s, x(s)) \, ds + \left[\int_{t+\Delta t}^T \varphi(s, x(s)) \, ds + N_{t+\Delta t}(u) \right].$$

Taking the infimum over the discrete measures $u = \sum_{i=1}^m \xi_i \delta(\theta_i) \in \mathcal{M}_{t+\Delta t}$, we deduce that

$$(79) \quad v(t, \xi) \leq \int_t^{t+\Delta t} \varphi(s, x(s)) \, ds + v(t+\Delta t, x(t+\Delta t)).$$

On using the Taylor expansion formula, this implies, as in the proof of Propositions 1 and 6, that

$$v(t, \xi) \leq v(t, \xi) + \Delta t \left[\varphi(t, \xi) + \frac{\partial v}{\partial t}(t, \xi) + \left\langle \frac{\partial v}{\partial \xi}(t, \xi), \psi(t, \xi) \right\rangle \right] + o(\Delta t).$$

Dividing by Δt and letting Δt converge to 0, we obtain (72(i)). It remains to prove (72(ii)), i.e. that

$$(80) \quad \frac{\partial v}{\partial t}(t, \xi) + \left\langle \frac{\partial v}{\partial \xi}(t, \xi), \psi(t, \xi) \right\rangle + \varphi(t, \xi) = 0,$$

whenever

$$(81) \quad v(t, \xi) < Rv(t, \xi) = 1 + \inf_{\eta \in \mathbb{R}_+^n} v(t, \xi + \eta).$$

Let $\bar{u} = \sum_{i=1}^m \bar{\xi}_i \delta(\bar{\theta}_i)$ be an optimal control. If (81) holds, then $t < \bar{\theta}_1$ necessarily. [If not, we would have $\bar{u} = \bar{\eta} \delta(t) + \bar{w}$ and $v(t, \xi) = 1 + v(t, \xi + \bar{\eta}) = Rv(t, \xi)$, which is impossible]. We choose Δt such that $t + \Delta t < \bar{\theta}_1$.

Then

$$\begin{aligned} v(t, \xi) &= \int_t^{t+\Delta t} \varphi(s, \bar{x}(s)) \, ds + \int_{t+\Delta t}^T \varphi(s, \bar{x}(s)) \, ds + N_t(\bar{v}) \\ &\geq \int_t^{t+\Delta t} \varphi(s, \bar{x}(s)) \, ds + v(t+\Delta t, \bar{x}(t+\Delta t)). \end{aligned}$$

The Taylor expansion formula implies that

$$v(t, \xi) \geq v(t, \xi) + \Delta t \left[\varphi(t, \xi) + \frac{\partial v}{\partial t}(t, \xi) + \left\langle \frac{\partial v}{\partial \xi}(t, \xi), \psi(t, \xi) \right\rangle \right] + o(\Delta t).$$

Dividing by Δt and letting Δt tend to 0, we deduce that

$$(82) \quad \frac{\partial v}{\partial t}(t, \xi) + \left\langle \frac{\partial v}{\partial \xi}(t, \xi), \psi(t, \xi) \right\rangle + \varphi(t, \xi) \leq 0.$$

Hence (82) and (72(ii)) imply (81). $\square$

14.4.8. Construction of the optimal impulsive control

We now show how to use the performance function v to construct the optimal control.

We define the “continuation set” $\mathcal{C} \subset [0, T] \times \mathbf{R}^n$ by

$$(83) \quad \mathcal{C} = \{ \{t, \xi\} \in [0, T] \times \mathbf{R}^n \text{ such that } v(t, \xi) < Rv(t, \xi) \}$$

and its complement $\mathcal{S}$, the “stopping set”

$$(84) \quad \mathcal{S} = \{ \{t, \xi\} \in [0, T] \times \mathbf{R}^n \text{ such that } v(t, \xi) = Rv(t, \xi) \}.$$

Theorem 1 (Bensoussan–Lions). *Suppose that the performance function satisfies (68).*

Then the optimal impulsive control $\bar{u} = \sum \bar{\xi}_i \delta(\bar{\theta}_i)$ is obtained by induction as follows:

$$(85) \quad \begin{cases} \text{(i) } \bar{\theta}_1 \text{ is the first time } \tau \in]t, T] \text{ such that } \{\tau, \xi\} \text{ belongs to the} \\ \text{stopping set } \mathcal{S}, \\ \text{(ii) } \bar{\xi}_1 \in \mathbf{R}_+^n \text{ minimizes } \xi \rightarrow v(\bar{\theta}_1, x(\bar{\theta}_1) + \xi) \text{ on } \mathbf{R}_+^n. \end{cases}$$

If $\{\bar{\theta}_i, \bar{\xi}_i\}$ are known for $k \ll i$, we construct $x_i(\cdot)$ as the solution of

$$(86) \quad \dot{x}(s) = \psi(s, x_i(s)) \quad \text{for } s > \bar{\theta}_i, \quad x_i(\bar{\theta}_i) = x_{i-1}(\bar{\theta}_i) + \bar{\xi}_i$$

and

$$(87) \quad \begin{cases} \text{(i) } \bar{\theta}_{i+1} \text{ is the first time } \tau \in]\bar{\theta}_i, T] \text{ such that} \\ \{\tau, x_{i-1}(\bar{\theta}_i) + \bar{\xi}_i\} \text{ belongs to the stopping set } \mathcal{S}. \\ \text{(ii) } \bar{\xi}_{i+1} \in \mathbf{R}_+^n \text{ minimizes } \xi \mapsto v(\bar{\theta}_{i+1}, x_i(\bar{\theta}_{i+1}) + \xi) \text{ on } \mathbf{R}_+^n. \end{cases}$$

Interpretation. Let ξ be the initial inventory at time t . If $\{t, \xi\} \in \mathcal{S}$, we order a stock $\bar{\eta}$ which minimizes $\eta \mapsto v(t, \xi + \eta)$. If not, we let the state $x(\cdot)$ of the stock evolve until a time $\bar{\theta}_1$ at which $\{\tau, \xi\}$ reaches the stopping set $\mathcal{S}$. At such a time $\bar{\theta}_1$, we order a new commodity $\bar{\xi}_1$ and let the state $x(\cdot)$ of the inventory evolves from the position $x(\bar{\theta}_1) + \bar{\xi}_1$ at time $\bar{\theta}_1$ and so on.

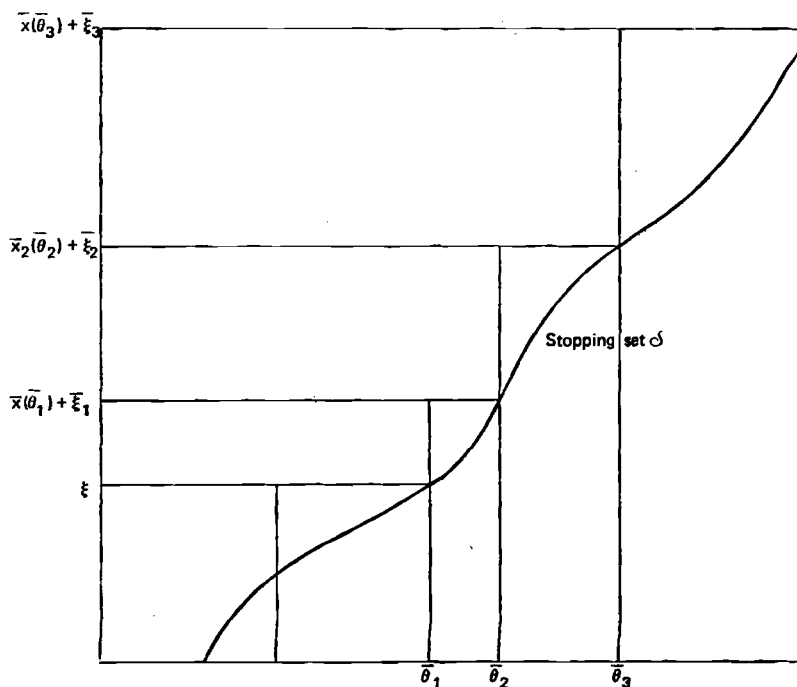


Fig. 16.

Proof. Consider the function $s \mapsto v(s, \bar{x}(s))$. Its derivative is equal to

$$(88) \quad \frac{d}{ds} v(s, \bar{x}(s)) = \frac{\partial v}{\partial s}(s, \bar{x}(s)) + \left\langle \frac{\partial v}{\partial \bar{x}}(s, \bar{x}(s)), \psi(s, \bar{x}(s)) \right\rangle.$$

Suppose that $\{t, \xi\}$ belongs to $\mathcal{C}$.

Let $\bar{\theta}_1 \in]t, T[$ be the smallest of the elements τ such that $\{\tau, \xi\}$ belongs to the stopping set $\mathcal{S}$. Then, for any $s \in]t, \bar{\theta}_1[$, we have $v(s, \xi) < Rv(s, \xi)$ and thus, by (88) and (80),

$$(89) \quad \frac{d}{ds} v(s, \bar{x}(s)) = -\varphi(s, \bar{x}(s)).$$

Integrating this equation from t to $\bar{\theta}_1$, we obtain

$$Rv(\bar{\theta}_1, \bar{x}(\bar{\theta}_1)) - v(t, \xi) = - \int_t^{\bar{\theta}_1} \varphi(s, \bar{x}(s)) ds.$$

If we write $Rv(\bar{\theta}_1, x(\bar{\theta}_1)) = 1 + v(\bar{\theta}_1, \bar{x}(\bar{\theta}_1) + \bar{\xi}_1)$, we obtain that

$$(90) \quad v(t, \xi) = \int_t^{\bar{\theta}_1} \varphi(s, x(s)) ds + 1 + v(\bar{\theta}_1, \bar{x}(\bar{\theta}_1) + \bar{\xi}_1).$$

Let $\bar{x}_1(s)$ be the solution of the differential equation

$$(91) \quad \dot{\bar{x}}_1(s) = \varphi(s, \bar{x}_1(s)) \quad \text{if } s > \bar{\theta}_1, \quad \bar{x}_1(\bar{\theta}_1) = \bar{x}(\bar{\theta}_1) + \bar{\xi}_1.$$

Let $\bar{\theta}_2 \in]\bar{\theta}_1, T]$ be the smallest of the elements $\tau \in]\bar{\theta}_1, T]$ that $\{\tau, \bar{x}(\bar{\theta}_1) + \bar{\xi}_1\}$ belongs to the stopping set $\mathcal{S}$. Then, for any $s \in]\bar{\theta}_1, \bar{\theta}_2[$, we have the relation (89), which, integrated from $\bar{\theta}_1$ to $\bar{\theta}_2$, yields

$$Rv(\bar{\theta}_2, \bar{x}_1(\bar{\theta}_2)) - v(\bar{\theta}_1, \bar{x}(\bar{\theta}_1) + \bar{\xi}_1) = - \int_{\bar{\theta}_1}^{\bar{\theta}_2} \varphi(s, \bar{x}_1(s)) \, ds.$$

Writing $Rv(\bar{\theta}_2, x_1(\bar{\theta}_2)) = 1 + v(\bar{\theta}_2, x_1(\bar{\theta}_2) + \bar{\xi}_2)$, we have that

$$(92) \quad v(\bar{\theta}_1, \bar{x}(\bar{\theta}_1) + \bar{\xi}_1) = \int_{\bar{\theta}_1}^{\bar{\theta}_2} \varphi(s, \bar{x}_1(s)) \, ds + 1 + v(\bar{\theta}_2, \bar{x}_1(\bar{\theta}_2) + \bar{\xi}_2).$$

Define by induction a sequence of times $\bar{\theta}_i$ and of vectors $\bar{\xi}_i$ such that

$$(93) \quad v(\bar{\theta}_i, \bar{x}_{i-1}(\bar{\theta}_i) + \bar{\xi}_i) = \int_{\bar{\theta}_i}^{\bar{\theta}_{i+1}} \varphi(s, \bar{x}_i(s)) \, ds + 1 + v(\bar{\theta}_{i+1}, \bar{x}_i(\bar{\theta}_{i+1}) + \bar{\xi}_{i+1}),$$

where $\bar{x}_i(\cdot)$ is the solution of the differential equation

$$(94) \quad \begin{cases} \text{(i) } \dot{\bar{x}}_i(s) = \varphi(s, \bar{x}_i(s)) \text{ if } s \geq \bar{\theta}_i, \quad \bar{x}_i(\bar{\theta}_i) = \bar{x}_{i-1}(\bar{\theta}_i) + \bar{\xi}_i, \\ \text{(ii) } \bar{\theta}_{i+1} \text{ is the smallest of the } \tau \in]\bar{\theta}_i, T] \text{ such that} \\ \quad \{\tau, \bar{x}_{i-1}(\bar{\theta}_i) + \bar{\xi}_i\} \text{ belongs to the stopping set } \mathcal{S}. \\ \text{(iii) } \bar{\xi}_{i+1} \in \mathbf{R}_+^n \text{ achieves the minimum of } \xi \mapsto v(\bar{\theta}_{i+1}, \bar{x}(\bar{\theta}_i) + \xi). \end{cases}$$

Adding eqs. (93), we obtain that

$$(95) \quad v(t, \xi) = \int_t^T \varphi(s, \bar{x}(s)) \, ds + N_t(\bar{u}),$$

where $u = \sum \bar{\xi}_i \delta(\bar{\theta}_i)$. $\square$

CHAPTER 15

FIXED POINT THEOREMS, QUASI-VARIATIONAL INEQUALITIES AND CORRESPONDENCES

In this chapter we prove some more fixed point theorems and some existence theorems for the critical points x of a correspondence S (i.e. the solutions of $0 \in S(x)$). We also study quasi-variational inequalities (introduced in Chapter 9) and complete our account of the properties of correspondences.

15.1. Fixed point and surjectivity theorems for correspondences

This section is devoted to the proof of several existence theorems for the critical points of a correspondence and their application to fixed point and surjectivity theorems. We obtain another proof of the Kakutani theorem (Theorem 9.2.3).

The results we present here are very useful in Mathematical Economics.

We choose to deduce all results from an adaptation (due to B. Cornet) of a theorem of Browder and Ky-Fan which states that, *if X is a convex compact subset of a locally convex space U and if S is an upper hemi-continuous correspondence with non-empty closed convex images satisfying the Ky-Fan boundary condition: $\forall x \in X, S(x) \cap T_x(x) \neq \emptyset$ (where $T_x(x) = N_x(x)^-$ is the "tangent cone" to X at x), then there exists a critical point x of the correspondence S .*

The proof of this theorem is based on the Ky-Fan inequality (Theorem 7.1.3).

In particular, we are led to introduce "inward" and "outward" correspondences S . If S has compact, convex images, these terms mean respectively that

$$\forall x \in X, S(x) \cap (x + T_x(x)) \neq \emptyset \quad \text{and} \quad S(x) \cap (x - T_x(x)) \neq \emptyset.$$

Let X be a compact, convex subset of U and let S be an upper hemi-continuous correspondence from X into U with non-empty closed convex images. If S is either inward or outward, then S has a fixed point. If S is outward, then for all $y \in X$ there exists $x \in X$ such that $y \in S(x)$.

We then consider a family of correspondences $S(\cdot, \lambda)$ depending on $\lambda \in [0, 1]$

We shall prove that if $S(\cdot, 0)$ satisfies the assumptions of the Browder–Ky-Fan theorem and if the critical points of $S(\cdot, \lambda)$ (if any) do not belong to the boundary of X , then $S(\cdot, 1)$ does have critical points (Granas's theorem). Finally, we mention several generalizations of the Browder–Ky-Fan theorem which may be useful.

15.1.1. The Browder–Ky-Fan existence theorem for critical points

Let U be a locally convex Hausdorff vector space and let U^* be its dual. We assume once and for all that

- $$(1) \quad \begin{cases} (i) & X \text{ is a convex compact subset of } U, \\ (ii) & S \text{ is an upper hemi-continuous correspondence from } X \text{ into } U \text{ with non-empty closed convex images (see Definition 2.5.2).} \end{cases}$$

Recall that upper semi-continuous correspondences from X into U supplied with the weak topology are upper hemi-continuous (see Proposition 2.5.1).

Definition 1. Let S be a correspondence from X into U . We shall say that x is a “critical point” of S if $0 \in S(x)$.

To prove the existence of a critical point of S , we have to add to assumption (1) a consistency assumption relating S and X .

Definition 2. Let X be a closed convex subset of U . We shall set

- $$(2) \quad \begin{cases} (i) & N_X(x) = \{p \in U^* \text{ such that } \langle p, x \rangle = \max_{y \in X} \langle p, y \rangle\}, \\ (ii) & \partial(X; p) = \{x \in X \text{ such that } \langle p, x \rangle = \min_{y \in X} \langle p, y \rangle\}, \\ (iii) & T_X(x) = N_X(x)^- = \{x \in U \text{ such that } \langle p, x \rangle \leq 0, \forall p \in N_X(x)\} \end{cases}$$

and say that $N_X(x)$ is the “normal cone” to X at x , $\partial(X, p)$ is a “supporting set” of X and that $T_X(x)$ is the “tangent cone” to X at x . We shall say that S satisfies the “Ky-Fan boundary-condition” if

- $$(3) \quad \forall p \in U^*, \forall x \in \partial(X; p), \quad \sigma^*(S(x), p) \geq 0.$$

We begin by proving a characterization of the Ky-Fan boundary condition.

Proposition 1. Suppose that S satisfies

- $$(4) \quad \forall x \in X, \quad S(x) \cap T_X(x) \neq \emptyset.$$

Then S satisfies by Ky-Fan boundary condition. The converse is true if the images $S(x)$ are compact.

Proof. We begin by noticing that

$$(5) \quad x \in \partial(X; p) \text{ if and only if } -p \in N_X(x).$$

Let $p \in U^*$ and $x \in \partial(X; p)$. Since there exist $y \in S(x) \cap T_X(x)$ and $-p \in N_X(x) = T_X(x)^-$, we deduce that $\sigma^*(S(x), p) \geq \langle p, y \rangle \geq 0$.

For the *converse*, we assume (3). Since $N_X(x)$ is convex and $S(x)$ is convex and compact, the minisup Theorem 7.1.5 implies that there exists $\bar{y} \in S(x)$ such that

$$0 \leq \inf_{-p \in N_X(x)} \sigma^*(S(x), p) = \inf_{-p \in N_X(x)} \sup_{y \in S(x)} \langle p, y \rangle = \inf_{-p \in N_X(x)} \langle p, \bar{y} \rangle$$

The latter inequality implies that $\bar{y} \in T_X(x)$, and thus, that (4) is satisfied. $\square$

Remark. Notice that, if $x \in \text{Int}(X)$ (for the initial topology of U), then $N_X(x) = 0$, $T_X(x) = U$.

Remark. If S is a (single valued) map from X into U , then the Ky-Fan boundary condition can be written:

$$\text{either } S(x) \in T_X(x) \text{ for all } x \in X \text{ or } \forall p \in U^*, \forall x \in \partial(X; p), \\ \langle p, S(x) \rangle \geq 0.$$

Theorem 1. Suppose that (1) holds. If S is upper hemi-continuous and satisfies the Ky-Fan boundary condition (4), then there exists $\bar{x} \in X$ such that $0 \in S(\bar{x})$.

Proof. Let us assume that the conclusion is false: we will contradict the Ky-Fan boundary condition (3). Since for all $x \in X$, we assume that $0 \notin S(x)$, we deduce that there exists $p \in U^*$ such that $\sigma^*(S(x); p) < 0 = \langle p, 0 \rangle$ (since $S(x)$ is closed and convex). If we set

$$(6) \quad V(p) = \{x \in X \text{ such that } \sigma^*(S(x), p) < 0\},$$

the contradiction of the conclusion amounts to writing

$$(7) \quad X \subset \bigcup_{p \in U^*} V(p).$$

Since the correspondence S is upper hemi-continuous, the subsets $V(p)$ are open. Since X is compact, this implies that

$$(8) \quad X \subset \bigcup_{i=1}^n V(p_i)$$

and that there exists a continuous partition of unity $\{\beta_1, \dots, \beta_n\}$ subordinate to this covering of X .

We introduce the function $a : X \times X \rightarrow \mathbb{R}$ defined by

$$(9) \quad a(x, y) = \sum_{i=1}^n \beta_i(x) \langle p_i, x - y \rangle.$$

It obviously satisfies the assumptions of the Ky-Fan Theorem 7.1.3. Hence there exists $\bar{x} \in X$ such that $\sup_{y \in X} a(\bar{x}, y) \leq 0$, i.e.

$$(10) \quad \langle \bar{p}, \bar{x} \rangle \leq \inf_{y \in X} \langle \bar{p}, y \rangle \quad \text{in which } \bar{p} = \sum_{i \in I} \beta_i(\bar{x}) p_i,$$

where $I = \{i = 1, \dots, n \text{ such that } \beta_i(\bar{x}) > 0\}$. This set is non-empty. Inequality (10) amounts to saying that $\bar{x} \in \partial(X, \bar{p})$. It therefore implies that $\sigma^*(S(\bar{x}), \bar{p}) \geq 0$. But, since $\beta_i(\bar{x}) > 0$ and $\sum_{i \in I} \beta_i(\bar{x}) = 1$, we obtain that

$$\sigma^*(S(\bar{x}), \bar{p}) \leq \sum_{i \in I} \beta_i(\bar{x}) \sigma^*(S(\bar{x}), p_i) < 0,$$

because if $i \in I$, then $\beta_i(\bar{x}) > 0$ and thus, $\bar{x} \in V(p_i)$, i.e. $\sigma^*(S(\bar{x}), p_i) < 0$. Therefore, we have obtained a contradiction. $\square$

Corollary 1. *Suppose that U is a Hilbert space (supplied with the weak topology) and that X is the unit ball of U . If S is an upper hemi-continuous correspondence from X into U with non-empty closed convex images satisfying*

$$(11) \quad \forall x \in X \text{ such that } \|x\| = 1, \text{ then } \sigma^*(S(x), x) \geq 0,$$

then S has a critical point $\bar{x}$.

Proof. Identify U with U^* . Then X is weakly compact and, $\forall p \in U^*$, $\partial(X; p) = 0$ if $p = 0$ and $\partial(X; p) = \{-p/\|p\|\}$ if $p \neq 0$. Therefore (11) obviously implies the Ky-Fan boundary condition (3) for $-S$. Hence $-S$ has a critical point (which is also a critical point of S)! $\square$

Corollary 2. (Altman). *Let U be a Hilbert space (supplied with the weak topology) and let X be the unit ball of U . Let S be an upper hemi-continuous correspondence from X into U with non-empty closed convex values satisfying*

$$(12) \quad \forall x \in X \text{ such that } \|x\| = 1, \exists u \in S(x), \quad \|u\|^2 \geq \|u - x\|^2 - \|x\|^2$$

Then S has a critical point.

Proof. Since

$$\langle u, x \rangle = \frac{1}{2} (\|u\|^2 + \|x\|^2 - \|u - x\|^2)$$

it is clear that condition (12) implies condition (11) of Corollary 1. Thus there exists a critical point. $\square$

Definition 3. We shall say that a correspondence S from X into U is “*semi-coercive*” if

$$(13) \quad \lim_{\substack{\|x\| \rightarrow \infty \\ x \in X}} \frac{\sigma^*(S(x), x)}{\|x\|} > 0.$$

It is clear that any coercive correspondence is semi-coercive (see Definition 7.2.1).

Corollary 3. Let U be a Hilbert space supplied with the weak topology. Let S be a correspondence from U into U with non-empty closed convex images satisfying

$$(14) \quad S \text{ is semi-coercive and upper hemi-continuous.}$$

Then S has a critical point.

Proof. Let $a = \lim_{x \rightarrow \infty} \sigma^*(S(x), x) / \|x\| > 0$. We take $\varepsilon = \frac{1}{2}a$. There exists $A > 0$ such that $\sigma^*(S(x), x) \geq \frac{1}{2}a\|x\|$ whenever $\|x\| > A$. In particular, if $\|x\| = A$, $\sigma^*(S(x), x) \geq \frac{1}{2}aA$. Hence the restriction of S to the ball of radius A satisfies the boundary condition (11) of Corollary 1, and thus, has a critical point in this ball. $\square$

Corollary 4 (Cornet). Let $X = [-1, +1]^n$ be the unit cube of $\mathbf{R}^n$ and $F = f_1, \dots, f_n$ be a continuous map from X into $\mathbf{R}^n$ satisfying the boundary condition

$$(15) \quad \begin{aligned} \forall i = 1, \dots, n, \quad f_i(x) &\leq 0 \text{ whenever } x_i = +1 \text{ and} \\ f_i(x) &\geq 0 \text{ whenever } x_i = -1. \end{aligned}$$

Then there exists $\bar{x} \in [-1, +1]^n$ such that $F(\bar{x}) = 0$.

Proof. Let $p \in \mathbf{R}^{n*}$. We set $I_p^+ = \{i \text{ such that } p_i > 0\}$ and $I_p^- = \{i \text{ such that } p_i < 0\}$. Then $\partial(X, p) = \{x \in [-1, +1]^n \text{ such that } x_i = +1 \text{ if } i \in I_p^- \text{ and } x_i = -1 \text{ if } i \in I_p^+\}$. Hence, if $x \in \partial(X, p)$,

$$\langle p, F(x) \rangle = \sum_{i \in I_p^+} p_i f_i(x) + \sum_{i \in I_p^-} p_i f_i(x) \geq 0$$

when (15) holds. Thus (15) is nothing other than the Ky-Fan boundary condition. $\square$

We shall consider now the case when X is a simplex.

Lemma 1. Let $x = \mathcal{M}^n = \{x \in \mathbf{R}_+^n \text{ such that } \sum_{i=1}^n x^i = 1\}$. If $\mathbf{1}$ denotes the vector $\{1, \dots, 1\}$, we have

$$(16) \quad \begin{cases} \text{(i) } \partial(X, \lambda \mathbf{1}) = \mathcal{M}^n \\ \text{(ii) } \forall p \neq \lambda \mathbf{1}, \partial(X, p) = \partial(X, q) = \{x \in \mathcal{M}^n \text{ such that } \langle q, x \rangle = 0\} \\ \text{where } q = p - (\min_{1 \leq j \leq n} p^j) \mathbf{1}. \end{cases}$$

Proof. Let $p = \{p^1, \dots, p^n\} \in \mathbf{R}^{n*}$. We set $I(p) = \{i \text{ such that } p_i > q = \min_{1 \leq j \leq n} p^j\}$. Hence if $X = \mathcal{M}^n$, we obtain that

$$(17) \quad \partial(X, p) = \{x \in \mathcal{M}^n \text{ such that } x^i = 0 \text{ for all } i \in I(p)\}.$$

To see this, let $\bar{x} \in \mathcal{M}^n$ minimize $\langle p, x \rangle$ over $\mathcal{M}^n$. Suppose that i belongs to $I(p)$ and $j \notin I(p)$. Take $y \in \mathcal{M}^n$ such that $y^i + y^j = 1 - \sum_{k \neq i, j} \bar{x}^k = a$ and $y^k = \bar{x}^k$ for $k \neq i, j$. Hence

$$(p^i - q) \bar{x}^i + qa = p^i \bar{x}^i + q \bar{x}^j \leq p^i y^i + q y^j = (p^i - q) y^i + qa$$

for any $y^i \in [0, 1]$. Since $p^i - q > 0$, this implies that $\bar{x}^i = 0$. Now,

$$(18) \quad \partial(X, p + \lambda \mathbf{1}) = \partial(X, p),$$

since $I(p + \lambda \mathbf{1}) = I(p)$. We can therefore consider only the linear forms p such that $\min_{1 \leq j \leq n} p^j = 0$. Hence, since $I(p) = \{i \text{ such that } p^i > 0\}$, we deduce that $x \in \mathcal{M}^n$ satisfies $\langle p, x \rangle = 0$ if and only if $x^i = 0$ for all $i \in I(p)$. Thus (17) and (18) imply (16). $\square$

Corollary 5 (Cornet). Let $X = \mathcal{M}^n$, let $H = \{x \in \mathbf{R}^n \text{ such that } \sum_{i=1}^n x_i = 0\}$ and let F be a continuous map from $\mathcal{M}^n$ into H . If F satisfies the boundary condition

$$(19) \quad f_i(x) \geq 0 \text{ whenever } x_i = 0,$$

then there exists $\bar{x} \in \mathcal{M}^n$ such that $F(\bar{x}) = 0$.

Proof. We apply Theorem 1 and Lemma 1. If $p = \lambda \mathbf{1}$, then $\partial(X, p) = \mathcal{M}^n$ and $\langle p, F(x) \rangle = \lambda \sum_{i=1}^n f_i(x) = 0$. If $p \neq \lambda \mathbf{1}$ and $x \in \partial(X, p)$, then $x^i = 0$

whenever $i \in I(p) = \{j \text{ such that } p^j > q = \min_k p^k\}$. Hence

$$\begin{aligned} \langle p, F(x) \rangle &= \sum_{i \in I(p)} p^i f_i(x) + q \sum_{i \notin I(p)} f_i(x) \\ &= \sum_{i \in I(p)} (p^i - q) f_i(x) + q \sum_{i=1}^p f_i(x) = \\ &= \sum_{i \in I(p)} (p^i - q) f_i(x) \geq 0 \end{aligned}$$

by (19).

Thus the Ky-Fan boundary condition is satisfied. $\square$

Corollary 6. Let $X = \mathcal{M}^n$ and let F be a continuous map from $\mathcal{M}^n$ into $\mathbf{R}$ satisfying the "Walras law", i.e.

$$(20) \quad \forall x \in \mathcal{M}^n, \quad \langle x, F(x) \rangle = \sum_{i=1}^n x_i f_i(x) = 0.$$

Then there exists $\bar{x} \in \mathcal{M}^n$ such that, for any $i = 1, \dots, n$, either $\bar{x} = 0$ or $f_i(\bar{x}) = 0$ or both.

Proof. We apply Corollary 5 to the map G defined by

$$(21) \quad \forall i = 1, \dots, n, \quad g_i(x) = x_i f_i(x).$$

Since $x_j = 0$ obviously implies that $g_j(x) = 0$, there exists $x \in \mathcal{M}^n$ such that

$$(22) \quad \forall j = 1, \dots, n, \quad g_j(\bar{x}) = 0. \quad \square$$

Remark. If $\bar{x} \in \mathring{\mathcal{M}}^n$, then we deduce that

$$(23) \quad \forall j = 1, \dots, n, \quad f_j(\bar{x}) = 0.$$

We obtain the same conclusion under other assumptions below.

Proposition 2 (Cornet-Lasry). Let F be a continuous map from $\mathcal{M}^n$ into $\mathbf{R}^n$ satisfying

$$(24) \quad \begin{cases} \text{(i) } \forall x \in \mathcal{M}^n, \langle x, F(x) \rangle = 0, \\ \text{(ii) } F(\mathring{\mathcal{M}}^n) \text{ is bounded below,} \\ \text{(iii) } \exists \alpha > 0 \text{ s.t., } \forall x \notin \mathring{\mathcal{M}}^n, \liminf_{y \rightarrow x} \sum_{i=1}^n f_i(x) \geq \alpha. \end{cases}$$

Then there exists $\bar{x} \in \mathcal{M}^n$ such that $F(\bar{x}) = 0$.

Remark. This result is the counter part of the Debreu Theorem 8.2.2.

Remark. Notice that the more familiar condition

$$(25) \quad \forall x \in \mathcal{M}^n \setminus \dot{\mathcal{M}}^n, \quad \liminf_{y \rightarrow x} \sum_{i=1}^n f_i(x) = +\infty$$

implies the "boundary condition" (41(iii)).

Proof. By (24(ii)) there exists a scalar $a < 0$ such that $\inf_{x \in \mathcal{M}^n} f_i(x) > a$ for all i .

We introduce the correspondence T mapping $\mathcal{M}^n$ into $\mathcal{M}^n$ defined by

$$T(x) = \begin{cases} \left\{ \frac{f_i(x) - a}{\sum_{i=1}^n f_i(x) - na} \right\}_{i=1, \dots, n} & \text{if } x \in \dot{\mathcal{M}}^n, \\ \left\{ y \in \mathcal{M}^n \text{ such that } \langle y, x \rangle \leq \frac{1}{n} - \varepsilon \right\} & \text{if } x \notin \dot{\mathcal{M}}^n. \end{cases}$$

where

$$\varepsilon = -2 \frac{\alpha}{n^2 a} > 0 \quad (\text{since } \alpha > 0 \text{ and } a < 0).$$

We shall prove that the correspondence S defined by

$$S(x) = T(x) - \frac{1}{n} \mathbf{1}$$

is upper semi-continuous (i.e. is closed), satisfies the Ky-Fan boundary condition (4) and that any $\bar{x}$ satisfying $0 \in S(\bar{x})$ belongs to $\dot{\mathcal{M}}^n$ and is a solution of $F(\bar{x}) = 0$.

(a) *The correspondence T is closed.* Let $\{y_p, x_p\}$ be a sequence of points of the graph of T converging to $\{y, x\}$. If $x \in \dot{\mathcal{M}}^n$, it is clear that $y = T(x)$. If $x \notin \dot{\mathcal{M}}^n$, we have to check that $\langle y, x \rangle \leq (1/n) - \varepsilon$. If there exists an infinity of integers p such that $x_p \notin \dot{\mathcal{M}}^n$, then inequalities $\langle y_p, x_p \rangle \leq (1/n) - \varepsilon$ imply that $\langle y, x \rangle \leq (1/n) - \varepsilon$, i.e. that $y \in T(x)$. If there exists an infinity of integers p such that $x_p \in \dot{\mathcal{M}}^n$, then

$$\langle y_p, x_p \rangle = \frac{\sum_{i=1}^n x_{i,p} f_i(x_p) - a \sum_{i=1}^n x_{i,p}}{\sum_{i=1}^n f_i(x_p) - na} = \frac{-a}{\sum_{i=1}^n f_i(x_p) - na}$$

by the Walras law (24(i)).

By assumption (24(iii)), we have, since $a < 0$,

$$\sum_{i=1}^n f_i(x_p) - na \geq \alpha - na$$

when p is large enough. Hence,

$$\langle y_p, x_p \rangle \leq \frac{-a}{\alpha - na} \leq \frac{1}{n} \left(\frac{1}{1 - \frac{\alpha}{na}} \right) \leq \frac{1}{n} + 2 \frac{\alpha}{n^2 a} = \frac{1}{n} - \varepsilon$$

since we can choose α such that

$$0 < \frac{\alpha}{na} < 1.$$

Thus, $\langle y, x \rangle \leq (1/n) - \varepsilon$. Hence the graph is closed and T and S are upper semi-continuous.

(b) S satisfies the Ky-Fan boundary condition. This is clear if $p = \lambda \mathbf{1}$, since $\partial(\mathcal{M}^n; \mathbf{1}) = \mathcal{M}^n$ and $\lambda \langle \mathbf{1}, S(x) \rangle = \lambda[1 - 1] = 0$ for $x \in \mathcal{M}^n$. Let $p \in \mathbf{R}^{n*}$ satisfy $\min p^j = 0$ and let $I(p) = \{i \text{ such that } p^i > 0\}$. Let $x \in \partial(\mathcal{M}^n; p)$. Then if $i \in I(p)$, $e^i = \{0, \dots, 1, 0, \dots, 0\}$ belongs to $T(x)$ since $\langle e^i, x \rangle = x_i = 0 \leq (1/n) - \varepsilon$ because $x_i = 0$ when $x \in \partial(\mathcal{M}^n; p)$. Since $T(x)$ is convex,

$$y = \frac{1}{|I(p)|} \sum_{i \in I(p)} e_i \text{ belongs to } T(x) \text{ and } z = y - \frac{1}{n} \mathbf{1}$$

belongs to $S(x)$ and satisfies

$$\langle p, z \rangle = \langle p, y \rangle - \frac{1}{n} \sum_{i=1}^n p_i = \left(\frac{1}{|I(p)|} - \frac{1}{n} \right) \sum_{i=1}^n p_i \geq 0$$

since $|I(p)| \leq n$.

(c) By Theorem 1, there exists $\bar{x}$ such that $0 \in S(\bar{x})$. Such an $\bar{x}$ belongs to $\mathcal{M}$. Otherwise $0 = z - (1/n)\mathbf{1}$ with $\sum_{i=1}^n z^i x_i \leq (1/n) - \varepsilon$, which is impossible. Hence $\bar{x} \in \mathcal{M}^n$. This implies that, for any i ,

$$f_i(\bar{x}) = a + \frac{1}{n} \left(\sum_{i=1}^n f_i(\bar{x}) - na \right) = b.$$

The Walras law (41(i)) implies that $\sum_{i=1}^n f_i(\bar{x}) \bar{x}_i = b \sum_{i=1}^n \bar{x}_i = b = 0$. Hence $f_i(\bar{x}) = b = 0$ for all $i = 1, \dots, n$. $\square$

Remark. The above Theorem 1 is equivalent to the following result due to Ky-Fan.

Theorem 2 (Ky-Fan). *Let S and T be two upper hemi-continuous correspondences satisfying*

$$(26) \quad \forall x \in X, \quad S(x) - T(x) \text{ is non-empty, convex, and closed}$$

and

$$(27) \quad \forall p \in U^*, \forall x \in \partial(X; p) \quad \exists u \in S(x) \text{ and } v \in T(x) \\ \text{such that } \langle p, u \rangle \geq \langle p, v \rangle.$$

Hence there exists $\bar{x} \in X$ such that $S(\bar{x}) \cap T(\bar{x}) \neq \emptyset$

Proof. Theorem 2 follows from Theorem 1 on replacing S by $S - T$. Theorem 1 follows from Theorem 2 on taking $T = 0$.

15.1.2. Properties of inward and outward correspondences

It will be convenient to use the following concepts introduced by Ky-Fan.

Definition 3. We shall say that a correspondence S is “outward” (resp. “inward”) if

$$(28) \quad \forall p \in U^*, \forall x \in \partial(X; p), \quad \langle p, x \rangle \geq \sigma^b(S(x), p) \\ (\text{resp. } \sigma^*(S(x), p) \geq \langle p, x \rangle)$$

“strongly inward” if

$$(29) \quad \forall p \in U^*, \forall x \in \partial(X; p) \quad \sigma^*(S(x), p) \geq \sigma^*(X, p).$$

It is clear that *any strongly inward correspondence is inward and that any correspondence mapping X into itself is inward.*

We also notice that S is *outward (resp. inward) if and only if $I-S$ (resp. $S-I$) satisfies Ky-Fan the boundary-condition (3).*

If a correspondence S satisfies

$$(30) \quad \forall x \in X, \quad S(x) \cap (x + T_x(x)) \neq \emptyset,$$

then S is inward. The converse is true if the images $S(x)$ are convex and compact.

In the same way, any correspondence S satisfying

$$(31) \quad \forall x \in X, \quad S(x) \cap (x - T_x(x)) \neq \emptyset$$

is outward, the converse being true if the images $S(x)$ of S are convex and compact (see Proposition 1).

We point out the following examples of outward and strongly inward correspondences.

Proposition 3. *The condition*

$$(32) \quad \forall p \in U^*, \forall x \in \partial(X; p), \quad S(x) \cap \partial(X; p) \neq \emptyset$$

implies that S is outward and the condition

$$(33) \quad \forall p \in U^*, \forall x \in \partial(X; p), \quad S(x) \cap \partial(X; -p) \neq \emptyset$$

implies that S is strongly inward.

Proof. We prove the first statement. Condition (32) implies that there exists $y \in S(x)$ such that $y \in \partial(X; p)$, i.e. such that $\langle p, y \rangle = \langle p, x \rangle = \min_{z \in X} \langle p, z \rangle$. Hence $\sigma^b(S(x), p) \leq \langle p, y \rangle = \langle p, x \rangle$, and (28) implies that S is outward. The proof of the second statement is similar. $\square$

By applying the Browder–Ky–Fan theorem to the correspondences $I-S$ and $S-I$, we obtain the following fixed point theorem.

Theorem 3 (Ky–Fan). *Suppose that (I) holds. If S is either inward or outward then there exists a fixed point of S .*

In particular, Theorem 3 implies the Kakutani fixed point Theorem 9.3.4. We now consider any fixed y in X . Applying the Browder–Ky–Fan theorem to the correspondences $y-S$ and $S-y$, we obtain the following result.

Theorem 4 (Rogalski–Cornet). *Suppose that (I) holds. If S is either outward or strongly inward, then, for any $y \in X$, there exists a solution $\bar{x} \in X$ to the multi-valued equation $y \in S(\bar{x})$.*

Proof. Notice that, if S is outward, the correspondence $y-S$ satisfies the Ky–Fan boundary condition since

$$(34) \quad \forall p \in U^*, \forall x \in \partial(X; p), \quad \sigma^*(y-S(x), p) = \langle p, y \rangle - \sigma^b(S(x), p) \geq \langle p, x \rangle - \sigma^b(S(x), p) \geq 0.$$

Thus there exists a solution $\bar{x}$ to the equation $0 \in y-S(\bar{x})$.

Consider now the case when S is strongly inward. Then the correspondence S - y satisfies the Ky-Fan boundary condition since

$$(35) \quad \begin{aligned} \forall p \in U^*, \forall x \in \partial(X; p), \quad \sigma^*(S(x) - y, p) &= \sigma^*(S(x), p) - \langle p, y \rangle \\ &\geq \sigma^*(S(x), p) - \sigma^*(X, p) \geq 0. \end{aligned}$$

Hence there exists a solution $\bar{x}$ to the equation $0 \in S(\bar{x}) - y$. $\square$

In particular, we deduce the following statement from Proposition 3.

Theorem 5. *Suppose that (I) holds and that S satisfies either condition (32) or condition (33). Then, for any $y \in X$, there exists $\bar{x}$ such that $y \in S(\bar{x})$.*

Proof. Obviously, condition (32) implies that S is outward and condition (33) implies that S is strongly inward. Hence Theorem 5 follows from Theorem 4. $\square$

We point out the following particular cases.

Proposition 4. *Suppose that (I) holds. If S is upper hemi-continuous and maps X into itself, then either the condition*

$$(36) \quad \forall p \in U^*, \quad S[\partial(X; p)] \subset \partial(X; p)$$

or the condition

$$(37) \quad \forall p \in U^*, \quad S[\partial(X; p)] \subset \partial(X; -p)$$

implies that S maps X onto X .

Proposition 5. *Suppose that X is the unit ball of a Hilbert space U and that F is a weakly continuous map from X into X satisfying*

$$(38) \quad \text{for every } x \text{ such that } \|x\| = 1, F(x) = \pm x.$$

Then $F(X) = X$.

Proof. We apply the above Proposition 4 since in this case,

$$\partial(X; \pm p) = \mp \left\{ \frac{p}{\|p\|} \right\}. \quad \square$$

Example. We take

$$(39) \quad U = \mathbf{R}^{n*}, \quad X = \mathcal{M}^n, \quad \partial X = \mathcal{M}^n \setminus \hat{\mathcal{M}}^n.$$

Then, for any $p \in \mathbb{R}^n$ such that $\max_j p^j > \min_j p^j = 0$,

$$\partial(X; p) = \{x \in \partial X \text{ such that } \langle p, x \rangle = 0\}$$

by Lemma 1. We obtain the following result.

Proposition 6 (Cornet). *Let S be an upper hemi-continuous correspondence from $\mathcal{M}^n$ into itself, with non-empty convex compact values. If S satisfies the boundary condition*

$$(40) \quad \forall x \in \partial X, \quad S(x) = \{y \in \mathcal{M}^n \text{ such that } \langle y, x \rangle = 0\},$$

then S maps $\mathcal{M}^n$ onto itself.

15.1.3. Critical points of homotopic correspondences

From Theorem 1 we can deduce other existence theorems using the Poincaré method. We adapt a result due to Granas.

Suppose that

$$(41) \quad \begin{cases} \text{(i) } U \text{ is a finite dimensional space,} \\ \text{(ii) } X \text{ is a convex compact subset of } U \text{ whose interior is non-} \\ \quad \text{empty.} \end{cases}$$

Then the boundary $\partial X = X \setminus \text{Int}(X)$ is distinct from X .

Theorem 6 (Granas). *Suppose that (41) holds. Let S be an upper hemi-continuous correspondence from $X \times [0, 1]$ into U with non-empty closed convex images. We assume that, for $t = 0$, the correspondence $x \mapsto S(x, 0)$ satisfies the Ky-Fan boundary condition (6).*

If we assume furthermore that

$$(42) \quad \forall t \in [0, 1[, \quad \text{all the critical points of } S(\cdot, t) \text{ (if any) belong to } \text{Int}(X),$$

then the correspondence $x \mapsto S(x, 1)$ has critical points.

Remark. The correspondences $S(\cdot, 0)$ and $S(\cdot, 1)$ are said to be “homotopic”. Theorem 6 states that under assumption (42), a correspondence homotopic to a correspondence satisfying the Ky-Fan boundary condition also has critical points.

Proof. We shall assume that $S(\cdot, 1)$ does not have critical points and obtain a contradiction. We write $A = \partial X$ which is a closed subset of X , and

$$(43) \quad B = \{x \in X \text{ such that } \exists t \in [0, 1] \text{ satisfying } 0 \in S(x, t)\}.$$

The set B is non-empty since it contains the critical points of $S(\cdot, 0)$, which exist by the Browder–Ky-Fan Theorem 1. It is closed, since S is an upper hemi-continuous correspondence from $X \times [0, 1]$ into U . Moreover, $A \cap B = \emptyset$ because, if $x \in A$ and if $t \in [0, 1]$, assumption (42) implies that $x \notin B$. If $x \in A$ and $t = 1$, $x \notin B$ since we assumed that $S(\cdot, 1)$ has no critical points.

If $d(x, A) = \inf_{y \in A} d(x, y)$ denotes the distance from x to A , then $d(x, A) + d(x, B) > 0$ for all $x \in X$. We introduce the continuous function f defined on X by

$$(44) \quad f(x) = \frac{d(x, A)}{d(x, A) + d(x, B)} \in [0, 1].$$

It is equal to 0 on A and to 1 on B . We define the correspondence T on X by

$$(45) \quad T(x) = S(x, f(x)),$$

which is obviously upper semi-continuous with non-empty closed convex images. If $x \in A$, $T(x) = S(x, 0)$. Therefore, T satisfies the Ky-Fan boundary condition (3) and thus, has a critical point $\bar{x}$ by the Browder–Ky-Fan Theorem 1. Therefore $0 \in T(\bar{x}) = S(\bar{x}, f(\bar{x}))$. By the very definition of B , this implies that $\bar{x} \in B$ and thus, that $f(\bar{x}) = 1$. Hence $0 \in S(\bar{x}, 1)$, which is a contradiction. $\square$

In particular, we obtain the following result.

Theorem 7. *Suppose that (41) holds. Let R and S be two upper hemi-continuous correspondences from X into U with non-empty closed convex images. Suppose that*

$$(46) \quad \begin{cases} \text{(i) } R \text{ satisfies the Ky-Fan boundary condition (3),} \\ \text{(ii) } \forall s \geq 0, \forall x \in \partial X, 0 \notin R(x) + sS(x). \end{cases}$$

Then the correspondence S has critical points.

Proof. Theorem 7 follows from Theorem 6 by setting $S(x, t) = (1-t)R(x) + tS(x)$. Assumption (46(i)) obviously implies assumption (42). $\square$

As a useful particular case, we mention the following result.

Theorem 8. *Suppose that (41) holds and let x_0 belong to the interior of X . Let S be an upper hemi-continuous correspondence from X into U with non-empty closed convex images. If*

$$(47) \quad \forall s > 0, \forall x \in \partial X, \quad x_0 \notin x + sS(x),$$

then S has critical points.

Proof. Theorem 8 follows from Theorem 7 by taking $R(x) = x - x_0$. $\square$

15.1.4. Other existence theorems for critical points

The Ky-Fan boundary condition is only one of the possible sufficient conditions for the existence of a critical point of a correspondence S . It is a particular case of a family of boundary conditions which we are about to describe.

Consider two locally Hausdorff vector spaces U and V , a subset X of U and a correspondence S from X into V .

The boundary conditions depend upon

$$(48) \quad \text{a continuous linear operator } L \in \mathcal{L}(U, V)$$

and a function f from $X \times X$ into $\mathbf{R}$ satisfying

$$(49) \quad \begin{cases} \text{(i) } \forall x \in X, y \mapsto f(x, y) \text{ is concave,} \\ \text{(ii) } \forall y \in X, x \mapsto f(x, y) \text{ is lower semi-continuous,} \\ \text{(iii) } \forall y \in X, f(y, y) \leq 0. \end{cases}$$

We assume that

$$(50) \quad X \text{ is a convex compact subset of } U.$$

The Ky-Fan Theorem 7.1.3 implies that the subsets

$$(51) \quad \partial_{L,f}(X; p) = \{x \in X \text{ such that } \sup_{y \in X} [f(x, y) + \langle L^*p, x - y \rangle] \leq 0\}$$

are non-empty. Notice that, when $U = V$, $L = I$ and $f = 0$, $\partial_{L,f}(X; p) = \partial(X; p)$. The Browder-Ky-Fan Theorem 1 can be extended as follows.

Theorem 9. *Suppose that assumptions (48), (49) and (50) holds. Let S be a correspondence from X into V which is upper hemi-continuous with non-empty closed convex images. Suppose moreover that*

$$(52) \quad \forall p \in U^*, \forall x \in \partial_{L,f}(X; p), \quad \sigma^*(S(x), p) \geq 0.$$

Then there exists a critical point of S .

Proof. This is the same as that of Theorem 1 except that the function a defined on $X \times X$ by (9) is replaced by the function

$$(53) \quad a(x, y) = f(x, y) - \sum_{i=1}^n \beta_i(x) \langle L^*p_i, x - y \rangle.$$

Property (10) becomes $\bar{x} \in \partial_{L,f}(X; \bar{p})$. Then $\sigma^*(S(\bar{x}), \bar{p}) \geq 0$ by assumption (52) and $\sigma^*(S(\bar{x}), \bar{p}) < 0$ by construction. Thus the non-existence of a critical point of S implies a contradiction. $\square$

Remark. An important example of functions f satisfying (49) is the function f defined by $f(x, y) = g(x) - g(y)$ where g is a lower semi-continuous convex function.

We shall now relax the assumption that X is convex compact.

Theorem 10. Let S be an upper hemi-continuous correspondence from a topological space X into a Hausdorff locally convex vector space V , with non-empty closed convex images. Suppose that

$$(54) \quad \begin{cases} \text{(i) either } X \text{ is compact,} \\ \text{(ii) or, more generally, that there exist } p_1, \dots, p_n \in V^* \text{ such that} \\ \quad \{x \in X \text{ such that } \max_{i=1, \dots, n} \sigma^b(S(x), p_i) \leq 0\} \text{ is compact.} \end{cases}$$

Finally, we assume that

$$(55) \quad \begin{cases} \forall \varepsilon > 0, \text{ there exists a continuous map } C_\varepsilon \text{ from } V^* \text{ supplied with} \\ \text{the finite topology into } X \text{ such that } \sup_{p \in V^*} \sigma^b(S(C_\varepsilon(p)), p) \leq \varepsilon. \end{cases}$$

Then there exists a critical point of S .

Proof. If we set

$$(56) \quad f(x, p) = \sigma^b(S(x), p),$$

then any critical point $\bar{x}$ of S is a solution to the variational inequalities $\sup_{p \in V^*} f(\bar{x}, p) \leq 0$. But such a solution exists by Theorem 7.1.2. To see this, observe that f is lower semi-continuous with respect to x (for S is upper hemi-continuous), f is concave with respect to p and the function $x \mapsto \max_{i=1, \dots, n} f(x, p_i)$ is lower semi-compact by assumption (54). Thus there exists a solution $\bar{x}$ to

$$\begin{aligned} \sup_{p \in V^*} f(\bar{x}, p) &\leq \inf_{C \in \mathcal{C}(V^*, X)} \sup_{p \in V^*} f(C(p), p) \\ &\leq \inf_{\varepsilon > 0} \sup_{p \in V^*} \sigma^b(S(C_\varepsilon(p)), p) \leq \inf_{\varepsilon > 0} \varepsilon = 0. \end{aligned}$$

(by assumption (55)). $\square$

In fact, when X is a subset of a reflexive Banach space, we can relax the compactness assumption (54).

Theorem 11. Let U and V be two reflexive Banach spaces, let X be a closed subset of U (supplied with the weak topology) and let S be an upper hemi-continuous correspondence from X into V with non-empty closed convex images. Suppose that

$$(57) \quad \text{there exists } M \in \mathcal{L}(V, U) \text{ such that } \bigcup_{x \in X} \{x - MS(x)\} \text{ is bounded}$$

and that

(58) $\forall \varepsilon > 0$, there exists a continuous map C_ε from the unit ball B^* of V^* into X such that $\sup_{p \in B^*} \sigma^b(S(C_\varepsilon(p)), p) \leq \varepsilon$.

Then there exists a critical point of S .

Proof. Consider the function f defined on $X \times B^*$ by (56). Any solution $\bar{x} \in X$ to $\sup_{p \in V^*} f(\bar{x}, p) \leq 0$ is a critical point of S . To prove that such a solution exists, we can apply Theorem 13.1.2, which is a generalization of Theorem 7.1.2. However, we give a direct proof. The functions f_ε defined on $B^* \times B^*$ by $f_\varepsilon(q, p) = f(C_\varepsilon(q), p)$ satisfy the assumptions of the Ky-Fan Theorem 7.1.3. Hence there exists $\bar{q} \in B^*$ such that $f_\varepsilon(\bar{q}, p) \leq \sup_{q \in B^*} f_\varepsilon(q, q) \leq \varepsilon$ (by (58)). Thus $v^* = \inf_{x \in X} \sup_{p \in B^*} f(x, p) \leq \sup_{p \in B^*} f_\varepsilon(p, p) \leq \varepsilon$ for all $\varepsilon > 0$. Thus $v^* = 0$.

Now, there exists a minimizing sequence of elements $x_n \in X$ such that $\sup_{p \in B^*} f(x_n, p) \leq 1/n$. By the Fenchel inequality, we obtain

$$\begin{aligned} \forall q \in B^*, \quad \langle q, x_n \rangle &\leq f(x_n, M^*q) + \sup_{y \in X} [\langle q, y \rangle - f(y, M^*q)] \\ &\leq \frac{1}{n} + \sup_{y \in X} \sup_{v \in S(y)} \langle q, y - Mv \rangle \\ &\leq \frac{1}{n} + \sup_{y \in X} \sup_{v \in S(y)} \|y - Mv\| < +\infty \quad (\text{by assumption (57)}). \end{aligned}$$

Hence the sequence of elements x_n is bounded, i.e. relatively compact in U for the weak topology. A generalized subsequence converges to an element $\bar{x} \in X$ which satisfies $f(\bar{x}, p) \leq \liminf_{x \rightarrow x_n} f(x_n, p) \leq 0$ since f is lower semi-continuous with respect to x . Therefore $\bar{x}$ is a critical point of S . $\square$

Remark. Taking $M = 0$, assumption (57) requires that X is bounded, and thus relatively compact for the weak topology.

Remark. In Theorem 11, we need assume only that U is the dual of a barreled space. We can also extend Theorem 11 to the case when V is a Fréchet space.

15.2. Quasi-variational inequalities

We shall extend existence Theorem 9.3.1 for a solution $\bar{x}$ to the quasi-variational inequalities

$$(*) \quad \begin{cases} \text{(i) } \bar{x} \in S(\bar{x}), \\ \text{(ii) } \sup_{y \in S(\bar{x})} \varphi(\bar{x}, y) \leq 0 \end{cases}$$

by replacing the assumption of lower semi-continuity of φ with respect to x by the assumption of pseudo-monotonicity introduced in Section 13.2. In particular, we can prove the existence of a solution $\bar{x}$ to

$$(**) \quad \begin{cases} \text{(i) } \bar{x} \in S(\bar{x}), \\ \text{(ii) } \forall y \in S(\bar{x}), \langle L(\bar{x}), \bar{x} - y \rangle \leq 0 \end{cases}$$

when X is a convex compact subset, L is *monotone*, bounded and finitely continuous map from X into U^* and S a *continuous* correspondence from X into X . We can relax the latter requirement by using another fixed point theory. Assume for instance that L is *linear and U-elliptic*. Let T be the map associating with any $x \in U$ the solution $T(x) \in S(x)$ to the variational inequalities

$$(***) \quad \langle L(T(x)), T(x) - y \rangle \leq 0 \quad \text{for any } y \in S(x).$$

Then any solution x to $(**)$ is a fixed point of T . In most instances, U is a dense subset of a Hilbert space V , ordered by a closed convex cone P of positive elements satisfying

$$P = \{x \in V \text{ such that } (x, y) \geq 0 \forall y \in P\}.$$

(For instance, $V = L^2(\Omega)$, $P = L_+^2(\Omega)$.)

Define a sub-solution x^b and a super-solution x^* of T by

$$x^b \leq T(x^b) \quad \text{and} \quad T(x^*) \leq x^*$$

respectively.

We shall prove that, when T is *increasing*, the existence of a sub-solution x^b and a super-solution x^* satisfying $x^b \leq x^*$ imply the existence of a fixed point $\bar{x}$ satisfying $x^b \leq \bar{x} \leq x^*$. We next prove that the map T defined by $(***)$ is increasing whenever L satisfies.

$$\langle L(\sup(x, 0)), \sup(-x, 0) \rangle \leq 0$$

and S satisfies the property

$$\begin{aligned} &\text{If } x_1 \leq x_2, y_1 \in S(x_1), y_2 \in S(x_2), \text{ then} \\ &\inf(y_1, y_2) \in S(x_1) \quad \text{and} \quad \sup(y_1, y_2) \in S(x_2). \end{aligned}$$

15.2.1. Selection of fixed point by pseudo-monotone functions

We shall extend Theorem 9.3.1 by replacing the continuity assumption by an assumption of pseudo-monotonicity (see Section 13.2).

Theorem 1. *Suppose that*

(1) *X is a convex compact subset*

that a function $\varphi : X \times X \rightarrow \mathbf{R}$ satisfies

$$(2) \quad \begin{cases} \text{(i) } \forall y, x \mapsto \varphi(x, y) \text{ is lower semi-continuous for the finite topology and pseudo-monotone,} \\ \text{(ii) } \forall x, y \mapsto \varphi(x, y) \text{ is concave} \\ \text{(iii) } \sup_{y \in X} \varphi(y, y) \leq 0 \end{cases}$$

and that a correspondence S from X into X satisfies

(3) *S is upper semi-continuous with non-empty closed convex images.*

We assume also that the function φ and the correspondence S are consistent in the sense that

(4) *$\{x \in X \text{ such that } \alpha(x) = \sup_{y \in S(x)} \varphi(x, y) \leq 0\}$ is closed.*

Then there exists $\bar{x} \in X$ such that

$$(5) \quad \begin{cases} \text{(i) } \bar{x} \in S(\bar{x}), \\ \text{(ii) } \sup_{y \in S(\bar{x})} \varphi(\bar{x}, y) \leq 0. \end{cases}$$

Proof. This is analogous to the proof of Theorem 9.3.1. By assuming the conclusion false, we deduced that the compact set X is contained in $V_0 \cup \bigcup_{i=1}^n V(p_i)$, where $V_0 = \{x \text{ such that } \alpha(x) > 0\}$ and $V(p) = \{x \in X \text{ such that } \langle p, x \rangle - \sigma^*(S(x), p) > 0\}$. We introduced a continuous partition of unity $\{\beta_0, \dots, \beta_n\}$ subordinate to this covering and the function a defined on $X \times X$ by

$$(6) \quad a(x, y) = \beta_0(x) \varphi(x, y) + \sum_{i=1}^n \beta_i(x) \langle p_i, x - y \rangle.$$

We continue by applying the Brézis-Nirenberg-Stampacchia Theorem 13.2.1 to deduce the existence of $\bar{x}$ such that

$$(7) \quad a(\bar{x}, y) = \beta_0(\bar{x}) \varphi(\bar{x}, y) + \sum_{i=1}^n \beta_i(\bar{x}) \langle p_i, \bar{x} - y \rangle \leq 0 \quad \forall y \in X.$$

[The function a is pseudo-monotone. Let a (compact) generalized sequence $\{x_\mu\}$ converge to $\bar{x}$ and satisfy $\limsup_\mu a(x_\mu, \bar{x}) \leq 0$. Since the functions $\beta_0, \beta_1, \dots, \beta_n$ are continuous, we deduce that

$$\limsup_\mu a(x_\mu, \bar{x}) = \beta_0(\bar{x}) \cdot \limsup_\mu \varphi(x_\mu, \bar{x}) \leq 0.$$

Hence either $\beta_0(\bar{x}) = 0$ or $\beta_0(\bar{x}) > 0$ and $\limsup_{\mu} \varphi(x_{\mu}, \bar{x}) \leq 0$. In this case, the pseudo-monotonicity of φ implies that $\forall y \in X$,

$$(8) \quad \varphi(\bar{x}, y) \leq \liminf_{\mu} \varphi(x_{\mu}, y).$$

In both cases, we deduce that $\alpha(\bar{x}, y) \leq \liminf_{\mu} \alpha(x_{\mu}, y) = \beta_0(\bar{x}) \liminf_{\mu} \varphi(x_{\mu}, y) + \lim_{\mu} \sum_{j=1}^n \beta_j(x_{\mu}) \langle p_j, x_{\mu} - y \rangle$.

We end the proof by contradicting (7) as in Theorem 9.3.1. $\square$

We now prove a result analogous to Theorem 9.3.2.

Theorem 2. (Joly–Mosco). *Let X be convex and compact. Suppose that a map L from X into U^* satisfies*

$$(9) \quad L \text{ is monotone, finitely continuous and bounded,}$$

and that

$$(10) \quad S \text{ is a continuous correspondence from } X \text{ into } X \text{ with non-empty closed convex images.}$$

Then there exists a solution $\bar{x}$ of the quasi-variational inequalities

$$(11) \quad \begin{cases} \text{(i) } \bar{x} \in S(\bar{x}), \\ \text{(ii) } \langle L(\bar{x}), \bar{x} - y \rangle \leq 0 \text{ for all } y \in S(\bar{x}). \end{cases}$$

Proof. We apply Theorem 1 to the case when φ is defined by $\varphi(x, y) = \langle L(x), x - y \rangle$. By Proposition 13.2.5, φ is pseudomonotone and finitely continuous with respect to x . Hence assumptions (1), (2) and (3) are satisfied. In order to apply Theorem 1, it remains to prove that property (4) holds.

Let $\{x_{\mu}\}$ be a generalized sequence satisfying $\alpha(x_{\mu}) \leq 0$ and converging to $\bar{x}$. We have to prove that $\alpha(\bar{x}) \leq 0$. We fix any $y \in S(\bar{x})$.

Since S is lower semi-continuous, there exists a generalized sequence $\{y_{\mu}\}$ of elements $y_{\mu} \in S(x_{\mu})$ which converges to $y \in S(\bar{x})$.

Since L is monotone, we deduce from the definition of α and the boundedness of L that

$$\begin{aligned} \langle L(y), x_{\mu} - y \rangle &\leq \langle L(x_{\mu}), x_{\mu} - y \rangle \\ &\leq \langle L(x_{\mu}), x_{\mu} - y_{\mu} \rangle + \langle L(x_{\mu}), y_{\mu} - y \rangle \\ &\leq \alpha(x_{\mu}) + \|L(x_{\mu})\|_* \|y_{\mu} - y\| \leq M \|y_{\mu} - y\|. \end{aligned}$$

Letting x_{μ} and y_{μ} converge to x and y strongly, we deduce that

$$(12) \quad \forall y \in S(\bar{x}), \quad \langle L(y), \bar{x} - y \rangle \leq 0.$$

Taking $y = \bar{x} + \theta(z - \bar{x}) \in S(\bar{x})$, we obtain that

$$(13) \quad \forall z \in S(\bar{x}), \quad \langle L(\bar{x} + \theta(z - \bar{x})), \bar{x} - z \rangle \leq 0.$$

Hence, by letting θ go to 0, we infer that

$$(14) \quad \alpha(\bar{x}) = \sup_{z \in S(\bar{x})} \langle L(\bar{x}), \bar{x} - z \rangle \leq 0.$$

15.2.2. Fixed point theorem for increasing maps

Let V be a Hilbert space (identified with its dual V^*) and P a closed convex cone of V satisfying

$$(15) \quad P = \{x \in V \text{ such that } (x, y) \geq 0 \forall y \in P\}.$$

We shall use P as the cone of non negative elements defining on V a vector ordering

$$(16) \quad x \geq y \text{ if and only if } x - y \in P.$$

Example. The main example is the case where $V = L^2(\Omega)$, $P = L^2_+(\Omega)$. Property (15) is trivially satisfied.

We can apply the Zorn lemma in V thanks to the following proposition.

Proposition 1. *Let V and P satisfy property (15). Any non-empty "complete" subset M bounded from above (for the ordering) has an upper bound.*

Proof. Since M is completely ordered, we can write M in the form $M = \{x_i\}_{i \in I}$ where I is a completely ordered set and where $i \leq j$ implies that $x_j - x_i \in P$. Let $a \in V$ be an upper bound: $M \subset a - P$. Hence the family of real numbers $(x_i, a - x_{i_0})$ is increasing and bounded. Hence, it converges: if $j \geq i \geq i_0$, we have

$$(17) \quad |x_j - x_i|^2 = (x_j - x_i, x_j - x_i) \leq (x_j - x_i, a - x_{i_0})$$

since $a - x_{i_0} - (x_j - x_i) = (a - x_j) + (x_i - x_{i_0}) \in P + P = P$. This implies that $|x_j - x_i|^2$ converges to 0, i.e., that $\{x_j\}$ is a Cauchy generalized sequence of V . Since V is a Hilbert space, x_j converges to $\bar{x}$.

For any $y \in P$, we have $(\bar{x}, y) = \lim_i (x_i, y) \geq (x_j, y)$ for every j . Hence $\bar{x} \geq x_j$ for any $j \in I$. On the other hand, if $z \in V$ bounds M above, i.e. if $z \geq x_j$ for all j , then $(z, y) \geq \lim_i (x_j, y) = (\bar{x}, y)$ for all $y \in P$. Hence $z \geq \bar{x}$. We have proved that $\bar{x}$ is the upper bound of M . $\square$

Proposition 2 (Birkhoff–Tartar). *Let V and P satisfy (15) and let T be an increasing map from V into itself.*

Suppose that there exists a sub-solution x^b and a super-solution x^ of the equation $T(x) = x$ defined by*

$$(18) \quad x^b \leq x^*, \quad x^b \leq T(x^b) \quad \text{and} \quad T(x^*) \leq x^*.$$

Then the subset of fixed points $\bar{x}$ of T satisfying $x^b \leq \bar{x} \leq x^$ is non-empty and has a smallest and a largest element (for the ordering).*

Proof. We write

$$X = \{x \in V \text{ such that } x^b \leq x \leq x^* \text{ and } x \leq T(x)\},$$

$$Y = \{y \in V \text{ such that } x^b \leq y \leq x^* \text{ and } y \geq T(y)\},$$

$$Z = \{x \in X \text{ such that } x \leq y \text{ for all } y \in Y\}.$$

Then Z is non-empty (since $x^b \in Z$). By the Zorn lemma, there exists a maximal element $x^* \in Z$. By the very definition of Z , $x^* \in X$. Hence $x^b \leq x^* \leq x^*$ and $x^* \leq T(x^*)$. Since T is increasing, $x^b \leq T(x^b) \leq T(x^*) \leq T(x^*) \leq x^*$ and $T(x^*) \leq T(T(x^*))$. This implies that $T(x^*)$ also belongs to X .

On the other hand, when $y \in Y$, it satisfies $x^* \leq y$. Thus $T(x^*) \leq T(y) \leq y$. Hence $T(x^*) \in Z$. Since $x^* \leq T(x^*)$ and x^* is maximal, this implies that $x^* = T(x^*)$ is a fixed point of T . If $\bar{x}$ is any other fixed point between x^b and x^* , it belongs to Y and thus, $x^* \leq \bar{x}$ by the definition of Z . Hence x^* is the smallest fixed point of S . We can prove the existence of the largest fixed point of S by replacing Z by $\tilde{Z} = \{y \in Y \text{ such that } x \leq y \text{ for all } x \in X\}$. $\square$

15.2.3. Quasi-variational inequalities for increasing correspondences

Consider a Hilbert space V and a cone $P \subset V$ satisfying (15) and another Hilbert space U contained and dense in V , supplied with a stronger topology.

Denote by $x \mapsto x_+$ the unique orthogonal projection of x onto P (see Section 2.2.1). Hence the map $x \mapsto -(x)_- = x - x_+$ is the orthogonal projection onto $-P$ for which any $x \in V$ can be split into the unique decomposition

$$(19) \quad x = x_+ - x_- \in P - P \text{ such that } (x_+, x_-) = 0$$

We assume that

$$(20) \quad \begin{aligned} &\text{there exists } c > 0 \text{ such that if } x \in U, \quad x_+ \in U \text{ and} \\ &\|x_+\|_U \leq c \|x\|_U. \end{aligned}$$

We introduce a continuous linear operator $L \in \mathcal{L}(U, U^*)$ which is U -elliptic, i.e.

$$(21) \quad \text{there exists } C > 0 \text{ such that } \forall x \in U, \quad \langle Lx, x \rangle \geq C \|x\|_U^2.$$

We assume furthermore that L satisfies the property

$$(22) \quad \forall x \in U, \quad \langle Lx_+, x_- \rangle \leq 0.$$

For simplicity, we use the notation

$$(23) \quad \begin{cases} \text{(i) } \inf(x_1, x_2) = x_1 - (x_1 - x_2)_+ = x_2 - (x_2 - x_1)_+, \\ \text{(ii) } \sup(x_1, x_2) = x_2 + (x_1 - x_2)_+ = x_1 + (x_2 - x_1)_+. \end{cases}$$

Finally, we introduce a correspondence S with non-empty closed convex values which satisfies the following *monotonicity property*.

$$(24) \quad \begin{aligned} &\text{If } x_1 \leq x_2, y_1 \in S(x_1), y_2 \in S(x_2), \text{ then} \\ &\inf(y_1, y_2) \in S(x_1) \text{ and } \sup(y_1, y_2) \in S(x_2). \end{aligned}$$

We consider the following problem.

Find $\bar{x} \in V$ satisfying

$$(25) \quad \begin{cases} \text{(i) } \bar{x} \in S(\bar{x}), \\ \text{(ii) } \forall y \in S(\bar{x}), \langle L(\bar{x}) - f, \bar{x} - y \rangle \leq 0, \end{cases}$$

where f is given in U^* .

By Theorem 9.2.2, there exists a unique solution $x = T(z)$ of the variational inequalities

$$(26) \quad \begin{cases} \text{(i) } x \in S(z), \\ \text{(ii) } \langle L(x) - f, x - y \rangle \leq 0 \text{ for any } y \in S(z) \end{cases}$$

thanks to the assumption of U -ellipticity of L . Thus, it is clear that a solution $\bar{x}$ of (25) is nothing other than a fixed point $\bar{x} = T(\bar{x})$ of the map T defined above.

We shall say that x^b is a sub-solution and x^* a super-solution if we have

$$(27) \quad x^b \leq T(x^b) \quad \text{and} \quad T(x^*) \leq x^*$$

respectively.

Theorem 3 (Tartar). Let V, P and U satisfy (15) and (20). Suppose that $L \in \mathcal{L}(U, U^*)$ is U -elliptic and satisfies (22) and that the correspondence S from V into U with non-empty closed convex values satisfies property (24). If x^b and x^* are respectively a sub-solution and a super-solution with $x^b \leq x^*$, then the subset of solutions of (25) satisfying $x^b \leq x \leq x^*$ is non-empty and has a smallest and a largest element.

Proof. We apply the Birkhoff-Tartar Theorem 2 to the map T defined by (26). We have to check that T is increasing.

Let $x_1 \leq x_2$ and let $T(x_1) \in S(x_1)$ and $T(x_2) \in S(x_2)$ be the solutions of the variational inequalities

$$(28) \quad \begin{cases} \text{(i)} \langle L(T(x_1)) - f, T(x_1) - y_1 \rangle \leq 0 \text{ for any } y_1 \in S(x_1), \\ \text{(ii)} \langle L(T(x_2)) - f, T(x_2) - y_2 \rangle \leq 0 \text{ for any } y_2 \in S(x_2). \end{cases}$$

Assumption (24) on S implies that

$$\begin{aligned} y_1 &= T(x_1) - (T(x_1) - T(x_2))_+ \in S(x_1), \\ y_2 &= T(x_2) + (T(x_1) - T(x_2))_+ \in S(x_2). \end{aligned}$$

Taking such elements y_1 and y_2 in inequalities (28) and adding, we obtain that

$$(29) \quad \langle L[T(x_1) - T(x_2)], (T(x_1) - T(x_2))_+ \rangle \leq 0.$$

Since

$$(T(x_1) - T(x_2)) = (T(x_1) - T(x_2))_+ - (T(x_1) - T(x_2))_-$$

we deduce that

$$\begin{aligned} &\langle L(T(x_1) - T(x_2))_+, (T(x_1) - T(x_2))_+ \rangle \\ &\leq \langle L(T(x_1) - T(x_2)), (T(x_1) - T(x_2))_+ \rangle \leq 0 \end{aligned}$$

by assumption (22).

Hence, the U -ellipticity condition (21) implies that

$$(T(x_1) - T(x_2))_+ = 0, \quad \text{i.e. that } T(x_1) - T(x_2) \in -P.$$

Thus the map T is increasing. $\square$

Example. Let Ω be a smooth open bounded subset of $\mathbb{R}^n$. We take

$$(30) \quad V = L^2(\Omega), \quad P = L_+^2(\Omega), \quad U = H^1(\Omega).$$

Assumptions (15) and (20) are satisfied. We take for the operator L the operator defined by

$$(31) \quad \langle Lx, y \rangle = \sum_{i,j=1}^n \int_{\Omega} a_{ij}(\omega) D_i x(\omega) D_j y(\omega) d\omega + \int_{\Omega} a_0(\omega) x(\omega) y(\omega) d\omega,$$

where $\sum_{i,j=1}^n a_{ij}(\omega) \xi_i \xi_j \geq c \|\xi\|^2$ and $a_0(\omega) \geq c \forall \xi \in \mathbb{R}^n$ and for almost all $\omega \in \Omega$. Hence L is U -elliptic. Furthermore, property (22) is satisfied since, in this case

$$x_+ = \sup(x, 0), \quad x_- = \sup(-x, 0).$$

Finally, consider the case of two *increasing* functions M_1 and M_2 mapping $L^2(\Omega)$ into $H^1(\Omega)$. Then the correspondence S defined by

$$(32) \quad S(x) = \{y \in H^1(\Omega) \text{ such that } M_1(x) \leq y \leq M_2(x)\}$$

satisfies the monotony conditions (24).

15.3. Other properties and examples of upper and lower semi-continuous correspondences

We have proved that the subdifferential ∂f of a continuous convex function is upper semi-continuous. Some other examples of semi-continuous correspondences are given in this section. For instance, we have often used the correspondence S defined by

$$S(y) = \{x \in R \text{ such that } Lx \in Y - y\}$$

where R and Y are closed subsets cones of Fréchet spaces U and V and $L \in \mathcal{L}(U, V)$. We shall prove that, if the constraint qualification assumption holds, i.e.

$$0 \in \text{Int}(L(R) - Y),$$

then the correspondence S is lower semi-continuous and closed (with non-empty closed convex values). We have also met correspondences S defined by $S(y) = \{x \in R \text{ such that } \forall p \in P, \gamma(x, p, y) \leq 0\}$ where γ maps $R \times P \times Y$ into $\mathbb{R}$. We shall prove that such a correspondence S lower semi-continuous and closed if, for instance,

$$\left\{ \begin{array}{l} \text{(i) } \gamma \text{ is continuous,} \\ \text{(ii) } \forall y \in Y, \forall p \in P, x \mapsto \gamma(x, y, p) \text{ is convex,} \\ \text{(iii) } P \text{ is compact,} \\ \text{(iv) } \forall y \in Y, \exists x_y \in R \text{ such that, } \forall p \in P, \gamma(x_y, p, y) < 0. \end{array} \right.$$

We also prove an important theorem due to Michael: if S is a lower semi-continuous correspondence with non-empty closed convex images mapping a compact set Y into a Fréchet space U , there exist a *continuous* selection s of S , i.e. a continuous map s such that $\forall x \in X, s(x) \in S(x)$.

Next we show how to supply the family $\mathcal{F}(U)$ of non-empty (weakly) closed convex subsets of U with a topology (the weak Hausdorff topology) in such a way that continuous correspondences from Y into U (supplied with the weak topology) are the continuous maps from Y into $\mathcal{F}(U)$.

Finally, we prove that upper hemi-continuous correspondences with *compact* convex images are upper semi-continuous and similarly for lower semi-continuous correspondences.

15.3.1. Lower semi-continuity of preimages of linear operators

In most examples, we used in optimization and game theory the subset X of the form.

$$(1) \quad X = \{x \in R \text{ such that } Lx \in Y\}$$

where

$$(2) \quad \begin{cases} \text{(i) } U \text{ and } V \text{ are Banach spaces,} \\ \text{(ii) } R \subset U \text{ and } Y \subset V \text{ are closed convex subsets,} \\ \text{(iii) } L \in \mathcal{L}(U, V). \end{cases}$$

This motivates the study of continuity properties of the correspondence S mapping V into U defined by

$$(3) \quad S(y) = \{x \in R \text{ such that } Lx \in Y - y\}.$$

Theorem 1 (Robinson). *Suppose that (2) holds. If the constraint qualification assumption*

$$(4) \quad 0 \in \text{Int}(L(R) - Y)$$

(i.e. $\exists \delta > 0$ such that $S(y) \neq \emptyset$ when $\|y\| \leq \delta$) holds, then the correspondence S has non-empty closed convex images and is closed. Furthermore, there exist $\gamma > 0$, $c \geq 0$ and $d \geq 0$ such that if $y \in B_V(\gamma)$ and if $x \in R$, there exists $x_y \in S(y)$ such that

$$(5) \quad \|x - x_y\| \leq cd_Y(Lx + y)(d + \|x\|)$$

Hence S is lower semi-continuous at 0.

Remark. In the case where $R = U$ and $Y = \{0\}$, the constraint qualification assumption amounts to saying that L is *surjective*. Then Theorem 1 implies the famous open mapping theorem of Banach.

Theorem 2 (Banach). *Let $L \in \mathcal{L}(U, V)$ be a surjective operator mapping a Fréchet space U onto a Fréchet space V . Then L is open (i.e. maps open subsets onto open subsets). If we assume furthermore that L is injective, then $L^{-1} \in \mathcal{L}(V, U)$ is continuous.*

Proof of Theorem 2. In this case, $S(y) = L^{-1}(y)$. It is clear that the lower semi-continuity of S amounts to saying that L maps open subsets onto open subsets.

□

Proof of Theorem 1. The proof is quite involved and so we introduce an intermediary lemma. We begin by indicating how the theorem follows from the lemma.

Lemma 1. Suppose that (2) and (4) hold. Then there exist a ball $B_U(a)$ of U and a ball $B_V(\eta)$ of V such that

$$(6) \quad B_V(\eta) \subset L[B_U(a) \cap R] - Y.$$

First step: Deduction of Theorem 1 from Lemma 1.

(a) We begin by observing that the constraint qualification assumption (4) implies that the images $S(y)$ are non-empty when $\|y\| < \delta$. They are obviously convex and closed.

(b) We prove that S is closed. Let $\{y_\mu, x_\mu\}$ be a generalized sequence of point of the graph of S (i.e. such that $Lx_\mu + y_\mu \in Y$) which converges to $\{y, x\}$. Since L is continuous and since R and Y are closed, we deduce that $Lx + y \in Y$.

(c) Let $\delta > 0$ be such that $B_V(\delta) \subset L(R) - Y$. We take $\gamma = \min(\delta, \frac{1}{2}\eta)$. We fix $x \in R$. If $Lx \in Y - y$, we take $x_y = x$. Otherwise, we choose $z \in Y - y$ such that $\|Lx - z\| = d_{Y-y}(Lx) = d_Y(Lx + y)$. We deduce from (6) that there exist $u \in B_V(a) \cap R$ and $v \in Y - y$ such that $-\gamma(Lx + z)/(\|Lx + z\|) = Lu - v$ because $B_V(\gamma) \subset L(B_U(a) \cap R) - Y + B_V(\gamma)$. We set $\lambda = (\|Lx - z\|)/(\gamma + \|Lx - z\|) \in]0, 1[$. Multiplying the latter equation by λ , we obtain.

$$L(\lambda u + (1 - \lambda)x) = \lambda v + (1 - \lambda)z \in Y - y,$$

since v and z belong to $Y - y$. We set $x_y = \lambda u + (1 - \lambda)x$, which belongs to R since both u and x belong to R . Hence $x_y \in S(y)$. Finally,

$$\begin{aligned} \|x - x_y\| &= \|\lambda(x - u)\| \leq \lambda(\|x\| + \|u\|) = \\ &= (d_Y(Lx + y))/(\gamma + d_Y(Lx + y)) (\|x\| + \|u\|) \\ &\leq (1/\gamma) d_Y(Lx + y) (\|x\| + a) \end{aligned}$$

since $\|u\| \leq a$. Hence (5) is proved.

(d) We prove that S is lower semi-continuous at 0. If $x \in S(0)$ and $\varepsilon > 0$ are fixed, we choose $\varrho = \min(\gamma, \varepsilon/c(d + \|x\|))$. If $y \in B_V(\varrho) \subset B_V(\gamma)$, we deduce from (5) that there exists $x_y \in S(y)$ such that $\|x - x_y\| \leq cd_Y(Lx + y)(d + \|x\|) \leq c\varrho(d + \|x\|) \leq \varepsilon$.

Second step: Proof of Lemma 1.

(a) We begin by proving that there exist $\eta > 0$ and $a > 0$ such that

$$(7) \quad 2B_V(\eta) \subset \overline{L[B_U(a) \cap R] - Y}.$$

Denote by F the convex set

$$(8) \quad F_n = L[B_U(n\epsilon) \cap R] - Y.$$

Because $\bigcup_{n=1}^{\infty} B_U(n\epsilon) \cap R = R$, it follows that $\bigcup_{n=1}^{\infty} F_n \supset L(R) - Y$ has a non-empty interior thanks to assumption (4).

We deduce from the Baire theorem (1) that one of the sets F_n has a non-empty interior. Let $u \in \text{Int } F_n$. Since $0 \in \text{Int}(L(R) - Y)$, there exist $\alpha > 0$, $x \in R$ and $y \in Y$ such that $-\alpha u = Lx - y \in L(B_V(\|x\|) \cap R) - Y$. We set $\lambda = 1/(\alpha + 1)$. So, $0 = (1 - \lambda)u + \lambda(-\alpha u)$ belongs to the interior of F_a , where $a = (1 - \lambda)n + \lambda\|x\|$. Hence there exists $\eta > 0$ such that (7) holds.

(b) We now prove that the conclusion (5) of Lemma 1 holds. We deduce from (7) that, for any integer n ,

$$(9) \quad 2B_V(\eta_n) \subset 2^{-n}F_a,$$

where $\eta_n = 2^{-n}\eta$.

Let y be any element of $B_V(\eta)$. Since $2y \in 2B_V(\eta) \subset F_a$, we can find $z_0 \in F_a$ such that $2y - z_0 \in B_V(\eta)$. Assume that we have constructed a sequence z_k ($0 \leq k \leq n-1$) such that

$$(10) \quad z_k \in F_a \quad \text{and} \quad 2y - \sum_{k=0}^{n-1} 2^{-k}z_k \in B_V(\eta_n).$$

Since $B_V(\eta_n) \subset 2^{-n}F$, we can find

$$(11) \quad z_n \in F_a \quad \text{such that} \quad 2y - \sum_{k=0}^{n-1} 2^{-k}z_k - 2^{-n}z_n \in B_V(\eta_{n+1}).$$

Therefore,

$$(12) \quad \lim_{n \rightarrow \infty} \left(y - \frac{1}{2} \sum_{k=0}^n 2^{-k}z_k \right) = 0$$

Since $z_k \in F_a = L[B_U(a) \cap R] - Y$, there exist $x_k \in B_U(a) \cap R$ and $y_k \in Y$ such that $z_k = L(x_k) - y_k$. Consider the sequence

$$(13) \quad u_n = \frac{1}{2} \sum_{k=0}^n 2^{-k}x_k.$$

¹ The Baire theorem states that in a complete metric space V , if a denumerable union of closed subsets F_n has a non-empty interior, then one of the F_n 's has a non-empty interior.

It is clear that $\{u_n\}$ is a Cauchy sequence of U since

$$\|u_n - u_m\| = \left\| \frac{1}{2} \sum_{k=n}^m 2^{-k} x_k \right\| \leq \frac{1}{2} \sum_{k=n}^m 2^{-k} \|x_k\| \leq \frac{1}{2} \sum_{k=n}^m 2^{-k} \varepsilon.$$

Therefore, since U is complete, this Cauchy sequence $\{u_n\}$ converges to u_* in U .

Since $B_V(a) \subset R$ is closed and convex, then

$$(14) \quad \frac{1}{1-2^{-n-1}} u_n = \frac{1-2^{-1}}{1-2^{-n-1}} \sum_{k=0}^n 2^{-k} x_k \in B_U(a) \cap R$$

and thus,

$$(15) \quad u_* = \lim_{n \rightarrow \infty} \left(\frac{1}{1-2^{-n-1}} u_n \right) \in B_U(a) \cap R.$$

Therefore, because

$$\lim_{n \rightarrow \infty} \left(y - Lu_n + \frac{1}{2} \sum_{k=0}^n 2^{-k} y_k \right) = 0,$$

it follows that

$$(16) \quad \lim_{n \rightarrow \infty} \frac{1}{2} \sum_{k=0}^n 2^{-k} y_k = L(u_*) - y.$$

Since Y is closed and convex, we deduce in the same way that

$$v_* = \lim_{n \rightarrow \infty} \frac{1}{2} \sum_{k=0}^n 2^{-k} y_k$$

belongs to Y and that

$$y = L(u_*) - v_* \in L[B_U(a) \cap R] - Y.$$

Thus (6) holds. $\square$

Remark. In the case when R and Y are cones, we can restate the theorem in the following way.

Theorem 3. Suppose that U and V are both Banach spaces and that R and Y are closed convex cones satisfying $L(R) - Y = V$.

Then there exists a constant M such that

$$(17) \quad \forall y \in V, \exists x \in S(y) \text{ such that } \|x\| \leq M\|y\|$$

and thus, S is lower semi-continuous on V .

Proof. By Lemma 1, we know that $B_V(\eta) \subset L[B_U(a) \cap R] - Y$. Let $y \in V$. Then $-\eta(y)/\|y\|$ belongs to $B_V(\eta)$ and thus, there exists $\tilde{x} \in B_U(a) \cap R$ such that $-\eta(y/\|y\|) \in L\tilde{x} - Y$. Then $x = (\|y\|/\eta)\tilde{x}$ belongs to $S(y)$ and $\|x\| = (\|y\|/\eta)\|\tilde{x}\| \leq (\|y\|/\eta)a$. Therefore, we have proved (17) with $M = a/\eta$. $\square$

15.3.2. Lower semi-continuity of correspondences defined by constraints

We consider now another correspondence S we used in optimization theory (see Section 5.1), defined by

$$(18) \quad \tilde{S}(y) = \{x \in R \text{ such that } \forall p \in P, \gamma(x, p, y) \leq 0\}$$

where

$$(19) \quad \begin{cases} \text{(i) } R \text{ is a closed convex subset of a topological vector space } U, \\ \text{(ii) } P \text{ is a closed convex subset of a topological vector space } V^*, \\ \text{(iii) } Y \text{ is a topological space,} \\ \text{(iv) } \gamma \text{ maps } R \times P \times Y \text{ into } \mathbf{R}. \end{cases}$$

Proposition 1. *Let assume that*

$$(20) \quad \forall p \in P, \text{ that function } \{x, y\} \mapsto \gamma(x, p, y) \text{ is lower semi-continuous.}$$

Then the correspondence S defined by (19) is closed.

Proof. The graph of S is the set of pairs $\{x, y\}$ such that $\gamma(x, p, y) \leq 0$ for all $p \in P$. It is obviously closed when (20) holds. $\square$

Theorem 4. *Suppose that*

$$(21) \quad \begin{cases} \text{(i) } \forall p \in P, y \in Y, x \mapsto \gamma(x, p, y) \text{ is convex,} \\ \text{(ii) } \forall x \in R, \{p, y\} \mapsto \gamma(x, p, y) \text{ is upper semi-continuous} \end{cases}$$

and that

$$(22) \quad \begin{cases} \text{(i) } P \text{ is compact,} \\ \text{(ii) } \forall y \in Y, \text{ we can associate } x_y \in R \text{ such that} \\ \quad \gamma(x_y, p, y) < 0 \quad \forall p \in P. \end{cases}$$

Then the correspondence S defined by (19) is lower semi-continuous.

Proof. The proof follows from the following two lemmas.

Lemma 2. *Given the assumptions of Theorem 4,*

$$(23) \quad \begin{aligned} &\forall x_0 \in S(y_0) \text{ and } \forall \theta \in]0, 1[, \text{ there exists a neighborhood } N_\theta(y_0) \\ &\text{such that} \\ &\gamma(\theta x_{y_0} + (1-\theta)x_0, p, y) < 0 \quad \forall p \in P, \forall y \in N_\theta(y_0). \end{aligned}$$

Proof. Write

$$(24) \quad -c_0 = \sup_{p \in P} \gamma(x_{y_0}, p, y_0).$$

The number c_0 is strictly positive by assumption (22). Let $x = \theta x_{y_0} + (1-\theta)x_0$ where $x_0 \in S(y_0)$. Since γ is convex with respect to x , we deduce that

$$(25) \quad \begin{aligned} \gamma(x, p, y_0) &\leq \theta \gamma(x_{y_0}, p, y_0) + (1-\theta) \gamma(x_0, p, y_0) \\ &\leq -\theta c_0 \quad \forall p \in P. \end{aligned}$$

Since γ is upper semi-continuous with respect to $\{p, y\}$, we can associate with any $p \in P$ open neighborhoods $N(p)$ and $N_p(y_0)$ of p and y_0 such that

$$(26) \quad \gamma(x, q, y) \leq \gamma(x, p, y_0) + \varepsilon \leq -\theta c_0 + \varepsilon < 0 \quad \forall q \in N(p), y \in N_p(y_0)$$

(where we take $\varepsilon < \theta c_0$).

Now, since P is compact, it can be covered by a finite union $\bigcup_{i=1}^n N(p_i)$ of neighborhoods $N(p_i)$. If we set

$$N_\theta(y_0) = \bigcap_{i=1}^n N_{p_i}(y_0),$$

we deduce from (26) that $\gamma(x, q, y) < 0$ whenever $y \in N_\theta(y_0)$ and q ranges over P . $\square$

Lemma 3. Suppose that (25) and property (23) of Lemma 5 hold. Then the correspondence S defined by (18) is lower semi-continuous.

Proof. Let $x_0 \in S(y_0)$ and let $N(x_0)$ be a neighborhood of x_0 . Let x_{y_0} be the element associated with y_0 by (22(ii)). Then $x = \theta x_{y_0} + (1-\theta)x_0 = x_0 + \theta(x_{y_0} - x_0)$ belongs to $N(x_0)$ when $\theta > 0$ is small enough. Thus, assumption (23) implies the existence of a neighborhood $N_\theta(y_0)$ of y_0 such that $\gamma(x, p, y) = \gamma(\theta x_{y_0} + (1-\theta)x_0, p, y) < 0$ for all $p \in P$ and $y \in N_\theta(y_0)$. This implies that $x \in S(y) \cap N(x_0)$ whenever $y \in N_\theta(y_0)$. $\square$

15.3.3. Continuous selection theorem

Let S be a correspondence mapping a topological space Y into a topological space U . A selection s of S is a map from Y into U satisfying

$$(27) \quad \forall y \in Y, \quad s(y) \in S(y).$$

The problem we shall solve is the following: Is there a continuous selection of a lower semi-continuous correspondence? Under convenient assumptions, the answer is positive.

Theorem 5 (Michael). *Suppose that*

$$(28) \quad \begin{cases} \text{(i) } Y \text{ is a compact space} \\ \text{(ii) } U \text{ is a Fréchet space} \end{cases}$$

and that

$$(29) \quad S \text{ is a lower semi-continuous correspondence with non-empty closed convex images } S(y).$$

Then there exists a continuous selection s of S .

Proof. Consider an increasing sequence $P_1 \leq P_2 \leq \dots \leq P_n \leq \dots$ of semi-norms P_n defining the topology of the Fréchet space U . We use the decreasing subsequence of neighborhoods B_n of 0 defined by

$$B_n = \{x \in U \text{ such that } P_n(x) \leq 2^{-(n+1)}\}.$$

We construct by induction a sequence of continuous functions $s_n : Y \rightarrow U$ satisfying

$$(H_1) \quad s_1(y) \in S(y) + B_1$$

and, for $n \geq 2$,

$$(H_n) \quad \begin{cases} \text{(i) } s_n(y) \in S(y) + B_n, \\ \text{(ii) } s_n(y) \in s_{n-1}(y) + 2B_{n-1}. \end{cases}$$

Once these functions are constructed, we shall prove that they converge to a continuous function s satisfying

$$(27) \quad s(y) \in S(y) \quad \forall y \in Y.$$

First step: construction of s_1 . Let $\sigma_1 : Y \rightarrow U$ be any (not necessarily continuous) selection of the correspondence S .

Since S is lower semi-continuous, there exists an open neighborhood $N_1(y)$ of y such that

$$\forall z \in N_1(y), \quad (\sigma_1(y) + B_1) \cap S(z) \neq \emptyset.$$

This can be rewritten as

$$(30) \quad \forall z \in N_1(y), \quad \sigma_1(y) \in S(z) + B_1.$$

Since Y is compact there exists a continuous "partition of unity" β_i^1 subordinate to a covering $\{N_1(y_i^1)\}$ by k_1 open neighborhoods $N_1(y_i^1)$.

We set

$$s_1(z) = \sum_{i=1}^{k_1} \beta_i^1(z) \sigma_1(y_i^1).$$

This is a *continuous function*. Let z be fixed. By (30),

$$\begin{aligned} s_1(z) &= \sum_{i=1}^{k_1} \beta_i^1(z) \sigma_1(y_i^1) \in \sum_{i=1}^{k_1} \beta_i^1(z) (S(z) + B_1) \\ &\subset S(z) + B_1 \end{aligned}$$

since $S(z) + B_1$ is a convex subset and $\beta_i^1(z) \geq 0$, $\sum_{i=1}^k \beta_i^1(z) = 1$. Thus the function s_1 satisfies property (H_1) .

Second step: construction of s_{n+1} . Suppose that the continuous functions s_k satisfying (H_k) are constructed for $2 \leq k \leq n$. We construct s_{n+1} satisfying (H_{n+1}) . For this purpose, we choose $\sigma_n(y)$ satisfying

$$\sigma_n(y) \in [s_n(y) + B_n] \cap S(y),$$

this set being non-empty by $(H_n(i))$.

Since S is lower semi-continuous, there exists an open neighborhood $N_{n+1}^{(1)}(y)$ of y such that

$$\forall z \in N_{n+1}^{(1)}(y), \quad \sigma_n(y) \in S(z) + B_{n+1}.$$

Since s_n is continuous, there exists an open neighborhood $N_{n+1}^{(2)}(y)$ of y such that

$$\forall z \in N_{n+1}^{(2)}(y), \quad s_n(z) \in s_n(y) + B_n.$$

Therefore, if $N_{n+1}(y) = N_{n+1}^{(1)}(y) \cap N_{n+1}^{(2)}(y)$, we obtain that

$$(31) \quad \forall z \in N_{n+1}(y), \quad \begin{cases} \sigma_n(y) \in S(z) + B_{n+1}, \\ s_n(z) \in s_n(y) + B_n. \end{cases}$$

Let β_i^{n+1} be a partition of unity subordinate to a finite covering $\{N_{n+1}(y_i^{n+1})\}$ of the compact set Y . Write

$$s_{n+1}(z) = \sum_{i=1}^{k_{n+1}} \beta_i^{n+1}(z) \sigma_n(y_i^{n+1}).$$

Then s_{n+1} is a continuous function. Let $z \in Y$ be fixed. We obtain that

$$\begin{aligned} s_{n+1}(z) &= \sum_{i=1}^{k_{n+1}} \beta_i^{n+1}(z) \sigma_n(y_i^{n+1}) \in \sum_{i=1}^{k_{n+1}} \beta_i^{n+1}(z) (S(z) + B_{n+1}) \\ &\subset S(z) + B_{n+1} \end{aligned}$$

since $S(z) + B_{n+1}$ is convex.

On the other hand, by (31) and $(H_n(ii))$, we have that

$$\begin{aligned} s_{n+1}(z) &\in \sum_{i=1}^{k_n+1} \beta_i^{n+1}(z) (s_n(y_i^{n+1}) + B_n) \\ &\subset \sum_{i=1}^{k_n+1} \beta_{y_i}^{n+1}(z) (s_n(z) + 2B_n) \subset s_n(z) + 2B_n. \end{aligned}$$

Therefore, we have proved that (H_{n+1}) is satisfied. This completes the inductive construction of the sequence $\{s_n\}$.

Third step: construction of s . The sequence $\{s_n\}$ is a Cauchy sequence in the space $\mathcal{C}(Y, U)$ of continuous functions, supplied with the topology of uniform convergence, because, for every n and every $z \in Y$,

$$\begin{aligned} p_n(s_{n+k+1}(z) - s_n(z)) &= \\ &= p_n\left(\sum_{j=0}^k s_{n+j+1}(z) - s_{n+j}(z)\right) \leq \sum_{j=0}^k p_n(s_{n+j+1}(z) - s_{n+j}(z)) \\ &\leq \sum_{j=0}^k p_{n+j}(s_{n+j+1}(z) - s_{n+j}(z)) \leq 2 \sum_{j=0}^k 2^{-(n+j+1)}. \end{aligned}$$

Since U is complete, the space $\mathcal{C}(Y, U)$ is also complete. Thus the sequence s_n converges to a continuous function $s \in \mathcal{C}(Y, U)$. Since $s_n(z) \in S(z) + B_n$, we deduce that $s(z) \in \bar{S}(z) = S(z)$ for any z (since $S(z)$ is closed by assumption).

Remark. The theorem remains true if we replace the assumption “ Y is compact” by the weaker assumption “ Y is paracompact”.

We used the compactness assumption only to construct continuous partition of unity associated with a covering $\{N_{n+1}(y)\}$ of Y . But “paracompact” spaces are (by definition) the topological spaces in which we can associate a continuous partition of unity with such coverings. Metric spaces are paracompact.

Application: Surjectivity of lower semi-continuous correspondences.

As an application, we deduce the following surjectivity theorem from the above selection theorem (and Proposition 1.4).

Theorem 6. Suppose X is a convex compact subset of U . If U is a Fréchet space and S is lower semi-continuous from X into itself and satisfies either the condition

$$\forall p \in U^*, \quad S[\partial(X; p)] \subset \partial(x; p)$$

or the condition

$$\forall p \in U^*, \quad S[\partial(x; p)] \subset \partial(x; -p),$$

then S maps X onto itself.

15.3.4. Weak Hausdorff topology on the family of closed subsets of topological vector spaces

Since a correspondence S with non-empty closed values from Y into a locally convex Hausdorff space can be regarded as a map from Y into the family

$$(32) \quad \mathcal{F}(U) \text{ of non-empty closed subsets of } U,$$

the question arises as to whether it is possible to endow $\mathcal{F}(U)$ with a topology for which continuous maps are nothing other than continuous correspondences. We have already met such a topology in Section 13.3.7.

Suppose that U is supplied with the weak topology $\sigma(U, U^*)$, defined by the semi-norms

$$(33) \quad p_K(x) = \sup_{p \in K} |\langle p, x \rangle| = \sup_{\cos(K)} |\langle p, x \rangle|$$

when K ranges over the family $\mathcal{S}$ of finite subsets of U^* (where $\cos(K)$ denotes the symmetric convex hull of K).

We thus define on $\mathcal{F}(U)$ the "semi-distances"

$$(34) \quad d_K(A, B) = \max(\delta_K(A, B), \delta_K(B, A))$$

where we set

$$(35) \quad \delta_K(A, B) = \sup_{y \in A} \inf_{x \in B} p_K(y - x).$$

Proposition 2. *The functions $d_K(A, B)$ are semi-distances. The family $\mathcal{F}(U)$ of closed subsets of U supplied with the topology defined by the semi-distances $d_K(A, B)$ when K ranges over the finite subsets of U^* is a Hausdorff space.*

Definition 1. We shall say that this topology is the "weak-Hausdorff topology".

Proof. By the very definition, the functions $d_K(A, B)$ are symmetric. If A, B, C are three subsets, $\delta_K(A, B) \leq \delta_K(A, C) + \delta_K(C, B)$. To see this, let y belong to A , x to B and z to C . Since $p_K(y - x) \leq p_K(y - z) + p_K(z - x)$, we deduce that

$$\inf_{x \in B} p_K(y - x) \leq p_K(y - z) + \inf_{x \in B} p_K(z - x) \leq p_K(y - z) + \delta_K(C, B).$$

Taking the infimum over C , we deduce that

$$\inf_{x \in B} p_K(y - x) \leq \inf_{z \in C} p_K(y - z) + \delta_K(C, B) \leq \delta_K(A, C) + \delta_K(C, B).$$

Finally, taking the supremum over A , we deduce that $\delta_K(A, B) \leq \delta_K(A, C) + \delta_K(C, B)$. To prove that $\mathcal{F}(U)$ is a Hausdorff space, we have to prove that, if $\mathbf{d}_K(A, B) = 0$ for all K 's, then $A = B$.

Let $y \in A$. Since

$$\delta_K(A, B) = 0 \text{ for any } K,$$

then $\inf_{x \in B} \mathbf{p}_K(y - x) = 0$ for any K . This implies that y belongs to the closure of B . Then $A \subset \bar{B} = B$, since B is assumed to be closed. We show in the same way that $B \subset \bar{A} = A$. Thus $A = B$. $\square$

Remark. It is clear that the two following statements are equivalent:

$$(36) \quad \begin{cases} (a) A \subset B + B_K(\varepsilon), \\ (b) \forall y \in A, (y + B_K(\varepsilon)) \cap B \neq \emptyset, \end{cases}$$

where we set

$$(37) \quad B_K(\varepsilon) = \{x \in U \text{ such that } \mathbf{p}_K(x) \leq \varepsilon\}.$$

We deduce the following result.

Proposition 3. *If $A \subset B + B_K(\varepsilon)$, then $\delta_K(A, B) \leq \varepsilon$. Conversely, if $\delta_K(A, B) \leq \varepsilon$, then $A \subset B + B_K(2\varepsilon)$.*

Proof. This is left as an exercise. $\square$

Proposition 4. *Let S be a correspondence with non-empty closed values mapping a topological space Y into a locally convex Hausdorff space U . If S is upper semi-continuous at y_0 , then*

$$(38) \quad \text{For any finite set } K, \text{ for any } \varepsilon > 0, \text{ there exists a neighborhood } N(y_0) \text{ of } y_0 \text{ such that } \delta_K(S(y), S(y_0)) \leq \varepsilon \text{ whenever } y \in N(y_0).$$

The converse is true if $S(y_0)$ is compact.

Proof. If S is upper semi-continuous at y_0 , then, for any finite subset K and any $\varepsilon > 0$, there exists a neighborhood $N(y_0)$ of y_0 such that

$$(39) \quad \forall y \in N(y_0), \quad S(y) \subset S(y_0) + B_K(\varepsilon).$$

This implies that $\delta_K(S(y), S(y_0)) \leq \varepsilon$ whenever $y \in N(y_0)$.

Conversely, if $\delta_K(S(y), S(y_0)) \leq \frac{1}{2}\varepsilon$ whenever $y \in N(y_0)$, we deduce from Proposition 3 that (39) holds. Thus, S is upper semi-continuous at y_0 , since any neighborhood N of the compact set $S(y_0)$ contains a subset $S(y_0) + B_K(\varepsilon)$. $\square$

Proposition 5. *Let S be a correspondence with non-empty closed images mapping a topological space Y into a locally convex space U .*

If

- (40) *For any finite subset K , $\forall \varepsilon > 0$, there exists a neighborhood $N(y_0)$ of y_0 such that $\delta_K(S(y_0), S(y)) \ll \varepsilon$ when $y \in N(y_0)$,*

then

- (41) *S is lower semi-continuous at y_0 .*

The converse is true if we also assume that

- (42) *$S(y_0)$ is compact.*

Proof. If $\delta_K(S(y_0), S(y)) \ll \frac{1}{2}\varepsilon$ when $y \in N(y_0)$, then $S(y_0) \subset S(y) + B_K(\varepsilon)$ (by Proposition 4). Thus, $\forall x \in S(y_0)$, $(x + B_K(\varepsilon)) \cap S(y) = \emptyset$ when $y \in N(y_0)$. This implies that S is lower semi-continuous at y_0 .

Conversely, suppose that S is lower semi-continuous at y_0 and that $S(y_0)$ is compact. Then we can cover $S(y_0)$ with a finite number of open neighborhoods $x_i + B_K(\varepsilon)$ where $x_i \in S(y_0)$, i.e.

$$(43) \quad S(y_0) \subset \bigcup_{i=1}^n (x_i + B_K(\varepsilon)).$$

Since S is lower semi-continuous, there exist n neighborhoods $N_{x_i}(y_0)$ of y_0 such that

$$(44) \quad y \in N_{x_i}(y_0) \text{ implies that } S(y) \cap (x_i + B_K(\varepsilon)) \neq \emptyset.$$

Let $N(y_0) = \bigcap_{i=1}^n N_{x_i}(y_0)$. Then any $x \in S(y_0)$ belongs to a set $x_i + B_K(\varepsilon)$. Furthermore, for any $y \in N(y_0)$, we know that $x_i \in S(y) + B_K(\varepsilon)$. Thus $x \in x_i + B_K(\varepsilon) \subset S(y) + 2B_K(\varepsilon)$ when $y \in N(y_0)$. Therefore, $S(y_0) \subset S(y) + 2B_K(\varepsilon)$, i.e. $\delta_K(S(y_0), S(y)) \ll 2\varepsilon$ when $y \in N(y_0)$. $\square$

15.3.5. Relations between hemi-continuity and semi-continuity

Let $\mathcal{X}(U)$ denote the family of non-empty closed convex subsets. In this case, we have a characterization of the Hausdorff semidistances.

Proposition 6. *The following inequality is always true:*

$$(45) \quad \sup_{p \in K} [\sigma^*(A, p) - \sigma^*(B, p)] \ll \delta_K(A, B).$$

If B is convex, it becomes an identity, i.e.

$$(46) \quad \sup_{p \in K} [\sigma^*(A, p) - \sigma^*(B, p)] = \delta_K(A, B).$$

Proof. We can write

$$\begin{aligned} a &= \sup_{p \in K} [\sigma^*(A, p) - \sigma^*(B, p)] = \sup_{p \in K} \left[\sup_{y \in A} \langle p, y \rangle - \sup_{x \in B} \langle p, x \rangle \right] \\ &= \sup_{p \in K} \sup_{y \in A} \inf_{x \in B} \langle p, y - x \rangle. \end{aligned}$$

Since

$$(47) \quad \sup_{p \in K} \inf_{x \in B} \langle p, y - x \rangle \leq \inf_{x \in B} \sup_{p \in K} \langle p, y - x \rangle,$$

we deduce that

$$a \leq \sup_{y \in A} \inf_{x \in B} \sup_{p \in K} \langle p, y - x \rangle = \sup_{y \in A} \inf_{x \in B} p_K(y - x) = \delta_K(A, B).$$

If we assume that B is convex, then inequality (47) is actually an identity by the minisup Theorem 7.1.5, since we can replace K by its convex hull, which is compact.

In this case, we have

$$\delta_K(A, B) = \sup_{p \in K} [\sigma^*(A, p) - \sigma^*(B, p)]. \quad \square$$

This characterization of the semi-distances implies a characterization of upper and lower semi-continuous correspondences.

Theorem 7 (Castaing). *Suppose that S has non-empty closed convex images. It is upper hemi-continuous whenever it is upper semi-continuous. The converse is true whenever*

$$(48) \quad \text{the images } S(y) \text{ are compact.}$$

Proof. The first statement has already been proved (Proposition 2.5.1). We prove the converse in the case of finite dimensional space only.

Associate with any pair $\{p, \varepsilon\} \in U^* \times]0, \infty[$, the subset

$$(49) \quad F_{p, \varepsilon} = \{y \in U \text{ such that } \langle p, y \rangle \leq \sigma^*(S(x_0), p) + \varepsilon\}$$

and with any finite sequence J of pairs $\{p, \varepsilon\}$ the subset

$$(50) \quad F_J = \bigcap_{\{p, \varepsilon\} \in J} F_{p, \varepsilon}.$$

To say that S is upper hemi-continuous at x_0 amounts to saying that we can associate with any finite sequence J a neighborhood $N_J(x_0)$ such that

$$(51) \quad S(N_J(x_0)) \subset F_J.$$

Now, let N be any fixed open neighborhood of $S(x_0)$. By (51), it is enough to prove that for some J , $F_J \subset N$. Let $G = \hat{N}$ be the complement of N and $G_J = G \cap F_J$. Suppose that $\forall J$, F_J is not contained in N , i.e. that $\forall J$, $G_J \neq \emptyset$. The subsets G_J have the finite intersection property (since $G_I \cap G_J \supset G_{I \cup J}$). Furthermore, there exists a finite sequence J_0 for which F_{J_0} is compact. Therefore, there exists $y \in \bigcap_{G_J \subset G_{J_0}} G_J$. Since $y \in G$ and since $S(x_0) \subset N$, then $y \notin S(x_0)$, i.e. there exists $p \in U^*$ such that $\langle p, y \rangle > \sigma^*(S(x_0), p)$. This is a contradiction since $y \in F_J$ where $J = J_0 \cup \{p, \varepsilon\}$ with $\varepsilon < \langle p, y \rangle - \sigma^*(S(x_0), p)$. Hence

$$(52) \quad \text{there exists } J \text{ such that } S(N_J(x_0)) \subset F_J \subset N. \quad \square$$

Theorem 8 (Valadier). *Let S be a correspondence with non-empty convex compact images from Y into a locally convex space U . If*

$$(53) \quad S \text{ is lower semi-continuous at } y_0$$

then

$$(54) \quad \forall p \in U^*, \text{ the function } y \mapsto \sigma^*(S(y), p) \text{ is lower semi-continuous at } y_0.$$

The converse is true if we also assume that

$$(55) \quad \text{the images } S(y) \text{ stay in a compact subset } C.$$

Proof. If S is lower semi-continuous at y_0 into U supplied with the weak topology, we deduce from Propositions 6 and 7 that for any finite subset $K \subset U$ and any $\varepsilon > 0$,

$$(56) \quad \sup_{p \in K} (\sigma^*(S(y_0); p) - \sigma^*(S(y); p)) \leq \delta_K(S(y_0), S(y)) \leq \varepsilon$$

whenever y ranges over a neighborhood $N(y_0)$ of y_0 .

Conversely, Suppose that S is not lower semi-continuous at y_0 , i.e. that there exist $\varepsilon > 0$ and a neighborhood $x_0 + K^*$ (where K is the convex symmetric hull of a finite subset) of $x_0 \in S(y_0)$ such that for any neighborhood $N(y_0)$ of y_0 , there exists $y_N \in N(y_0)$ such that

$$(57) \quad \begin{aligned} \varepsilon &\leq \delta_K(S(y_0), S(y_N)) = \max_{p \in K} [\sigma^*(S(y_0); p) - \sigma^*(S(y_N); p)] \\ &= \sigma^*(S(y_0); p_N) - \sigma^*(S(y_N); p_N) \end{aligned}$$

the maximum is achieved at $p_N \in K$ since K is compact and the support functions $\sigma^*(S(y), p)$ are continuous with respect to p .

Since K is compact, a subsequence (also denoted by) p_N converges to $p^* \in K$. We shall prove that $y \mapsto \sigma^*(S(y); p^*)$ is not lower semi-continuous, i.e. that

$$(58) \quad \frac{\varepsilon}{2} \leq \sigma^*(S(y_0); p^*) - \sigma^*(S(y_N); p^*) \quad \text{for } N \text{ large enough.}$$

We have that,

$$\sigma^*(S(y_0); p_N) \leq \sigma^*(S(y_0); p_N - p^*) + \sigma^*(S(y_0); p^*).$$

Let C denote the compact set containing images $S(y)$. Then

$$\begin{aligned} \sigma^*(S(y_N); p^*) &\leq \sigma^*(S(y_N); p^* - p_N) + \sigma^*(S(y_N); p_N) \\ &\leq \sigma^*(C; p^* - p_N) + \sigma^*(S(y_N); p_N). \end{aligned}$$

Therefore,

$$(59) \quad \begin{aligned} \varepsilon &\leq \sigma^*(S(y_0); p_N) - \sigma^*(S(y_N); p_N) \leq \sigma^*(S(y_0); p^*) - \sigma^*(S(y_N); p^*) \\ &\quad + \sigma^*(S(y_0); p_N - p^*) + \sigma^*(C; p^* - p_N). \end{aligned}$$

Now, since K is an equicontinuous subset of U^* , the weak topology coincides on K with the topology of uniform convergence over the compact subsets. Therefore, since $S(y_0)$ and C are compact,

$$(60) \quad \sigma^*(S(y_0); p_N - p^*) + \sigma^*(C; p^* - p_N) \leq \frac{1}{2} \varepsilon \quad \text{for } p_N \text{ close enough to } p^*.$$

Hence (58) holds and the proposition is proved. $\square$

APPENDIX A

SUMMARY OF LINEAR FUNCTIONAL ANALYSIS

In this appendix we summarize the main results of linear functional analysis which are used in this book.

We recall (without proof) the statements of the results known as the Hahn–Banach theorems. We proceed by proving that if two spaces U and F are paired, then F is the dual of U and U the dual of F when they are supplied with the weak topologies $\sigma(U, F)$ and $\sigma(F, U)$. We use this result to prove that the space of discrete measures is the dual of the space of all functions. We then study the topologies of uniform convergence over subsets of a covering defined on spaces of functions and give sufficient conditions implying that the duals coincide.

We employ these results to construct stronger topologies on U and F , using the fact that U and F are spaces of continuous linear forms on F and U respectively. In particular, we prove the Mackey theorem stating that the duals of U supplied with the Mackey topology and the weak topology coincide.

Finally, we show that when U is a Hausdorff locally convex vector space and $F = U^*$, then U^* is also the dual of U supplied with the weak and the Mackey topology. Furthermore, we prove the Banach–Steinhaus theorem and deduce that the initial and the strong topologies coincide when U is barreled. Recall that we have proved the open mapping principle in Section 15.3 (see Theorem 15.3.2) and that other results of functional analysis are scattered throughout the book (the study of gauges and support functions in Section 1.4, polarity in Section 1.4, Hilbert spaces in Sections 2.2 and 2.3 and the continuity of convex functions in Sections 3.3 and 3.4).

1. Hahn–Banach theorems

Several connected results are known under the name “Hahn–Banach theorem”. We *recall* several of these statements.

(a) If U is a Hilbert space, it is easy to deduce from the projection theorem (see Proposition 2.2.1) the following result.

Theorem 1. *If K is a non-empty closed convex subset of a Hilbert space and if $x \notin K$, then there exists $p \in U^*$, $p \neq 0$, such that $\sigma^*(K, p) < \langle p, x \rangle$.*

(b) If $U = \mathbb{R}^n$ is a finite dimensional space. Theorem 1 implies the following separation theorem.

Theorem 2. *If K is a non-empty convex subset of a finite-dimensional space and if $x \notin K$, then there exists $p \in \mathbb{R}^{n*}$, $p \neq 0$ such that $\sigma^*(K, p) \leq \langle p, x \rangle$.*

(The proof of Theorem 2 uses the fact that the unit sphere is compact.)

Note that the minisup theorems of Chapters 7 and 13 used only Theorem 2 and not the more sophisticated results we are about to state.

(c) *Case of Hausdorff locally convex spaces.* The separation theorems follow from the following extension theorems.

Theorem 3. *Let U be a vector space and let p be a positively homogeneous convex function defined on U . Let M be a vector subspace of U and $f : M \rightarrow \mathbb{R}$ be a linear form satisfying $f(x) \leq p(x)$ for all $x \in M$. Then there exists a linear form $f : U \rightarrow \mathbb{R}$ defined on U which extends f and which satisfies $f(x) \leq p(x)$ for all $x \in U$.*

This result depends upon Zorn's lemma. Notice that we did not assume that U is endowed with a topology. When U is a Hausdorff locally convex space, Theorem 3 implies the following.

Theorem 4. *Let M be a vector subspace of a Hausdorff locally convex space U . Let f be a continuous linear form defined on M . Then there exists a continuous linear form defined on U which extends f .*

Theorem 3 also implies the following separation theorem.

Theorem 5. *Suppose that K is a non-empty open convex subset of a Hausdorff locally convex space U . If $x \notin K$, then there exists $p \in U^*$, $p \neq 0$, such that $\sigma^*(K; p) \leq \langle p, x \rangle$.*

Finally, Theorem 5 implies the following generalization of Theorem 1.

Theorem 1 bis. *Let K be a non-empty closed convex subset of a Hausdorff locally convex space U . If $x \notin K$, there exists $p \in U^*$ such that $\sigma^*(K; p) < \langle p, x \rangle$.*

2. Paired spaces

We begin by showing that the dual U^* is not $\{0\}$.

Proposition 1. *Let U be an Hausdorff locally convex vector space. Then there exist non-zero continuous linear forms. Furthermore, the bilinear form $\{p, x\} \in U^* \times U \rightarrow p(x)$ is not degenerate.*

Proof. Let $x_0 \neq 0$. Let $M = \mathbb{R}x_0$ be the one-dimensional vector space spanned by x_0 and let $f: M \rightarrow \mathbb{R}$ be the continuous linear form defined by $f(\lambda x_0) = \lambda$. Then Theorem 4 implies the existence of continuous linear form p on U which extends f and is not equal to 0 (since $p(x_0) = f(x_0) = 1$). Now, if $p(x) = 0$ for all x , then $p = 0$ by the very definition of a zero form. On the other hand, the equations $p(x_0) = 0$ for all $p \in U^*$ imply that $x_0 = 0$. (If not, if $x_0 \neq 0$, there would exist $p \in U^*$ such that $p(x_0) = 1 \neq 0$.) $\square$

Now, let F be a vector space isomorphic to V^* . Then there exists a bijective linear operator j from U^* onto F . Write $\langle p, x \rangle = j^{-1}(p)(x)$. Proposition 1 implies that

- (1) the bilinear form $\{p, x\} \in F \times U \rightarrow \langle p, x \rangle$ is not degenerate.

It is remarkable that (1) is also a sufficient condition for F to be isomorphic to U^* . To see this, associate with a bilinear form $\langle \cdot, \cdot \rangle$ on $F \times U$ the weak topologies $\sigma(F, U)$ and $\sigma(U, F)$ on F and U defined by the families of seminorms

$$(2) \quad p_M(p) = \sup_{x \in M} |\langle p, x \rangle|, \quad M \subset U$$

and

$$(3) \quad p_K(x) = \sup_{p \in K} |\langle p, x \rangle|, \quad K \subset F$$

obtained when M and K range over the families $\mathcal{S}$ of finite subsets of U and F .

Denote by F_σ and U_σ the vector spaces F and U supplied with the weak topologies $\sigma(F, U)$ and $\sigma(U, F)$.

Theorem 6. *Let $\{p, x\} \rightarrow \langle p, x \rangle$ be a non-degenerate bilinear form on the product $F \times U$ of two vector spaces. Then there exists a bijective linear operator j mapping F onto the dual $(U_\sigma)^*$ of U_σ and a bijective linear operator k mapping U onto the dual $(F_\sigma)^*$ of F_σ .*

Proof. It suffices to prove the first statement. We define j from F into $(U_\sigma)^*$ by

$$(4) \quad \forall p \in F, \quad jp: x \rightarrow jp(x) = \langle p, x \rangle.$$

It is clear that j is linear. It is injective because, if $j(p) = 0$, then $jp(x) = \langle p, x \rangle = 0$ for all $x \in U$. Thus, $p = 0$ since the duality pairing is non-degen-

erate. It remains to prove that j is *surjective*. Let $f: x \rightarrow f(x)$ be a continuous linear form on V . We prove that it can be written $f(x) = \langle p, x \rangle$ where $p \in F$. Since f is continuous, there exists a semi-norm $\mathbf{p}_K$ associated with a finite subset $K = \{p_1, \dots, p_n\} \subset F$ such that

$$|f(x)| \leq c \mathbf{p}_K(x) \quad \text{where } c > 0.$$

Therefore the equalities $\langle p_i, x \rangle = 0$ imply that $f(x) = 0$. Introduce the linear operator L from U onto $\mathbf{R}^n$ defined by $Lx = \{\langle p_i, x \rangle\}_{i=1, \dots, n}$, and its image $T = L(U) \subset \mathbf{R}^n$. We define the linear form g on T by $g(t) = f(x)$ whenever $t = Lx$. It is clear that $g(t)$ does not depend upon the choice of the solution x of $Lx = t$, since $f(x) = 0$ whenever $Lx = 0$. Since $T \subset \mathbf{R}^n$, g can be written $g(t) = \sum_{i=1}^n \lambda^i t_i$. Therefore, we have proved that

$$f(x) = g(Lx) = \sum_{i=1}^n \lambda^i \langle p_i, x \rangle = \langle p, x \rangle$$

where $p = \sum_{i=1}^n \lambda^i p_i$. $\square$

This result allows us to prove the following theorem.

Theorem 7. *The space $\mathcal{S}^*(X)$ of discrete measures on a set X is (isomorphic to) the dual of the space $\mathcal{S}(X) = \mathbf{R}^X$ of functions defined on X , supplied with the topology of pointwise convergence, defined by the family of semi-norms*

$$(5) \quad \mathbf{p}_L(f) = \sup_{x \in L} |f(x)|$$

obtained when L ranges over the family $\mathcal{S}$ of finite subsets of X .

Proof. It is clear that the bilinear form defined by $\langle m, f \rangle = \sum_{i=1}^n \alpha^i f(x_i)$ is non-degenerate. On the other hand, it is easy to check that the semi-norms $\mathbf{p}_K$ (see (5)) of the topology of pointwise convergence and the semi-norms $\mathbf{p}_K$ (see (3)) of the weak topology $\sigma(\mathcal{S}(X), \mathcal{S}^*(X))$ define the same topology. Hence $\mathcal{S}^*(X)$ is isomorphic to $\mathcal{S}(X)^*$ by Theorem 6. $\square$

3. Topologies of uniform convergence

More generally, let $\mathcal{A}$ be a *covering* of the set X satisfying

(6) if K and L belong to $\mathcal{A}$, then $K \cup L$ belongs to $\mathcal{A}$.

We denote by $\mathcal{A}(X)$ the *vector space of functions* defined on X which are *bounded on every subset L in the covering $\mathcal{A}$* . We can then define the semi-norms $\mathbf{p}_L(\cdot)$ by (5) as L ranges over $\mathcal{A}$. We shall supply $\mathcal{A}(X)$ with the family

of semi-norms $\mathbf{p}_L$ with L ranging over $\mathcal{A}$. This topology is called the “*topology of uniform convergence over the subsets of $\mathcal{A}$* ”. The space $\mathcal{A}(X)$ is complete.

Let $\mathcal{U}$ be the family of all subsets of X . Then $\mathcal{U}(X)$ is the Banach space of bounded functions on X . Clearly, we have $\mathcal{U}(X) \subset \mathcal{A}(X) \subset \mathcal{S}(X)$, the canonical injections being continuous. Since $\mathcal{U}(X)$ is dense in $\mathcal{S}(X)$, any space $\mathcal{A}(X)$ is dense in $\mathcal{S}(X)$.

Since the canonical injection j from $\mathcal{A}(X)$ into $\mathcal{S}(X)$ has a dense image and is injective, its transpose j^* is injective and has a dense image. We will identify j^* with the canonical injection from $\mathcal{S}^*(X)$ into $\mathcal{A}^*(X)$.

Now, let $\mathcal{H}(X) \subset \mathcal{A}(X)$ be a subspace of functions. We shall denote by $\mathcal{H}_\alpha(X)$ this space supplied with the topology induced by $\mathcal{A}$, called the *topology of uniform convergence over the subsets of $\mathcal{A}$* . In particular, any space $\mathcal{H}(X)$ of functions on X can be supplied with the topology of pointwise convergence. It will be denoted by $\mathcal{H}_o(X)$. The transpose $j^* \in \mathcal{L}(\mathcal{H}_o^*(X), \mathcal{H}_\alpha^*(X))$ of the identity map j (which is continuous from $\mathcal{H}_\alpha(X)$ into $\mathcal{H}_o(X)$), is injective with a dense image. Therefore, we will identify $\mathcal{H}_o^*(X)$ with a dense subspace of $\mathcal{H}_\alpha^*(X)$. The question arises as to whether the duals $\mathcal{H}_o^*(X)$ and $\mathcal{H}_\alpha^*(X)$ coincide. For this purpose, we denote by $\Psi : X \rightarrow \mathcal{H}_\alpha^*(X)$ the map defined by

$$(6) \quad \Psi(x) : f \in \mathcal{H}(X) \rightarrow \langle \Psi(x), f \rangle = f(x).$$

[If $\mathcal{H}(X) = \mathcal{A}(X)$, Ψ is the Dirac operator. Otherwise, $\Psi(x)$ is the restriction of the Dirac measure $\delta(x)$ to $\mathcal{H}(x)$.]

Theorem 8. Let $\mathcal{H}(X)$ be a subspace of $\mathcal{A}(X)$. Suppose that

$$(7) \quad \forall L \in \mathcal{A}, \quad \Psi(L) \text{ is a compact subset of } \mathcal{H}_o^*(X)$$

when $\mathcal{H}_o^*(X)$ is supplied with the weak-topology $\sigma(\mathcal{H}_o^*(X), \mathcal{H}_o(X))$. Then $\mathcal{H}_o^*(X) = \mathcal{H}_\alpha^*(X)$.

Proof. We have to prove that any $\mu_0 \in \mathcal{H}_\alpha^*(X)$ belongs to $\mathcal{H}_o^*(X)$. Since μ_0 is continuous on $\mathcal{H}_\alpha(X)$, there exists a semi-norm $\mathbf{p}_L$ such that

$$(8) \quad |\langle \mu_0, f \rangle| \leq \mathbf{p}_L(f) \text{ for any } f \in \mathcal{H}_\alpha(X).$$

But we can write, by (6),

$$(9) \quad \begin{aligned} \mathbf{p}_L(f) &= \sup_{x \in L} |f(x)| = \sup_{x \in L} |\langle \Psi(x), f \rangle| \\ &= \sup_{\mu \in \overline{\cos \Psi(L)}} |\langle \mu, f \rangle| \end{aligned}$$

where $\overline{\cos \Psi(L)}$ is the closed convex symmetric hull of $\Psi(L)$ in $\mathcal{H}_o^*(X)$. Since $\Psi(L)$ is compact by assumption, $\overline{\cos \Psi(L)}$ is compact on $\mathcal{H}_o^*(X)$ and thus, is also a compact subset of $\mathcal{H}_\alpha^*(X)$.

Now, inequalities (8) and (9) show that if $\sup_{\mu \in \overline{\cos \Psi(L)}} |\langle \mu, f \rangle| \leq 1$ (i.e. if $f \in [\overline{\cos \Psi(L)}]^*$), then $\langle \mu_0, f \rangle \leq 1$. Therefore $\mu_0 \in [\overline{\cos \Psi(L)}]^{**}$. Since $\overline{\cos \Psi(L)}$ contains 0 and is convex and closed in $\mathcal{H}_\alpha^*(X)$ (because $\overline{\cos \Psi(x)}$ is compact in $\mathcal{H}_\alpha^*(X)$), we deduce that $[\overline{\cos \Psi(L)}]^{**} = \overline{\cos \Psi(L)}$. Hence $\mu_0 \in \overline{\cos \Psi(L)} \subset \mathcal{H}_\alpha^*(X)$. $\square$

Corollary 1. *Let X be a topological space. Let $\mathcal{C}_x(X)$ and $\mathcal{C}_o(X)$ be the spaces of continuous functions on X supplied with the topology of uniform convergence over compact subsets and the topology of pointwise convergence respectively. Then $\mathcal{C}_x^*(X) = \mathcal{C}_o^*(X)$.*

Proof. It is clear that the operator Ψ from X into $\mathcal{C}_o^*(X)$ is continuous (since, if x_μ converges to x , then, for any $f \in \mathcal{C}(X)$, $\langle \Psi(x_\mu), f \rangle = f(x_\mu)$ converges to $\langle \Psi(x), f \rangle = f(x)$, for f is continuous). Then $\Psi(L)$ is compact for any compact subset L of X . Assumption (5) is satisfied when we take the covering $\mathcal{X}$ of X by all the compact subsets. $\square$

Corollary 2. *Let X be a compact space. Let $\mathcal{C}_u(X)$ and $\mathcal{C}_o(X)$ be the spaces of continuous functions supplied with the topology of uniform convergence and the topology of pointwise convergence respectively. Then $\mathcal{C}_u^*(X) = \mathcal{C}_o^*(X)$.*

4. Topologies associated with a duality pairing

Consider two paired spaces U and F . By Theorem 6, we can identify U with the space $\mathcal{L}(F)$ of continuous linear forms on F supplied with the weak topology $\sigma(F, U)$. Therefore, since any continuous linear form is bounded over bounded subsets, $U = \mathcal{L}(F) \subset \mathcal{A}(F)$ whenever $\mathcal{A}$ is a covering formed by bounded subsets of F_σ . We shall denote by $\alpha(U, F)$ the topology induced by $\mathcal{A}(F)$. It is given by the family of semi-norms $\mathbf{p}_K$ defined by (3) as K ranges over $\mathcal{A}$. We distinguish in particular the following two cases.

(a) Case $\mathcal{A} = \mathcal{B}$ is the family of all bounded subsets of F_σ . The topology $\beta(U, F)$ is called the *strong topology* on U .

(b) Case $\mathcal{A} = \mathcal{C}$ is the family of compact subsets of F_σ . The topology $\tau(U, F)$ is called the *Mackey topology* on U .

We will denote by $U_\beta = \mathcal{L}_\beta(F)$ and $U_\tau = \mathcal{L}_\tau(F)$ the space U supplied with the topologies $\beta(U, F)$ and $\tau(U, F)$ respectively. Theorem 8 implies the following.

Theorem 9 (Mackey). *Let U and F be two paired spaces. Then $F = (U_o)^*$ is also equal to $(U_\tau)^*$, i.e. the dual of U supplied with the Mackey topology.*

Proof. We take $X = F$ and we consider the spaces $U_\sigma = \mathcal{L}_\sigma(F)$ and $U_\tau = \mathcal{L}_\tau(F)$. The map Ψ from F into $\mathcal{L}_\sigma^*(F)$ is the identity because, by Theorem 6, $\mathcal{L}_\sigma^*(F) = (U_\sigma)^*$ is equal to F and $\Psi(x) : p \in F \rightarrow \langle \Psi(x), p \rangle = \langle p, x \rangle$ is nothing other than the continuous linear form $x \in U = (F_\sigma)^*$ (always by Theorem 6). Thus assumption (7) of Theorem 8 is satisfied when $\mathcal{A} = \mathcal{C}$ is the covering of F_σ by compact subsets. $\square$

In particular, *convex subsets that are closed for the Mackey topology remain closed for the weak topology.*

The question arises as to whether the topology of an Hausdorff locally convex space U can be defined in this way and what its properties might be.

If $\mathbf{p}$ is a semi-norm of the initial topology, denote by $B_{\mathbf{p}} = \{x \in U \text{ such that } p(x) \leq 1\}$ its associated unit ball.

Then

$$\mathbf{p}(x) = \pi(B_{\mathbf{p}}; x) = \sigma^*(B_{\mathbf{p}}^*; x).$$

Hence, we can write

$$\mathbf{p}(x) = \mathbf{p}_{B_{\mathbf{p}}^*}(x) = \sup_{p \in B_{\mathbf{p}}^*} |\langle p, x \rangle|.$$

If we denote by $\mathcal{C}$ the covering of the dual U^* of U by the subsets $\lambda B_{\mathbf{p}}^*$ when $\mathbf{p}$ ranges over the family of semi-norms and $\lambda \in \mathbf{R}_+$, we see that the *initial topology is nothing other than the topology $\varepsilon(U, U^*)$* . We leave the proof of the following lemma as an exercise.

Lemma 1. *A closed convex symmetric subset of U^* is equicontinuous if and only if it is equal to $\lambda B_{\mathbf{p}}^*$ for some semi-norm and some $\lambda > 0$.*

Then the initial topology is the topology $\varepsilon(U, U^*)$ associated with the covering $\mathcal{C}$ of U^* by the equicontinuous subsets of linear forms on U . We check that *equicontinuous subsets are relatively weakly compact*. Indeed, if $K \subset U^*$ is an equicontinuous subset of linear forms, its closure $\bar{K}$ in $\mathcal{S}(U)$ is also an equicontinuous set of linear forms (and thus, $\bar{K} \subset U^*$). Furthermore, $\bar{K}$ is bounded. Since $\mathcal{S}(U) = \mathbf{R}^U$, the Tychonov compactness theorem shows that $\bar{K}$ is compact.

We have the inclusions $\mathcal{S} \subset \mathcal{C} \subset \mathcal{T} \subset \mathcal{B}$. This means that *the injections $U_\beta \rightarrow U_\tau \rightarrow U = U_\varepsilon \rightarrow U_\sigma$ are continuous*. The Mackey Theorem 9 implies that $U_\sigma^* = U^* = U_\tau^* \subset U_\beta^*$.

Remark. Since U is a topological vector subspace of $\mathcal{L}(U^*)$, which is complete, the closure of U in $\mathcal{L}(U^*)$ supplies a completion of U . $\square$

The following question remains to be answered. Under what conditions does the initial topology $\varepsilon(U, U^*)$ coincide with the strong topology?

5. The Banach–Steinhaus theorem

This amounts to giving sufficient conditions for a bounded subset of continuous linear forms to be equicontinuous. The following result (sometimes called the Banach–Steinhaus theorem) answers this question.

Theorem 10. *Let U be a barreled space and let V be a Hausdorff locally convex space. Then any pointwise bounded set of continuous linear operators from U into V is equicontinuous.*

Proof. Let $\mathcal{H} \subset \mathcal{L}(U, V)$ be a pointwise bounded set of linear operators B . This implies that, for any semi-norm q defining the topology of V , we have that $\pi(x) = \sup_{B \in \mathcal{H}} q(Bx) < +\infty$. Therefore, the function $x \rightarrow \pi(x)$ is clearly a lower semi-continuous semi-norm. Since U is barreled, this implies that π is actually continuous (by definition of a barreled space). Then there exists a semi-norm p on U and $c > 0$ such that

$$\pi(x) = \sup_{B \in \mathcal{H}} q(Bx) \leq cp(x)$$

This means that $\mathcal{H}$ is equicontinuous. $\square$

Corollary 3. *Let U be a barreled space. Then the initial topology, the strong topology and the Mackey topology coincide, i.e. $U_\beta = U_\tau = U_\epsilon$.*

Remark. Recall that any Fréchet space is barreled, because by Proposition 3.3.4, any lower semi-continuous semi-norm is continuous.

In particular, we mention the following consequence which we have often used.

Corollary 4. *Let U be a barreled space. Any (weakly) bounded subset of U^* is weakly compact.*

A bounded subset of a locally convex space U is not necessarily weakly compact. If $U = L^1(-1, +1)$, we saw that the bounded set of functions $x \rightarrow nf(nx)$ (where $f \in L^1(-1, +1)$) is bounded but that it is not weakly compact.

The result is true if and only if U is *reflexive* (i.e., barreled and satisfying $U = (U^*)^*$).

APPENDIX B

THE KNASTER-KURATOWSKI-MAZURKIEWICZ LEMMA

We devote this appendix to the proof of the Knaster-Kuratowski-Mazurkiewicz lemma, which we used to prove the Ky-Fan inequality (see Theorems 7.1.2 and 7.1.3) and, consequently, the Brouwer and Kakutani fixed-point theorems (see Theorems 9.3.5 and 9.3.4). This lemma will be deduced from the Sperner lemma, which is of interest in itself. For this purpose, we break up a simplex into a sequence of "barycentric subdivisions" which yield coverings of the initial simplex by simplexes whose diameters converge to 0. By studying the structure of these interlocking simplexes, we will prove the Sperner lemma.

For the sake of completeness, we shall deduce the Brouwer fixed point theorem from the Knaster-Kuratowski-Mazurkiewicz lemma. Recall that the Brouwer theorem implies the Ky-Fan inequality (using the Lasry Theorem 7.1.4; see remark ending Section 9.3).

1. Barycentric subdivision of simplexes

Consider $(n+1)$ -affinely independent points $x^0, x^1, \dots, x^n$ of a vector space. We shall say that $\text{co}(x^0, x^1, \dots, x^n)$ is a n -simplex and that $\text{co}(x^{j_0}, x^{j_1}, \dots, x^{j_k})$ is a k -dimensional face of the simplex.

Notice that there are $n+1$ faces of dimension $n-1$ of a n -simplex.

We shall construct the *barycentric subdivision* of an n -simplex into $(n+1)!$ smaller n -simplexes by induction. The construction involves taking the barycenter

$$y^n = \frac{1}{n+1} (x^0 + x^1 + \dots + x^n)$$

of the simplex as the common vertex for all new smaller n -simplexes of the subdivision. Thus, barycentric subdivision is carried out by induction from simplexes of lower dimensions to those of higher dimensions.

For $n = 1$, the 1-simplex $\text{co}\{x^0, x^1\}$ is subdivided into the two 1-simplexes $\text{co}\{x^0, y^1\}$ and $\text{co}\{y^1, x^1\}$ where $y^1 = \frac{1}{2}(x^0 + x^1)$ is its barycenter.

Suppose that we can perform the *barycentric subdivision* of a k -simplex into $(k+1)!$ simplexes of dimension k for any $k < n$ and let us prove that this property holds for $k = n$.

Consider the $n+1$ faces $S_i^{(n-1)}$ of dimension $n-1$ of the n -simplex $\text{co}\{x^0, x^1, \dots, x^n\}$. They can be subdivided in $n! = (n-1+1)!$ simplexes $T_{i,j}^{(n-1)}$ of dimension $n-1$ by assumption. There are $(n+1)! = (n+1)n!$ such $(n-1)$ -simplexes $T_{i,j}^{(n-1)}$. Put

$$(1) \quad T_{i,j}^{(n)} = \text{co}(T_{i,j}^{(n-1)} \cup y^{(n)})$$

where

$$y^n = \frac{1}{n+1} (x^0 + x^1 + \dots + x^n)$$

is the barycenter. It is clear that it is a n -simplex. Since any point of the n -simplex which is neither the barycenter y^n nor lies in any proper faces belongs to a segment joining the barycenter y^n to some simplex $T_{i,j}^{(n-1)}$, we have covered $\text{co}(x^0, \dots, x^n)$ by $(n+1)!$ simplexes of dimension n having the barycenter y^n as common vertex. For $n = 2$, we obtain the $6 = (2+1)!$ following simplexes.

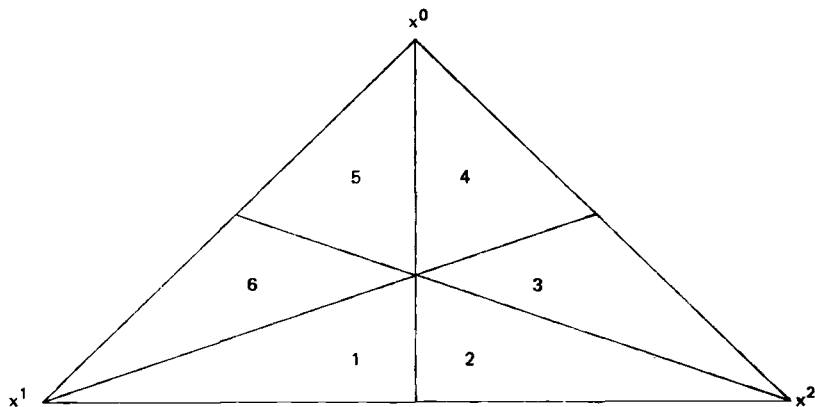


Fig. 17.

By iteration of formula (1), we see that any n -simplex $T^{(n)}$ of the barycentric subdivision can be written in the form

$$(2) \quad T^{(n)} = \text{co}(y^0, y^1, \dots, y^n)$$

where, $y^0 = x^{i_0}$, $y^1 = \frac{1}{2}(x^{i_0} + x^{i_1})$, ...,

$$(3) \quad y^k = \frac{1}{k+1} (x^{i_0} + x^{i_1} + \dots + x^{i_k}), \dots,$$

$$y^n = \frac{1}{n+1} (x^{i_0} + x^{i_1} + \dots + x^{i_n})$$

for a convenient permutation $\{i_0, i_1, \dots, i_n\}$ of the $(n+1)$ -indexes.

In other words, any n -simplex of the barycentric subdivision is the convex hull of the barycenters of a strictly increasing sequence $\{x^{j_0}\}, \{x^{j_0}, x^{j_1}\}, \dots, \{x^{j_0}, x^{j_1}, \dots, x^{j_k}\}, \dots, \{x^{j_0}, \dots, x^{j_n}\}$ of subsets of $\{x^0, \dots, x^n\}$.

Now, let $T^{(n)} = \text{co}(y^0, y^1, \dots, y^n)$ be any n -simplex of the barycentric subdivision and

$$(4) \quad T^{(n-1)} = \text{co}(y^{j_0}, y^{j_1}, \dots, y^{j_{n-1}})$$

be any $(n-1)$ -face of $T^{(n)}$.

Then we can write either

$$(5) \quad T^{(n-1)} = \text{co}(y^0, y^1, \dots, y^{n-1})$$

or

$$(6) \quad T^{(n-1)} = \text{co}(y^0, \dots, y^{k-1}, y^{k+1}, \dots, y^n)$$

with $0 \leq k \leq n-1$.

In the first case, we see that $T^{(n-1)}$ is the face of the only simplex $T^{(n)} = \text{co}(y^0, y^1, \dots, y^{n-1}, y^n)$. In the second case, we have the following situation: y^{k-1} is the barycenter of $\{x^0, \dots, x^{j_{k-1}}\}$ and y^{k+1} is the barycenter of $\{x^0, \dots, x^{j_{k-1}}, x^{j_k}, x^{j_{k+1}}\}$. Let $\hat{y}^k$ be the barycenter of $\{x^0, \dots, x^{j_{k-1}}, x^{j_k}\}$ and $\check{y}^k$ be the barycenter of $\{x^0, \dots, x^{j_{k-1}}, x^{j_{k+1}}\}$. Then

$$(7) \quad \hat{T}^{(n)} = \text{co}(y^j, \dots, y^{k-1}, \hat{y}^k, \dots, y^n)$$

and

$$(8) \quad \check{T}^{(n)} = \text{co}(y^0, \dots, y^{k-1}, \check{y}^k, \dots, y^n)$$

are the only two n -simplexes of the subdivision having $T^{(n-1)}$ as a face.

Now, if we notice that case (5) happens if and only if $T^{(n-1)}$ lies in some $(n-1)$ -face of the initial n -simplex $\text{co}(x^0, \dots, x^n)$, we have proved the following result.

Proposition 1. *Any $(n-1)$ -face $T^{(n-1)}$ of any n -simplex of the barycentric subdivision of $\text{co}(x^0, \dots, x^n)$ is*

(i) *either a face of exactly one n -simplex of the subdivision if $T^{(n-1)}$ lies on a $(n-1)$ -face of $\text{co}(x^0, x^1, \dots, x^n)$,*

(ii) *or a common face of exactly two n -simplex of the subdivision otherwise.*

Example. If $n = 2$,

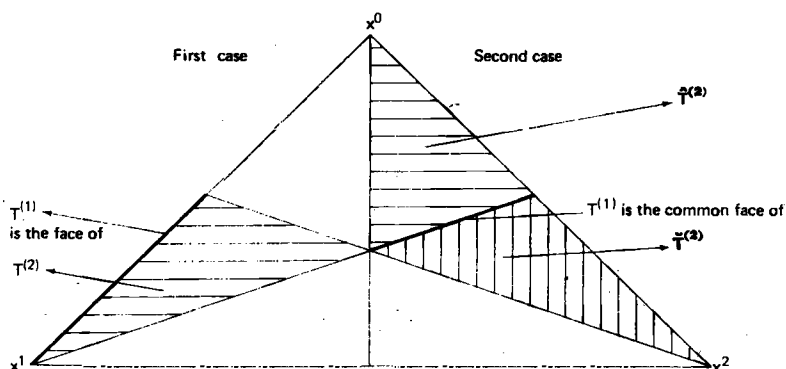


Fig. 18.

2. Sequence of barycentric subdivisions

By performing the barycentric subdivision for each of the $(n+1)!$ simplexes of dimension n of a n -simplex $\text{co}(x^0, \dots, x^n)$, we obtain a covering by smaller n -simplexes. By iterating this process ν -times, we obtain the so-called ν^{th} -barycentric subdivision of the simplex. Let us state and prove the counterpart of Proposition 1.

Proposition 2. Let $T_{\nu}^{(n-1)}$ be any $(n-1)$ -face of any n -simplex of the ν^{th} -barycentric subdivision of $\text{co}(x^0, x^1, \dots, x^n)$. Then it is

- (i) either a face of exactly one n -simplex of the ν^{th} -subdivision if $T_{\nu}^{(n-1)}$ lies in a $(n-1)$ -face of $\text{co}(x^0, x^1, \dots, x^n)$,
- (ii) Or a common face of exactly two n -simplexes of the ν^{th} -subdivision otherwise.

Proof. We proceed by induction of ν . Proposition 1 proves the case where $\nu = 1$. Suppose that Proposition 2 holds for $\nu - 1$.

(a) *First case.* The simplex $T_{\nu}^{(n-1)}$ has been obtained by subdividing a $(n-1)$ -simplex $T_{\nu-1}^{(n-1)}$ of the $(\nu-1)^{\text{th}}$ subdivision. Since $T_{\nu}^{(n-1)}$ lies in a $(n-1)$ -face of $\text{co}(x^0, \dots, x^n)$, $T_{\nu-1}^{(n-1)}$ also lies in such a face. Then, by the induction assumption $T_{\nu-1}^{(n-1)}$ is the face of a unique n -simplex $T_{\nu-1}^{(n)}$. Moreover, $T_{\nu}^{(n-1)}$ is a $(n-1)$ -simplex of the barycentric subdivision of $T_{\nu-1}^{(n)}$, and it lies on its $(n-1)$ -face $T_{\nu-1}^{(n-1)}$.

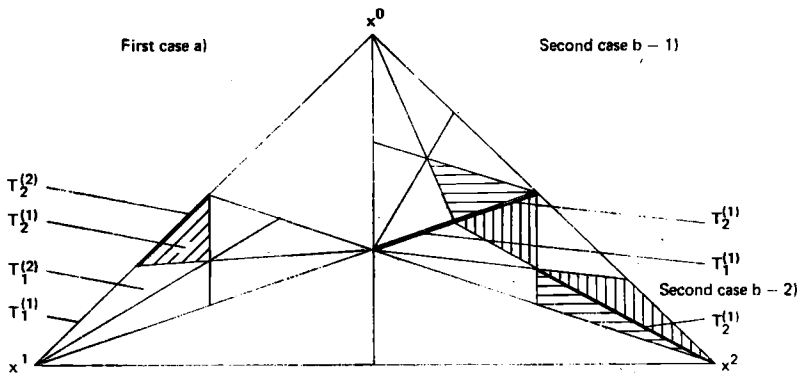


Fig. 19.

Then Proposition 1 implies that $T_v^{(n-1)}$ is a face of exactly one n -simplex $T_v^{(n)}$ of the barycentric subdivision of $T_{v-1}^{(n)}$. On the other hand, if the subdivision of any n -simplex $S_{v-1}^{(n)}$ entails $T_v^{(n-1)}$, then $S_{v-1}^{(n)}$ must have $T_v^{(n-1)}$ as a face and thus, it equals $T_{v-1}^{(n)}$. Therefore, $T_v^{(n-1)}$ is a face of exactly $T_v^{(n)}$.

(b) *Second case.*

(b-1) Firstly, assume that $T_v^{(n-1)}$ has been obtained by subdividing $T_{v-1}^{(n-1)}$ which does not lie on a face of $\text{co}(x^0, \dots, x^n)$. Then $T_v^{(n-1)}$ can only be a face of a n -simplex $T_v^{(n)}$ of the barycentric subdivision of a n -simplex $T_{v-1}^{(n)}$ having $T_{v-1}^{(n-1)}$ as a face. By the induction assumption, *there are exactly two such n -simplexes $\check{T}_{v-1}^{(n)}$ and $\hat{T}_{v-1}^{(n)}$* . The first part of Proposition 1 implies that each of these simplexes, when subdivided, entails *exactly one n -simplex $\check{T}_v^{(n)}$ (and $\hat{T}_v^{(n)}$)* having $T_v^{(n-1)}$ as a face.

(b-2) Assume that $T_v^{(n-1)}$ cannot be obtained by subdividing any $(n-1)$ -simplex of the $(v-1)$ th subdivision. Let $T_{v-1}^{(n)}$ be the unique n -simplex whose subdivision entails $T_v^{(n-1)}$. Then $T_v^{(n-1)}$ cannot be on any $(n-1)$ face of $T_{v-1}^{(n)}$.

Proposition 1 implies that $T_v^{(n-1)}$ is a common face of exactly two n -simplexes of the subdivision of $T_{v-1}^{(n)}$ (which then belong to the v th subdivision). Any other n -simplex of the $(v-1)$ th subdivision, apart from $T_{v-1}^{(n)}$, does not include $T_v^{(n-1)}$ and thus, never yields a n -simplex having $T_v^{(n-1)}$ as a face in the subdivision. $\square$

3. The Sperner lemma

Theorem 1 (Sperner's lemma). *Let $\mathcal{U}_v$ be the v th barycentric subdivision of a n -simplex $S = \text{co}(x^0, x^1, \dots, x^n)$. Suppose that σ is an application mapping each*

vertex $y_v = \sum_{i=0}^n \lambda_v^i x^i$ of a n -simplex $T_v^{(n)}$ of $\mathcal{T}_v$, onto a vertex $x^i = \sigma y_v$, such that

$$(9) \quad \text{the index } i \text{ satisfies } \lambda_v^i > 0.$$

Then there exists a n -simplex $T_v^{(n)} = \text{co}(y_v^0, \dots, y_v^n)$ of $\mathcal{T}_v$, such that

$$(10) \quad \forall i = 1, \dots, n, \quad \sigma y_v^i = x^i.$$

Proof. Let $\mathcal{T}_v$ be the family of the n -simplexes of the v^{th} barycentric subdivision and $\mathcal{F}_v$ be the sub-family of simplexes $T_v^{(n)} = \text{co}(y_v^0, y_v^1, \dots, y_v^n)$ satisfying

$$(11) \quad \sigma\{y_v^0, y_v^1, \dots, y_v^n\} = \{x^0, x^1, \dots, x^n\}.$$

We shall prove by induction that the number q_n of elements of $\mathcal{F}_v$ is *odd* (and thus, that there exists at least one simplex $T_v^{(n)}$ satisfying (11)). For $n = 0$, this is obviously true. Suppose that the result holds for $n-1$. We prove it true for n .

For this purpose, we shall say that an $(n-1)$ -face of a simplex $T_v^{(n)}$ of $\mathcal{T}_v$ is *regular* if its vertices have $x^0, x^1, \dots, x^{n-1}$ as images. Let us denote by $T_1, T_2, \dots, T_m$ the n -simplexes of the v^{th} -subdivision $\mathcal{T}_v$ of $\text{co}(x^0, x^1, \dots, x^n)$ and τ_i the number of regular faces of T_i . We shall determine the sum $\tau_1 + \tau_2 + \dots + \tau_m$ in two different ways.

(a) *the sum $\tau_1 + \tau_2 + \dots + \tau_m$ is odd.* Indeed let q_{n-1} be the number of $(n-1)$ -simplexes $T_v^{(n-1)} = \text{co}(y_v^0, \dots, y_v^{n-1})$ contained in $\text{co}(x^0, x^1, \dots, x^{n-1})$ and which satisfy $\sigma\{y_v^0, y_v^1, \dots, y_v^{n-1}\} = \{x^0, x^1, \dots, x^{n-1}\}$. By the induction hypothesis, this number q_{n-1} is odd.

Now, we compute the number of $(n-1)$ -simplexes which are regular faces $T_v^{(n-1)}$ of n -simplexes of $\mathcal{T}_v$. The following possibilities can occur.

(α) $T_v^{(n-1)}$ is contained in $\text{co}(x^0, x^1, \dots, x^{n-1})$. If $\sigma\{y_v^0, y_v^1, \dots, y_v^{n-1}\} = \{x^0, x^1, \dots, x^{n-1}\}$, it is the regular face of exactly one n -simplex $T_v^{(n)}$ by the first statement of Proposition 2. Then, there are q_{n-1} such simplexes, by the very definition of q_{n-1} . If $\sigma\{y_v^0, y_v^1, \dots, y_v^{n-1}\} \neq \{x^0, x^1, \dots, x^{n-1}\}$ it is not a regular face.

(β) $T_v^{(n-1)}$ is contained in a $(n-1)$ -face of $\text{co}(x^0, x^1, \dots, x^n)$, but not in the face $\text{co}(x^0, x^1, \dots, x^{n-1})$. In this case, $T_v^{(n-1)}$ cannot be the regular face of a n -simplex $T_v^{(n)}$ because of assumption (9).

(γ) $T_v^{(n-1)}$ is not contained in any $(n-1)$ -face of $\text{co}(x^0, x^1, \dots, x^n)$. In this case, the second statement of Proposition 2 implies that $T_v^{(n-1)}$ is the regular face of exactly two n -simplexes of $\mathcal{T}_v$ (which have $T_v^{(n-1)}$ as a common face).

Therefore,

$$(12) \quad \tau_1 + \tau_2 + \dots + \tau_m = q_{n-1} + 2k \text{ is odd.}$$

(b) *Computation of q_n .* We prove now that

$$(13) \quad \tau_1 + \tau_2 + \dots + \tau_m = q_n + 2l.$$

If T_i belongs to $\mathcal{S}_v$, it obviously has only one regular face ($\tau_i = 1$). There are q_n such simplexes. If $T_i = \text{co}(y_v^0, y_v^1, \dots, y_v^n)$ does not belong to $\mathcal{S}_v$, then either $\{y_v^0, \dots, y_v^n\}$ does not contain $\{x^0, x^1, \dots, x^{n-1}\}$ and thus, has no regular face ($\tau_i = 0$) or $\{y_v^0, \dots, y_v^n\}$ does contain $\{x^0, x^1, \dots, x^{n-1}\}$. In this case, $\tau_i = 2$. To see this, suppose that T_i does not belong to $\mathcal{S}_v$ and that $T_i = \text{co}\{y_v^0, \dots, y_v^n\}$ is such that $\sigma y_v^0 = x^0, \sigma y_v^1 = x^1, \dots, \sigma y_v^{n-1} = x^{n-1}$. Then we have $\sigma y_v^n \neq x^n$ (because $T_i \notin \mathcal{S}_v$). If, for example, $\sigma y_v^n = x^0$, then T_i has two regular faces: $\text{co}(y_v^0, y_v^1, \dots, y_v^{n-1})$ and $\text{co}(y_v^n, y_v^1, \dots, y_v^{n-1})$.

It follows from these considerations that (13) holds. Hence, we deduce from (12) and (13) that

$$(14) \quad q_n = q_{n-1} + 2(k-l) \text{ is odd. } \square$$

4. The Knaster-Kuratowski-Mazurkiewicz lemma

We state this result.

Theorem 2 (Knaster-Kuratowski-Mazurkiewicz lemma). *Consider $n+1$ closed subsets F_i of an-simplex $\text{co}\{x^0, x^1, \dots, x^n\}$ such that any face $\text{co}\{x_{i_0}, x_{i_1}, \dots, x_{i_i}\}$ is contained in the union $F_{i_0} \cup F_{i_1} \cup \dots \cup F_{i_i}$. Then the intersection $\bigcap_{i=0}^n F_i \neq \emptyset$.*

To prove this, we use the Sperner lemma to construct n simplexes $T_v^{(n)} = \text{co}(y_v^0, \dots, y_v^n)$ of the v^{th} -barycentric subdivision such that $y_v^i \in F_i$ for all i . Intuition suggests as v increases, the n -simplexes of the v^{th} -subdivision will decrease in such a way that the sequences y_v^i will all converge to the same point y_* which will necessarily belong to the intersection $\bigcap_{i=1}^n F_i$. We begin by proving the following proposition.

Proposition 3. *Let $T_v^{(n)}$ be any n -simplex of the v^{th} subdivision of $S = \text{co}(x^0, x^1, \dots, x^n)$. Then*

$$(15) \quad \text{diameter}(T_v^{(n)}) \leq \left(\frac{n}{n+1}\right)^v \text{diameter}(S).$$

Proof. This inequality follows by iteration from the case when $\nu = 1$. This case will be proved by induction on n . The case $n = 1$ is obvious. Suppose that it is true for $n-1$. We deduce that it holds for n . Let $T = T_1^{(n)}$ be the convex hull of $y^0, y^1, \dots, y^n$ where $y^0 = x^0, y^1 = \frac{1}{2}(x^0 + x^1), \dots, y^{(k)} = (1/(k+1))(x^0 + \dots + x^k), \dots$. We must prove that

$$(16) \quad \text{diameter}(T) \leq \frac{n}{n+1} \text{diameter}(S).$$

We have that

$$\begin{aligned} \|x^j - y^n\| &= \left\| \frac{1}{n+1} \sum_{i \neq j} (x^i - x^j) \right\| \\ &\leq \frac{1}{n+1} \sum_{i \neq j} \|x^i - x^j\| \leq \frac{n}{n+1} \text{diameter}(S). \end{aligned}$$

Therefore, for any $x = \sum_{j=0}^n \lambda^j x^j \in S$,

$$\|x - y^n\| = \left\| \sum_{j=0}^n \lambda^j (x^j - y^n) \right\| \leq \frac{n}{n+1} \text{diameter}(S).$$

In particular, taking $x = y^j$, we deduce that

$$(17) \quad \max_{0 \leq j \leq n-1} \|y^j - y^n\| \leq \frac{n}{n+1} \text{diameter}(S).$$

On the other hand, by the induction assumption,

$$\begin{aligned} (18) \quad \max_{0 \leq i, j \leq n-1} \|y^{(i)} - y^{(j)}\| &\leq \frac{n-1}{n} \text{diameter}(R) \leq \\ &\leq \frac{n-1}{n} \text{diameter}(S) \leq \frac{n}{n+1} \text{diameter}(S) \end{aligned}$$

where $R = \text{co}(x^0, \dots, x^{n-1})$. Then (17) and (18) imply that

$$(19) \quad \max_{0 \leq i, j \leq n-1} \|y^{(i)} - y^{(j)}\| \leq \frac{n}{n+1} \text{diameter}(S).$$

It remains to prove that

$$(20) \quad \text{diameter}(T) \leq \max_{0 \leq i, j \leq n} \|y^{(i)} - y^{(j)}\|.$$

(Since (16) follows from (19) and (20).) For this purpose, take $a = \sum_{i=0}^n \alpha_i y^i$ and $b = \sum_{j=0}^n \beta_j y^j$ belonging to T . We have

$$\begin{aligned} \|a-b\| &= \left\| \sum_{i=0}^n \sum_{j=0}^n \alpha_i \beta_j y^i - \sum_{i=0}^n \sum_{j=0}^n \alpha_i \beta_j y^j \right\| \\ &\quad \left(\text{because } \sum_{i=0}^n \alpha_i = \sum_{j=0}^n \beta_j = 1 \right) \\ &= \left\| \sum_{i,j=0}^n \alpha_i \beta_j (y^i - y^j) \right\| \leq \sum_{i,j=0}^n \alpha_i \beta_j \|y^i - y^j\| \\ &\leq \max_{0 \leq i, j \leq n} \|y^i - y^j\| \sum_{i,j=0}^n \alpha_i \beta_j = \max_{0 \leq i, j \leq n} \|y^i - y^j\| \end{aligned}$$

(because $\sum_{i,j=0}^n \alpha_i \beta_j = 1$). $\square$

Proof of Theorem 2. Let $\mathcal{C}_v$ be the v^{th} -barycentric subdivision of the simplex $S = \text{co}(x^0, x^1, \dots, x^n)$. We construct the map σ as follows. If y_v is the vertex of a n -simplex $T_v^{(n)}$ of $\mathcal{C}_v$, we have

$$(21) \quad y_v = \sum_{i=0}^n \lambda_i^v x^i = \sum_{k=0}^s \lambda_{i_k}^v x^{i_k}$$

where $\{i_0, \dots, i_s\}$ is the set of indexes j such that $\lambda_j^v > 0$. Then $\text{co}(x^{i_0}, x^{i_1}, \dots, x^{i_s})$ is contained in $F_{i_0} \cup \dots \cup F_{i_s}$ by assumption. Thus y_v belongs to at least one closed subset F_i where $i \in \{i_0, \dots, i_s\}$.

Write

$$(22) \quad \sigma y_v = x^i \quad \text{for such an index } i.$$

Therefore, the assumption of the Sperner lemma is satisfied. Hence there exists at least a simplex $T_v^{(n)} = \text{co}\{y_v^0, \dots, y_v^n\}$ such that $\sigma y_v^i = x^i$. By the very construction of σ , this means that $y_v^i \in F_i$ for all $i = 0, \dots, n$. Now, by Proposition 3, we have that

$$(23) \quad \max_{0 \leq i, j \leq n} \|y_v^i - y_v^j\| \leq \left(\frac{n}{n+1} \right)^v \text{diameter}(S).$$

Since S is compact, we can extract subsequences (again denoted by) y_v^i which converge to the same element y_* (by (23)). Since $y_* = \lim y_v^i$ and F_i being closed, $y_* \in F_i$ for all i . $\square$

We state the following corollary.

Corollary 1. Let $F_i (i = 0, \dots, n)$ be n closed subsets covering an n -simplex $S = \text{co} \{x^0, x^1, \dots, x^n\}$ such that, for all i , F_i contains x^j and does not meet the face $T_i = \text{co} \{x^0, x^1, \dots, x^{j-1}, x^{j+1}, \dots, x^n\}$. Then the F_i have a non-empty intersection.

Proof. If $R = \text{co} \{x^{i_0}, \dots, x^{i_s}\}$ is a face of S and if $i \neq i_0, i_1, \dots, i_s$, we have $R \subset T_i$. Since $F_i \cap T_i = \emptyset$, we deduce that $F_i \cap R = \emptyset$.

This is true for all $i \neq i_0, \dots, i_s$, and so

$$R = \text{co} \{x^{i_0}, \dots, x^{i_s}\} \subset F_{i_0} \cup \dots \cup F_{i_s}.$$

Therefore $\bigcap_{i=0}^n F_i \neq \emptyset$ by Theorem 2. $\square$

5. The Brouwer theorem

We can deduce the Brouwer theorem from the Knaster-Kuratowski-Mazurkiewicz lemma.

Theorem 3 (Brouwer). Any continuous map f from S into S has a fixed point.

Proof. Denote by x_i and $f_i(x)$ the barycentric coordinates of x and $f(x) \in S$. Then $f_i(x) \geq 0$ for all i and $\sum_{i=0}^n f_i(x) = 1$. We define the $(n+1)$ subsets

$$(24) \quad F_i = \{x \in S \text{ such that } x_i \geq f_i(x)\}$$

They are closed since f is continuous. Let $T = \text{co} \{x^{i_0}, \dots, x^{i_s}\}$ be any face of S . Then any $x \in T$ satisfies $x_{i_0} + \dots + x_{i_s} = 1$. If x were not contained in any of $F_{i_0}, F_{i_1}, \dots, F_{i_s}$, we would have $w_{i_k} < f_{i_k}(x)$ for all i_k . Adding these inequalities, we would obtain the contradiction

$$1 = \sum_{k=0}^s x_{i_k} < \sum_{k=0}^s f_{i_k}(x) \leq \sum_{i=0}^n f_i(x) = 1.$$

Thus the assumptions of the Knaster-Kuratowski-Mazurkiewicz lemma are satisfied and so there exists $\bar{x} \in \bigcap_{i=0}^n F_i$, i.e. $\bar{x}$ satisfying $f_i(\bar{x}) \leq \bar{x}_i$ for all i . We cannot have $\bar{x}_i > f_i(\bar{x})$ for at least one i since such an inequality yields the contradiction $1 < 1$. Hence $\bar{x}_i = f_i(\bar{x})$ for all i , i.e. $\bar{x}$ is a fixed point of f . $\square$

Corollary 2. If we assume that

$$(25) \quad X \text{ is homeomorphic to an } n\text{-simplex } S,$$

any continuous map f mapping X into itself has a fixed point.

Proof. If $\varphi : X \rightarrow S$ is the homeomorphism, the map $g = \varphi f \varphi^{-1}$ is a continuous application from S into itself, which has a fixed point $\bar{y}$. Since $\bar{y} = \varphi f \varphi^{-1}(\bar{y})$, we see that $\bar{x} = \varphi^{-1}(\bar{y})$ is a fixed point of f . $\square$

In particular, one can prove that convex compact subsets of finite-dimensional vector spaces are homeomorphic to simplexes. This implies the existence of a fixed point of continuous maps from a convex compact subset of $\mathbb{R}^n$ into itself.

APPENDIX C

LYAPUNOV'S THEOREM ON THE RANGE OF A VECTOR VALUED MEASURE

Recall the statement of Lyapunov's theorem.

Theorem. Let $\mathcal{A}$ be a σ -algebra on a set Ω and μ_i ($i = 1, \dots, n$) be n non-atomic measures. Define

$$\mu(A) = \{\mu_1(A), \dots, \mu_n(A)\} \in \mathbf{R}^n.$$

Then $\mu(\mathcal{A})$ is a convex compact subset of $\mathbf{R}^n$.

Proof. We associate with μ the continuous linear operator $L \in \mathcal{L}(L^\infty(\Omega), \mathbf{R}^n)$ defined by

$$L\varphi = \left\{ \int \varphi \, d\mu_1, \dots, \int \varphi \, d\mu_n \right\}$$

where we set $L^\infty(\Omega) = L^\infty(\Omega, \mathcal{A}, d\sigma)$ with $\sigma = |\mu_1| + \dots + |\mu_n|$. Each μ_i is absolutely continuous with respect to σ . The Radon-Nikodym theorem implies that there exist n functions $f_i \in L^1(\Omega)$ such that $d\mu_i = f_i \, d\sigma$. Then L can be written in the form

$$L\varphi = \left\{ \int_{\Omega} \varphi f_1 \, d\sigma, \dots, \int_{\Omega} \varphi f_n \, d\sigma \right\}.$$

Consider the set

$$\mathcal{T} = \{\varphi \in L^\infty(\Omega) \text{ such that } \varphi(\omega) \in [0, 1] \text{ a.e.}\}.$$

(This is the set of fuzzy coalitions: see Section 10.4.4.)

We have seen that $\mathcal{T}$ is convex and weakly compact. Thus $L(\mathcal{T})$ is a convex compact subset of $\mathbf{R}^n$ (since L is obviously continuous on L^∞ supplied with the weak topology $\sigma(L^\infty, L^1)$).

We shall conclude by proving that $\mu(\mathcal{A}) = L(\mathcal{T})$. Since $\mu(\mathcal{A})$ is obviously contained in $L(\mathcal{T})$, it remains to prove that $L(\mathcal{T}) \subset \mu(\mathcal{A})$. Let $x \in L(\mathcal{T})$ and define

$$\mathcal{T}_x = \{\tau \in \mathcal{T} \text{ such that } L\tau = x\}$$

which is a weakly closed convex subset of $\mathcal{C}$. Hence $\mathcal{C}_x$ is weakly compact. By the Krein–Milman theorem (see Proposition 3.4.9) there exists an extremal point τ_0 . If we prove that any extremal point τ_0 of $\mathcal{C}_x$ is a characteristic function, we thus deduce that any $u \in L(\mathcal{C})$ is the image $\mu(\mathcal{A}) = L\tau_A$ of a characteristic function.

Therefore, suppose that τ_0 belongs to $\mathcal{C}_x$ and is not a characteristic function. Then there exists $\theta \in]0, \frac{1}{2}[$ and $A \in \mathcal{A}$ (such that $\sigma(A) > 0$) satisfying $\theta \leq \tau_0(\omega) \leq 1 - \theta$ for almost all $\omega \in A$.

Consider the subspace

$$V = \{\tau_A \varphi\}_{\varphi \in L^\infty(\Omega)}$$

Since $\sigma(A) > 0$ and σ is non-atomic, the dimension of V is greater than n . Therefore, the kernel of the restriction $L|_V$ of L to V is non-zero. Thus there exists $\varphi \in V$ such that $L\varphi = 0$. We can choose such a φ satisfying $-\theta \leq \varphi(\omega) \leq \theta$ for almost all ω . We see that $\tau_0 = \frac{1}{2}(\tau_0 - \varphi) + \frac{1}{2}(\tau_0 + \varphi)$ and that both $\tau_0 - \varphi$ and $\tau_0 + \varphi$ belong to $\mathcal{C}$ since $0 \leq \tau_0(\omega) \pm \varphi(\omega) \leq 1$ a.e.) and satisfy $L(\tau_0 - \varphi) = L(\tau_0 + \varphi) = x$. Hence $\tau_0 - \varphi$ and $\tau_0 + \varphi$ both belong to $\mathcal{C}_x$. This is a contradiction. $\square$

COMMENTS

Chapter 1

The results of this chapter must be regarded as classical. Material on convexity in finite or infinite dimensional vector spaces can be found in almost any text on functional analysis (e.g. Treves, Dunford and Schwartz, Schwartz and Yosida).

The foundations of the general theory of convex sets and functions is mainly due to Minkowski, who introduced the idea of characterizing closed convex sets by gauge functions and support functions. In the case of infinite dimensional spaces, see Hormander. For other results on the separation of convex sets, we refer to the papers of Klee. The introduction of indicators is due to Fenchel.

Chapter 2

The results of Sections 2.1 and 2.2 are classical. The general conjugacy correspondence defined in Section 2.4 was discovered by Fenchel. A comprehensive study of the consequences of the conjugacy theorem can be found in Rockafellar [6] in finite dimensional spaces and in Moreau [1] in infinite dimensional spaces. We refer to these books for further comments. Finally, the results of Section 2.5 are taken from the book of Berge [1].

Chapter 3

The results of Section 3.1 are classical, as well as the definition of proper maps (see Bourbaki [2]). The results of Section 3.2.2 are due to Arrow and Debreu. The results on continuity of convex functions must be considered as classical. The fact that a compact convex subset is the convex hull of its extremal points was first proved by Minkowski and generalized by Krein and Milman for infinite dimensional spaces. See also the papers of Aggeri [1] and Bronstedt [2].

Chapter 4

The explicit development of the theory of subdifferentiability is relatively recent. The reader should refer to Moreau [1] and Rockafellar [6] for a general review of the literature.

For further generalizations, see the papers of Clarke who defines generalized gradients for locally Lipschitz functions. For a review, see Aubin [13]. The results on differentiability and variational inequalities are classical. The relationship between the Legendre transform and the conjugate function was noted by Fenchel. See also the papers of Ioffe, and Ekeland [7].

The relationship between subdifferentiability and differentiability from the right is due to Moreau. The differentiability properties of a pointwise supremum are due to Valadier. Section 4.4 is essentially the papers of Ekeland [3] and Ekeland and Lebourg. The order relation used in the proof is a device due to Bishop and Phelps and used by Bronstedt [3] and Browder in different settings. See the papers of Edelstein, Asplund, Asplund and Rockafellar, and Lebourg for further applications. A review of these applications appears in Ekeland [8].

Chapter 5

The duality theory for constrained minimization problems is historically based on the famous paper of Kuhn and Tucker, who used the differential calculus approach. Slater seems to have been the first to use a "constraint qualification assumption" of the type $0 \in \text{Int}(L(R) - Y)$. Theorem 5.3.1 is due to Fenchel and Proposition 5.3.4 to Uzawa (see Arrow et al.).

The marginal properties of Lagrange multipliers (Proposition 5.1.4) was first discovered by Gale in the context of economics. See also the book of Karlin.

An extensive literature has been devoted to duality theory. The most general results seem to be the theorems of Clarke [8] and Aubin-Clarke [2].

Chapter 6

We note only that the main concepts of game theory we present in this chapter were defined early by economists. The model of duopoly was introduced in 1838 by Cournot. Edgeworth, in 1881, considered the two-person game defined

in Section 6.4. He assumed that the players could make any trade they wished. He defined the core (called the contract curve) and showed that it was non-empty.

Earlier, in 1874, Walras defined the general equilibrium concept of an economic system made up of households and firms. Edgeworth recognized that the Walras equilibrium did belong to the core. In 1909, Pareto defined an “optimal resource allocation”.

For the study of stability of two-person games, see Rockafellar [6], and McLinden [1].

Chapter 7

The first fundamental theorem of game theory was the minimax theorem of Von Neumann in 1928. He deduced it from the Brouwer theorem. Later, in 1941, Kakutani presented a simplified version of Von Neumann’s theorem and proved his famous fixed point theorem.

In 1958, Sion [1], [2] proved Theorem 7.1.7. Actually, only separation theorems are needed to prove the existence of the minisup, as was recognized by Nikaido [2], [3], Moreau [1] and Rockafellar [6].

Theorem 7.1.4 is due to Lasry (unpublished) and is used in Lasry–Robert for the study of topological degree for correspondences. It is also used to prove that the Brouwer theorem implies the Ky-Fan inequality, proved in Ky-Fan [12]. We deduce it from Theorem 7.1.2, due to Aubin [8]. See also the papers of Koenig, and Simons.

The next two sections on extensions of a game are a summary of the thesis of Moulin [3], to which we refer for further comments.

The theory of repeated games, introduced by Aumann and Mashler, was continued by Mertens, and Zamir. For games with imperfect information, see for instance, Ponsard, and Ponsard and Zamir. We refer to a forth-coming book of Mertens and Zamir.

Chapter 8

The study of the Walras model can now be considered as classical. See the fundamental book of Debreu [1], in 1959. The first proof of the existence of an equilibrium in a series of alternative models are due to Wald (1933) and later, by Arrow–Debreu (1954) and McKenzie [1] (1954), Nikaido [5]. The most general version is in Debreu [1].

We also refer to the books of Nikaido [5], Ekeland [4], [9] and Arrow and Hahn and, in particular, to Chapter 1 of the latter. Notice that we have left aside the problem of utility theory. We have chosen to use loss functions (minus utility functions) instead of preference orderings. For the relationship between these two concepts, we refer to Debreu [1]. See also Kannai [6] for the characterization of convex loss functions.

A comprehensive study in these models in the framework of quadratic optimization with explicit computation of solutions is made in Aubin [16].

We did not develop in this book further extensions of the Walrasian model such as finiteness properties Debreu [1] and uniqueness properties Balasko [1]. Stability is studied in Arrow and Hahn, for further developments see Balasko [3].

The global requirement of uniqueness has revealed itself to be excessively strong and was replaced by that of local uniqueness and the stability of the set of equilibria. For such an approach, using differential topology, we refer to the papers of Balasko [2], [3], Dierker, and Smale [1]. See the papers of Fuchs for the study of structural stability.

Broader interpretations and further extensions of the Walras model have been proposed. They include infinite-dimensional commodity spaces (T. Bewley [1], [2]), public goods (Milleron, Champsaur [2], [2], [4], Foley [1], [2], Rosenthal), fiscality (Guesnerie [1]), uncertainty (Arrow [4], Debreu [1], Radner, Dreze [1], [5], Guesnerie and De Montbrial, Malinvaud [3], Stigum, Starett, etc.), transaction costs (Hahn [3], Kurz, Starr [1], [2]), money (Grandmont [1], [2], Grandmont-Laroque [1], Starr, [2]). Time dependent economies are studied in Grandmont [1], [2], Green [1], De Montbrial [1]. Recently, the study of micro-economic foundations of macroeconomies is initiated by Malinvaud and Younes. For labour-managed and participatory economies, see Dreze [6].

"Barter curves" of exchanges economies and planification models can be studied in the framework of differential inclusions. See Antosiewicz and Cellina, Aubin [11], [14], Aubin, Cellina and Nohel, Aubin and Clarke [1], Aubin and Day, Champsaur, Dreze and Henry, Cornet, Dreze and de la Vallée and Poussin, Malinvaud [4], Smale [3], [4], etc.

Chapter 9

Theorem 9.1.1 on the existence of a non-cooperative equilibrium is due to Nash [1]. Another proof using the associated variational inequalities was given by Rosen. The existence of solutions of variational inequalities for quadratic functionals was proved by Stampacchia. The proof given here is due to Lions

and Stampacchia. The concept of Walras–Cournot equilibrium was introduced by Gabsewicz and Vial.

Theorem 9.3.3 can be found in Arrow and Debreu. The fixed point theorems are due to Brouwer and Kakutani. Section 9.4 is the first part of the fundamental paper of Arrow and Debreu.

Chapter 10

For the extension of the results of Section 10.1 to infinite dimensional spaces, we refer to Kurz Majumbar [1], [2], Radner [5]. See also Peleg [2], [3], [4] for properties of Pareto optima, Guesnerie [2] for the non-convex case. The selection procedure of Section 10.2.7 is due to Nash [2], it was improved by Harsanyi [1], [2]. For the concept of “second best”, we refer to Guesnerie [1].

The concept of the core was introduced by Gillies [1], [2]. We refer to Aumann [6] for a survey. See also the papers of Billera [1], [2], Scarf [5], Shapley [9], [10], [11], [12], [13], [14] etc. In the case of indivisibility, we refer to Henry [1] and Shapley and Scarf. The core of an economy with public goods is studied in Champsaur et al. The concept of fuzzy set is due to Zadeh.

Other concepts of solutions of games have been devised. For instance, the concept of “Von Neumann–Morgenstern solution” of a game with side-payments (see Von Neumann–Morgenstern) and the concept of bargaining set, introduced in Aumann–Maschler [1] and extended to games without side-payments by Assher [1], [2].

The use of Debreu–Scarf coalitions, i.e., replicated economies or games, was introduced by Debreu and Scarf in the case of economies.

Proposition 10.4.1 is due to Castaing [1]. The idea of using a continuum of players is due to Aumann [2]. An extensive study of economies and games with continuum of players lies in the books of Hildenbrand [6] and of Aumann and Shapley. Another approach uses non-standard analysis (see Brown and Robinson [1], [2] and Ali-Kahn). The idea of canonical cooperative equilibrium can be found in Shapley [4] (λ -transfer-value) and Baudier.

The concept of nucleolus was introduced by Schmeidler [1].

Fuzzy games were introduced by Aubin [9], [10], but the idea was dormant in former papers (Ekeland [4], for instance). Further generalizations of fuzzy games can be applied in sociology (see Aubin, Louis-Guérin and Zavalloni): each player is described by a vector of $\mathbb{R}^k$ (behavioral profile) and fuzzy coalitions are replaced by “groups of participation”, described by matrices whose entries belong to $[0, 1]$.

Chapter 11

An extensive literature is devoted to games with side-payments. We refer to the book of Owen [3] for general comments. For market games and relationships between economies and games with side-payments, see Shapley and Shubik [1], [4], [5] and Billera [3]. The concept of balanced games was introduced by Bondareva [1], [2] for games with side-payments and by Scarf [2] for games without side-payments. Theorem 11.2.2 is due to Scarf [5]. The concept of Shapley-value was introduced in Shapley [2]. The results of Section 11.3 are analogous to the results of the book of Aumann and Shapley dealing with non-atomic games. The idea of fuzzy-game with side-payments also lies in Owen [5], who introduced the linear extension operator (and in E. Baudier, personal communication). See Cornet [3] for the study of his extension operator.

Chapter 12

The equivalence Theorem 12.3.1 between the fuzzy core and the set of equilibria of an economy is analogous to the theorems of Debreu and Scarf, Aumann [2], and Hildenbrand [6]. We refer to the book of Hildenbrand [6].

The proof of non-emptiness of the fuzzy core (see Aubin [5]) implies the theorem of Scarf [2]. Analogous ideas can be found in Shapley [3] and Baudier.

Chapter 13

Theorem 13.1.1 is related to the minimax theorems of Rockafellar (see [6]). Theorem 13.2.1 was stated in a slightly more general form in Brezis et al.

The concept of monotonicity was introduced by Minty [1] and Zarantonello. The concept of pseudo-monotonicity is due to Brezis [1]. Several versions of Theorem 13.2.2 are due to Browder, Lions, Stampacchia, etc. (See the books of Lions [3], Brézis [3], Browder [3], Pascali for further comments.) The paper of Brøndstedt [4] is closely related to the use of γ -convex functions, introduced in Aubin [3]. See also Aggeri [2]. Other generalization of convexity can be found in the papers of Avriel, and Berge [1].

Chapter 14

Other proofs of Theorems 14.1.1 and 14.1.2 can be found in Rockafellar [11]. See also the book of Laurent and the papers of Joly and Laurent. The

characterization of the dual of L^∞ is due to Ioffe and Levin. The example 10.1.3. of “augmented Lagrangian” is due to Rockafellar [13]. See also the paper of Goldstein [1], [2] and Goldstein and Tretiakov [1], [2], Berbekas. The presentation of duality theory via the perturbations of the minimization problem is due to Rockafellar (see his book [6] and [11] for instance).

The Shapley–Folkman theorem appeared first in a paper of Starr [1]. See also Appendix B of the book of Arrow and Hahn. The proof we give in Section 14.2.3 is due to Ekeland. See also another proof in Arstein [5]. The Shapley–Folkman theorem was used by Hildebrand et al. to prove the convergence of the core to the set of equilibria. The results of Section 14.2 can be found in Ekeland [6] and Aubin–Ekeland. Theorem 14.2.4 implies the results of Aumann and Perles (see also Arkin and Levin).

An extensive literature deals with the calculus of variations and control theory. We mention in particular the books of Ekeland and Temam, Lions [4] Ioffe and Ticomirov [4], Young [4].

For the maximum principle in control theory, we refer to the book of Pontryagin. For applications of control theory to management problems, see the book of Bensoussan et al.

For control problems with infinite horizon, see Halkin [3] and Aubin and Clarke [3]. The latter paper deals also with constraints on the state and locally Lipschitz cost functionals.

Problems of calculus of variations involving convex Hamiltonians (instead of concave-convex Hamiltonians as in this book) are studied in Aubin and Ekeland [2].

We chose an approach which follows closely the papers of Rockafellar [8], [9], [14] and Berliocchi and Lasry [1], [2], [3] using the abstract Green formula proved by Aubin [1] (see [2]). For the non-convex case, see the papers of Clarke [1], [3], [5], [6], [7] and Aubin and Clarke [2]. The dynamic programming approach was introduced by Bellman in the framework of control problems. It was used by Bensoussan and Lions for stopping time problems and impulsive controls. We refer to forthcoming books of Bensoussan and Lions for further comments.

Chapter 15

The redaction of Section 15.1 is due to Cornet, who used a version of a theorem of Browder [2] and Ky Fan [12] to prove several surjectivity and fixed point theorems.

Corollaries 15.1.3 and 15.1.4 are due to Cornet, Proposition 15.1.2 to

Cornet and Lasry. It generalizes a classical existence theorem in Mathematical Economics, see McKenzie [1] and Debreu [1]. The concepts of inward and outward correspondences were introduced by Browder [1], [2] the version used here is due to Ky-Fan [12]; the concept of strongly inward correspondence is due to Cornet. Theorem 15.1.4 is due to Rogalski for outward correspondences, to Cornet [4] for strongly inward correspondences, Theorem 15.1.3 is due to Browder [2] and Ky-Fan [2]. Theorem 15.1.6 is a version of a theorem due to Granas: it extends the Leray-Schauder theorem to correspondences. Other fixed point theorems for dissipative maps are found in Aubin and Siegel and Ekeland [8]. The selection theorem 15.3.5 was proved by Michael [1], [2], [3]. See other selection theorems in Cellina and Antosiewicz related to the study of multivalued differential equations.

We refer to Lasry and Robert for the study of the topological degree for correspondences. For other references on non-linear functional analysis we refer to the books of Cronin, Lions [3], Krasnoselski [1], [2], Pascali, J.T. Schwartz [1], Vainberg.

Theorem 15.2.1 is analogous to the results of Joly and Mosco. The results of Sections 15.2.2 and 15.2.3 are due to Tartar [1], [3].

The results of Section 15.3.1 are due to S. Robinson [3] and others.

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